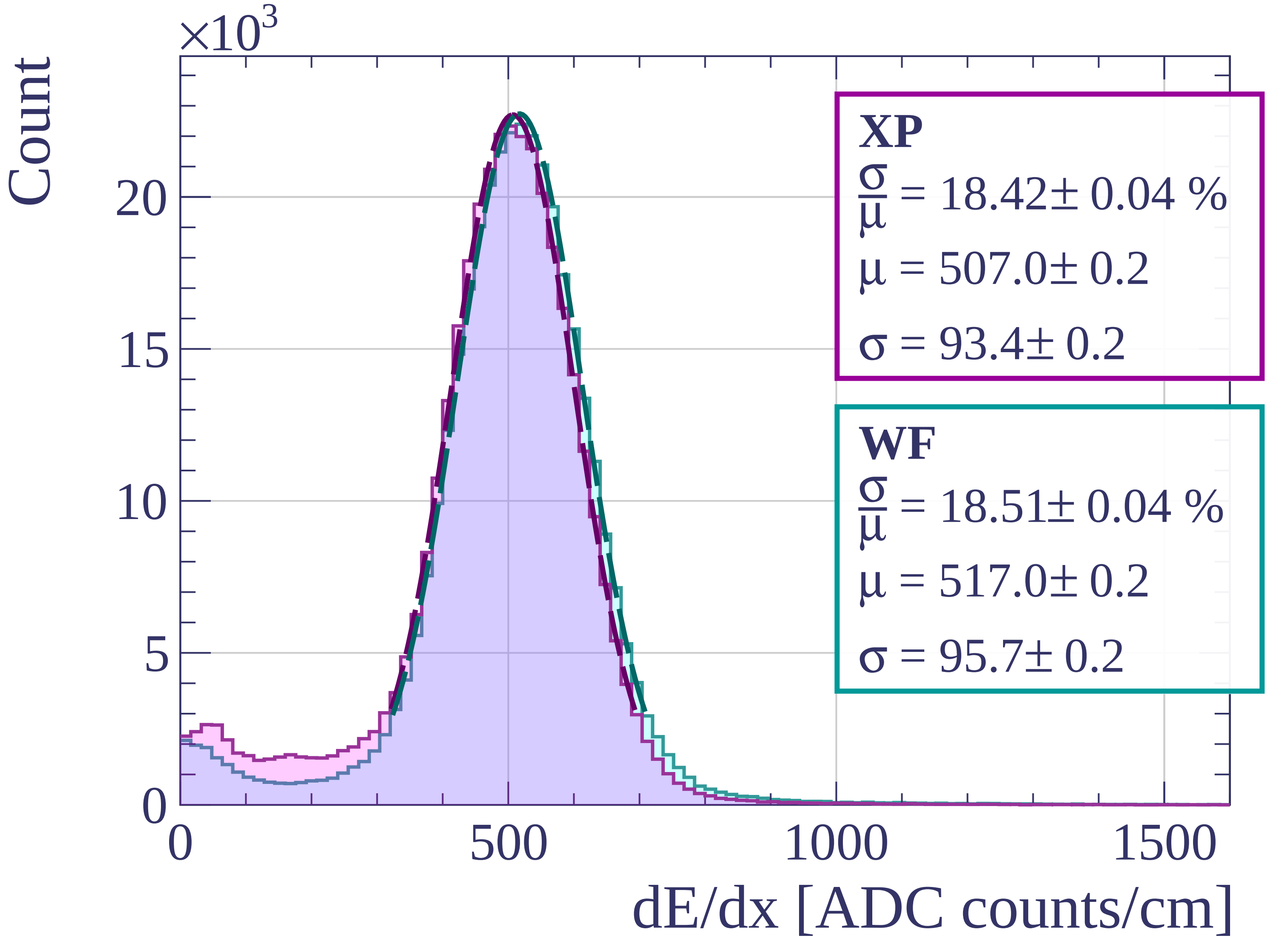
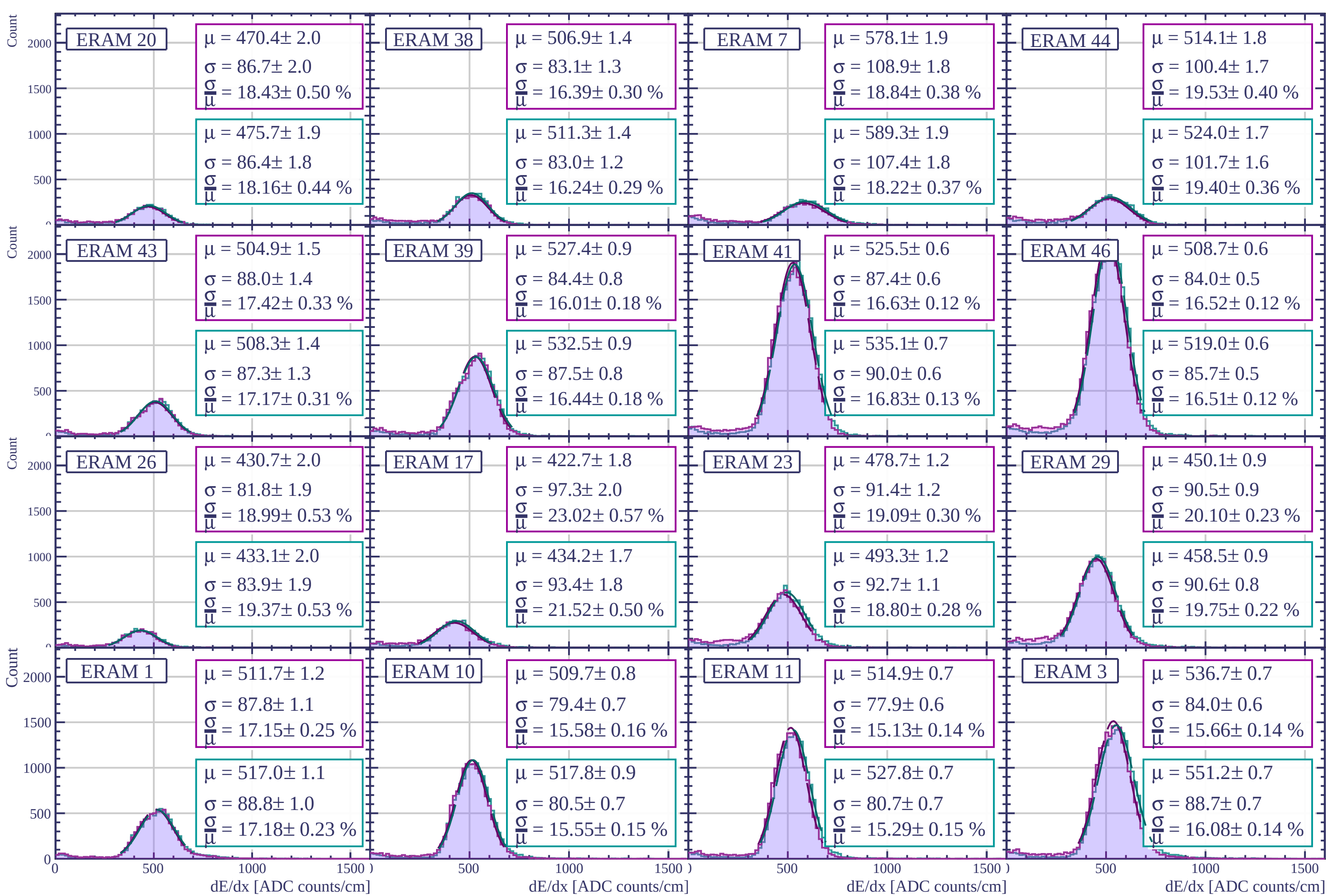
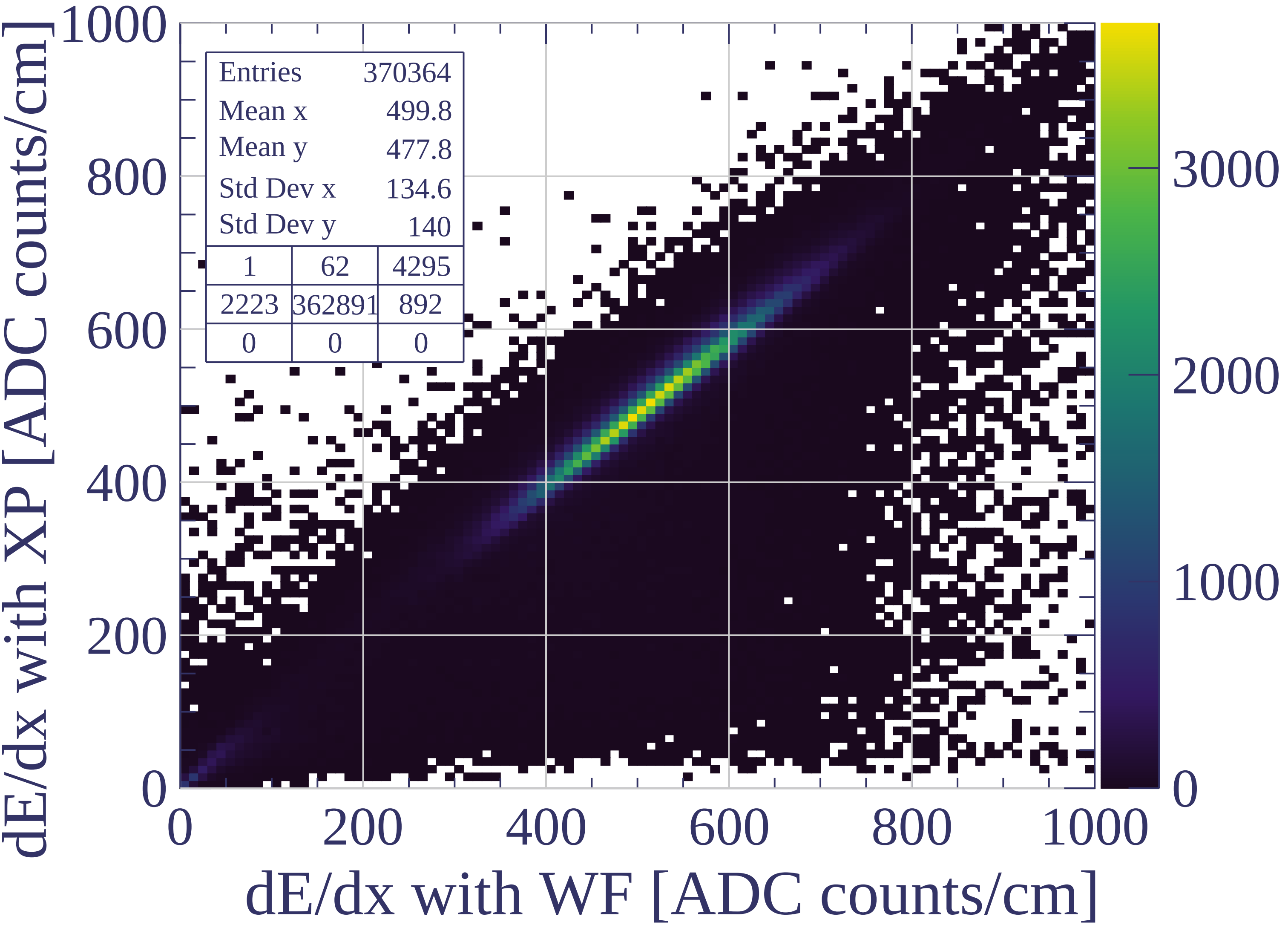
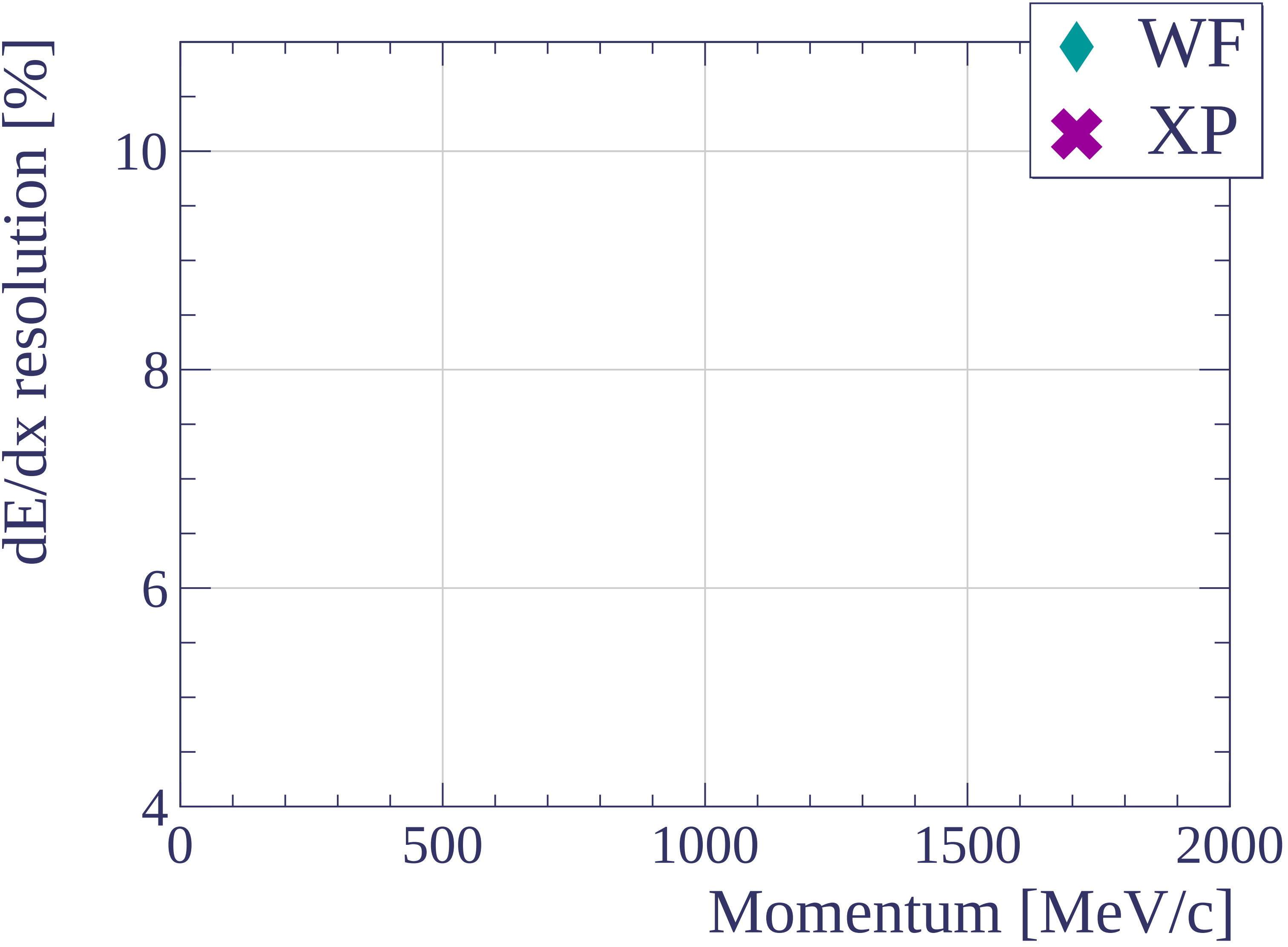


×10^3
Count
20
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 18.42± 0.04 %
μ = 507.0± 0.2
σ = 93.4± 0.2
WF
σ/μ = 18.51± 0.04 %
μ = 517.0± 0.2
σ = 95.7± 0.2

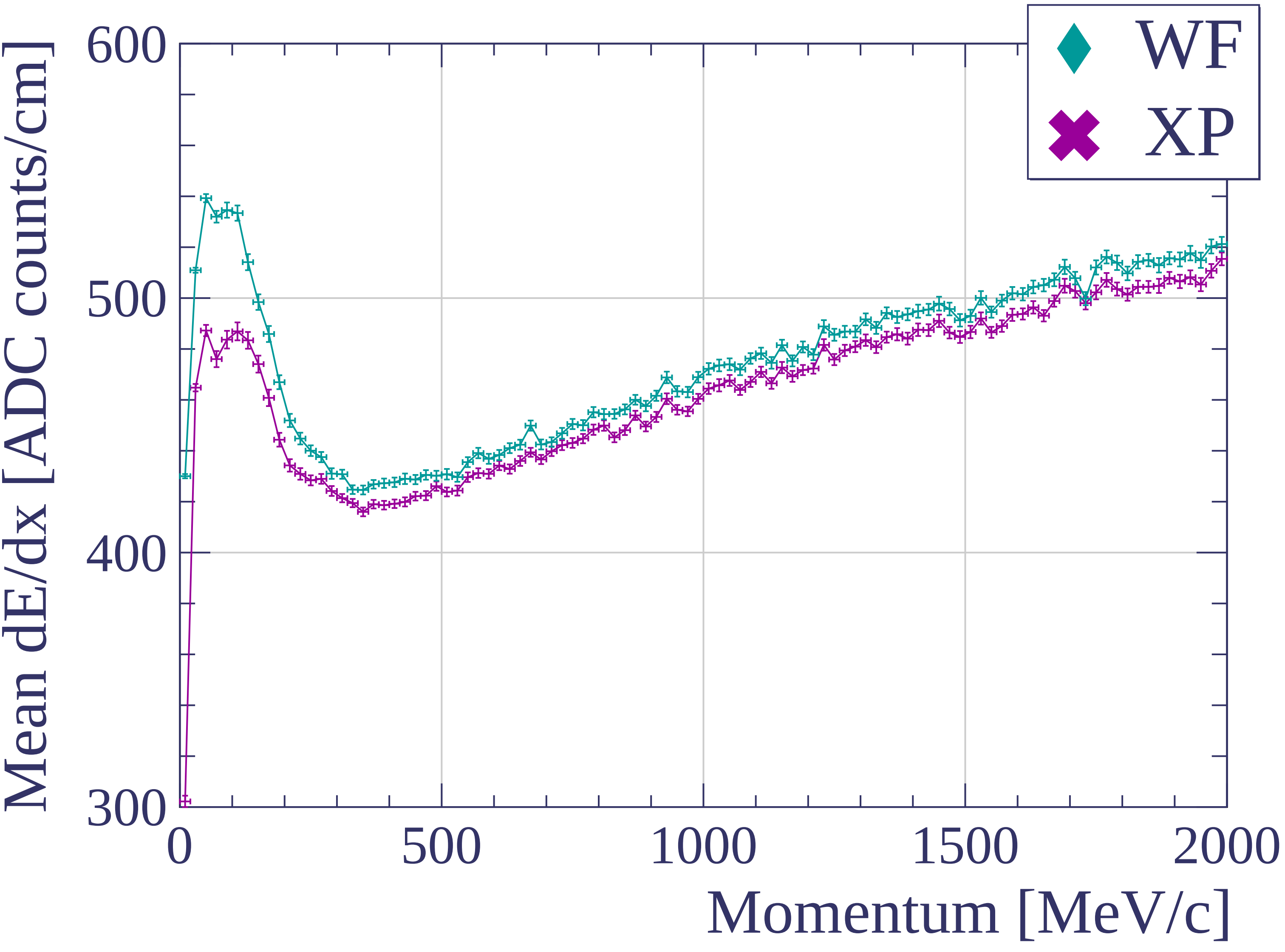
Count
ERAM 47
μ = 456.7± 4.0
σ = 112.2± 3.8
σ/μ = 24.58± 1.05 %
μ = 473.5± 3.2
σ = 96.7± 3.6
σ/μ = 20.43± 0.89 %
ERAM 16
μ = 550.6± 4.1
σ = 118.7± 3.7
σ/μ = 21.57± 0.84 %
μ = 572.4± 3.2
σ = 111.9± 3.3
σ/μ = 19.56± 0.69 %
ERAM 14
μ = 456.5± 4.0
σ = 123.7± 3.9
σ/μ = 27.09± 1.09 %
μ = 491.2± 2.8
σ = 105.1± 3.0
σ/μ = 21.39± 0.74 %
ERAM 15
μ = 568.1± 2.6
σ = 117.3± 2.6
σ/μ = 20.64± 0.56 %
μ = 585.3± 2.4
σ = 113.3± 2.6
σ/μ = 19.35± 0.52 %
Count
ERAM 42
μ = 483.0± 1.1
σ = 82.5± 1.1
σ/μ = 17.08± 0.26 %
μ = 487.1± 1.1
σ = 84.1± 1.1
σ/μ = 17.27± 0.27 %
ERAM 45
μ = 528.1± 0.8
σ = 84.6± 0.7
σ/μ = 16.02± 0.17 %
μ = 537.0± 0.9
σ = 87.8± 0.8
σ/μ = 16.35± 0.18 %
ERAM 37
μ = 540.5± 0.6
σ = 87.8± 0.6
σ/μ = 16.25± 0.13 %
μ = 551.9± 0.6
σ = 87.3± 0.5
σ/μ = 15.82± 0.12 %
ERAM 36
μ = 554.7± 0.7
σ = 92.8± 0.6
σ/μ = 16.72± 0.13 %
μ = 567.9± 0.7
σ = 97.1± 0.6
σ/μ = 17.11± 0.13 %
Count
ERAM 24
μ = 428.1± 1.4
σ = 82.4± 1.3
σ/μ = 19.26± 0.38 %
μ = 430.5± 1.4
σ = 80.7± 1.2
σ/μ = 18.75± 0.35 %
ERAM 30
μ = 406.4± 1.5
σ = 78.7± 1.5
σ/μ = 19.37± 0.43 %
μ = 417.1± 1.5
σ = 82.3± 1.5
σ/μ = 19.74± 0.43 %
ERAM 28
μ = 406.0± 1.3
σ = 80.2± 1.4
σ/μ = 19.74± 0.40 %
μ = 420.2± 1.2
σ = 78.1± 1.2
σ/μ = 18.59± 0.35 %
ERAM 19
μ = 444.1± 1.0
σ = 88.6± 1.0
σ/μ = 19.96± 0.27 %
μ = 449.0± 1.0
σ = 87.0± 0.9
σ/μ = 19.37± 0.25 %
Count
ERAM 21
μ = 444.9± 0.9
σ = 69.5± 0.8
σ/μ = 15.62± 0.22 %
μ = 452.9± 0.9
σ = 69.3± 0.8
σ/μ = 15.29± 0.21 %
ERAM 13
μ = 495.2± 0.8
σ = 78.2± 0.8
σ/μ = 15.79± 0.18 %
μ = 504.0± 0.8
σ = 79.7± 0.7
σ/μ = 15.81± 0.18 %
ERAM 9
μ = 484.2± 0.7
σ = 74.7± 0.6
σ/μ = 15.43± 0.15 %
μ = 497.0± 0.7
σ = 76.6± 0.6
σ/μ = 15.41± 0.15 %
ERAM 2
μ = 520.9± 0.7
σ = 85.1± 0.7
σ/μ = 16.34± 0.15 %
μ = 535.6± 0.8
σ = 89.2± 0.7
σ/μ = 16.65± 0.16 %
dE/dx [ADC counts/cm]
dE/dx [ADC counts/cm]
dE/dx [ADC counts/cm]
dE/dx [ADC counts/cm]


ERAM 20
μ = 470.4± 2.0
σ = 86.7± 2.0
σ/μ = 18.43± 0.50 %
μ = 475.7± 1.9
σ = 86.4± 1.8
σ/μ = 18.16± 0.44 %
ERAM 38
μ = 506.9± 1.4
σ = 83.1± 1.3
σ/μ = 16.39± 0.30 %
μ = 511.3± 1.4
σ = 83.0± 1.2
σ/μ = 16.24± 0.29 %
ERAM 7
μ = 578.1± 1.9
σ = 108.9± 1.8
σ/μ = 18.84± 0.38 %
μ = 589.3± 1.9
σ = 107.4± 1.8
σ/μ = 18.22± 0.37 %
ERAM 44
μ = 514.1± 1.8
σ = 100.4± 1.7
σ/μ = 19.53± 0.40 %
μ = 524.0± 1.7
σ = 101.7± 1.6
σ/μ = 19.40± 0.36 %
ERAM 43
μ = 504.9± 1.5
σ = 88.0± 1.4
σ/μ = 17.42± 0.33 %
μ = 508.3± 1.4
σ = 87.3± 1.3
σ/μ = 17.17± 0.31 %
ERAM 39
μ = 527.4± 0.9
σ = 84.4± 0.8
σ/μ = 16.01± 0.18 %
μ = 532.5± 0.9
σ = 87.5± 0.8
σ/μ = 16.44± 0.18 %
ERAM 41
μ = 525.5± 0.6
σ = 87.4± 0.6
σ/μ = 16.63± 0.12 %
μ = 535.1± 0.7
σ = 90.0± 0.6
σ/μ = 16.83± 0.13 %
ERAM 46
μ = 508.7± 0.6
σ = 84.0± 0.5
σ/μ = 16.52± 0.12 %
μ = 519.0± 0.6
σ = 85.7± 0.5
σ/μ = 16.51± 0.12 %
ERAM 26
μ = 430.7± 2.0
σ = 81.8± 1.9
σ/μ = 18.99± 0.53 %
μ = 433.1± 2.0
σ = 83.9± 1.9
σ/μ = 19.37± 0.53 %
ERAM 17
μ = 422.7± 1.8
σ = 97.3± 2.0
σ/μ = 23.02± 0.57 %
μ = 434.2± 1.7
σ = 93.4± 1.8
σ/μ = 21.52± 0.50 %
ERAM 23
μ = 478.7± 1.2
σ = 91.4± 1.2
σ/μ = 19.09± 0.30 %
μ = 493.3± 1.2
σ = 92.7± 1.1
σ/μ = 18.80± 0.28 %
ERAM 29
μ = 450.1± 0.9
σ = 90.5± 0.9
σ/μ = 20.10± 0.23 %
μ = 458.5± 0.9
σ = 90.6± 0.8
σ/μ = 19.75± 0.22 %
ERAM 1
μ = 511.7± 1.2
σ = 87.8± 1.1
σ/μ = 17.15± 0.25 %
μ = 517.0± 1.1
σ = 88.8± 1.0
σ/μ = 17.18± 0.23 %
ERAM 10
μ = 509.7± 0.8
σ = 79.4± 0.7
σ/μ = 15.58± 0.16 %
μ = 517.8± 0.9
σ = 80.5± 0.7
σ/μ = 15.55± 0.15 %
ERAM 11
μ = 514.9± 0.7
σ = 77.9± 0.6
σ/μ = 15.13± 0.14 %
μ = 527.8± 0.7
σ = 80.7± 0.7
σ/μ = 15.29± 0.15 %
ERAM 3
μ = 536.7± 0.7
σ = 84.0± 0.6
σ/μ = 15.66± 0.14 %
μ = 551.2± 0.7
σ = 88.7± 0.7
σ/μ = 16.08± 0.14 %
Count
dE/dx [ADC counts/cm]

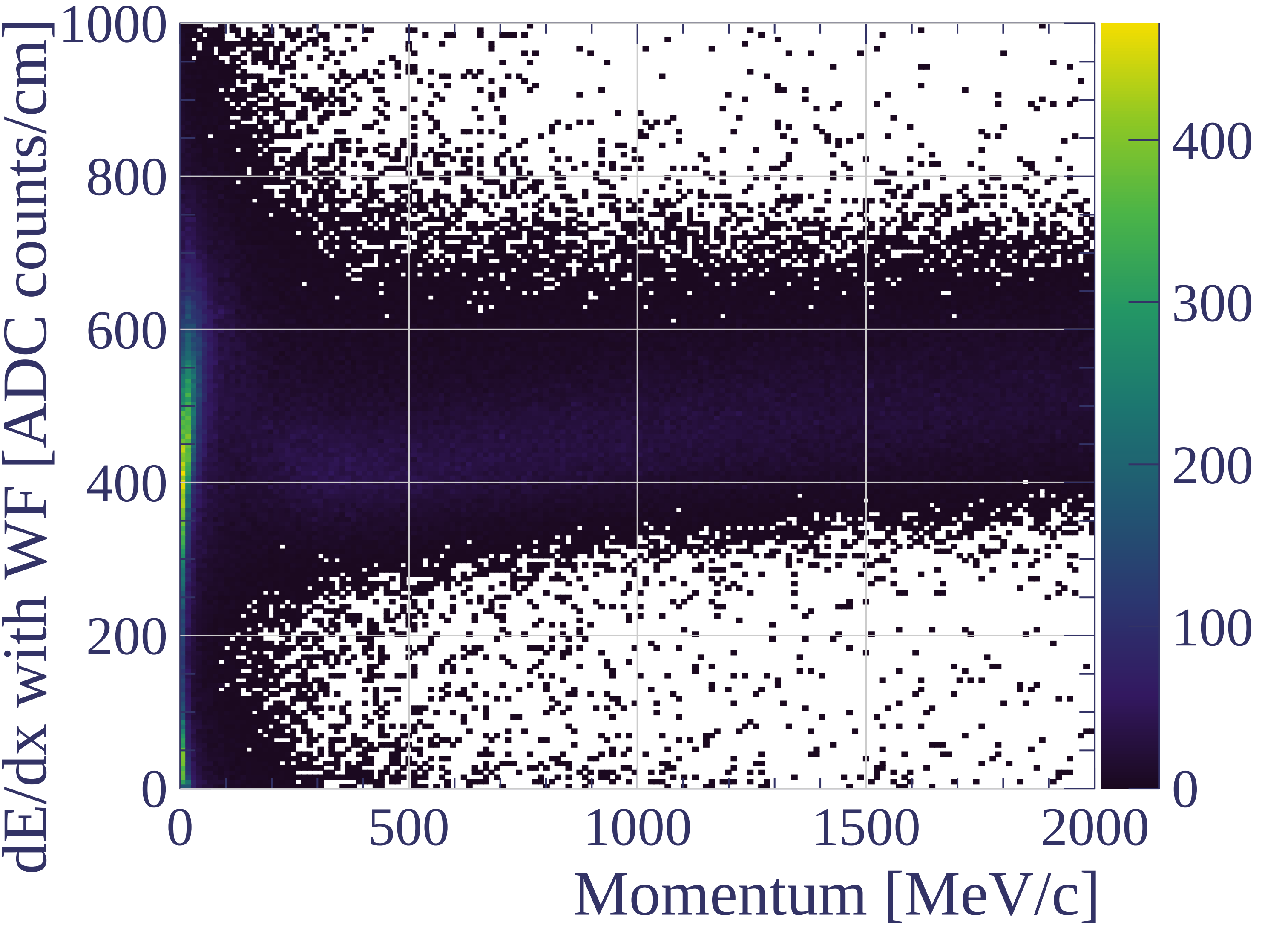

Entries 370364
Mean x 499.8
Mean y 477.8
Std Dev x 134.6
Std Dev y 140
1 62 4295
2223 362891 892
0 0 0
dE/dx with XP [ADC counts/cm]
dE/dx with WF [ADC counts/cm]
0
200
400
600
800
1000
0
1000
2000
3000

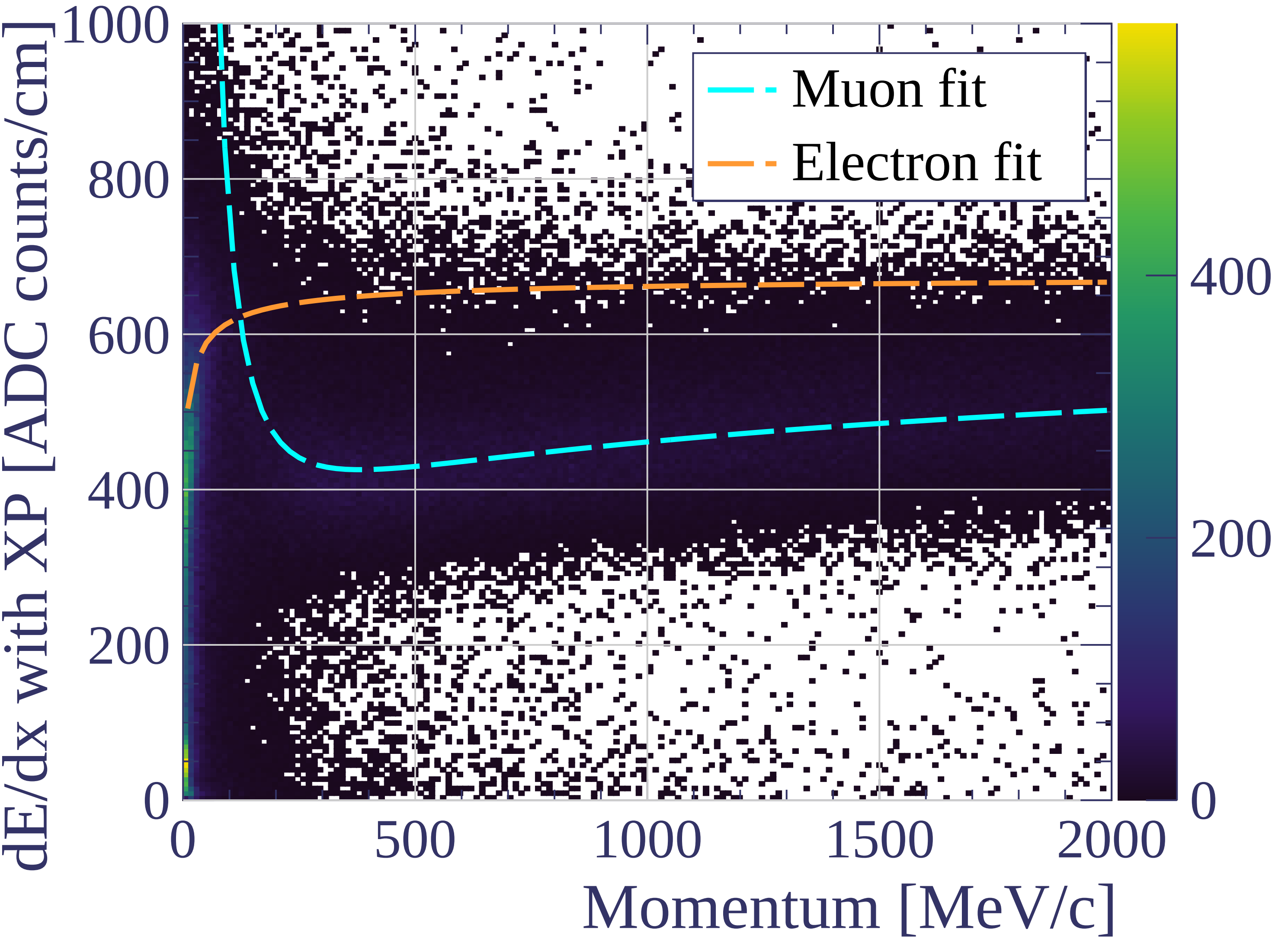


dE/dx resolution [%]
10
8
6
4
0
500
1000
1500
2000
Momentum [MeV/c]
WF
XP

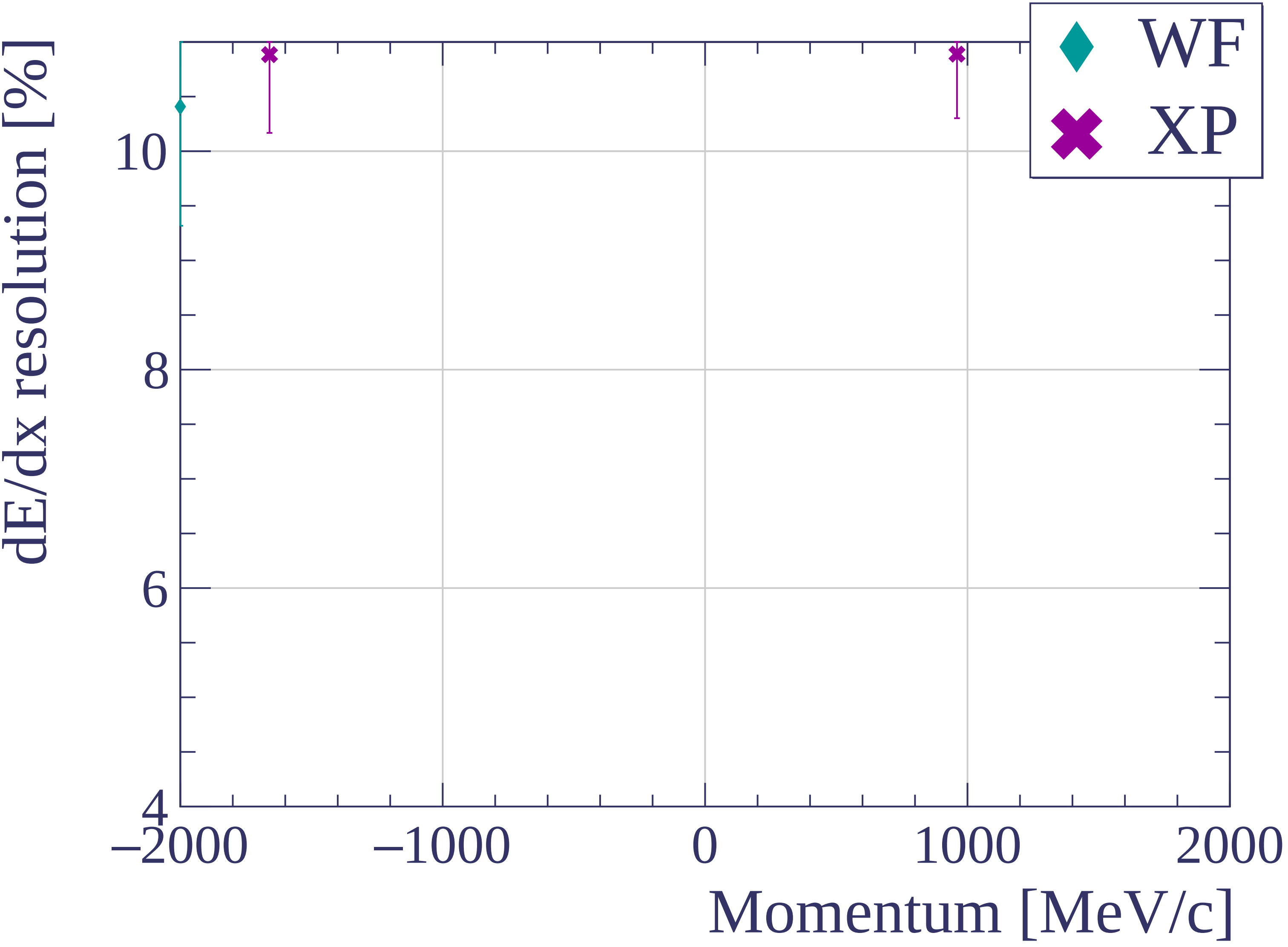


WF
XP
600
500
400
300
Mean dE/dx [ADC counts/cm]
0
500
1000
1500
2000
Momentum [MeV/c]

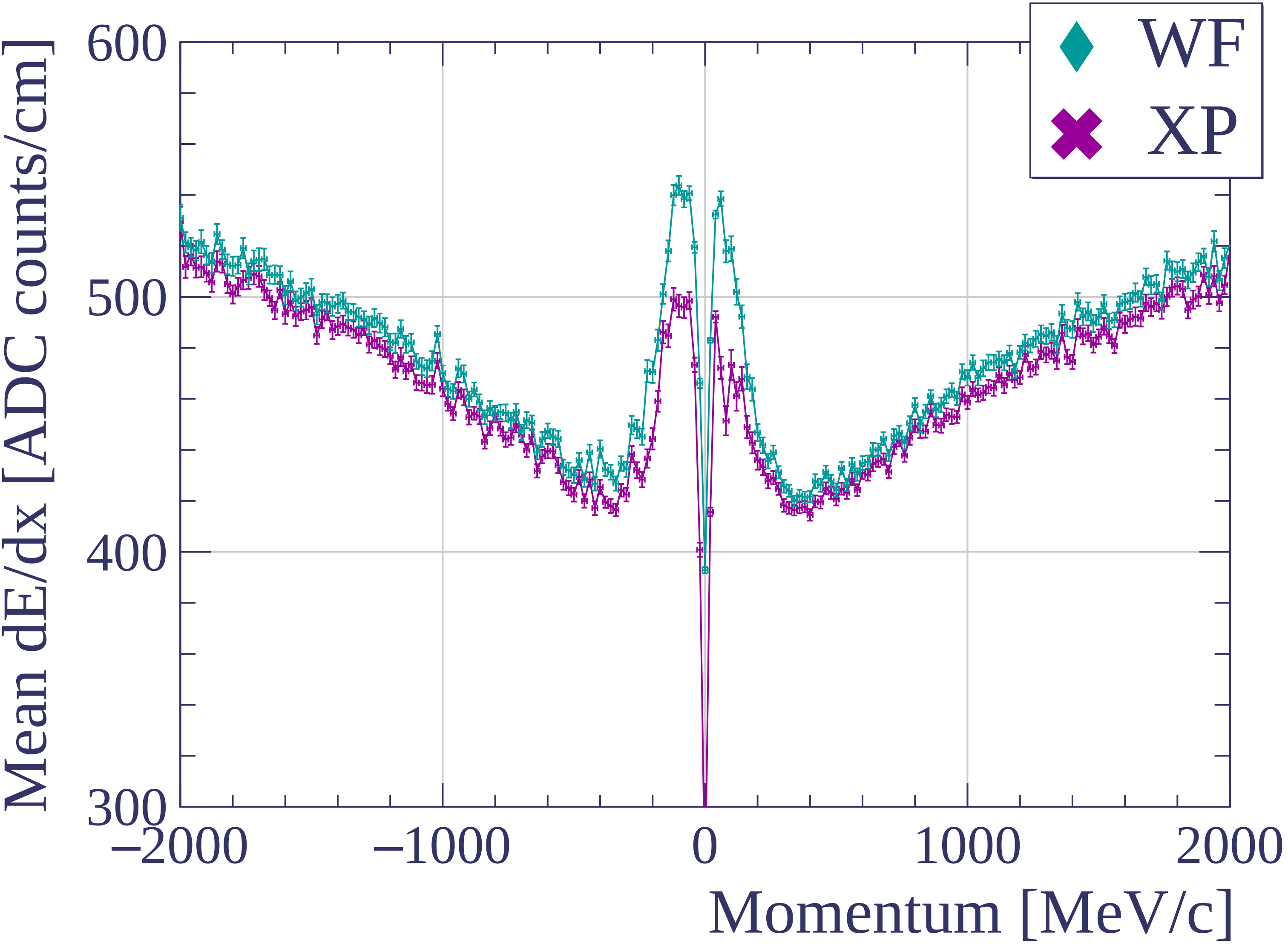

dE/dx with WF [ADC counts/cm]
1000
800
600
400
200
0
0
500
1000
1500
2000
Momentum [MeV/c]
400
300
200
100
0



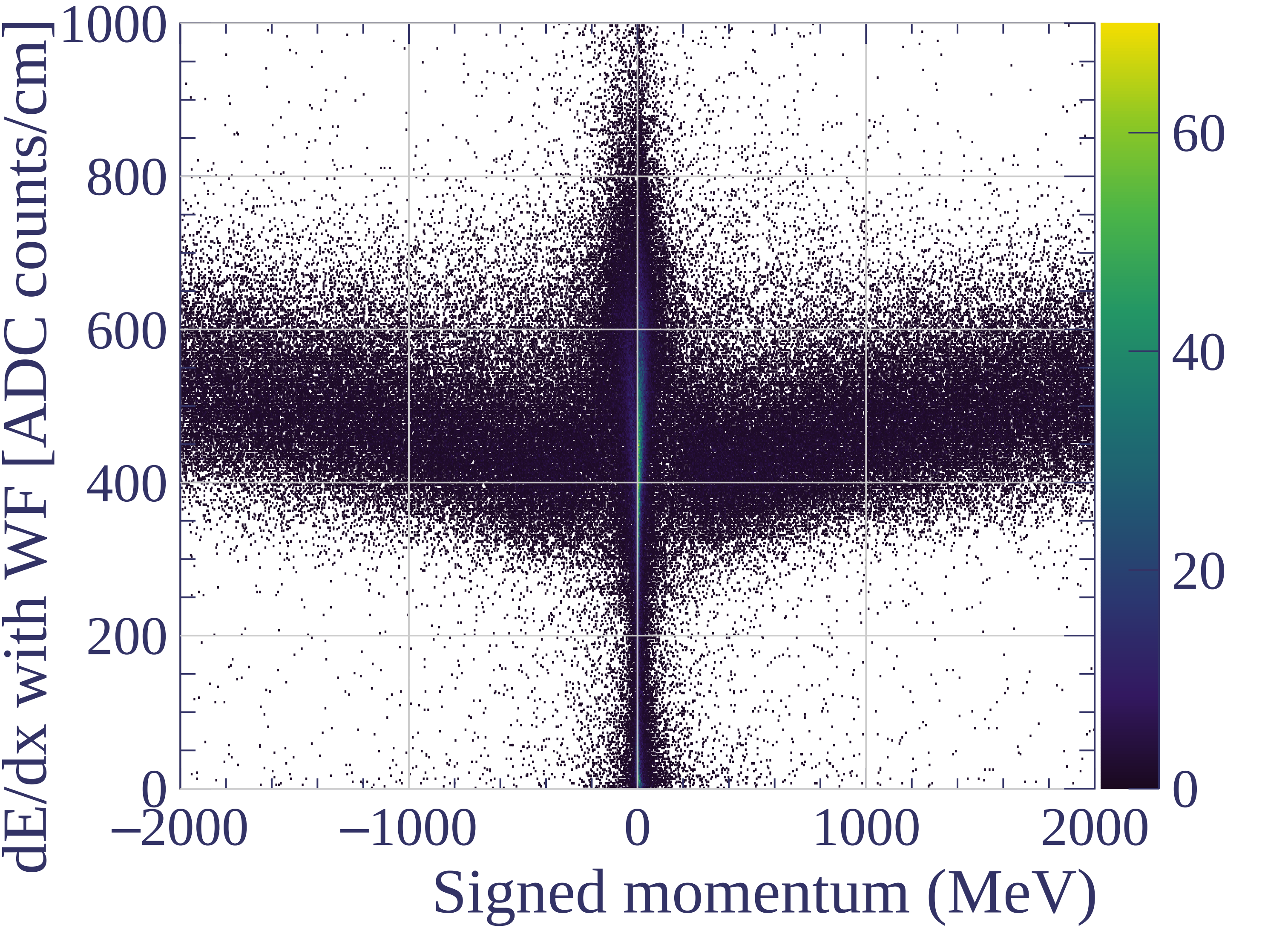
Muon fit
Electron fit
dE/dx with XP [ADC counts/cm]
Momentum [MeV/c]
0
200
400
600
800
1000
500
1500
2000

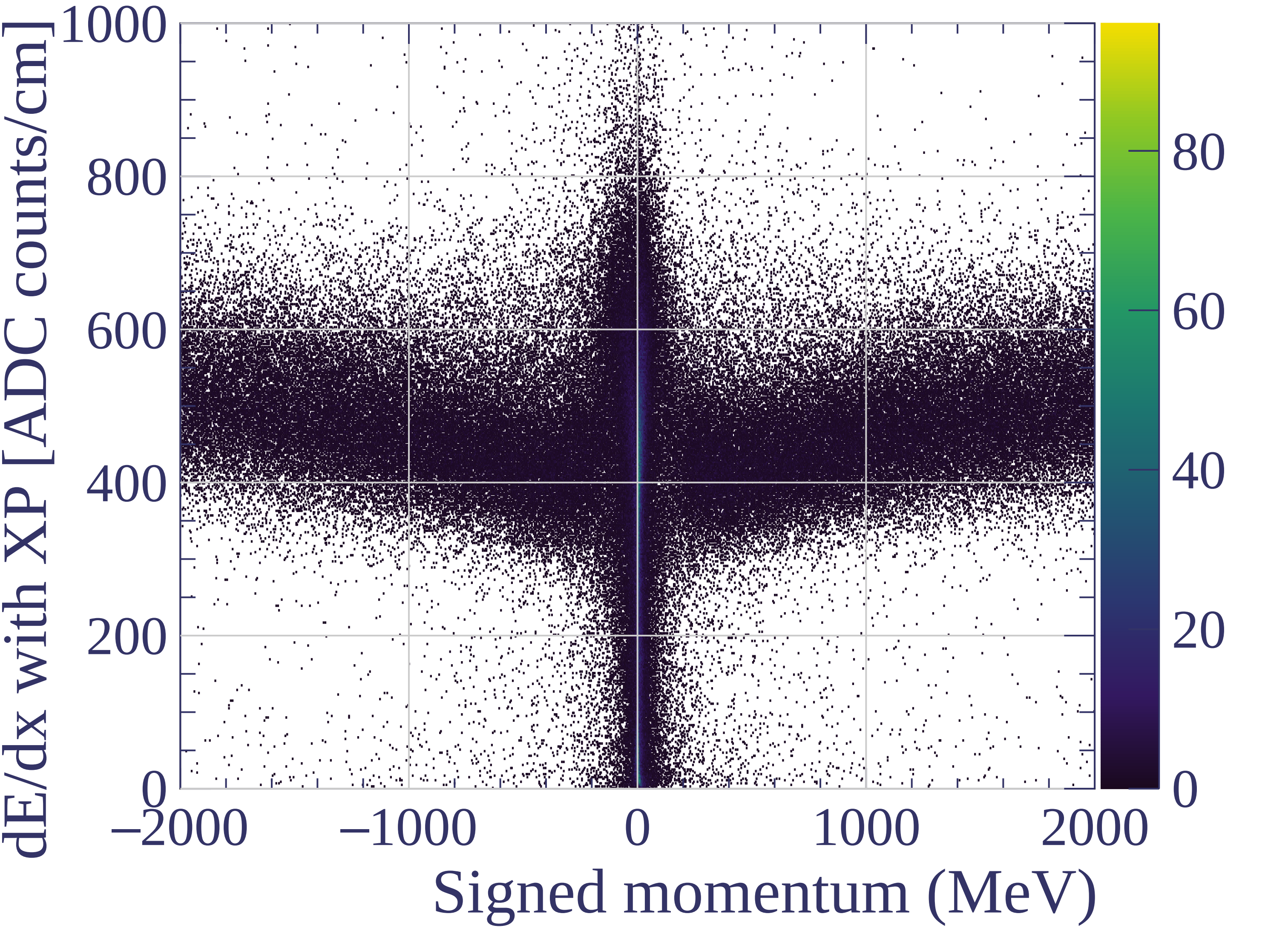


WF
XP
dE/dx resolution [%]
10
8
6
4
−2000
−1000
0
1000
2000
Momentum [MeV/c]

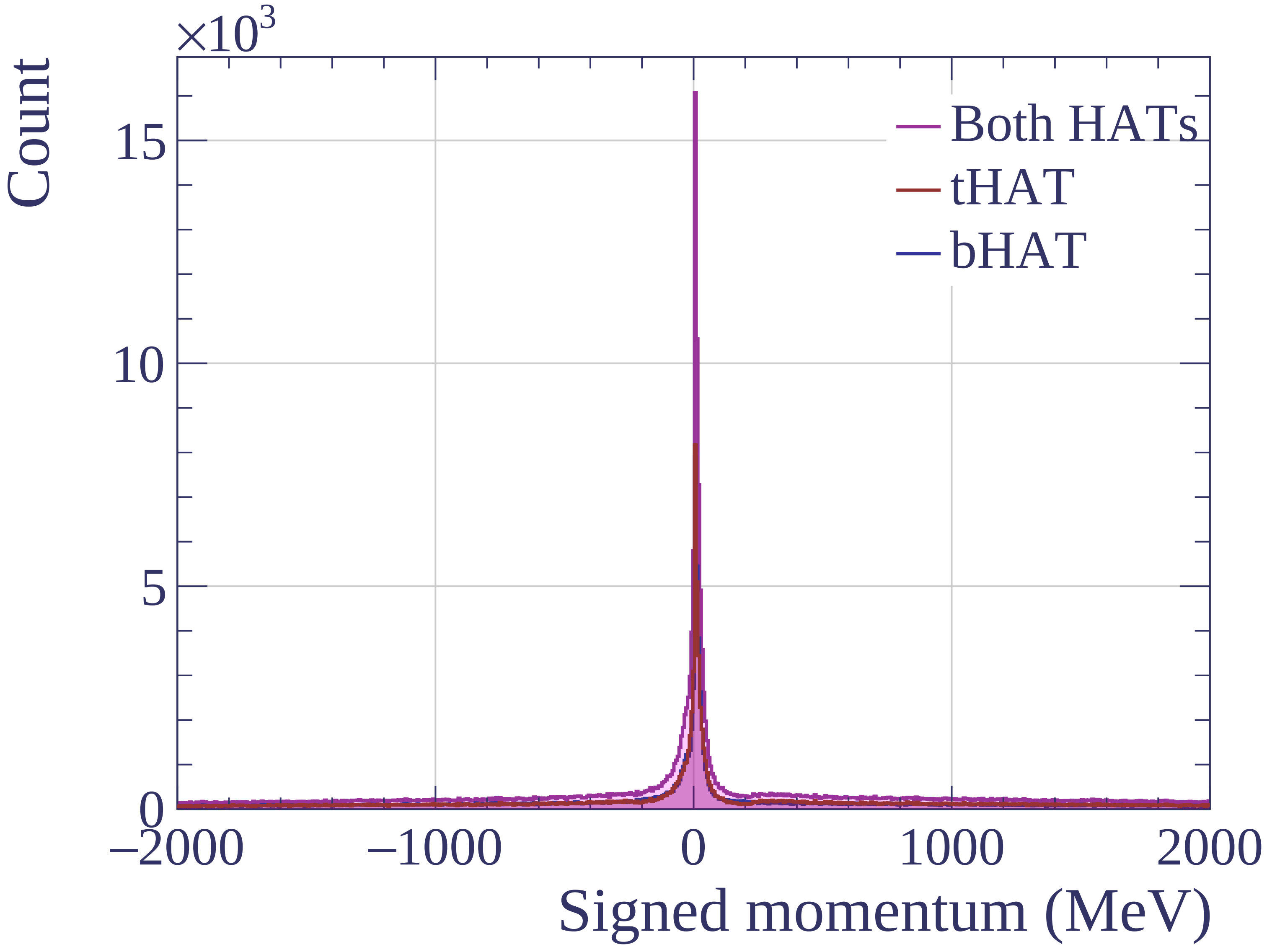


WF
XP
Mean dE/dx [ADC counts/cm]
600
500
400
300
−2000
−1000
0
1000
2000
Momentum [MeV/c]

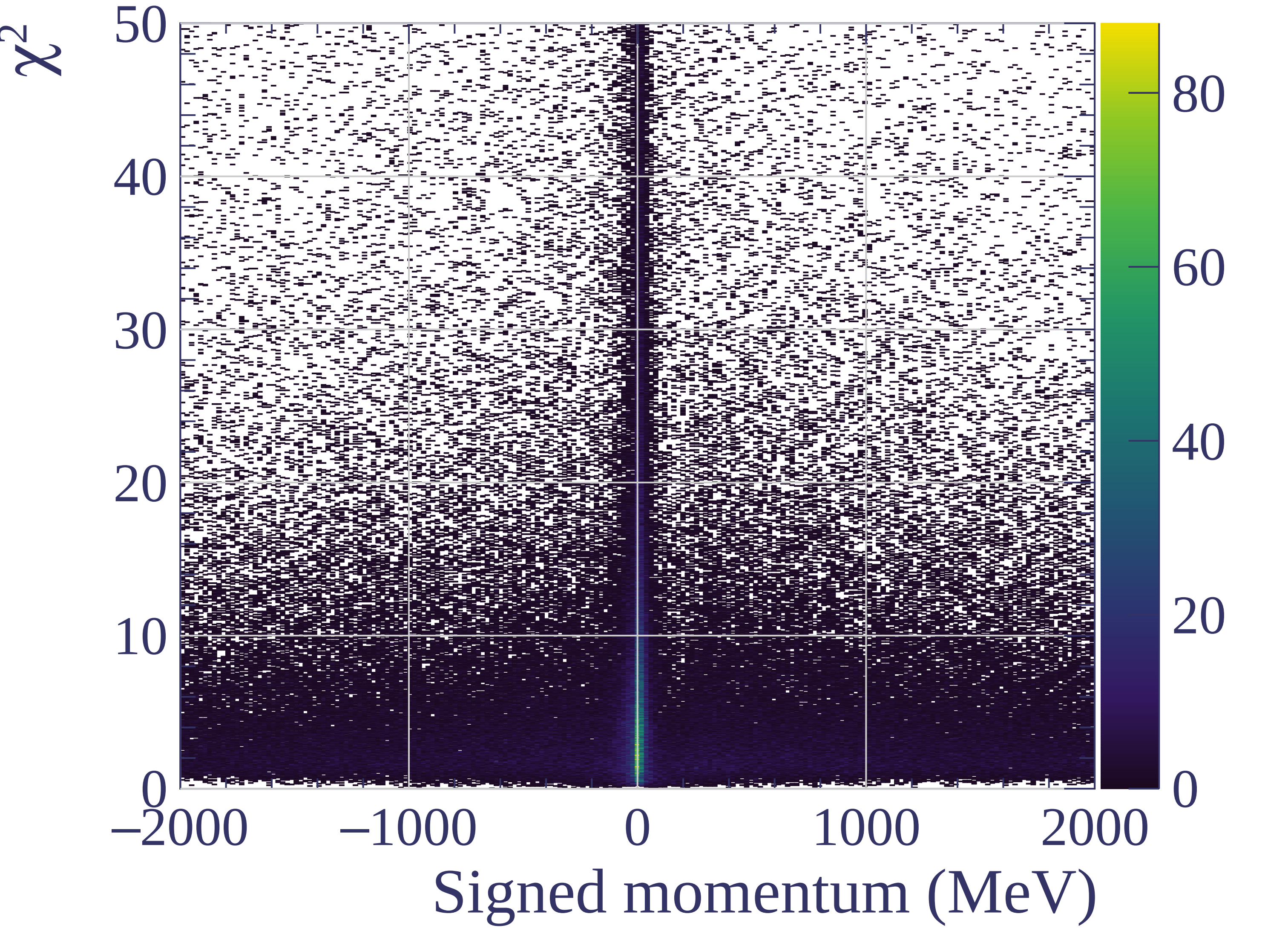

dE/dx with WF [ADC counts/cm]
Signed momentum (MeV)
1000
800
600
400
200
0
−2000
−1000
0
1000
2000
60
40
20
0



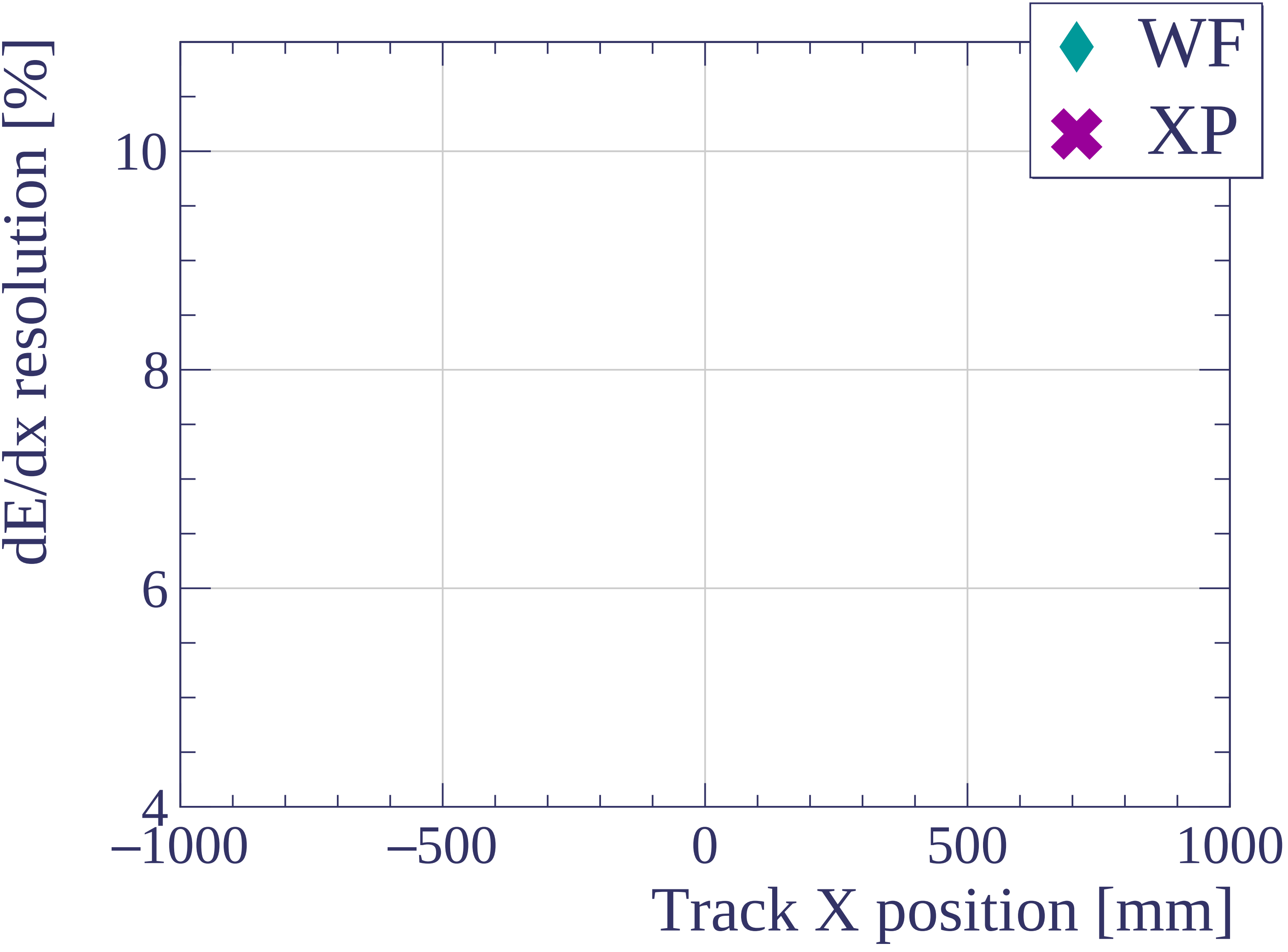
dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
−2000
−1000
0
1000
2000
Signed momentum (MeV)
80
60
40
20
0



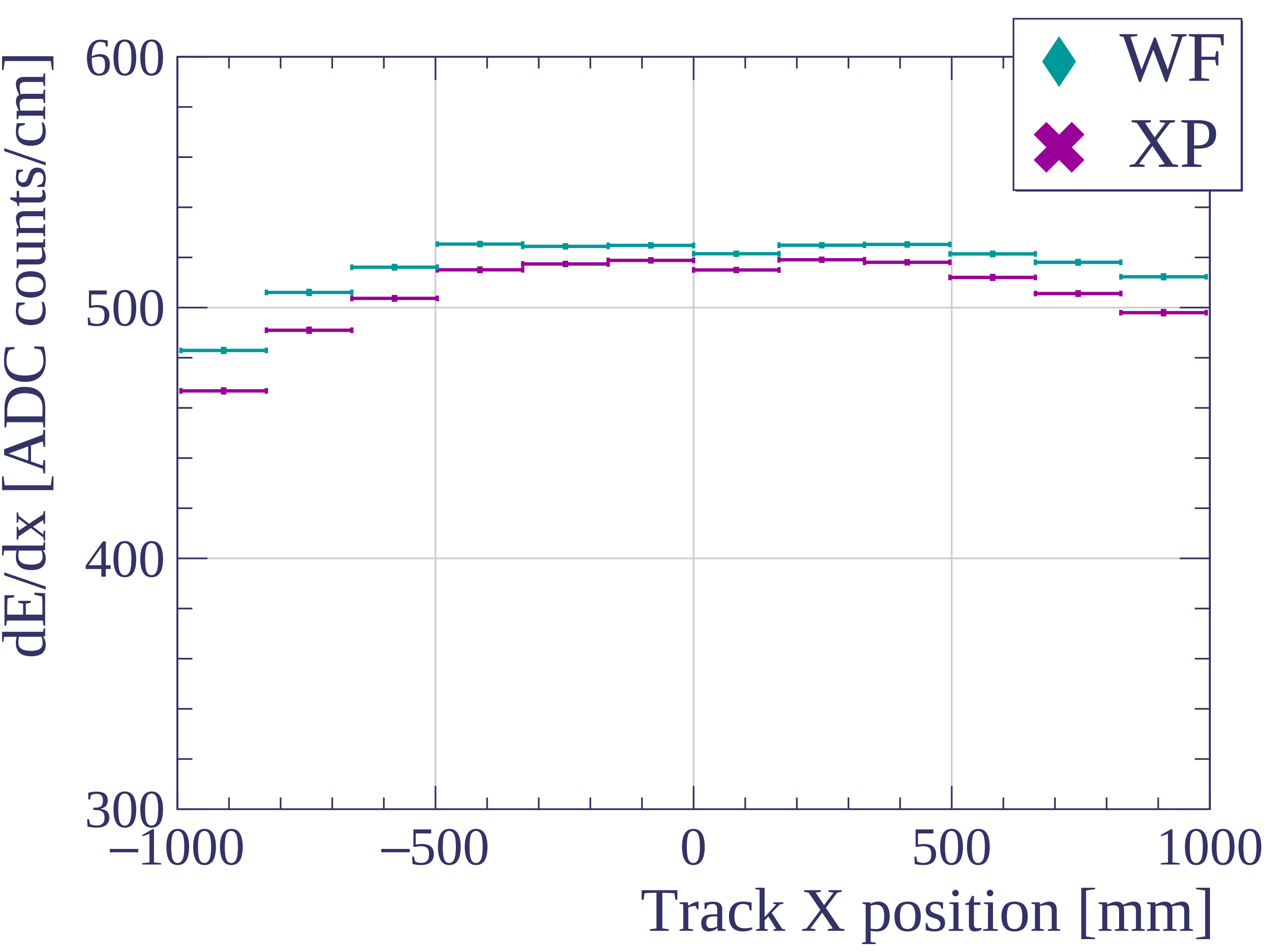
×10^3
Count
Both HATs
tHAT
bHAT
15
10
5
0
−2000
−1000
0
1000
2000
Signed momentum (MeV)



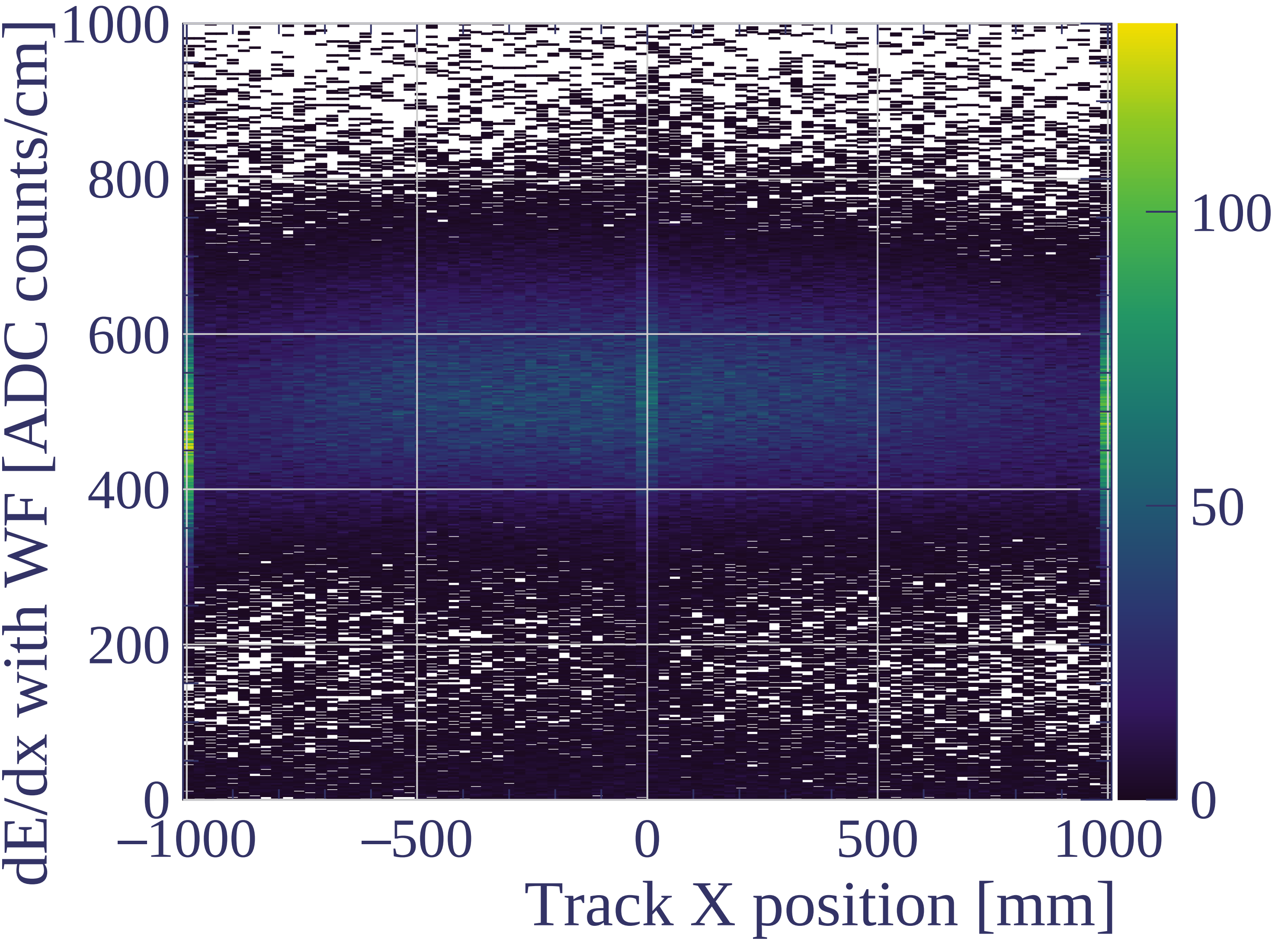
χ^2
50
40
30
20
10
0
−2000
−1000
0
1000
2000
Signed momentum (MeV)
80
60
40
20
0



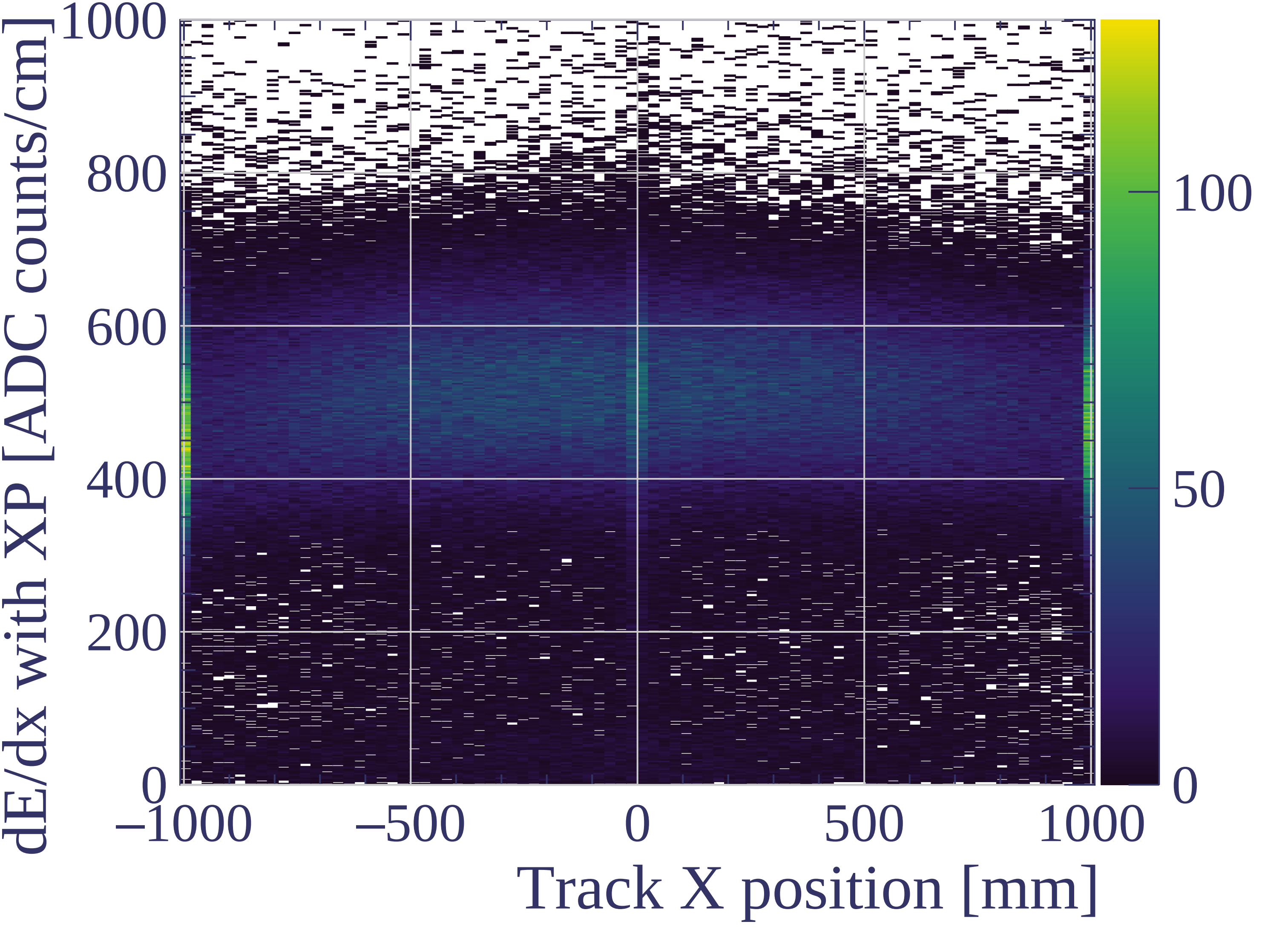
WF
XP
dE/dx resolution [%]
10
8
6
4
−1000
−500
0
500
1000
Track X position [mm]

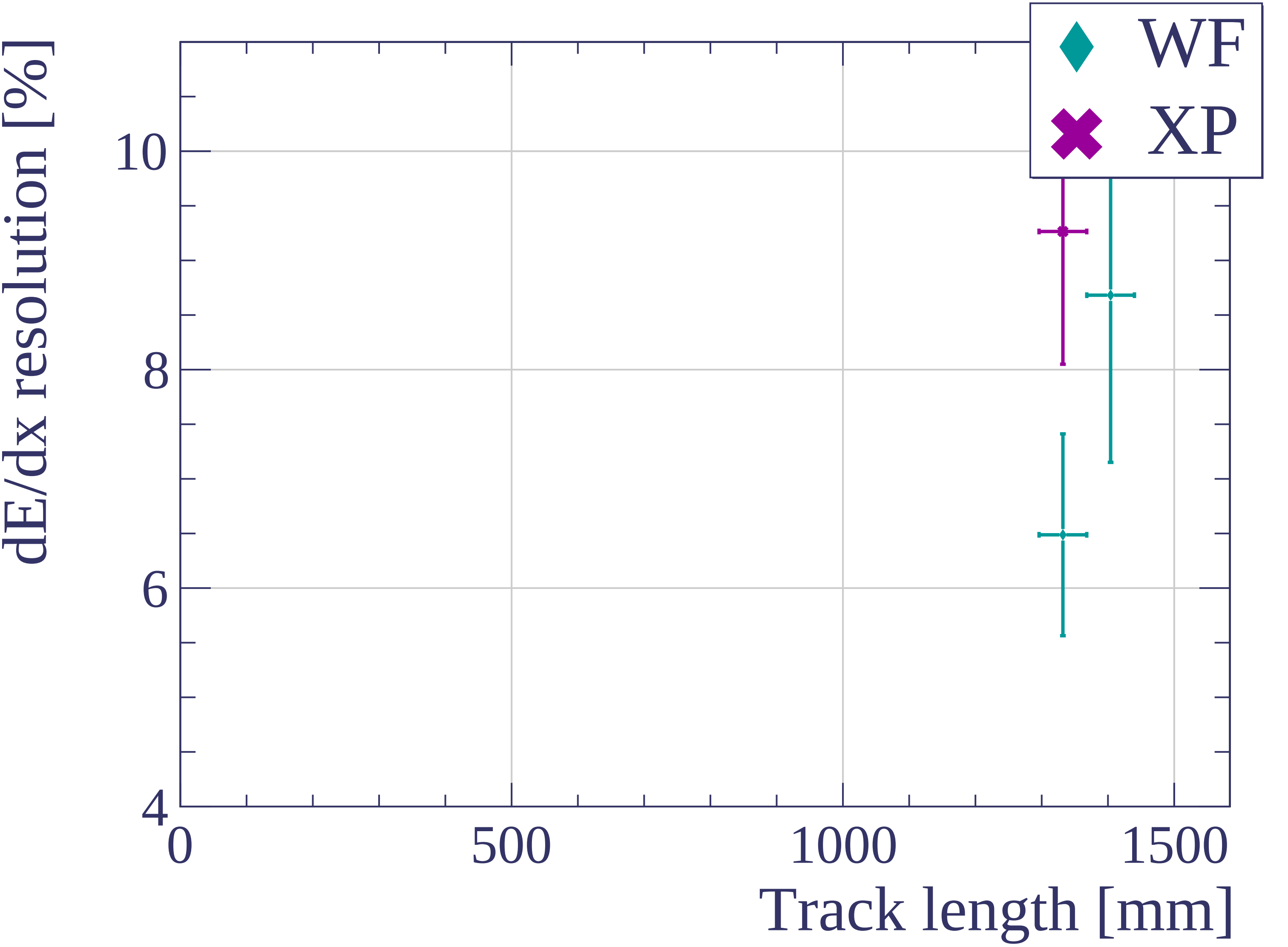


WF
XP
600
500
400
300
dE/dx [ADC counts/cm]
−1000
−500
0
500
1000
Track X position [mm]

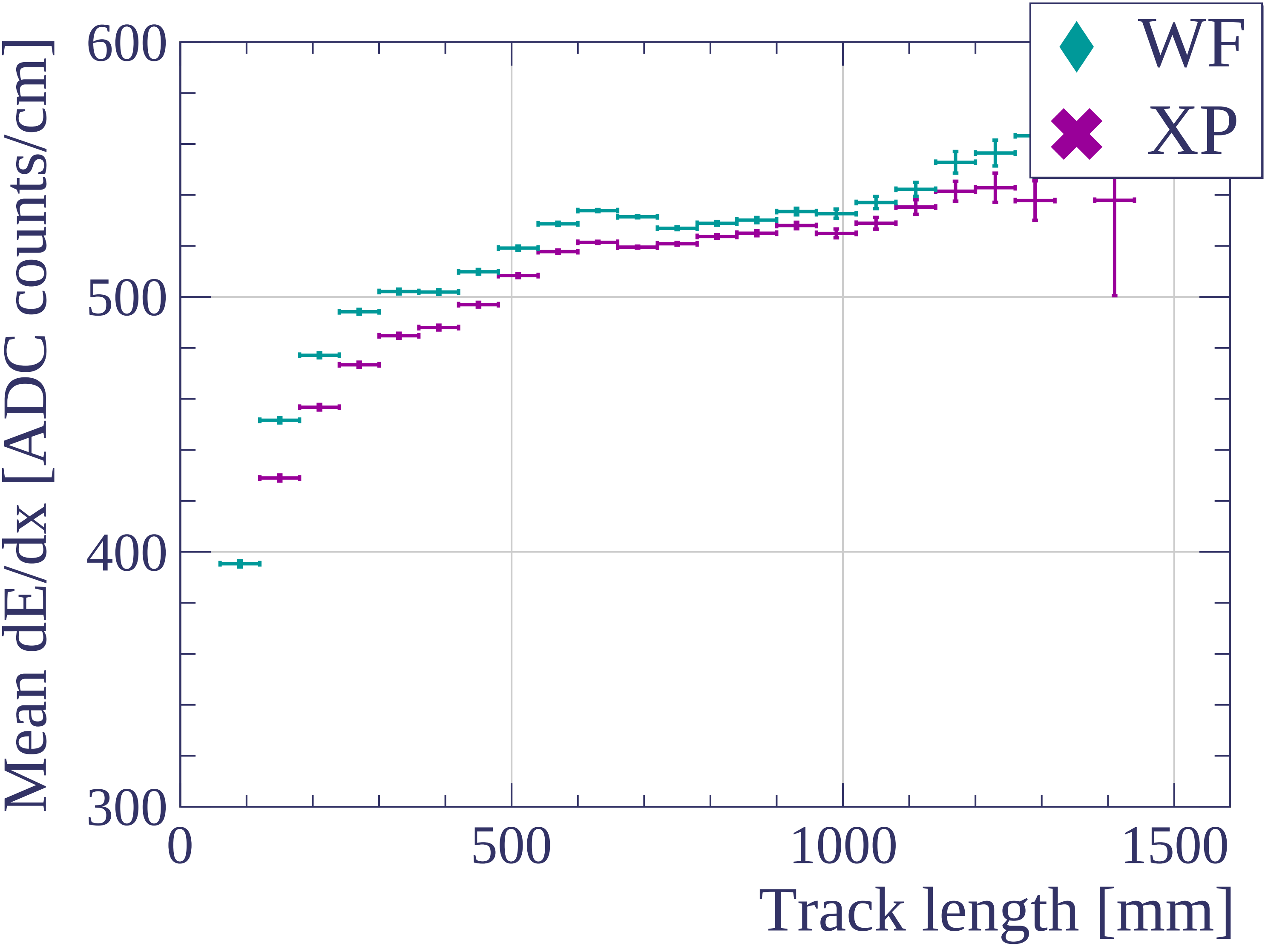


dE/dx with WF [ADC counts/cm]
1000
800
600
400
200
0
−1000
−500
0
500
1000
Track X position [mm]
100
50
0

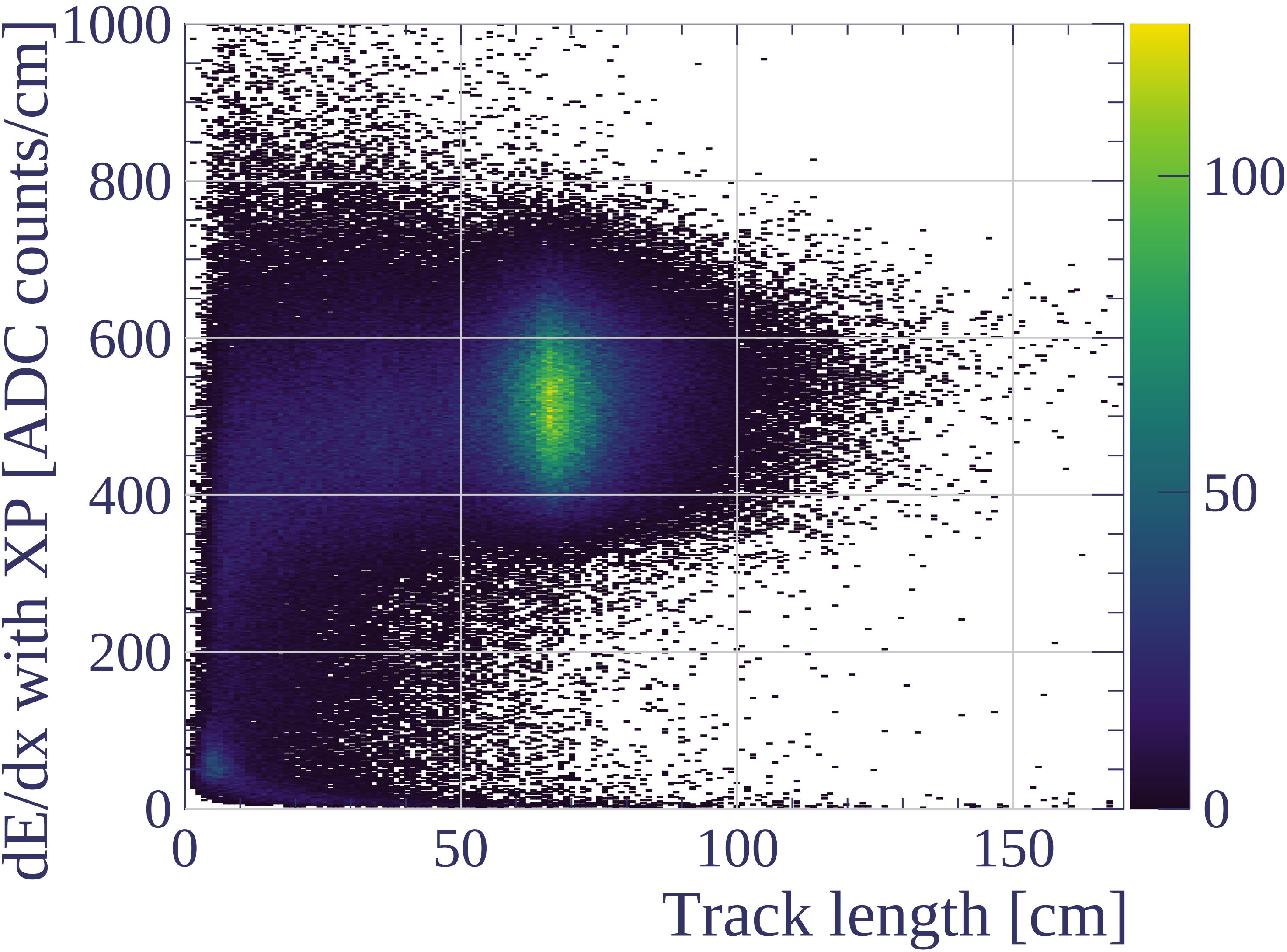

dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
−1000
−500
0
500
1000
Track X position [mm]
100
50
0



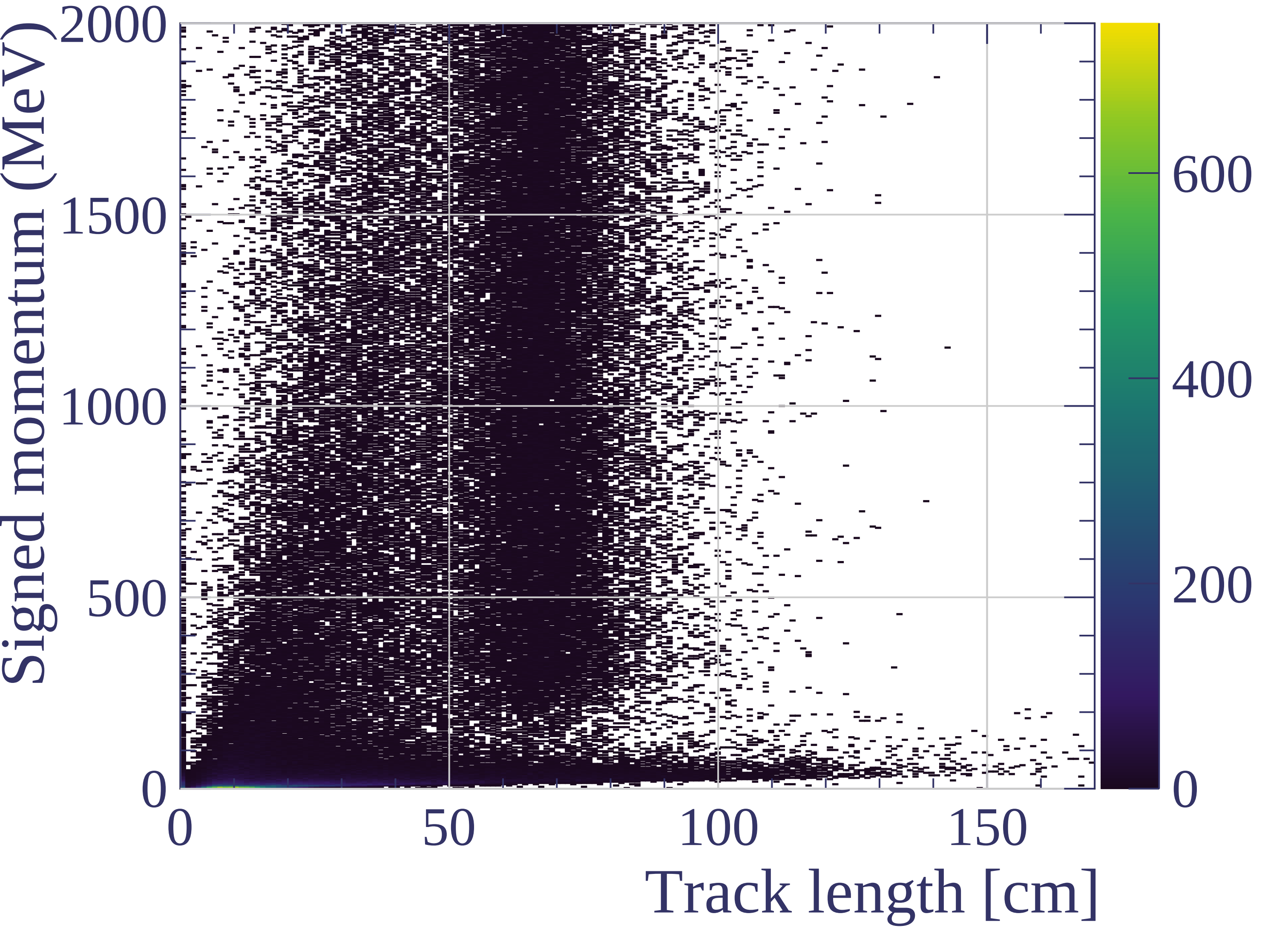
dE/dx resolution [%]
4
6
8
10
0
500
1000
1500
Track length [mm]
WF
XP

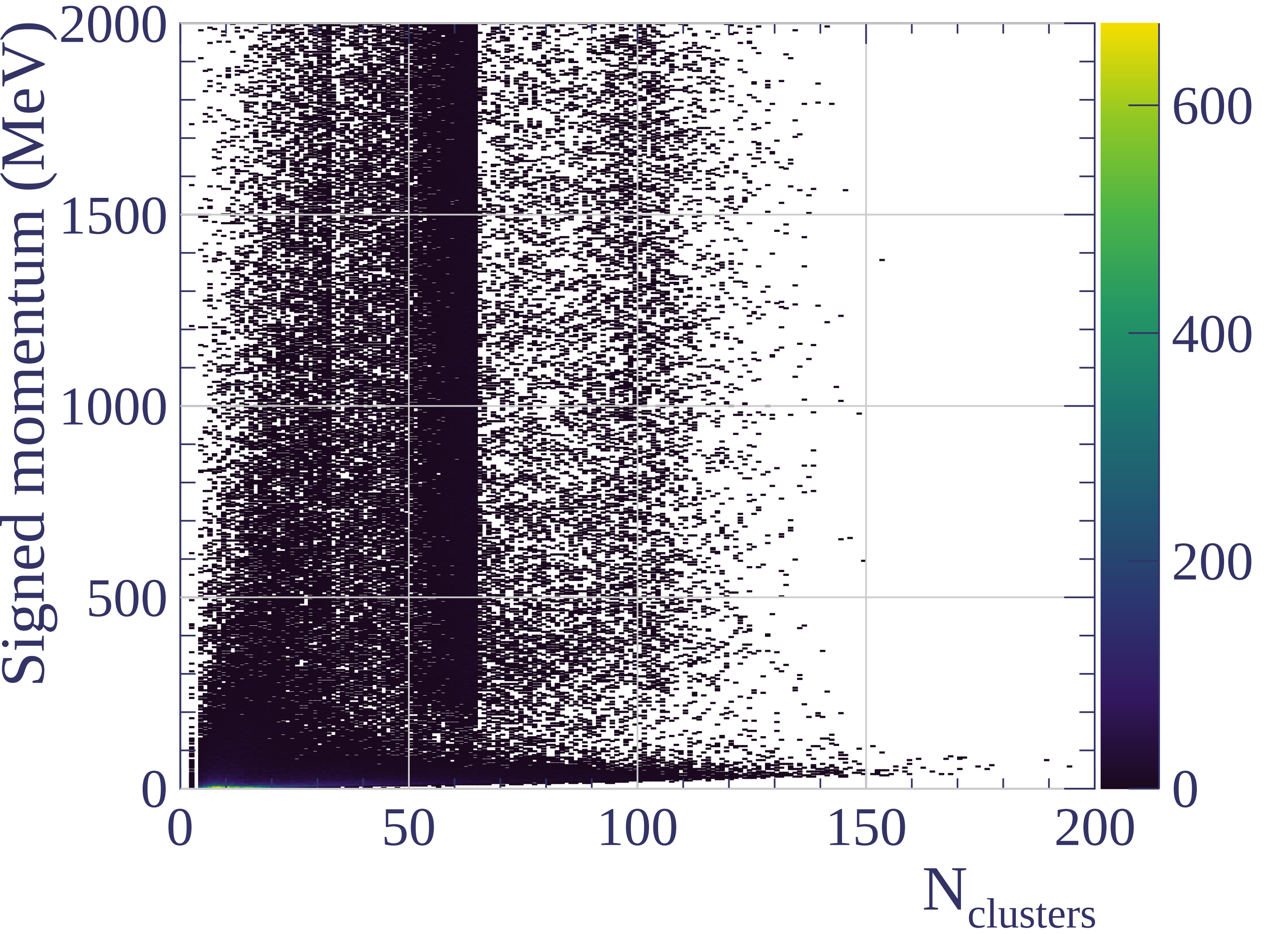


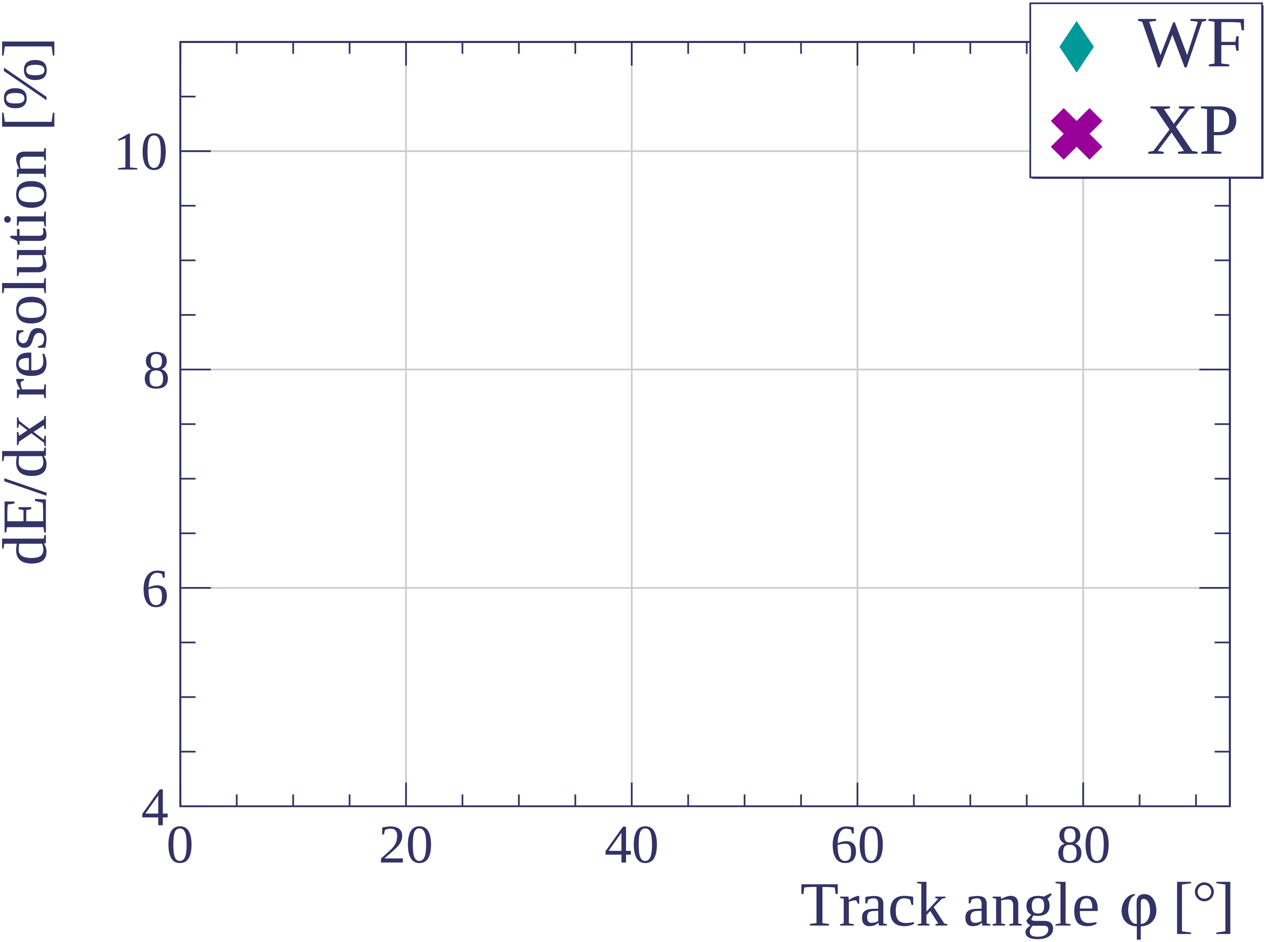
WF
XP
Mean dE/dx [ADC counts/cm]
600
500
400
300
0
500
1000
1500
Track length [mm]



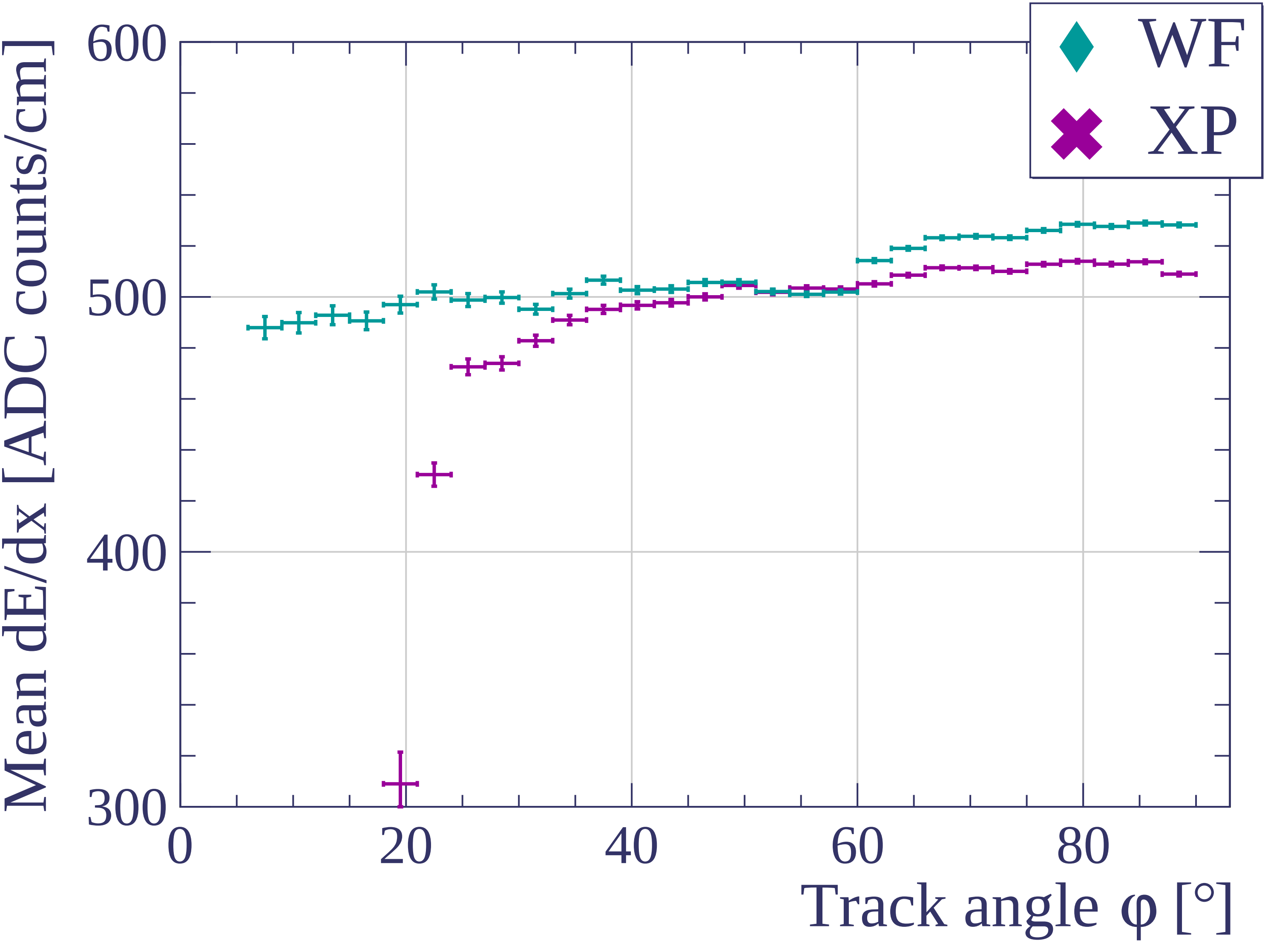
dE/dx with XP [ADC counts/cm]
Track length [cm]
0
200
400
600
800
1000
0
50
100
150
0
50
100


Signed momentum (MeV)
2000
1500
1000
500
0
0
50
100
150
Track length [cm]
600
400
200
0

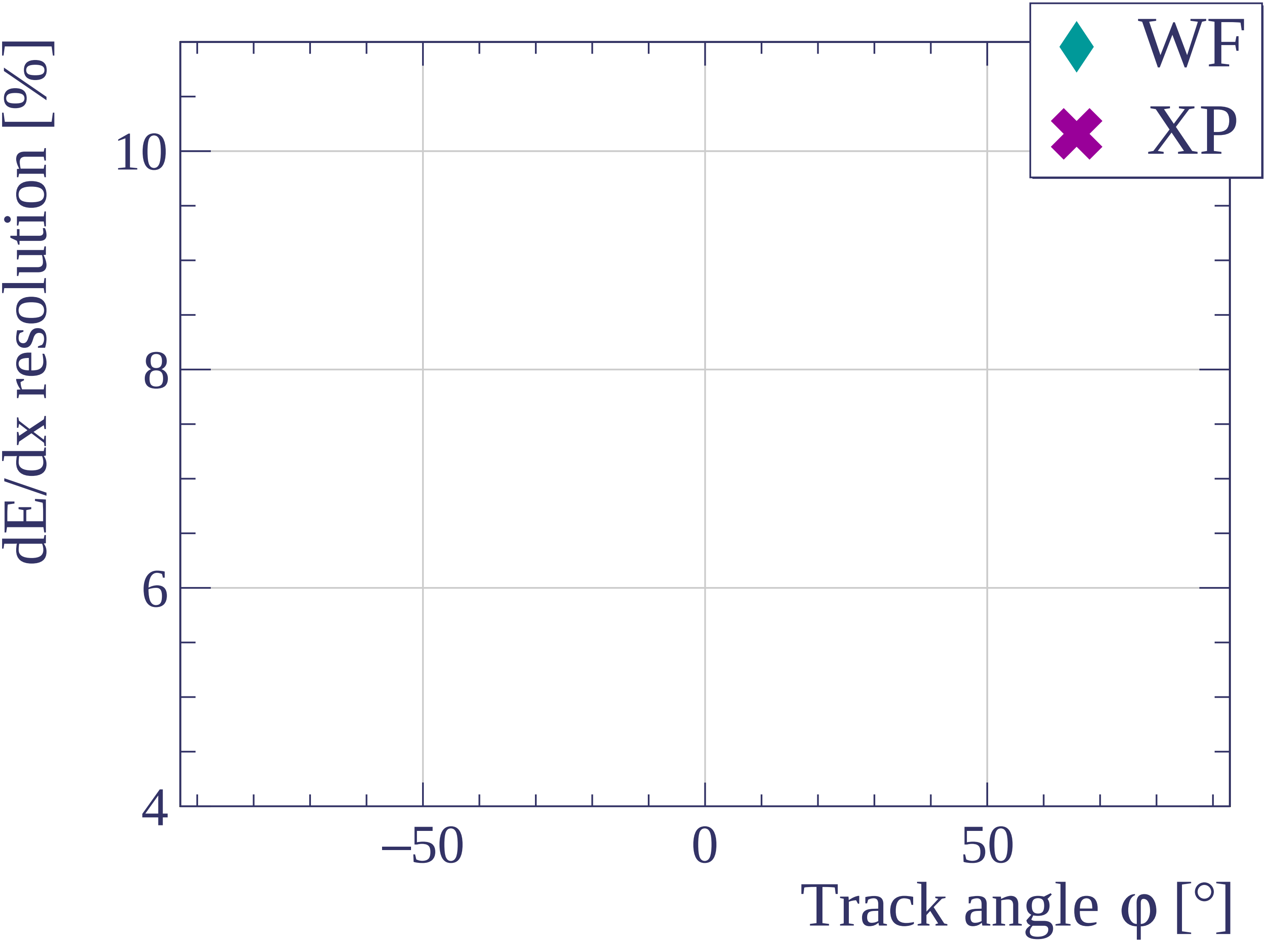

Signed momentum (MeV)
2000
1500
1000
500
0
0
50
100
150
200
$N_{clusters}$
600
400
200
0

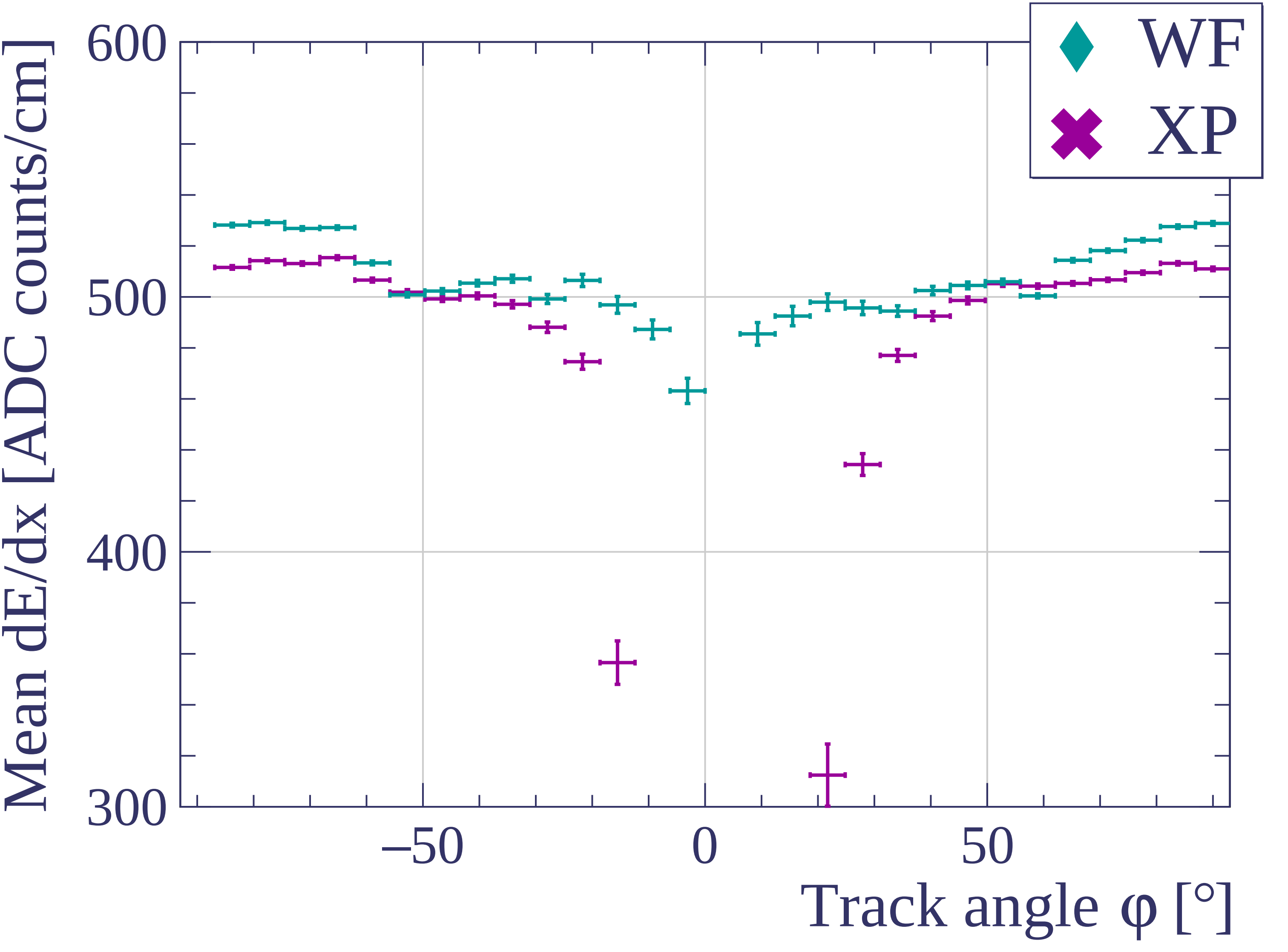


dE/dx resolution [%]
10
8
6
4
0
20
40
60
80
Track angle φ [°]
WF
XP

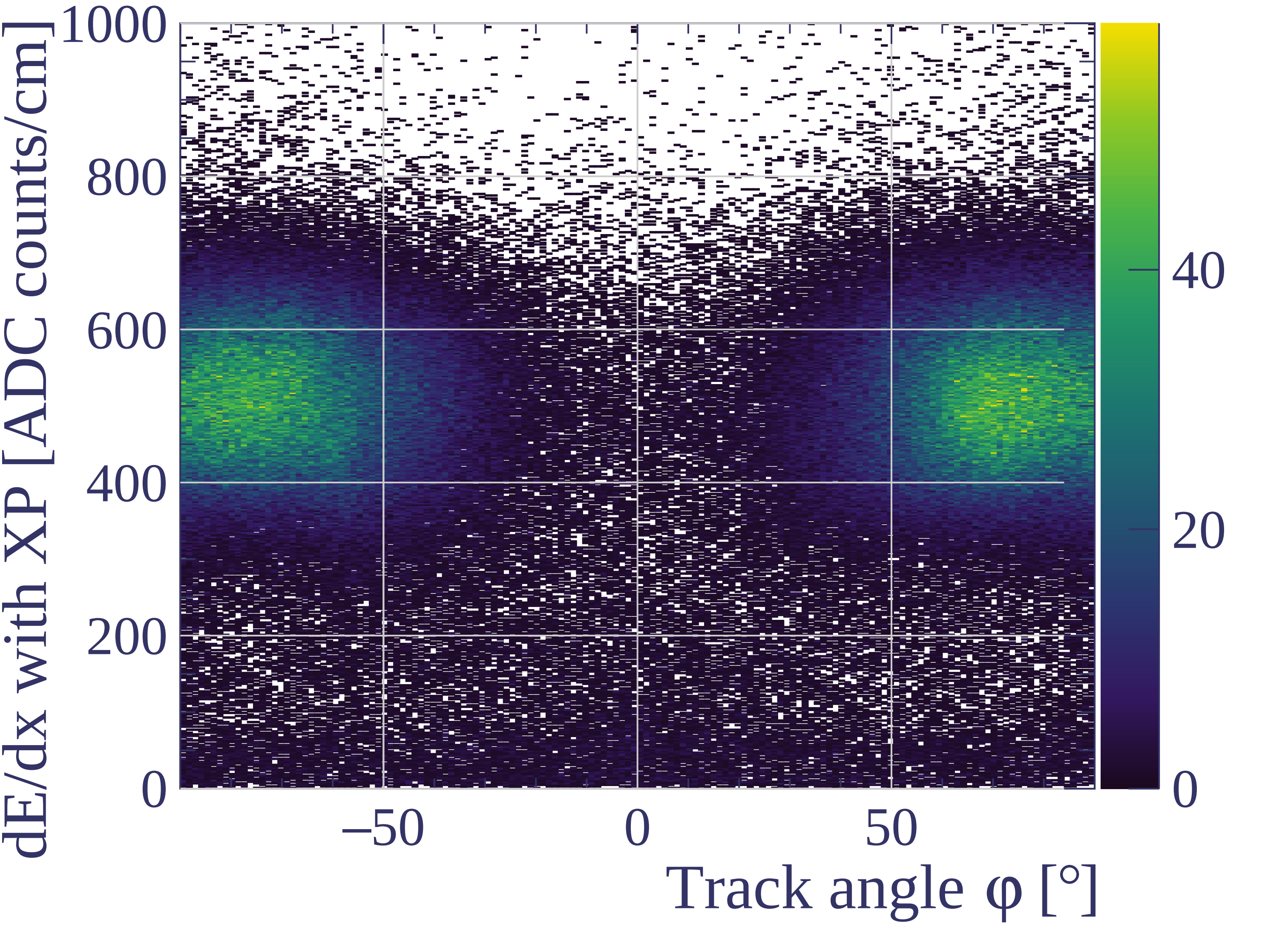


WF
XP
Mean dE/dx [ADC counts/cm]
300
400
500
600
0
20
40
60
80
Track angle φ [°]

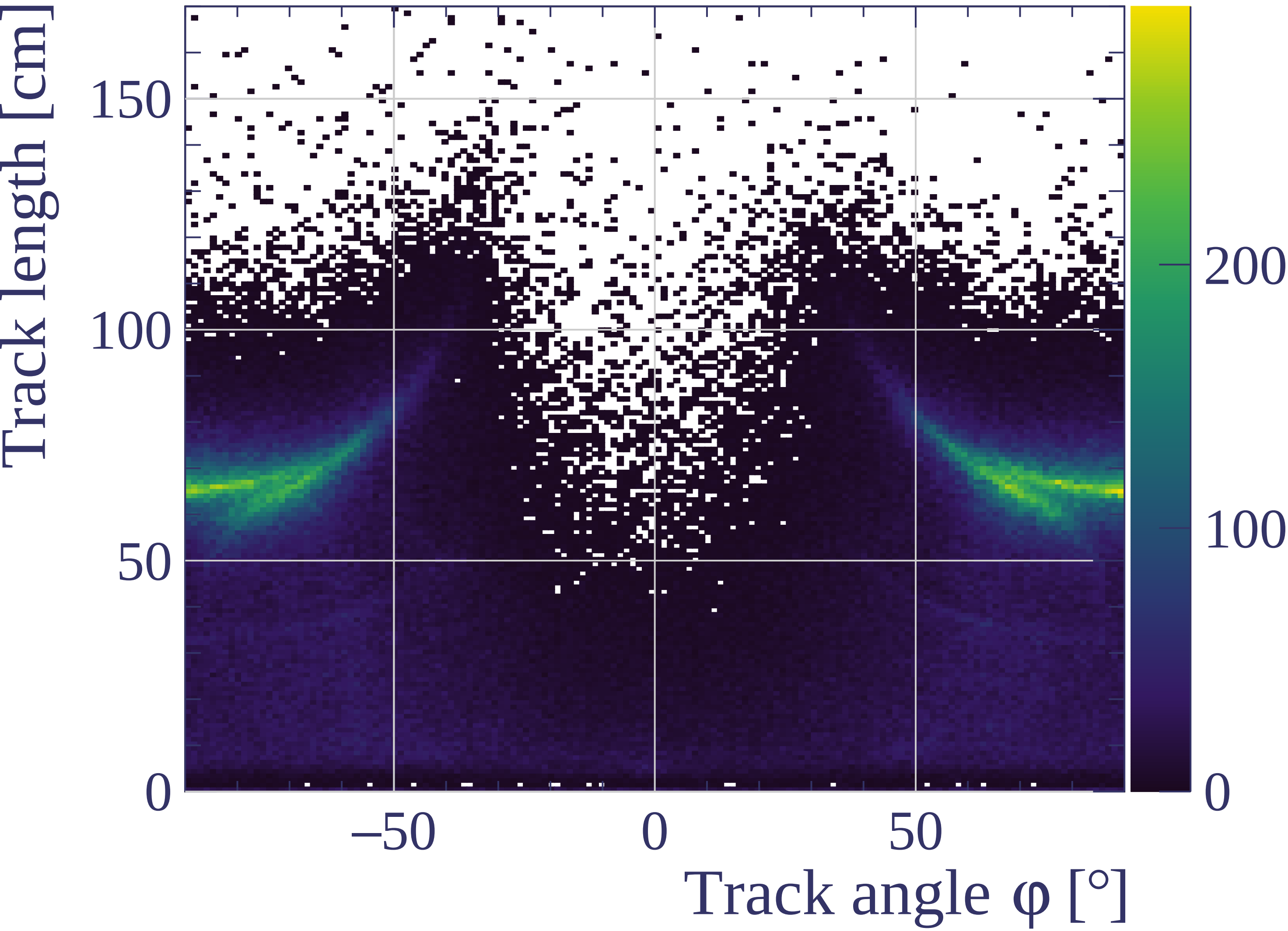

WF
XP
dE/dx resolution [%]
10
8
6
4
−50
0
50
Track angle φ [°]

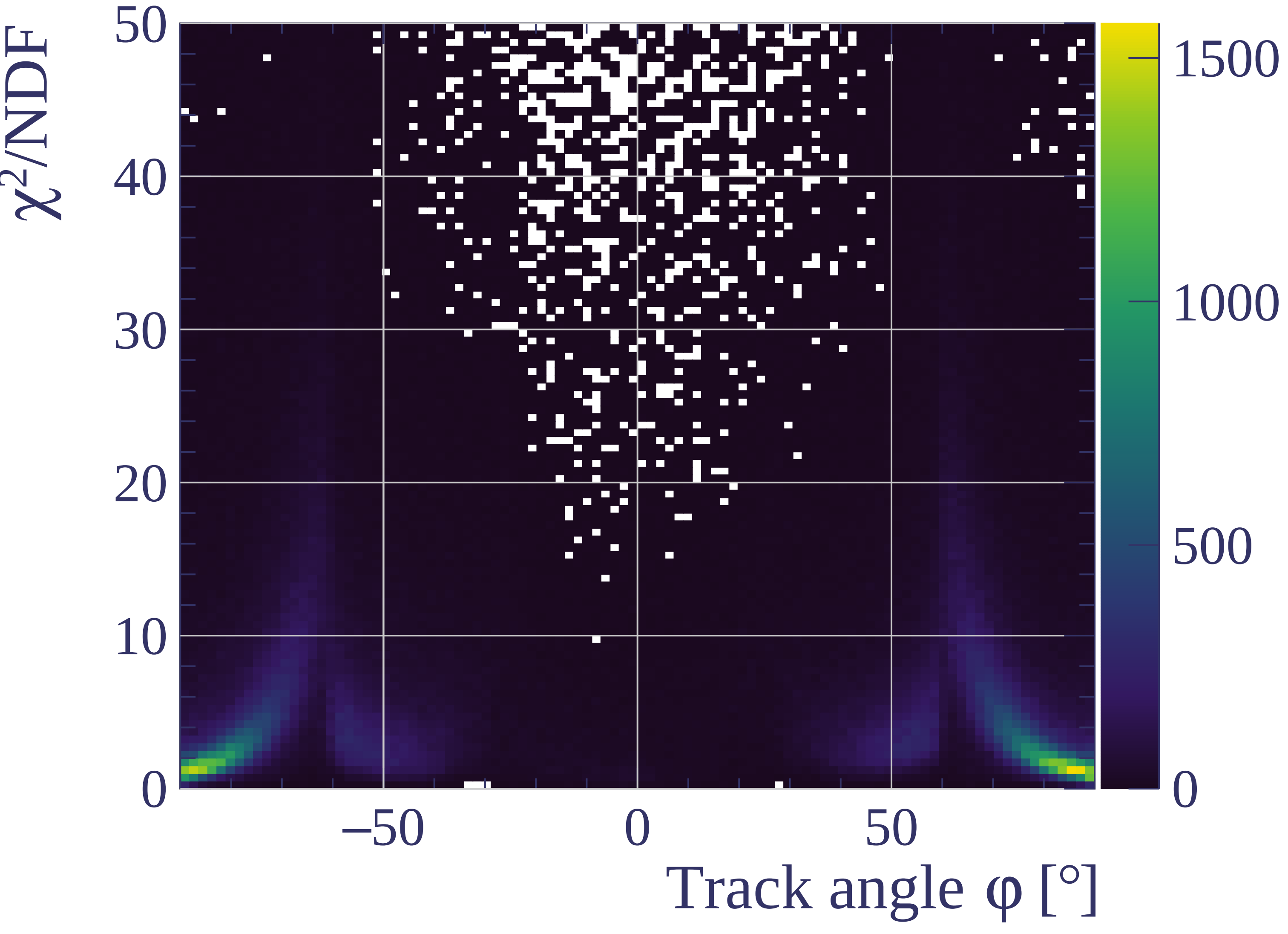


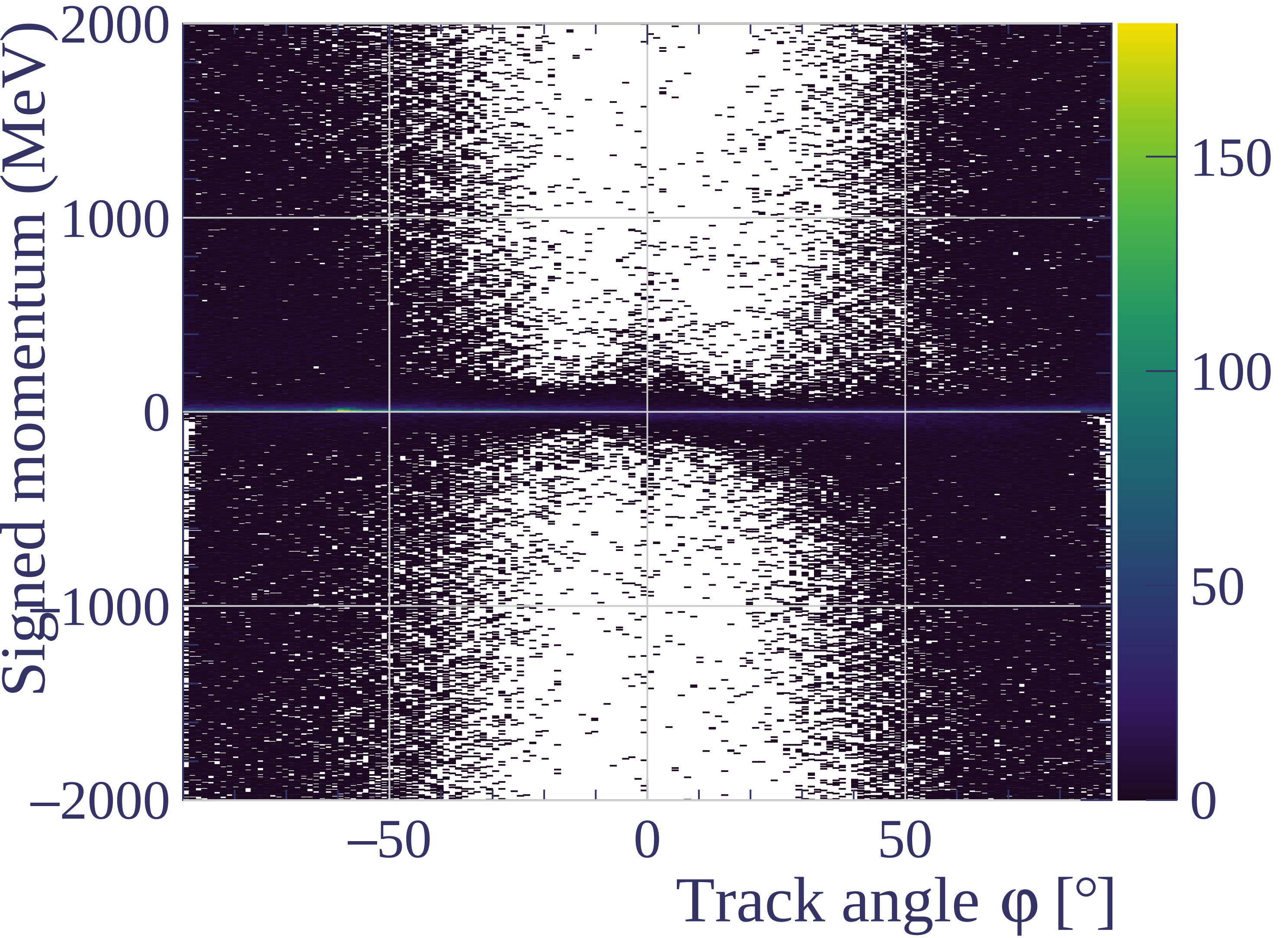
WF
XP
600
500
400
300
Mean dE/dx [ADC counts/cm]
−50
0
50
Track angle φ [°]

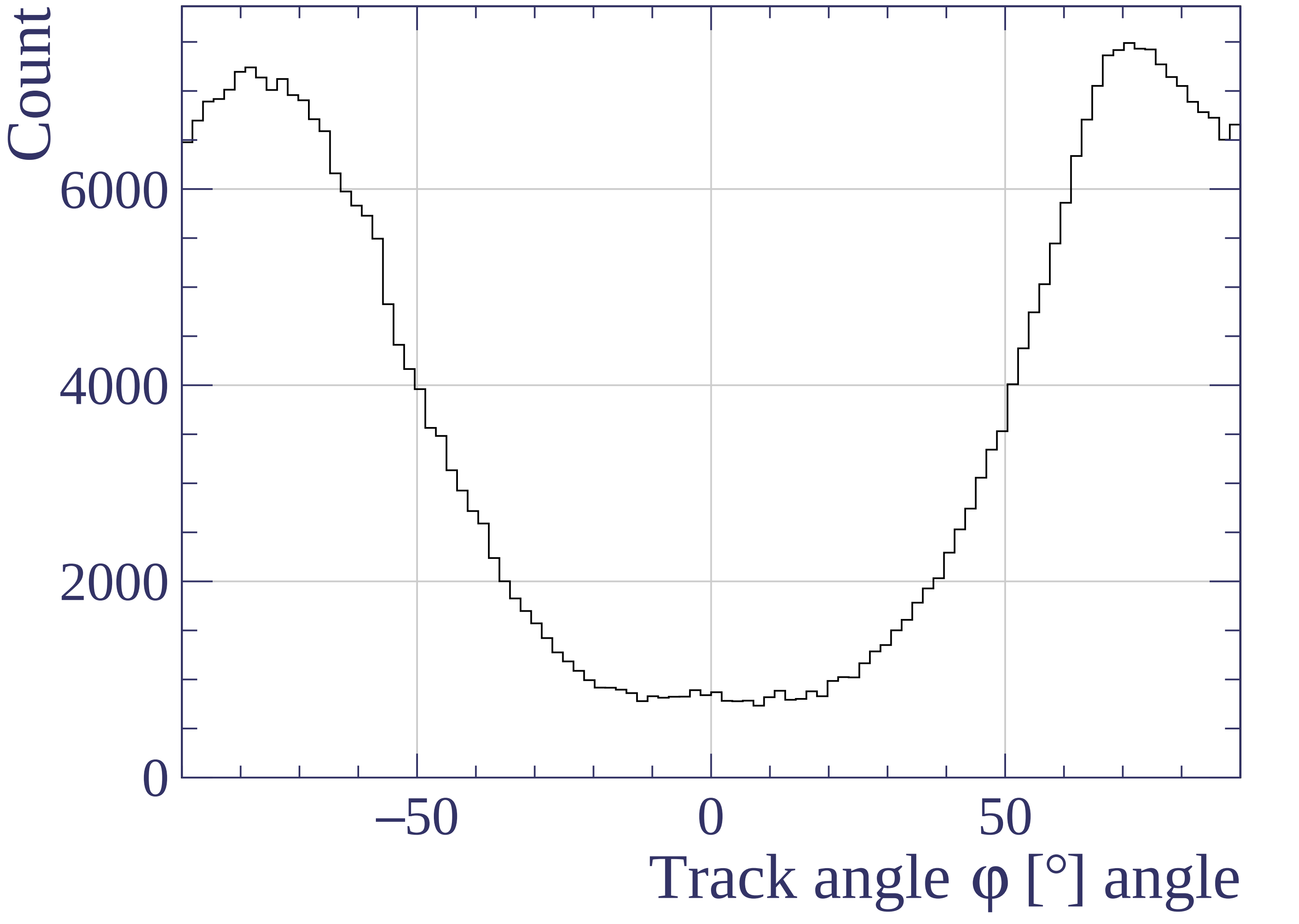


dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
−50
0
50
Track angle φ [°]
40
20
0

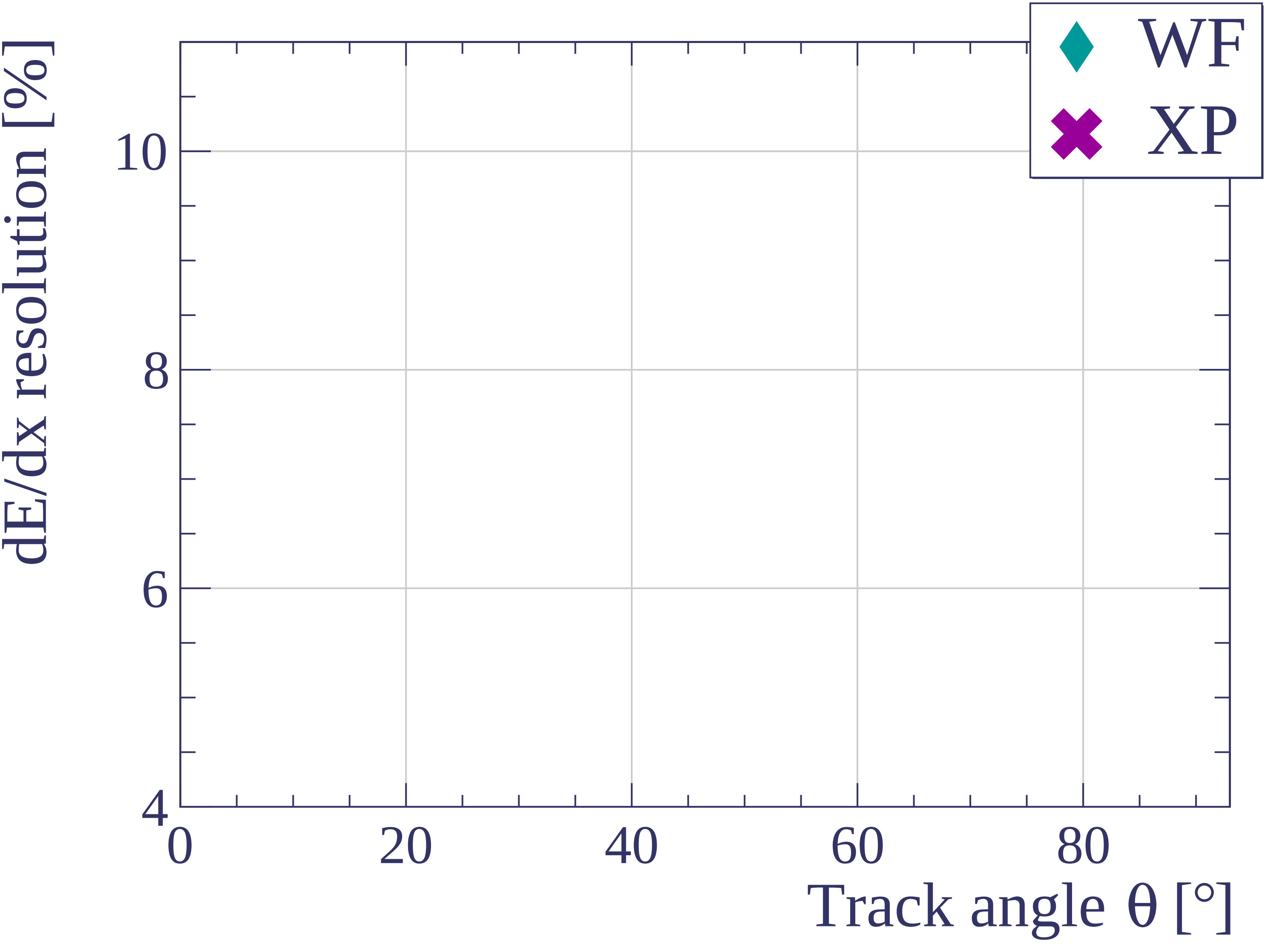

Track length [cm]
150
100
50
0
−50
0
50
Track angle φ [°]
200
100
0


χ^2/NDF
50
40
30
20
10
0
−50
0
50
Track angle φ [°]
1500
1000
500
0

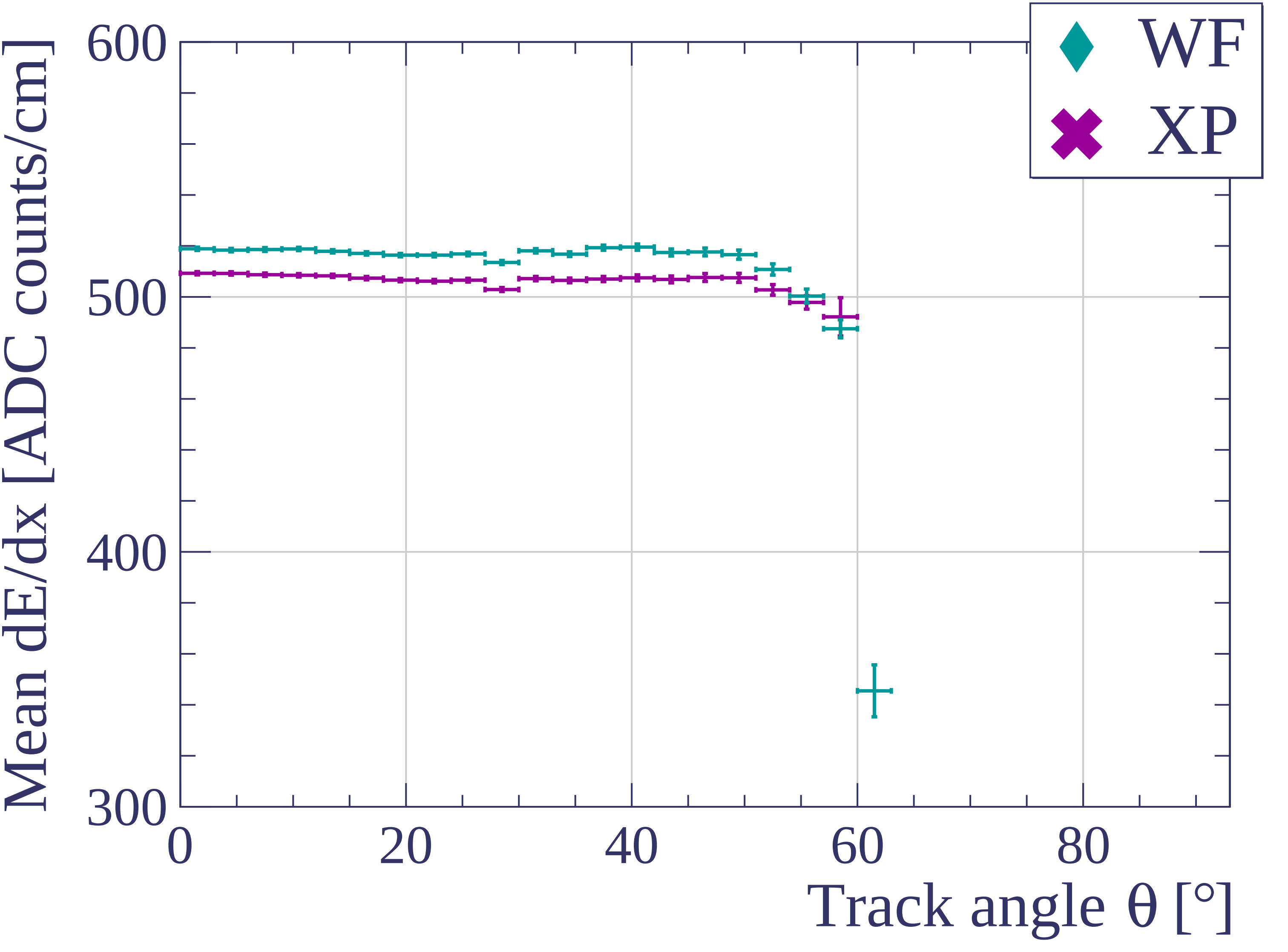

Signed momentum (MeV)
2000
1000
0
−1000
−2000
−50
0
50
Track angle φ [°]
150
100
50
0

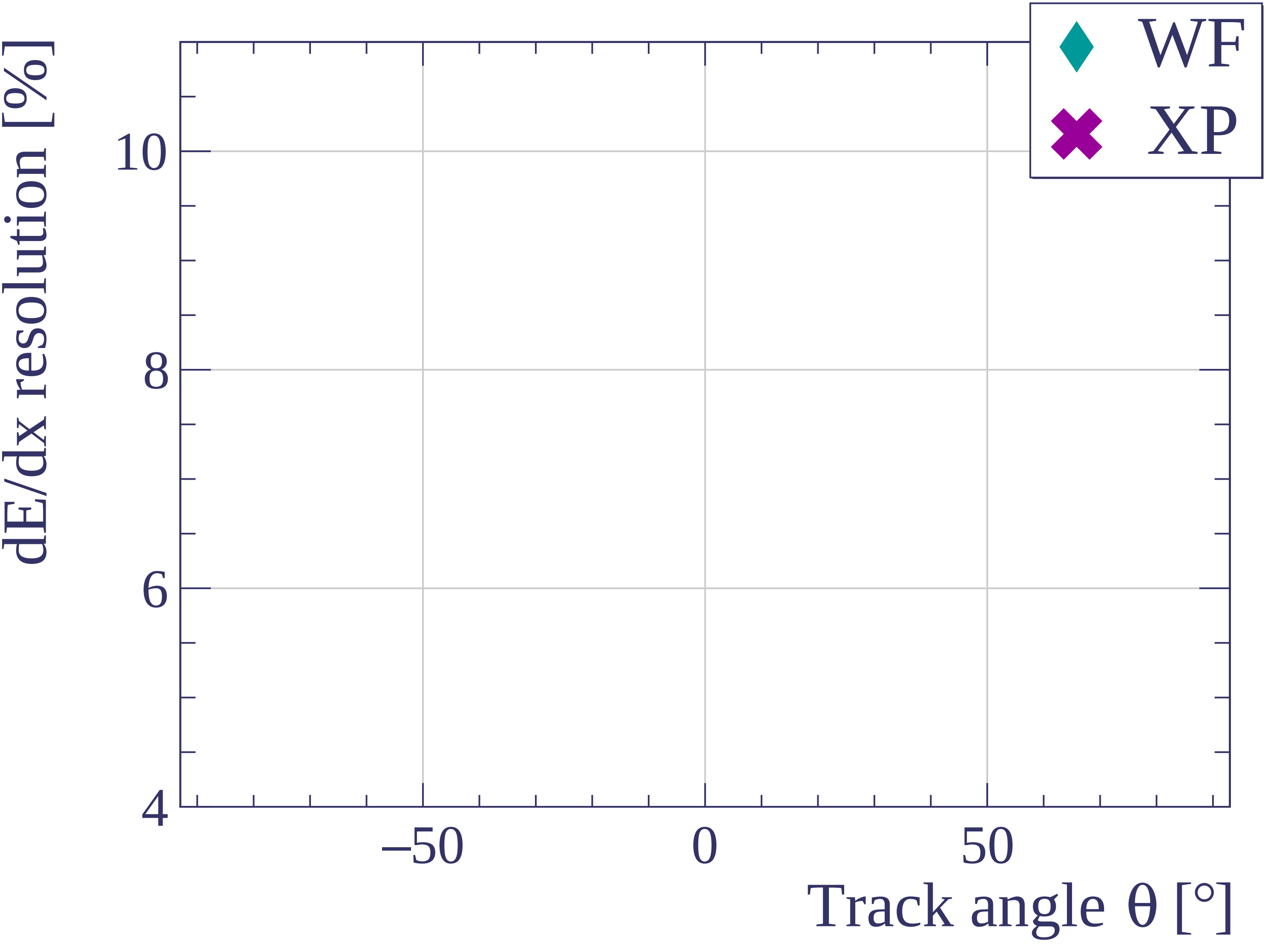


Count
6000
4000
2000
0
−50
0
50
Track angle φ [°] angle

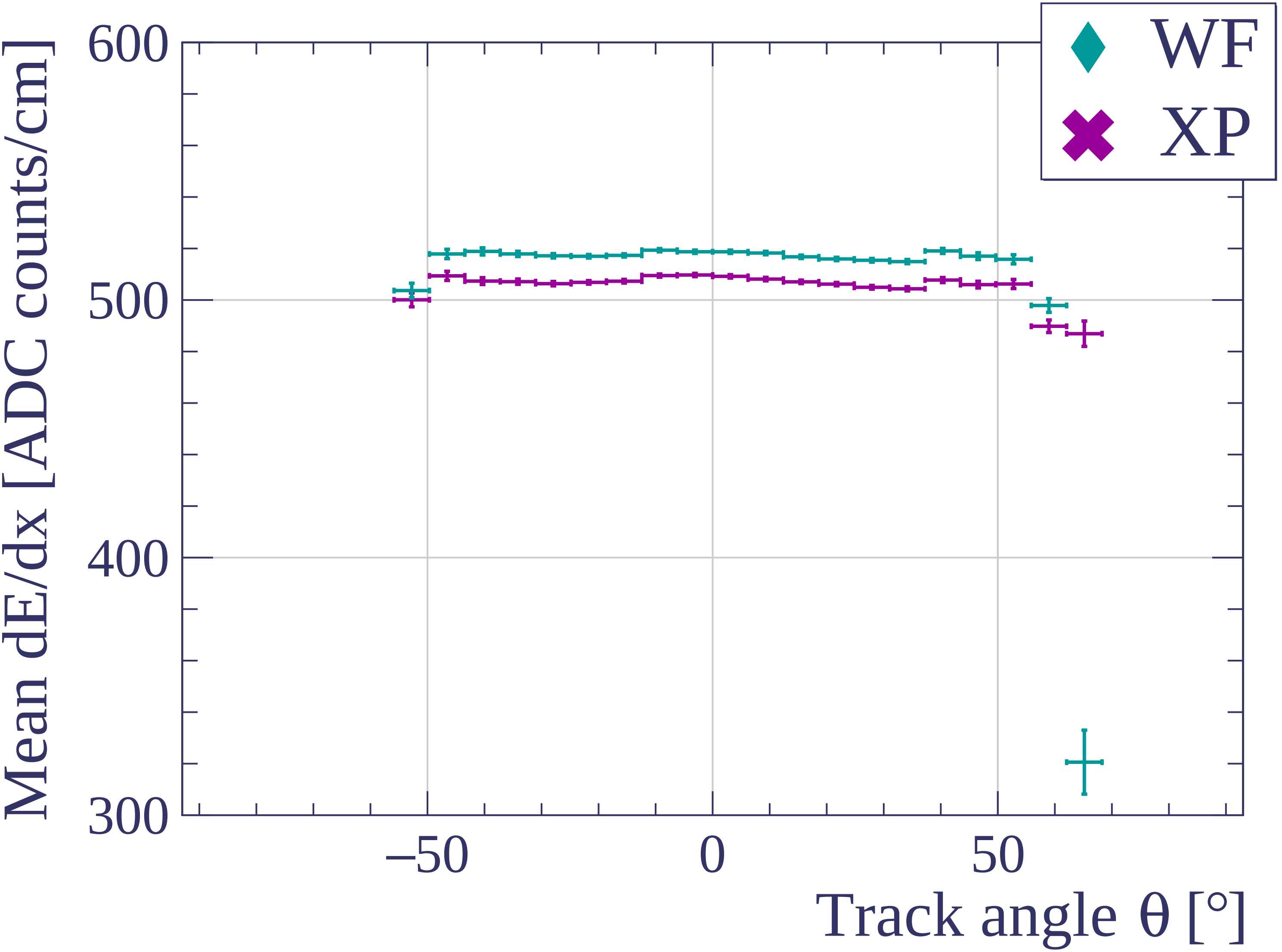


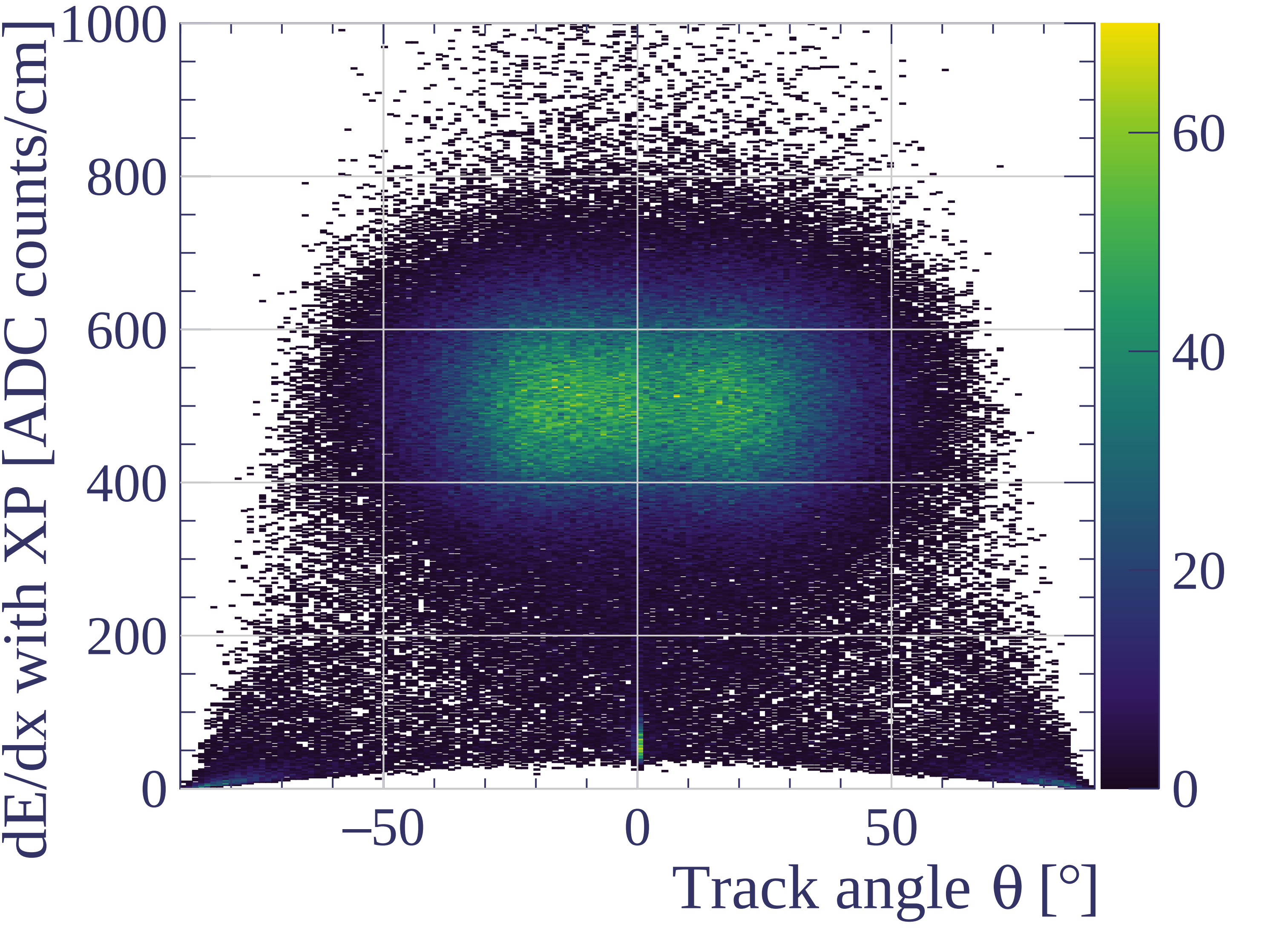
dE/dx resolution [%]
10
8
6
4
0
20
40
60
80
Track angle θ [°]
WF
XP


WF
XP
Mean dE/dx [ADC counts/cm]
Track angle θ [°]
600
500
400
300
0
20
40
60
80

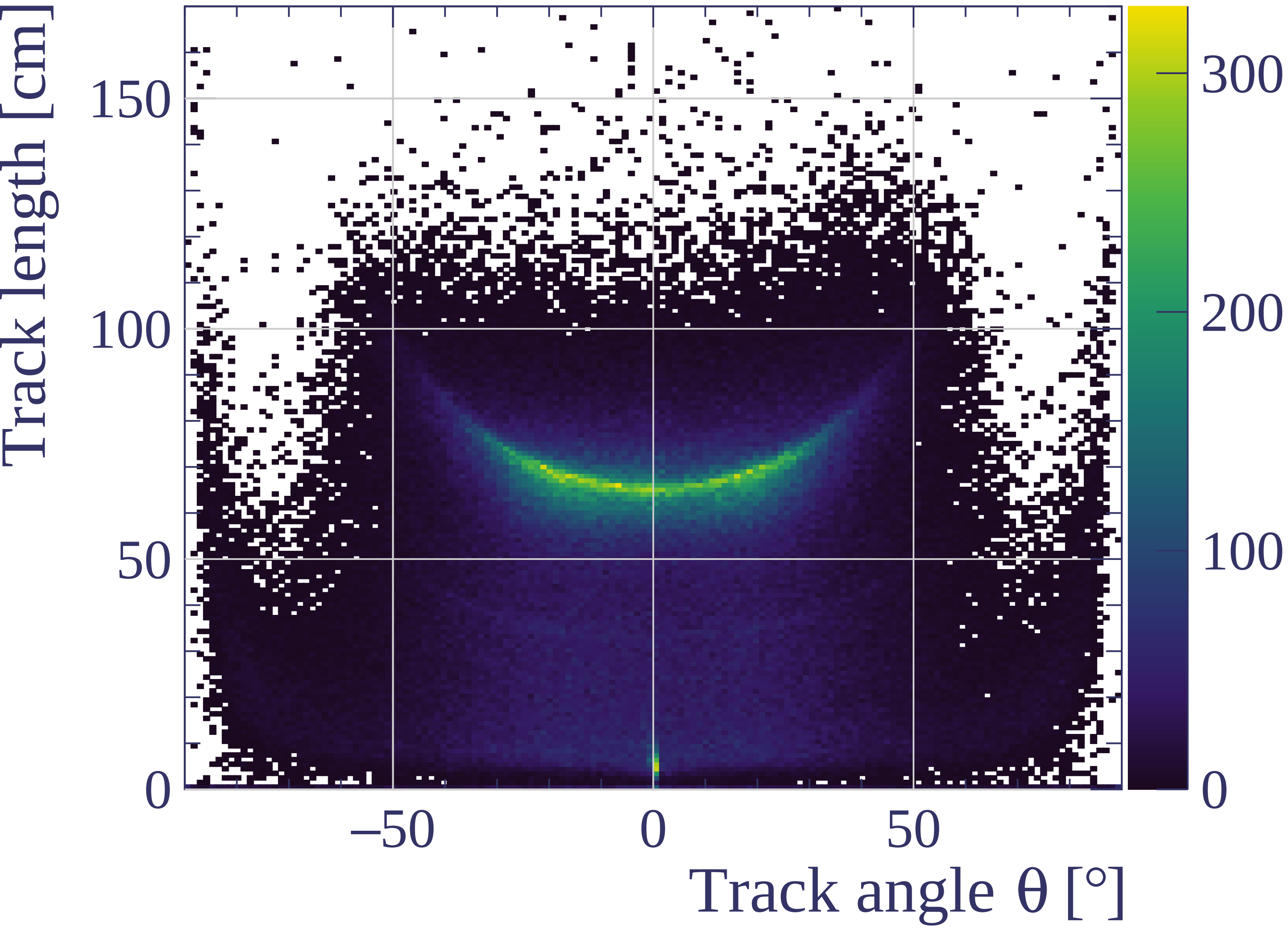


WF
XP
dE/dx resolution [%]
10
8
6
4
−50
0
50
Track angle θ [°]

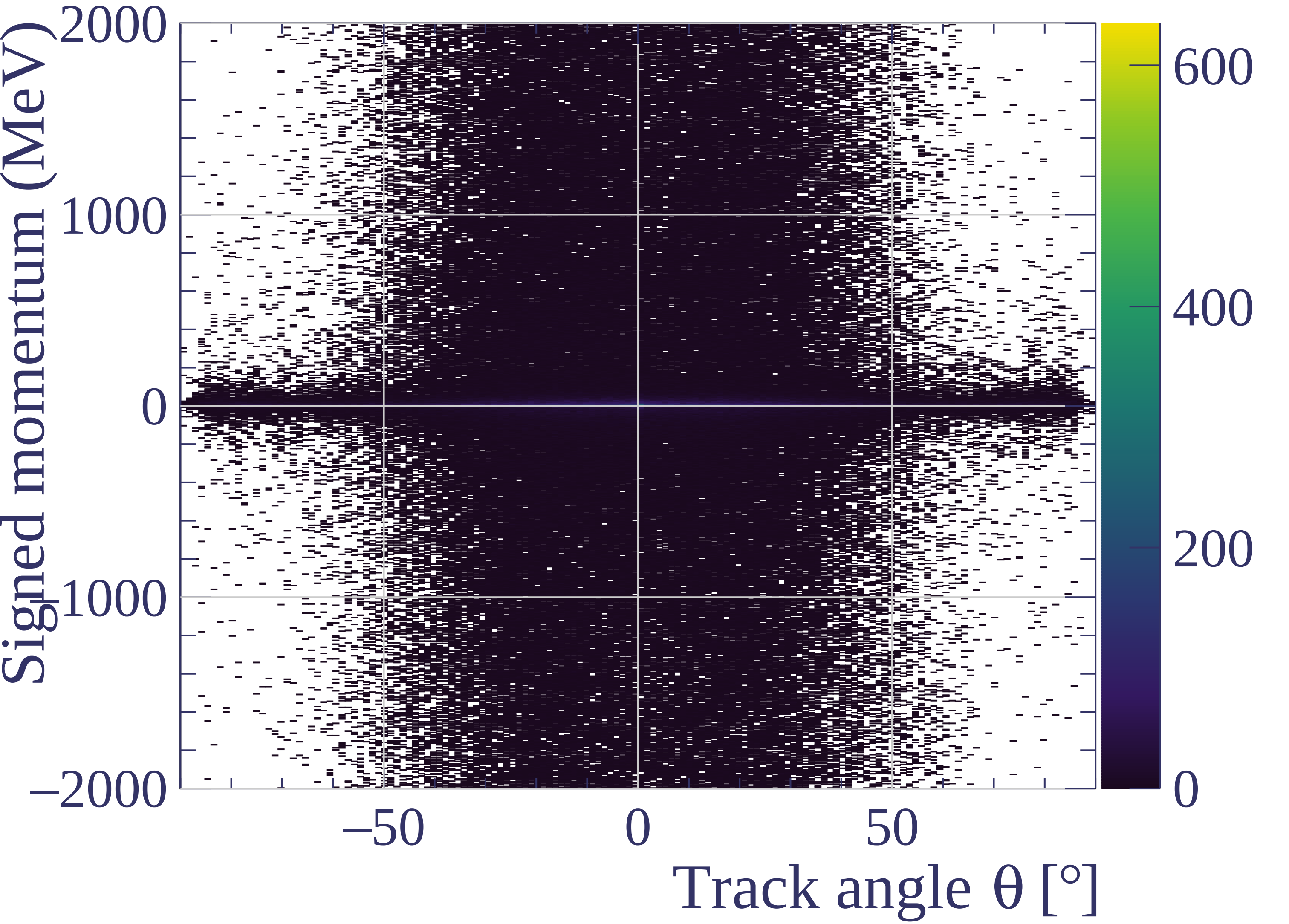

WF
XP
Mean dE/dx [ADC counts/cm]
600
500
400
300
−50
0
50
Track angle θ [°]



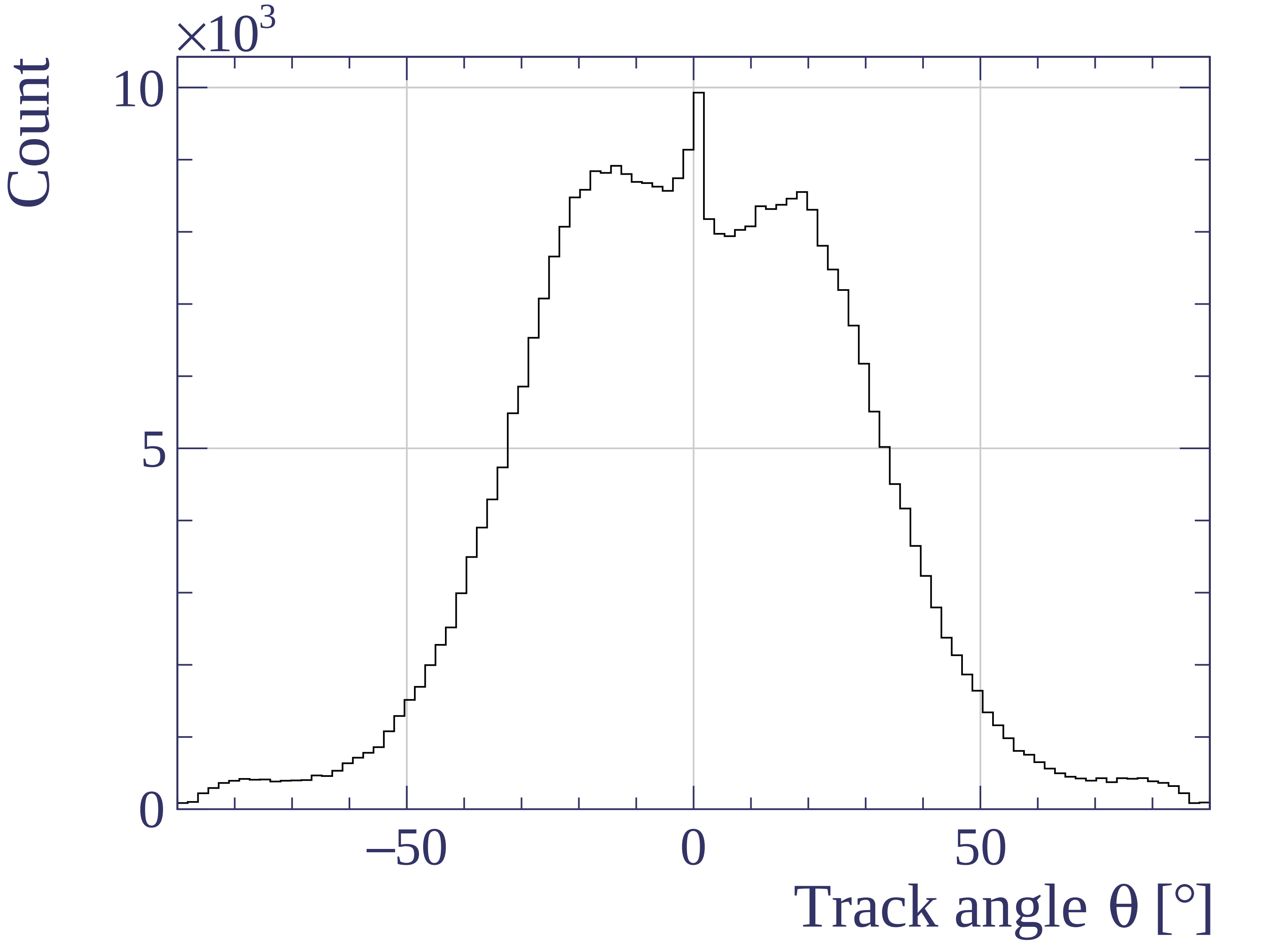
dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
−50
0
50
Track angle θ [°]
60
40
20
0



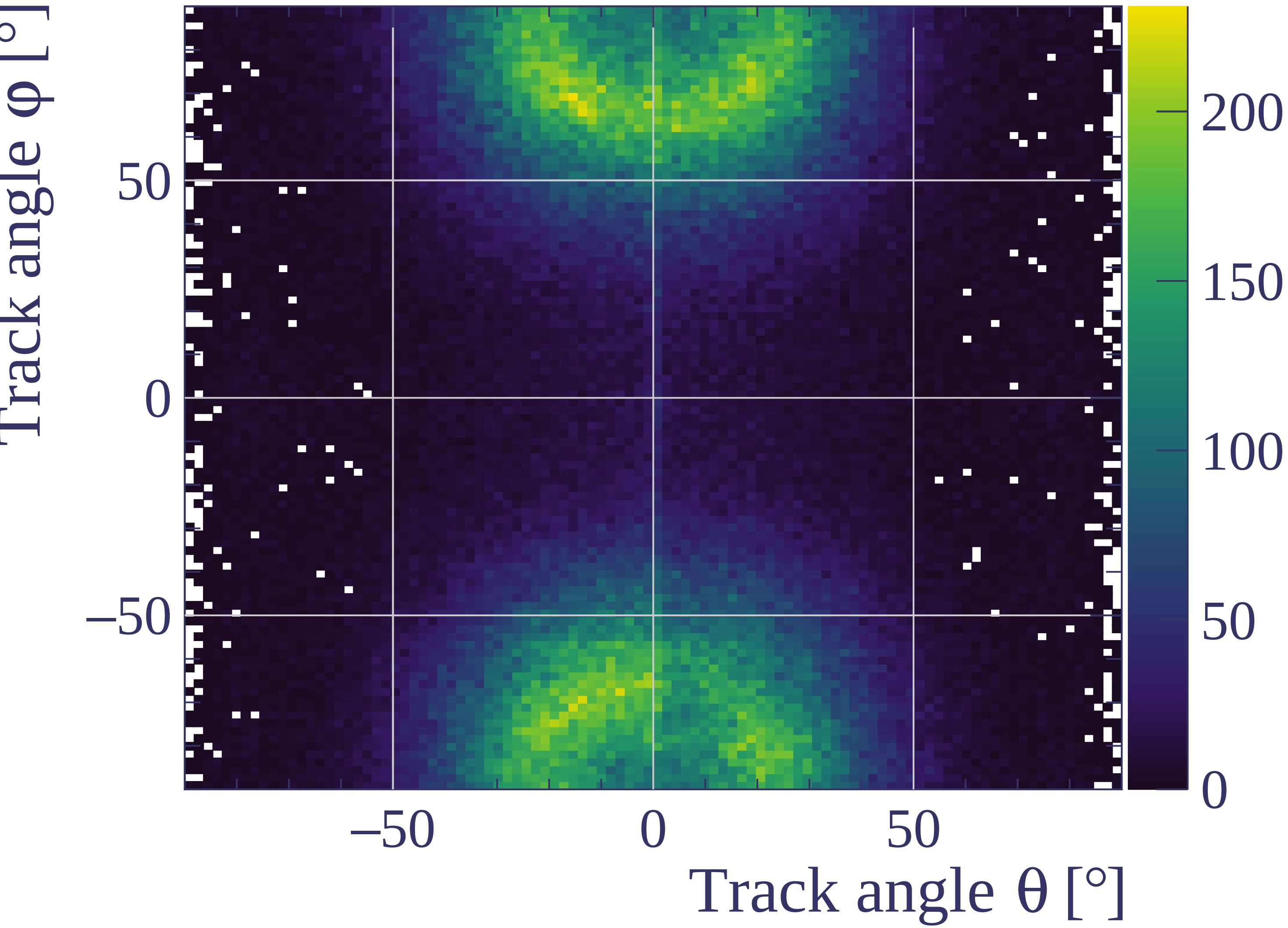
Track length [cm]
150
100
50
0
−50
0
50
Track angle θ [°]
300
200
100
0



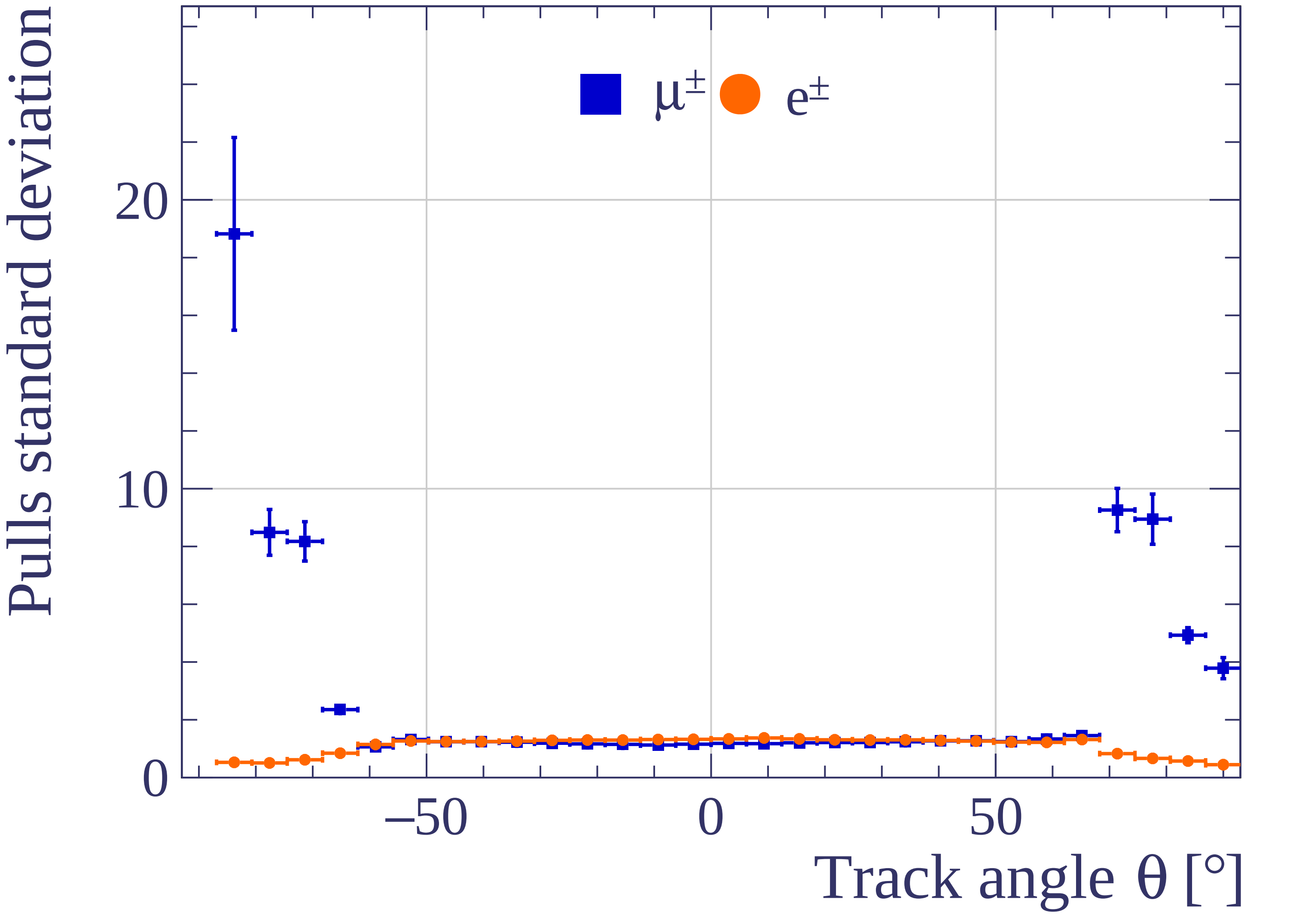
Signed momentum (MeV)
2000
1000
0
−1000
−2000
−50
0
50
Track angle θ [°]
600
400
200
0



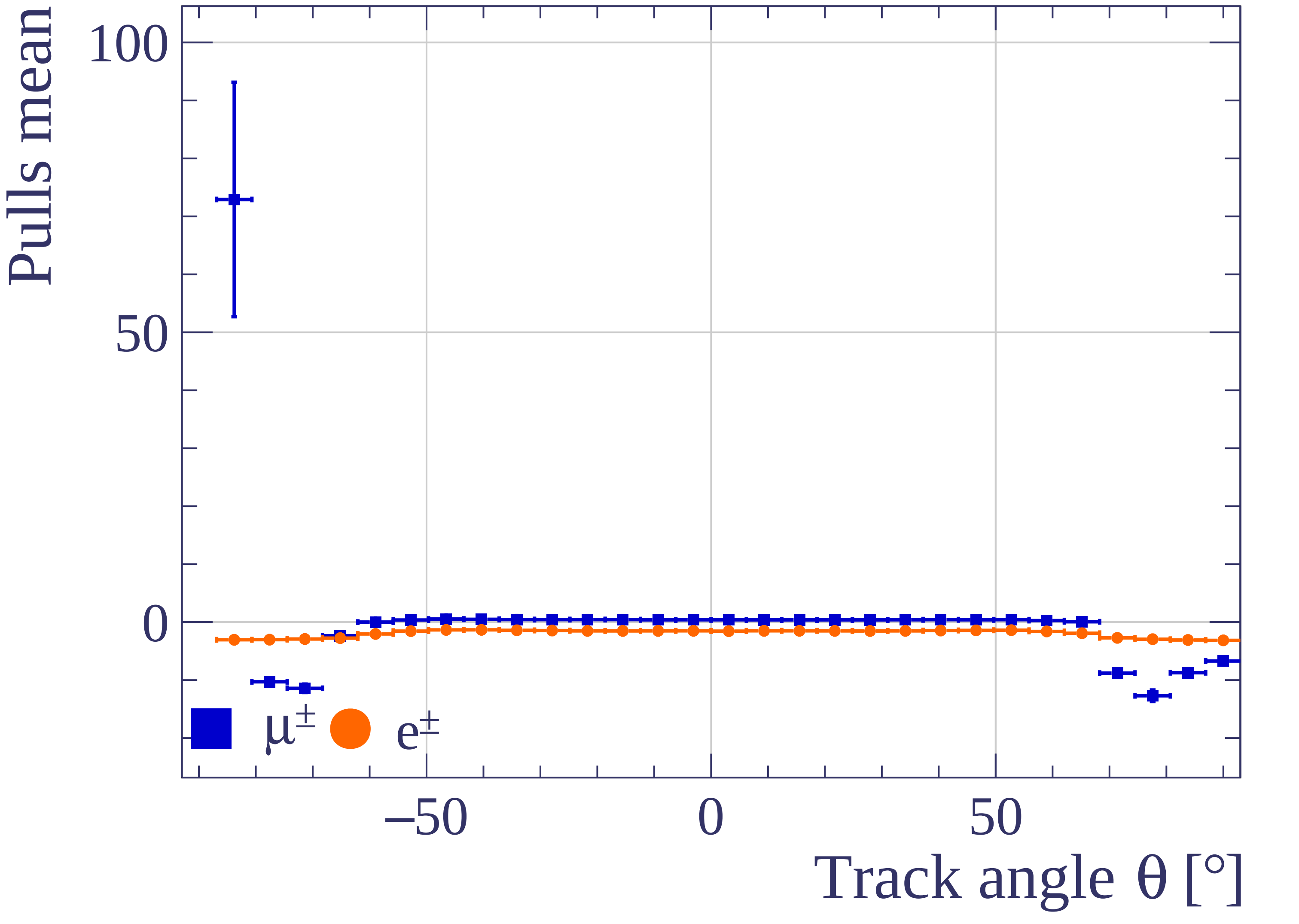
×10^3
Count
10
5
0
−50
0
50
Track angle θ [°]

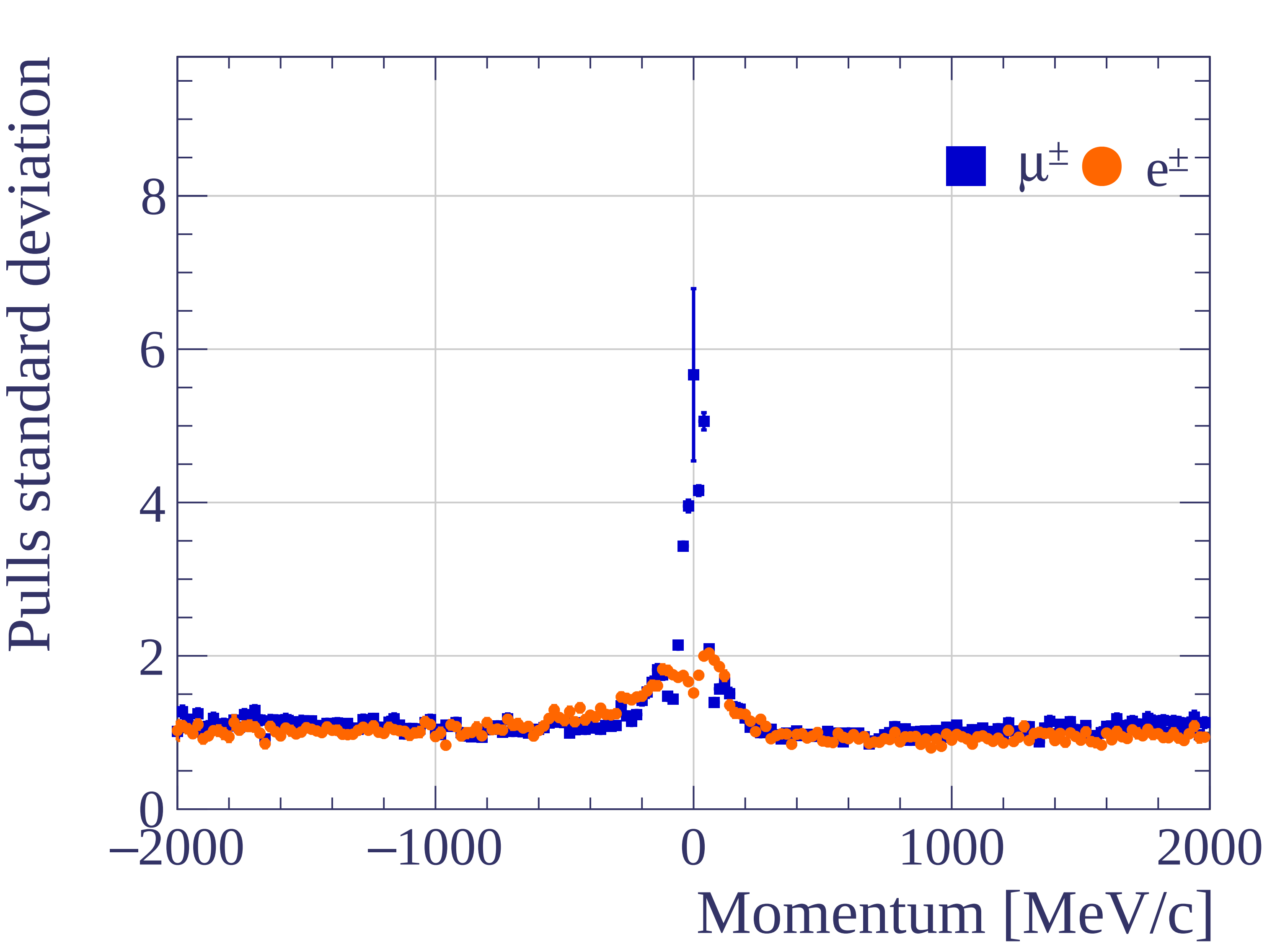


Track angle φ [°]
50
0
−50
−50
0
50
Track angle θ [°]
200
150
100
50
0

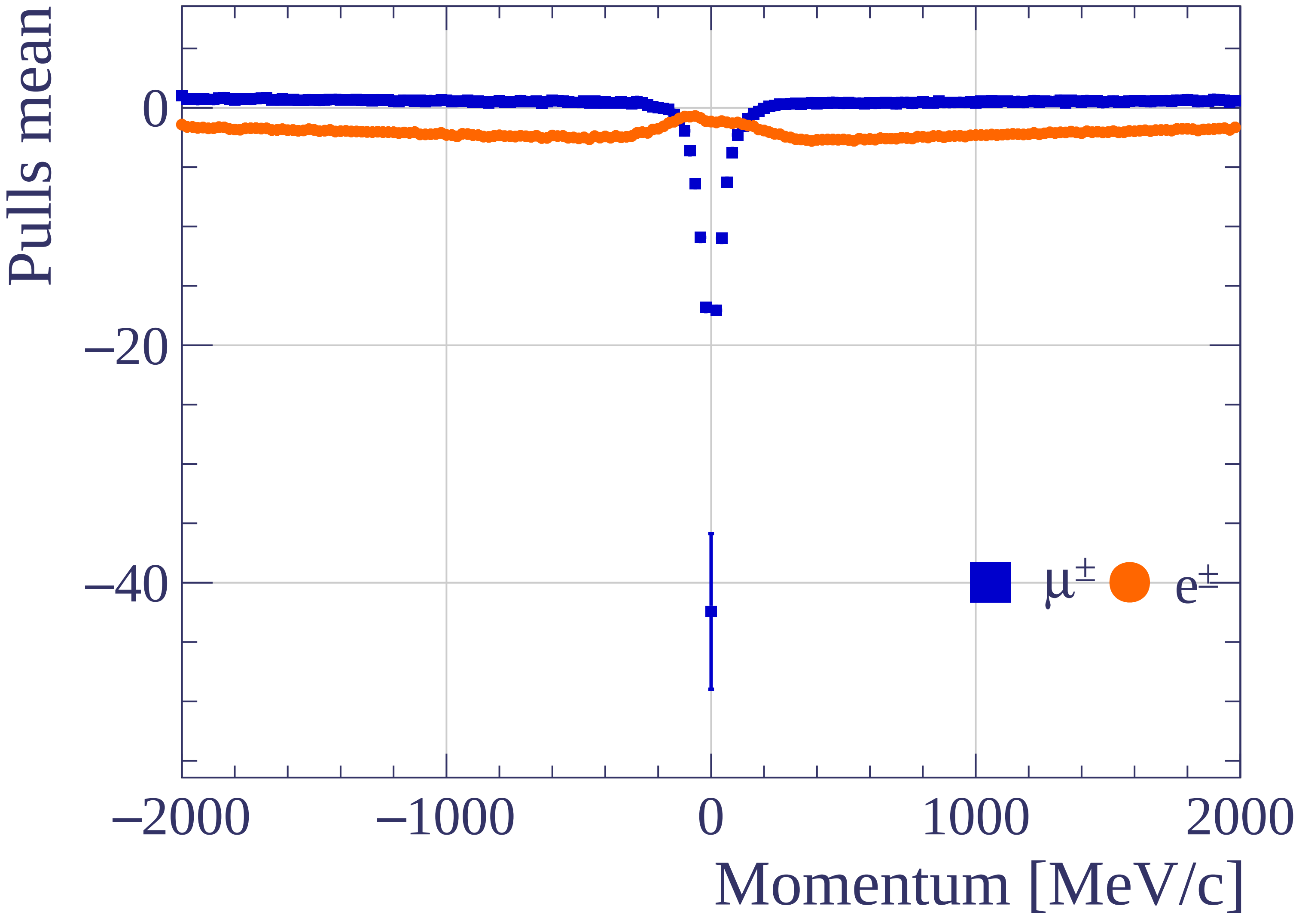


Pulls standard deviation
0
10
20
−50
0
50
Track angle θ [°]
μ±
e±

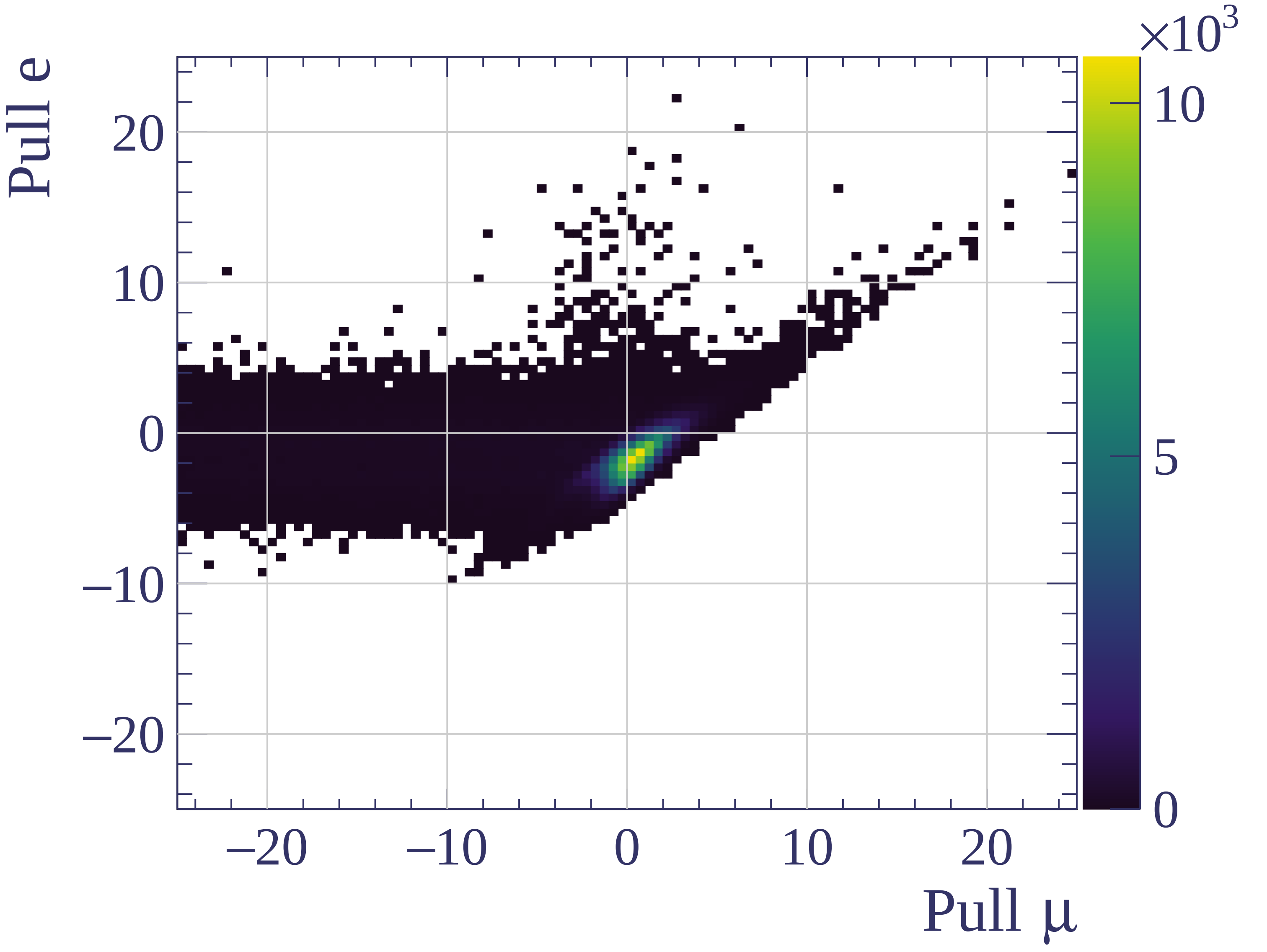

Pulls mean
100
50
0
−50
0
50
Track angle θ [°]
μ±
e±

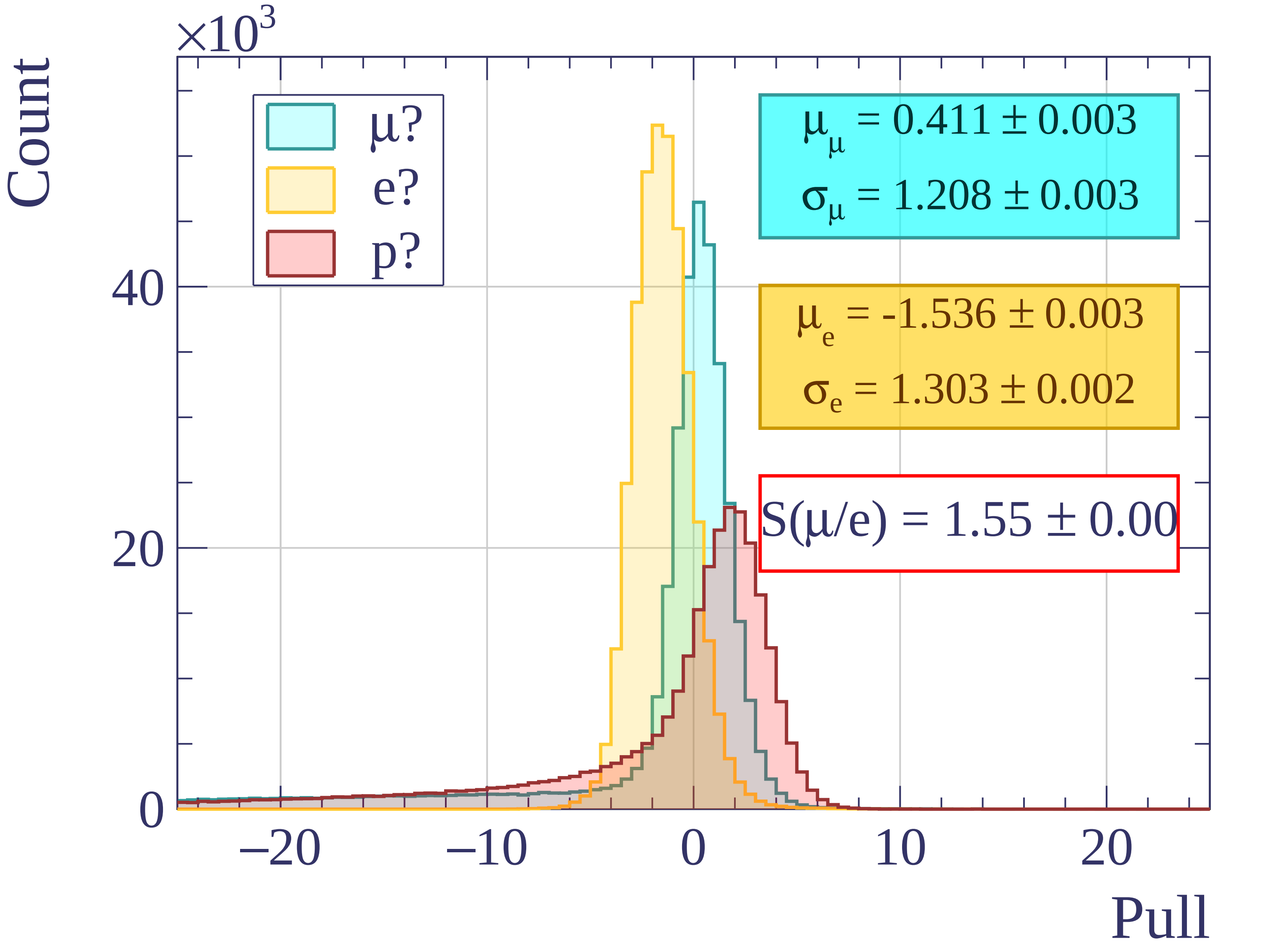


Pulls standard deviation
Momentum [MeV/c]
μ±
e±
0
2
4
6
8
−2000
−1000
0
1000
2000

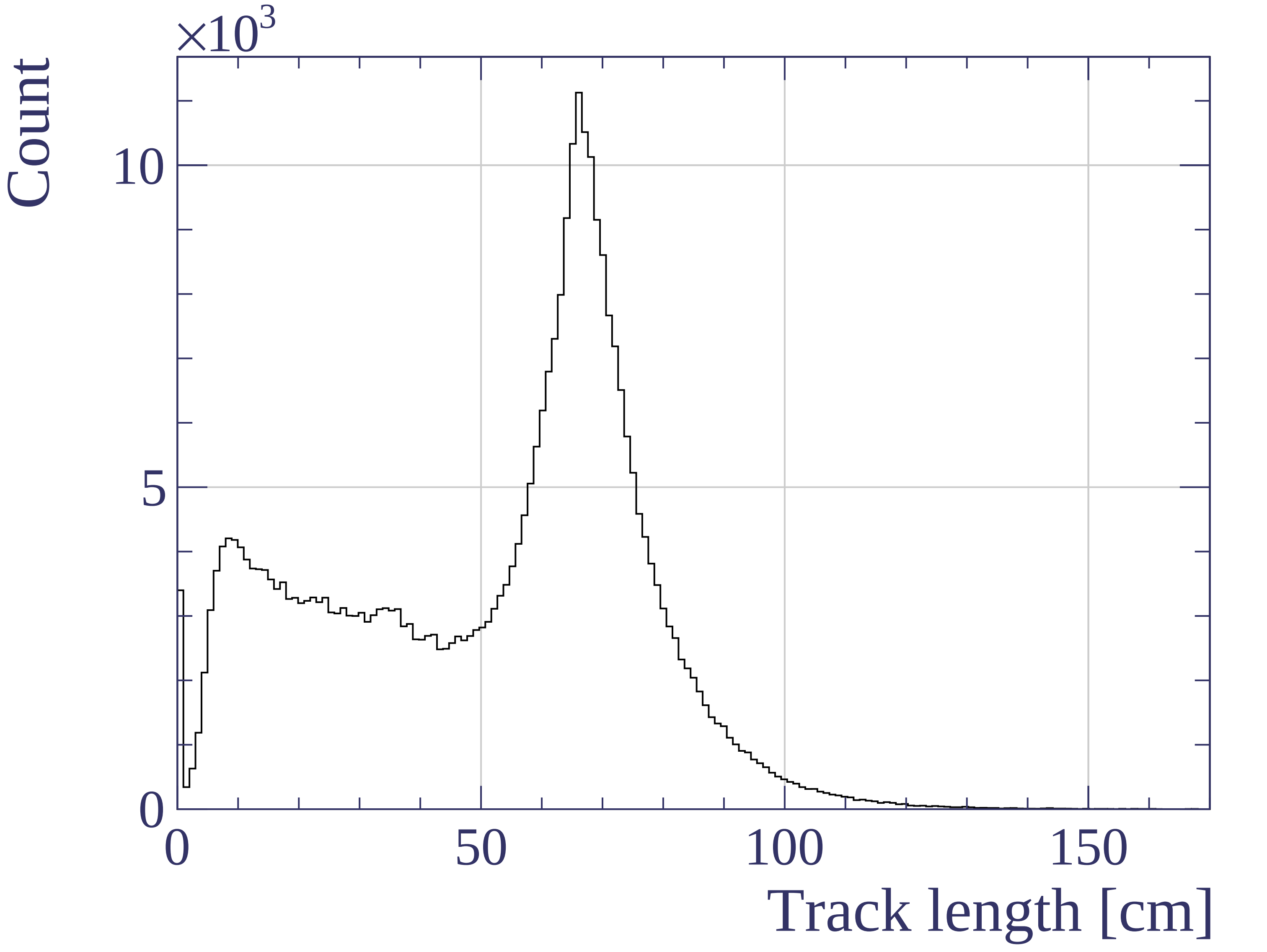


Pulls mean
0
−20
−40
−2000
−1000
0
1000
2000
Momentum [MeV/c]
μ±
e±

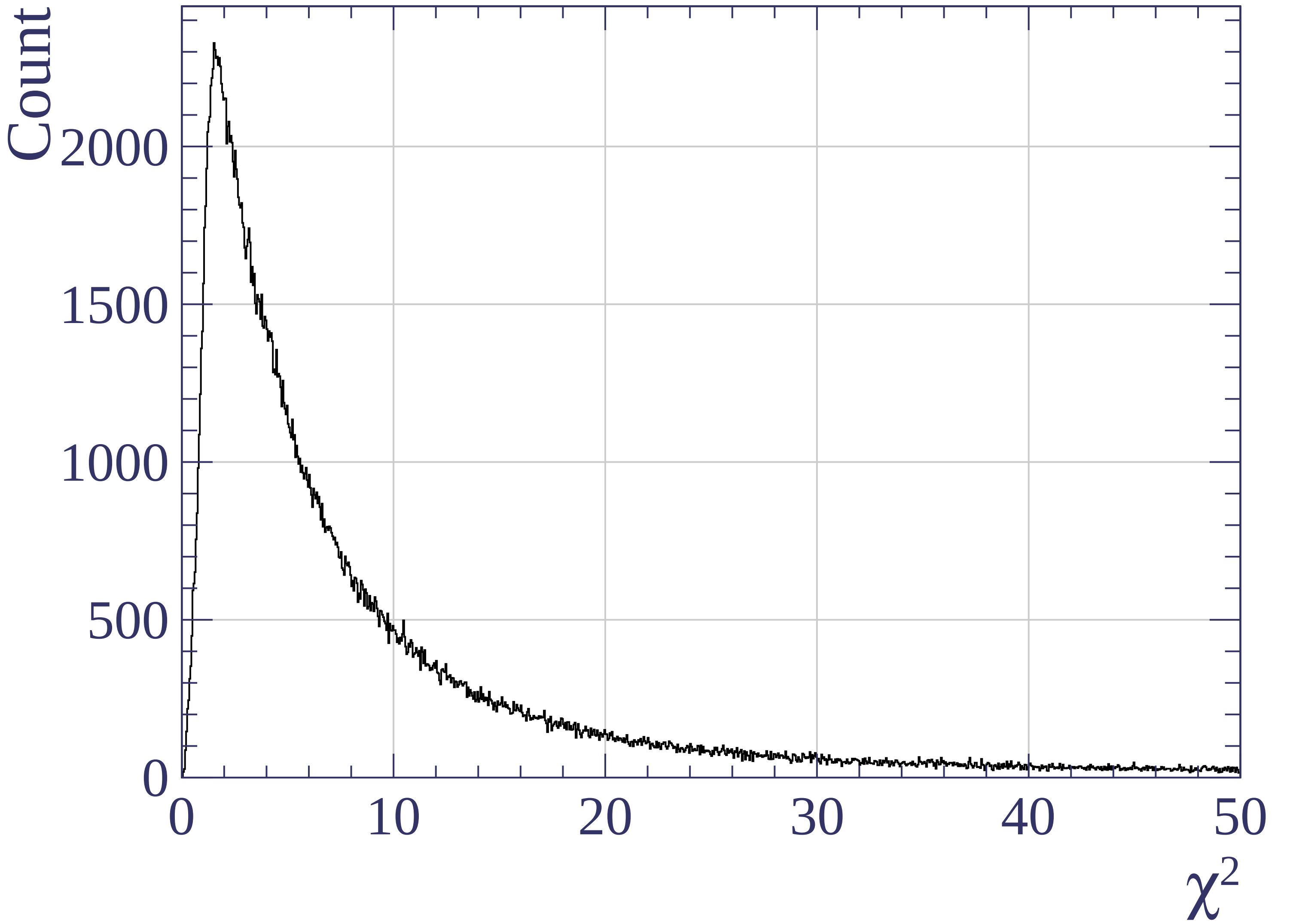

Pull e
Pull μ
20
10
0
−10
−20
−20
−10
0
10
20
×10³
10
5
0



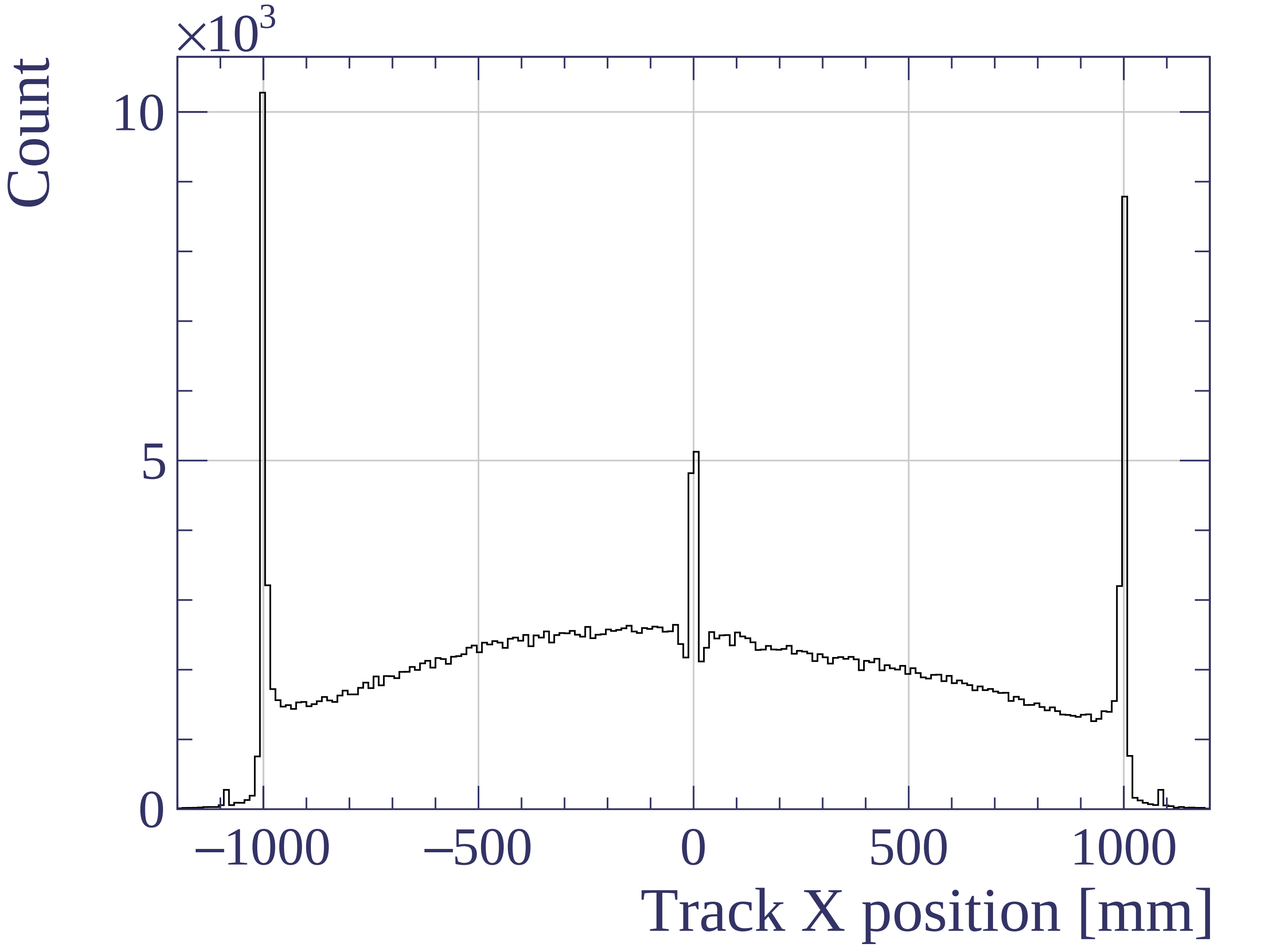
×10^3
Count
μ?
e?
p?
μ_μ = 0.411 ± 0.003
σ_μ = 1.208 ± 0.003
μ_e = -1.536 ± 0.003
σ_e = 1.303 ± 0.002
S(μ/e) = 1.55 ± 0.00
40
20
0
−20
−10
0
10
20
Pull



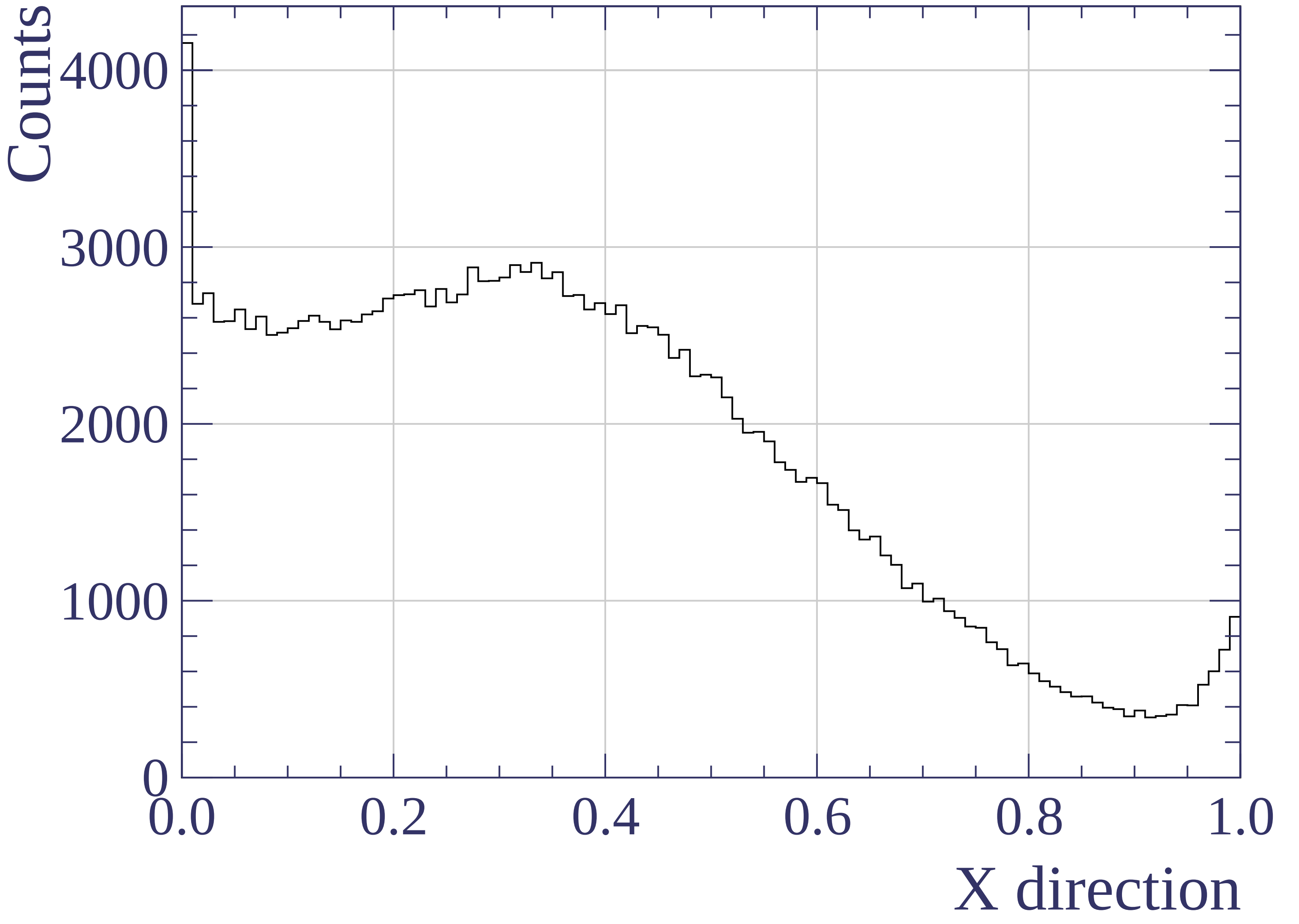
×10^3
Count
10
5
0
0
50
100
150
Track length [cm]



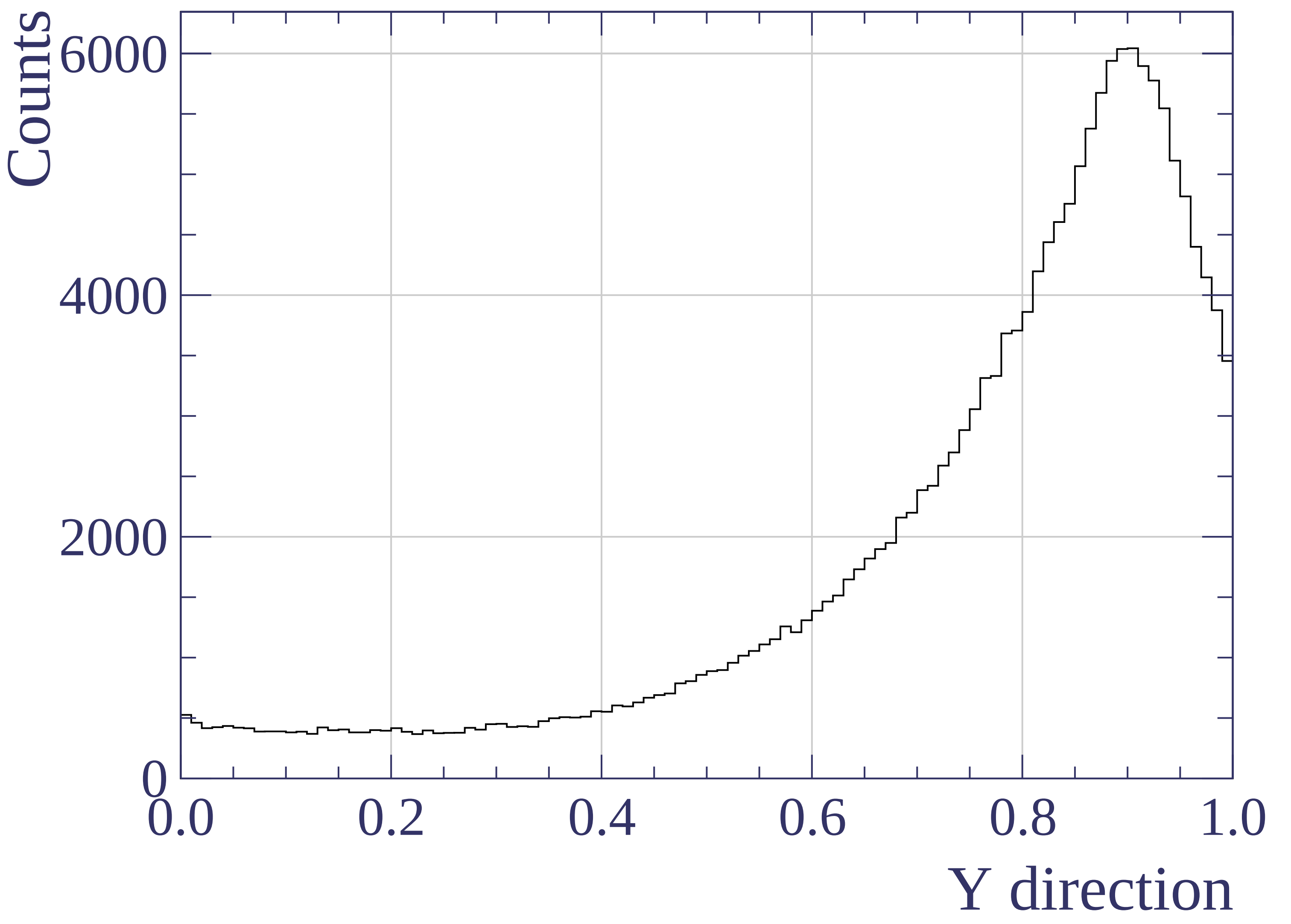
Count
2000
1500
1000
500
0
0
10
20
30
40
50
χ^2



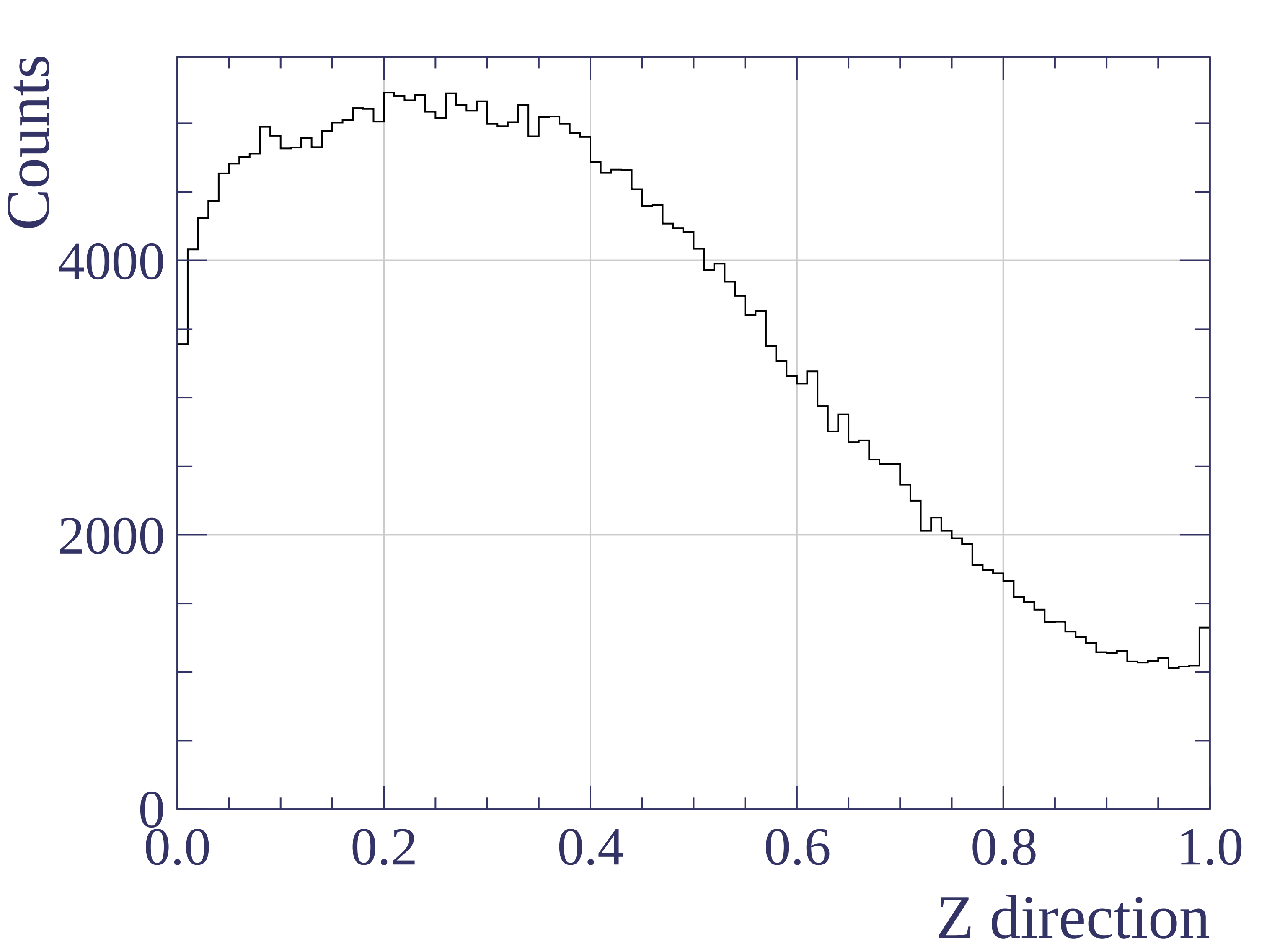
Count
×10^3
10
5
0
−1000
−500
0
500
1000
Track X position [mm]



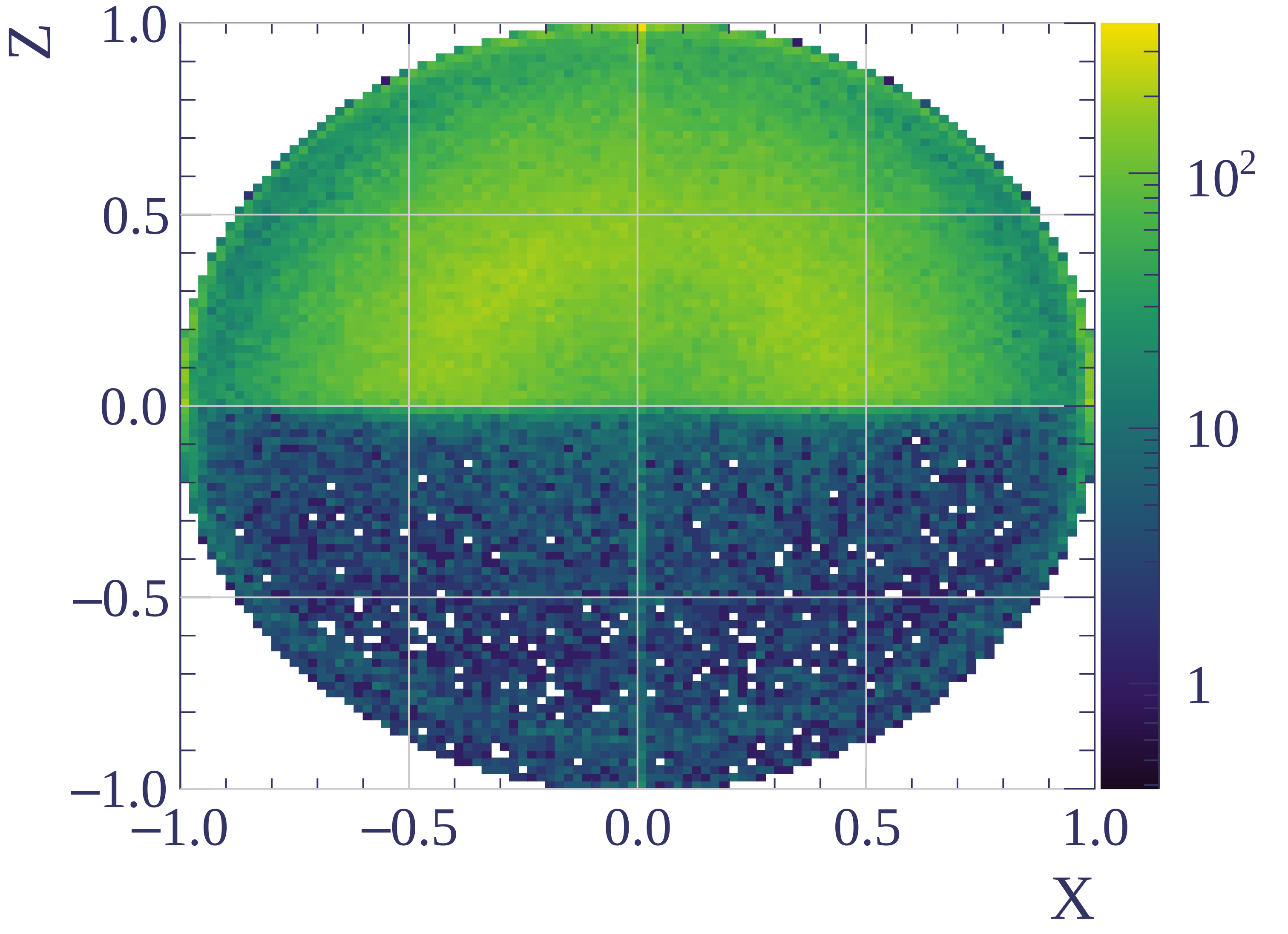
Counts
4000
3000
2000
1000
0
0.0
0.2
0.4
0.6
0.8
1.0
X direction



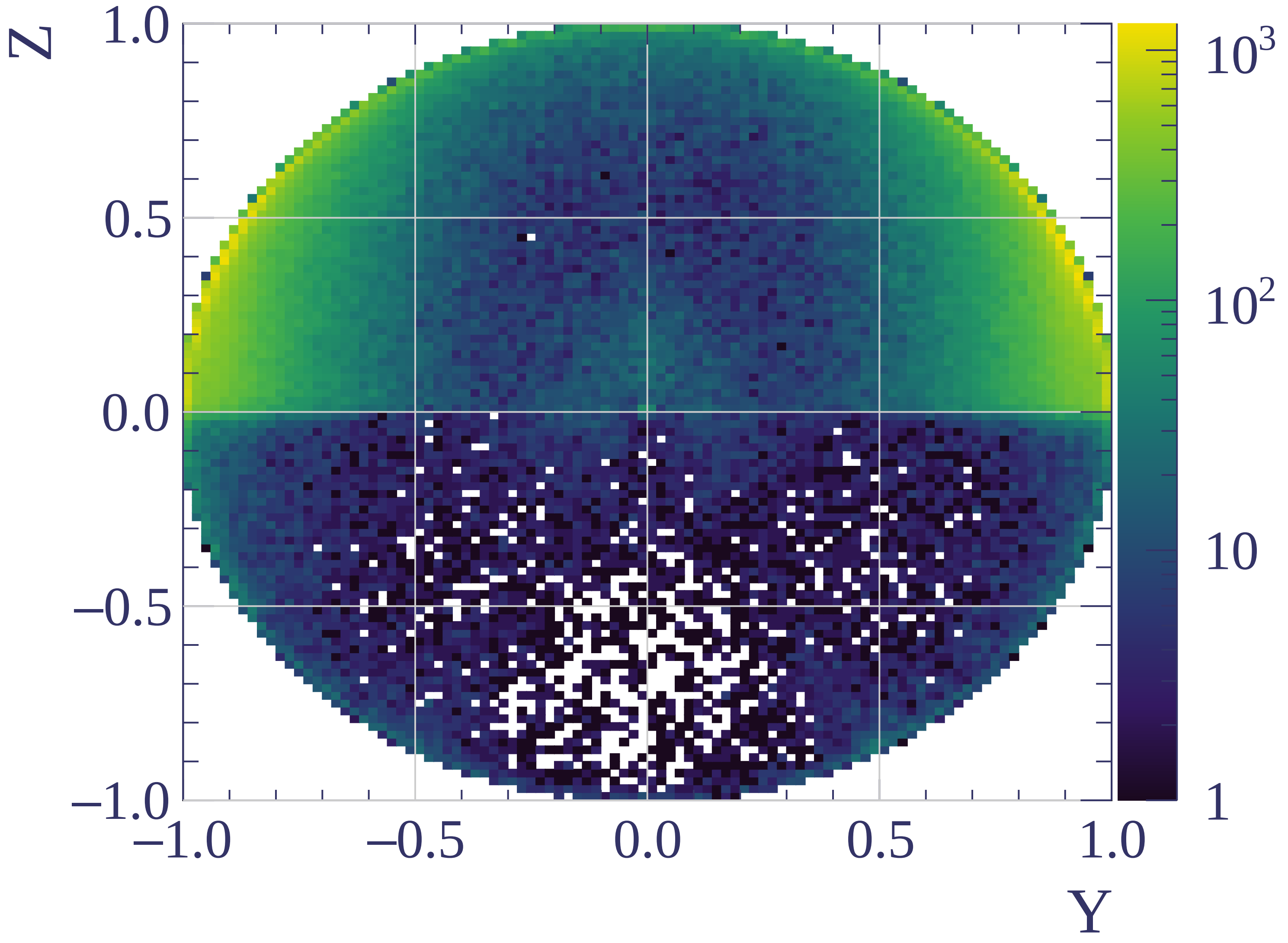
Counts
6000
4000
2000
0
0.0
0.2
0.4
0.6
0.8
1.0
Y direction



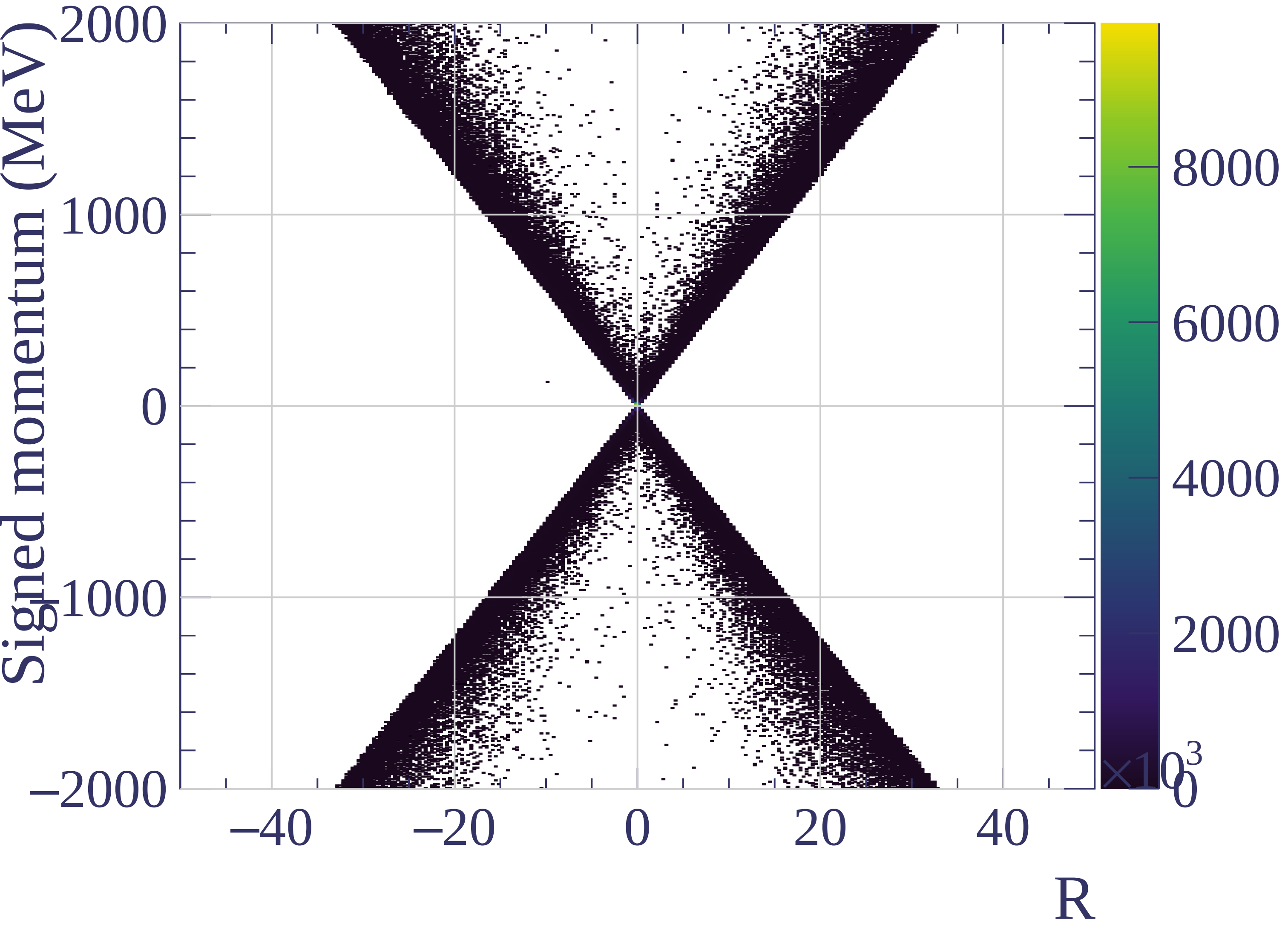
Counts
4000
2000
0
0.0
0.2
0.4
0.6
0.8
1.0
Z direction



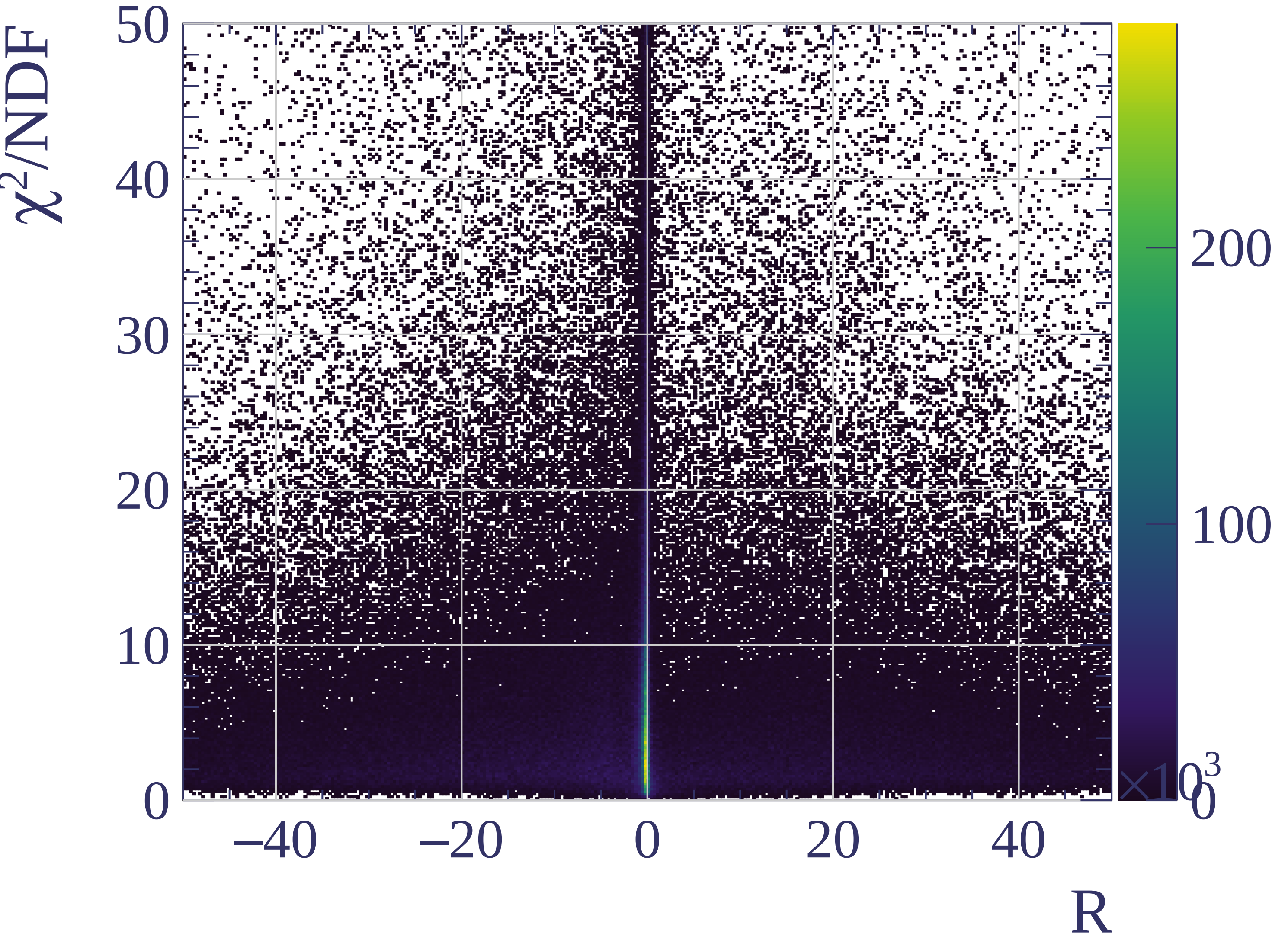
Z
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
X
10^2
10
1



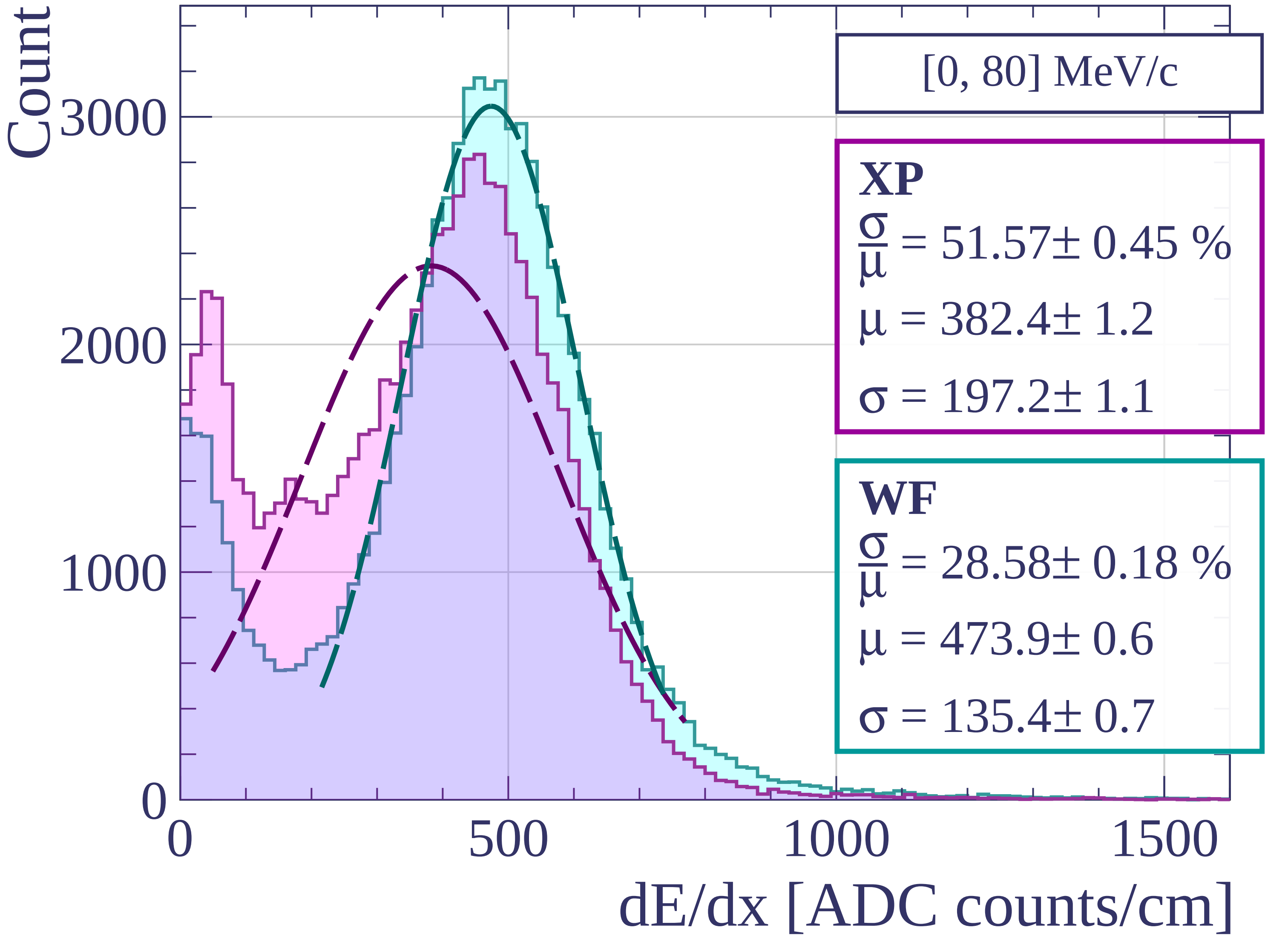
Z
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
Y
10^3
10^2
10
1



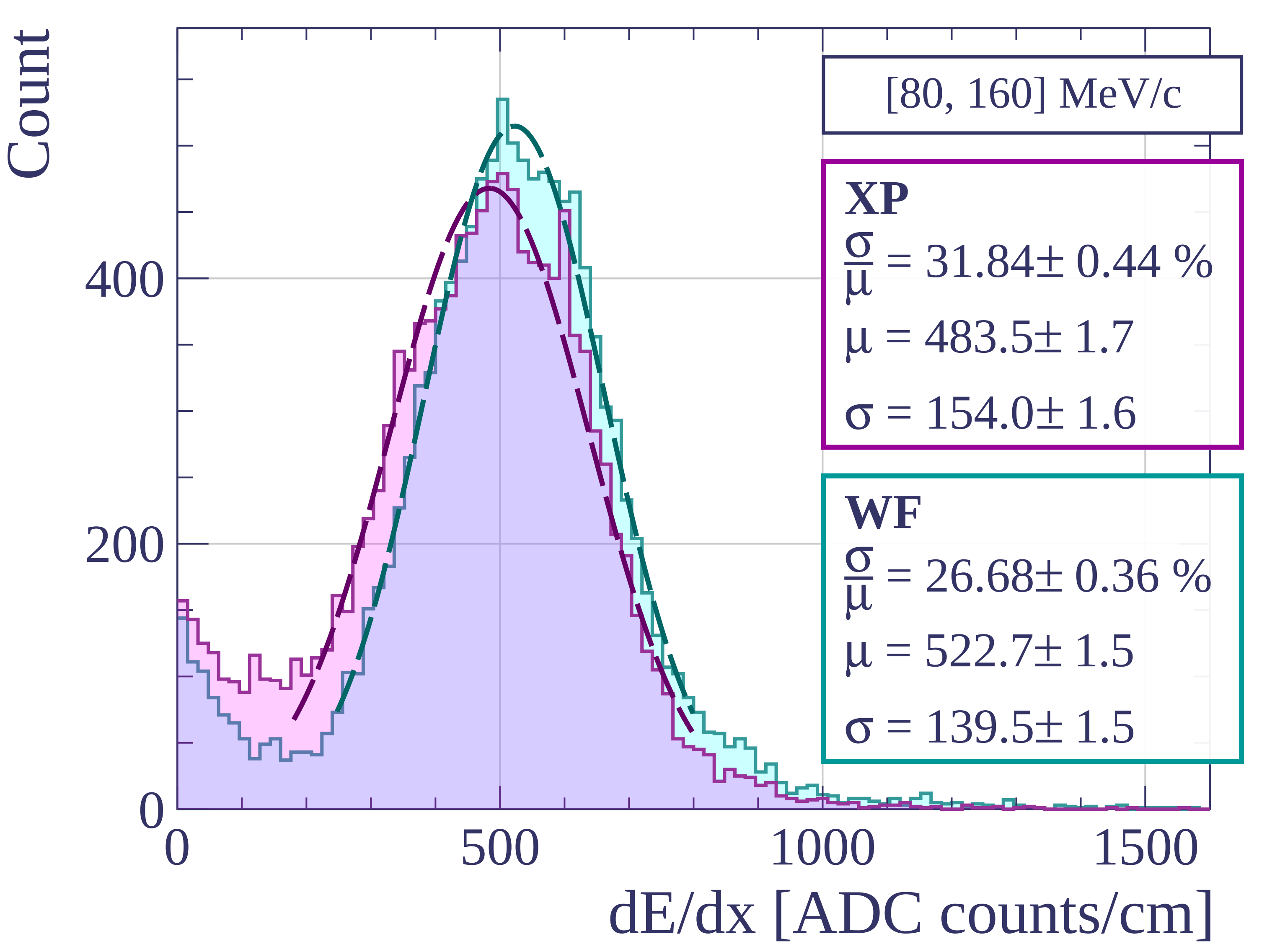
Signed momentum (MeV)
2000
1000
0
−1000
−2000
−40
−20
0
20
40
R
8000
6000
4000
2000
0
×10³



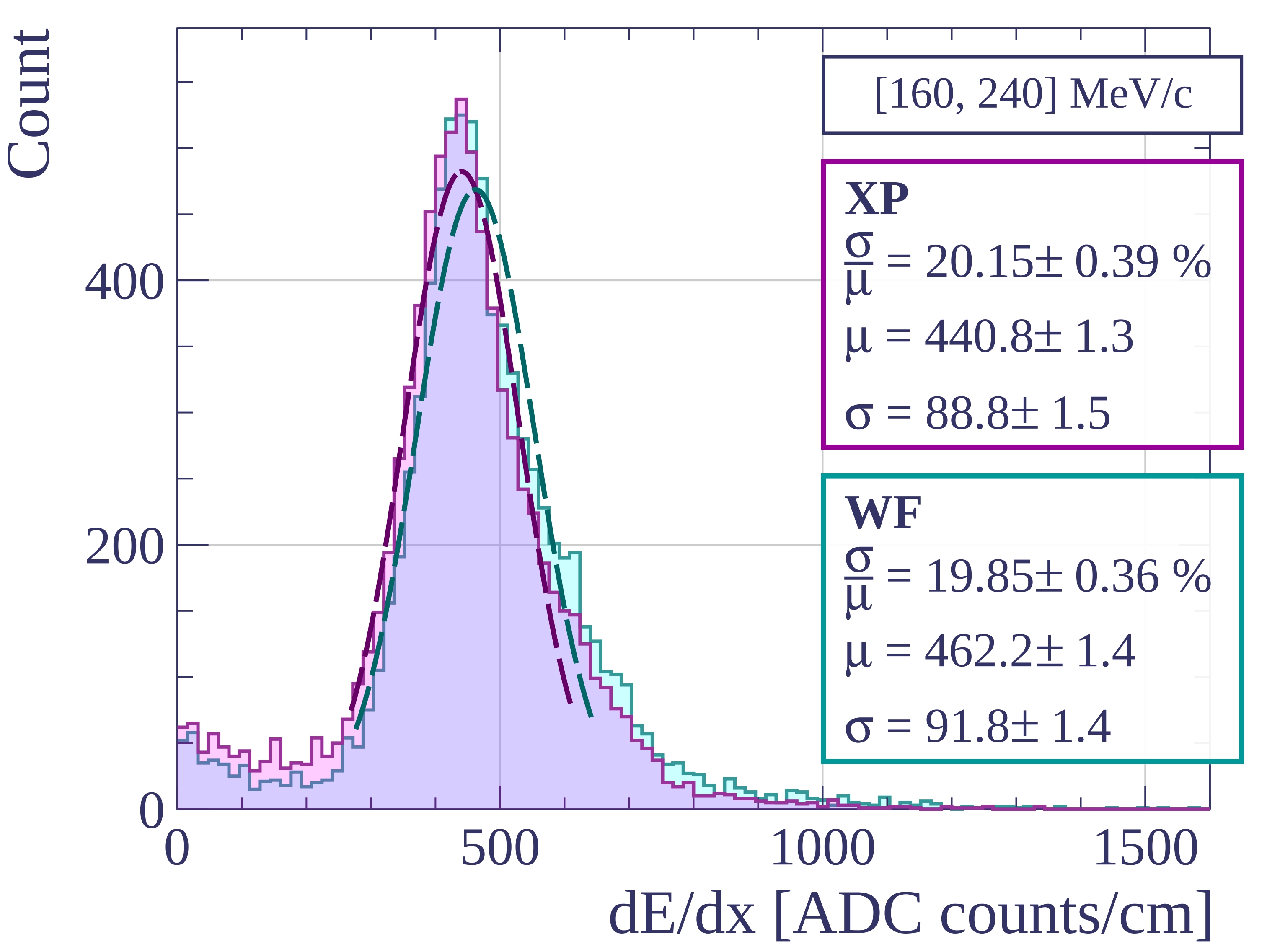
χ^2/NDF
50
40
30
20
10
0
−40
−20
0
20
40
R
200
100
0
$\times 10^3$



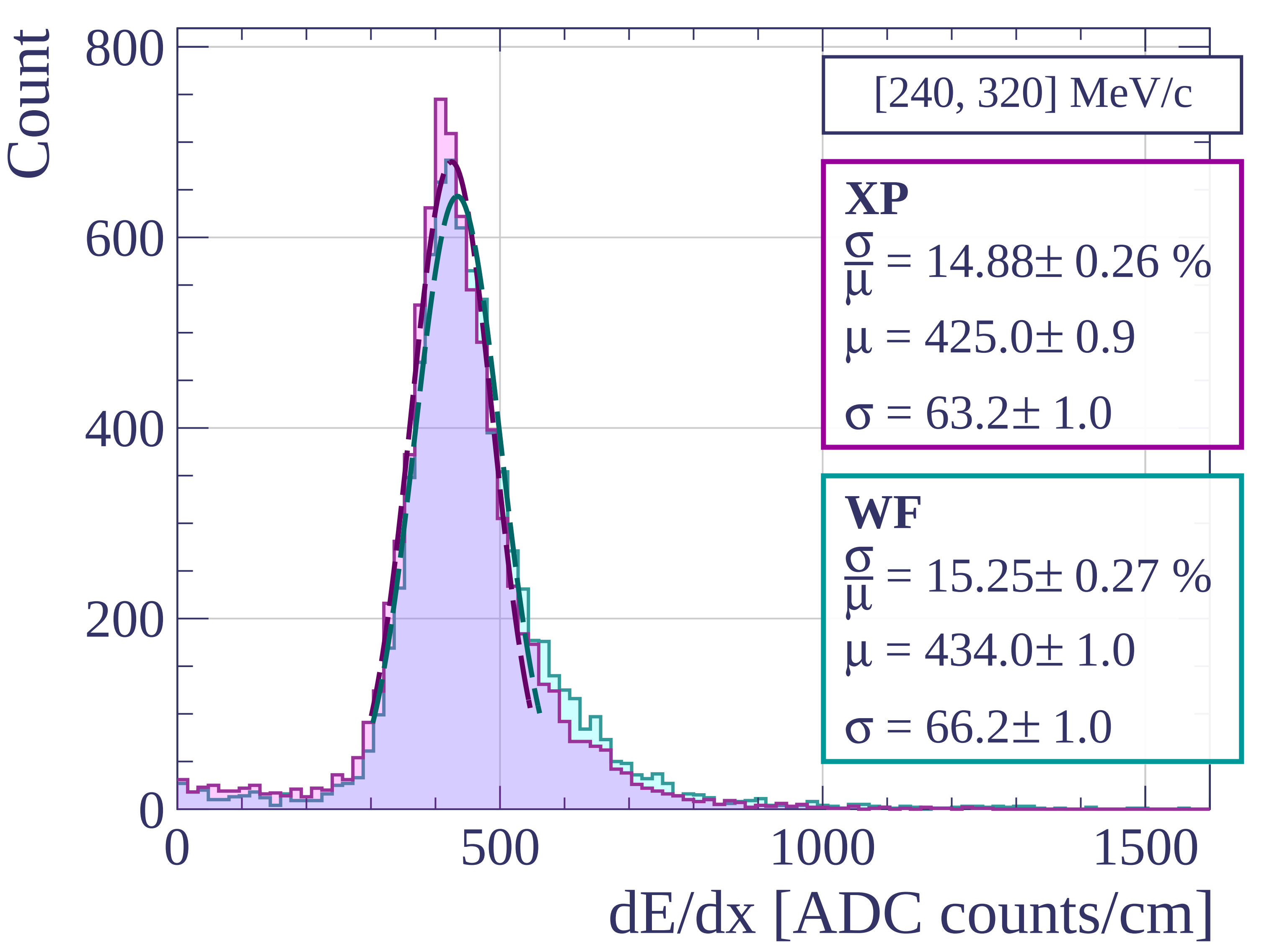
Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[0, 80] MeV/c
XP
$\frac{\sigma}{\mu}$ = 51.57± 0.45 %
μ = 382.4± 1.2
σ = 197.2± 1.1
WF
$\frac{\sigma}{\mu}$ = 28.58± 0.18 %
μ = 473.9± 0.6
σ = 135.4± 0.7

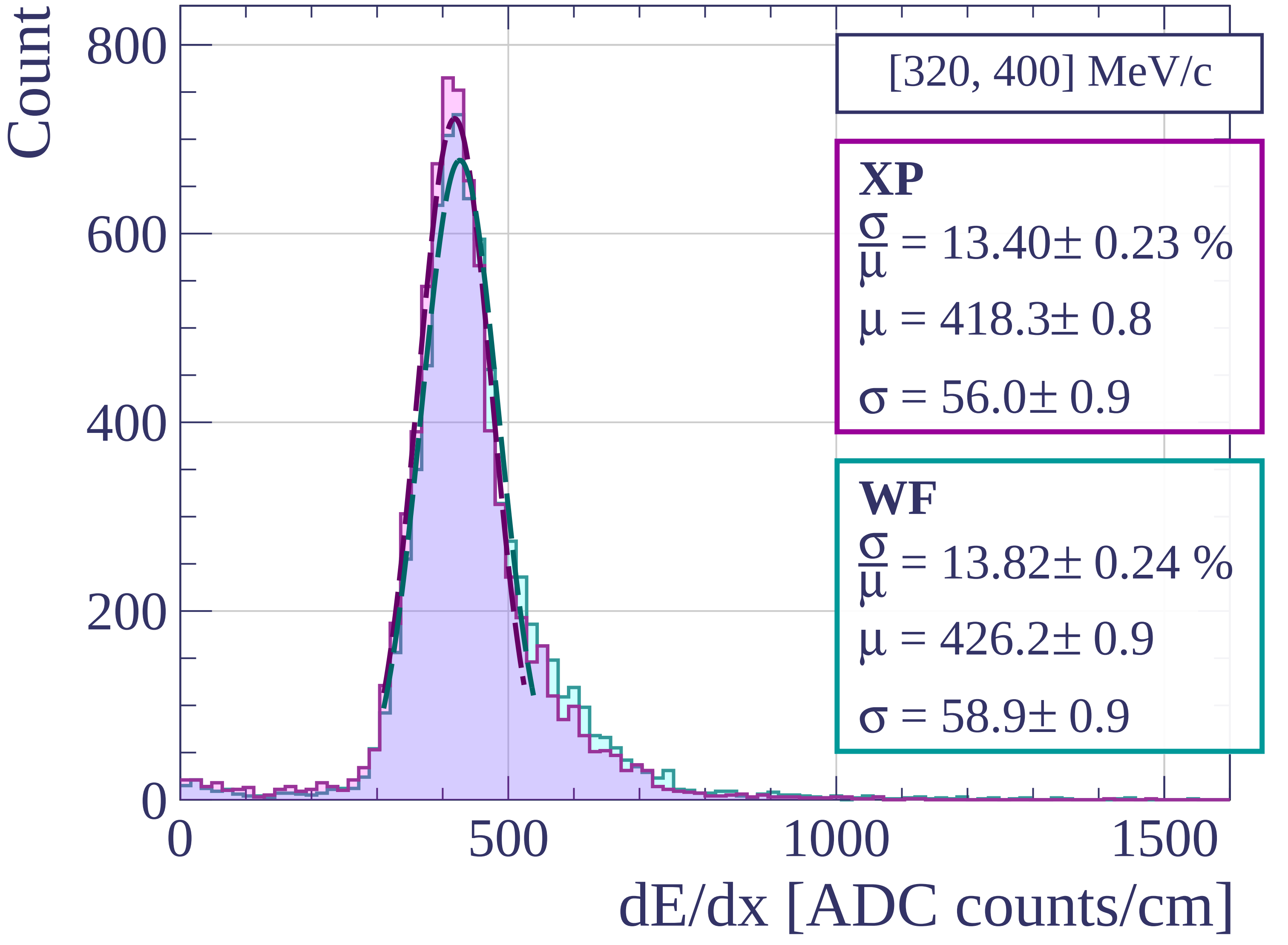


Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[80, 160] MeV/c
XP
$\frac{\sigma}{\mu}$ = 31.84± 0.44 %
μ = 483.5± 1.7
σ = 154.0± 1.6
WF
$\frac{\sigma}{\mu}$ = 26.68± 0.36 %
μ = 522.7± 1.5
σ = 139.5± 1.5

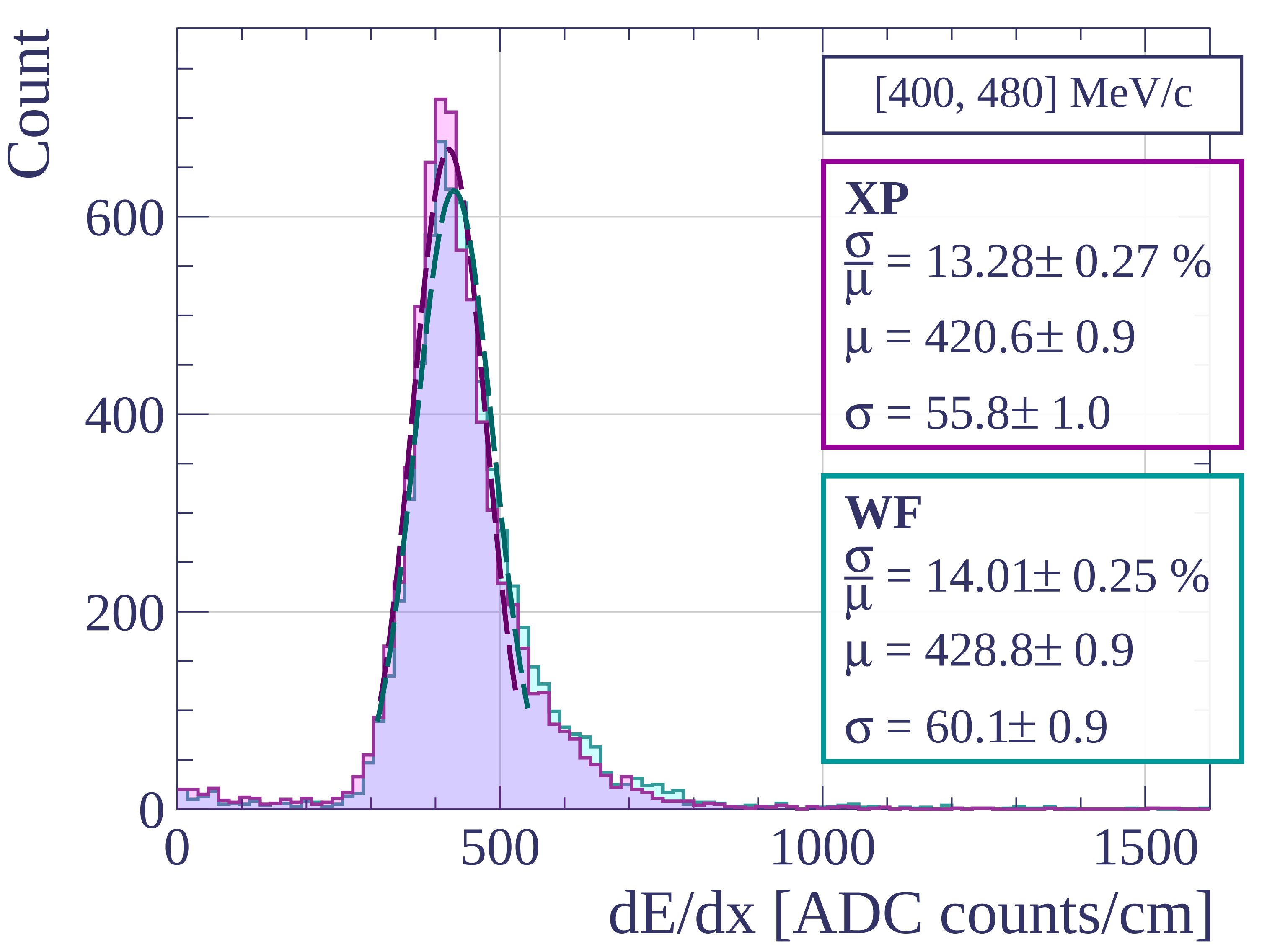


Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[160, 240] MeV/c
XP
$\frac{\sigma}{\mu}$ = 20.15± 0.39 %
μ = 440.8± 1.3
σ = 88.8± 1.5
WF
$\frac{\sigma}{\mu}$ = 19.85± 0.36 %
μ = 462.2± 1.4
σ = 91.8± 1.4

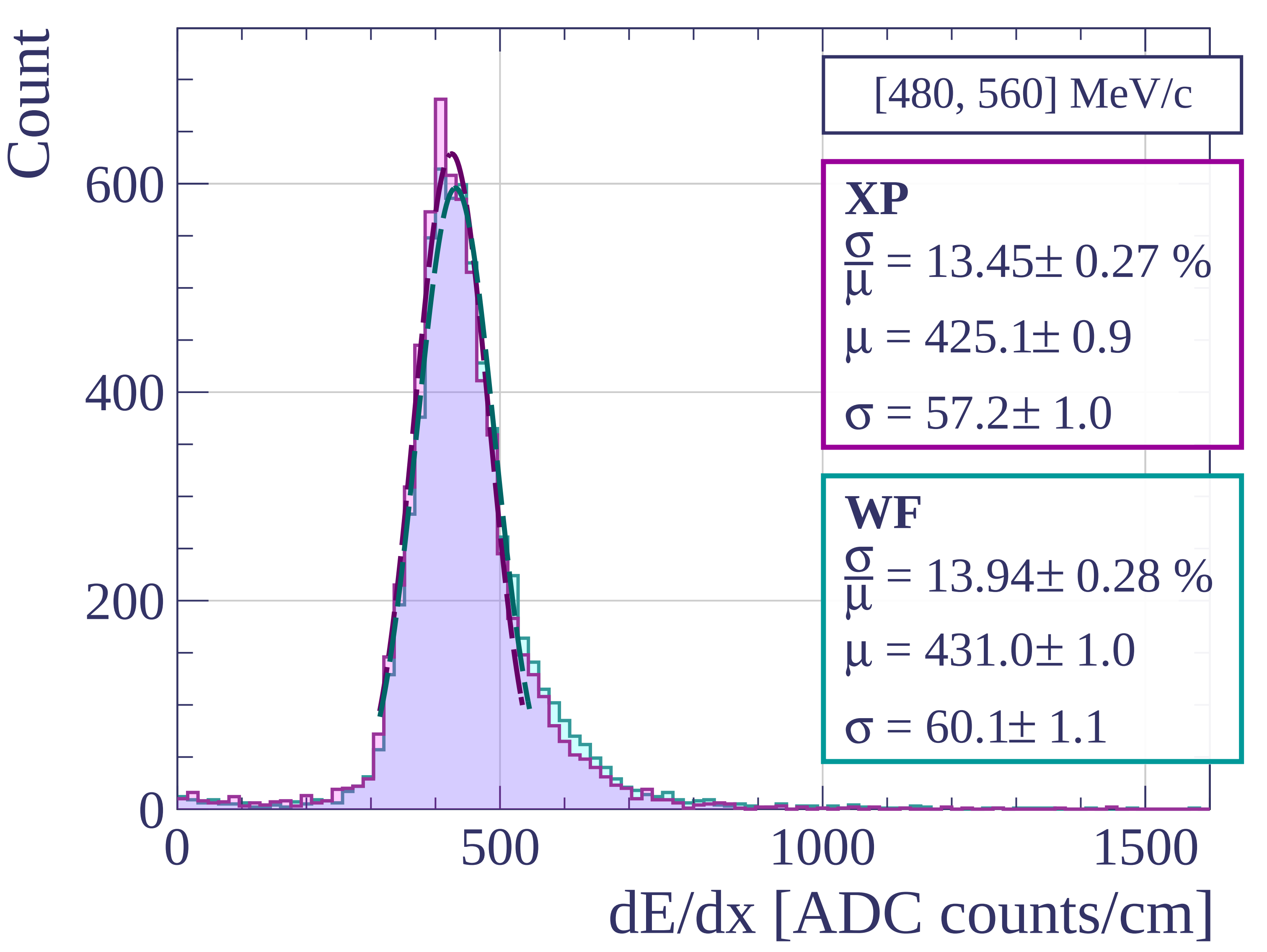

Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[240, 320] MeV/c
XP
σ/μ = 14.88± 0.26 %
μ = 425.0± 0.9
σ = 63.2± 1.0
WF
σ/μ = 15.25± 0.27 %
μ = 434.0± 1.0
σ = 66.2± 1.0



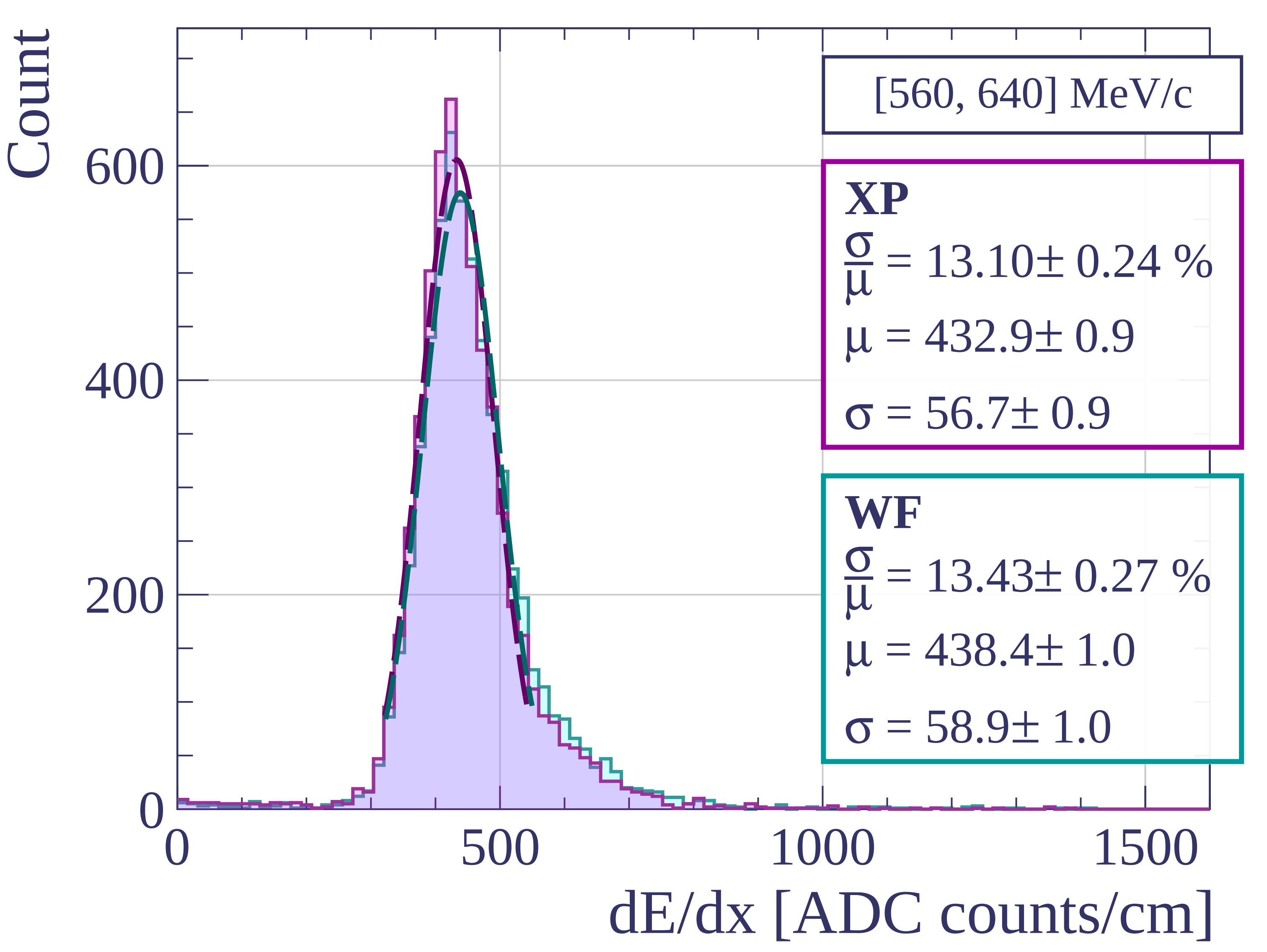
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[320, 400] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.40± 0.23 %
μ = 418.3± 0.8
σ = 56.0± 0.9
WF
$\frac{\sigma}{\mu}$ = 13.82± 0.24 %
μ = 426.2± 0.9
σ = 58.9± 0.9



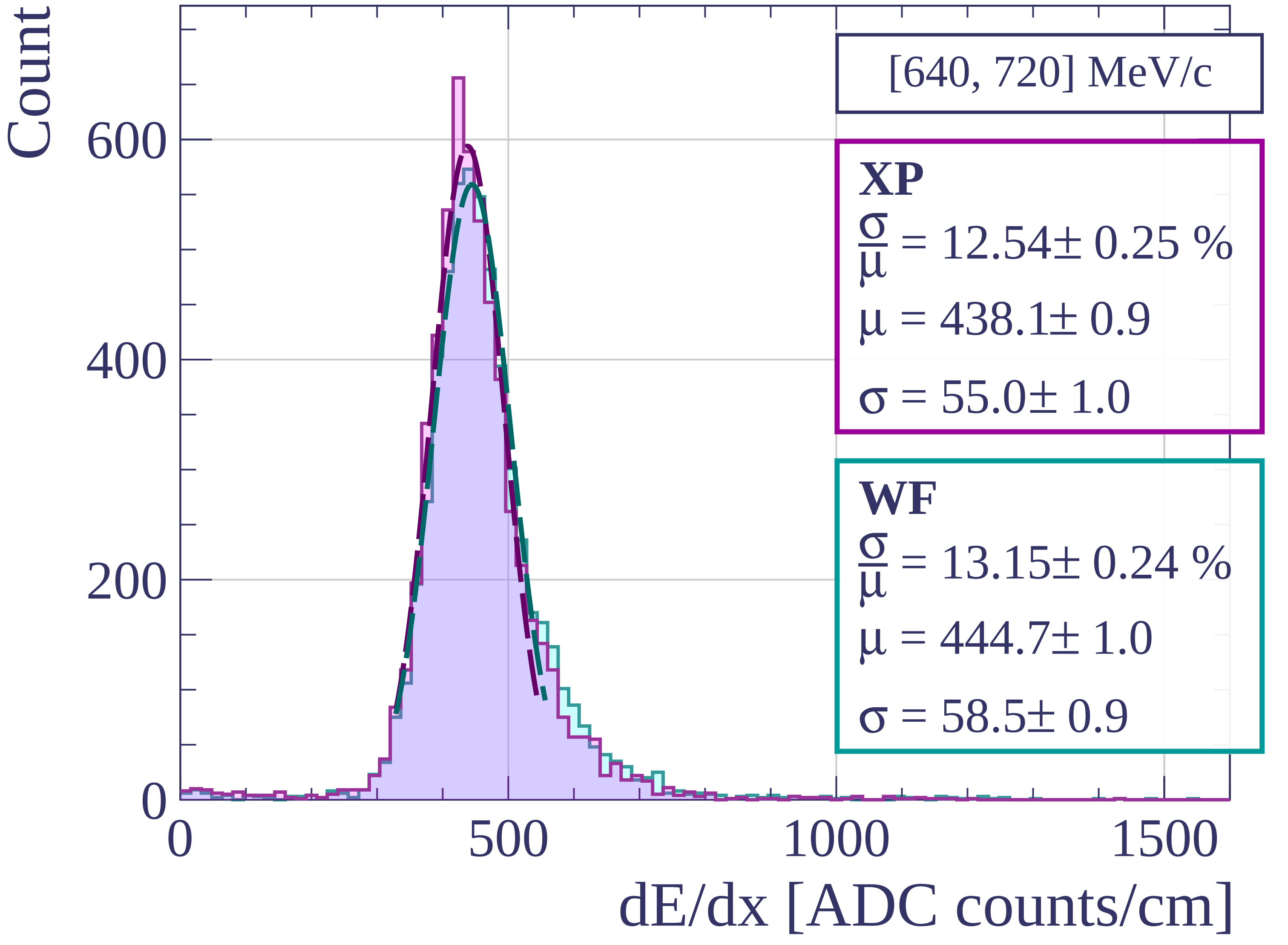
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[400, 480] MeV/c
XP
σ/μ = 13.28± 0.27 %
μ = 420.6± 0.9
σ = 55.8± 1.0
WF
σ/μ = 14.01± 0.25 %
μ = 428.8± 0.9
σ = 60.1± 0.9



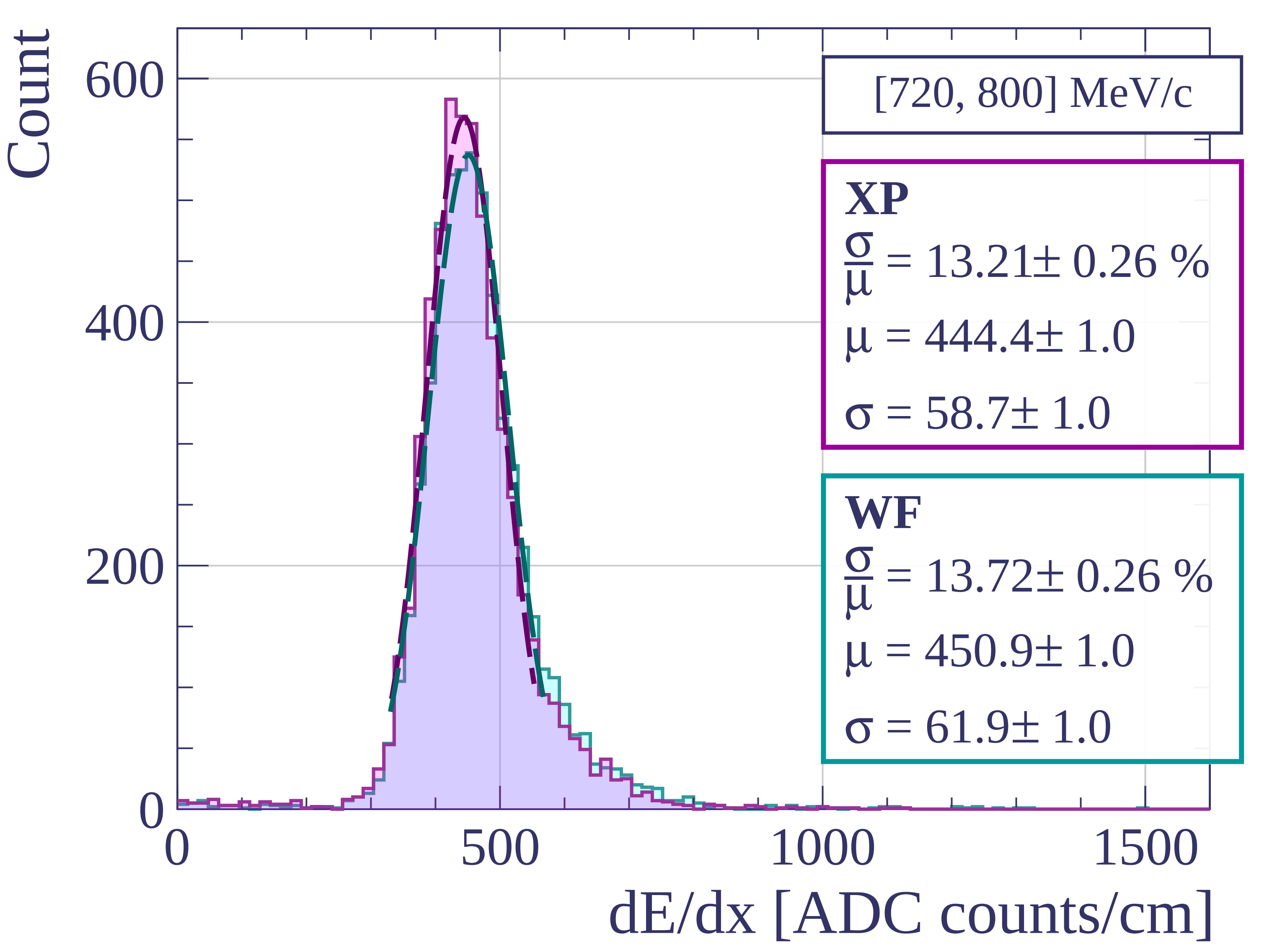
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[480, 560] MeV/c
XP
σ/μ = 13.45± 0.27 %
μ = 425.1± 0.9
σ = 57.2± 1.0
WF
σ/μ = 13.94± 0.28 %
μ = 431.0± 1.0
σ = 60.1± 1.1

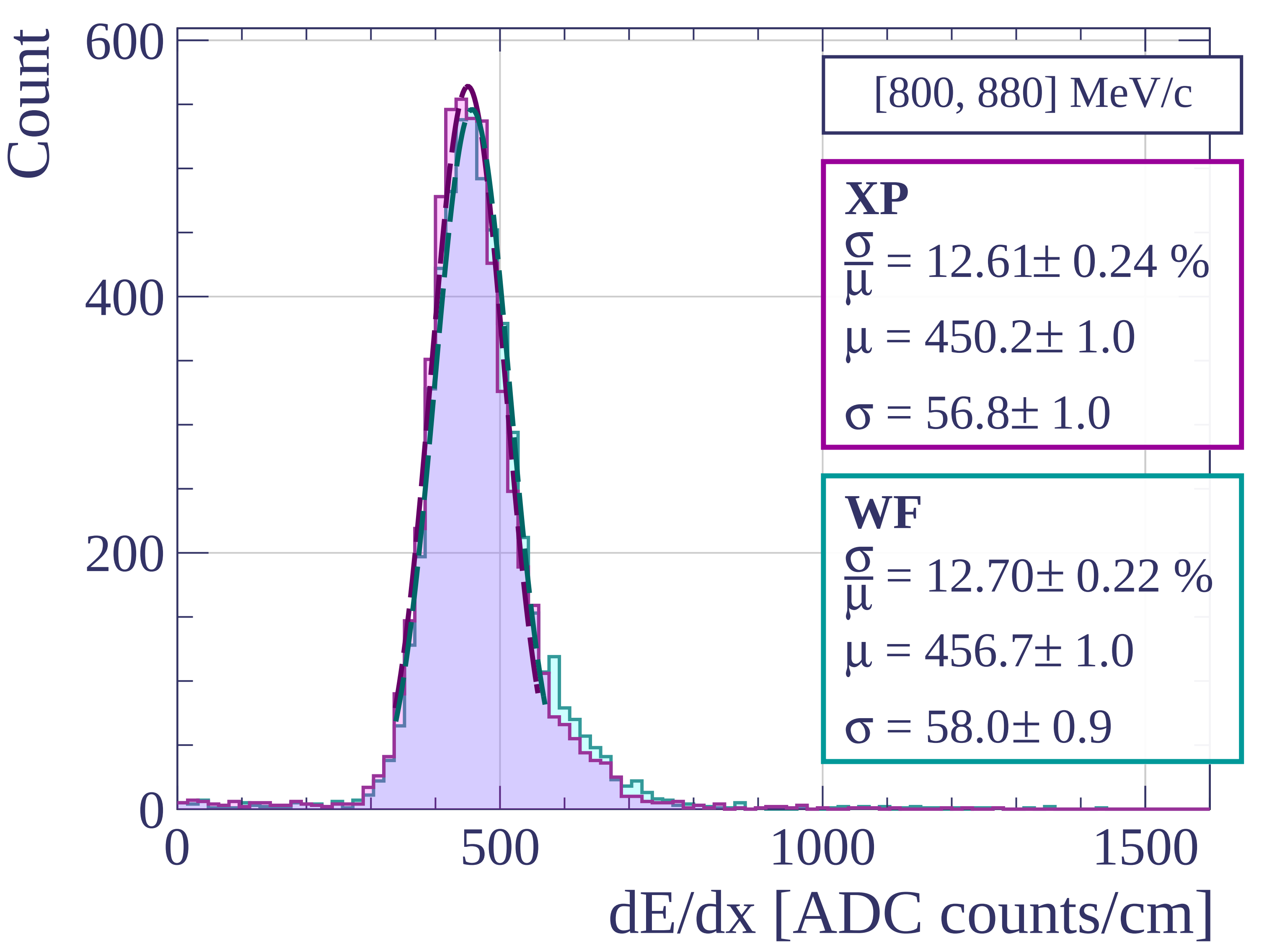


Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[560, 640] MeV/c
XP
σ/μ = 13.10± 0.24 %
μ = 432.9± 0.9
σ = 56.7± 0.9
WF
σ/μ = 13.43± 0.27 %
μ = 438.4± 1.0
σ = 58.9± 1.0

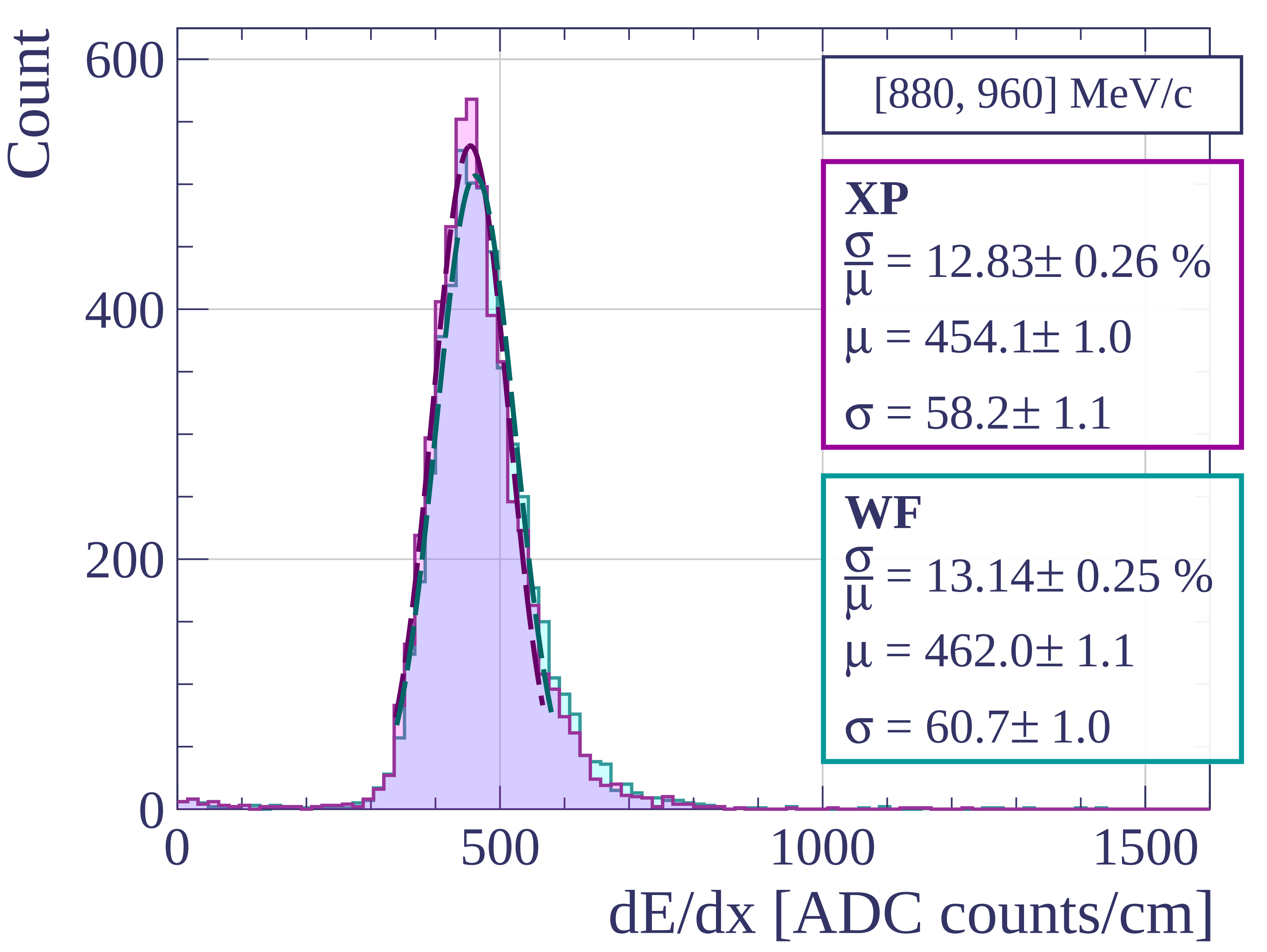


Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[640, 720] MeV/c
XP
σ/μ = 12.54± 0.25 %
μ = 438.1± 0.9
σ = 55.0± 1.0
WF
σ/μ = 13.15± 0.24 %
μ = 444.7± 1.0
σ = 58.5± 0.9

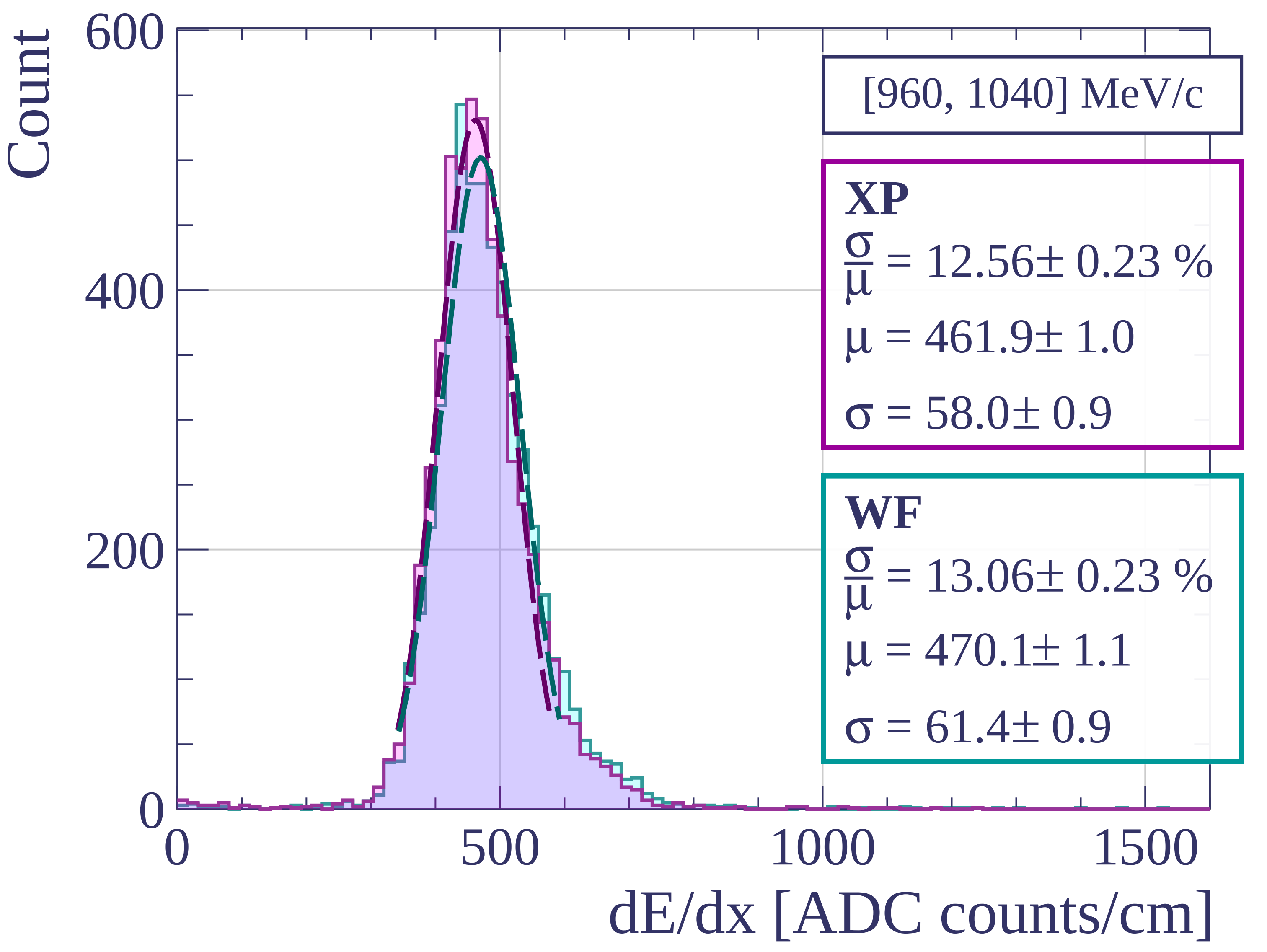

Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[720, 800] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.21± 0.26 %
μ = 444.4± 1.0
σ = 58.7± 1.0
WF
$\frac{\sigma}{\mu}$ = 13.72± 0.26 %
μ = 450.9± 1.0
σ = 61.9± 1.0



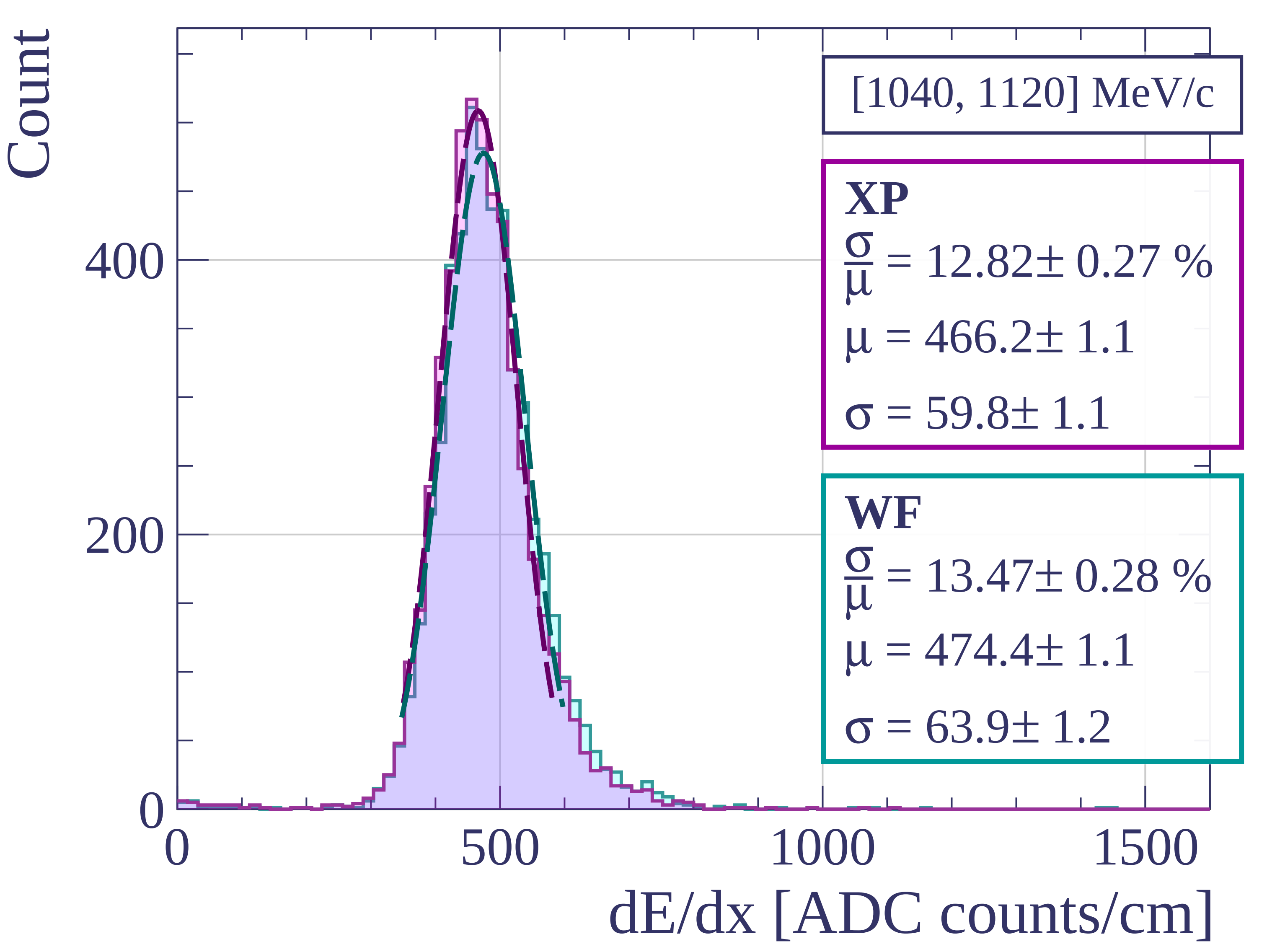
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[800, 880] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.61± 0.24 %
μ = 450.2± 1.0
σ = 56.8± 1.0
WF
$\frac{\sigma}{\mu}$ = 12.70± 0.22 %
μ = 456.7± 1.0
σ = 58.0± 0.9



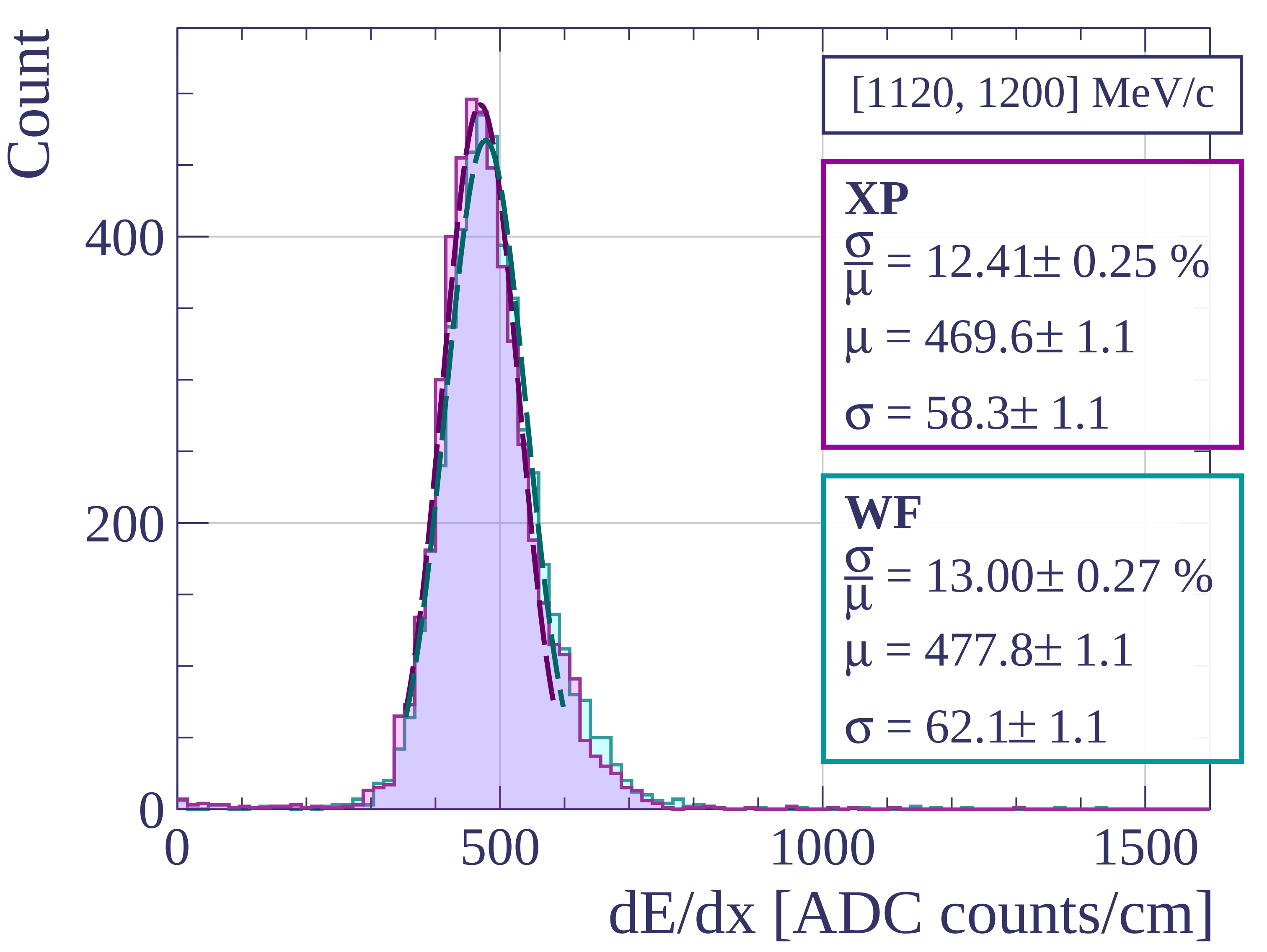
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[880, 960] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.83± 0.26 %
μ = 454.1± 1.0
σ = 58.2± 1.1
WF
$\frac{\sigma}{\mu}$ = 13.14± 0.25 %
μ = 462.0± 1.1
σ = 60.7± 1.0

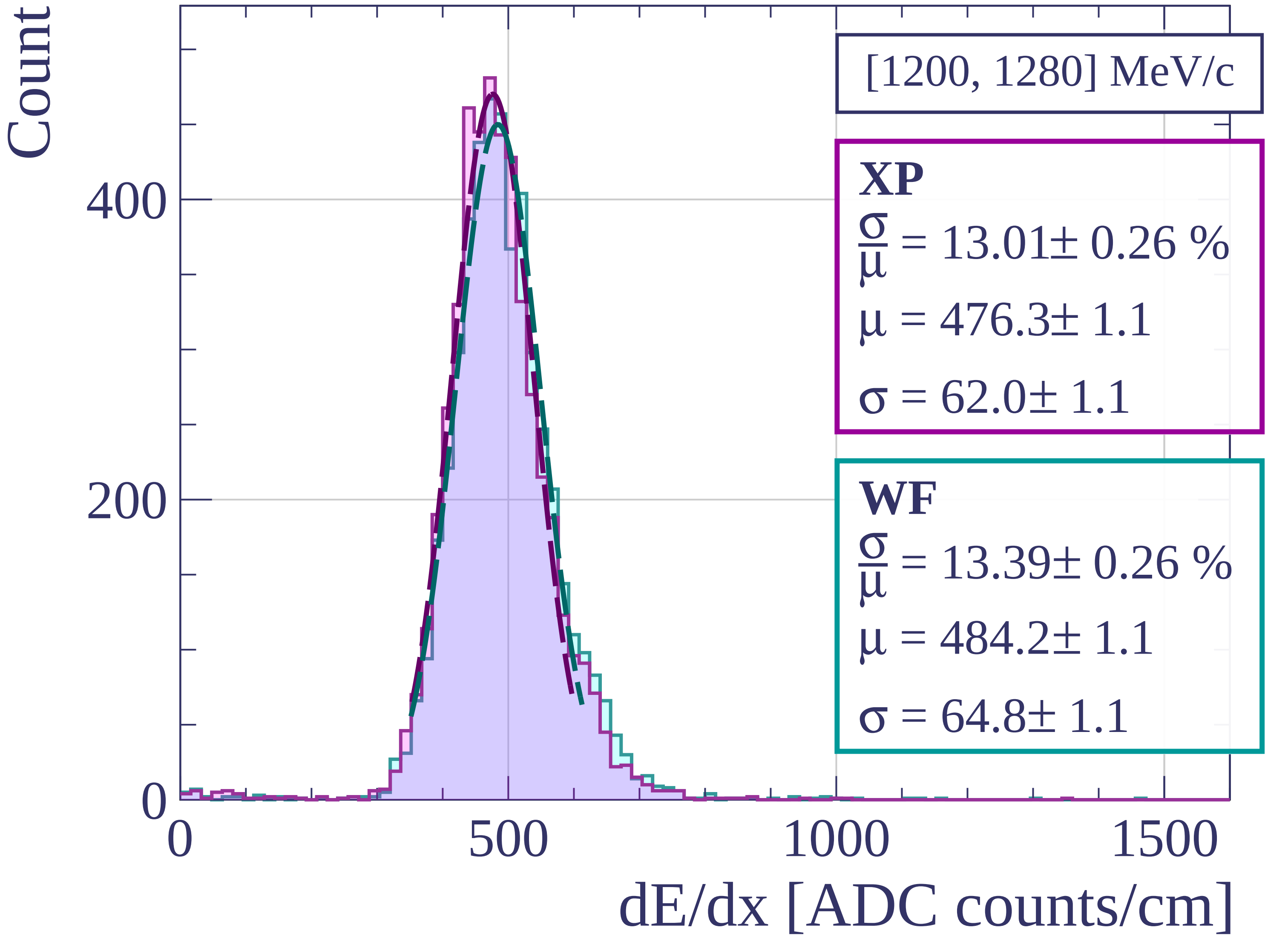


Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[960, 1040] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.56± 0.23 %
μ = 461.9± 1.0
σ = 58.0± 0.9
WF
$\frac{\sigma}{\mu}$ = 13.06± 0.23 %
μ = 470.1± 1.1
σ = 61.4± 0.9

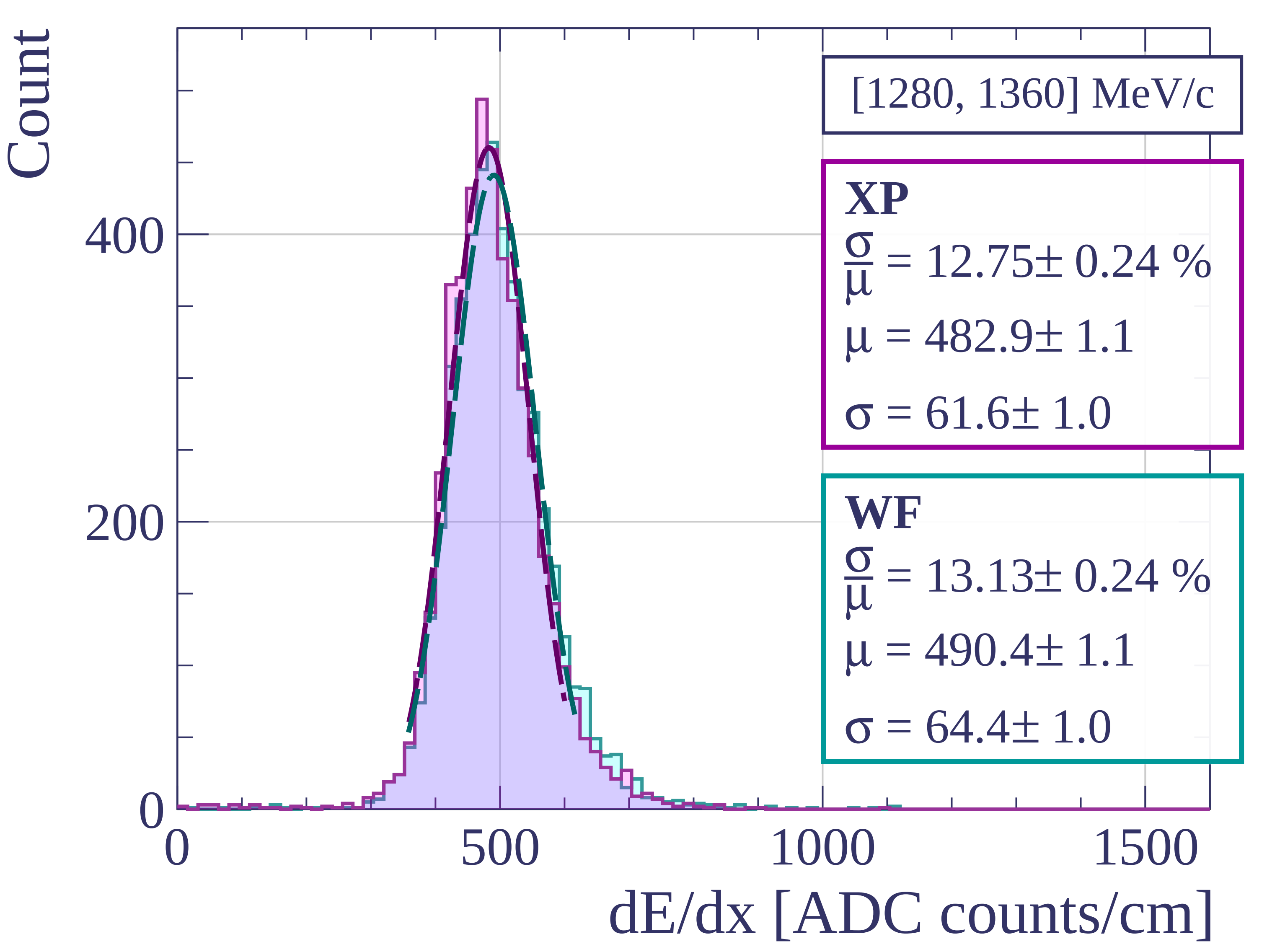


Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1040, 1120] MeV/c
XP
σ/μ = 12.82± 0.27 %
μ = 466.2± 1.1
σ = 59.8± 1.1
WF
σ/μ = 13.47± 0.28 %
μ = 474.4± 1.1
σ = 63.9± 1.2

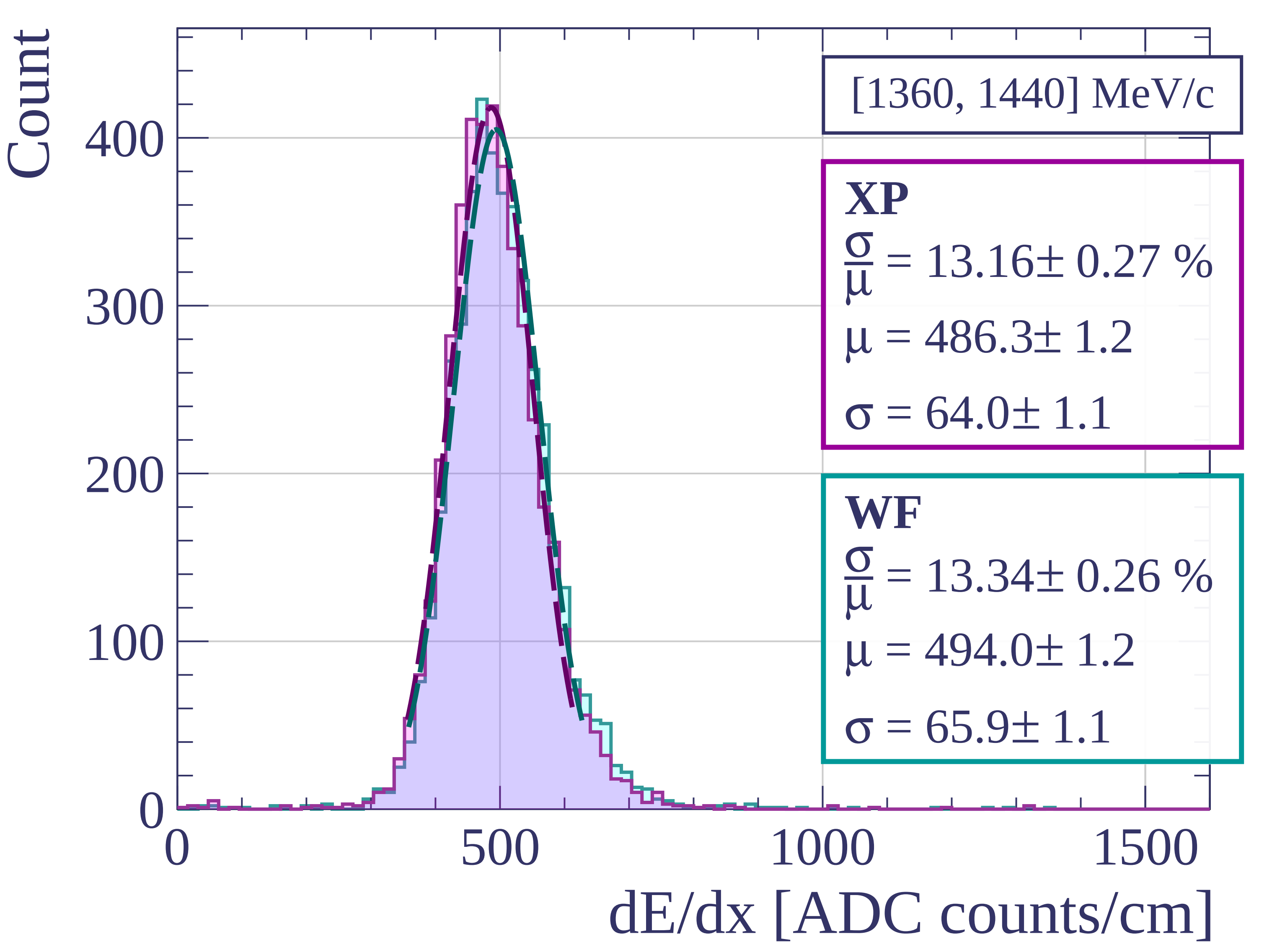

Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1120, 1200] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.41± 0.25 %
μ = 469.6± 1.1
σ = 58.3± 1.1
WF
$\frac{\sigma}{\mu}$ = 13.00± 0.27 %
μ = 477.8± 1.1
σ = 62.1± 1.1



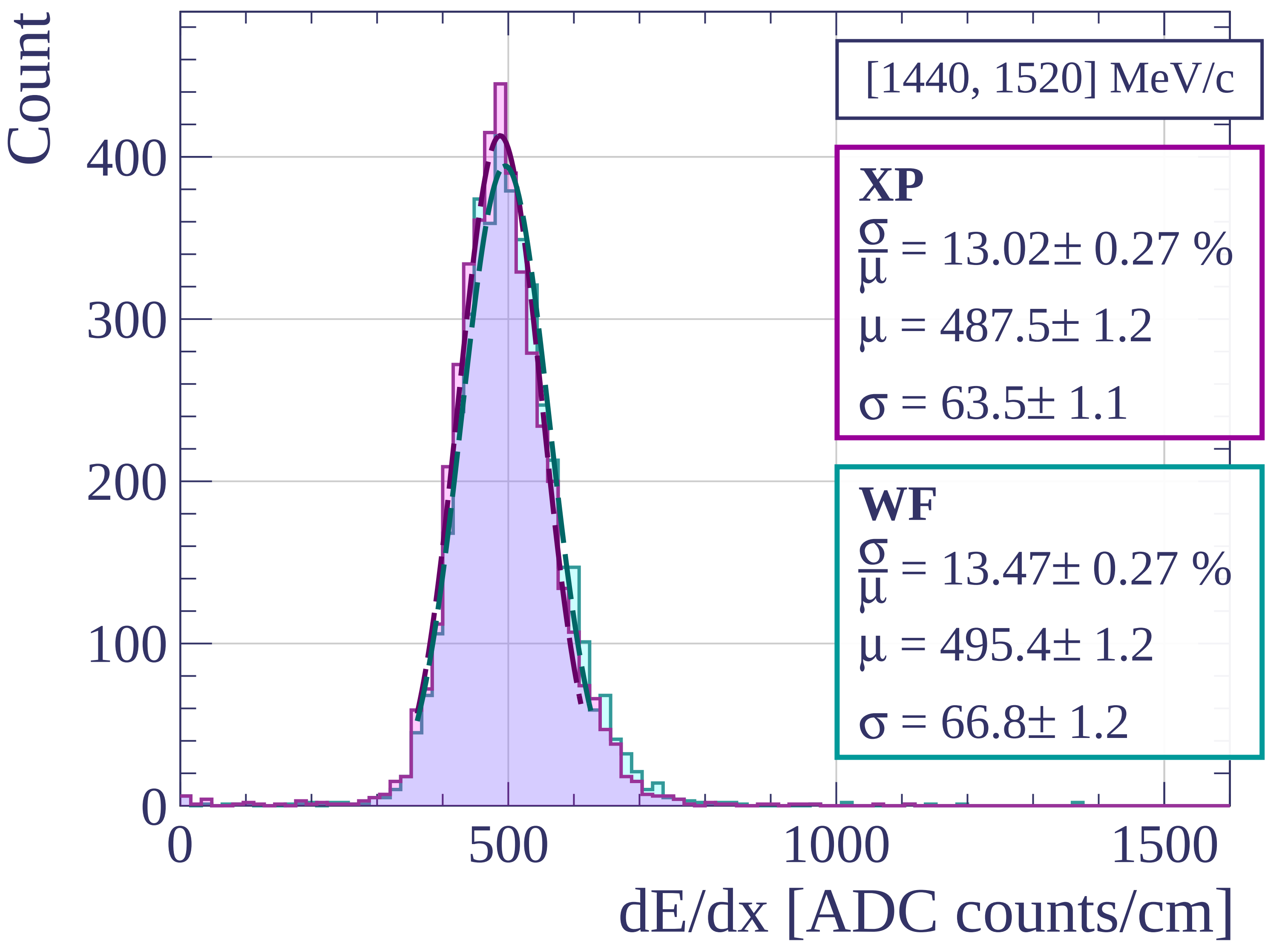
Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1200, 1280] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.01± 0.26 %
μ = 476.3± 1.1
σ = 62.0± 1.1
WF
$\frac{\sigma}{\mu}$ = 13.39± 0.26 %
μ = 484.2± 1.1
σ = 64.8± 1.1



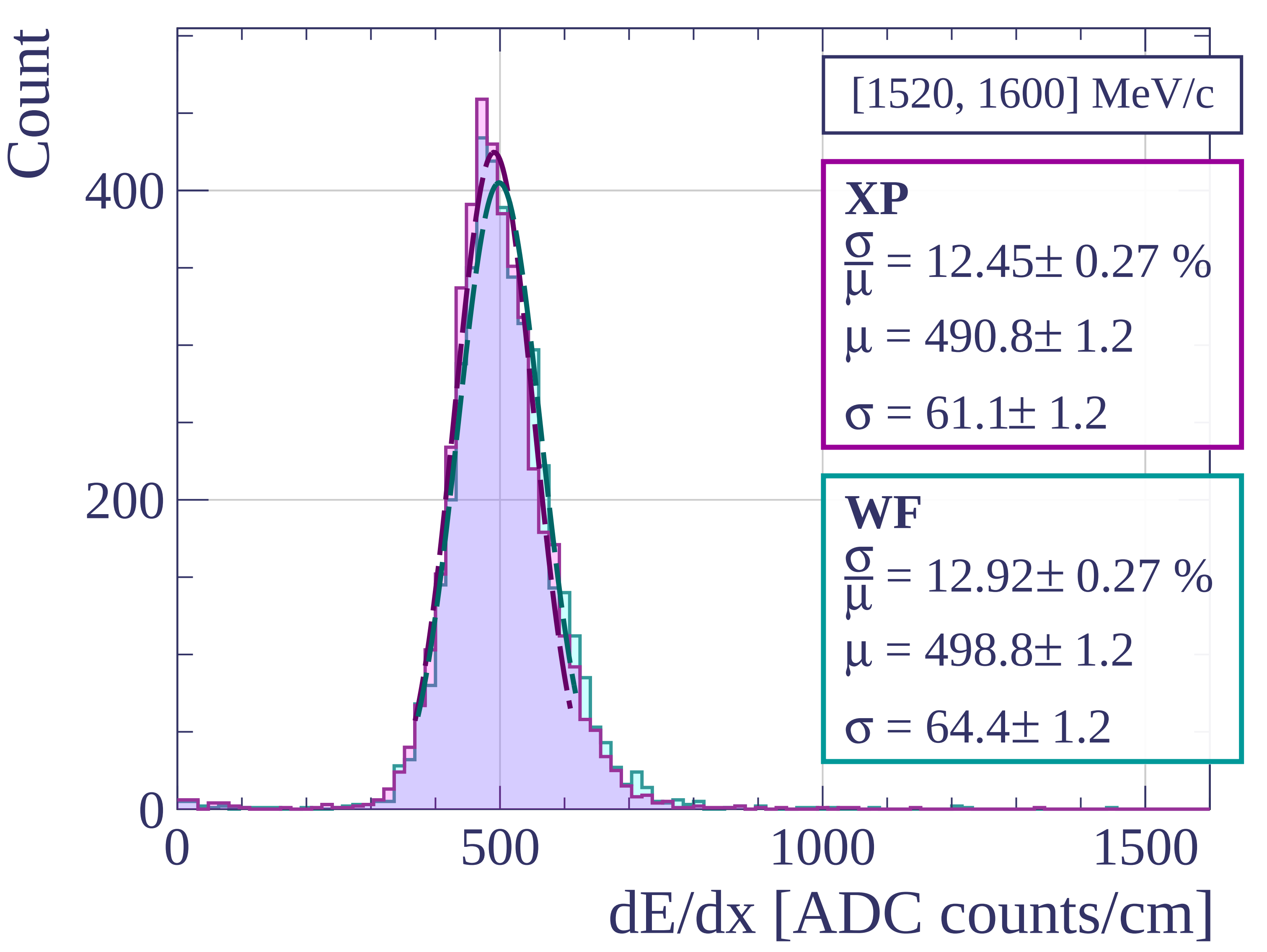
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
200
400
[1280, 1360] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.75± 0.24 %
μ = 482.9± 1.1
σ = 61.6± 1.0
WF
$\frac{\sigma}{\mu}$ = 13.13± 0.24 %
μ = 490.4± 1.1
σ = 64.4± 1.0



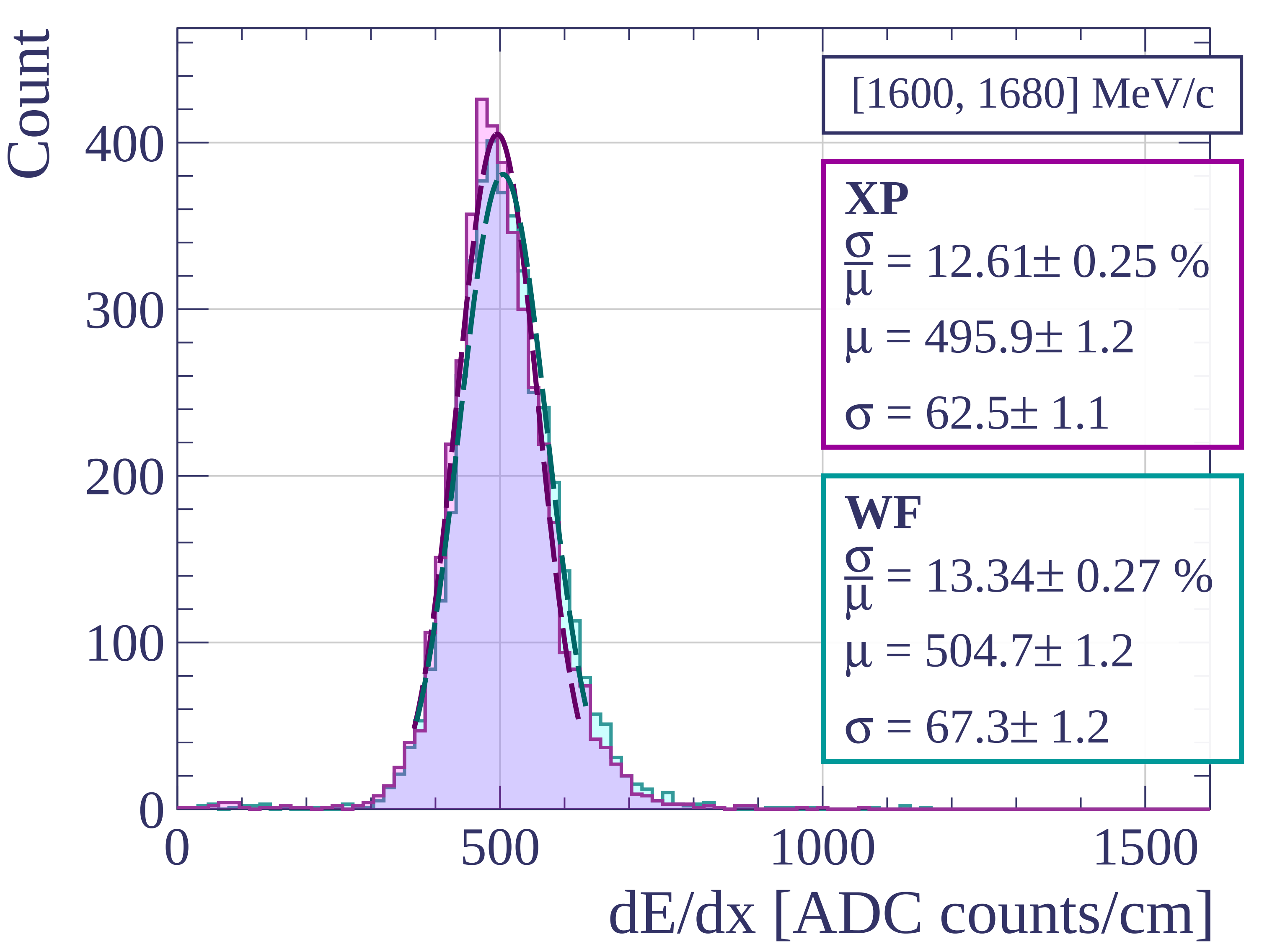
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1360, 1440] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.16± 0.27 %
μ = 486.3± 1.2
σ = 64.0± 1.1
WF
$\frac{\sigma}{\mu}$ = 13.34± 0.26 %
μ = 494.0± 1.2
σ = 65.9± 1.1

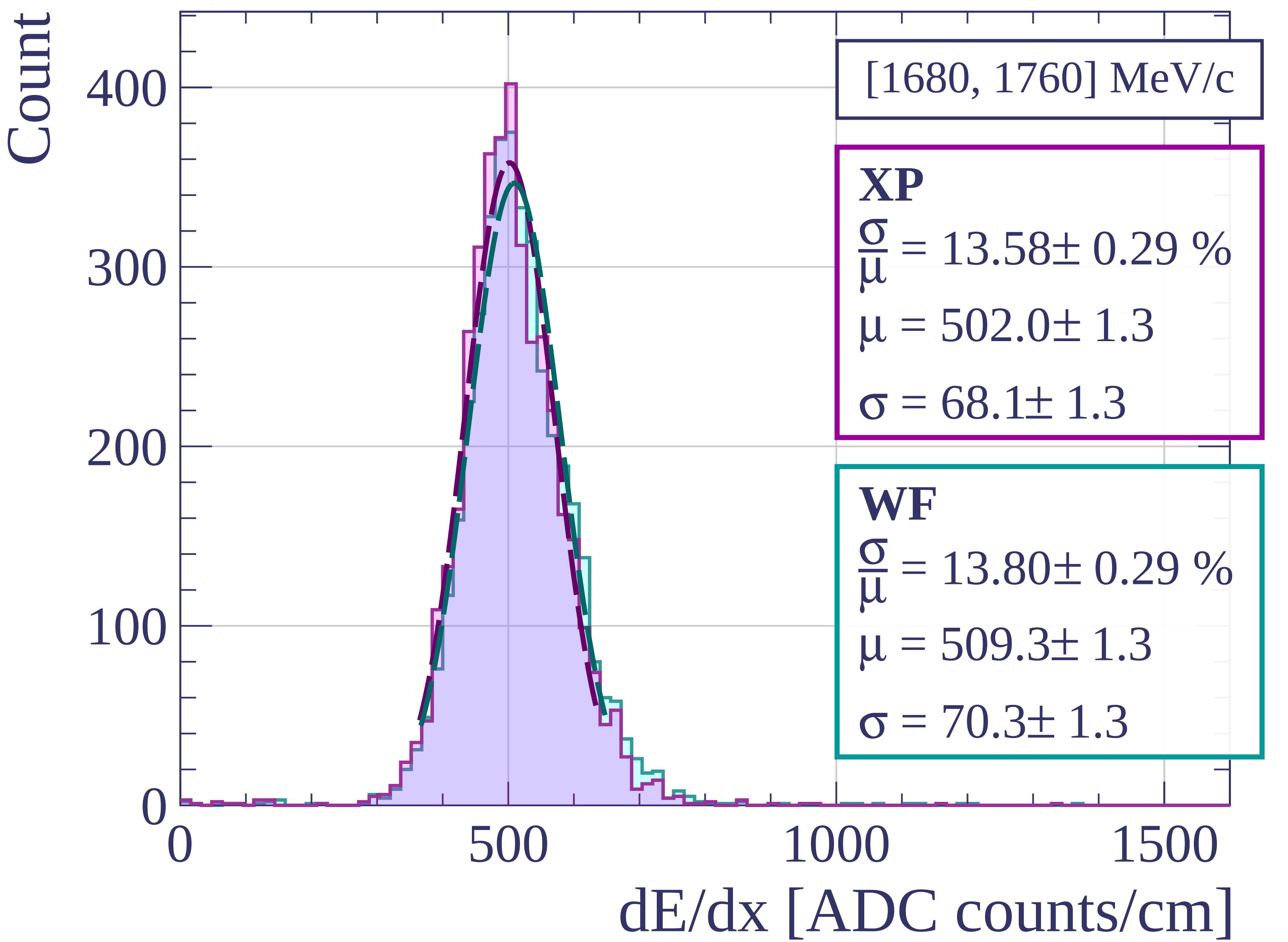


Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1440, 1520] MeV/c
XP
σ/μ = 13.02± 0.27 %
μ = 487.5± 1.2
σ = 63.5± 1.1
WF
σ/μ = 13.47± 0.27 %
μ = 495.4± 1.2
σ = 66.8± 1.2

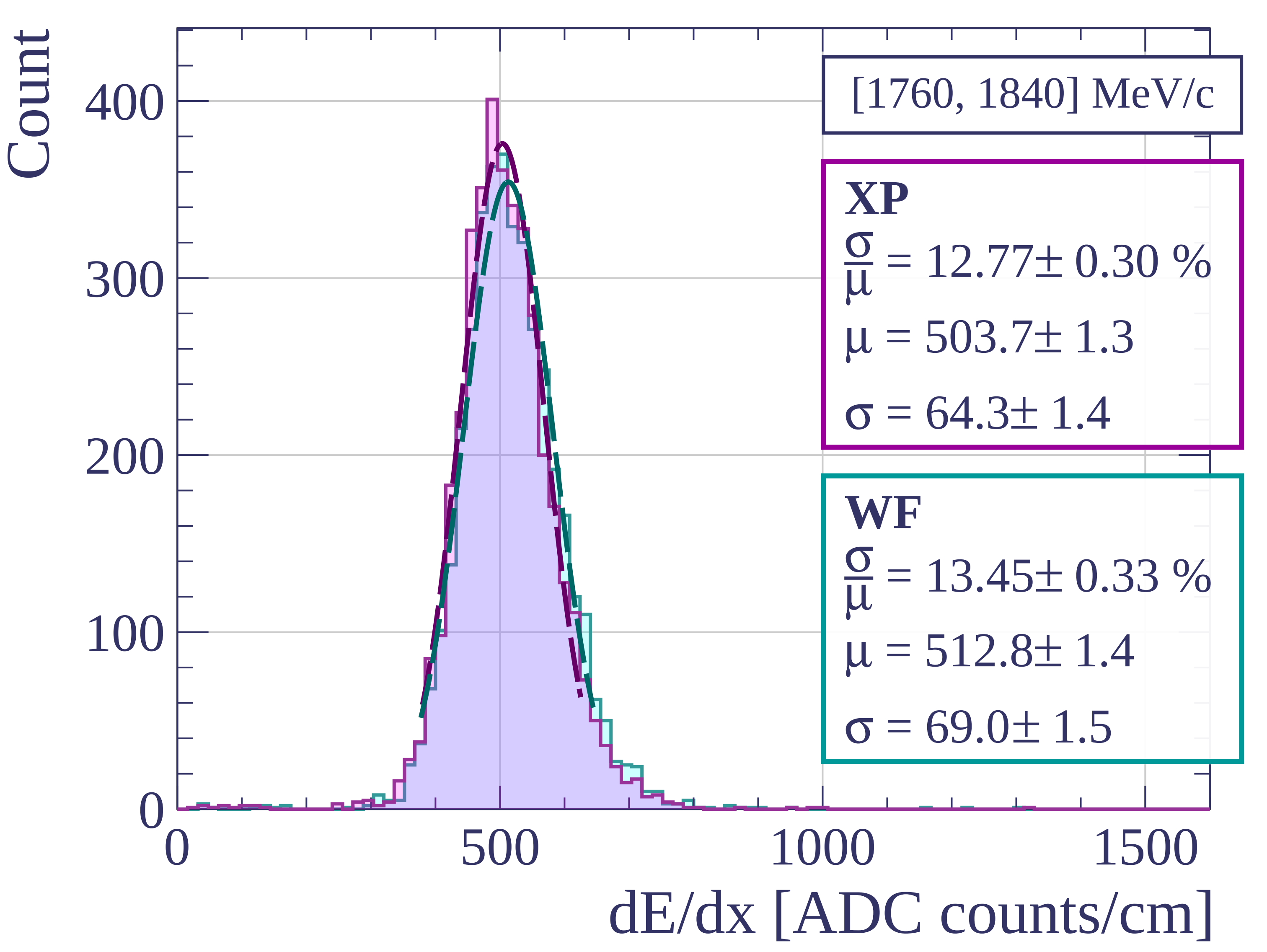


Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1520, 1600] MeV/c
XP
σ/μ = 12.45± 0.27 %
μ = 490.8± 1.2
σ = 61.1± 1.2
WF
σ/μ = 12.92± 0.27 %
μ = 498.8± 1.2
σ = 64.4± 1.2

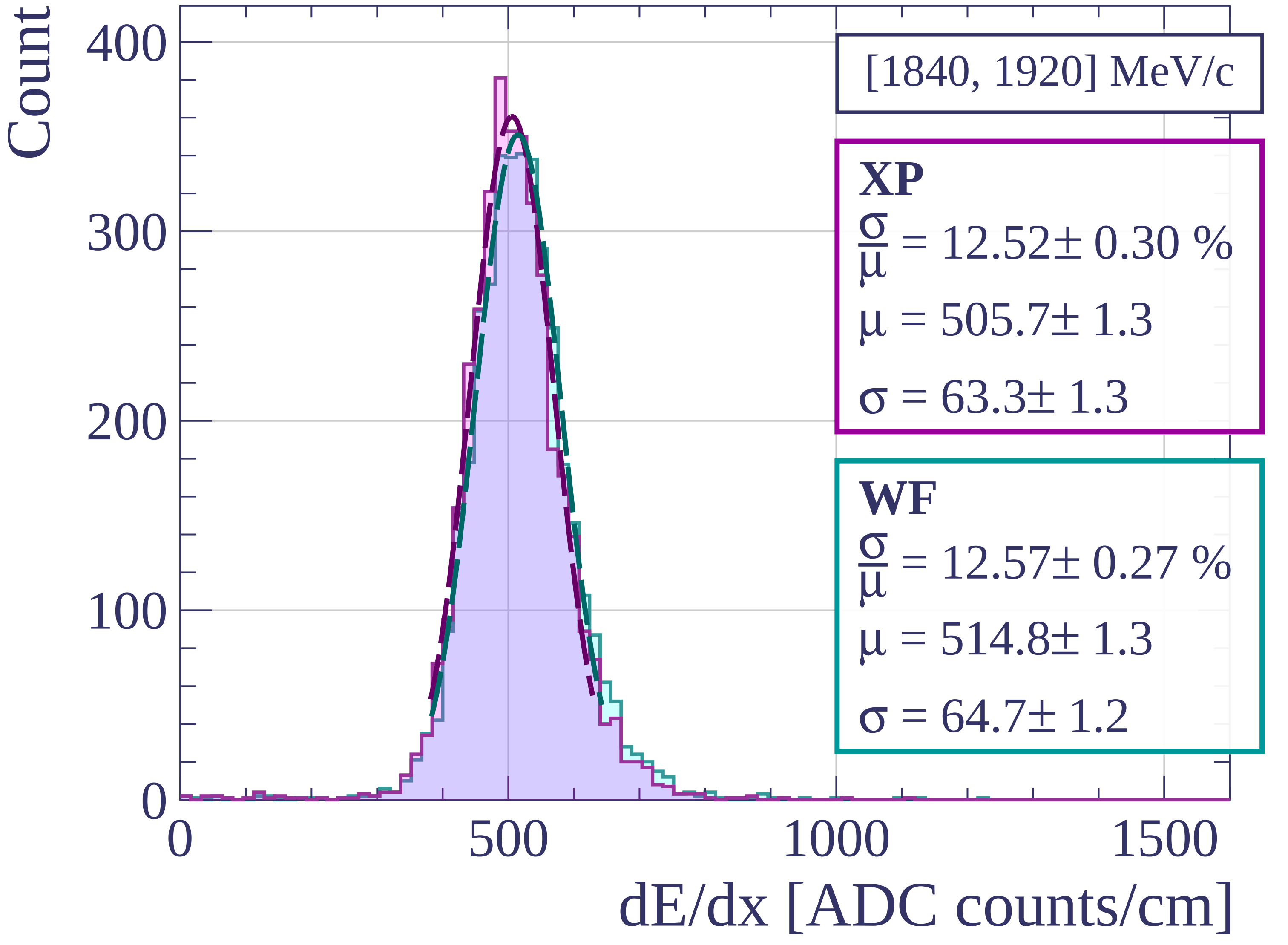

Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1600, 1680] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.61± 0.25 %
μ = 495.9± 1.2
σ = 62.5± 1.1
WF
$\frac{\sigma}{\mu}$ = 13.34± 0.27 %
μ = 504.7± 1.2
σ = 67.3± 1.2



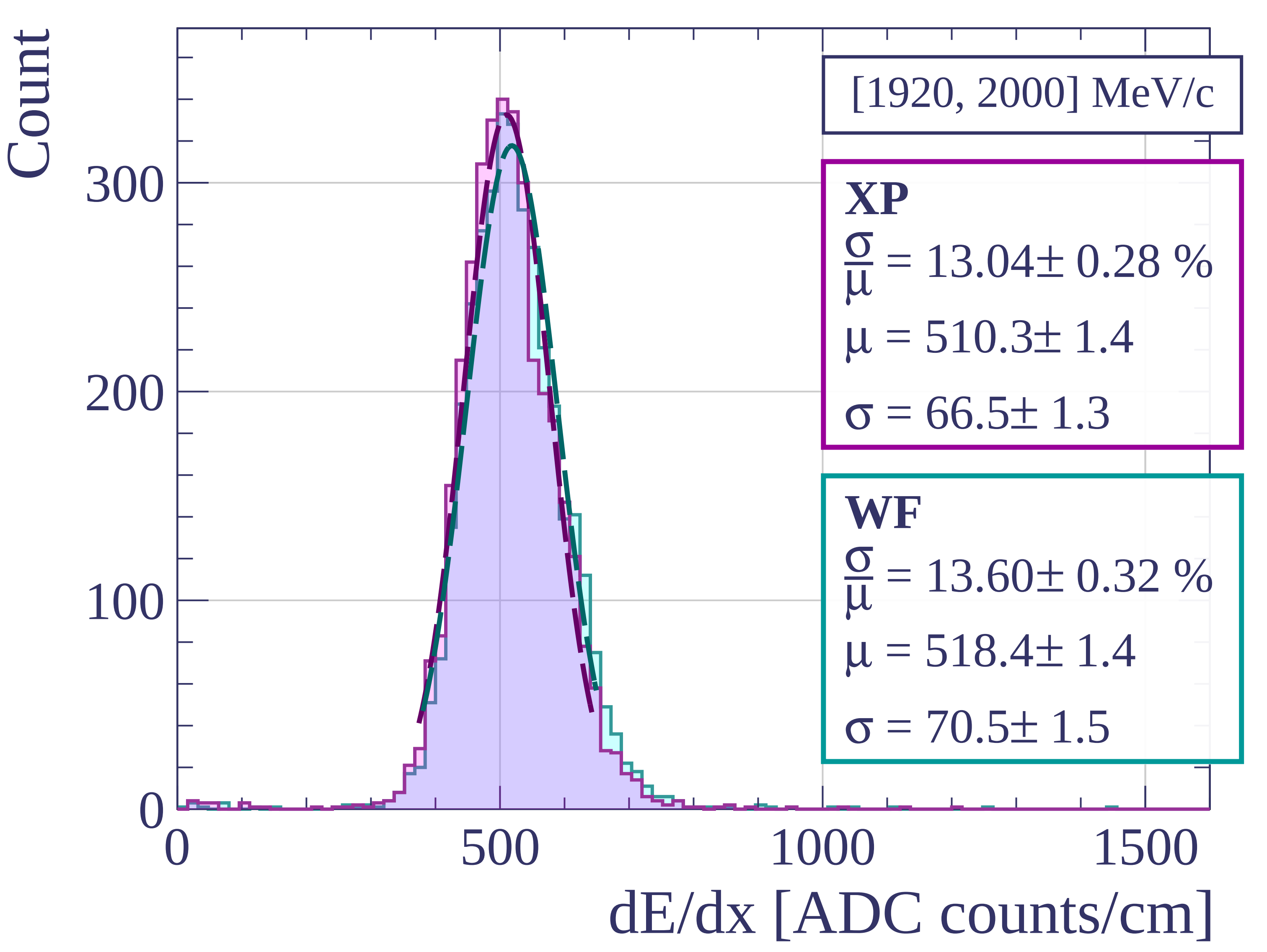
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1680, 1760] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.58± 0.29 %
μ = 502.0± 1.3
σ = 68.1± 1.3
WF
$\frac{\sigma}{\mu}$ = 13.80± 0.29 %
μ = 509.3± 1.3
σ = 70.3± 1.3



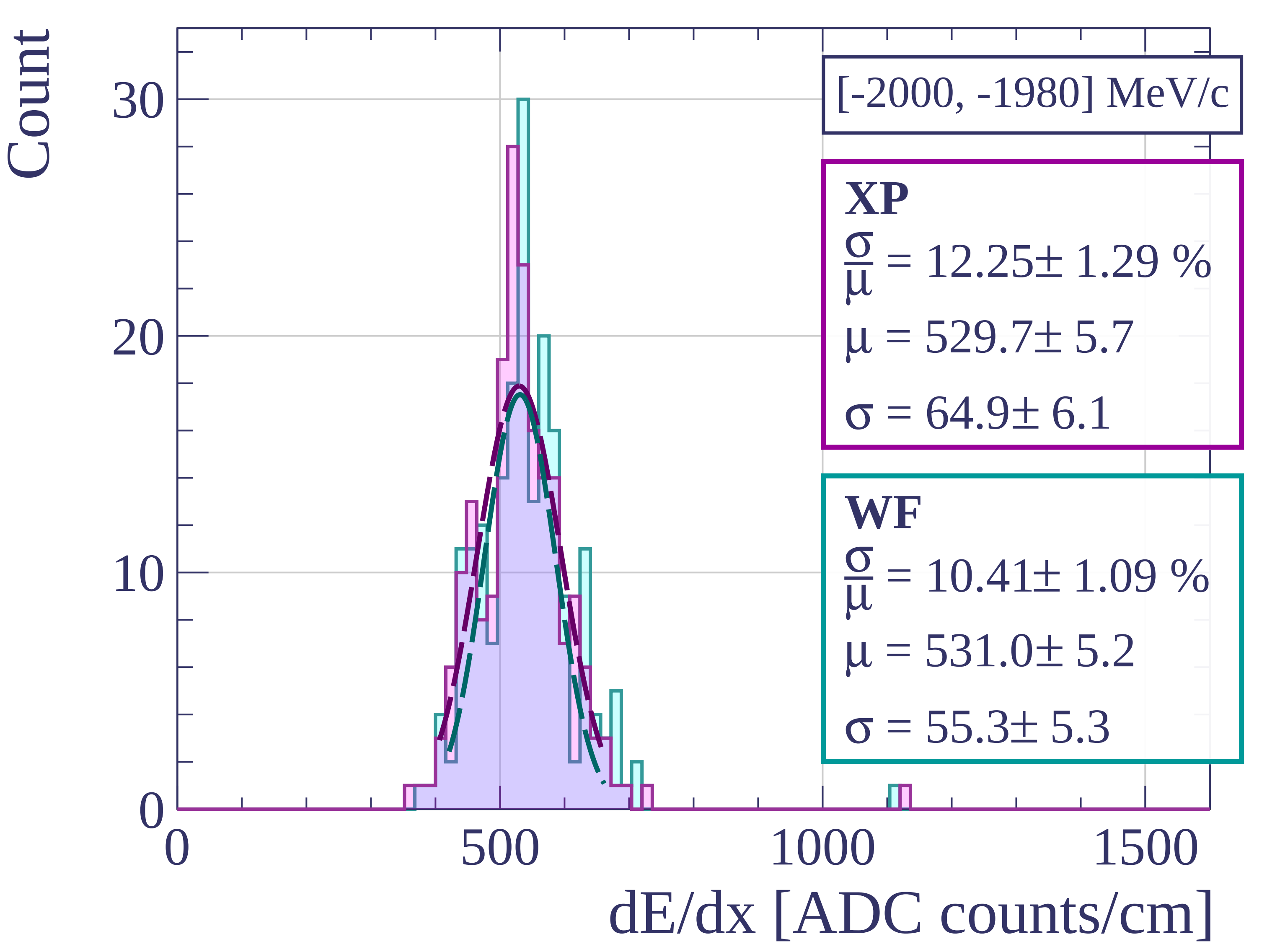
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1760, 1840] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.77± 0.30 %
μ = 503.7± 1.3
σ = 64.3± 1.4
WF
$\frac{\sigma}{\mu}$ = 13.45± 0.33 %
μ = 512.8± 1.4
σ = 69.0± 1.5



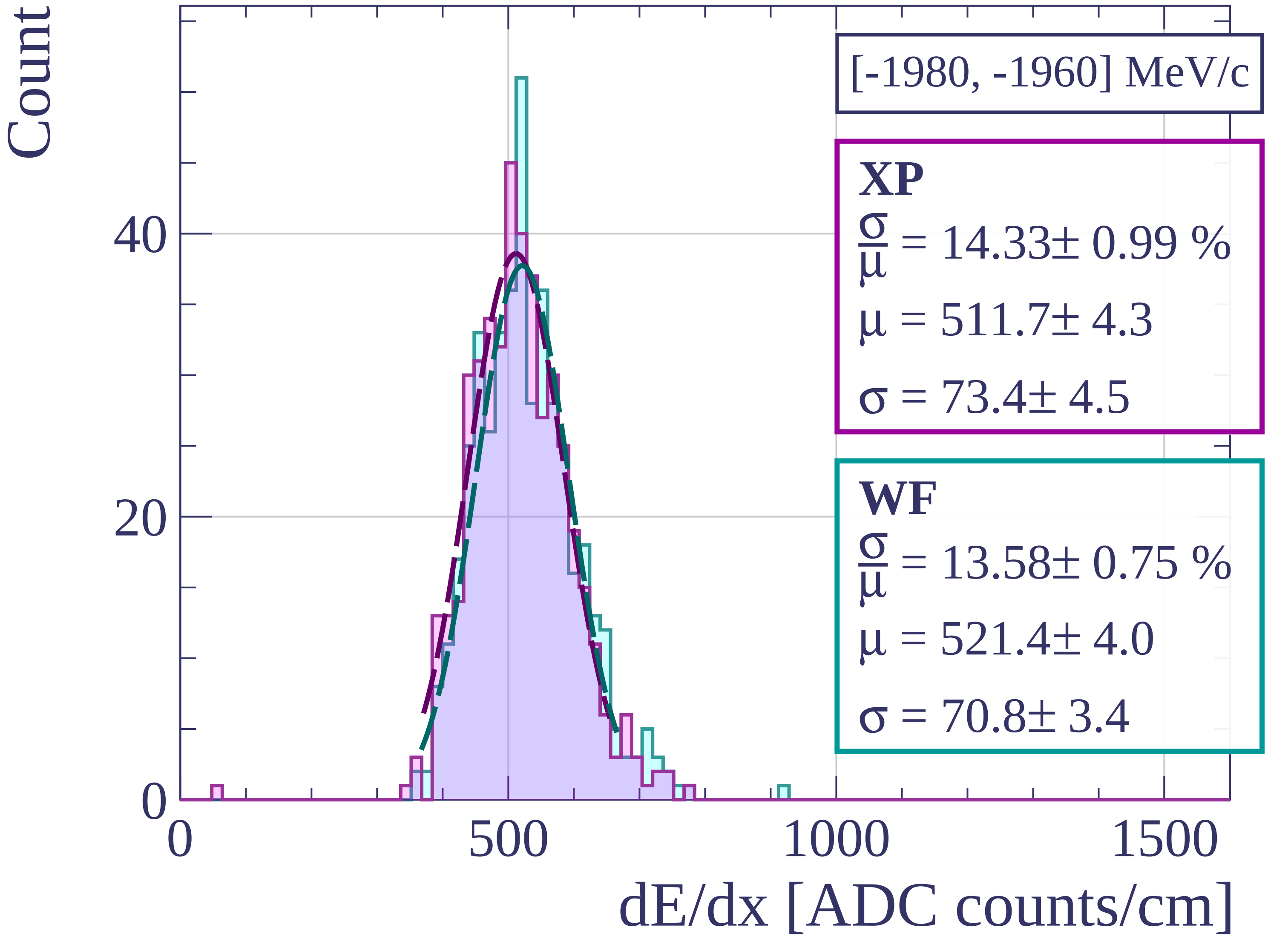
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1840, 1920] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.52± 0.30 %
μ = 505.7± 1.3
σ = 63.3± 1.3
WF
$\frac{\sigma}{\mu}$ = 12.57± 0.27 %
μ = 514.8± 1.3
σ = 64.7± 1.2



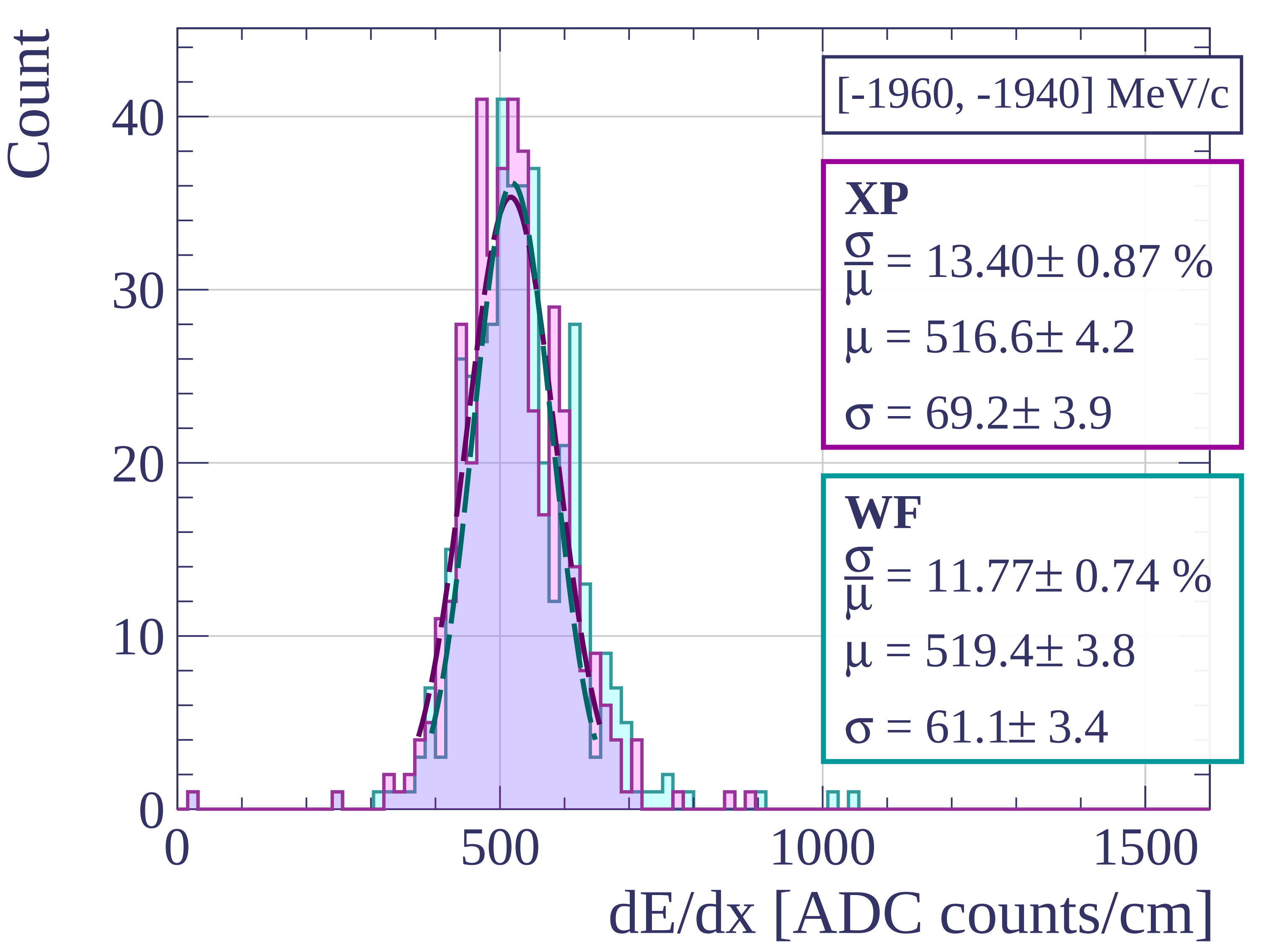
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1920, 2000] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.04± 0.28 %
μ = 510.3± 1.4
σ = 66.5± 1.3
WF
$\frac{\sigma}{\mu}$ = 13.60± 0.32 %
μ = 518.4± 1.4
σ = 70.5± 1.5

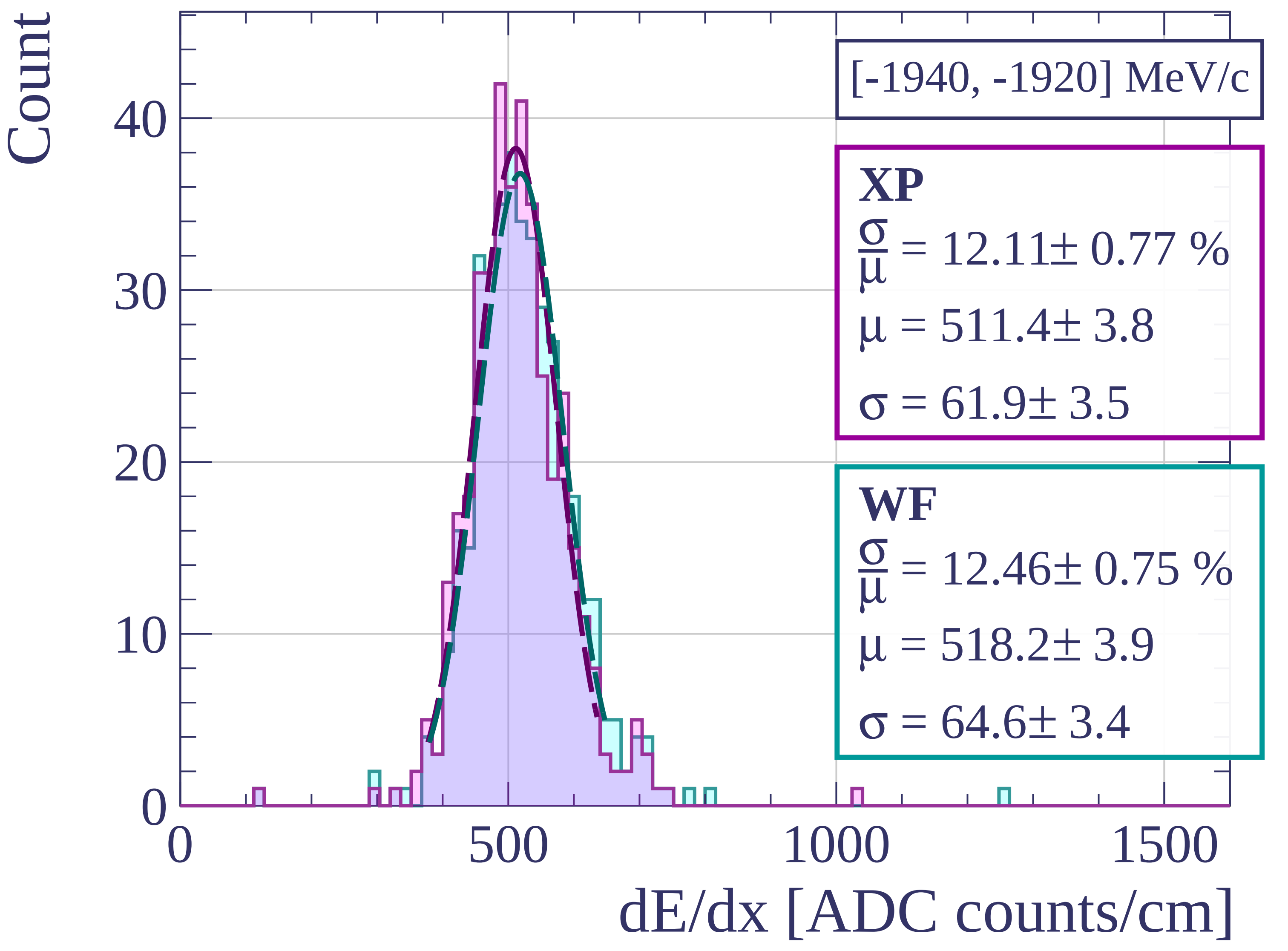


Count
dE/dx [ADC counts/cm]
[-2000, -1980] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.25± 1.29 %
μ = 529.7± 5.7
σ = 64.9± 6.1
WF
$\frac{\sigma}{\mu}$ = 10.41± 1.09 %
μ = 531.0± 5.2
σ = 55.3± 5.3

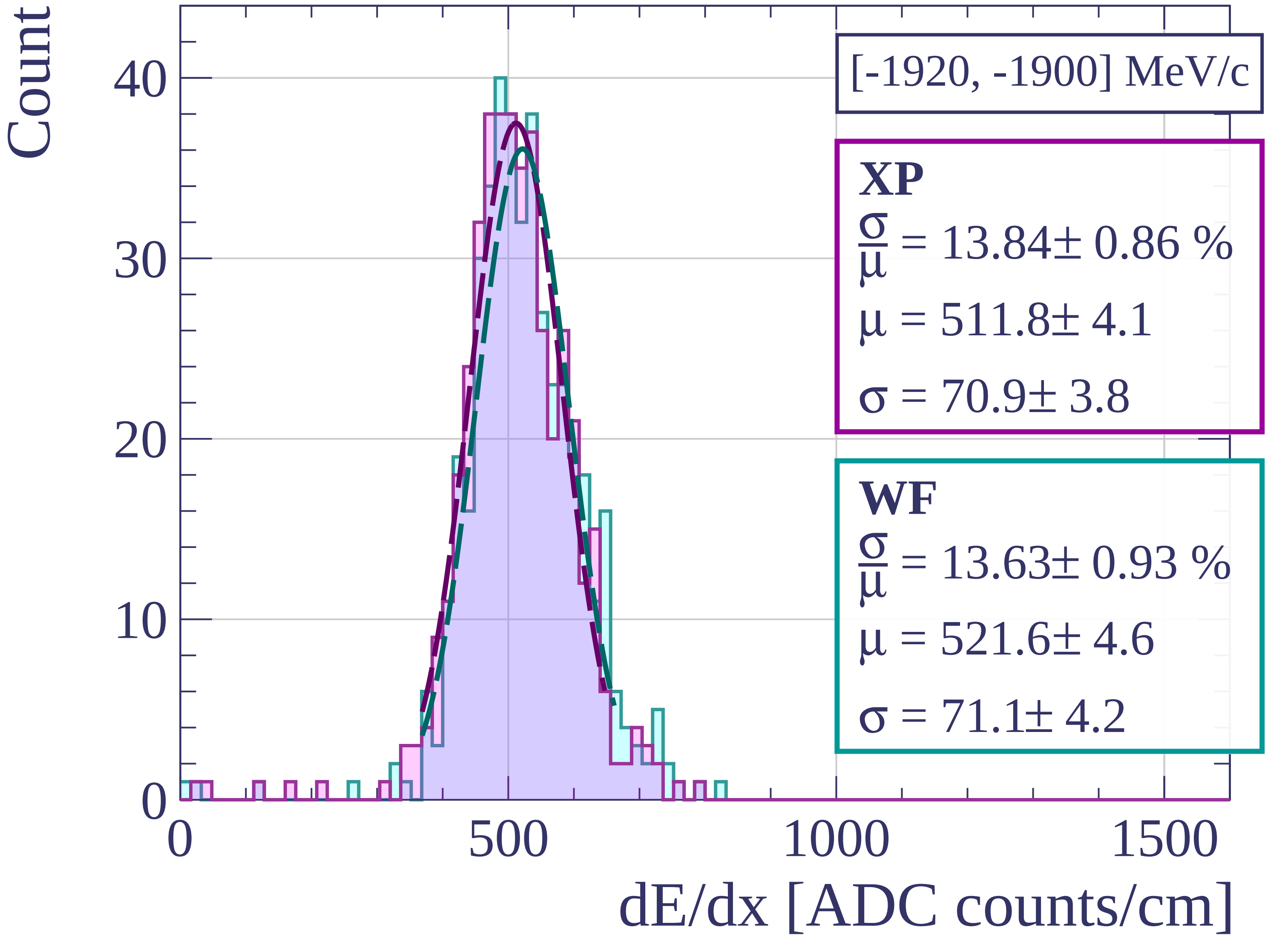


Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1980, -1960] MeV/c
XP
$\frac{\sigma}{\mu}$ = 14.33± 0.99 %
μ = 511.7± 4.3
σ = 73.4± 4.5
WF
$\frac{\sigma}{\mu}$ = 13.58± 0.75 %
μ = 521.4± 4.0
σ = 70.8± 3.4

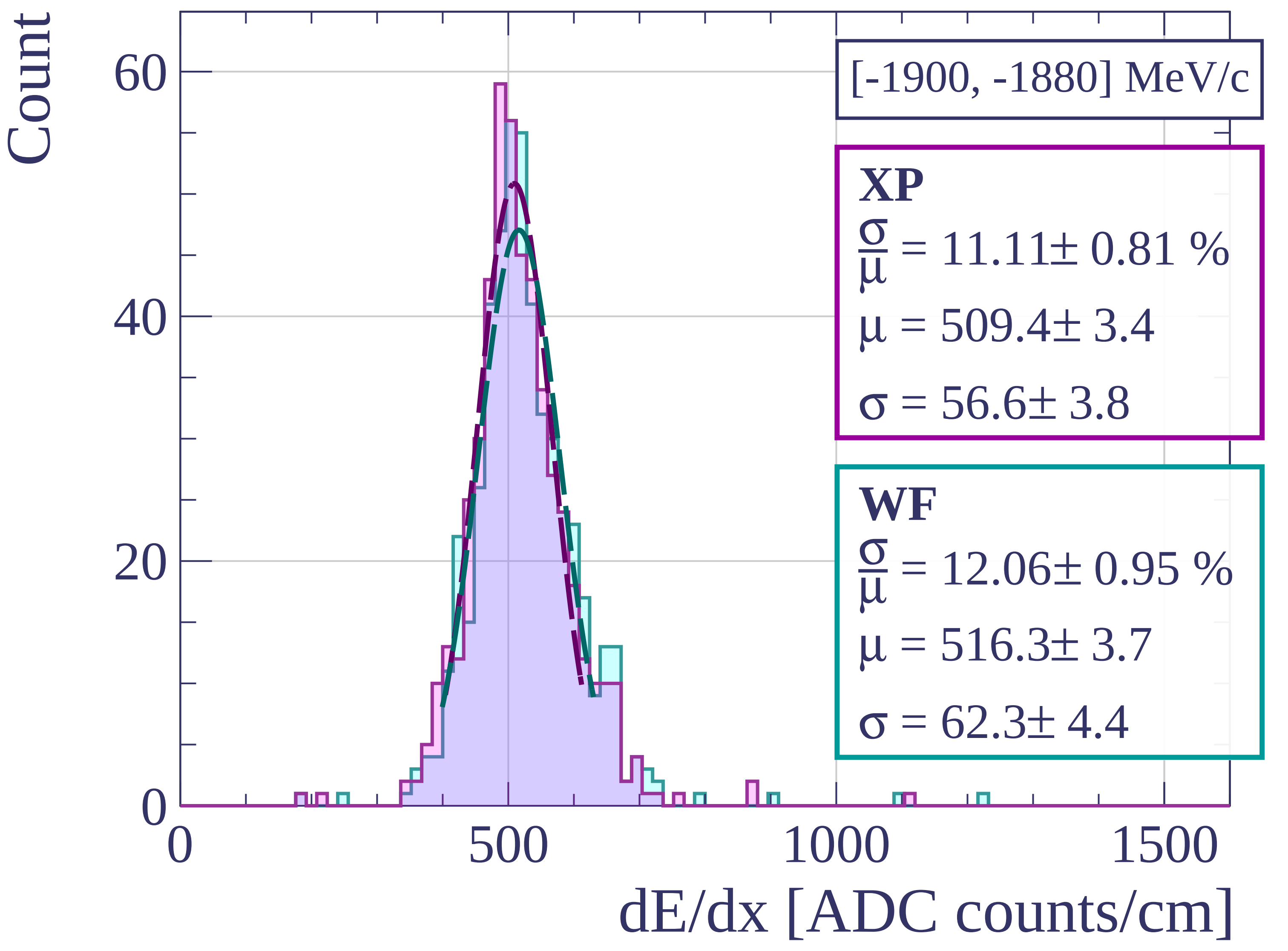

Count
dE/dx [ADC counts/cm]
0
10
20
30
40
0
500
1000
1500
[-1960, -1940] MeV/c
XP
σ/μ = 13.40± 0.87 %
μ = 516.6± 4.2
σ = 69.2± 3.9
WF
σ/μ = 11.77± 0.74 %
μ = 519.4± 3.8
σ = 61.1± 3.4

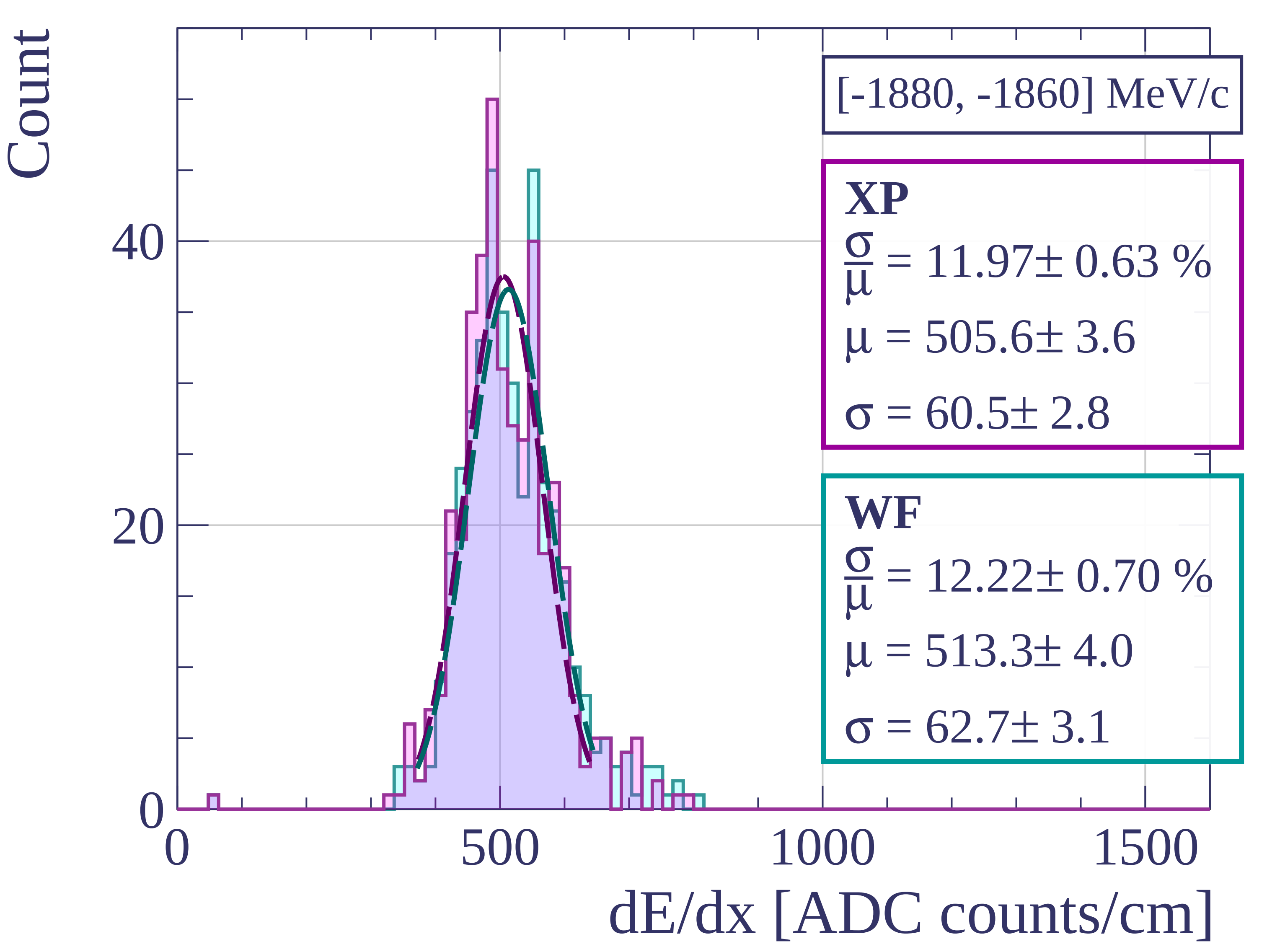


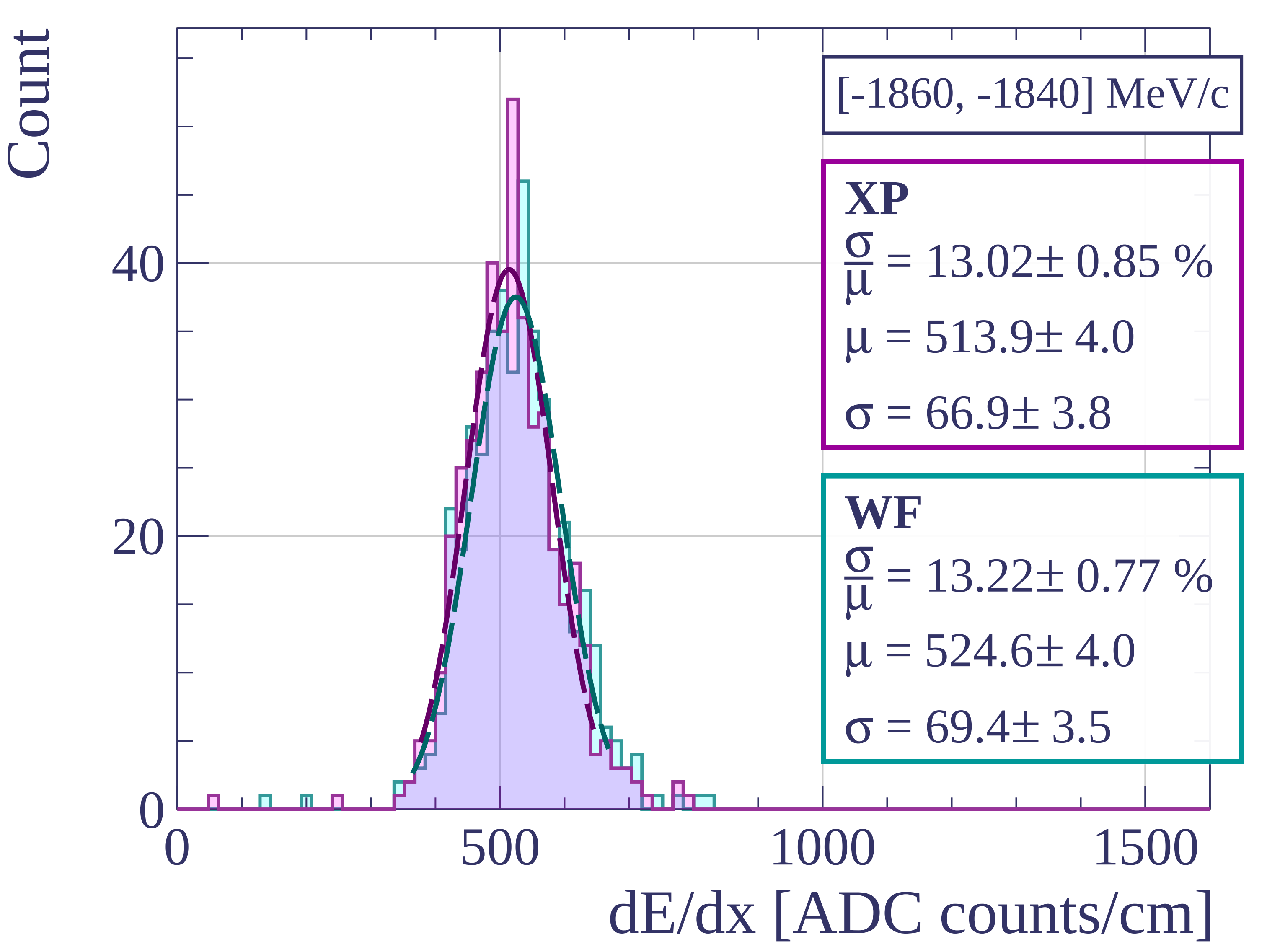
Count
dE/dx [ADC counts/cm]
0
10
20
30
40
0
500
1000
1500
[-1940, -1920] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.11± 0.77 %
μ = 511.4± 3.8
σ = 61.9± 3.5
WF
$\frac{\sigma}{\mu}$ = 12.46± 0.75 %
μ = 518.2± 3.9
σ = 64.6± 3.4



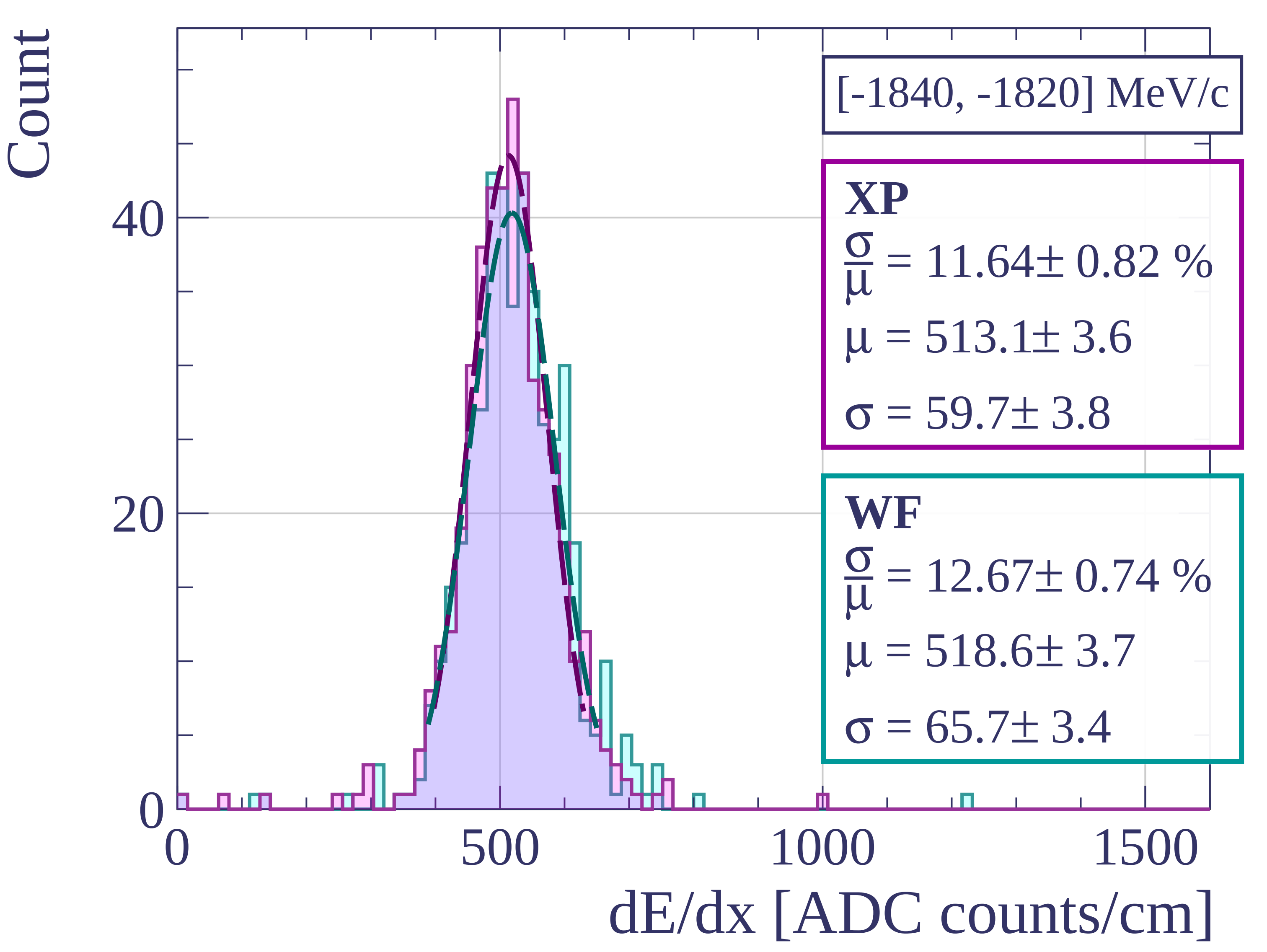
Count
40
30
20
10
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1920, -1900] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.84± 0.86 %
μ = 511.8± 4.1
σ = 70.9± 3.8
WF
$\frac{\sigma}{\mu}$ = 13.63± 0.93 %
μ = 521.6± 4.6
σ = 71.1± 4.2


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1900, -1880] MeV/c
XP
σ/μ = 11.11± 0.81 %
μ = 509.4± 3.4
σ = 56.6± 3.8
WF
σ/μ = 12.06± 0.95 %
μ = 516.3± 3.7
σ = 62.3± 4.4

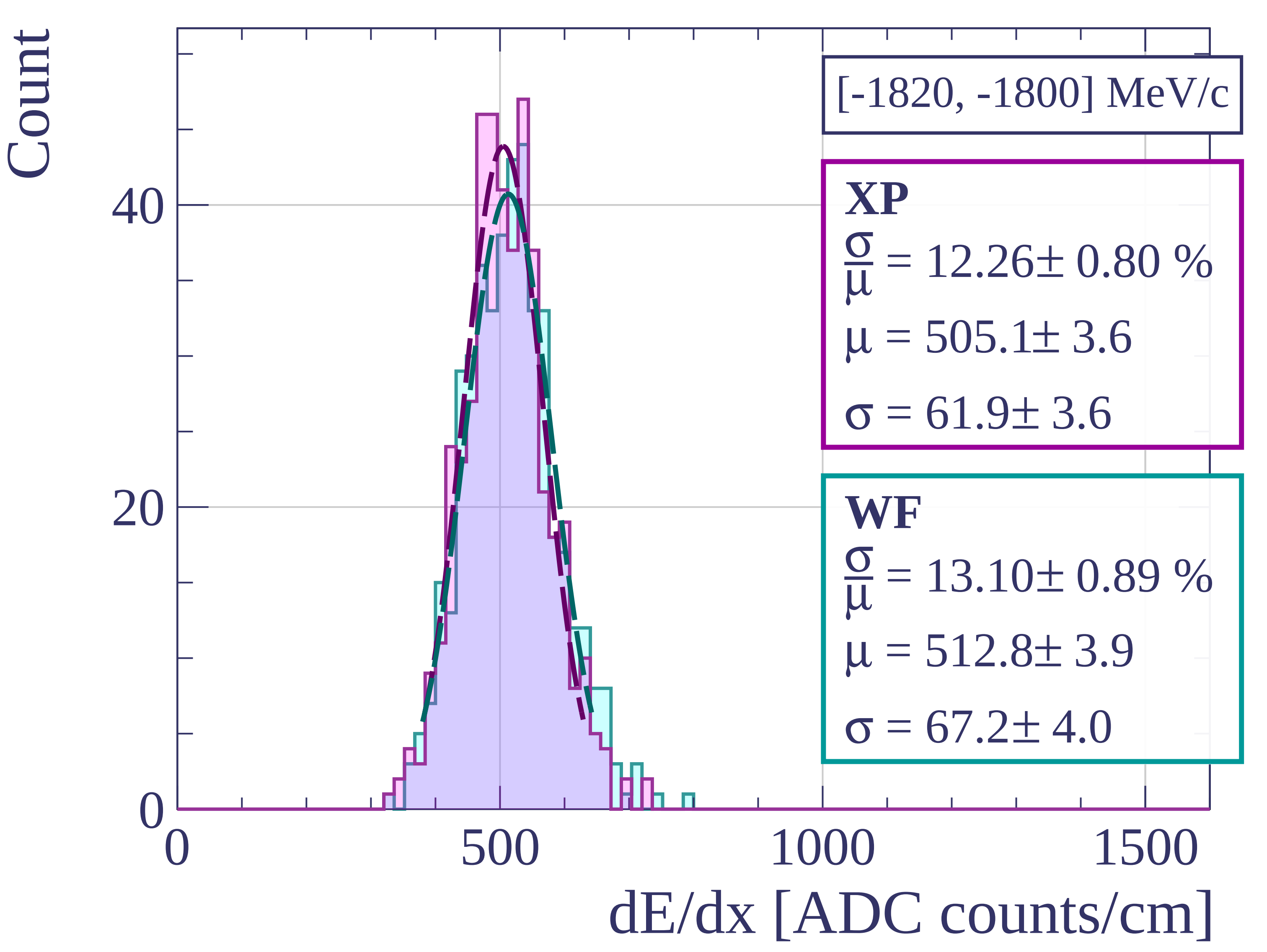

Count
0
20
40
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1880, -1860] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.97± 0.63 %
μ = 505.6± 3.6
σ = 60.5± 2.8
WF
$\frac{\sigma}{\mu}$ = 12.22± 0.70 %
μ = 513.3± 4.0
σ = 62.7± 3.1



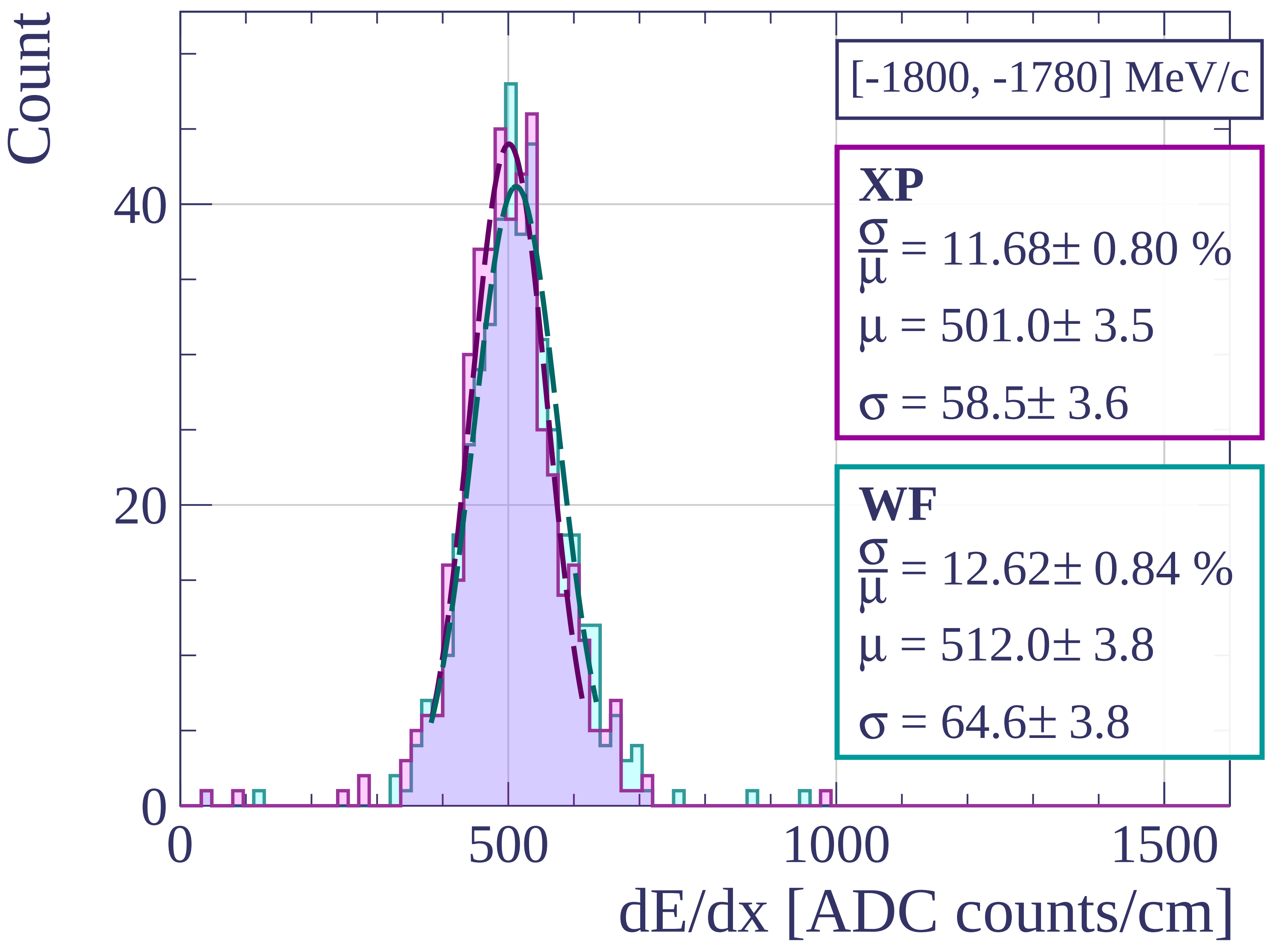
Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1860, -1840] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.02± 0.85 %
μ = 513.9± 4.0
σ = 66.9± 3.8
WF
$\frac{\sigma}{\mu}$ = 13.22± 0.77 %
μ = 524.6± 4.0
σ = 69.4± 3.5



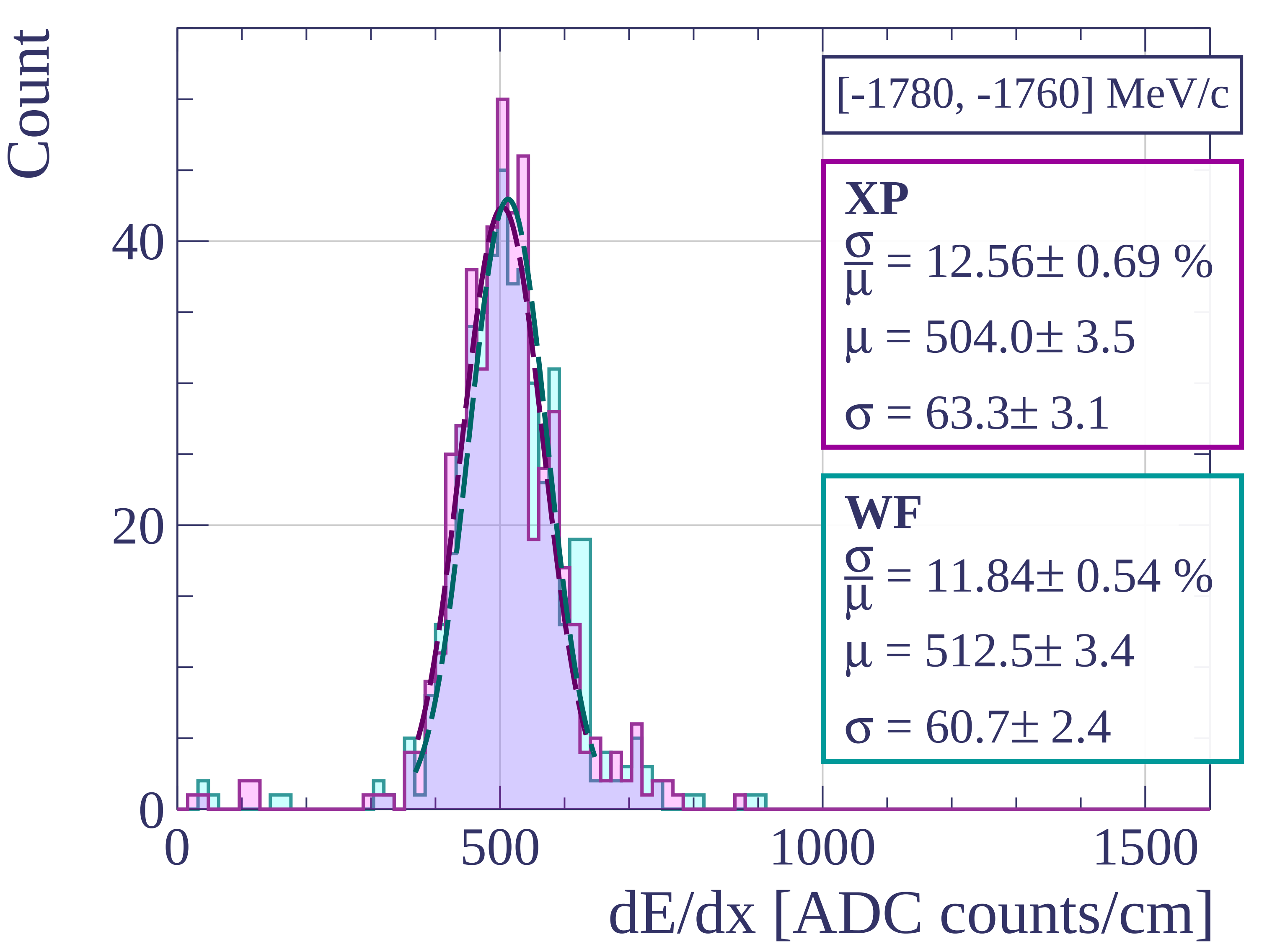
Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1840, -1820] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.64± 0.82 %
μ = 513.1± 3.6
σ = 59.7± 3.8
WF
$\frac{\sigma}{\mu}$ = 12.67± 0.74 %
μ = 518.6± 3.7
σ = 65.7± 3.4

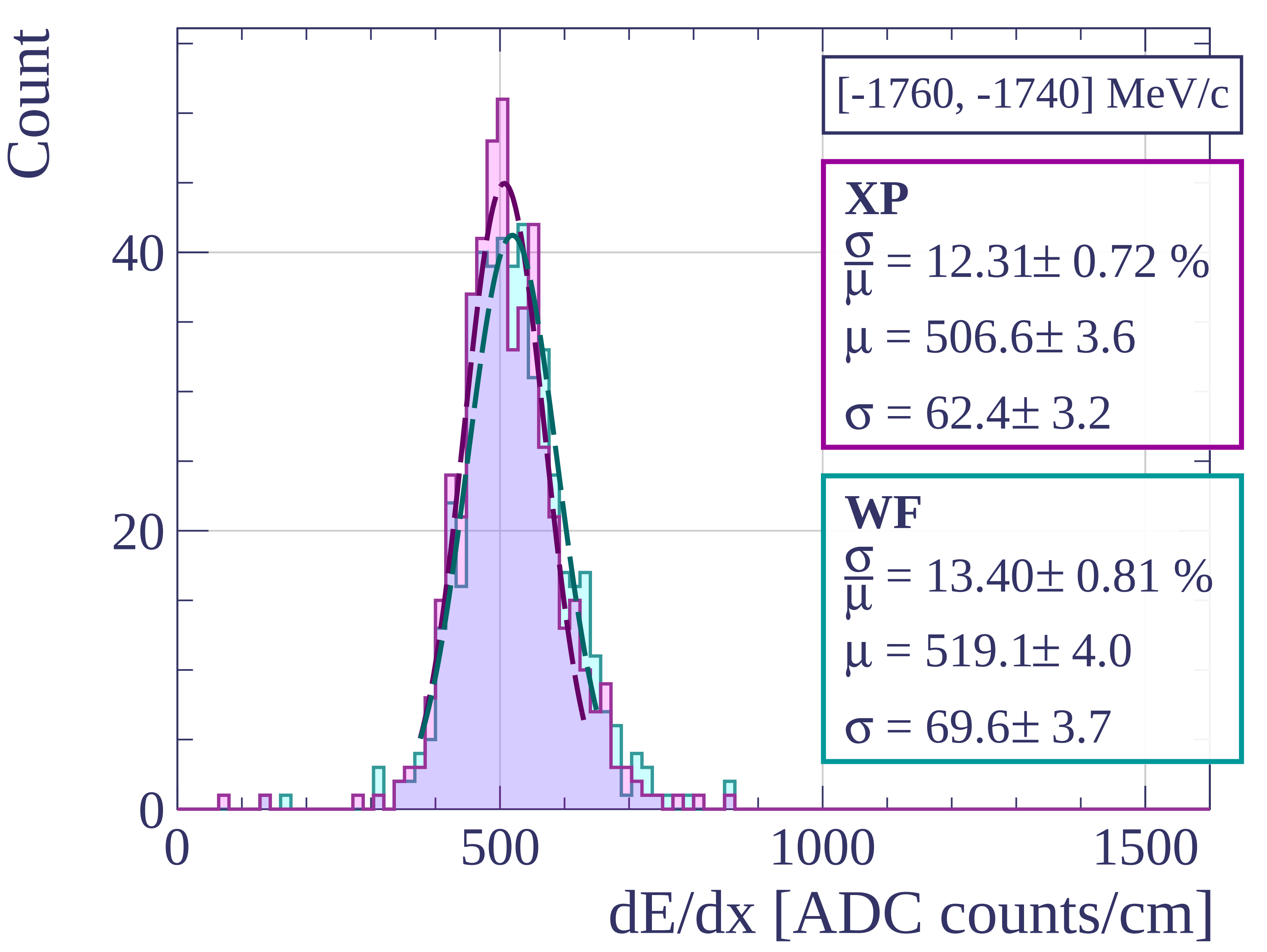


Count
0
20
40
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1820, -1800] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.26± 0.80 %
μ = 505.1± 3.6
σ = 61.9± 3.6
WF
$\frac{\sigma}{\mu}$ = 13.10± 0.89 %
μ = 512.8± 3.9
σ = 67.2± 4.0

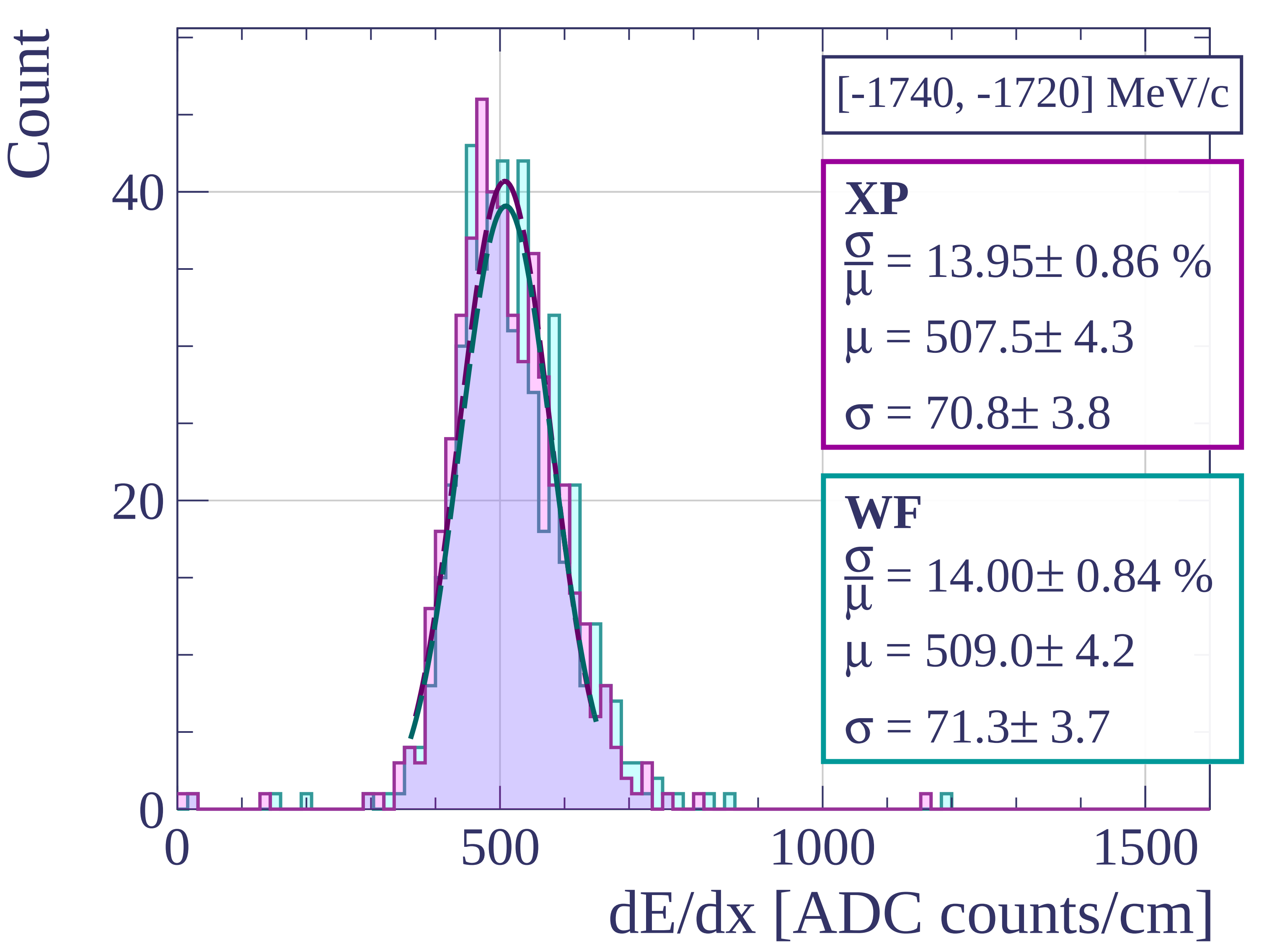


Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1800, -1780] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.68± 0.80 %
μ = 501.0± 3.5
σ = 58.5± 3.6
WF
$\frac{\sigma}{\mu}$ = 12.62± 0.84 %
μ = 512.0± 3.8
σ = 64.6± 3.8

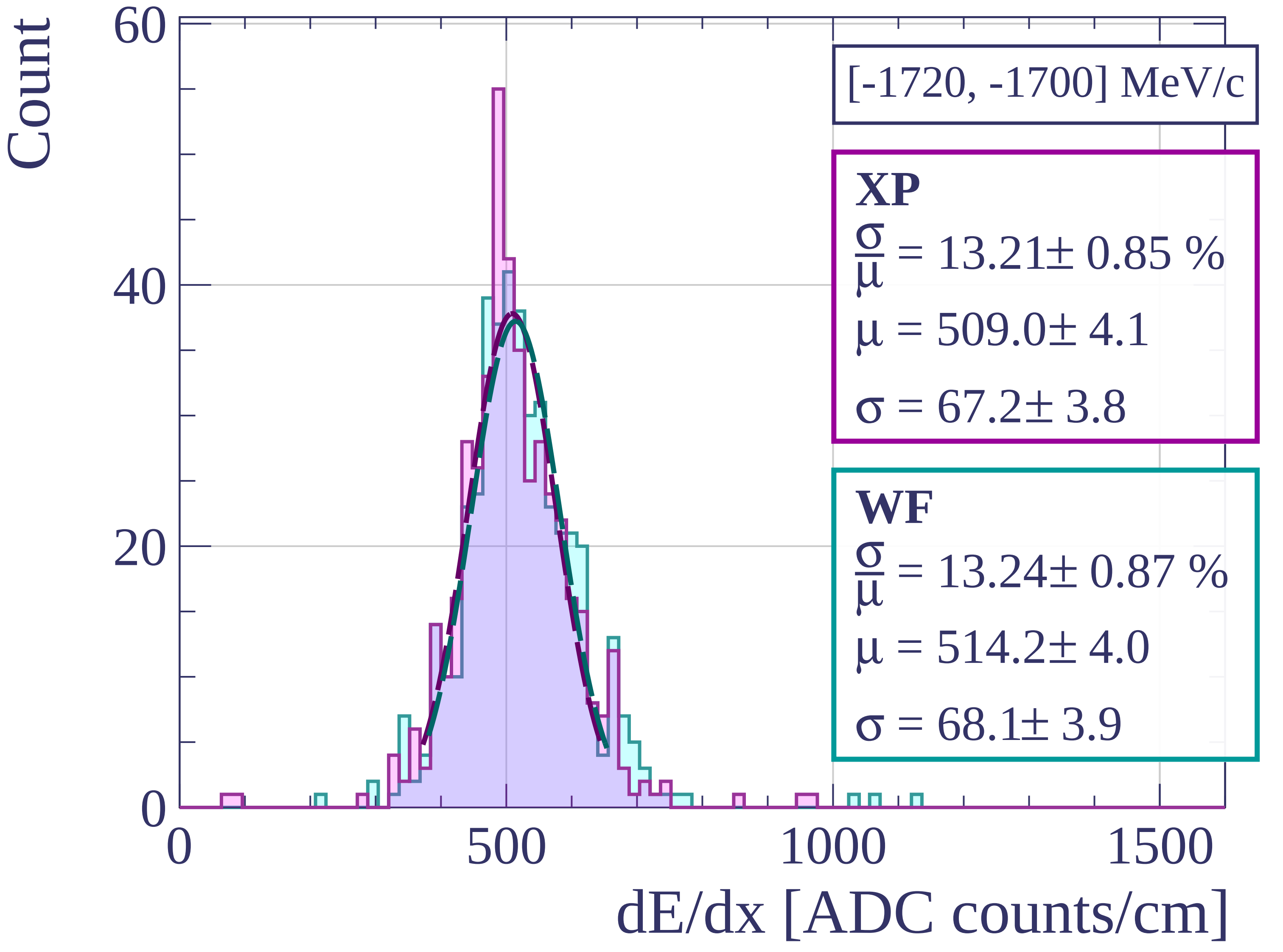

Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1780, -1760] MeV/c
XP
σ/μ = 12.56± 0.69 %
μ = 504.0± 3.5
σ = 63.3± 3.1
WF
σ/μ = 11.84± 0.54 %
μ = 512.5± 3.4
σ = 60.7± 2.4

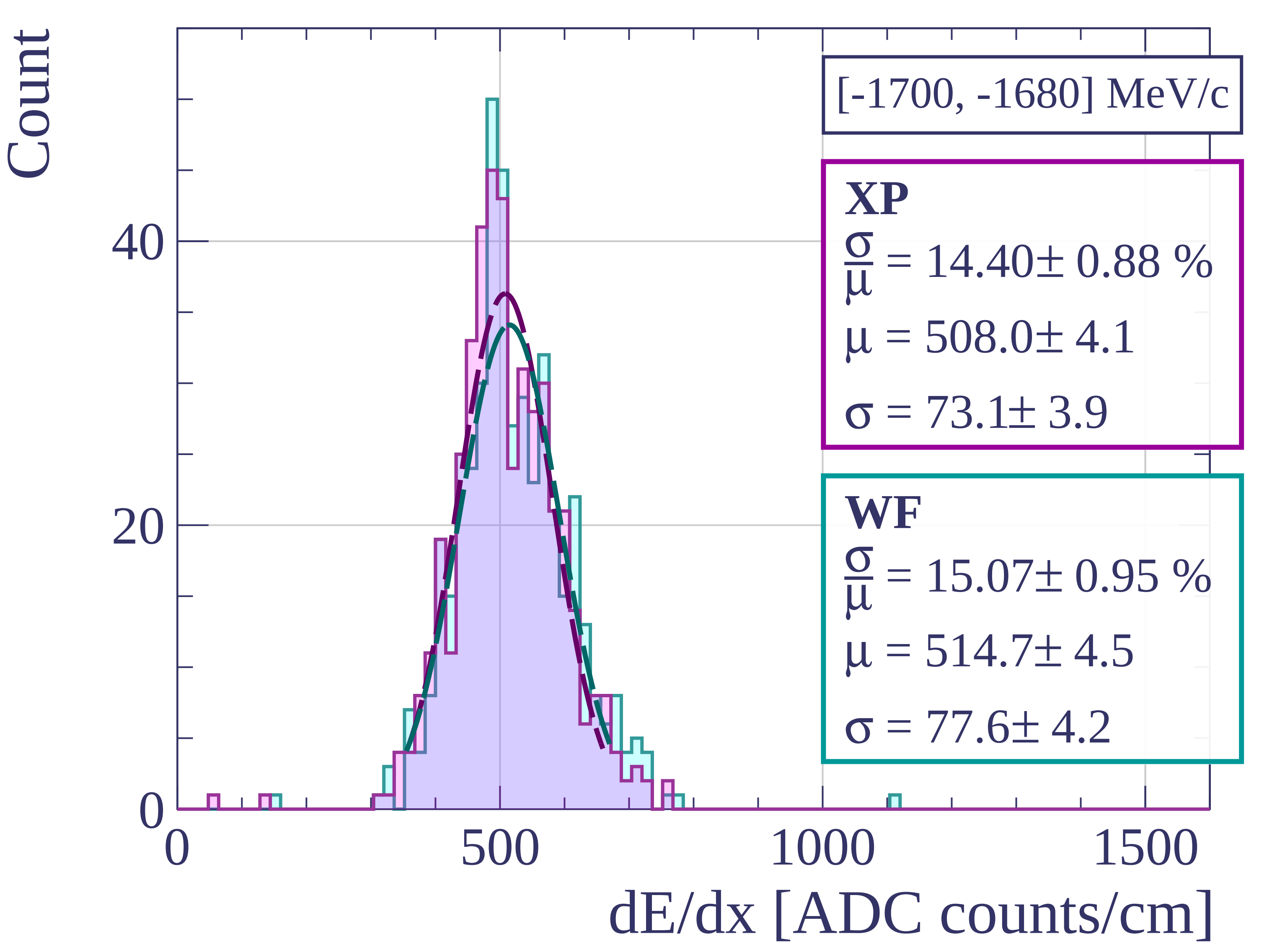


Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1760, -1740] MeV/c
XP
σ/μ = 12.31± 0.72 %
μ = 506.6± 3.6
σ = 62.4± 3.2
WF
σ/μ = 13.40± 0.81 %
μ = 519.1± 4.0
σ = 69.6± 3.7

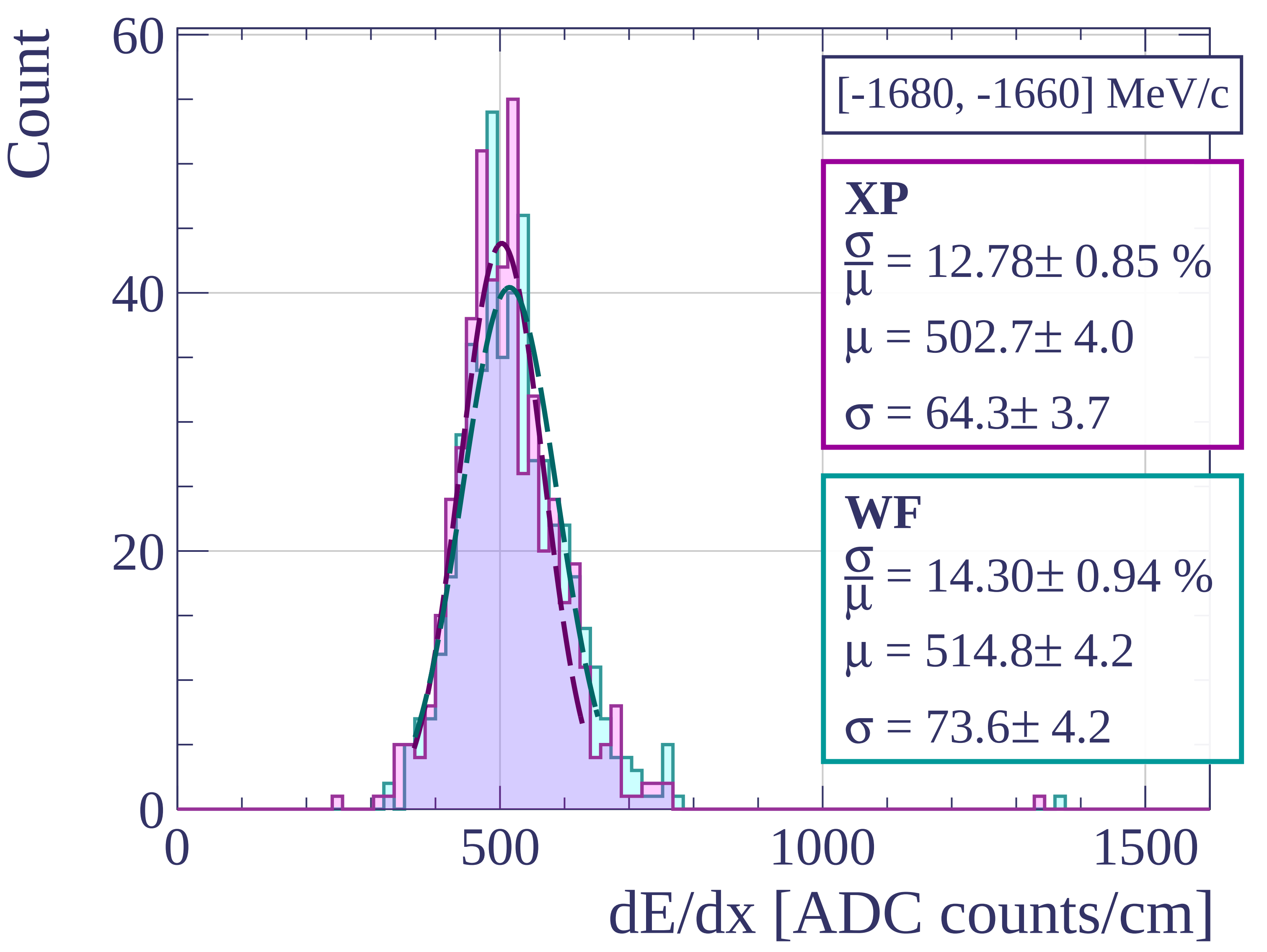


Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1740, -1720] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.95± 0.86 %
μ = 507.5± 4.3
σ = 70.8± 3.8
WF
$\frac{\sigma}{\mu}$ = 14.00± 0.84 %
μ = 509.0± 4.2
σ = 71.3± 3.7

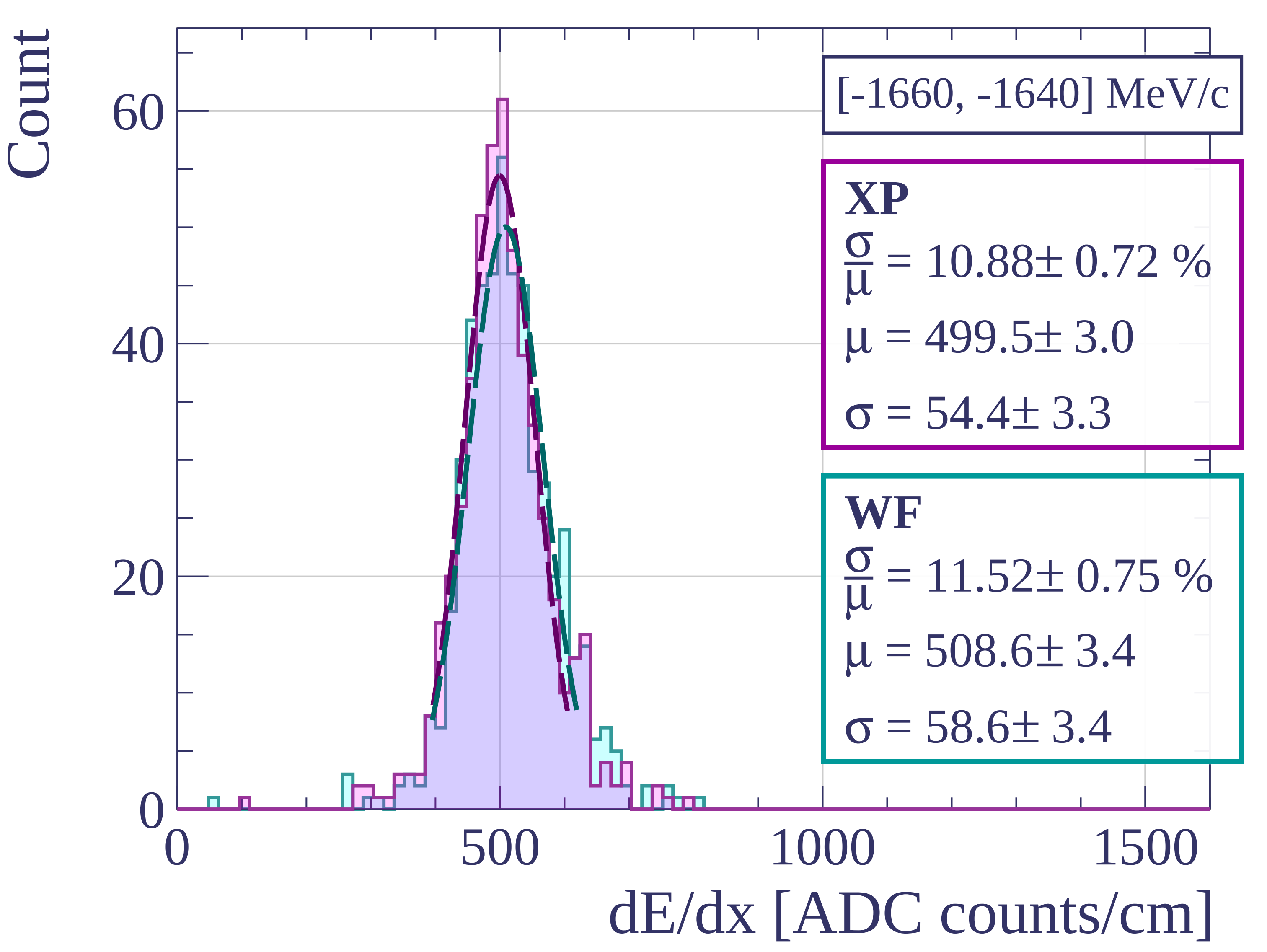

Count
dE/dx [ADC counts/cm]
0
20
40
60
0
500
1000
1500
[-1720, -1700] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.21± 0.85 %
μ = 509.0± 4.1
σ = 67.2± 3.8
WF
$\frac{\sigma}{\mu}$ = 13.24± 0.87 %
μ = 514.2± 4.0
σ = 68.1± 3.9



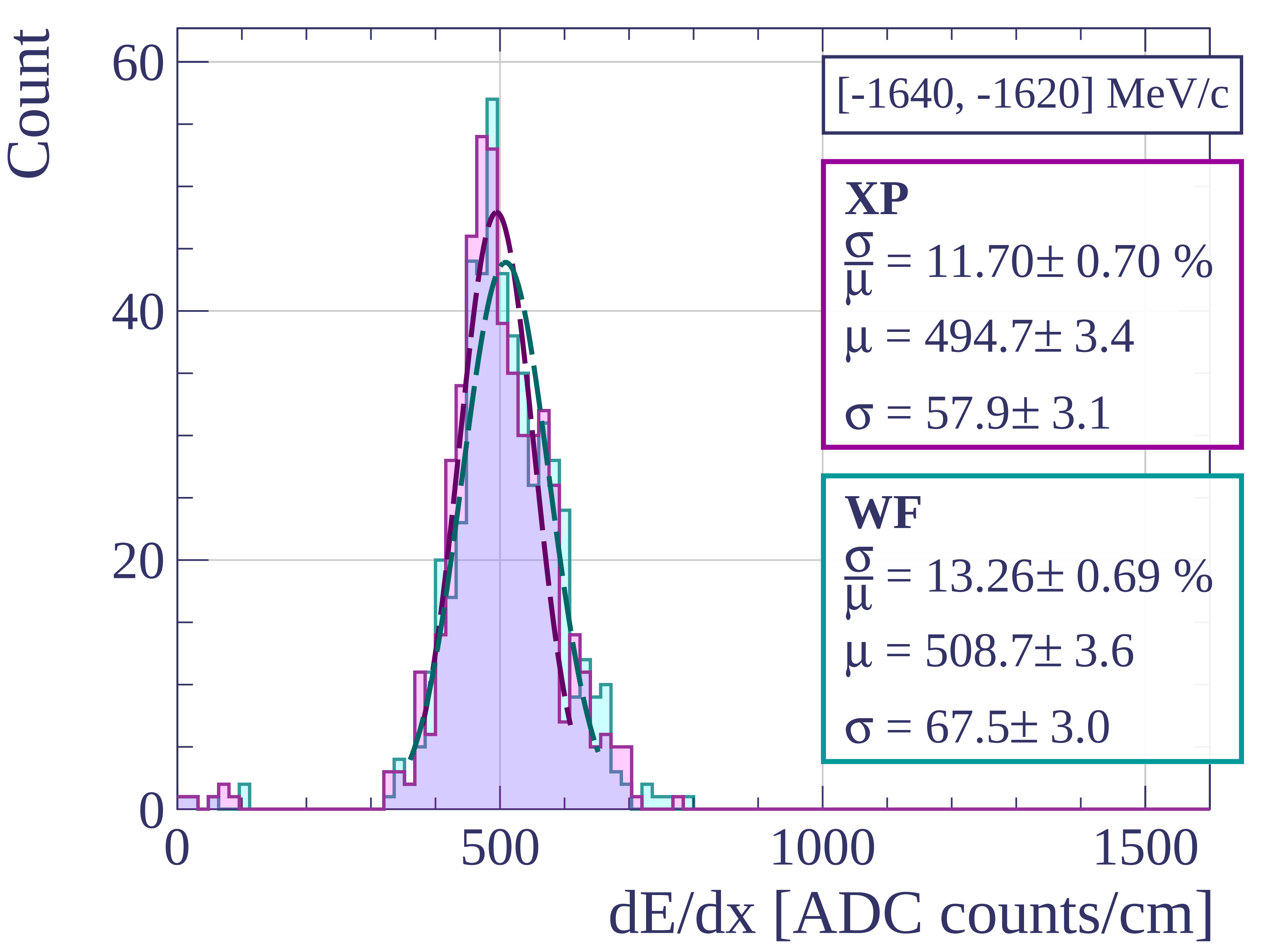
Count
dE/dx [ADC counts/cm]
[-1700, -1680] MeV/c
XP
$\frac{\sigma}{\mu}$ = 14.40± 0.88 %
μ = 508.0± 4.1
σ = 73.1± 3.9
WF
$\frac{\sigma}{\mu}$ = 15.07± 0.95 %
μ = 514.7± 4.5
σ = 77.6± 4.2
0
20
40
0
500
1000
1500



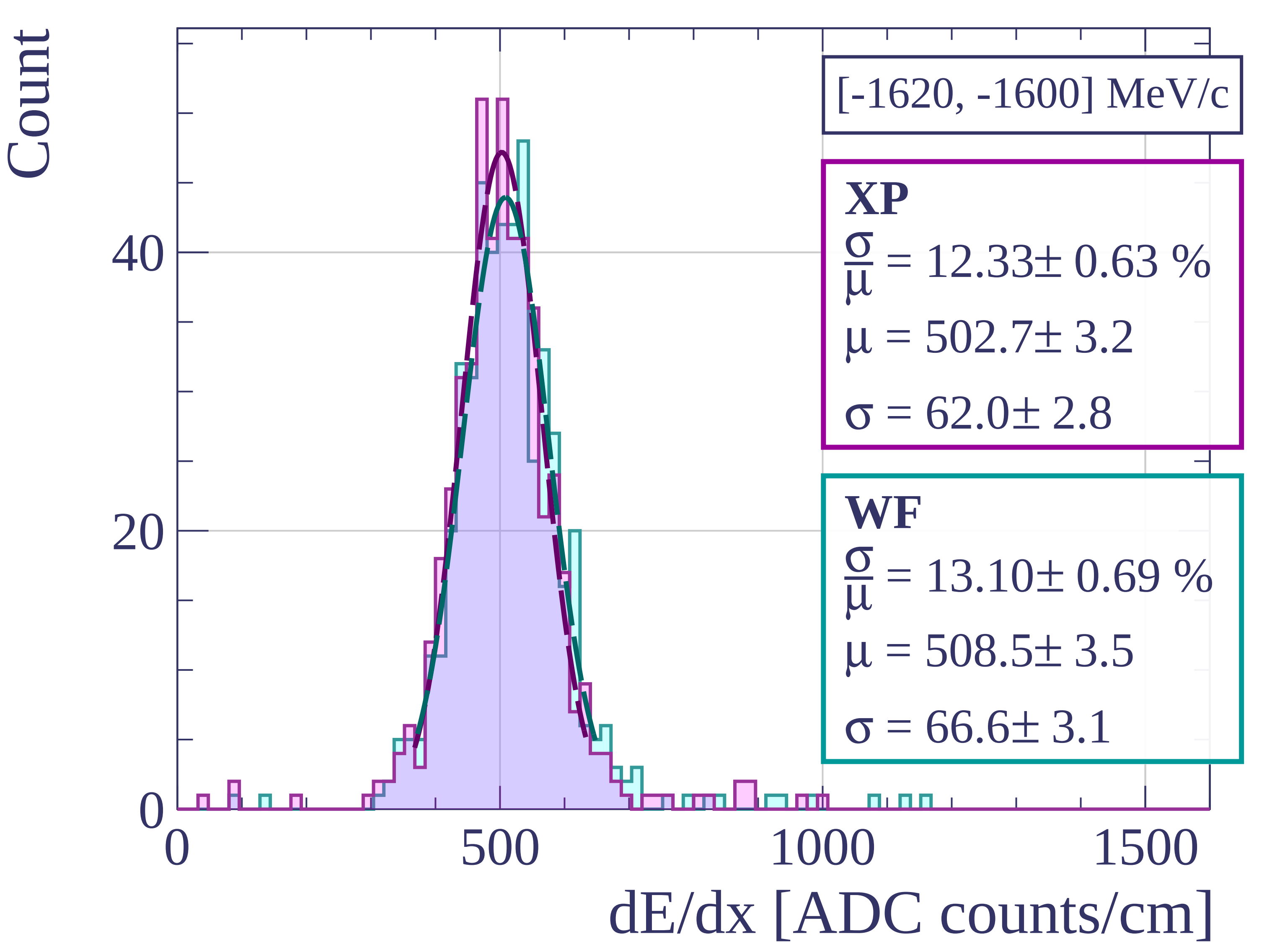
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1680, -1660] MeV/c
XP
σ/μ = 12.78± 0.85 %
μ = 502.7± 4.0
σ = 64.3± 3.7
WF
σ/μ = 14.30± 0.94 %
μ = 514.8± 4.2
σ = 73.6± 4.2

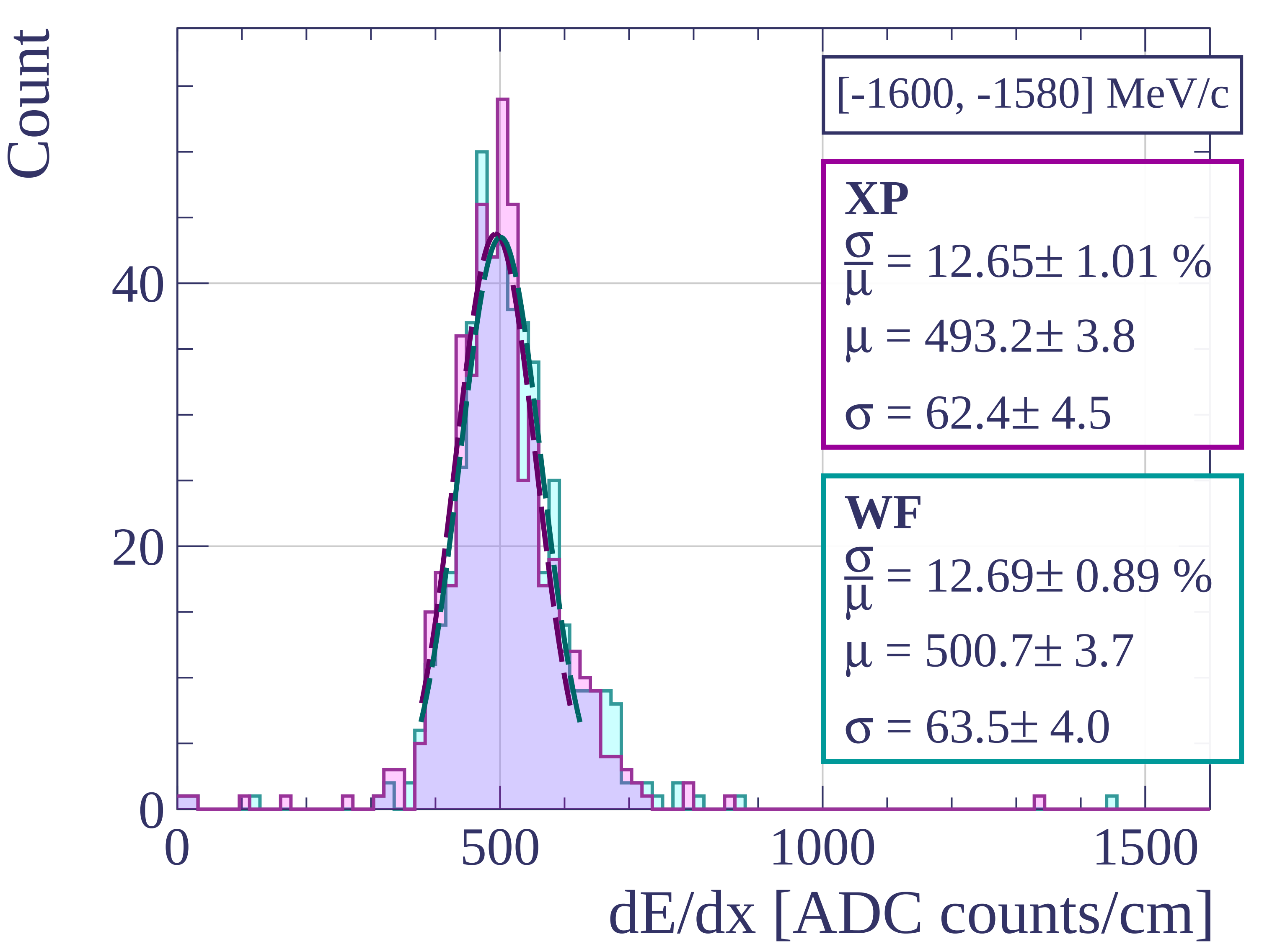


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1660, -1640] MeV/c
XP
$\frac{\sigma}{\mu}$ = 10.88± 0.72 %
μ = 499.5± 3.0
σ = 54.4± 3.3
WF
$\frac{\sigma}{\mu}$ = 11.52± 0.75 %
μ = 508.6± 3.4
σ = 58.6± 3.4

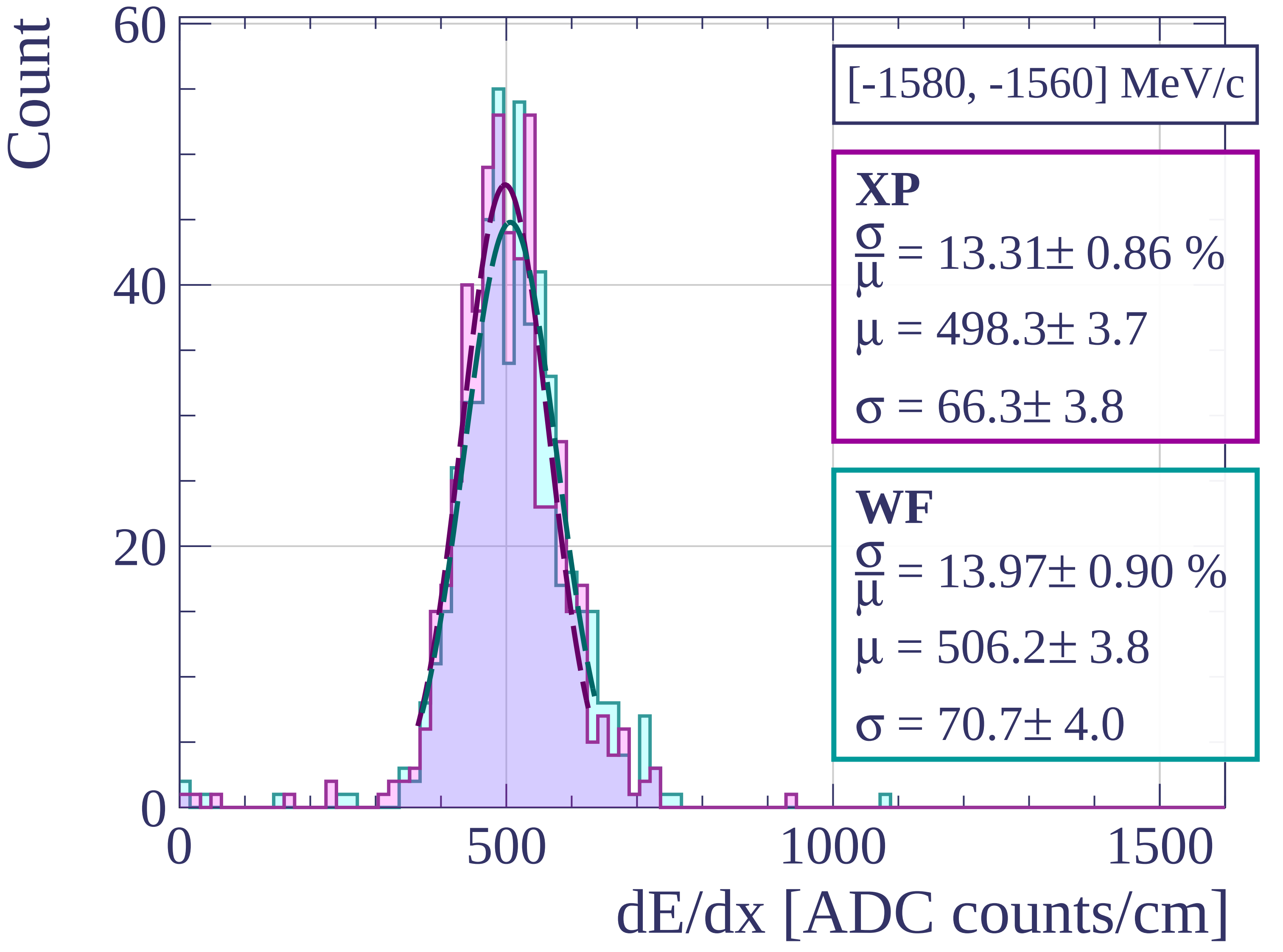


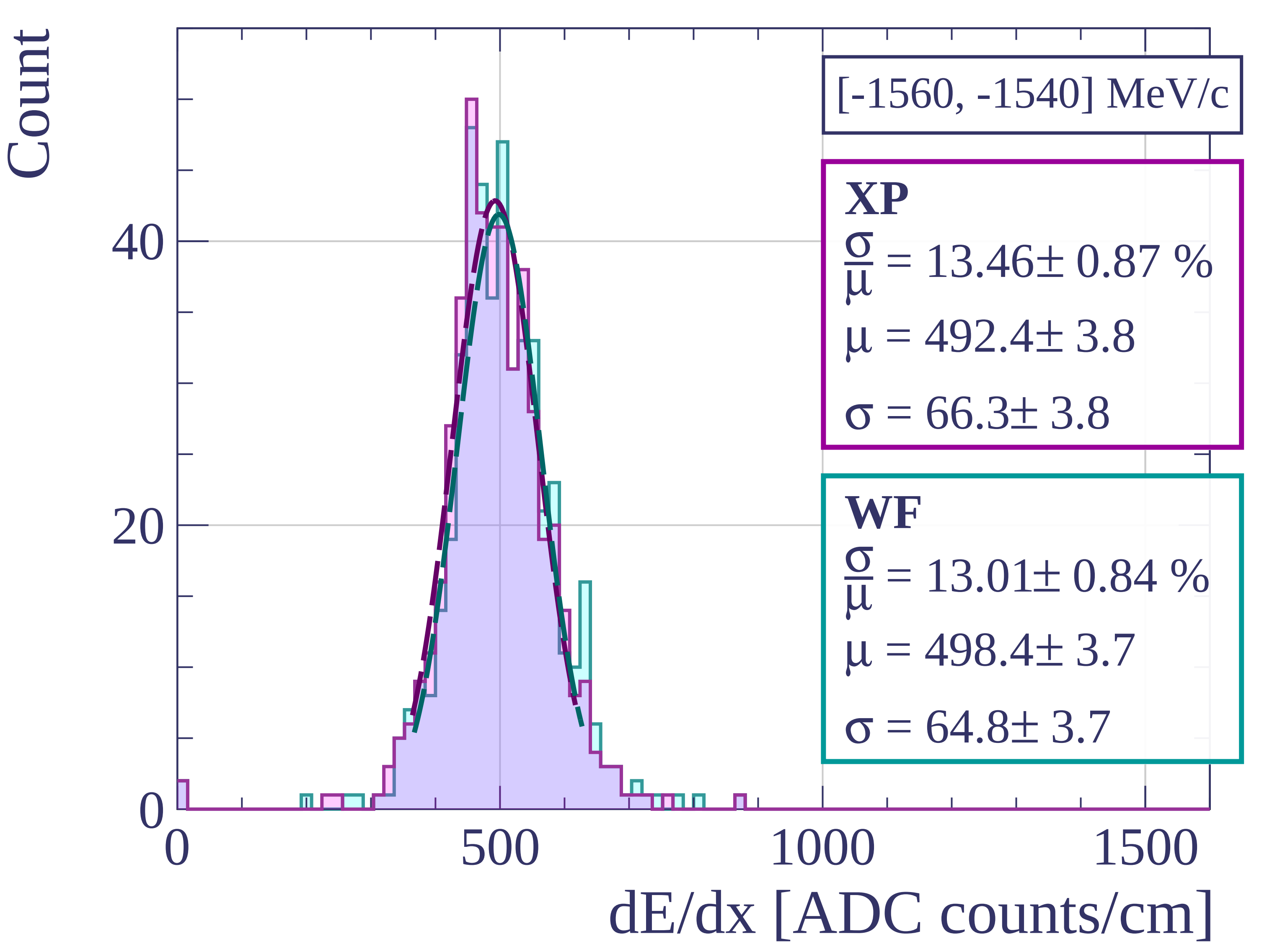
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1640, -1620] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.70± 0.70 %
μ = 494.7± 3.4
σ = 57.9± 3.1
WF
$\frac{\sigma}{\mu}$ = 13.26± 0.69 %
μ = 508.7± 3.6
σ = 67.5± 3.0


Count
dE/dx [ADC counts/cm]
0
20
40
0
500
1000
1500
[-1620, -1600] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.33± 0.63 %
μ = 502.7± 3.2
σ = 62.0± 2.8
WF
$\frac{\sigma}{\mu}$ = 13.10± 0.69 %
μ = 508.5± 3.5
σ = 66.6± 3.1

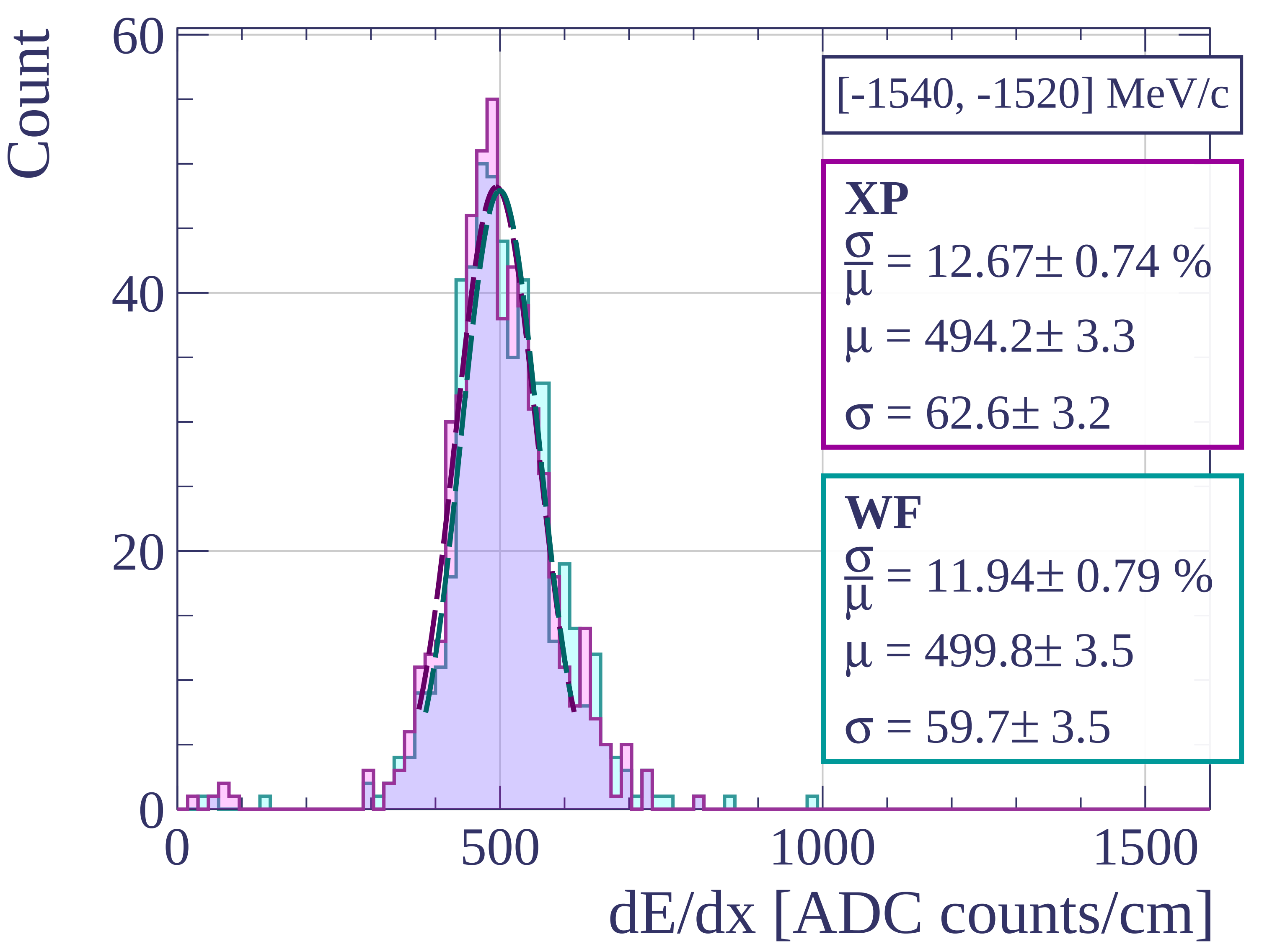


Count
[-1600, -1580] MeV/c
XP
σ/μ = 12.65± 1.01 %
μ = 493.2± 3.8
σ = 62.4± 4.5
WF
σ/μ = 12.69± 0.89 %
μ = 500.7± 3.7
σ = 63.5± 4.0
0
20
40
0
500
1000
1500
dE/dx [ADC counts/cm]

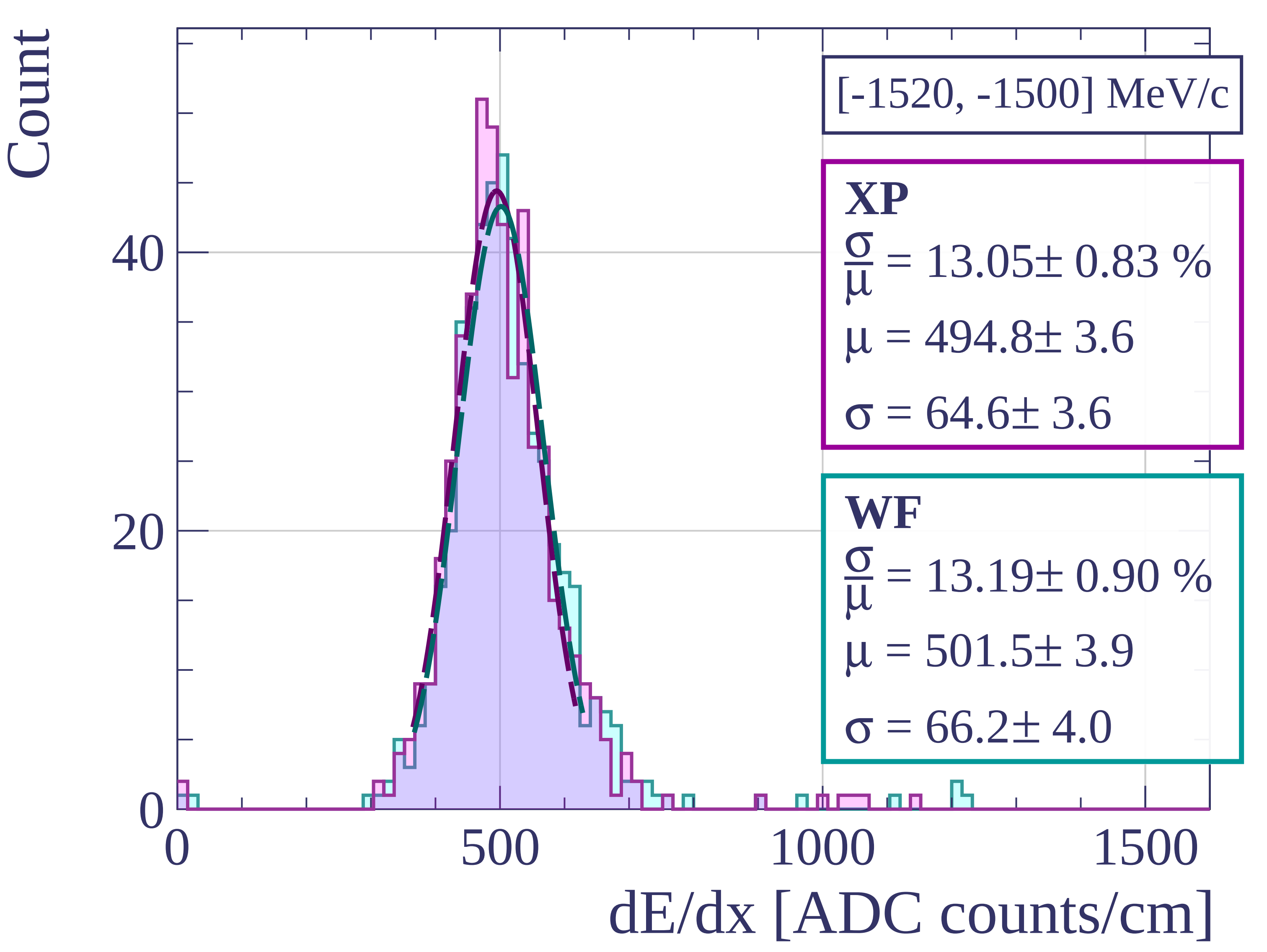

Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1580, -1560] MeV/c
XP
σ/μ = 13.31± 0.86 %
μ = 498.3± 3.7
σ = 66.3± 3.8
WF
σ/μ = 13.97± 0.90 %
μ = 506.2± 3.8
σ = 70.7± 4.0



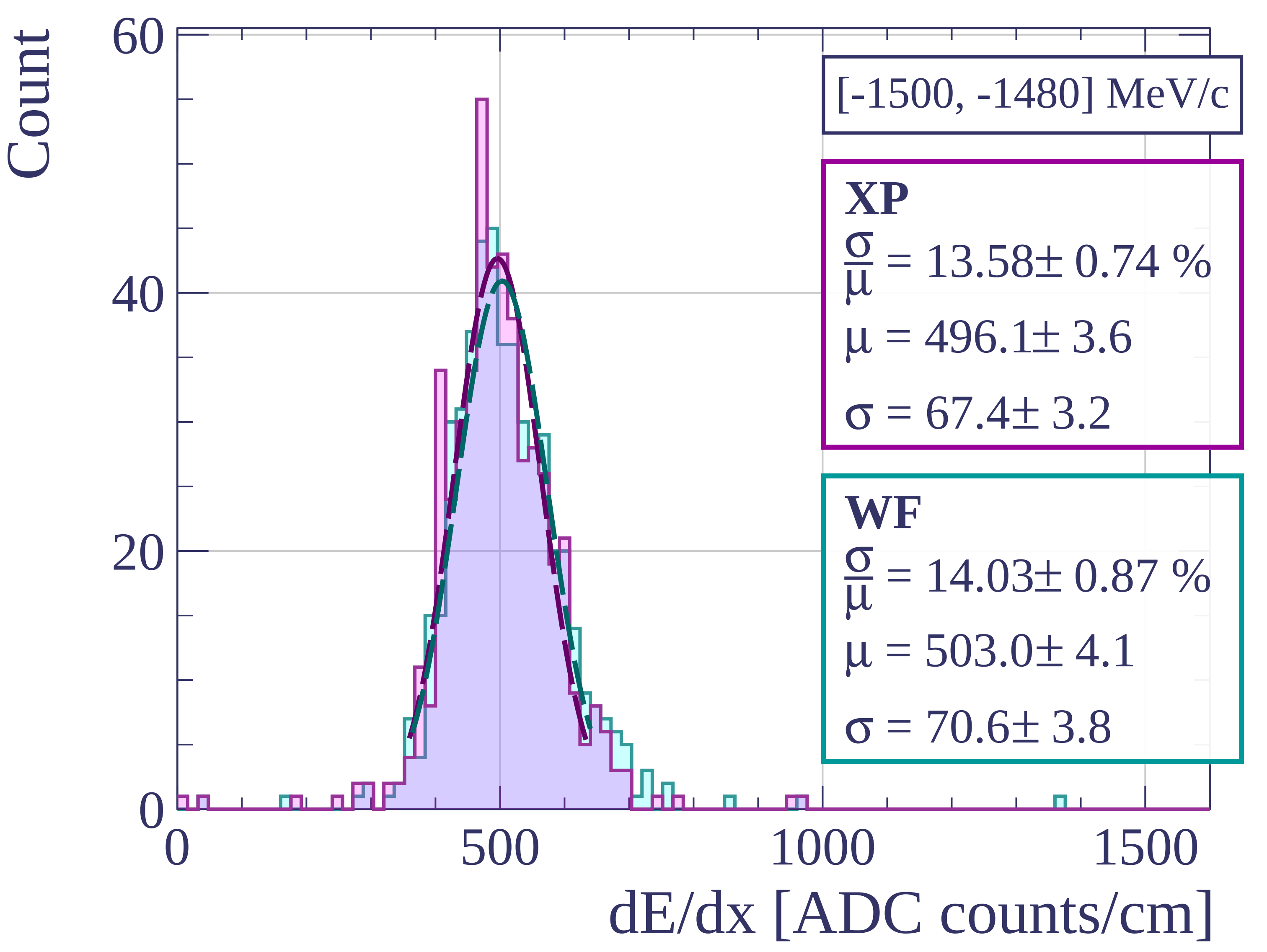
Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1560, -1540] MeV/c
XP
σ/μ = 13.46± 0.87 %
μ = 492.4± 3.8
σ = 66.3± 3.8
WF
σ/μ = 13.01± 0.84 %
μ = 498.4± 3.7
σ = 64.8± 3.7



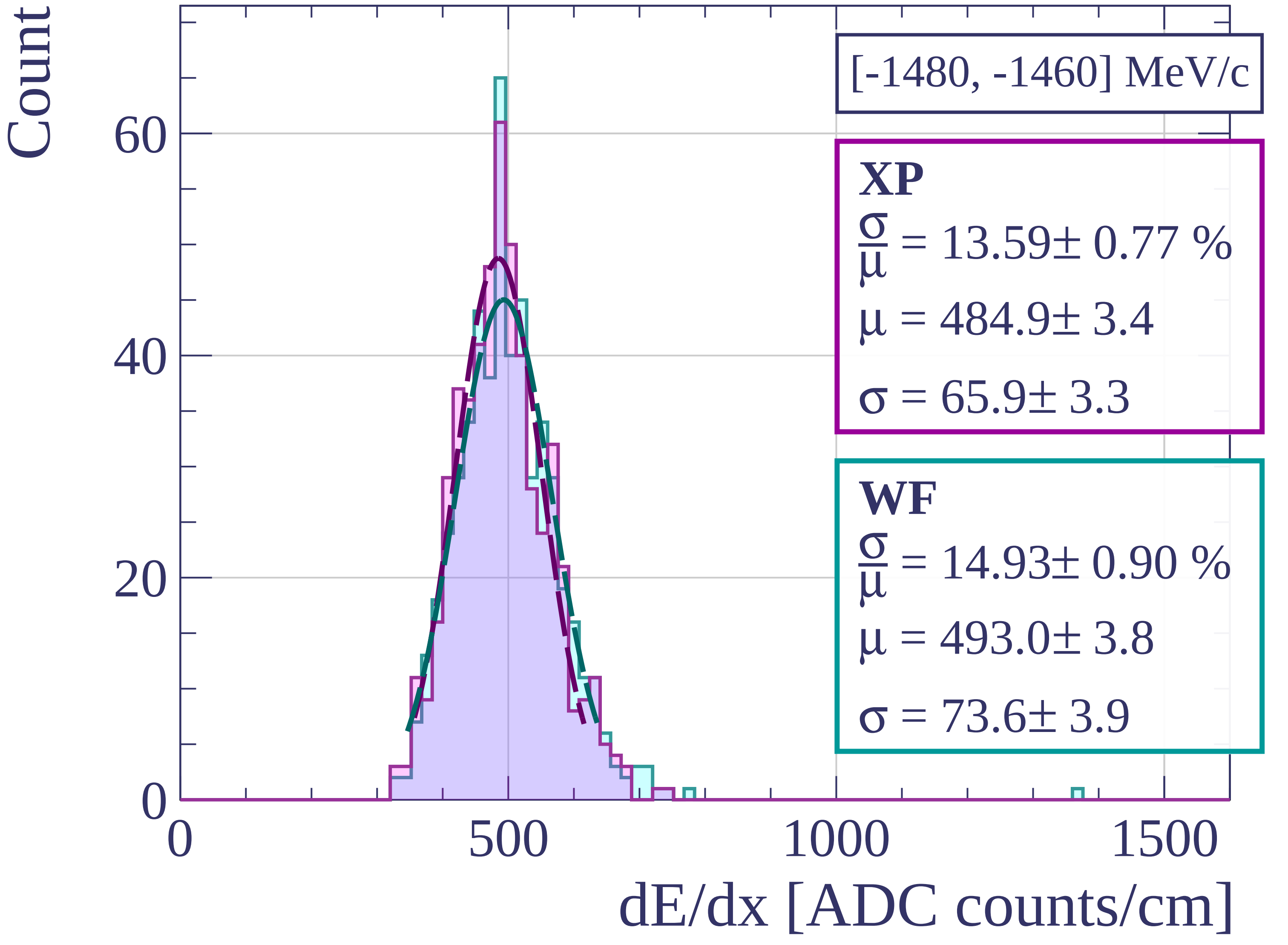
Count
0
20
40
60
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1540, -1520] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.67± 0.74 %
μ = 494.2± 3.3
σ = 62.6± 3.2
WF
$\frac{\sigma}{\mu}$ = 11.94± 0.79 %
μ = 499.8± 3.5
σ = 59.7± 3.5



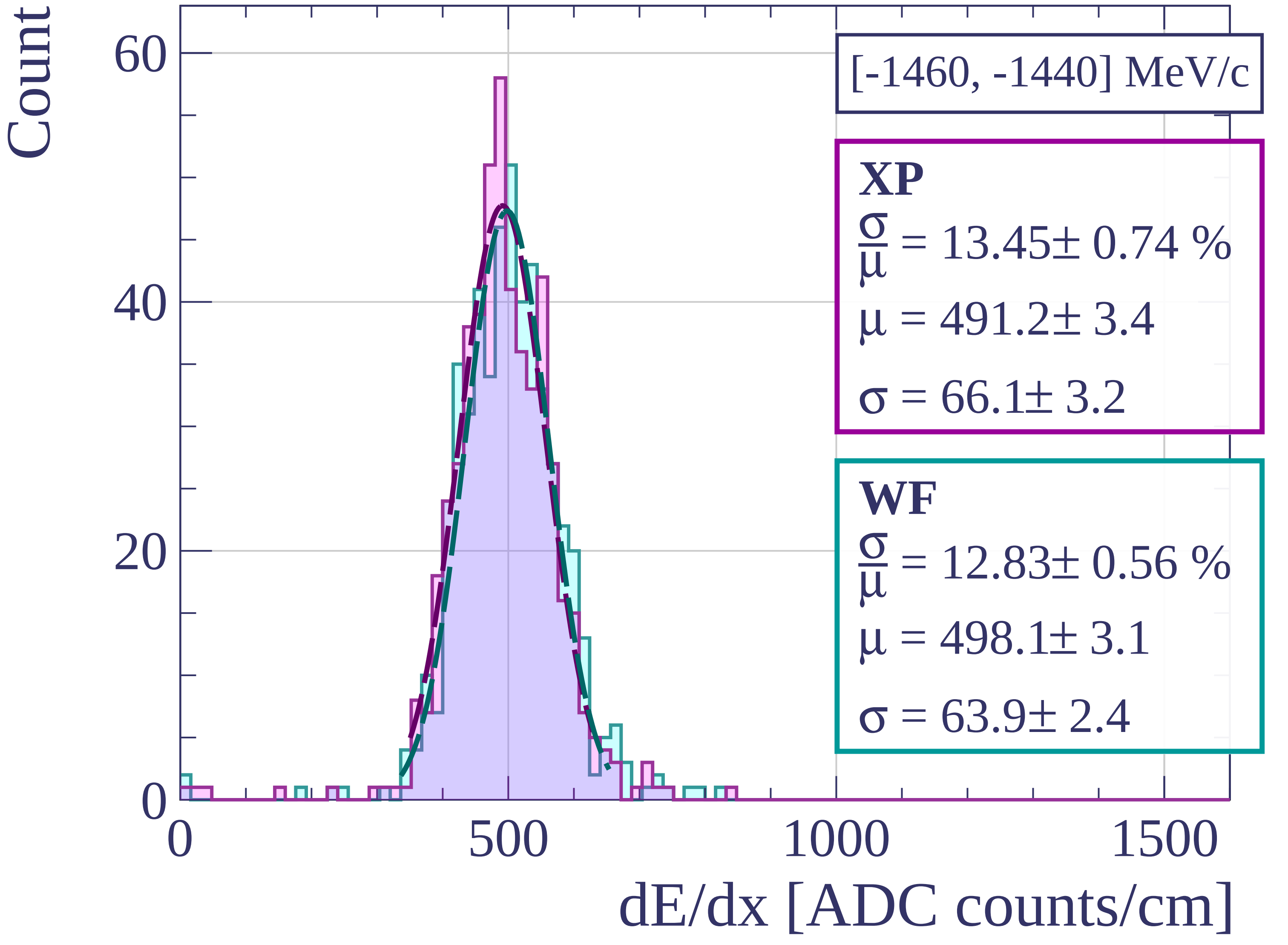
Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1520, -1500] MeV/c
XP
σ/μ = 13.05± 0.83 %
μ = 494.8± 3.6
σ = 64.6± 3.6
WF
σ/μ = 13.19± 0.90 %
μ = 501.5± 3.9
σ = 66.2± 4.0

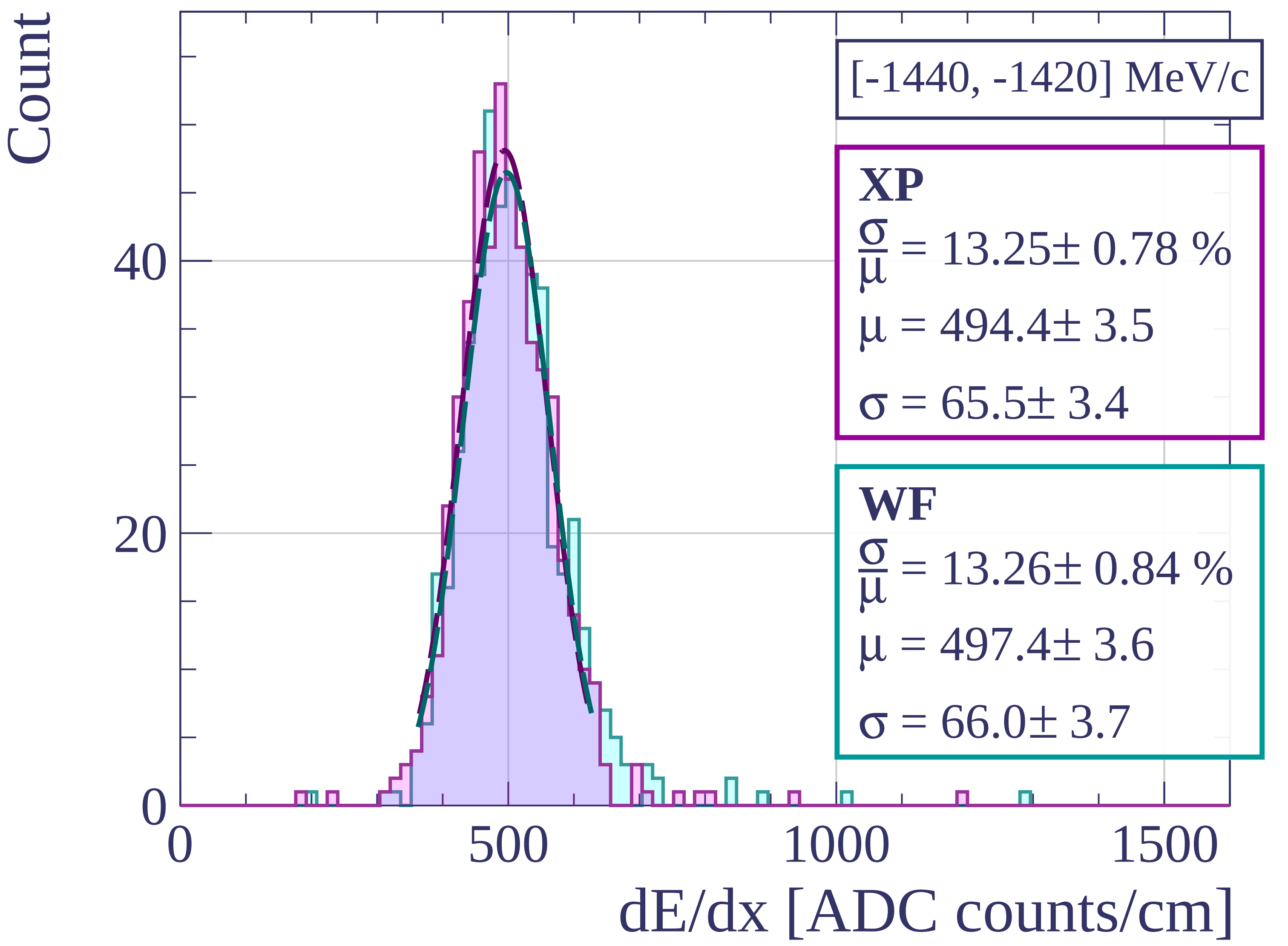


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1500, -1480] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.58± 0.74 %
μ = 496.1± 3.6
σ = 67.4± 3.2
WF
$\frac{\sigma}{\mu}$ = 14.03± 0.87 %
μ = 503.0± 4.1
σ = 70.6± 3.8

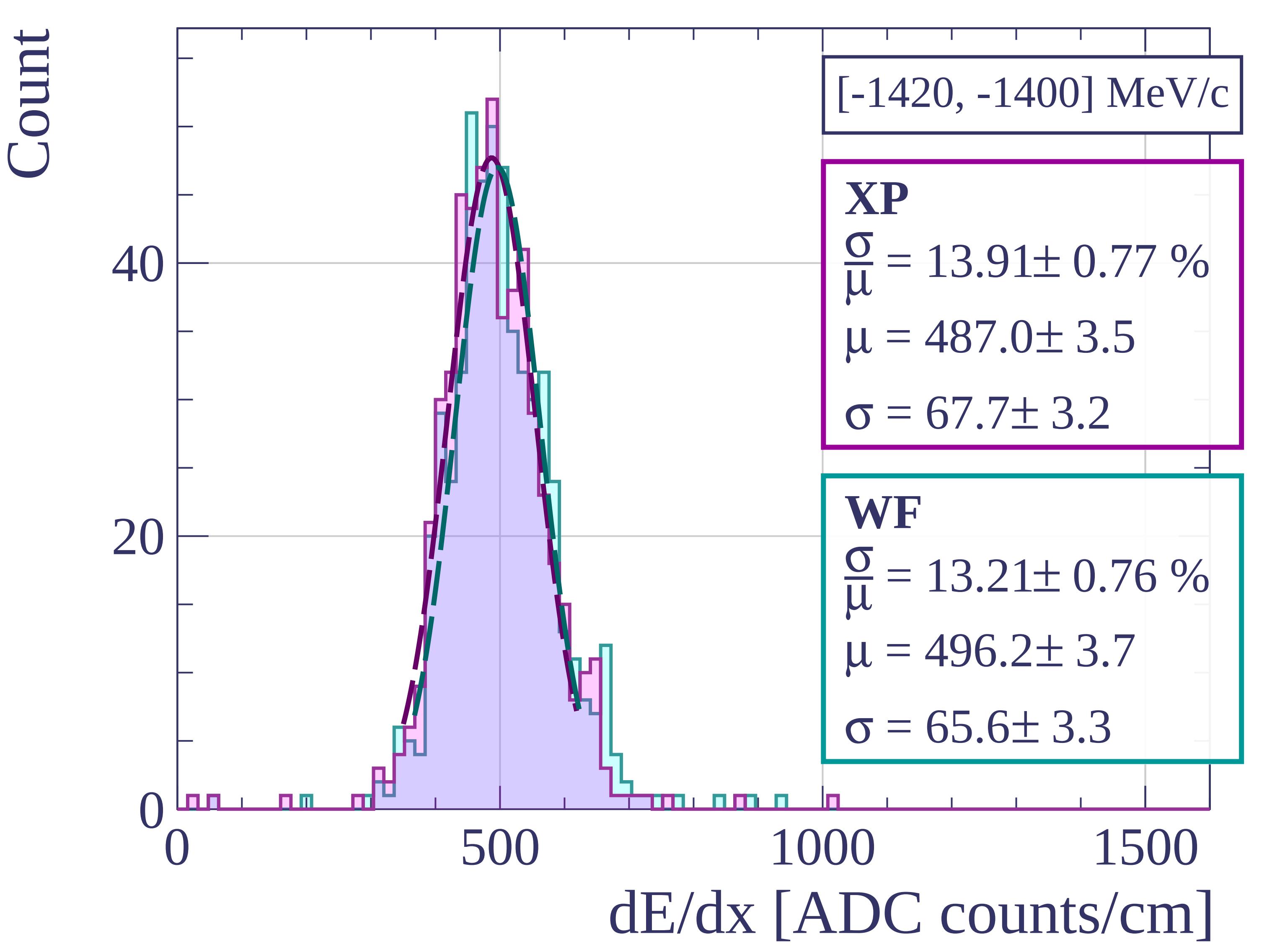


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1480, -1460] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.59± 0.77 %
μ = 484.9± 3.4
σ = 65.9± 3.3
WF
$\frac{\sigma}{\mu}$ = 14.93± 0.90 %
μ = 493.0± 3.8
σ = 73.6± 3.9

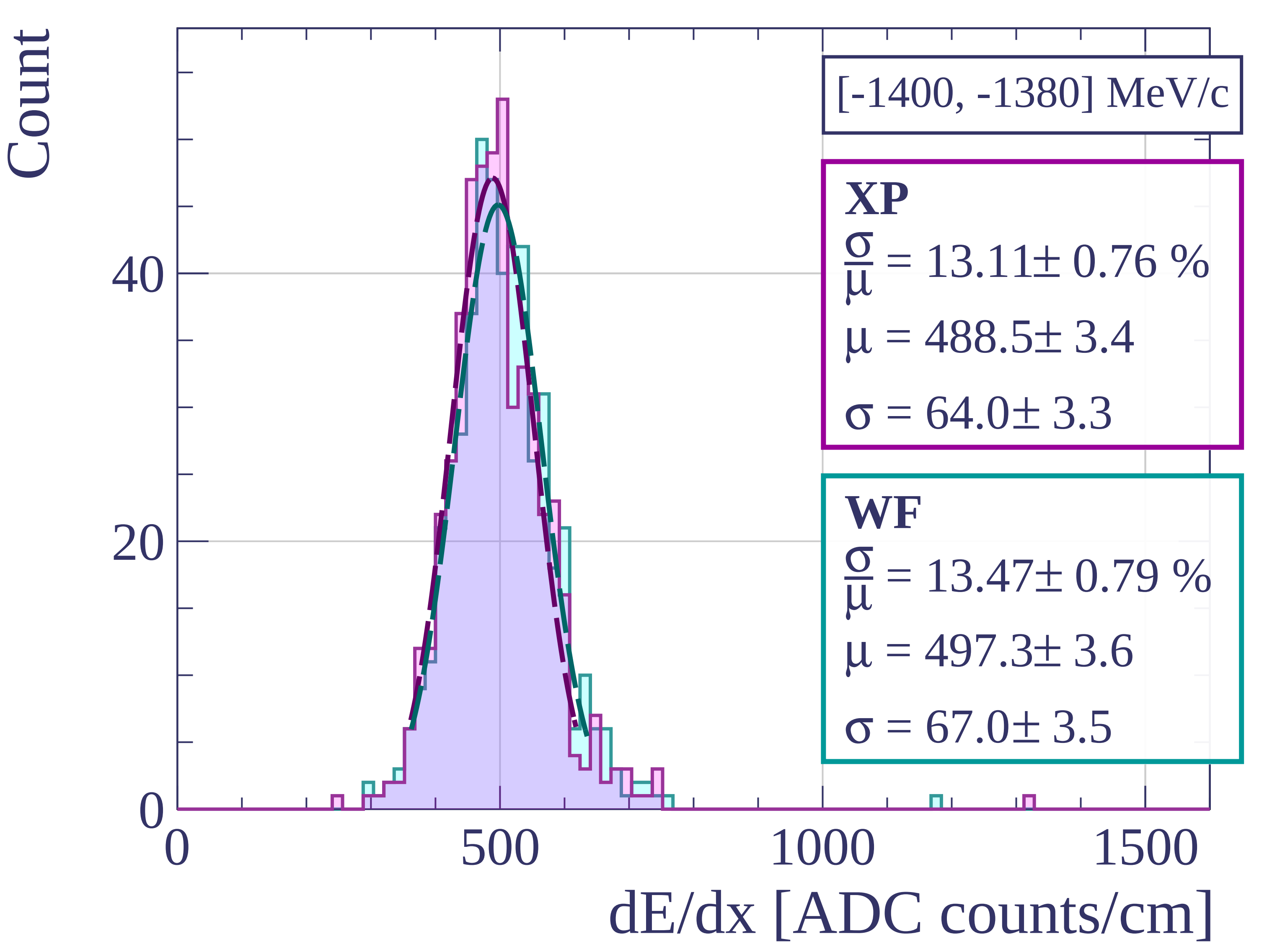

Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1460, -1440] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.45± 0.74 %
μ = 491.2± 3.4
σ = 66.1± 3.2
WF
$\frac{\sigma}{\mu}$ = 12.83± 0.56 %
μ = 498.1± 3.1
σ = 63.9± 2.4



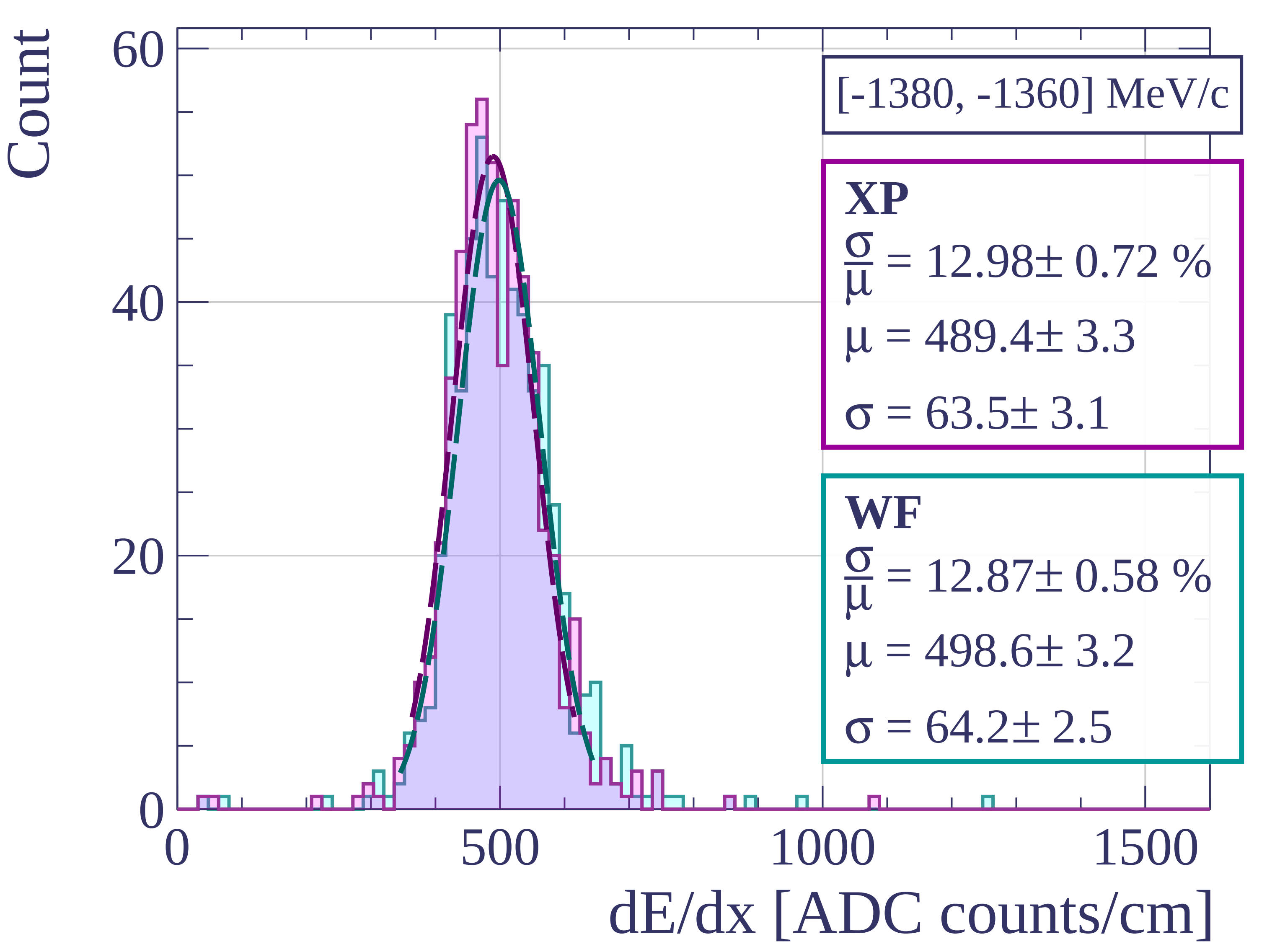
Count
dE/dx [ADC counts/cm]
0
20
40
0
500
1000
1500
[-1440, -1420] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.25± 0.78 %
μ = 494.4± 3.5
σ = 65.5± 3.4
WF
$\frac{\sigma}{\mu}$ = 13.26± 0.84 %
μ = 497.4± 3.6
σ = 66.0± 3.7



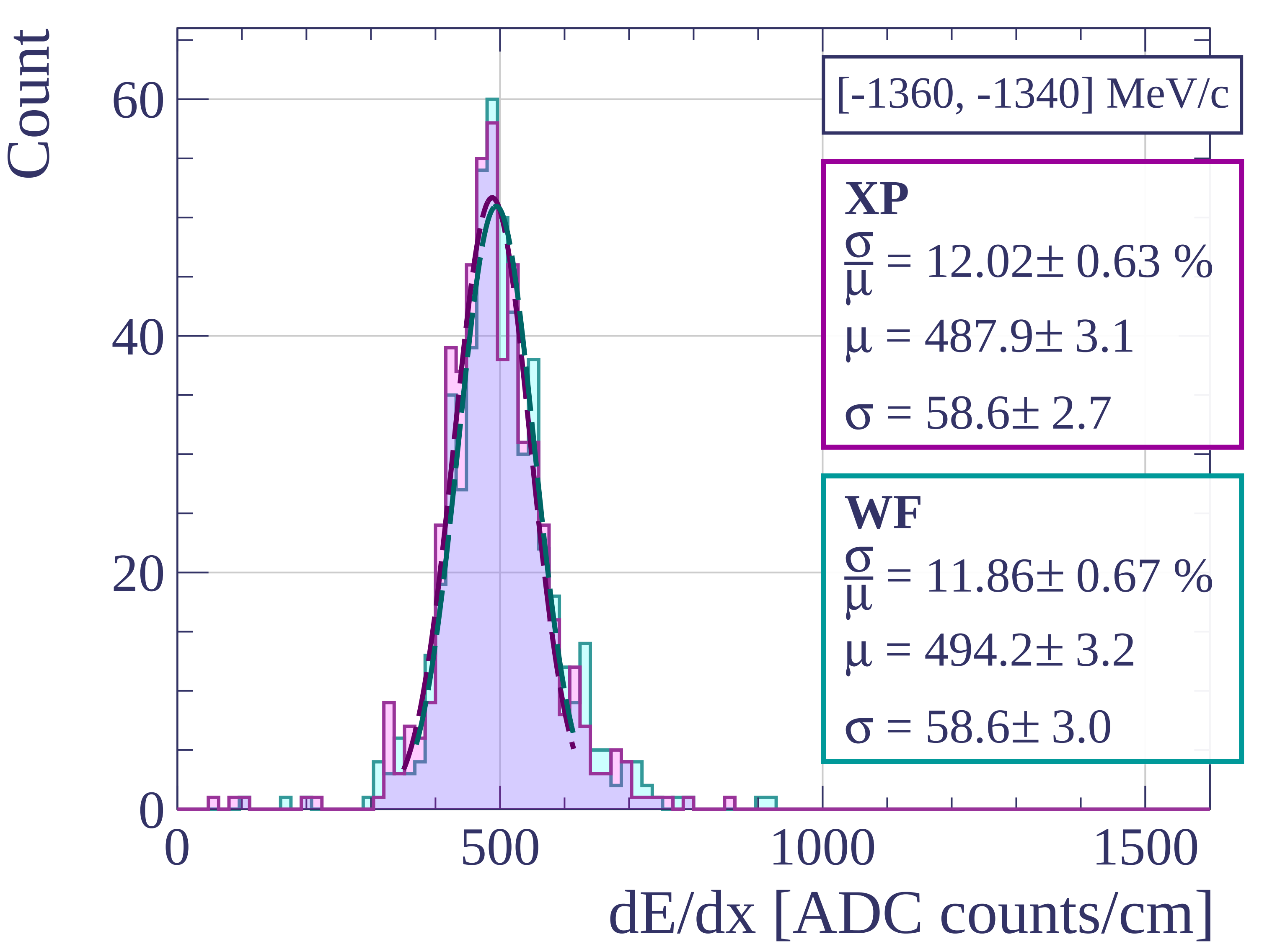
Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1420, -1400] MeV/c
XP
σ/μ = 13.91± 0.77 %
μ = 487.0± 3.5
σ = 67.7± 3.2
WF
σ/μ = 13.21± 0.76 %
μ = 496.2± 3.7
σ = 65.6± 3.3



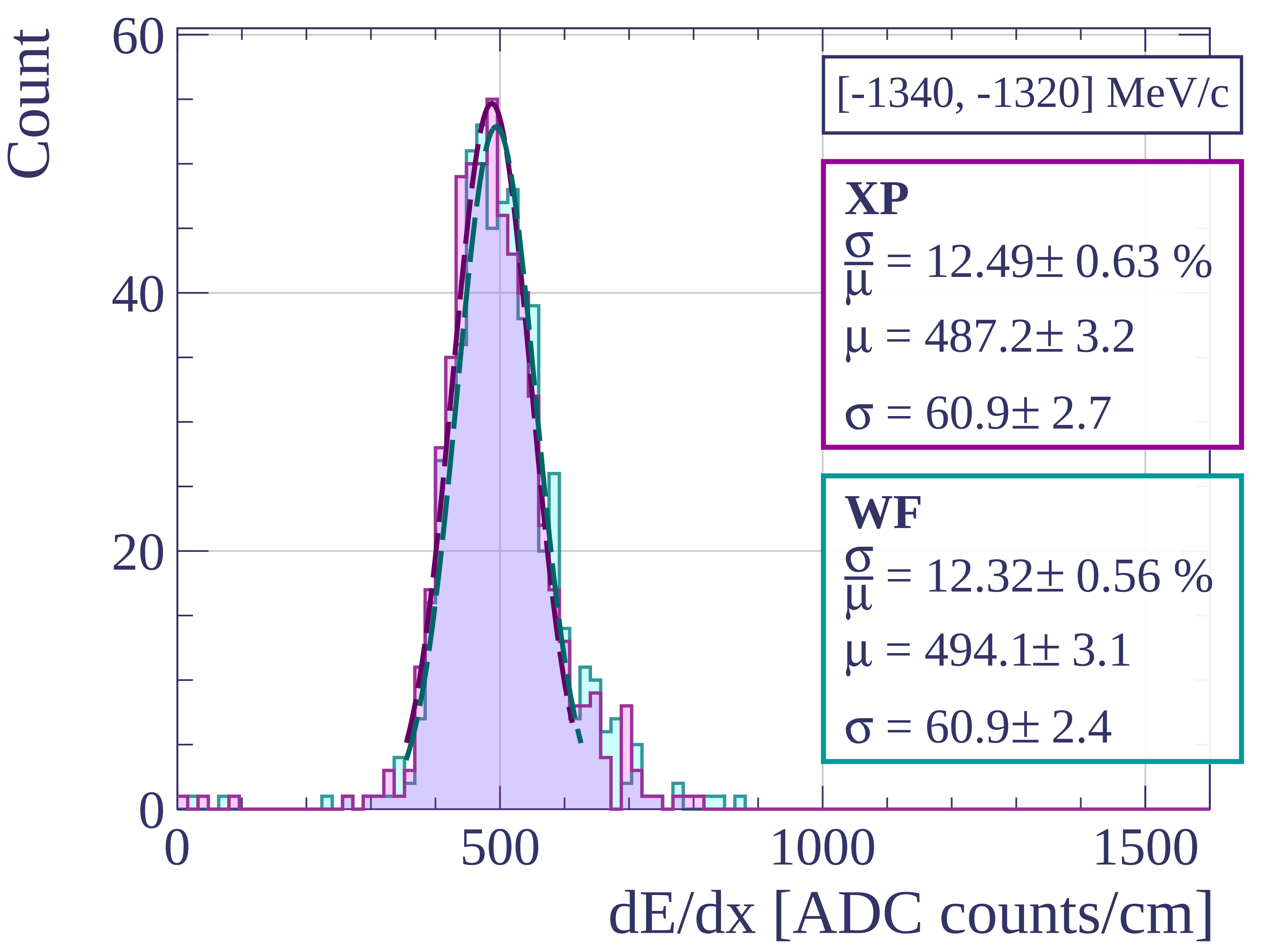
Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1400, -1380] MeV/c
XP
σ/μ = 13.11± 0.76 %
μ = 488.5± 3.4
σ = 64.0± 3.3
WF
σ/μ = 13.47± 0.79 %
μ = 497.3± 3.6
σ = 67.0± 3.5



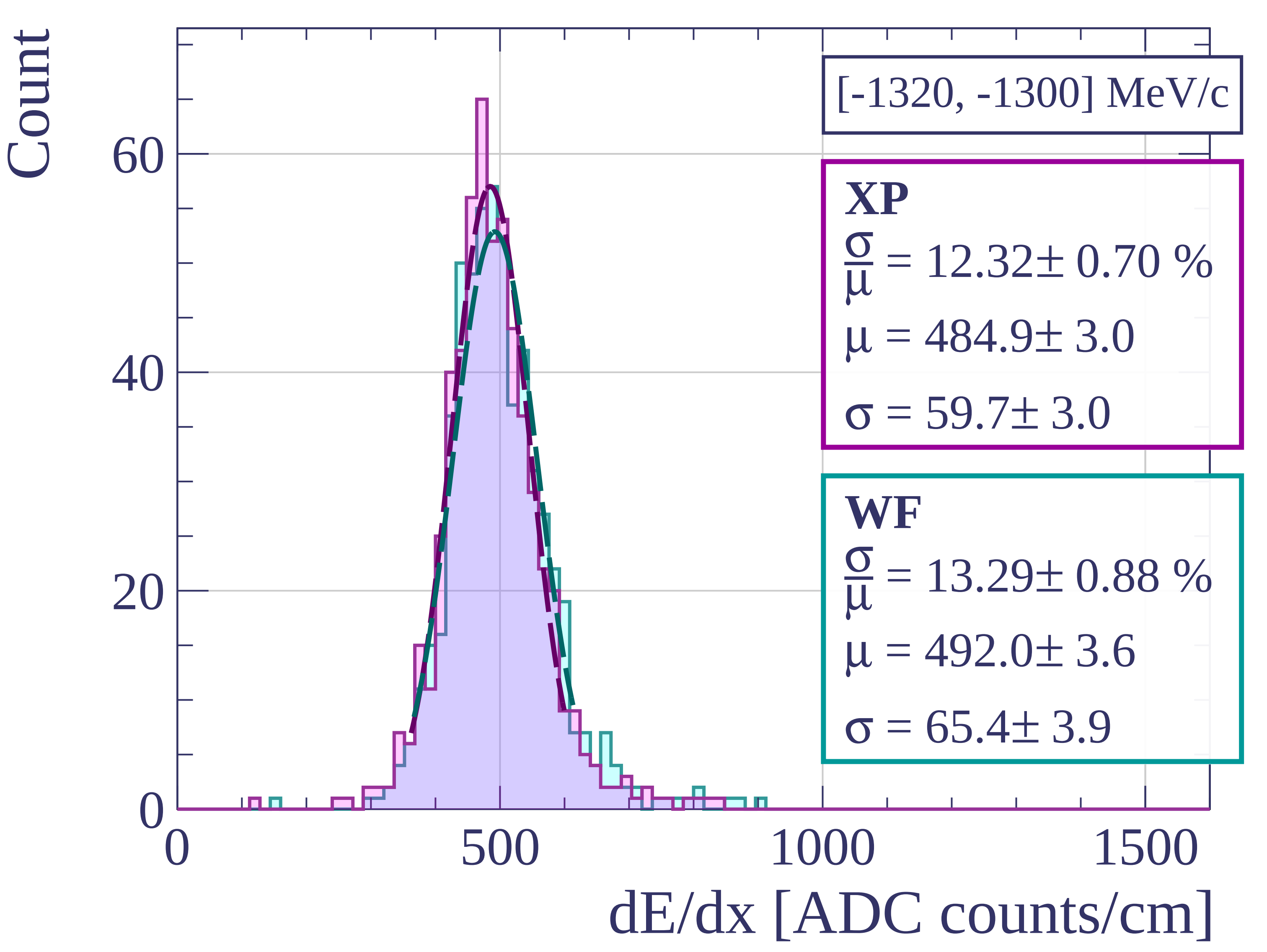
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1380, -1360] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.98± 0.72 %
μ = 489.4± 3.3
σ = 63.5± 3.1
WF
$\frac{\sigma}{\mu}$ = 12.87± 0.58 %
μ = 498.6± 3.2
σ = 64.2± 2.5



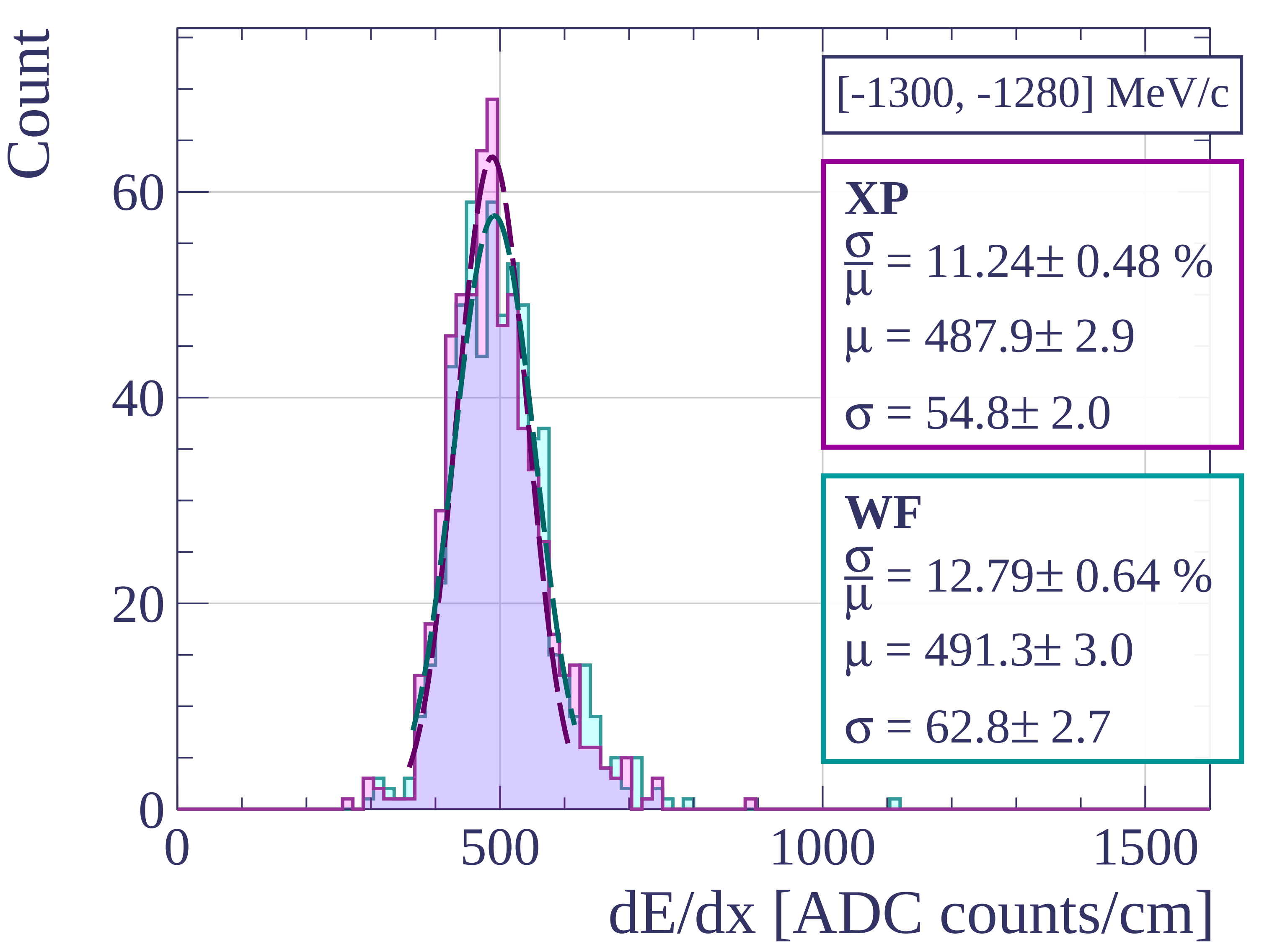
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1360, -1340] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.02± 0.63 %
μ = 487.9± 3.1
σ = 58.6± 2.7
WF
$\frac{\sigma}{\mu}$ = 11.86± 0.67 %
μ = 494.2± 3.2
σ = 58.6± 3.0

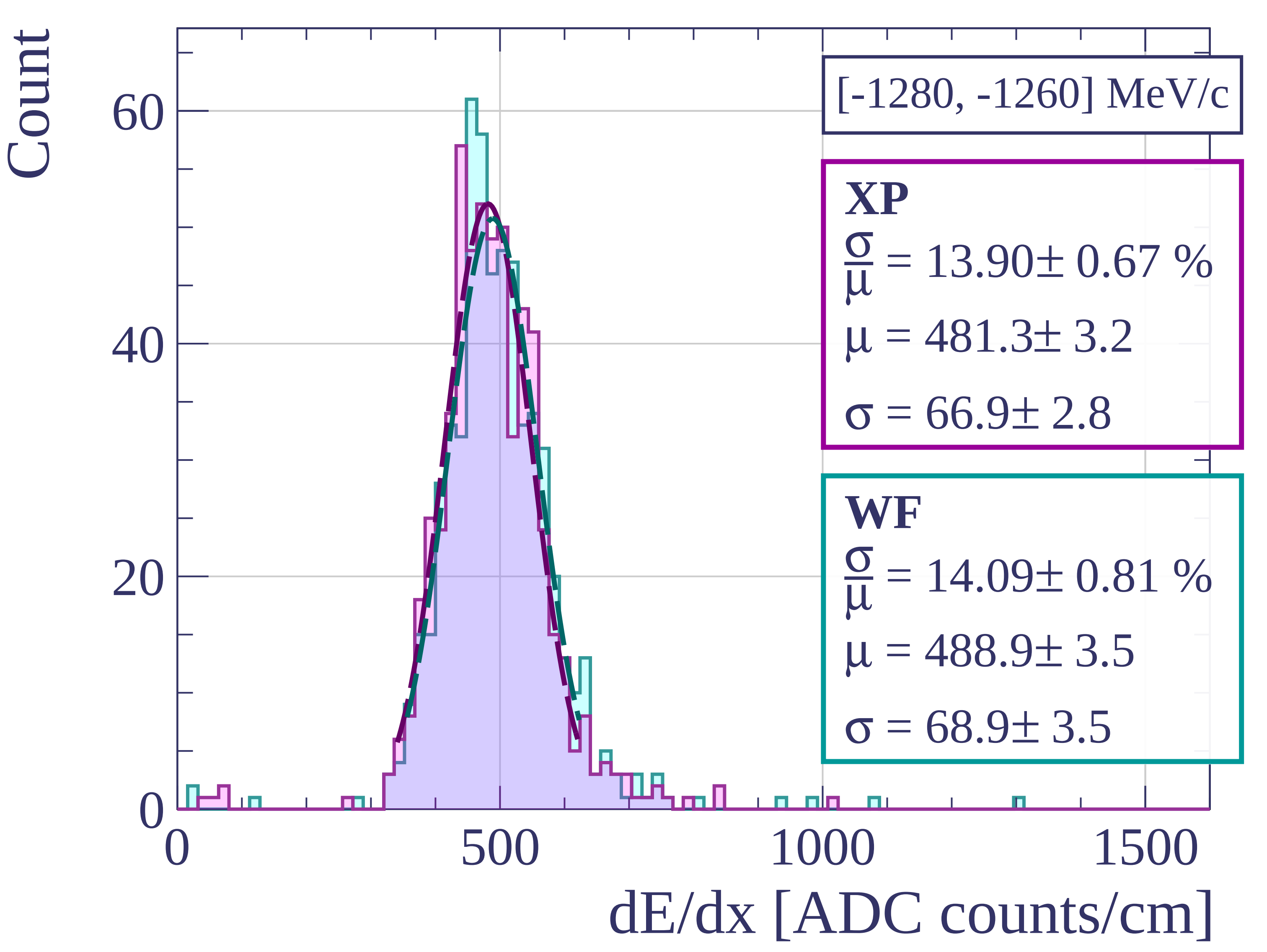


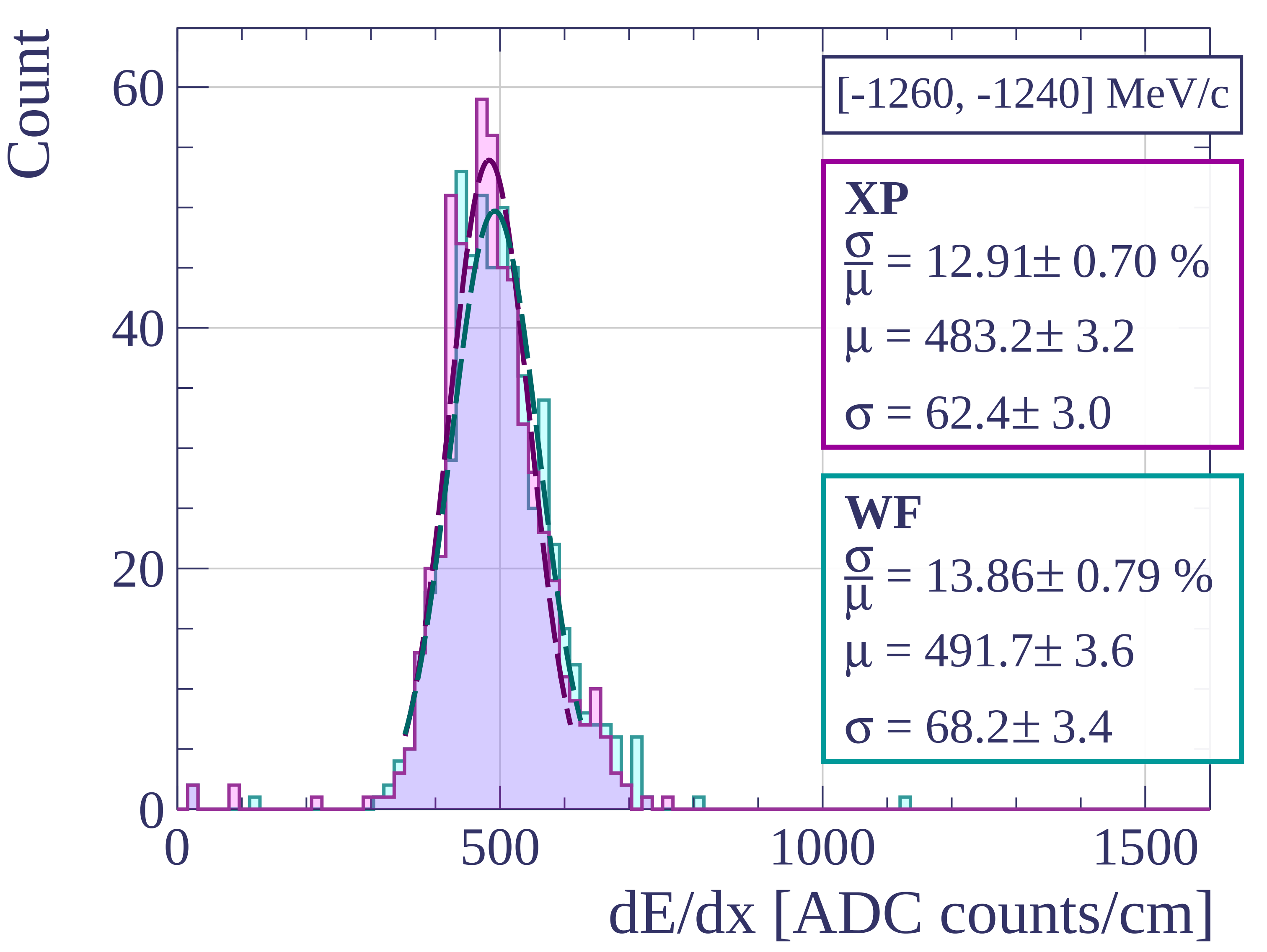
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1340, -1320] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.49± 0.63 %
μ = 487.2± 3.2
σ = 60.9± 2.7
WF
$\frac{\sigma}{\mu}$ = 12.32± 0.56 %
μ = 494.1± 3.1
σ = 60.9± 2.4

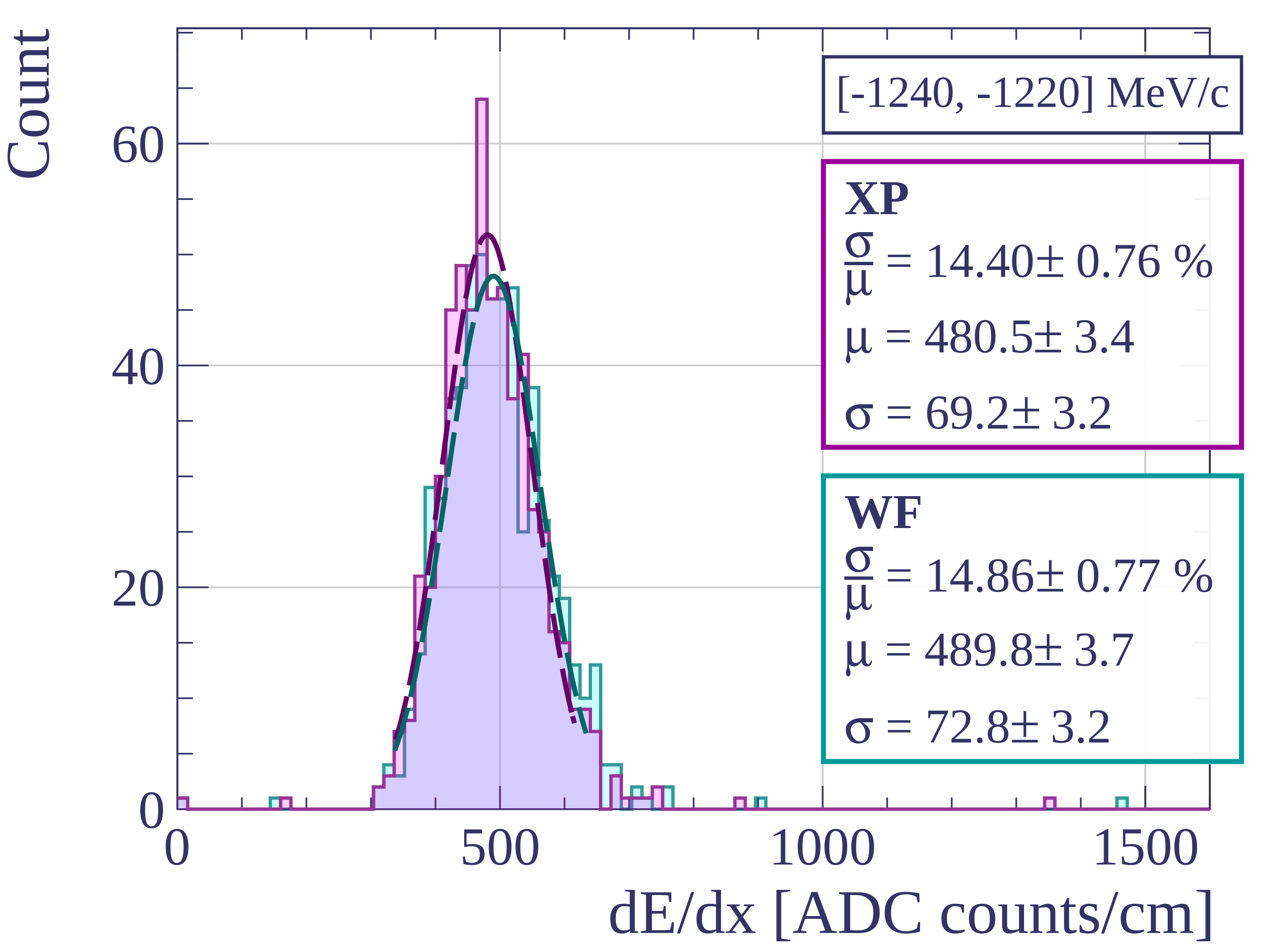


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1320, -1300] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.32± 0.70 %
μ = 484.9± 3.0
σ = 59.7± 3.0
WF
$\frac{\sigma}{\mu}$ = 13.29± 0.88 %
μ = 492.0± 3.6
σ = 65.4± 3.9

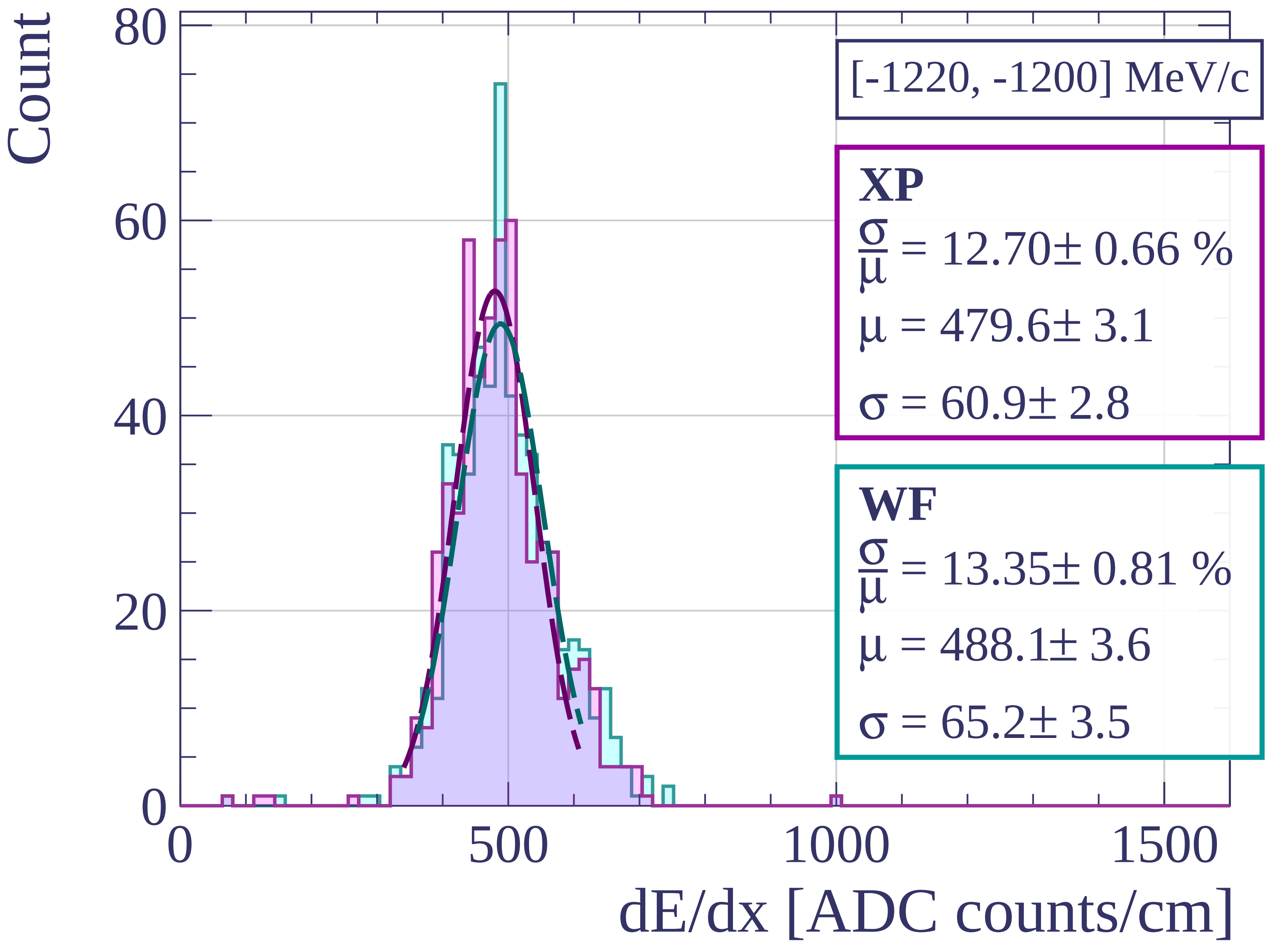

Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1300, -1280] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.24± 0.48 %
μ = 487.9± 2.9
σ = 54.8± 2.0
WF
$\frac{\sigma}{\mu}$ = 12.79± 0.64 %
μ = 491.3± 3.0
σ = 62.8± 2.7

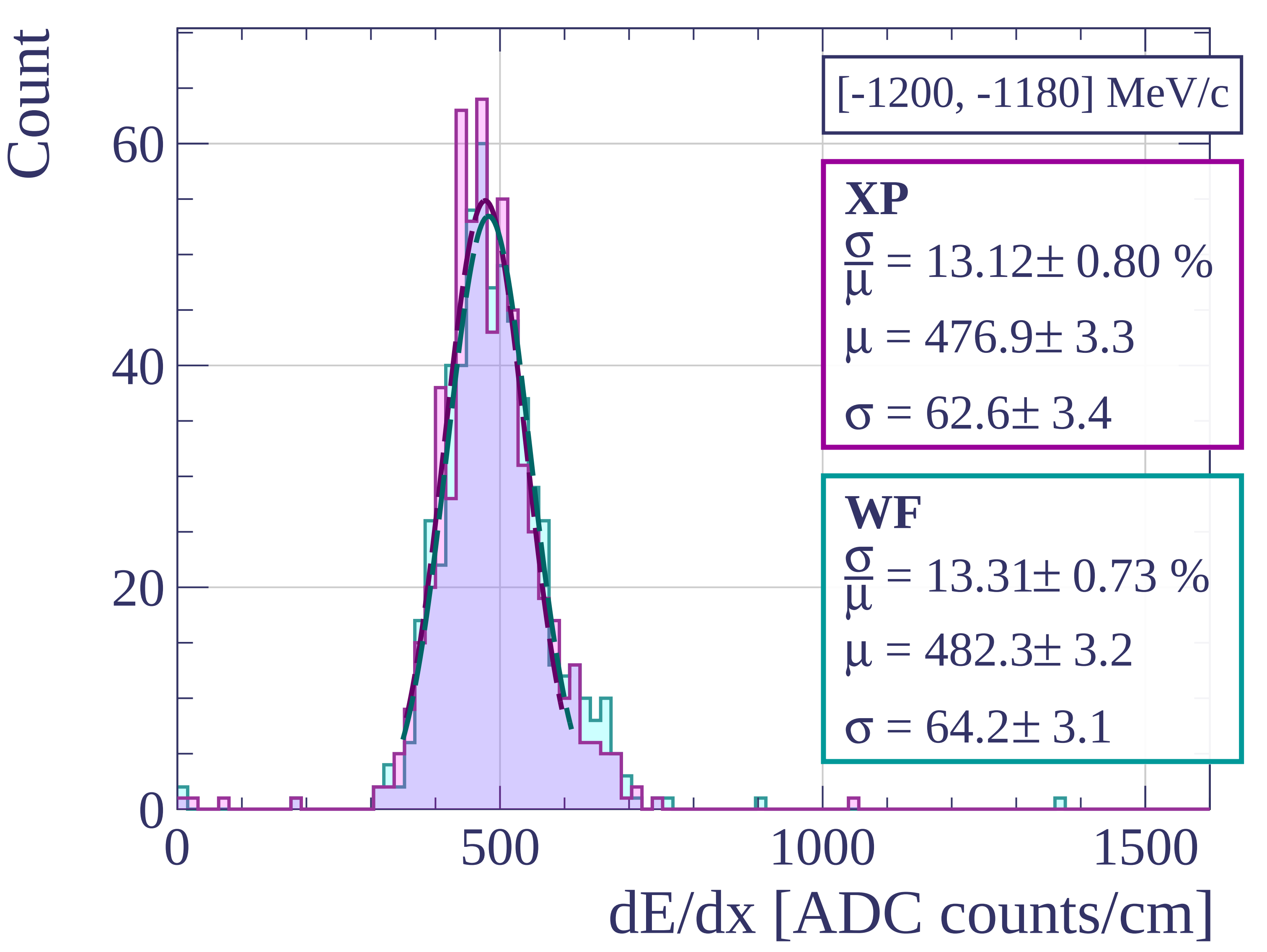

Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1280, -1260] MeV/c
XP
σ/μ = 13.90± 0.67 %
μ = 481.3± 3.2
σ = 66.9± 2.8
WF
σ/μ = 14.09± 0.81 %
μ = 488.9± 3.5
σ = 68.9± 3.5


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1260, -1240] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.91± 0.70 %
μ = 483.2± 3.2
σ = 62.4± 3.0
WF
$\frac{\sigma}{\mu}$ = 13.86± 0.79 %
μ = 491.7± 3.6
σ = 68.2± 3.4

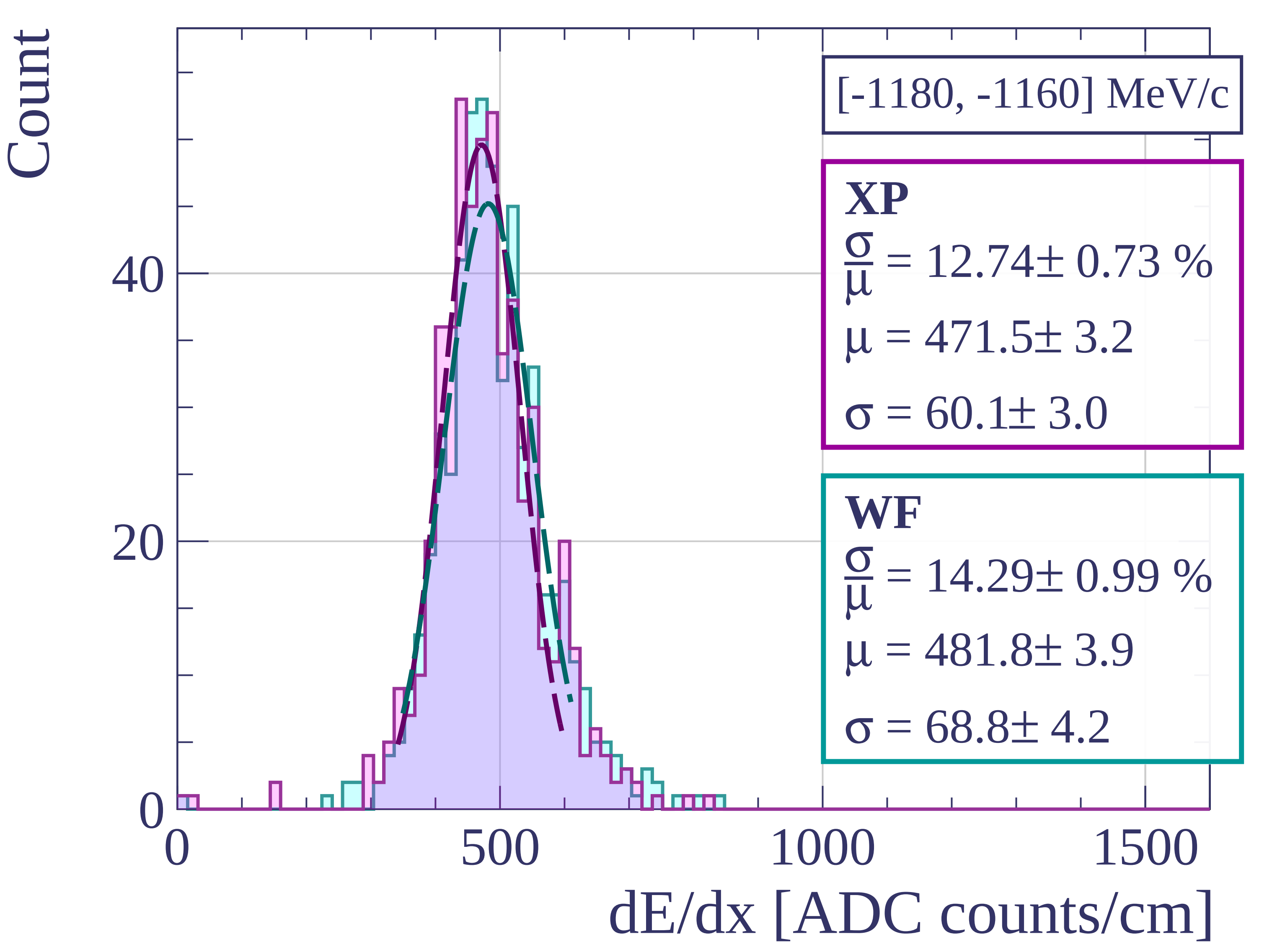


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1240, -1220] MeV/c
XP
$\frac{\sigma}{\mu}$ = 14.40± 0.76 %
μ = 480.5± 3.4
σ = 69.2± 3.2
WF
$\frac{\sigma}{\mu}$ = 14.86± 0.77 %
μ = 489.8± 3.7
σ = 72.8± 3.2

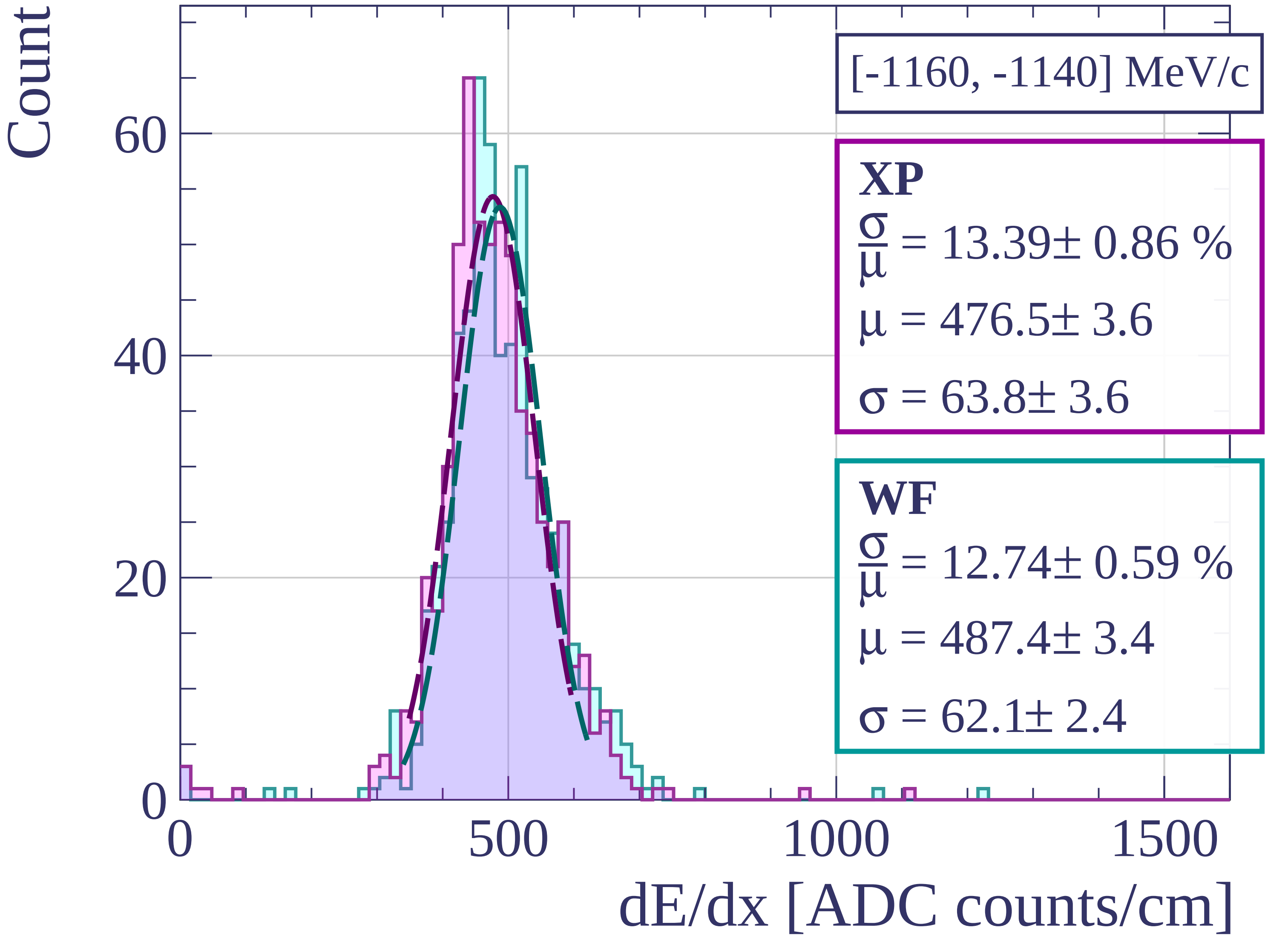

Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1220, -1200] MeV/c
XP
σ/μ = 12.70± 0.66 %
μ = 479.6± 3.1
σ = 60.9± 2.8
WF
σ/μ = 13.35± 0.81 %
μ = 488.1± 3.6
σ = 65.2± 3.5



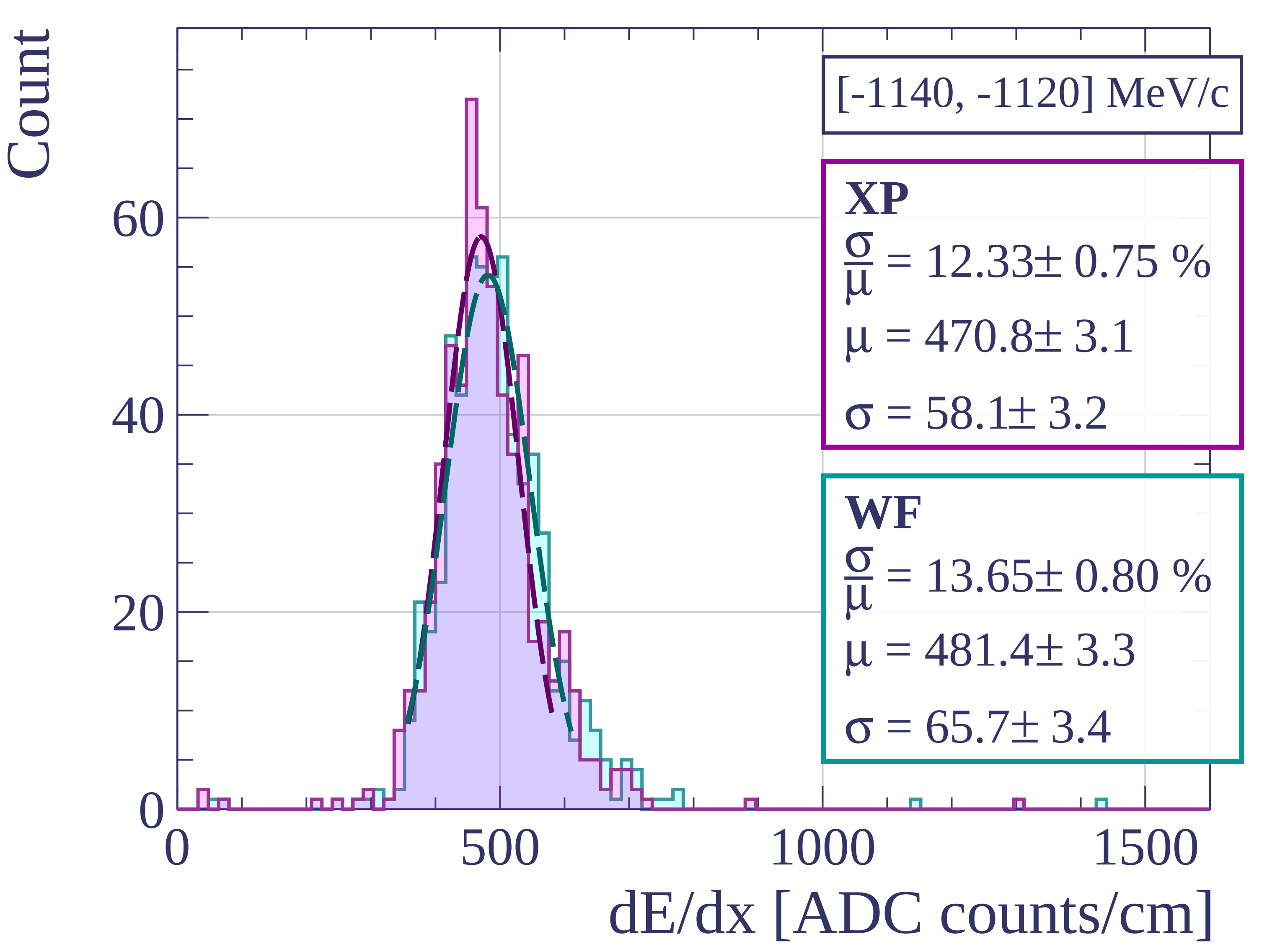
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1200, -1180] MeV/c
XP
σ/μ = 13.12± 0.80 %
μ = 476.9± 3.3
σ = 62.6± 3.4
WF
σ/μ = 13.31± 0.73 %
μ = 482.3± 3.2
σ = 64.2± 3.1



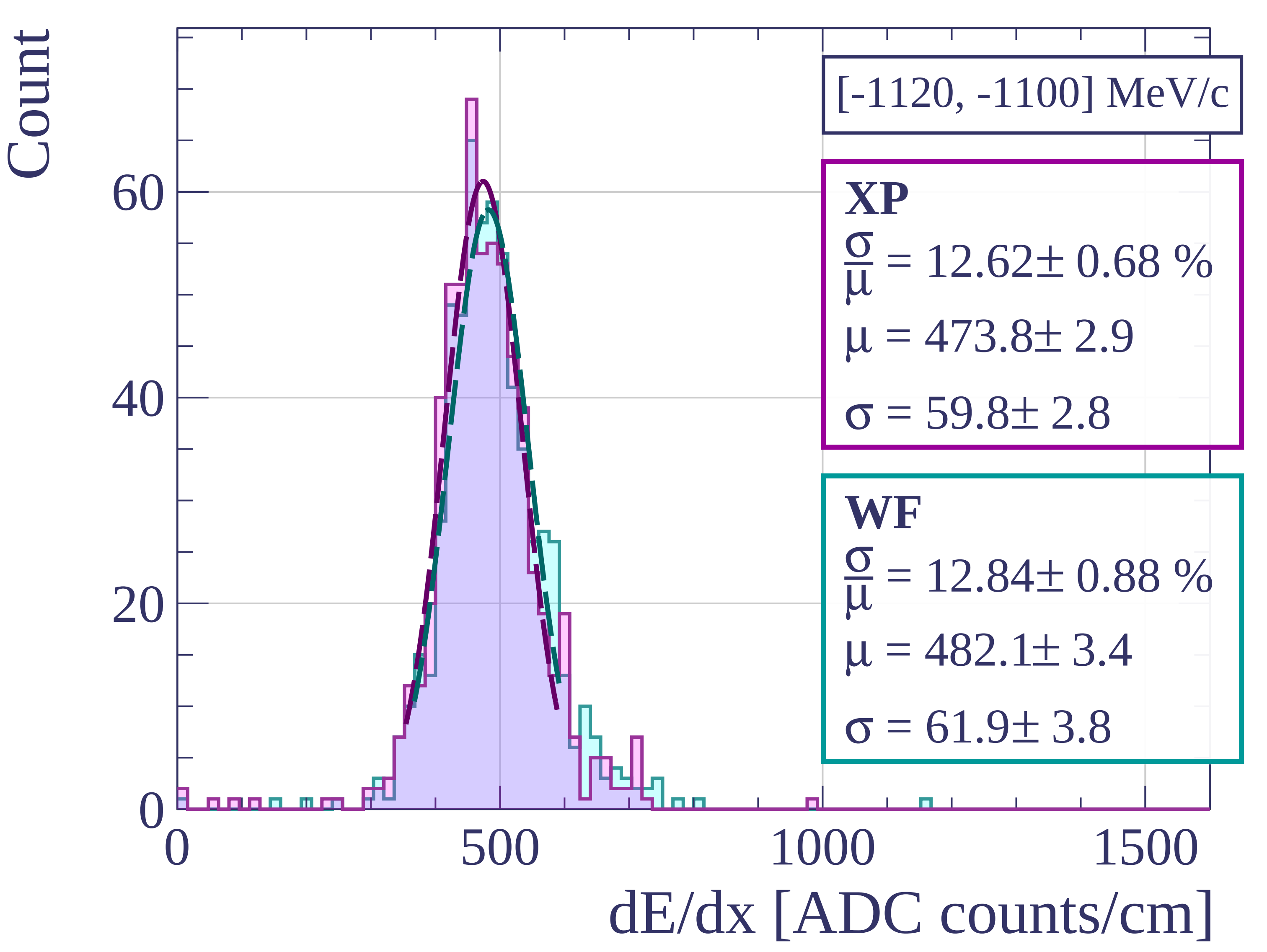
Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1180, -1160] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.74± 0.73 %
μ = 471.5± 3.2
σ = 60.1± 3.0
WF
$\frac{\sigma}{\mu}$ = 14.29± 0.99 %
μ = 481.8± 3.9
σ = 68.8± 4.2



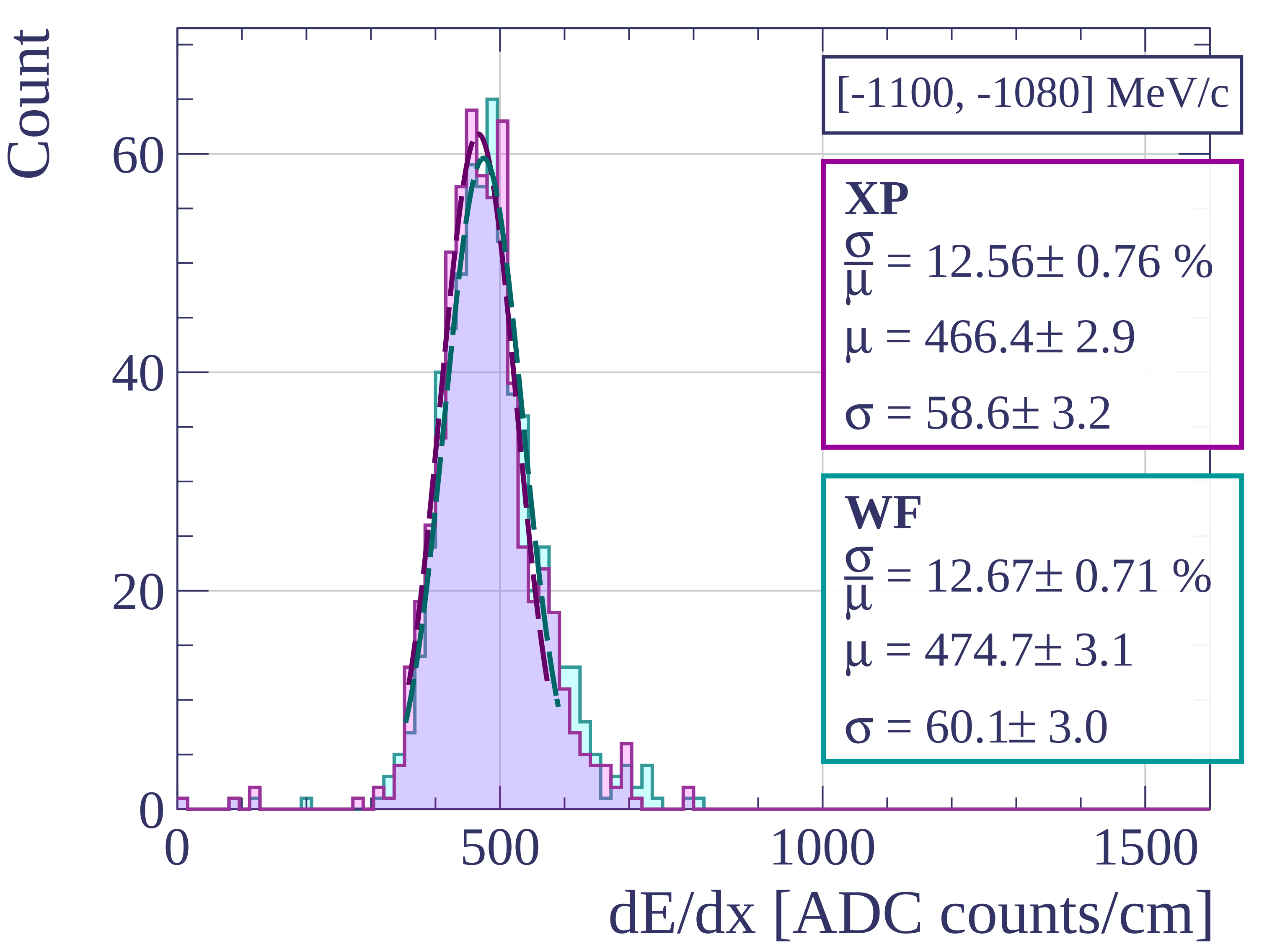
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1160, -1140] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.39± 0.86 %
μ = 476.5± 3.6
σ = 63.8± 3.6
WF
$\frac{\sigma}{\mu}$ = 12.74± 0.59 %
μ = 487.4± 3.4
σ = 62.1± 2.4



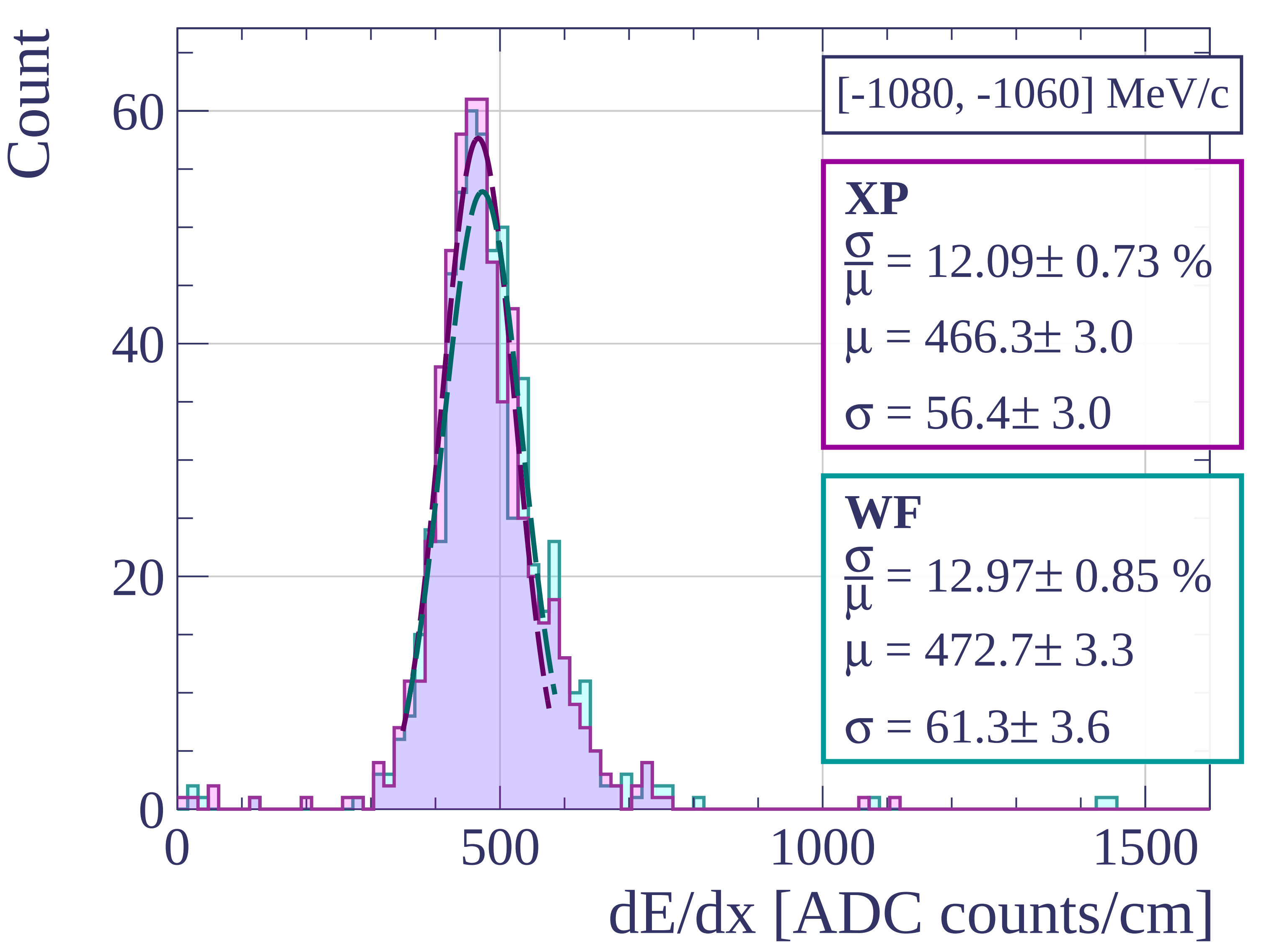
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1140, -1120] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.33± 0.75 %
μ = 470.8± 3.1
σ = 58.1± 3.2
WF
$\frac{\sigma}{\mu}$ = 13.65± 0.80 %
μ = 481.4± 3.3
σ = 65.7± 3.4

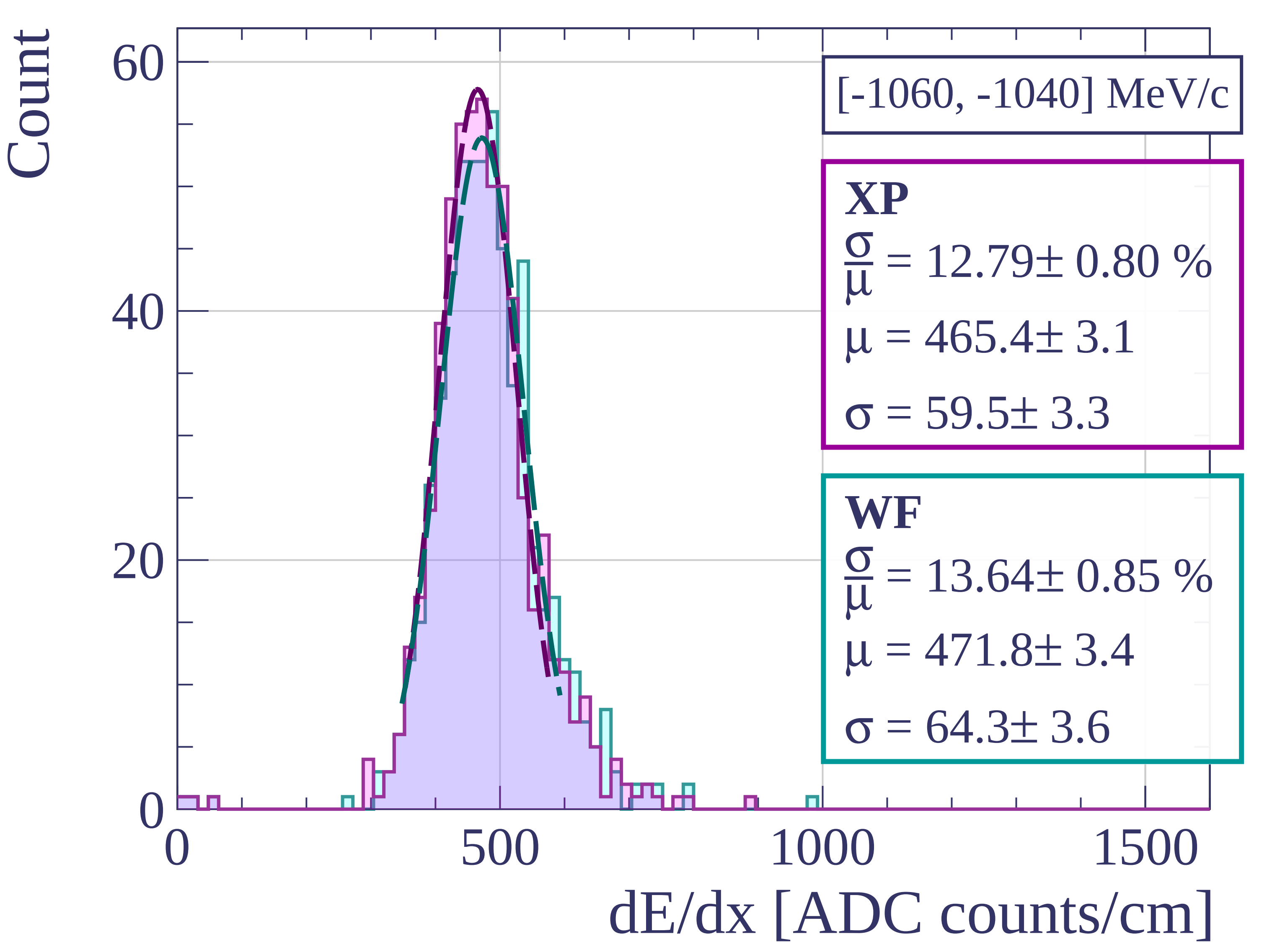


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1120, -1100] MeV/c
XP
σ/μ = 12.62± 0.68 %
μ = 473.8± 2.9
σ = 59.8± 2.8
WF
σ/μ = 12.84± 0.88 %
μ = 482.1± 3.4
σ = 61.9± 3.8

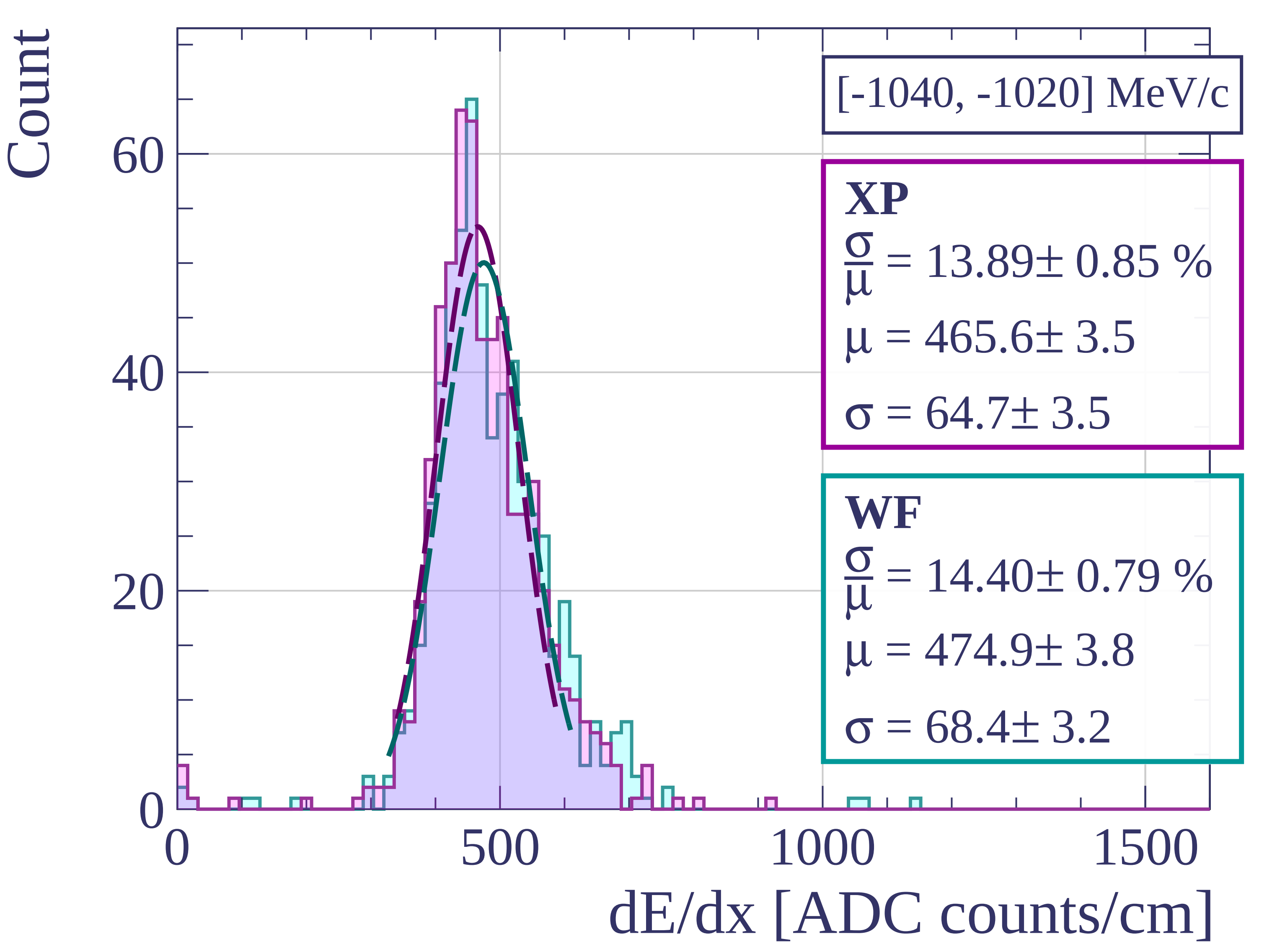


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1100, -1080] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.56± 0.76 %
μ = 466.4± 2.9
σ = 58.6± 3.2
WF
$\frac{\sigma}{\mu}$ = 12.67± 0.71 %
μ = 474.7± 3.1
σ = 60.1± 3.0

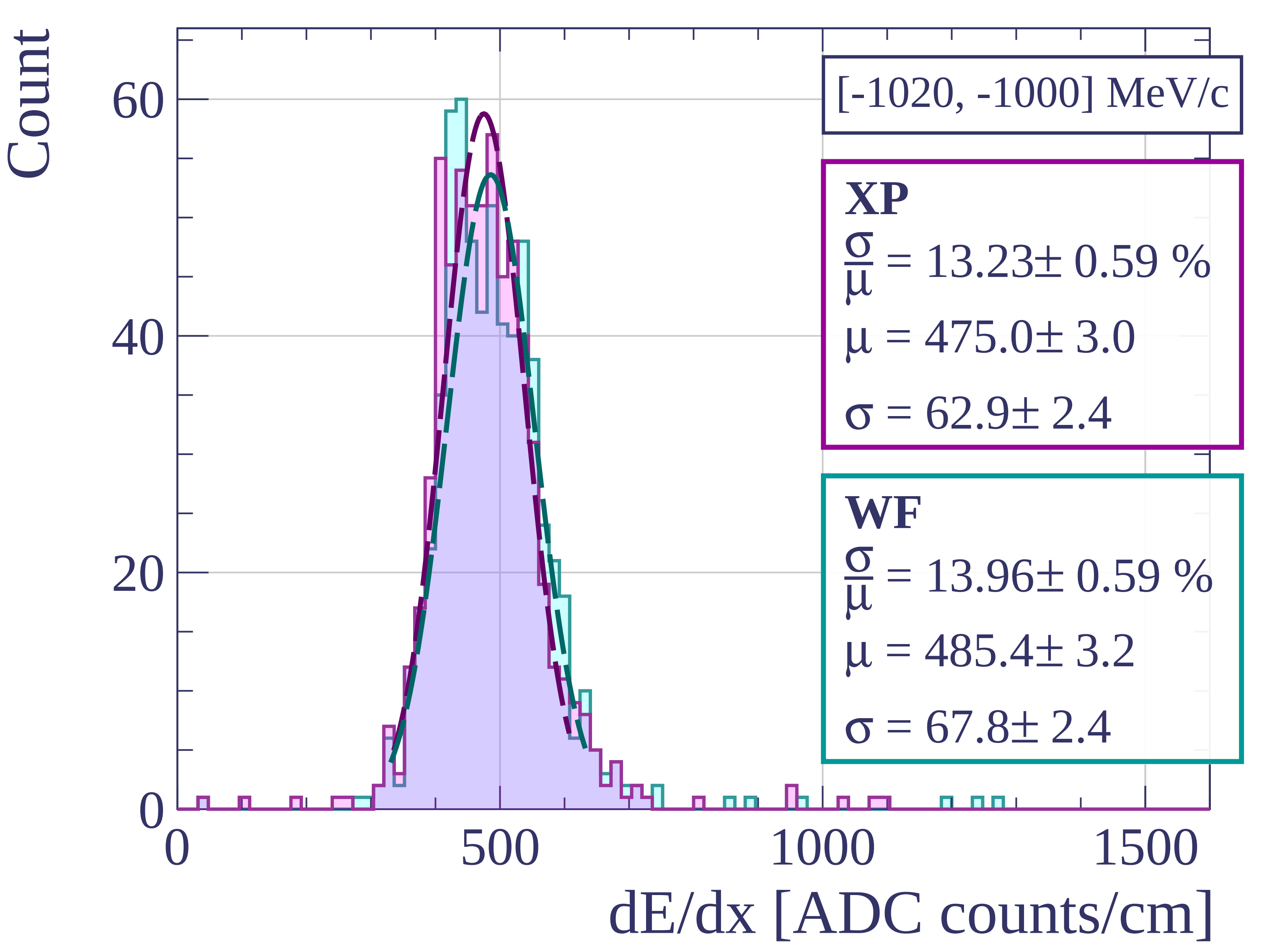

Count
dE/dx [ADC counts/cm]
0
20
40
60
0
500
1000
1500
[-1080, -1060] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.09± 0.73 %
μ = 466.3± 3.0
σ = 56.4± 3.0
WF
$\frac{\sigma}{\mu}$ = 12.97± 0.85 %
μ = 472.7± 3.3
σ = 61.3± 3.6



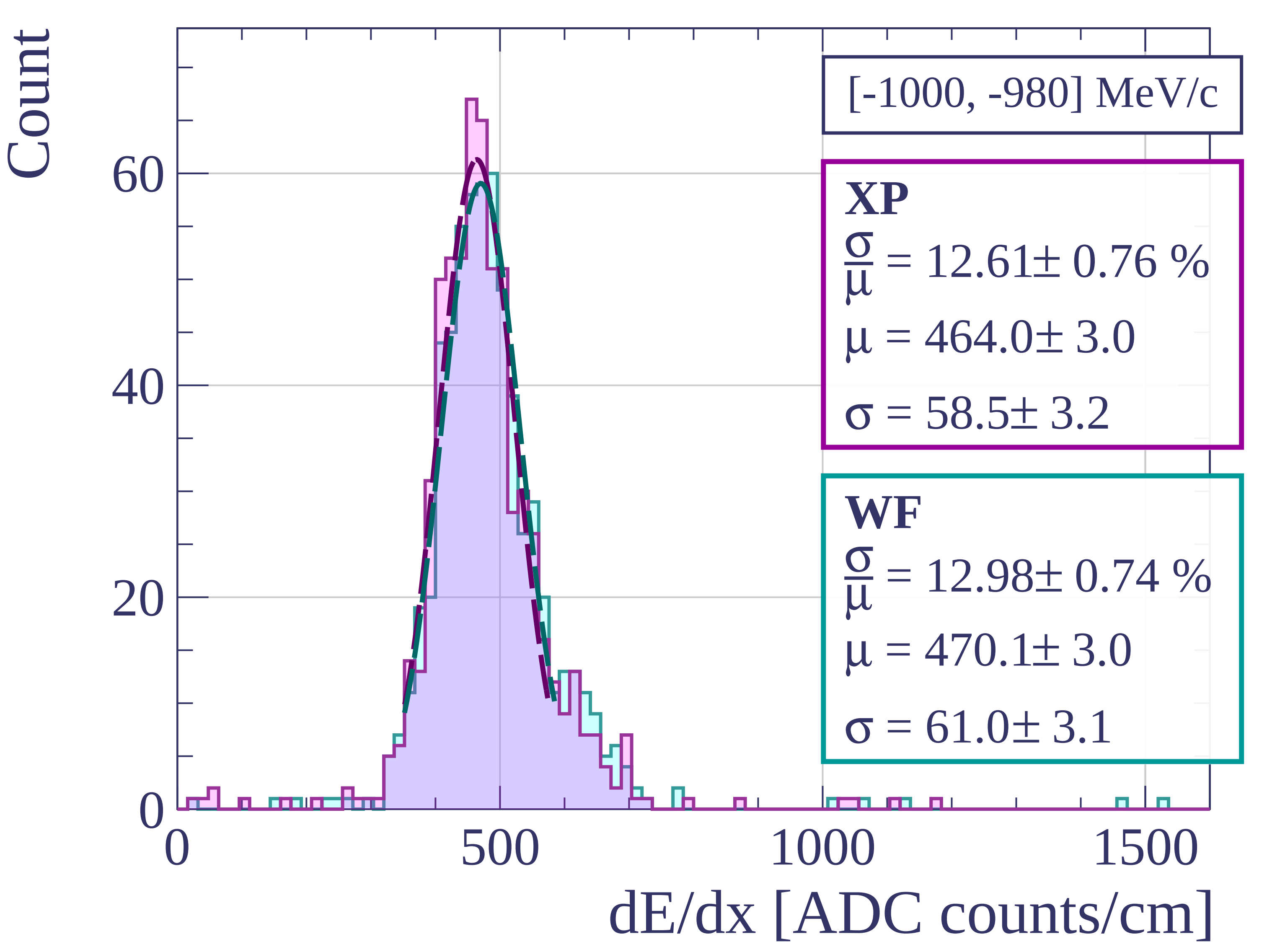
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1060, -1040] MeV/c
XP
σ/μ = 12.79± 0.80 %
μ = 465.4± 3.1
σ = 59.5± 3.3
WF
σ/μ = 13.64± 0.85 %
μ = 471.8± 3.4
σ = 64.3± 3.6



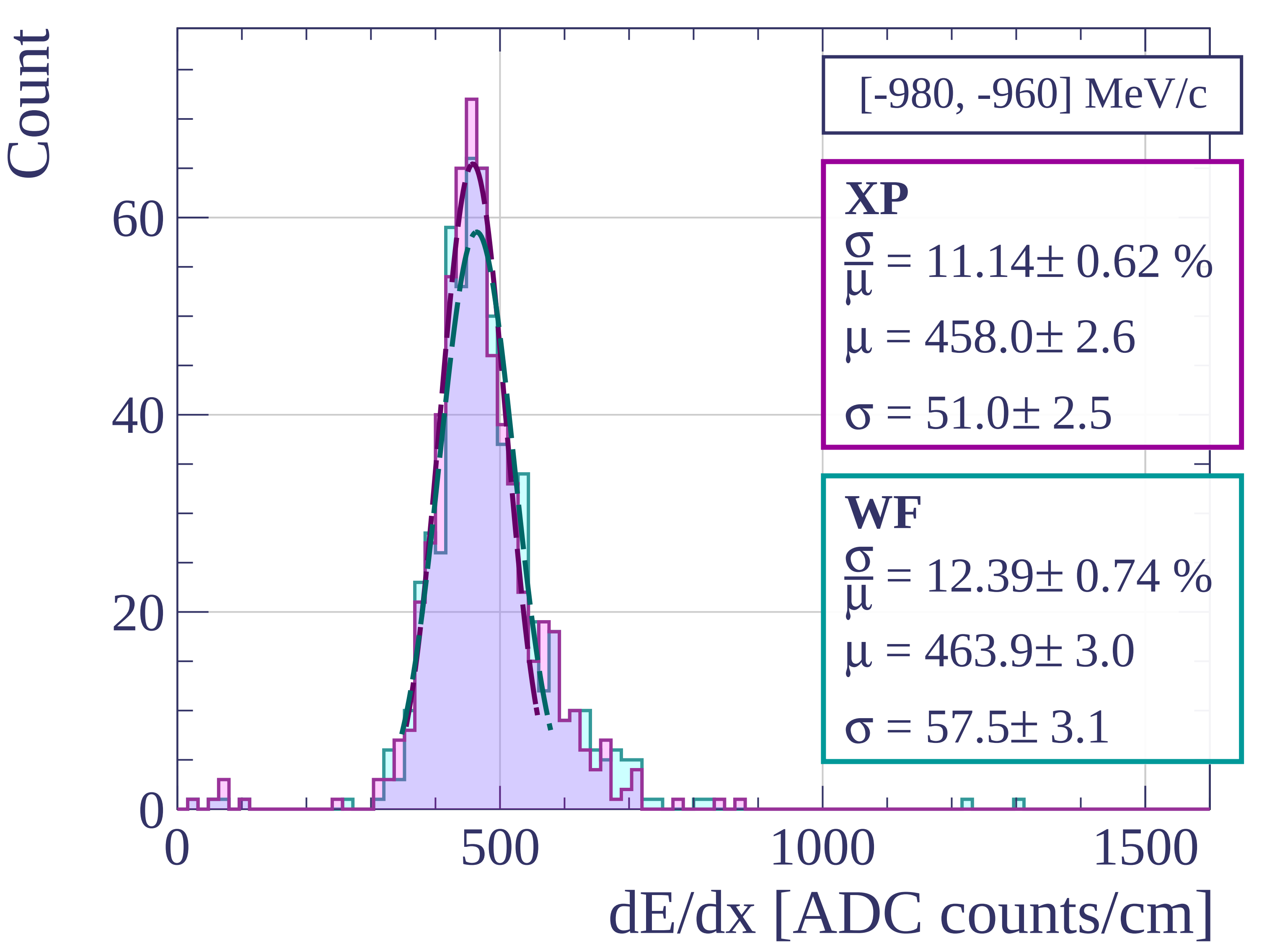
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1040, -1020] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.89± 0.85 %
μ = 465.6± 3.5
σ = 64.7± 3.5
WF
$\frac{\sigma}{\mu}$ = 14.40± 0.79 %
μ = 474.9± 3.8
σ = 68.4± 3.2



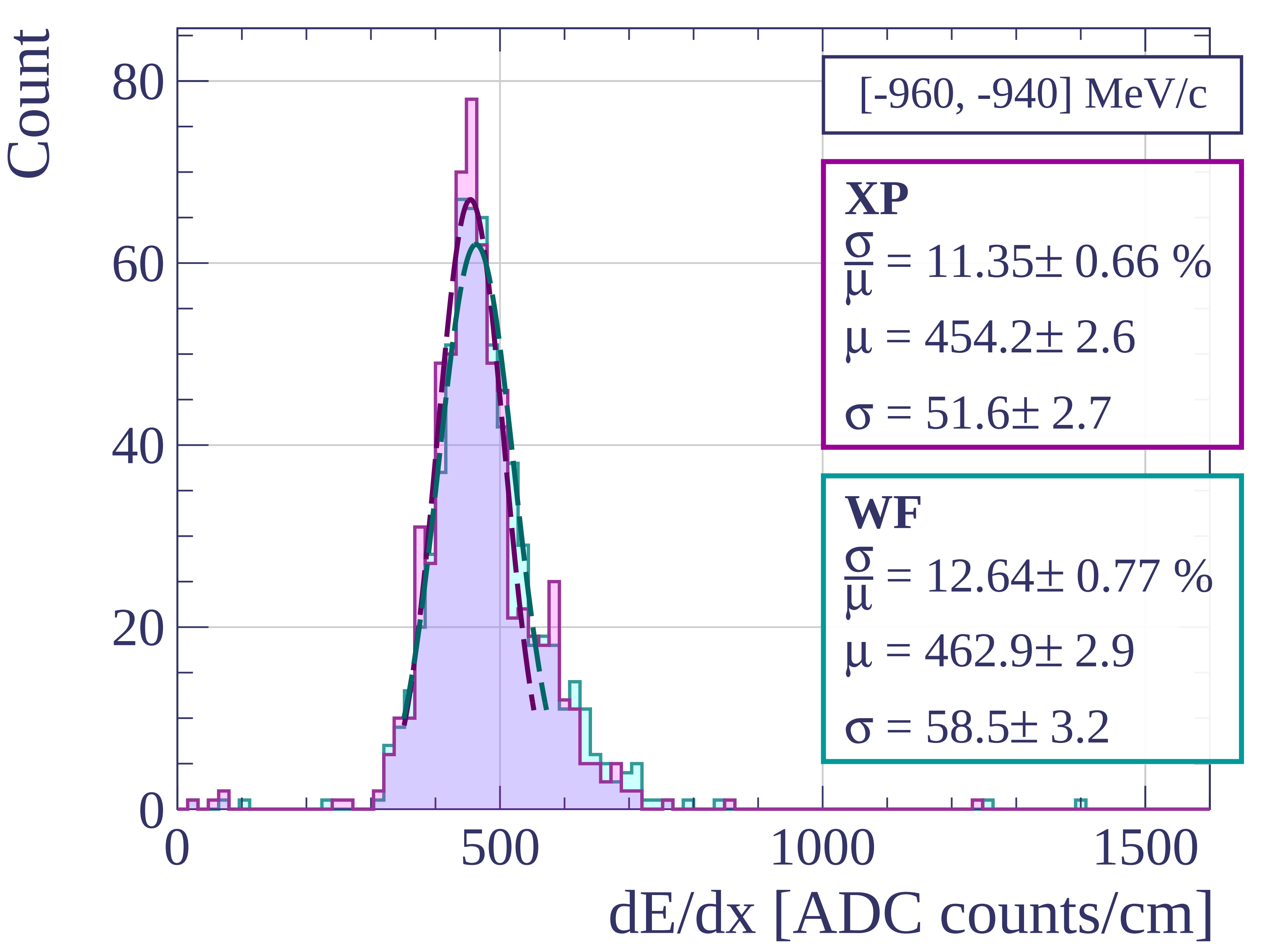
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-1020, -1000] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.23± 0.59 %
μ = 475.0± 3.0
σ = 62.9± 2.4
WF
$\frac{\sigma}{\mu}$ = 13.96± 0.59 %
μ = 485.4± 3.2
σ = 67.8± 2.4

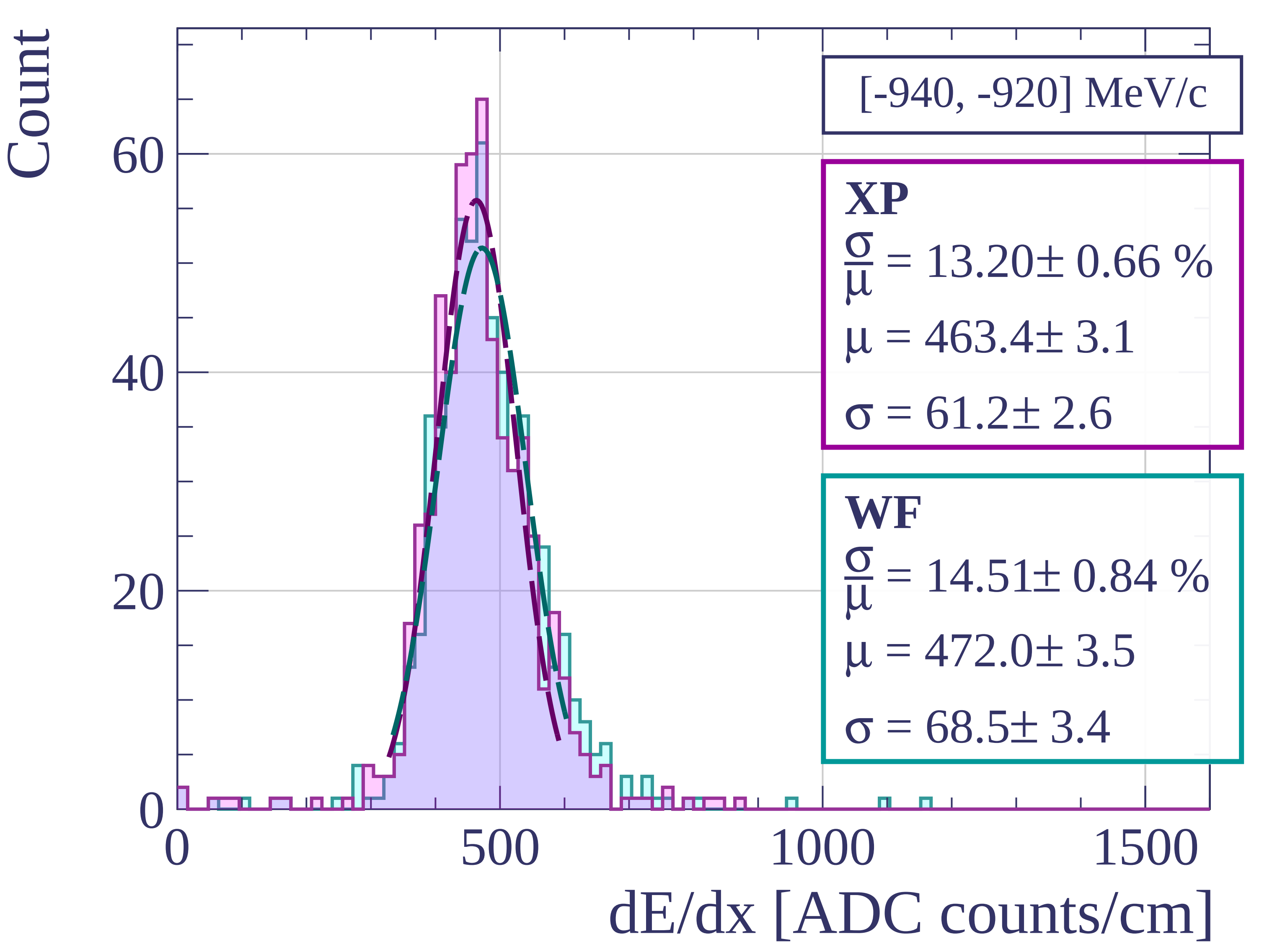


Count
dE/dx [ADC counts/cm]
0
20
40
60
0
500
1000
1500
[-1000, -980] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.61± 0.76 %
μ = 464.0± 3.0
σ = 58.5± 3.2
WF
$\frac{\sigma}{\mu}$ = 12.98± 0.74 %
μ = 470.1± 3.0
σ = 61.0± 3.1

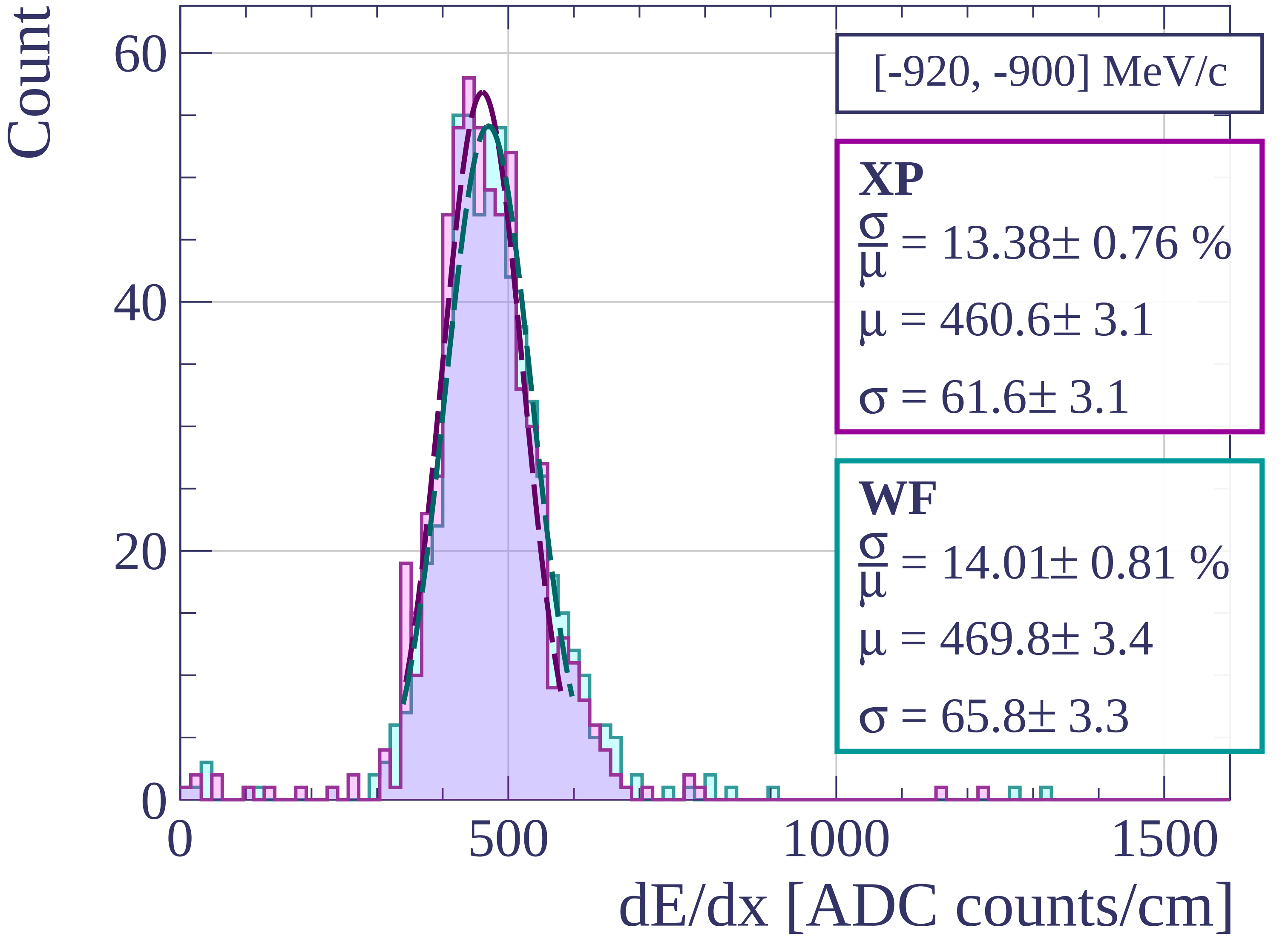


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-980, -960] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.14± 0.62 %
μ = 458.0± 2.6
σ = 51.0± 2.5
WF
$\frac{\sigma}{\mu}$ = 12.39± 0.74 %
μ = 463.9± 3.0
σ = 57.5± 3.1

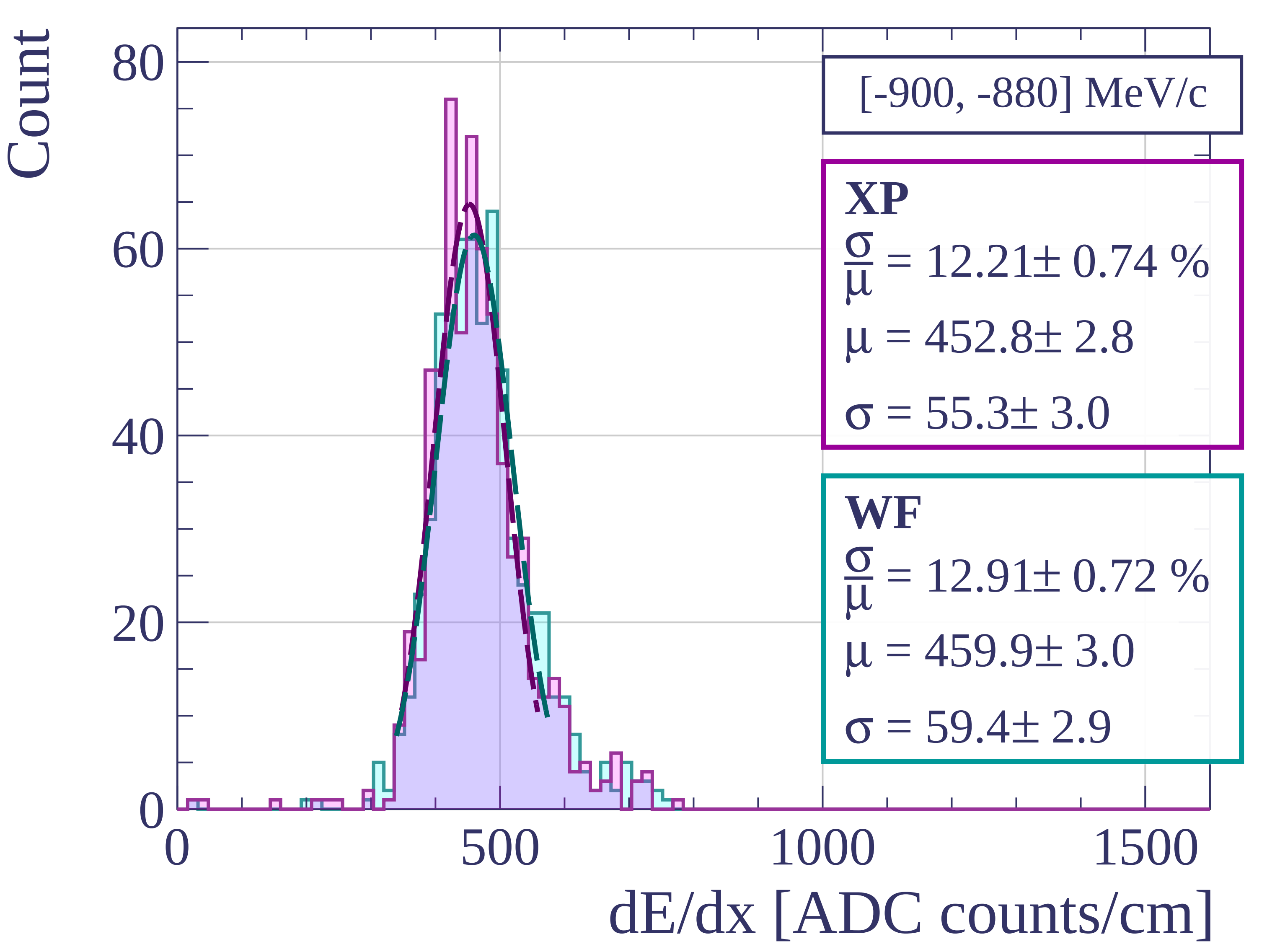

Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-960, -940] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.35± 0.66 %
μ = 454.2± 2.6
σ = 51.6± 2.7
WF
$\frac{\sigma}{\mu}$ = 12.64± 0.77 %
μ = 462.9± 2.9
σ = 58.5± 3.2

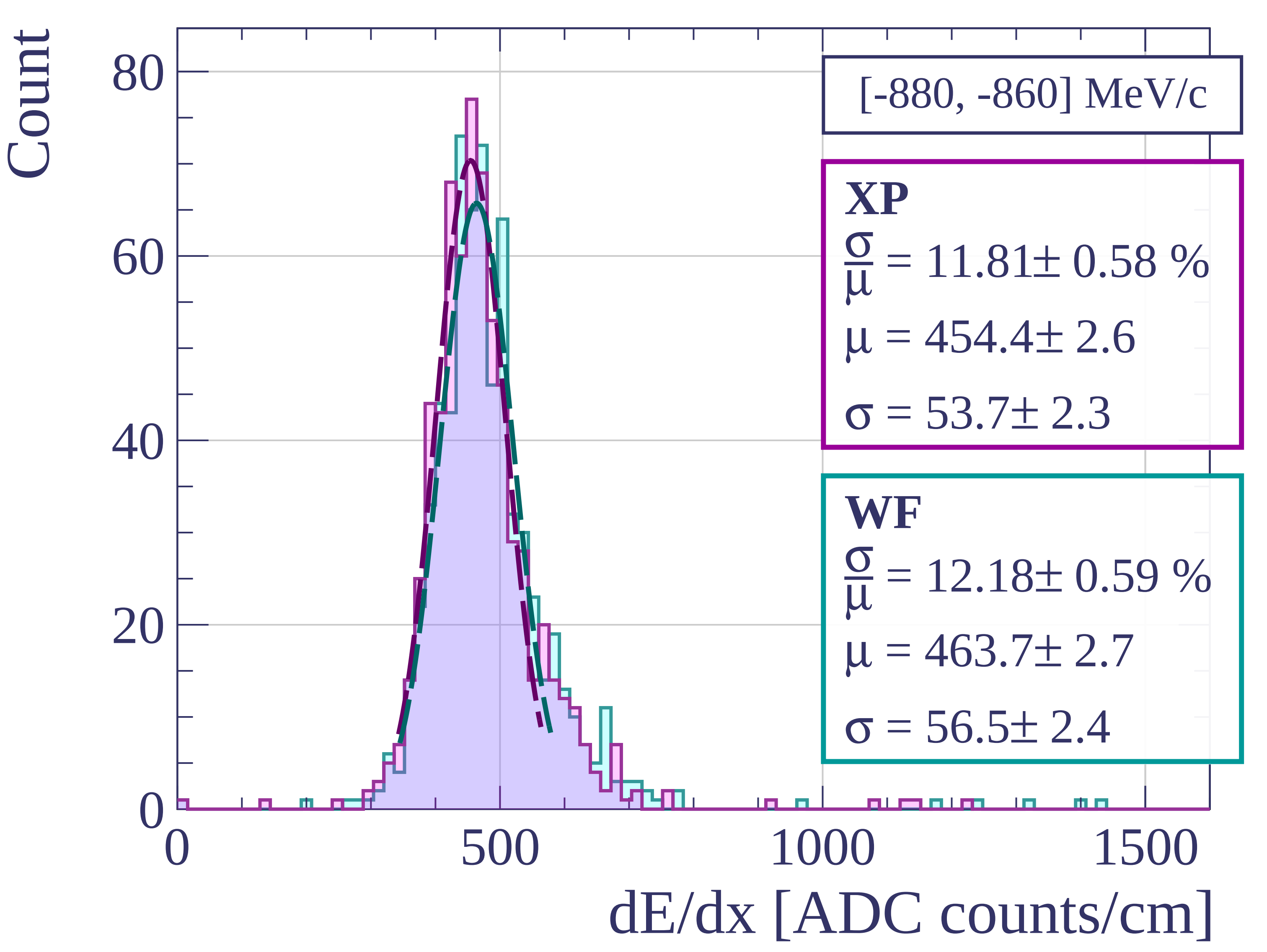


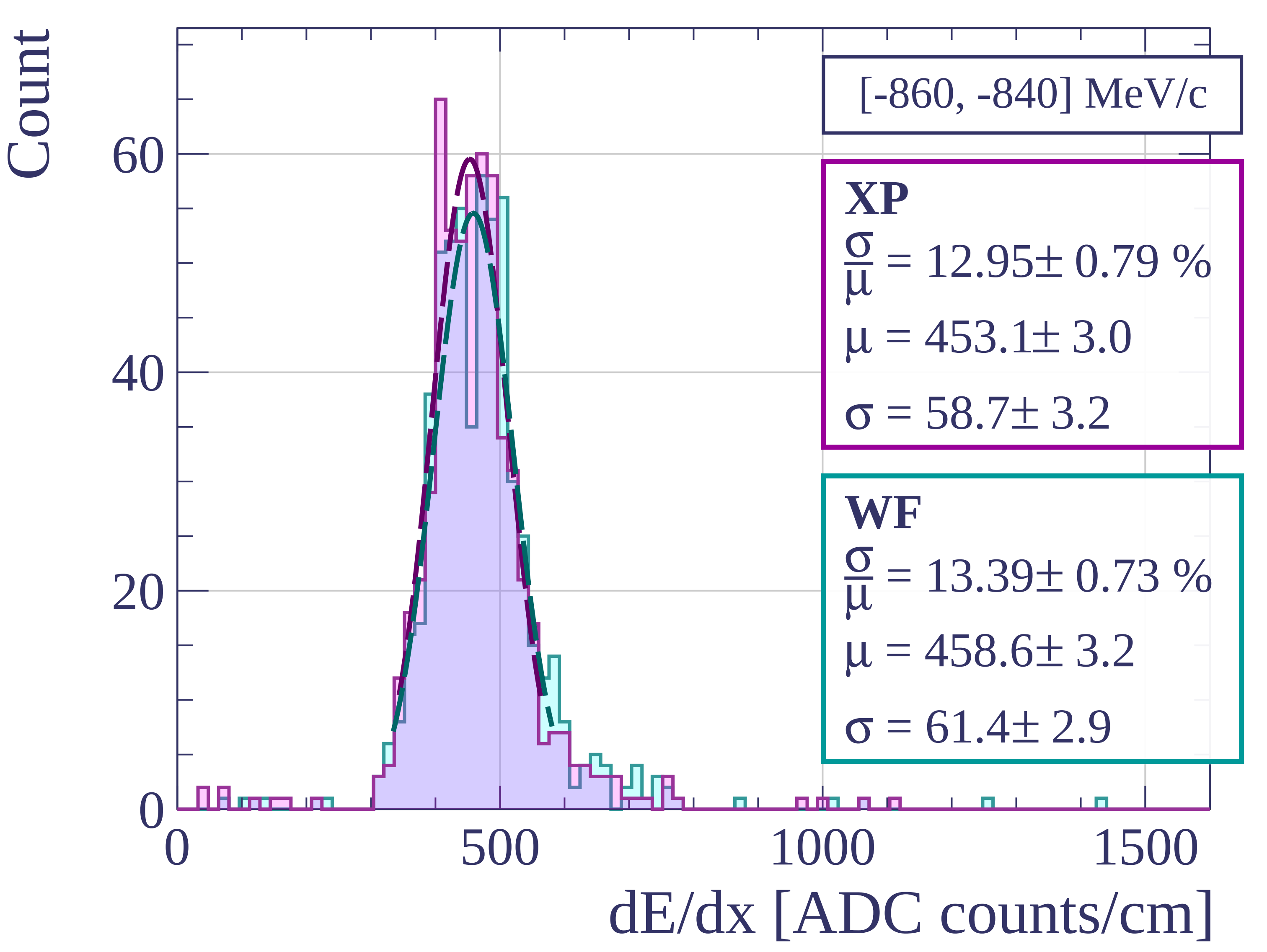
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-940, -920] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.20± 0.66 %
μ = 463.4± 3.1
σ = 61.2± 2.6
WF
$\frac{\sigma}{\mu}$ = 14.51± 0.84 %
μ = 472.0± 3.5
σ = 68.5± 3.4



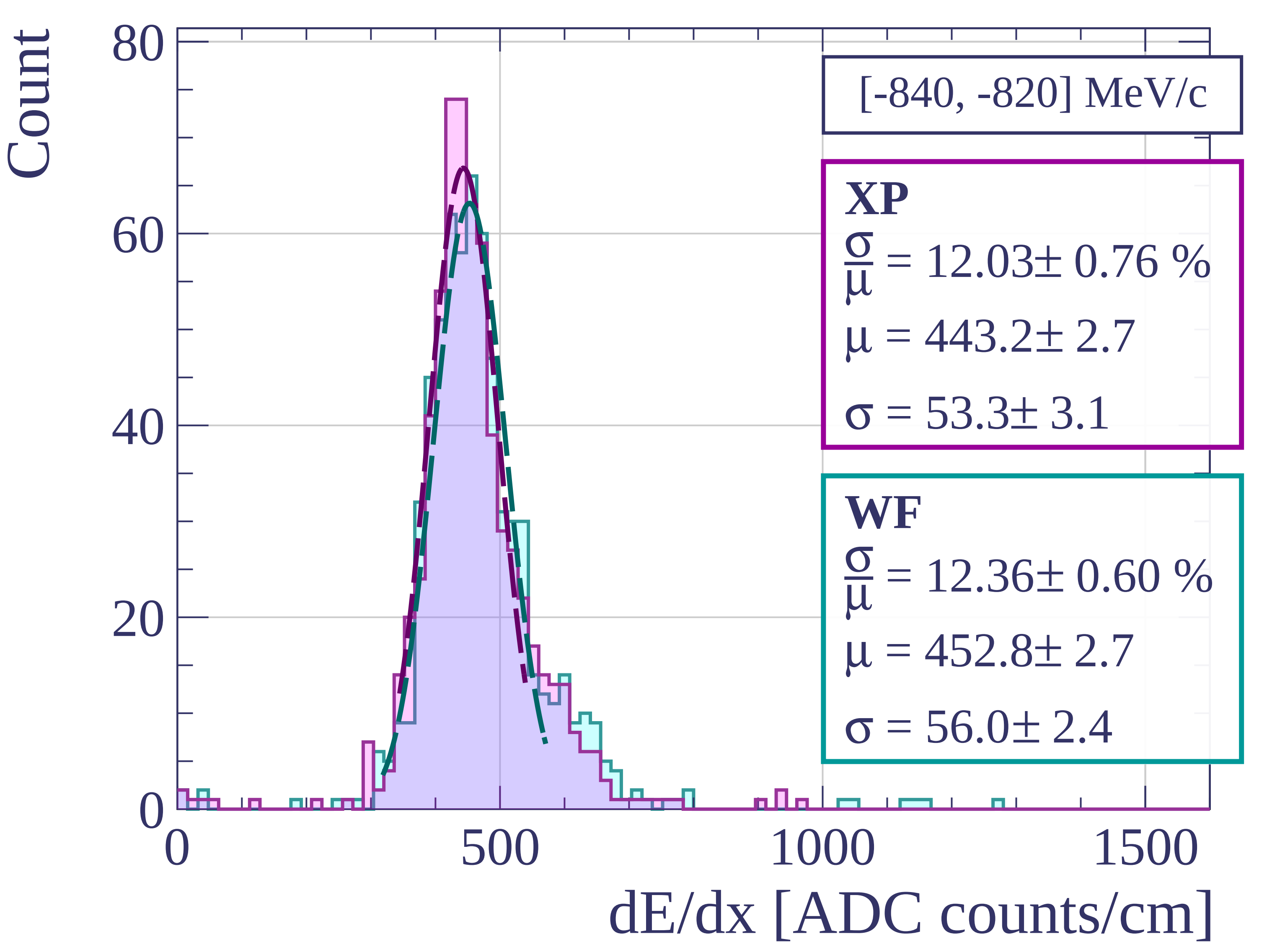
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-920, -900] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.38± 0.76 %
μ = 460.6± 3.1
σ = 61.6± 3.1
WF
$\frac{\sigma}{\mu}$ = 14.01± 0.81 %
μ = 469.8± 3.4
σ = 65.8± 3.3


Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-900, -880] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.21± 0.74 %
μ = 452.8± 2.8
σ = 55.3± 3.0
WF
$\frac{\sigma}{\mu}$ = 12.91± 0.72 %
μ = 459.9± 3.0
σ = 59.4± 2.9

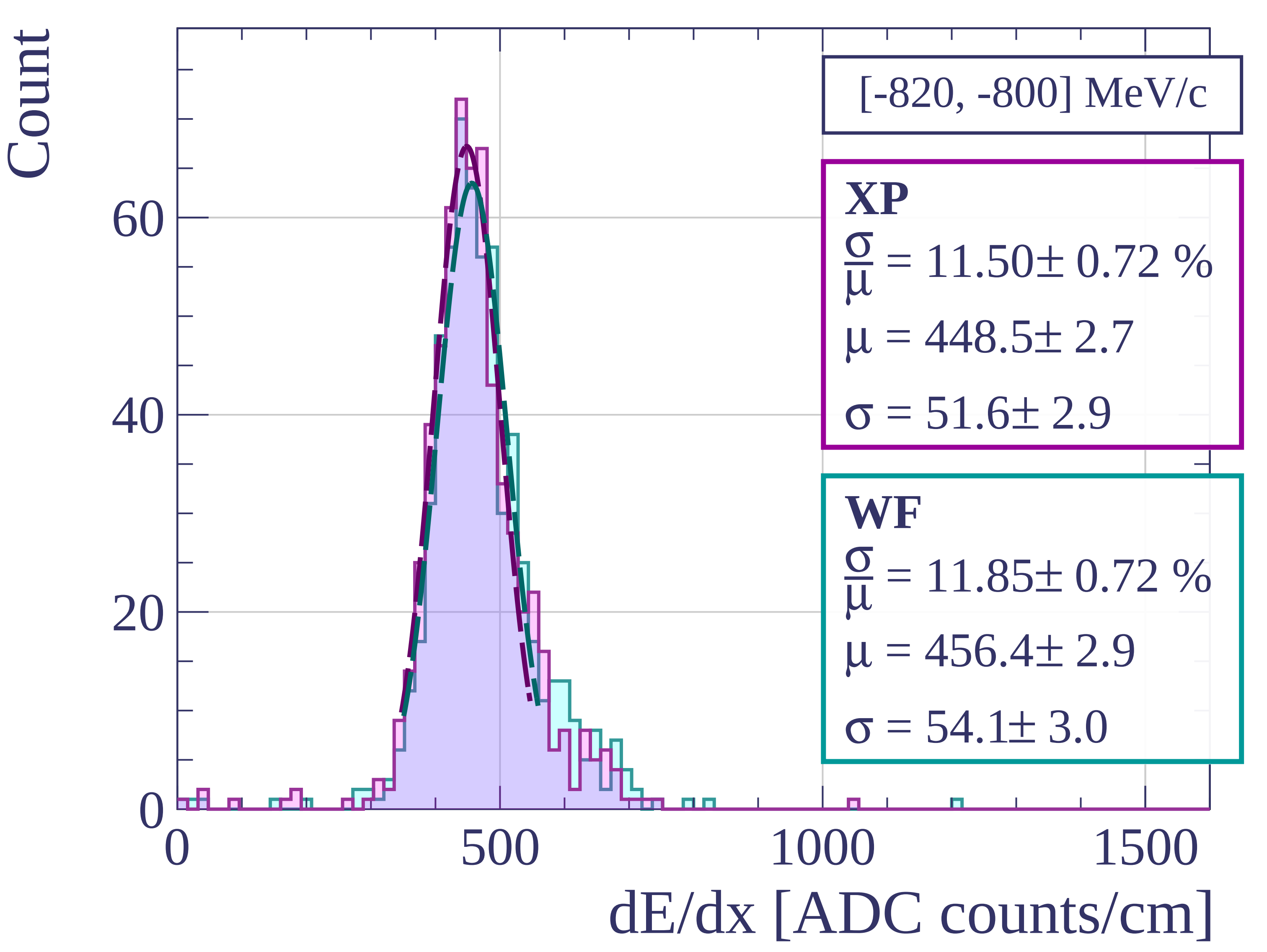

Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-880, -860] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.81± 0.58 %
μ = 454.4± 2.6
σ = 53.7± 2.3
WF
$\frac{\sigma}{\mu}$ = 12.18± 0.59 %
μ = 463.7± 2.7
σ = 56.5± 2.4



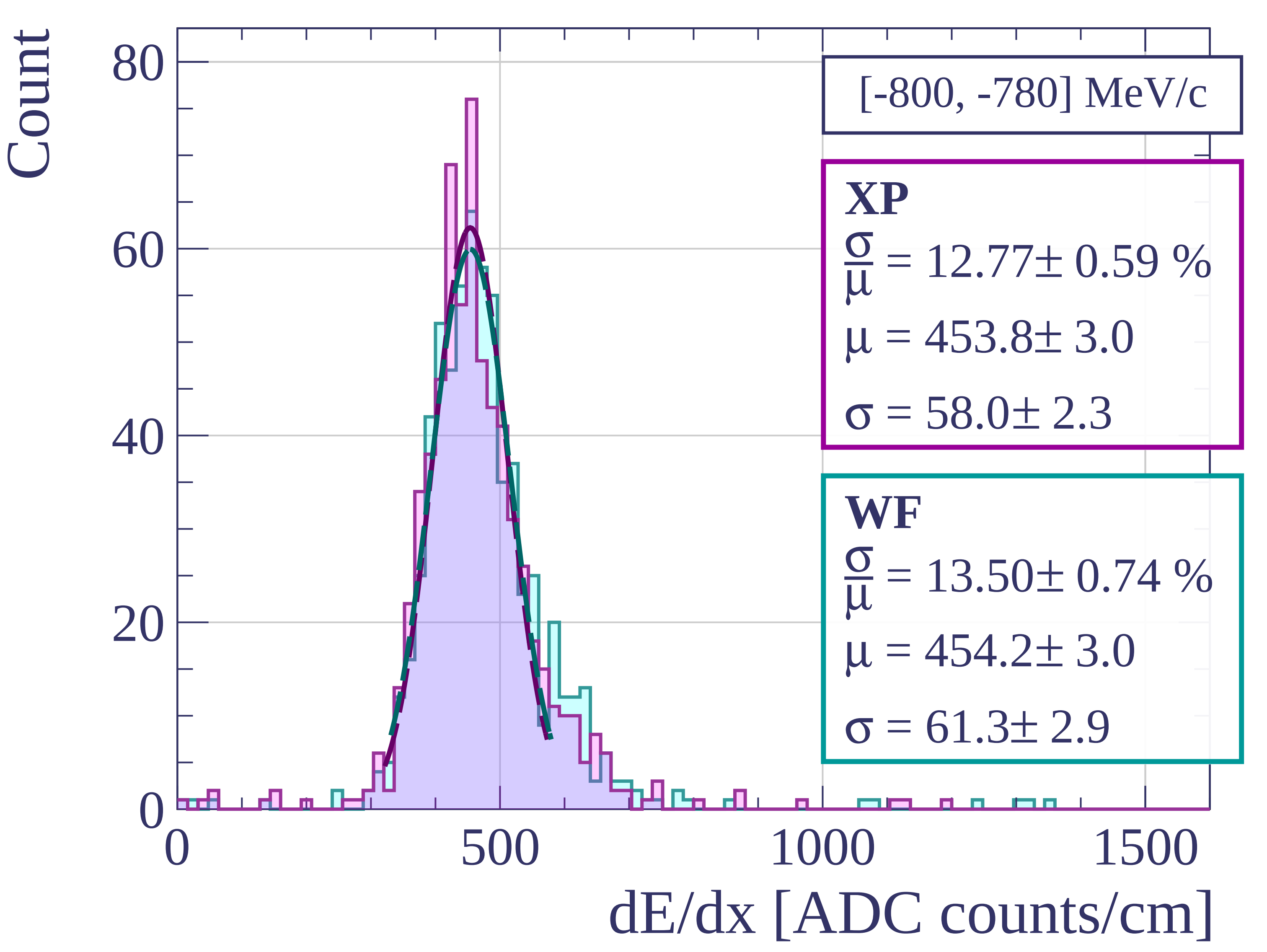
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-860, -840] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.95± 0.79 %
μ = 453.1± 3.0
σ = 58.7± 3.2
WF
$\frac{\sigma}{\mu}$ = 13.39± 0.73 %
μ = 458.6± 3.2
σ = 61.4± 2.9



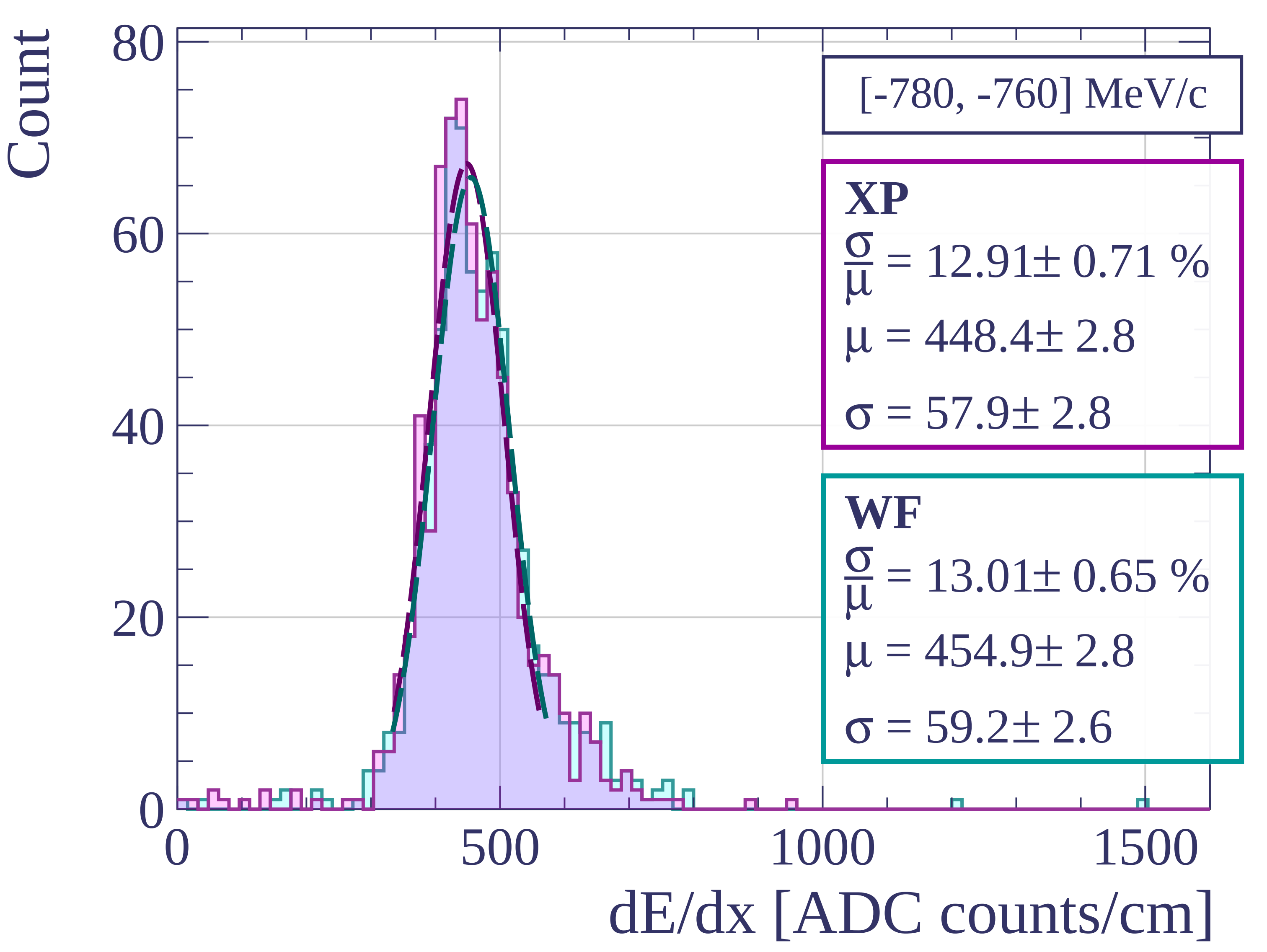
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-840, -820] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.03± 0.76 %
μ = 443.2± 2.7
σ = 53.3± 3.1
WF
$\frac{\sigma}{\mu}$ = 12.36± 0.60 %
μ = 452.8± 2.7
σ = 56.0± 2.4



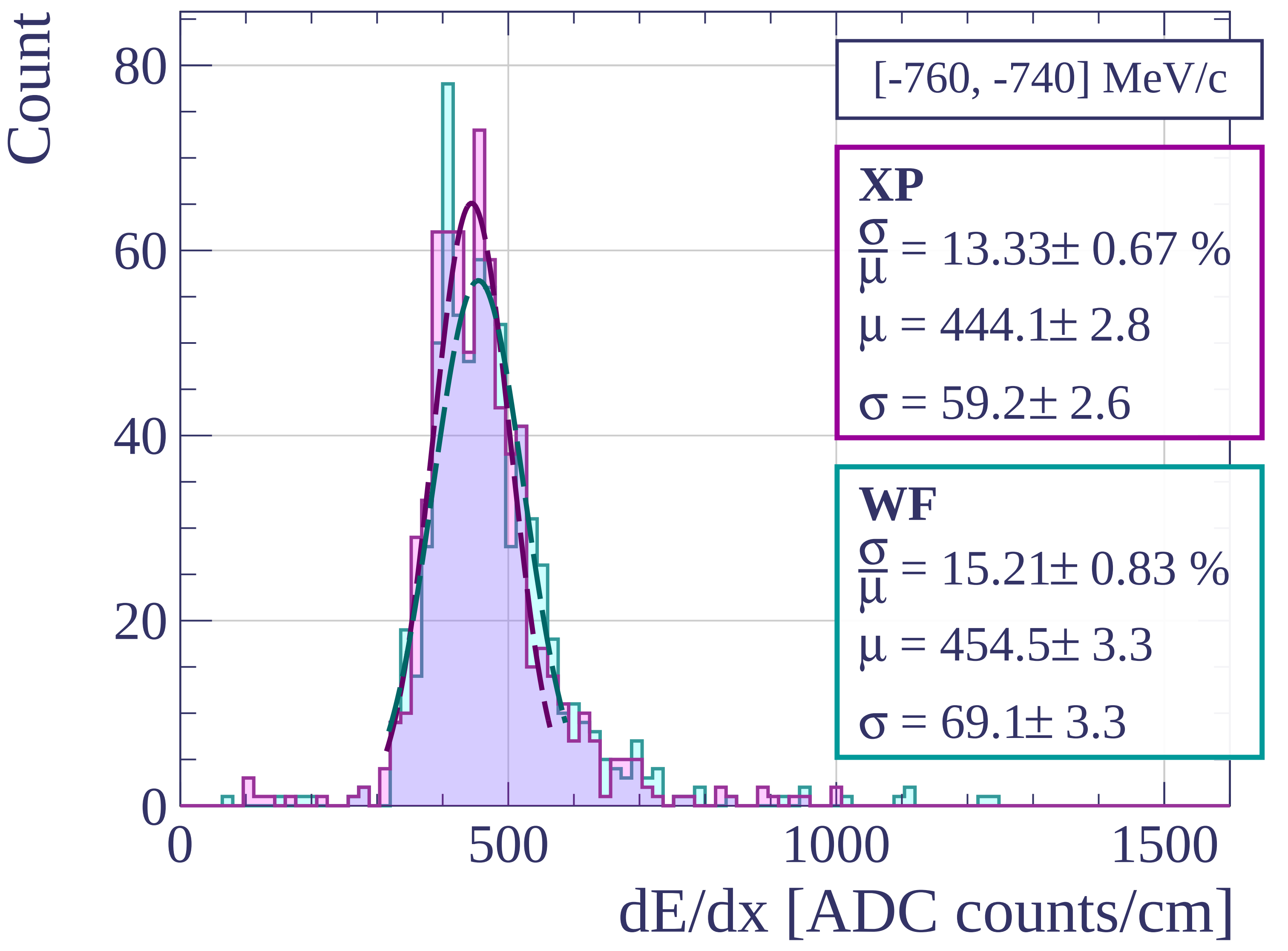
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-820, -800] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.50± 0.72 %
μ = 448.5± 2.7
σ = 51.6± 2.9
WF
$\frac{\sigma}{\mu}$ = 11.85± 0.72 %
μ = 456.4± 2.9
σ = 54.1± 3.0



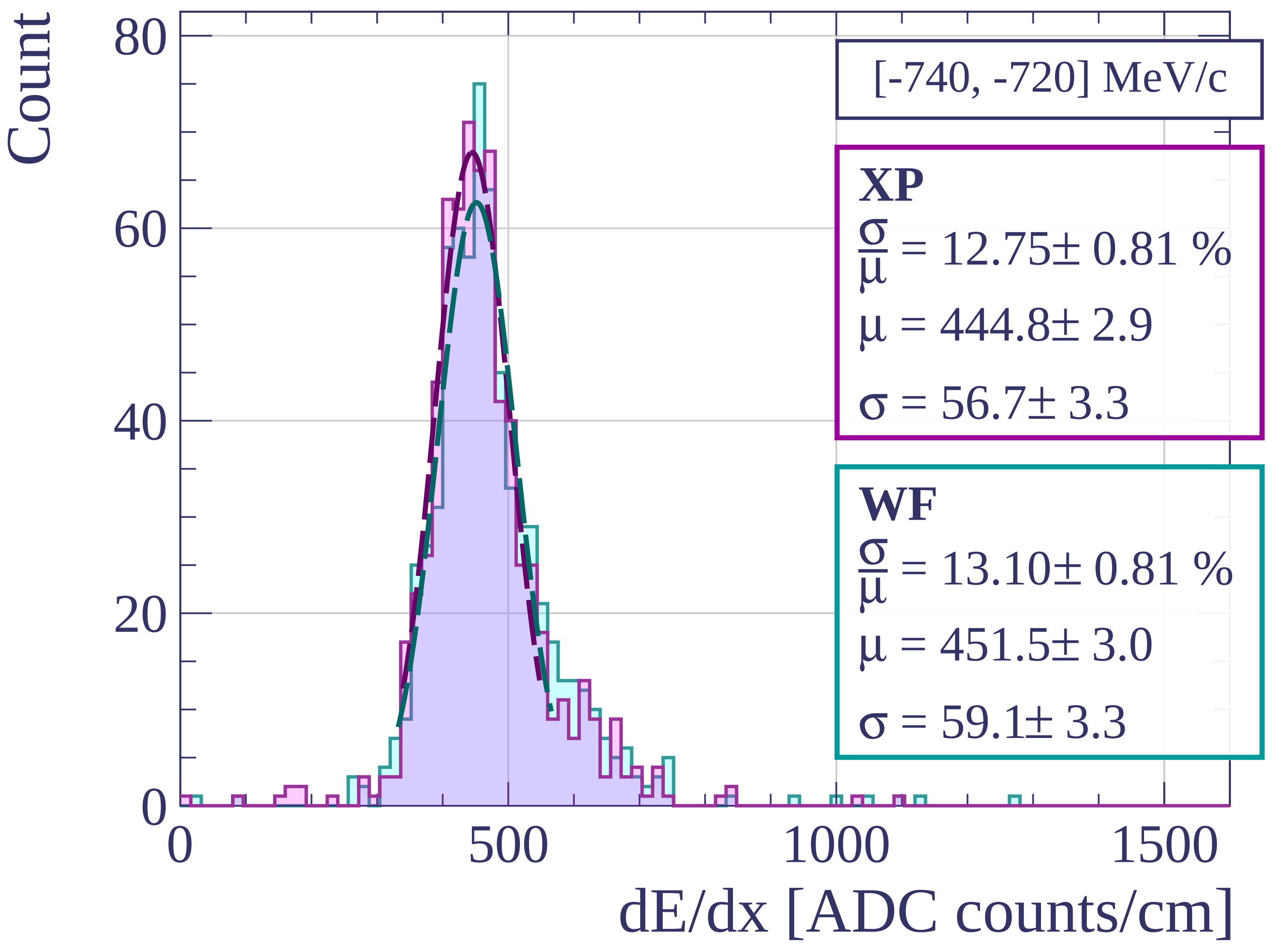
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-800, -780] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.77± 0.59 %
μ = 453.8± 3.0
σ = 58.0± 2.3
WF
$\frac{\sigma}{\mu}$ = 13.50± 0.74 %
μ = 454.2± 3.0
σ = 61.3± 2.9



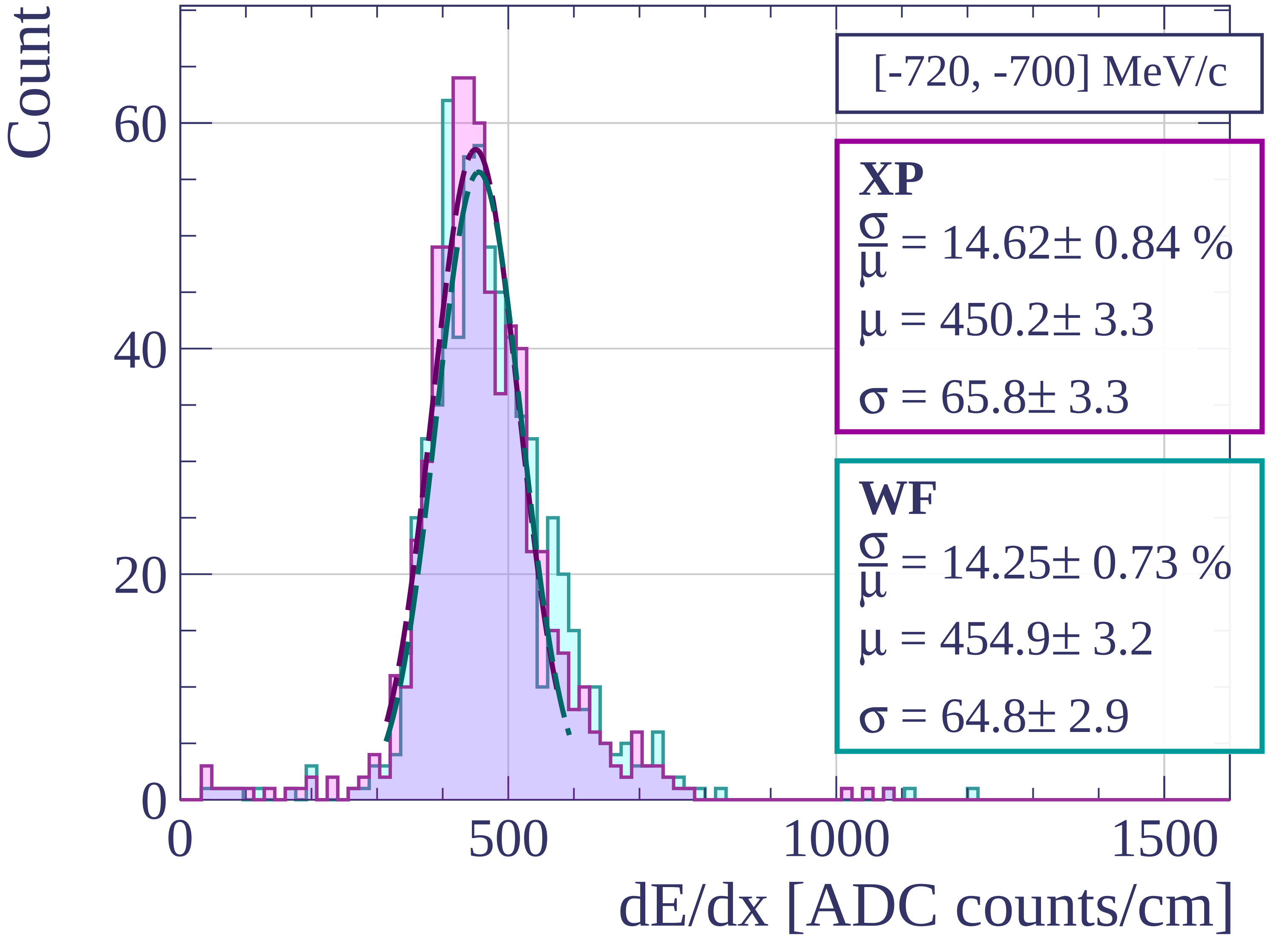
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-780, -760] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.91± 0.71 %
μ = 448.4± 2.8
σ = 57.9± 2.8
WF
$\frac{\sigma}{\mu}$ = 13.01± 0.65 %
μ = 454.9± 2.8
σ = 59.2± 2.6



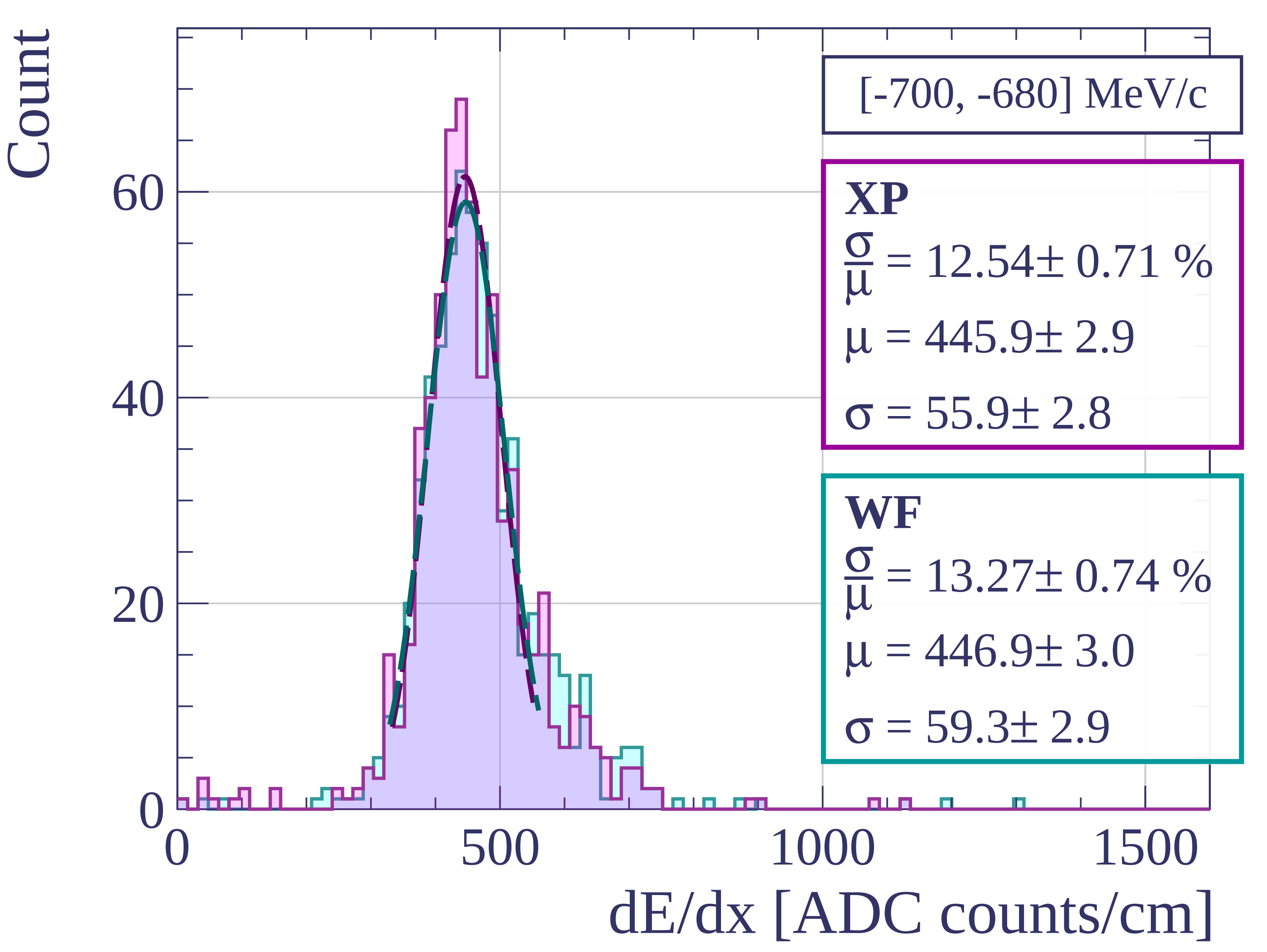
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-760, -740] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.33± 0.67 %
μ = 444.1± 2.8
σ = 59.2± 2.6
WF
$\frac{\sigma}{\mu}$ = 15.21± 0.83 %
μ = 454.5± 3.3
σ = 69.1± 3.3

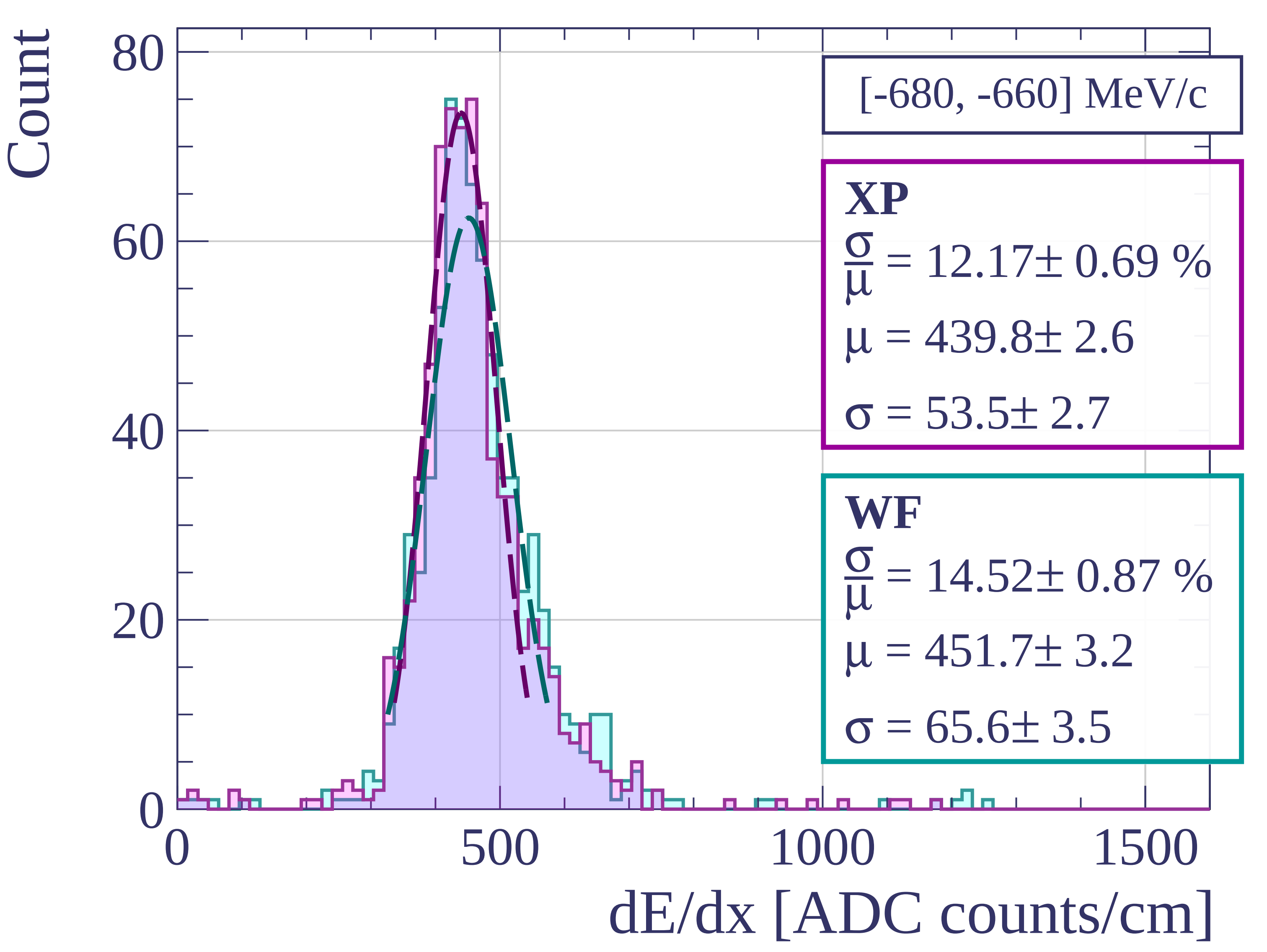


Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-740, -720] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.75± 0.81 %
μ = 444.8± 2.9
σ = 56.7± 3.3
WF
$\frac{\sigma}{\mu}$ = 13.10± 0.81 %
μ = 451.5± 3.0
σ = 59.1± 3.3

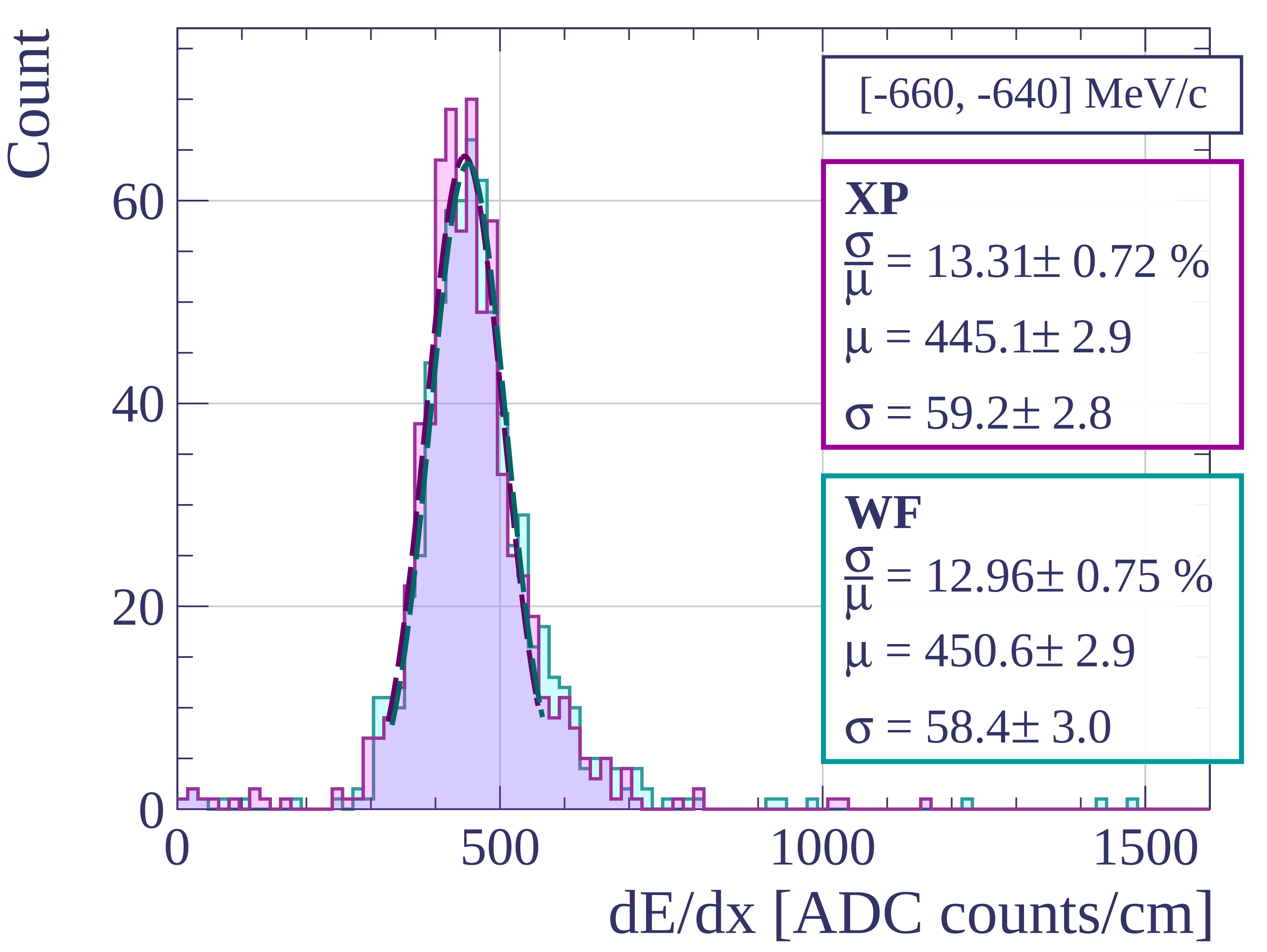


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-720, -700] MeV/c
XP
$\frac{\sigma}{\mu}$ = 14.62± 0.84 %
μ = 450.2± 3.3
σ = 65.8± 3.3
WF
$\frac{\sigma}{\mu}$ = 14.25± 0.73 %
μ = 454.9± 3.2
σ = 64.8± 2.9

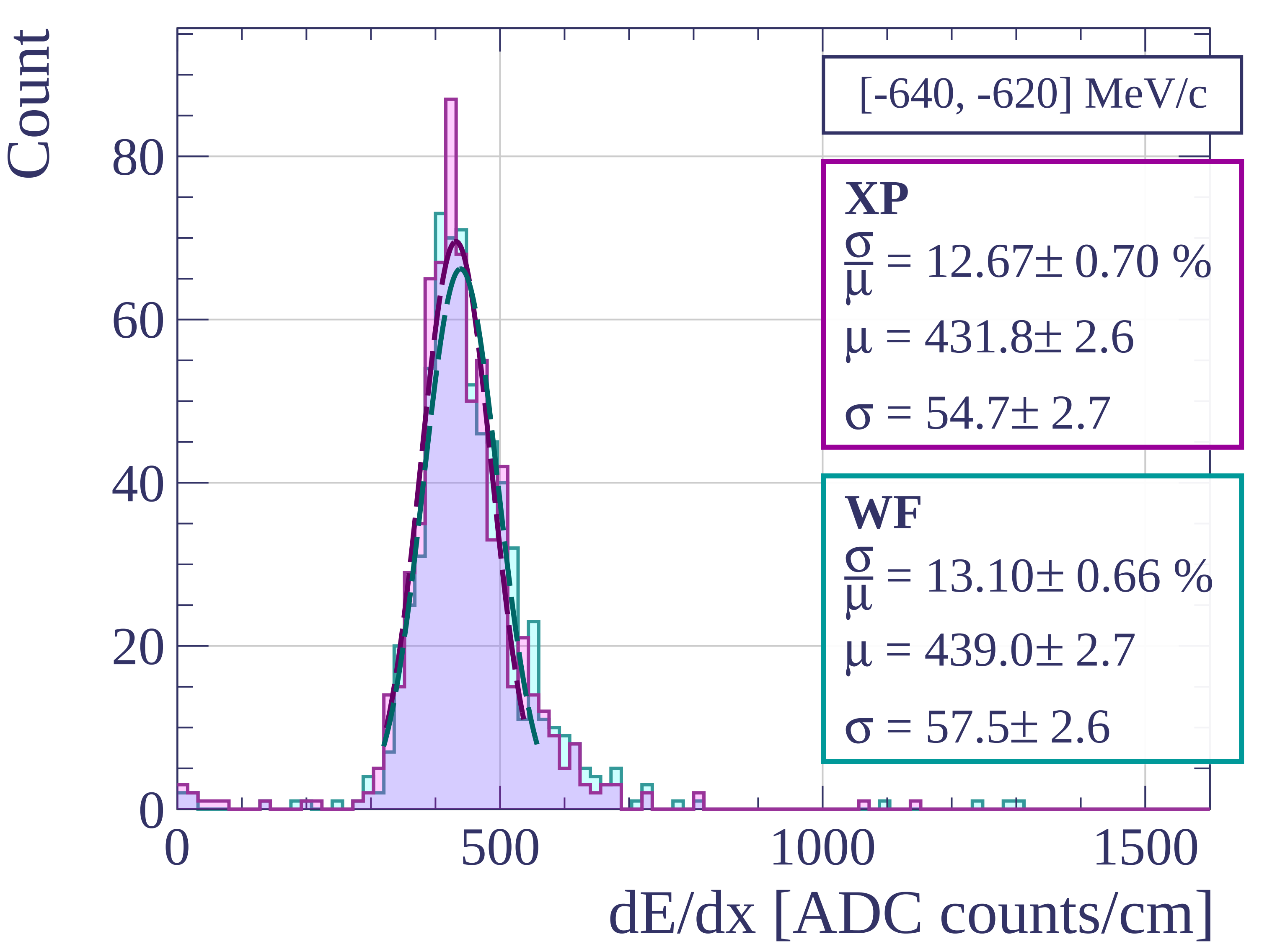

Count
0
20
40
60
0
500
1000
1500
dE/dx [ADC counts/cm]
[-700, -680] MeV/c
XP
σ/μ = 12.54± 0.71 %
μ = 445.9± 2.9
σ = 55.9± 2.8
WF
σ/μ = 13.27± 0.74 %
μ = 446.9± 3.0
σ = 59.3± 2.9



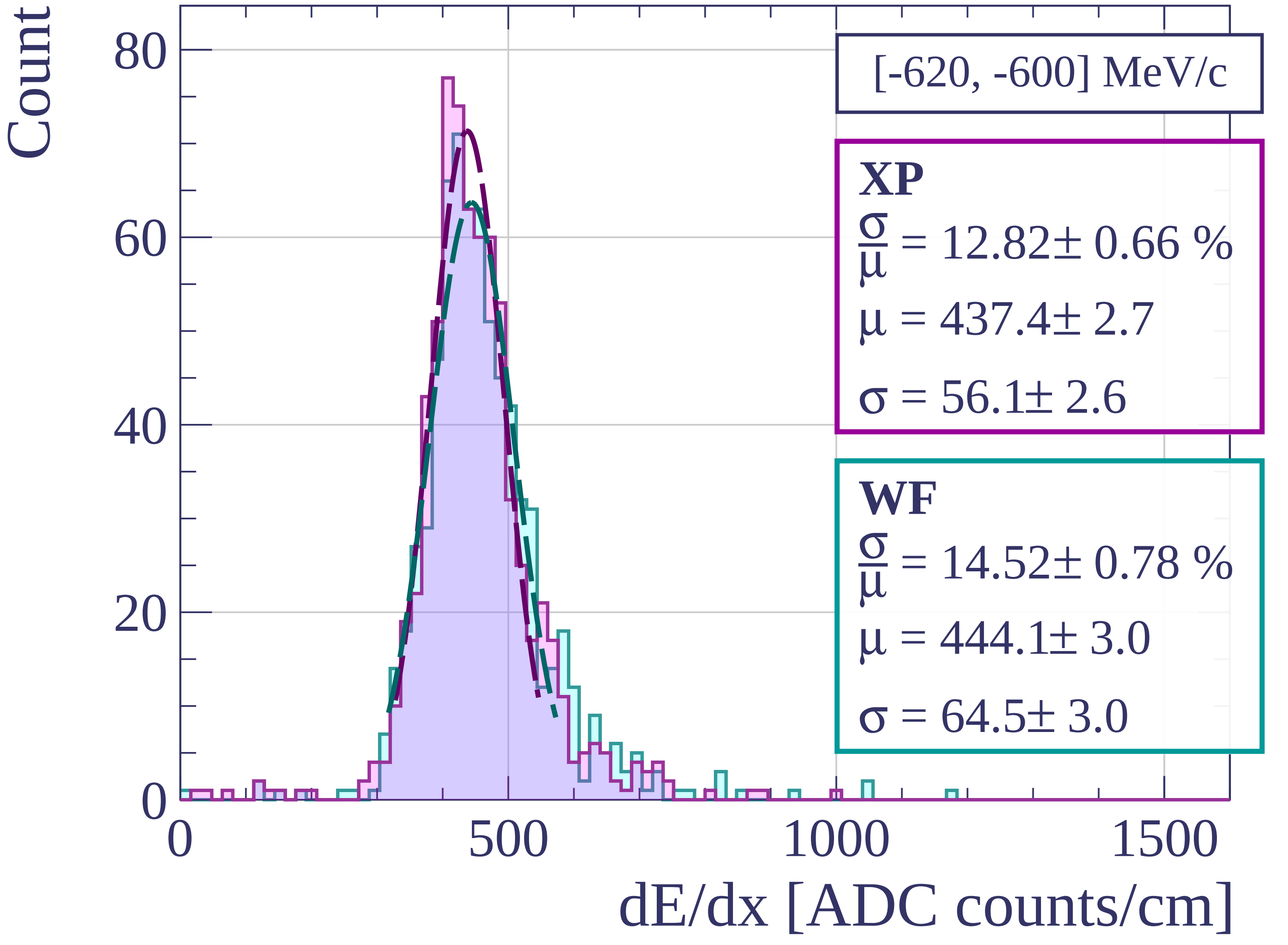
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-680, -660] MeV/c
XP
σ/μ = 12.17± 0.69 %
μ = 439.8± 2.6
σ = 53.5± 2.7
WF
σ/μ = 14.52± 0.87 %
μ = 451.7± 3.2
σ = 65.6± 3.5



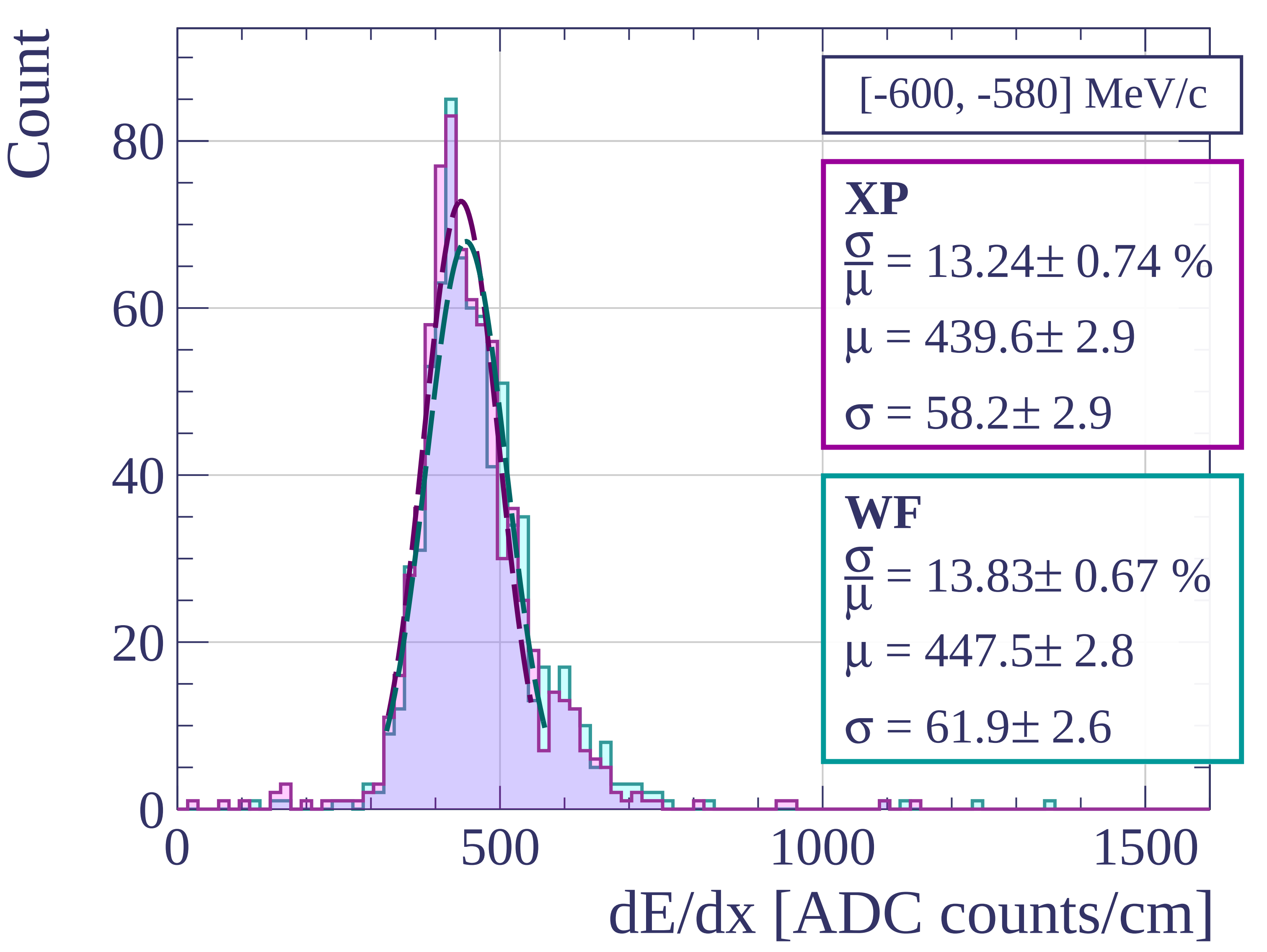
Count
0
20
40
60
0
500
1000
1500
dE/dx [ADC counts/cm]
[-660, -640] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.31± 0.72 %
μ = 445.1± 2.9
σ = 59.2± 2.8
WF
$\frac{\sigma}{\mu}$ = 12.96± 0.75 %
μ = 450.6± 2.9
σ = 58.4± 3.0



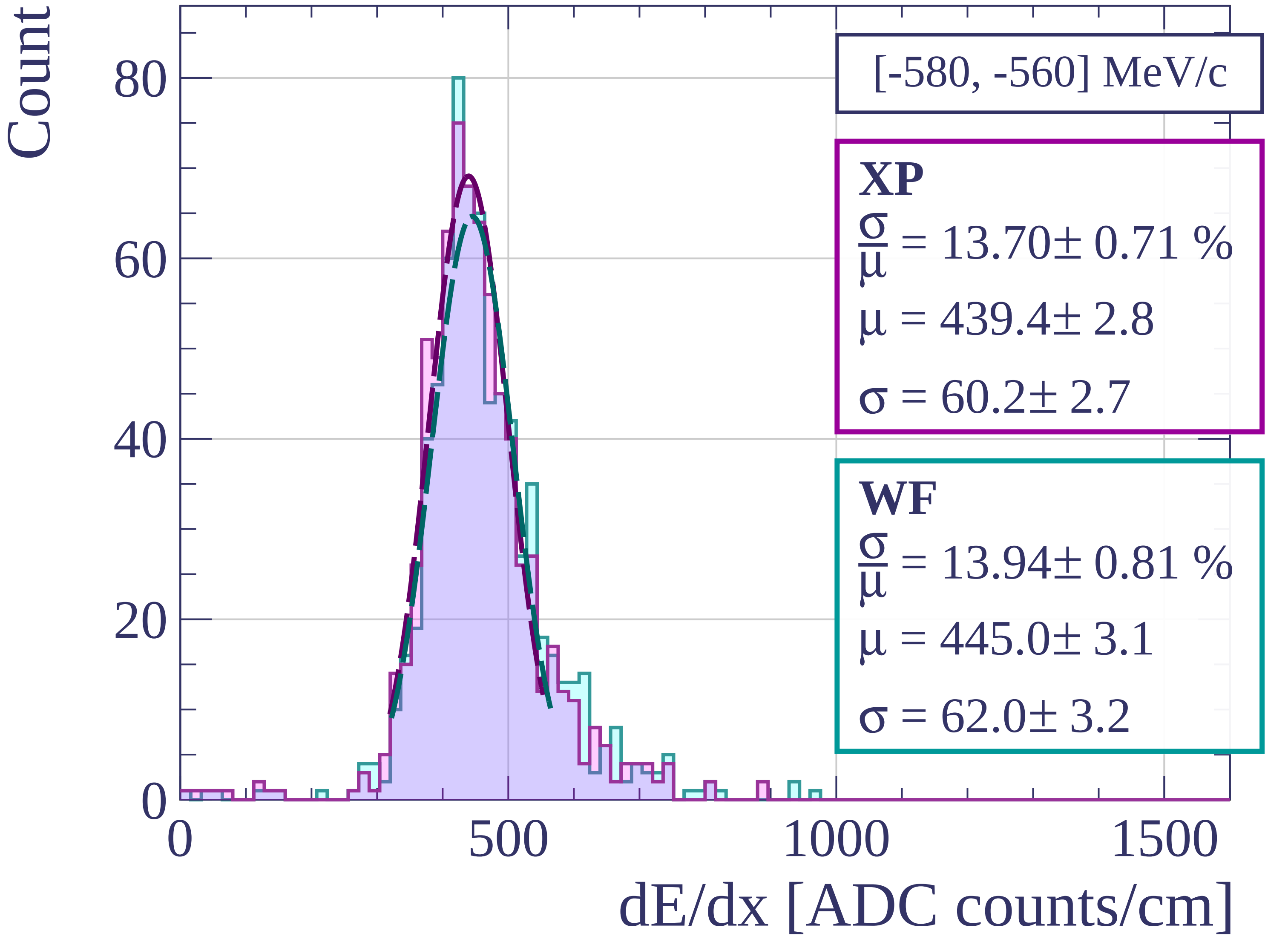
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-640, -620] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.67± 0.70 %
μ = 431.8± 2.6
σ = 54.7± 2.7
WF
$\frac{\sigma}{\mu}$ = 13.10± 0.66 %
μ = 439.0± 2.7
σ = 57.5± 2.6



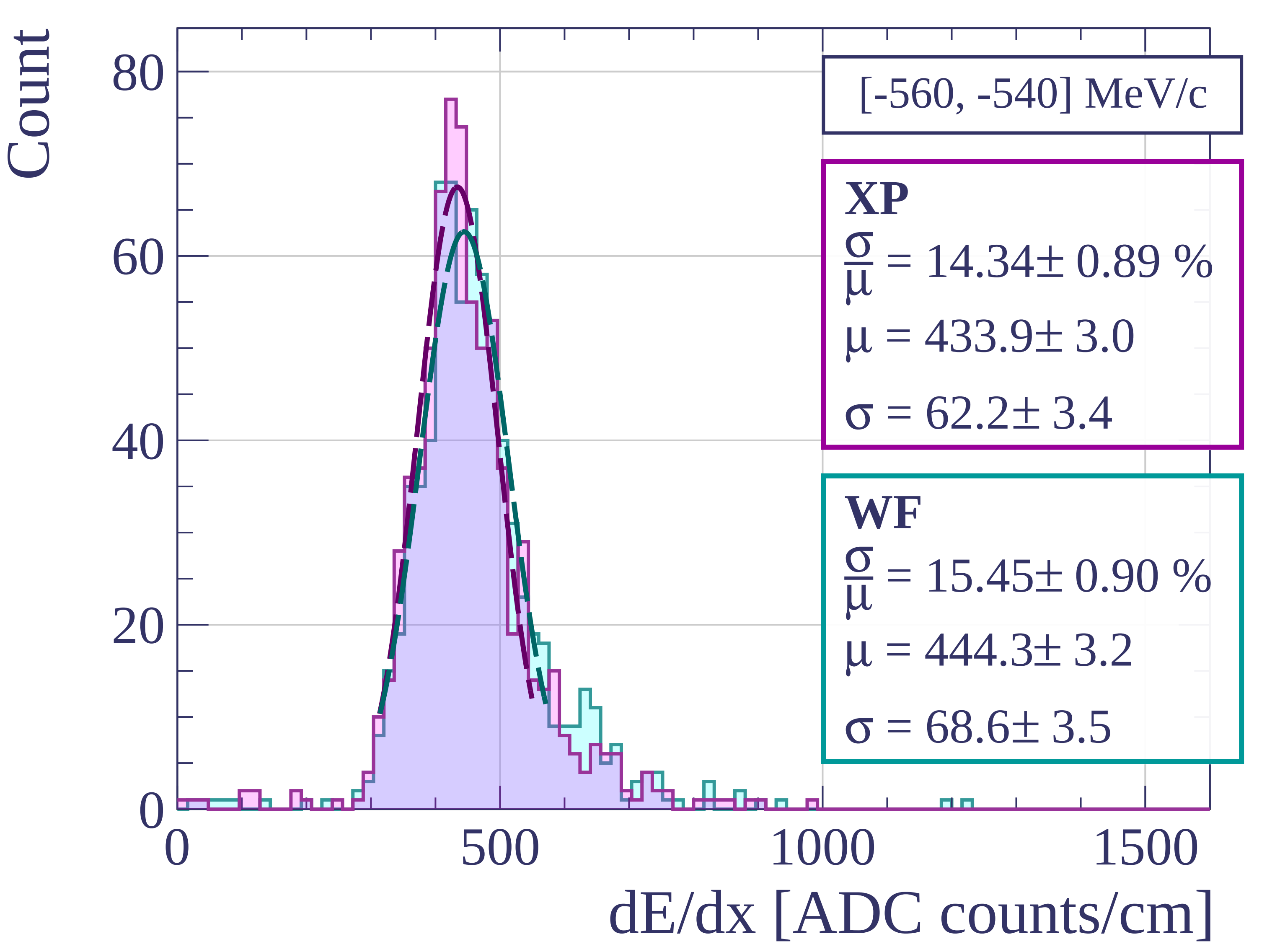
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-620, -600] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.82± 0.66 %
μ = 437.4± 2.7
σ = 56.1± 2.6
WF
$\frac{\sigma}{\mu}$ = 14.52± 0.78 %
μ = 444.1± 3.0
σ = 64.5± 3.0



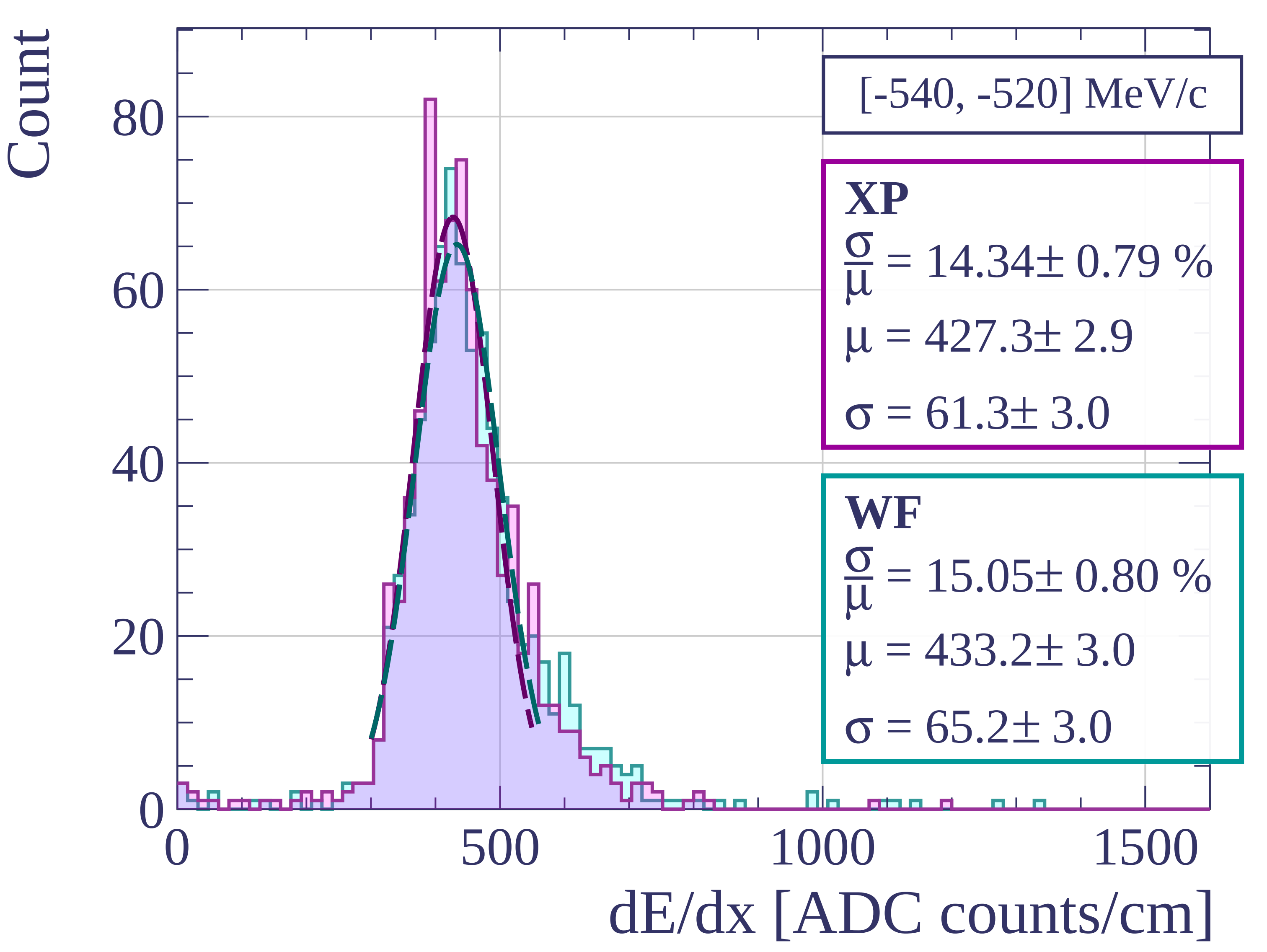
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-600, -580] MeV/c
XP
σ/μ = 13.24± 0.74 %
μ = 439.6± 2.9
σ = 58.2± 2.9
WF
σ/μ = 13.83± 0.67 %
μ = 447.5± 2.8
σ = 61.9± 2.6

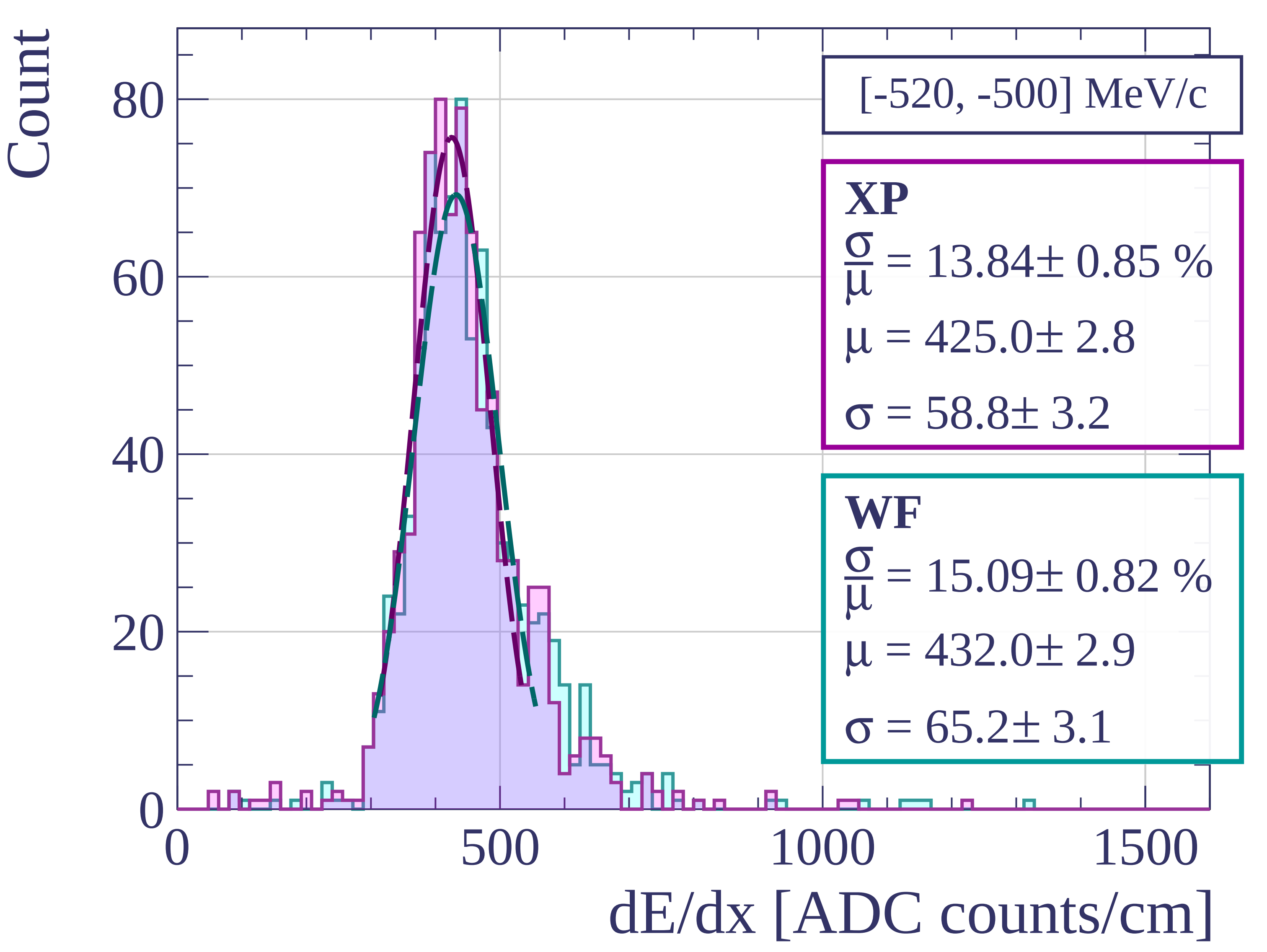


Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
20
40
60
80
[-580, -560] MeV/c
XP
σ/μ = 13.70± 0.71 %
μ = 439.4± 2.8
σ = 60.2± 2.7
WF
σ/μ = 13.94± 0.81 %
μ = 445.0± 3.1
σ = 62.0± 3.2

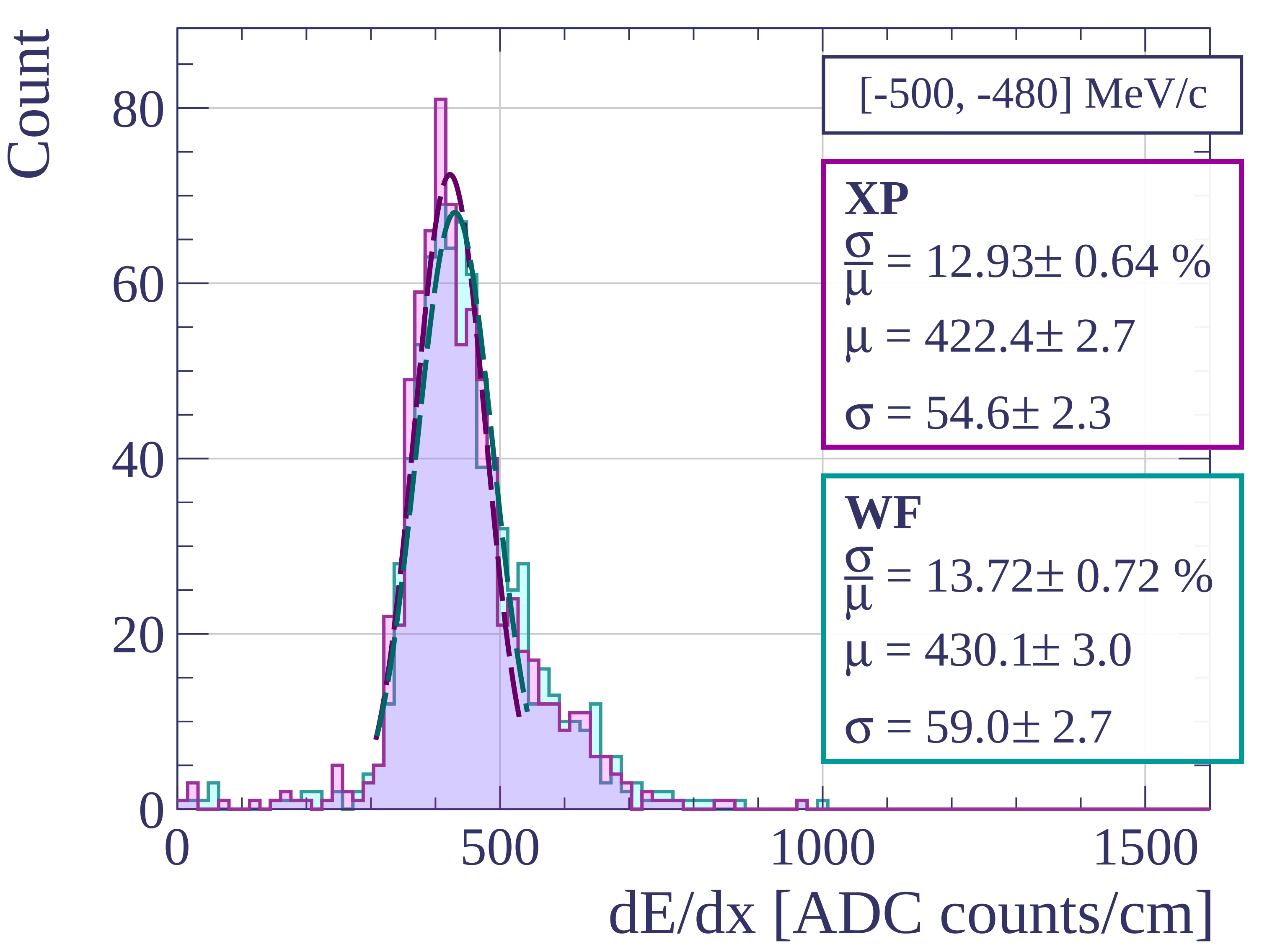


Count
0
20
40
60
80
0
500
1000
1500
dE/dx [ADC counts/cm]
[-560, -540] MeV/c
XP
σ/μ = 14.34± 0.89 %
μ = 433.9± 3.0
σ = 62.2± 3.4
WF
σ/μ = 15.45± 0.90 %
μ = 444.3± 3.2
σ = 68.6± 3.5

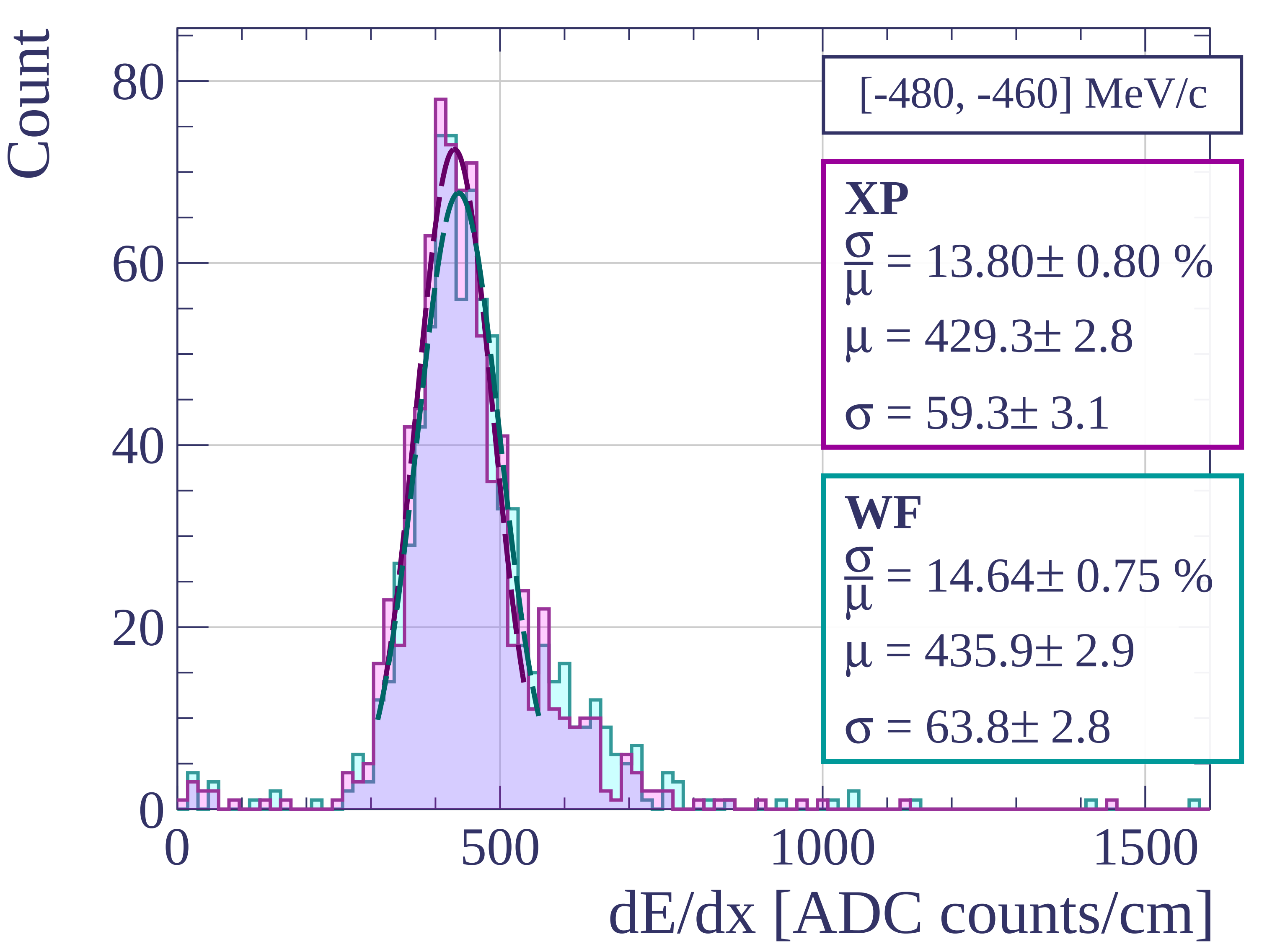

Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-540, -520] MeV/c
XP
$\frac{\sigma}{\mu}$ = 14.34± 0.79 %
μ = 427.3± 2.9
σ = 61.3± 3.0
WF
$\frac{\sigma}{\mu}$ = 15.05± 0.80 %
μ = 433.2± 3.0
σ = 65.2± 3.0



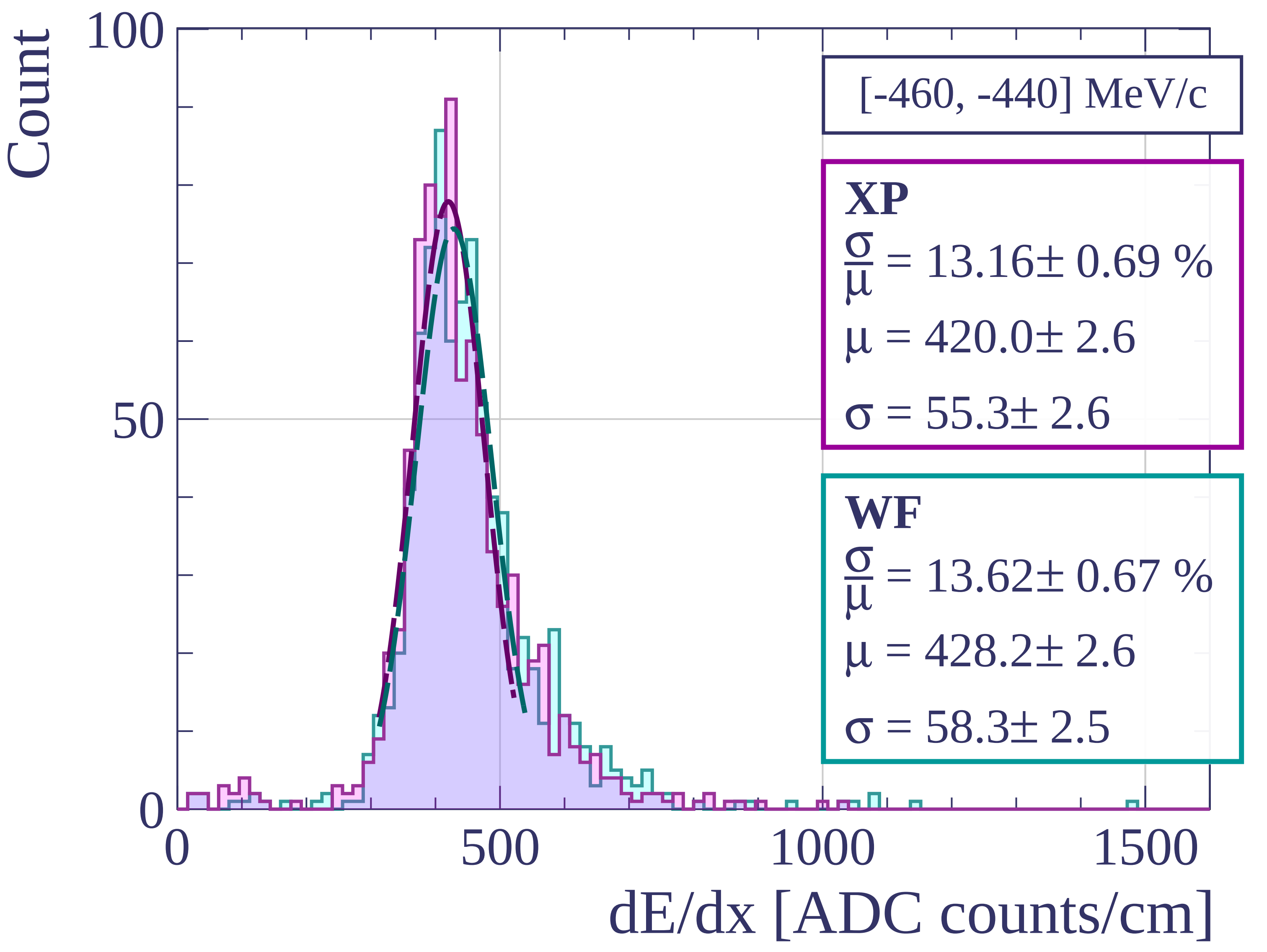
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-520, -500] MeV/c
XP
σ/μ = 13.84± 0.85 %
μ = 425.0± 2.8
σ = 58.8± 3.2
WF
σ/μ = 15.09± 0.82 %
μ = 432.0± 2.9
σ = 65.2± 3.1



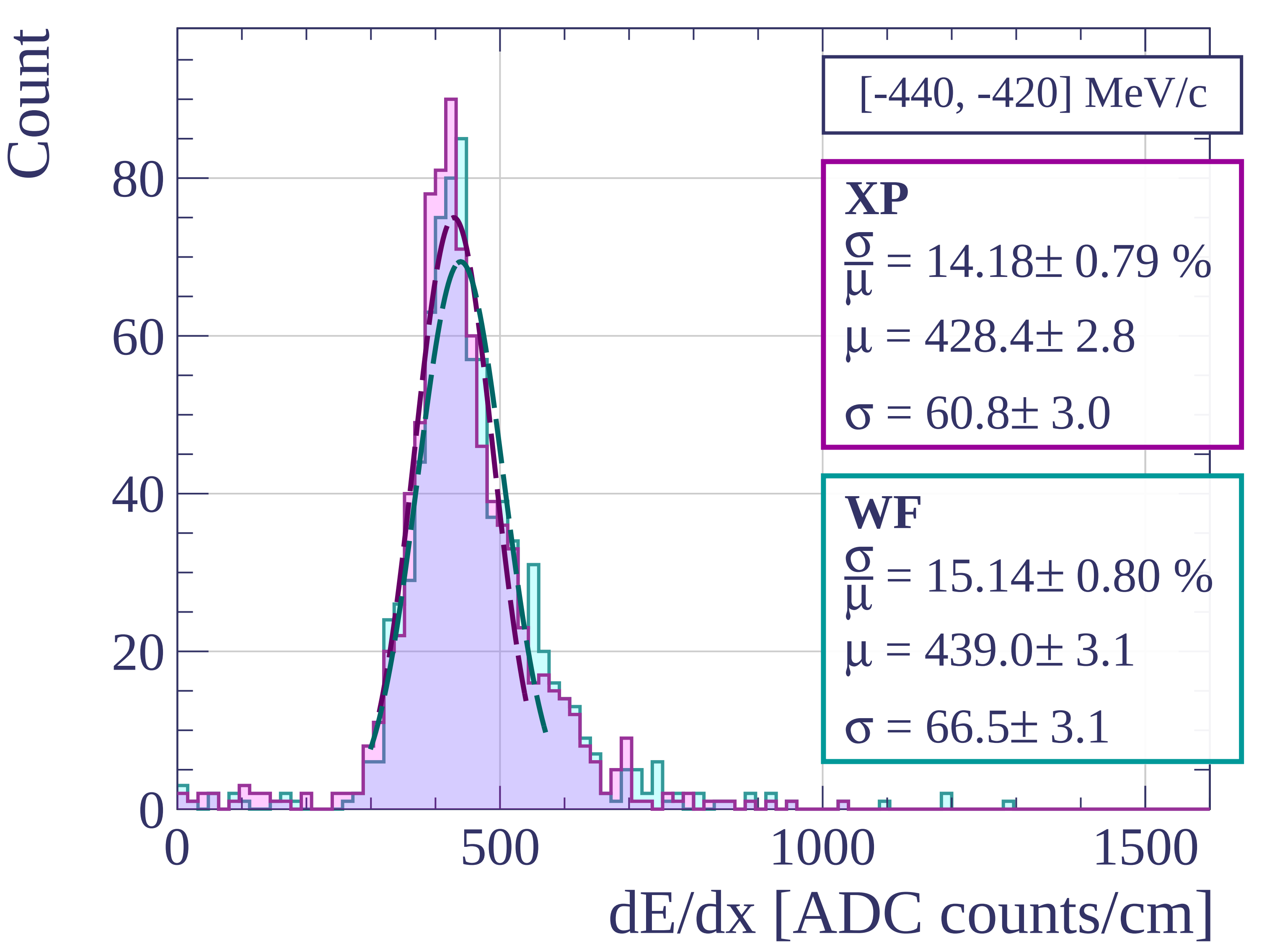
Count
0
20
40
60
80
0
500
1000
1500
dE/dx [ADC counts/cm]
[-500, -480] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.93± 0.64 %
μ = 422.4± 2.7
σ = 54.6± 2.3
WF
$\frac{\sigma}{\mu}$ = 13.72± 0.72 %
μ = 430.1± 3.0
σ = 59.0± 2.7



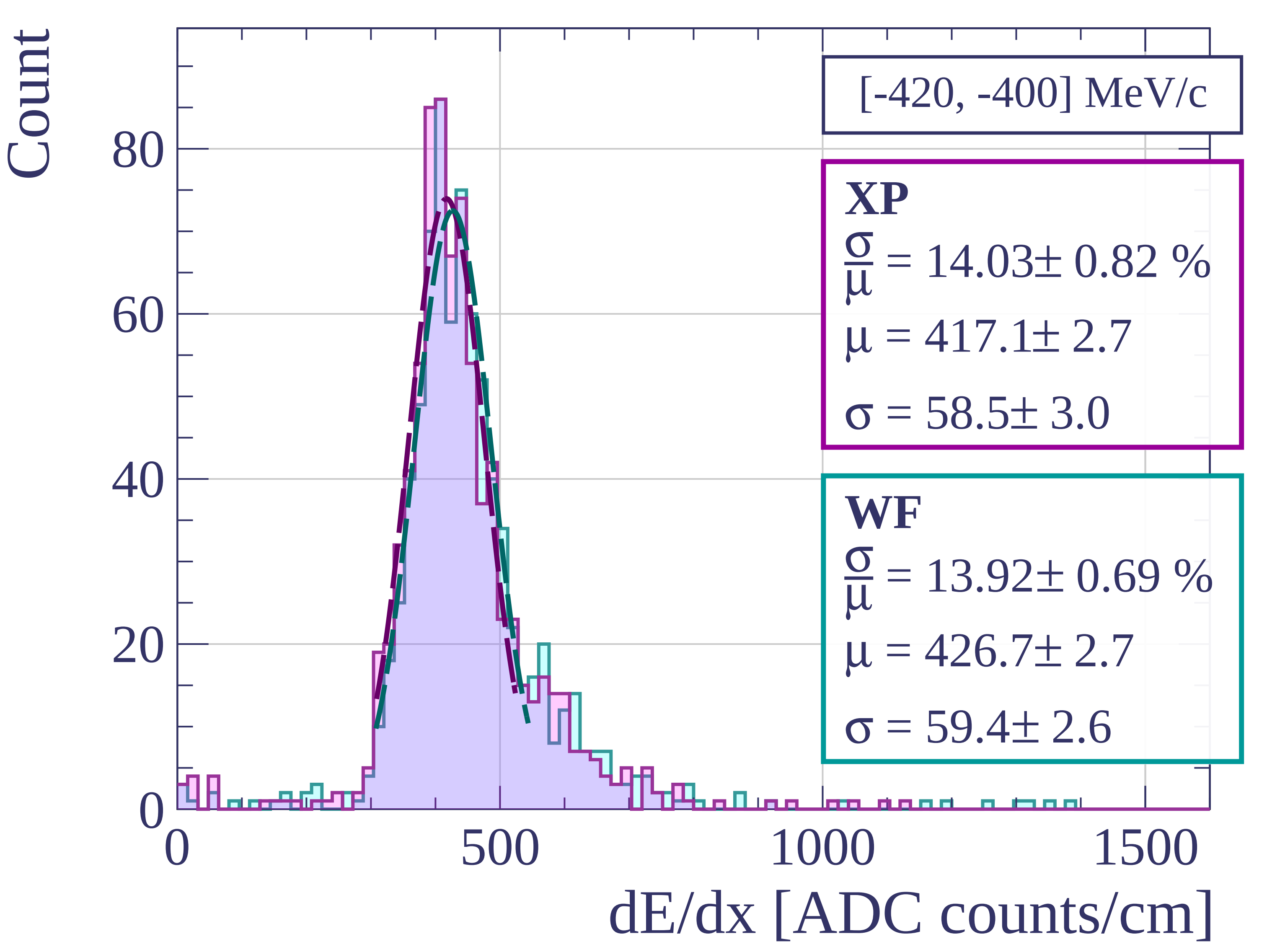
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-480, -460] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.80± 0.80 %
μ = 429.3± 2.8
σ = 59.3± 3.1
WF
$\frac{\sigma}{\mu}$ = 14.64± 0.75 %
μ = 435.9± 2.9
σ = 63.8± 2.8

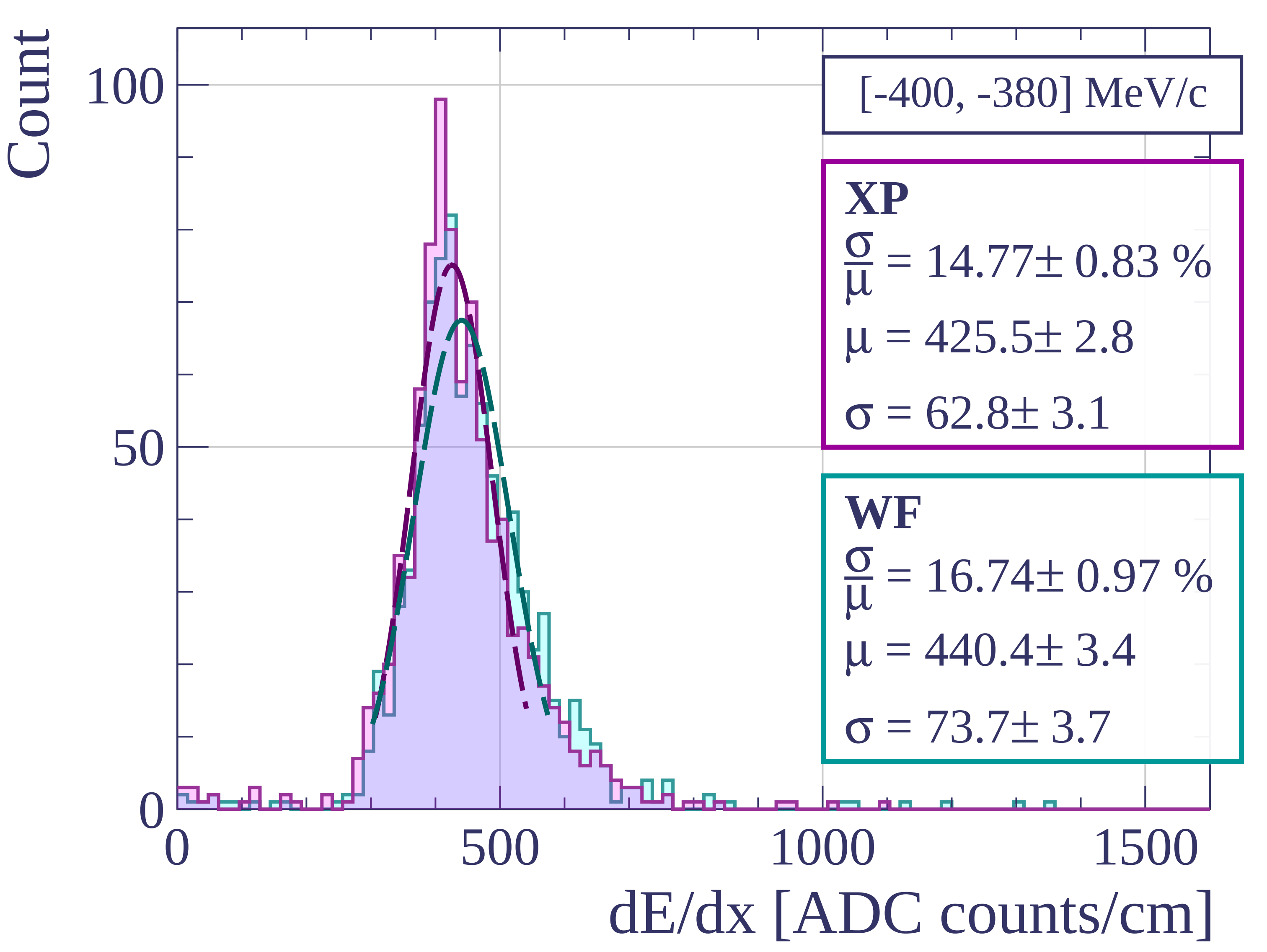


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-460, -440] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.16± 0.69 %
μ = 420.0± 2.6
σ = 55.3± 2.6
WF
$\frac{\sigma}{\mu}$ = 13.62± 0.67 %
μ = 428.2± 2.6
σ = 58.3± 2.5

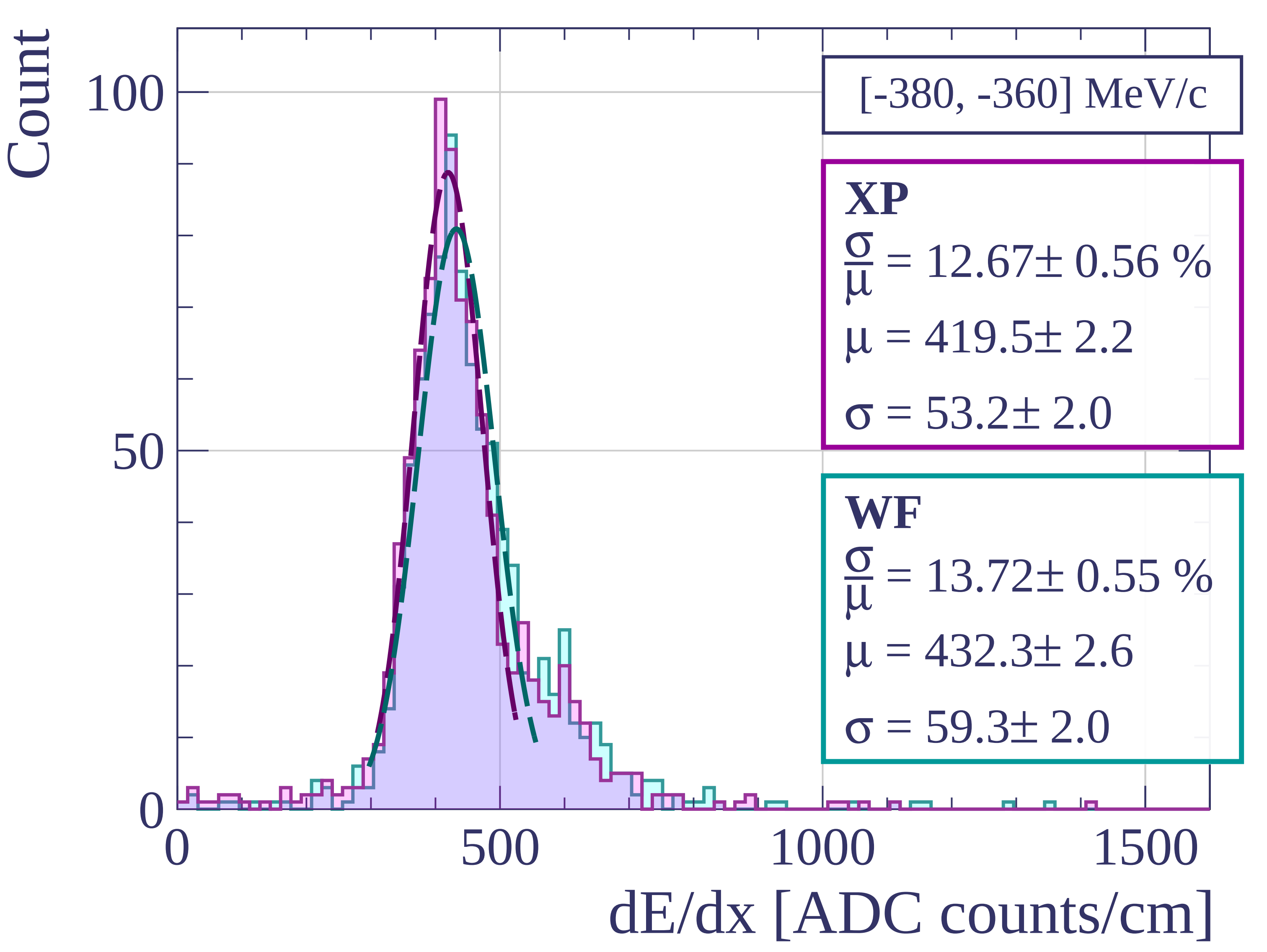


Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-440, -420] MeV/c
XP
$\frac{\sigma}{\mu}$ = 14.18± 0.79 %
μ = 428.4± 2.8
σ = 60.8± 3.0
WF
$\frac{\sigma}{\mu}$ = 15.14± 0.80 %
μ = 439.0± 3.1
σ = 66.5± 3.1

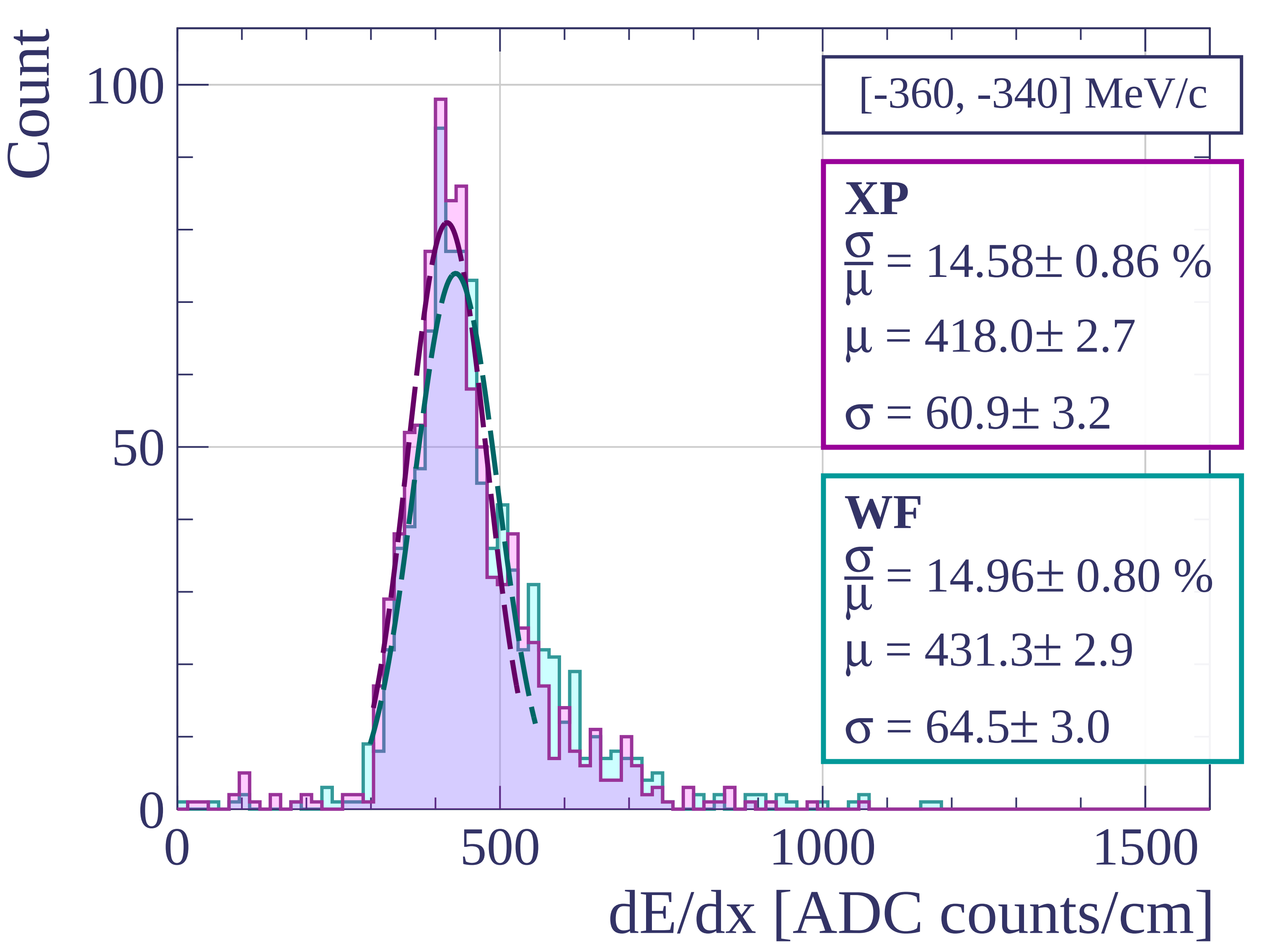

Count
0
20
40
60
80
0
500
1000
1500
dE/dx [ADC counts/cm]
[-420, -400] MeV/c
XP
σ/μ = 14.03± 0.82 %
μ = 417.1± 2.7
σ = 58.5± 3.0
WF
σ/μ = 13.92± 0.69 %
μ = 426.7± 2.7
σ = 59.4± 2.6



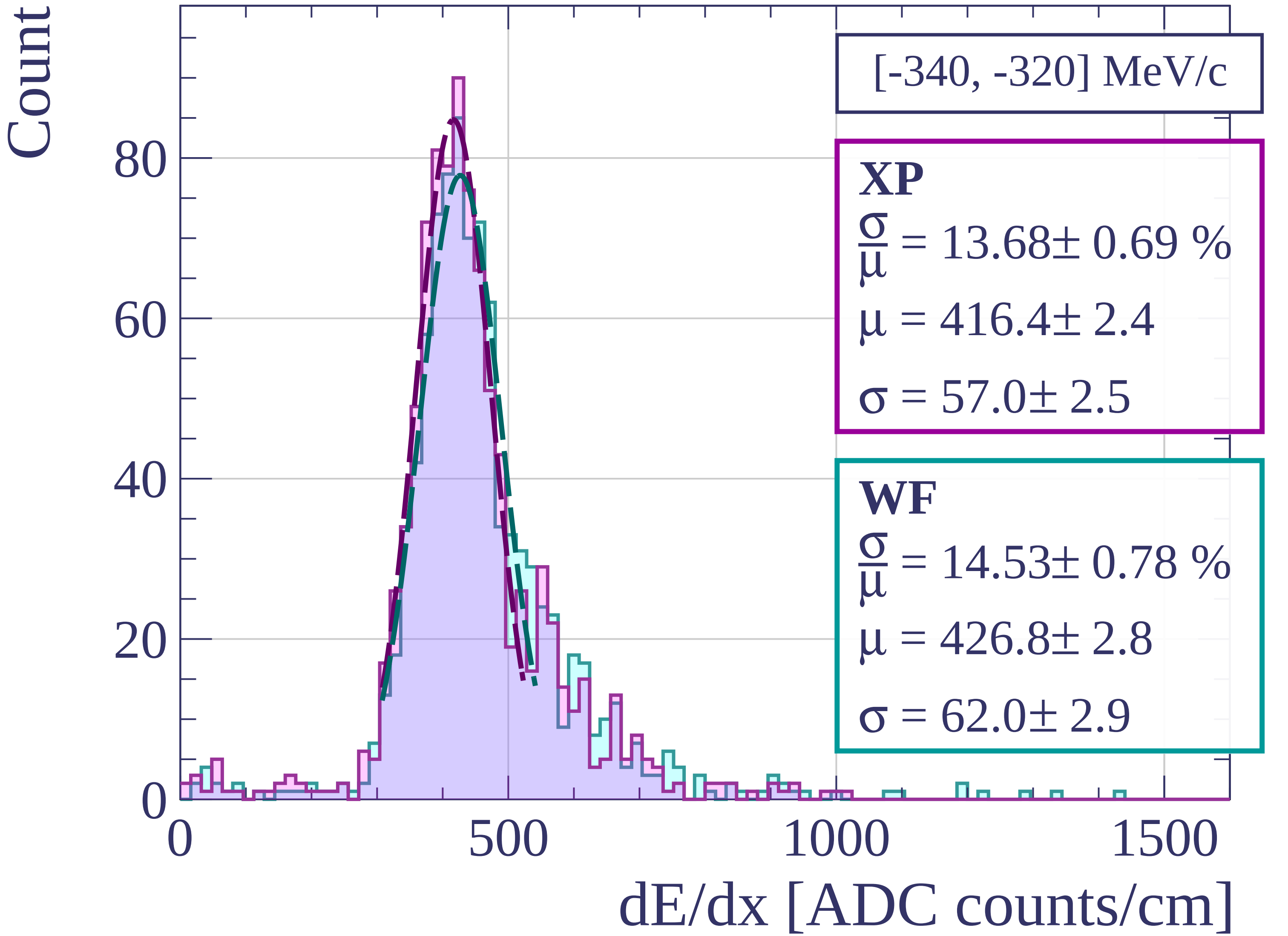
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-400, -380] MeV/c
XP
$\frac{\sigma}{\mu}$ = 14.77± 0.83 %
μ = 425.5± 2.8
σ = 62.8± 3.1
WF
$\frac{\sigma}{\mu}$ = 16.74± 0.97 %
μ = 440.4± 3.4
σ = 73.7± 3.7



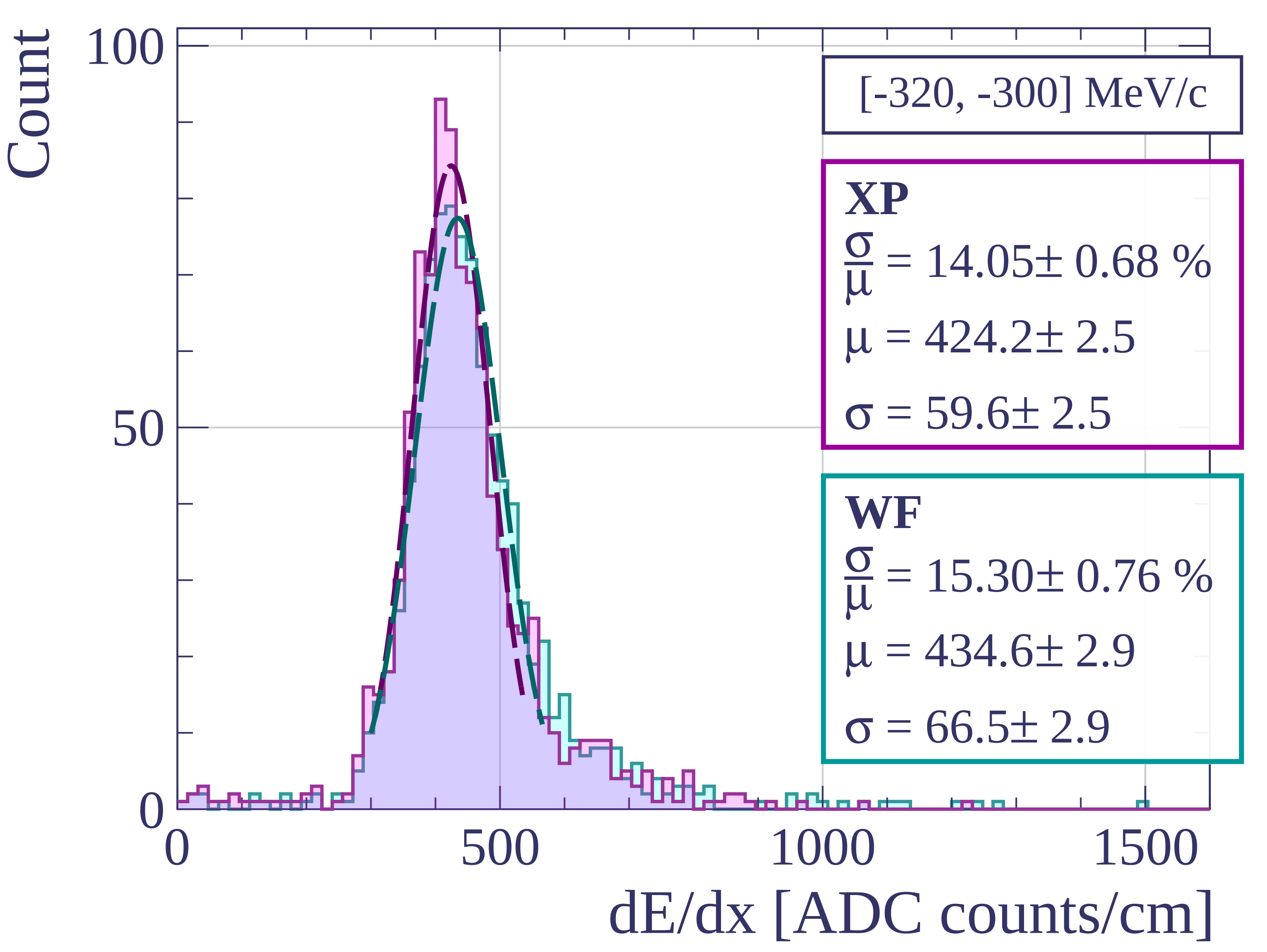
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-380, -360] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.67± 0.56 %
μ = 419.5± 2.2
σ = 53.2± 2.0
WF
$\frac{\sigma}{\mu}$ = 13.72± 0.55 %
μ = 432.3± 2.6
σ = 59.3± 2.0



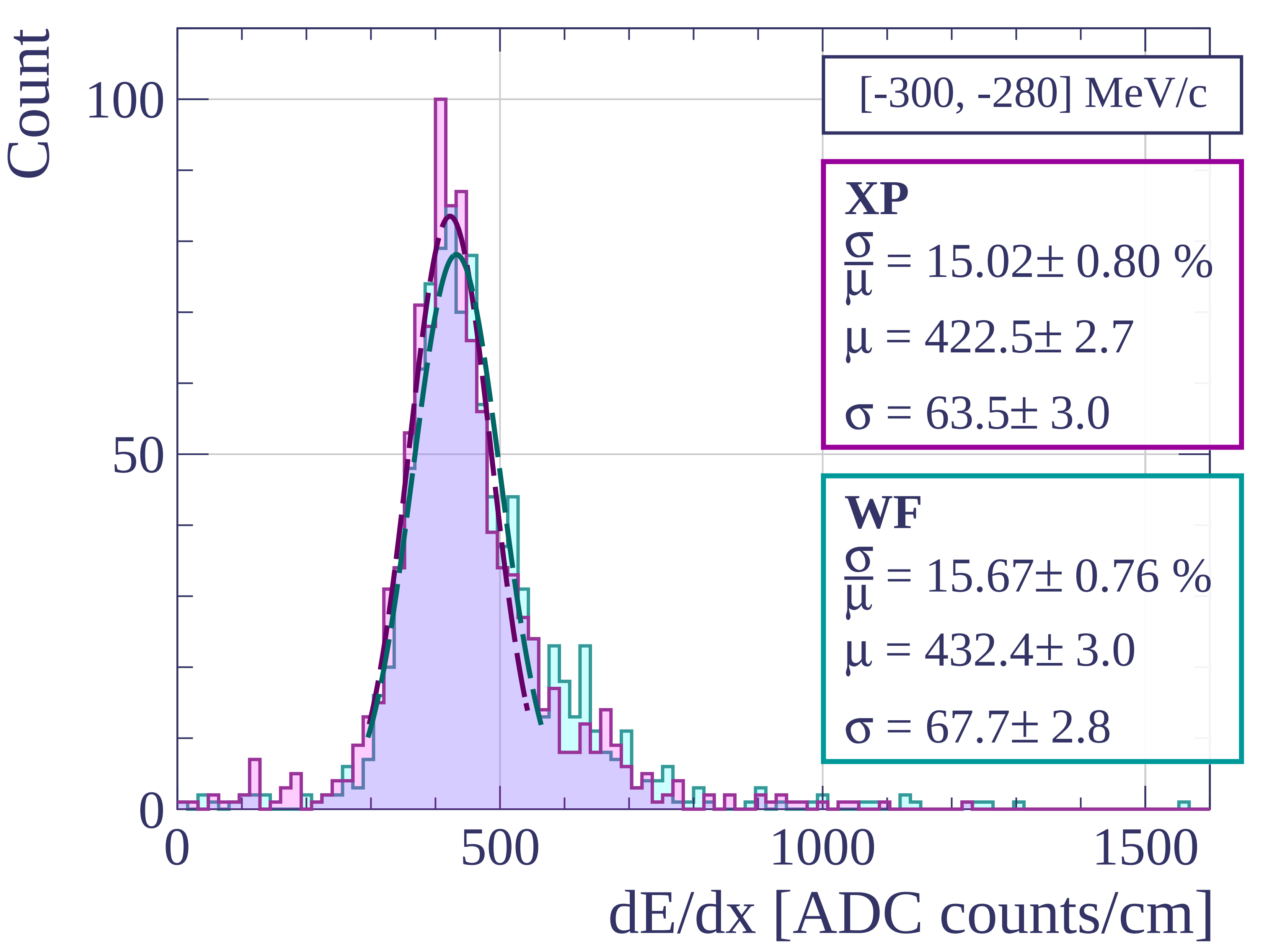
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-360, -340] MeV/c
XP
$\frac{\sigma}{\mu}$ = 14.58± 0.86 %
μ = 418.0± 2.7
σ = 60.9± 3.2
WF
$\frac{\sigma}{\mu}$ = 14.96± 0.80 %
μ = 431.3± 2.9
σ = 64.5± 3.0



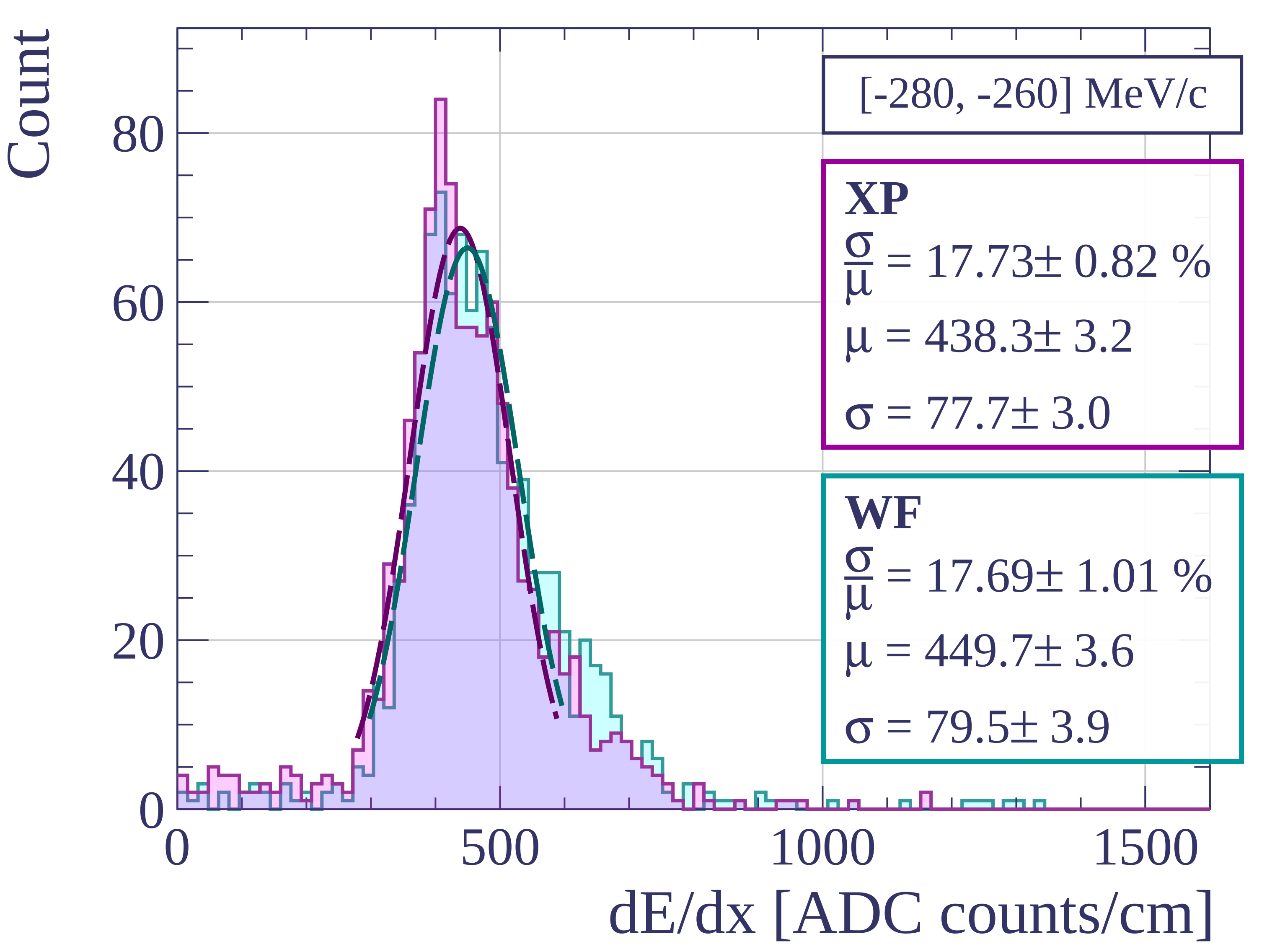
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-340, -320] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.68± 0.69 %
μ = 416.4± 2.4
σ = 57.0± 2.5
WF
$\frac{\sigma}{\mu}$ = 14.53± 0.78 %
μ = 426.8± 2.8
σ = 62.0± 2.9



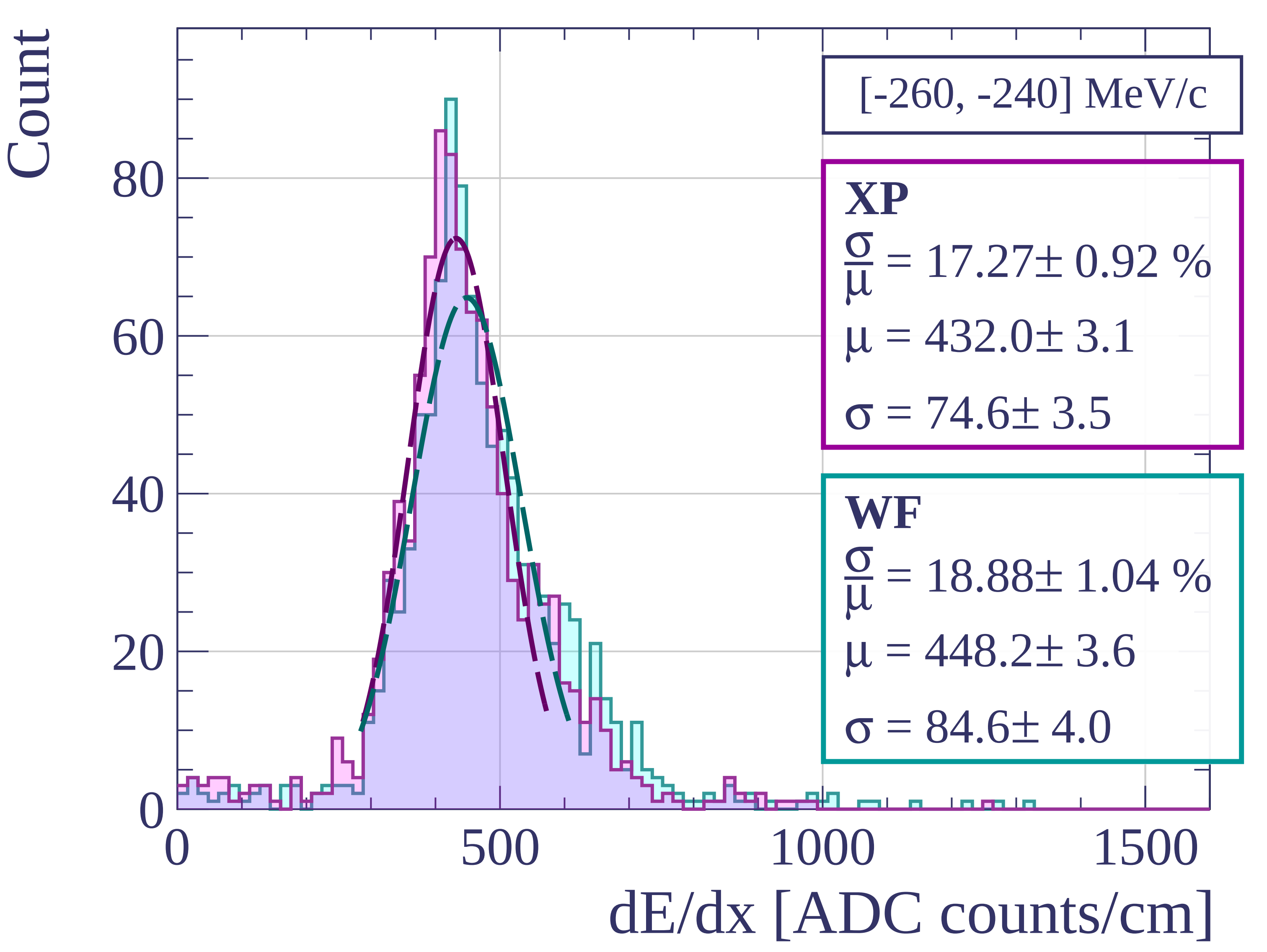
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-320, -300] MeV/c
XP
σ/μ = 14.05± 0.68 %
μ = 424.2± 2.5
σ = 59.6± 2.5
WF
σ/μ = 15.30± 0.76 %
μ = 434.6± 2.9
σ = 66.5± 2.9



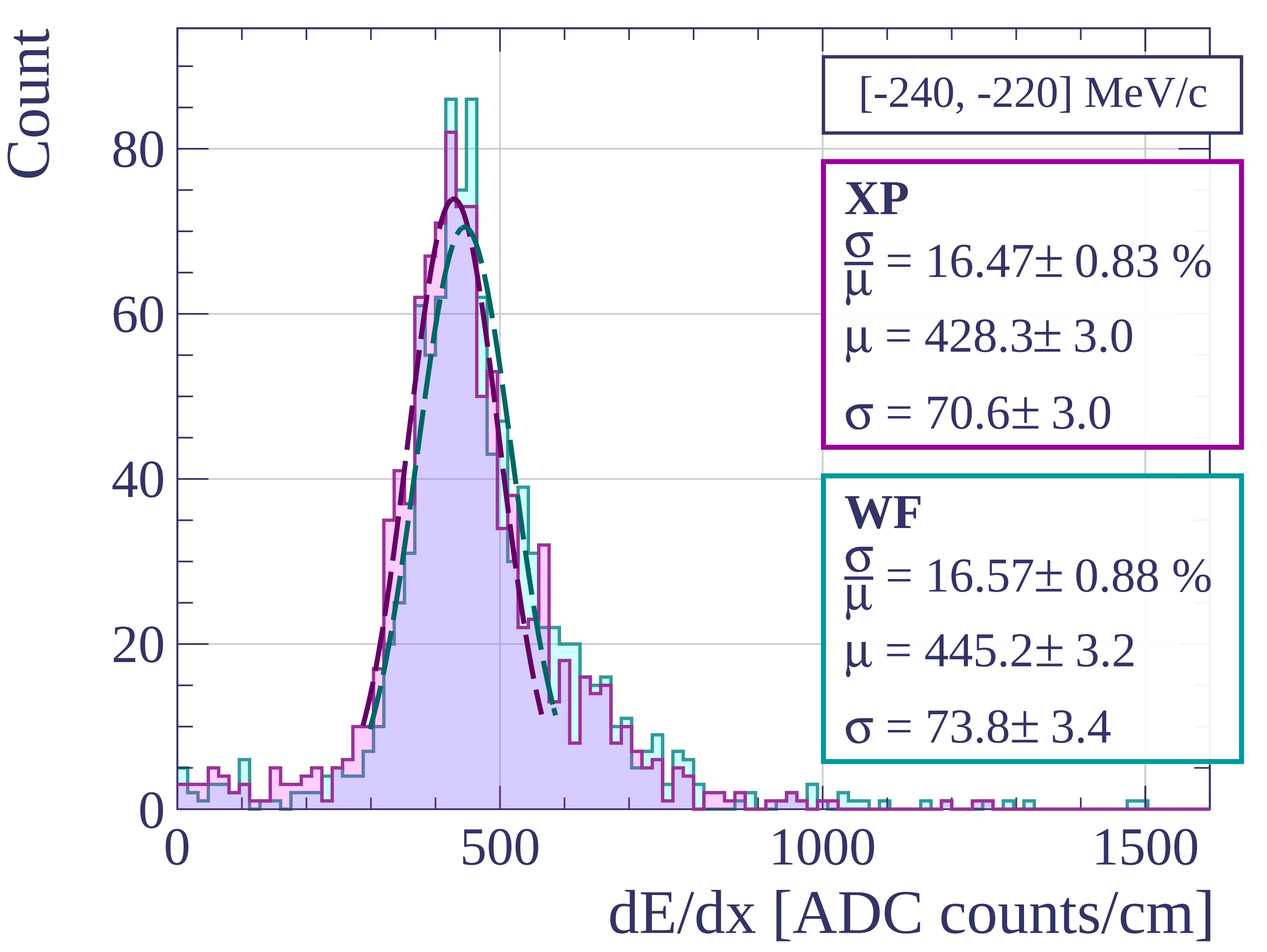
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-300, -280] MeV/c
XP
σ/μ = 15.02± 0.80 %
μ = 422.5± 2.7
σ = 63.5± 3.0
WF
σ/μ = 15.67± 0.76 %
μ = 432.4± 3.0
σ = 67.7± 2.8



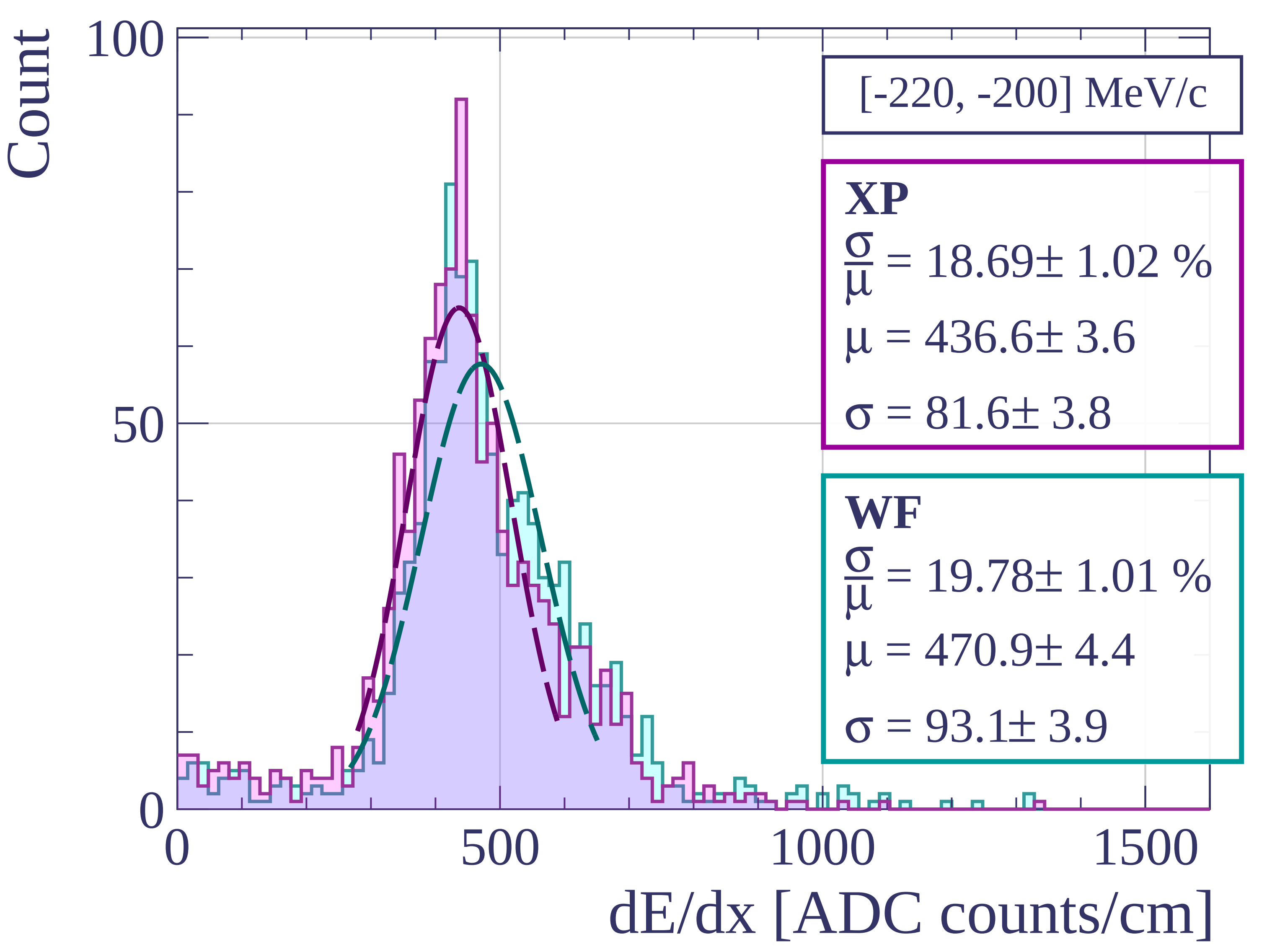
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-280, -260] MeV/c
XP
$\frac{\sigma}{\mu}$ = 17.73± 0.82 %
μ = 438.3± 3.2
σ = 77.7± 3.0
WF
$\frac{\sigma}{\mu}$ = 17.69± 1.01 %
μ = 449.7± 3.6
σ = 79.5± 3.9

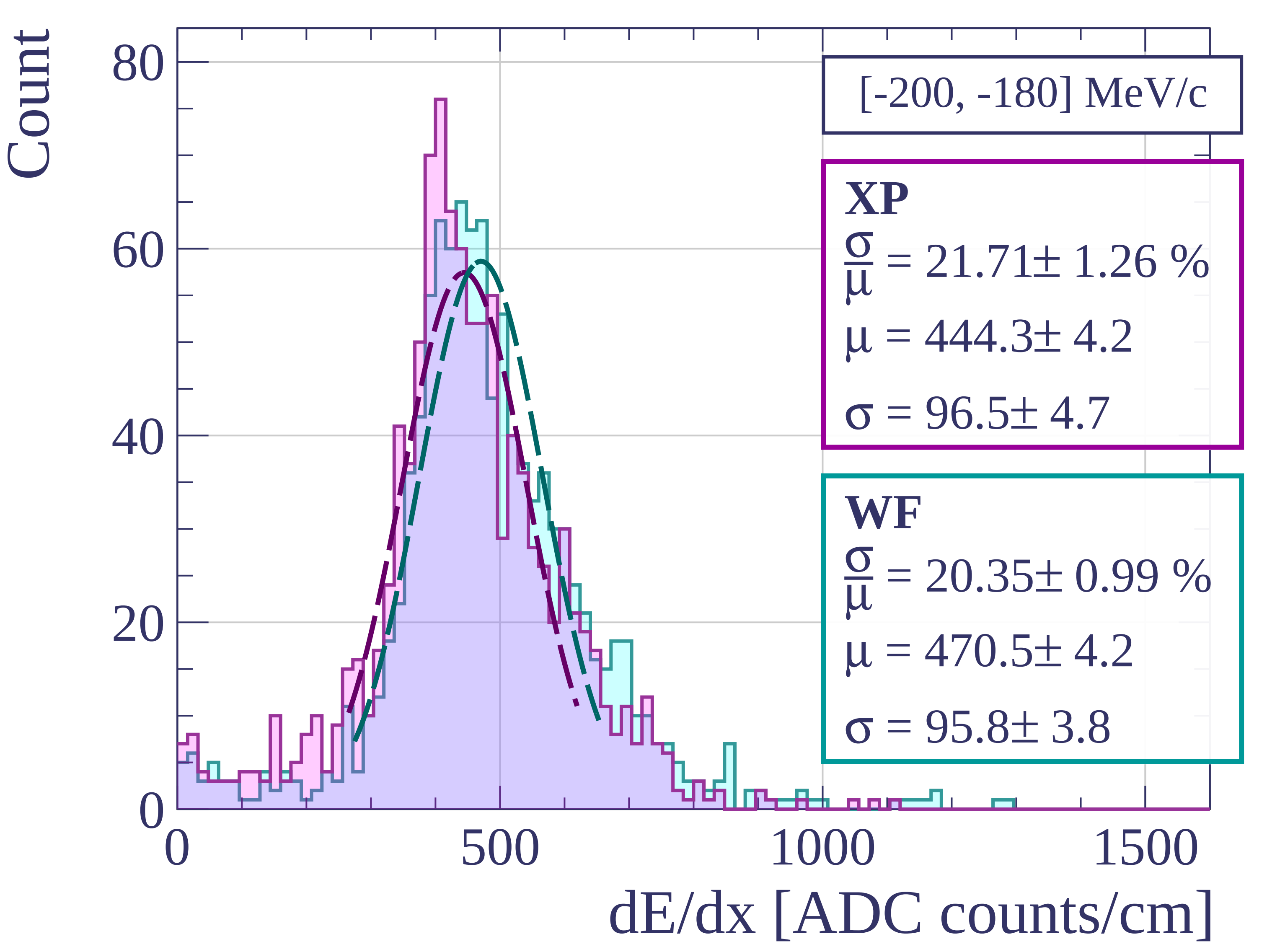


Count
dE/dx [ADC counts/cm]
0
20
40
60
80
0
500
1000
1500
[-260, -240] MeV/c
XP
$\frac{\sigma}{\mu}$ = 17.27± 0.92 %
μ = 432.0± 3.1
σ = 74.6± 3.5
WF
$\frac{\sigma}{\mu}$ = 18.88± 1.04 %
μ = 448.2± 3.6
σ = 84.6± 4.0

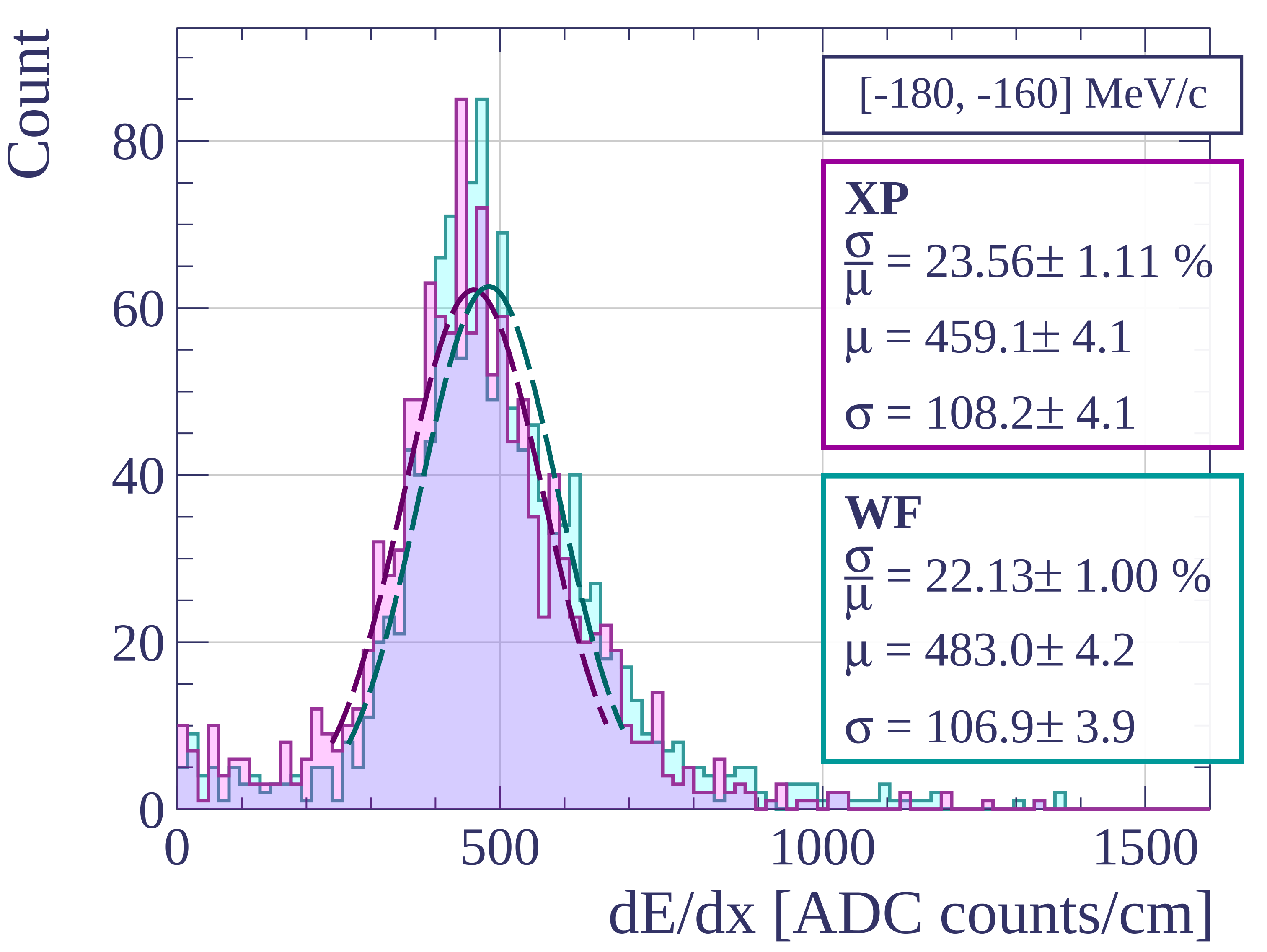


Count
dE/dx [ADC counts/cm]
0
20
40
60
80
0
500
1000
1500
[-240, -220] MeV/c
XP
σ/μ = 16.47± 0.83 %
μ = 428.3± 3.0
σ = 70.6± 3.0
WF
σ/μ = 16.57± 0.88 %
μ = 445.2± 3.2
σ = 73.8± 3.4

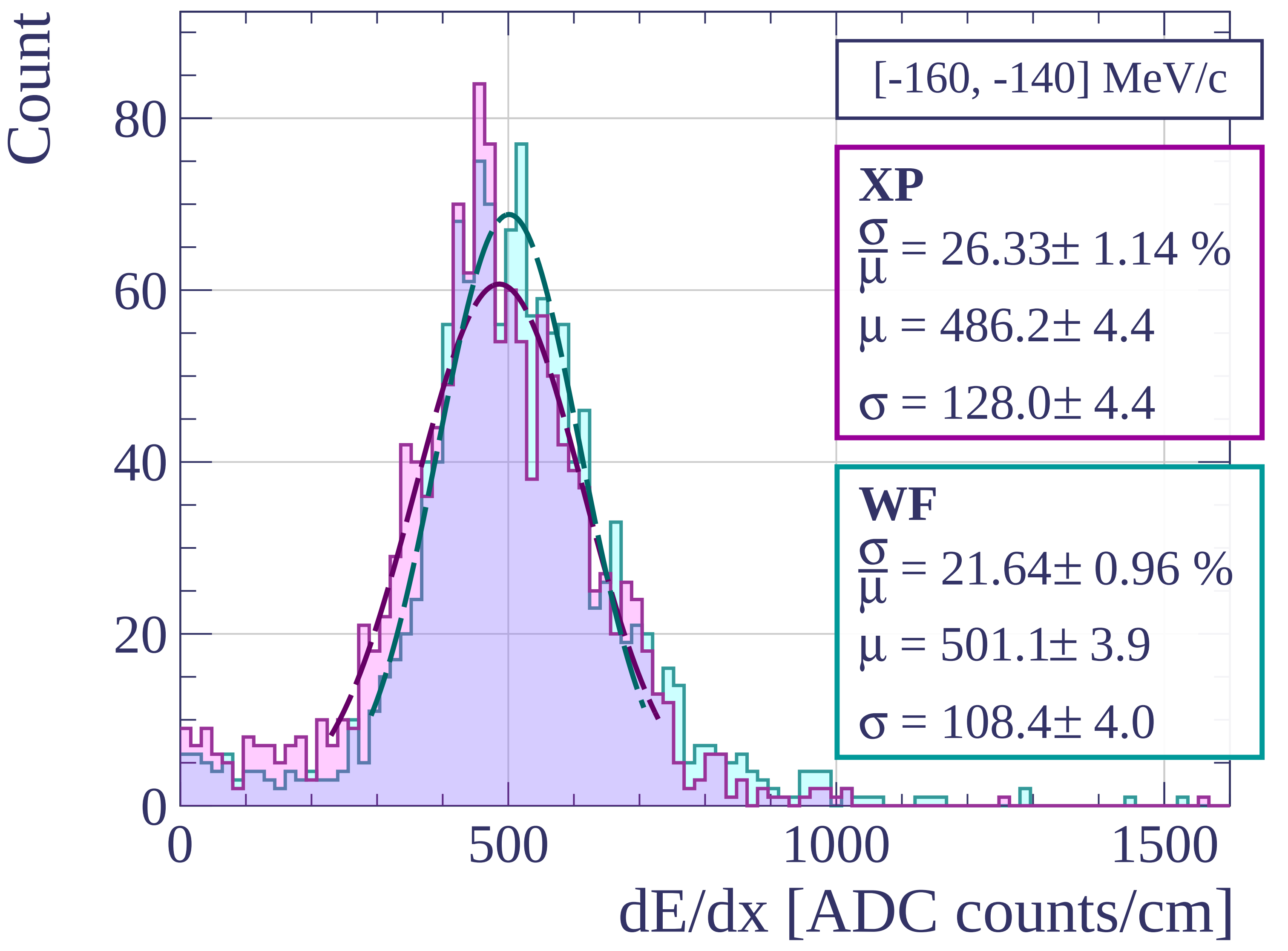

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-220, -200] MeV/c
XP
$\frac{\sigma}{\mu}$ = 18.69± 1.02 %
μ = 436.6± 3.6
σ = 81.6± 3.8
WF
$\frac{\sigma}{\mu}$ = 19.78± 1.01 %
μ = 470.9± 4.4
σ = 93.1± 3.9

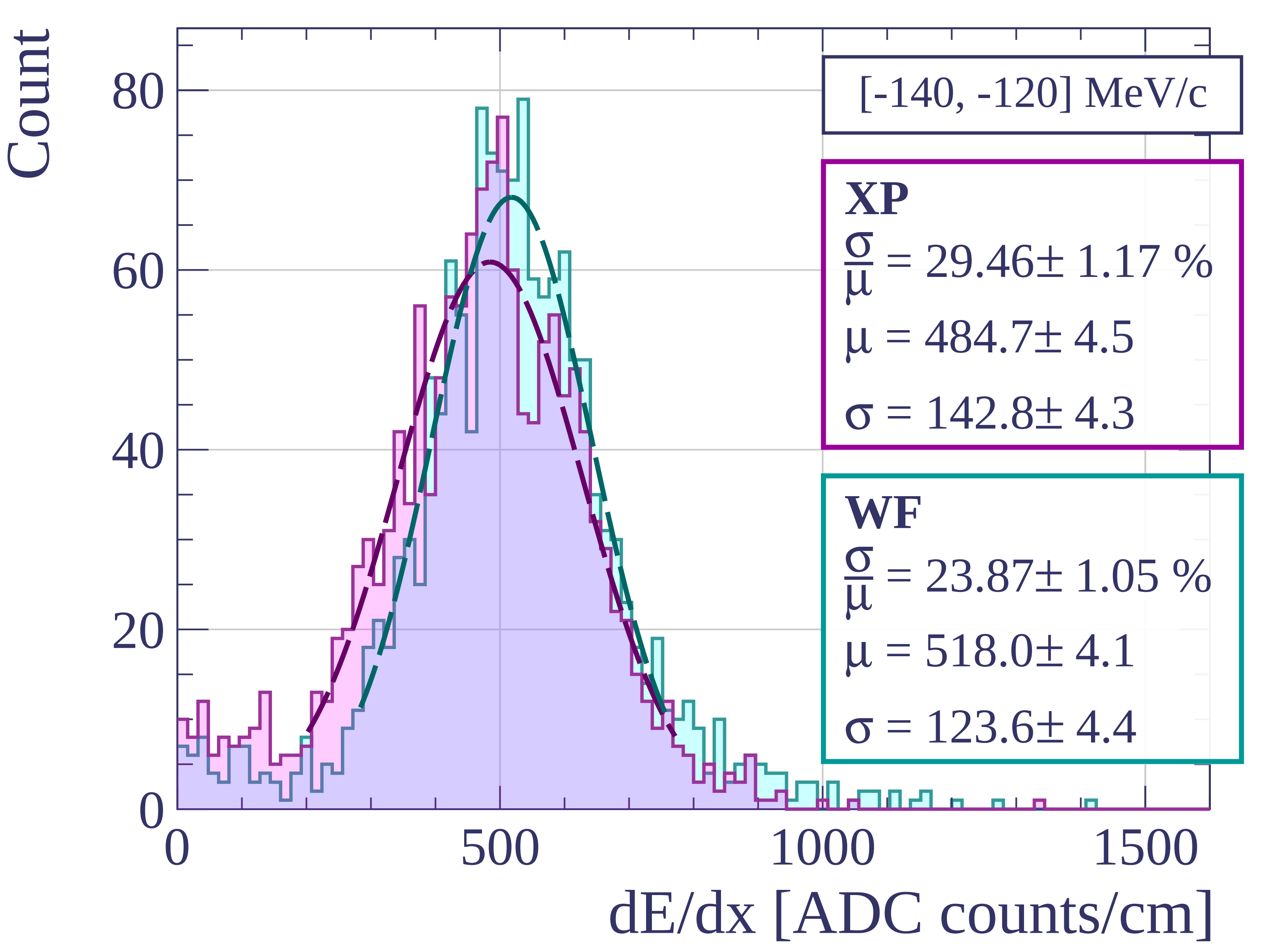


Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-200, -180] MeV/c
XP
σ/μ = 21.71± 1.26 %
μ = 444.3± 4.2
σ = 96.5± 4.7
WF
σ/μ = 20.35± 0.99 %
μ = 470.5± 4.2
σ = 95.8± 3.8

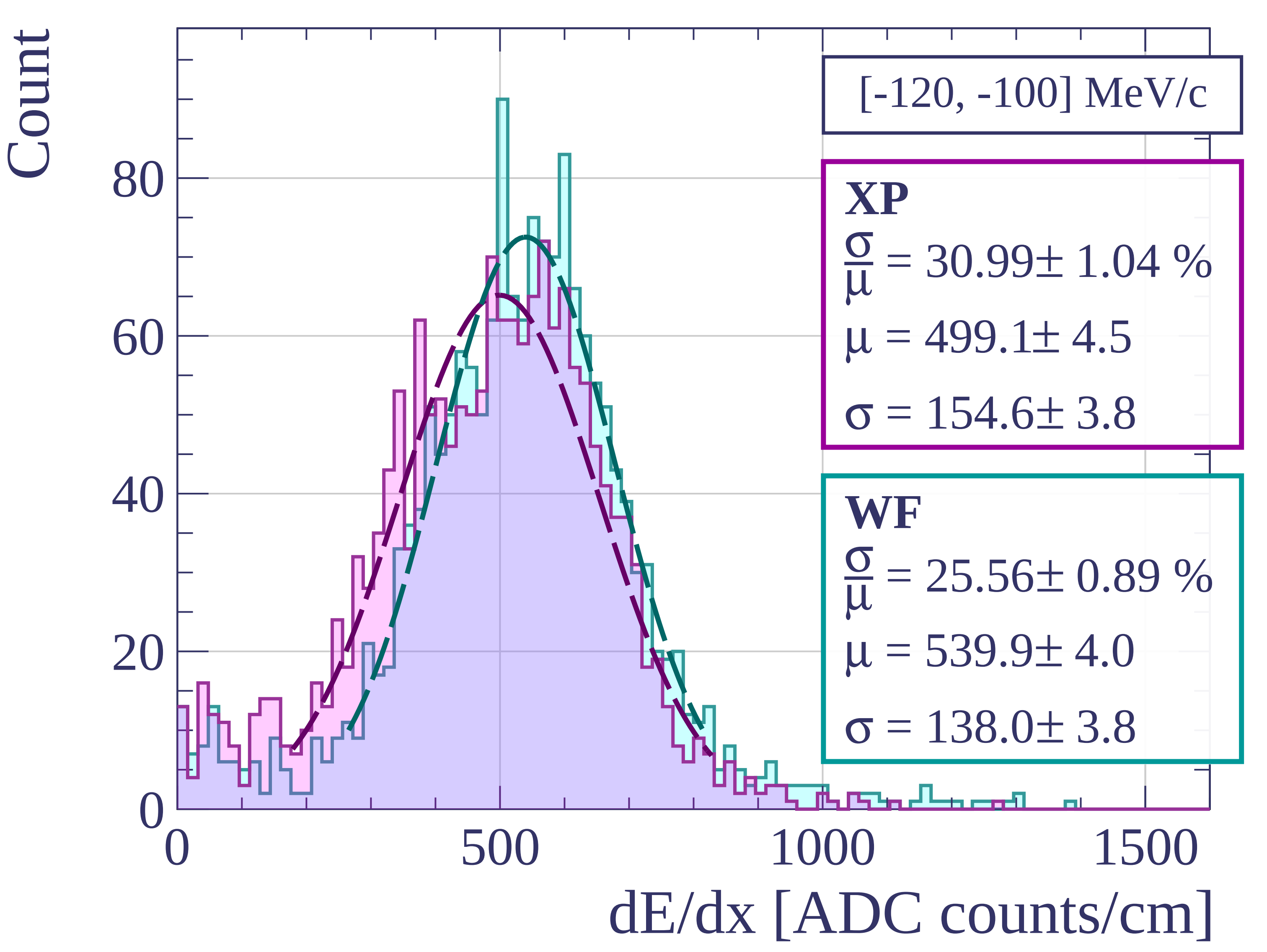


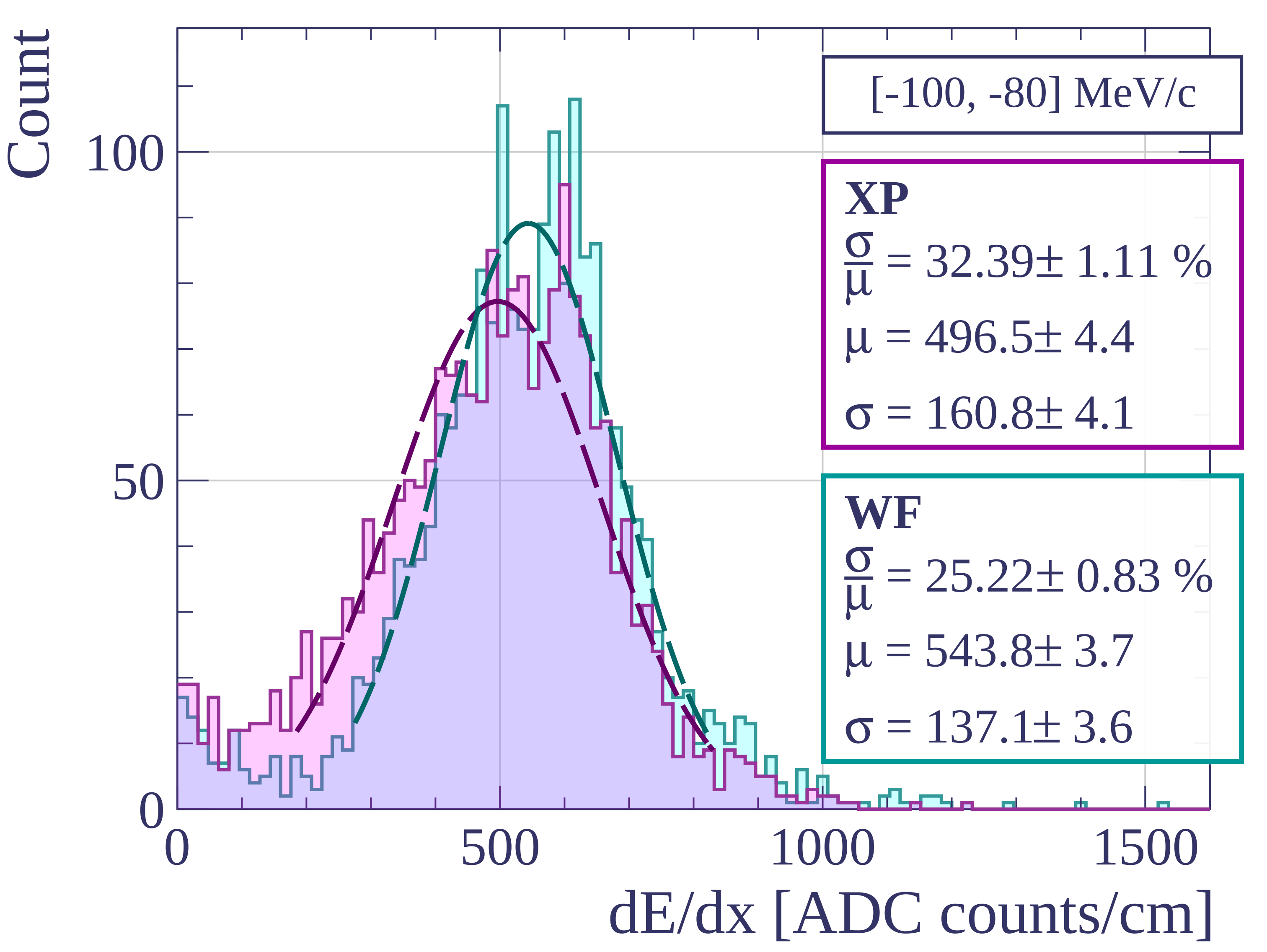
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-180, -160] MeV/c
XP
$\frac{\sigma}{\mu}$ = 23.56± 1.11 %
μ = 459.1± 4.1
σ = 108.2± 4.1
WF
$\frac{\sigma}{\mu}$ = 22.13± 1.00 %
μ = 483.0± 4.2
σ = 106.9± 3.9

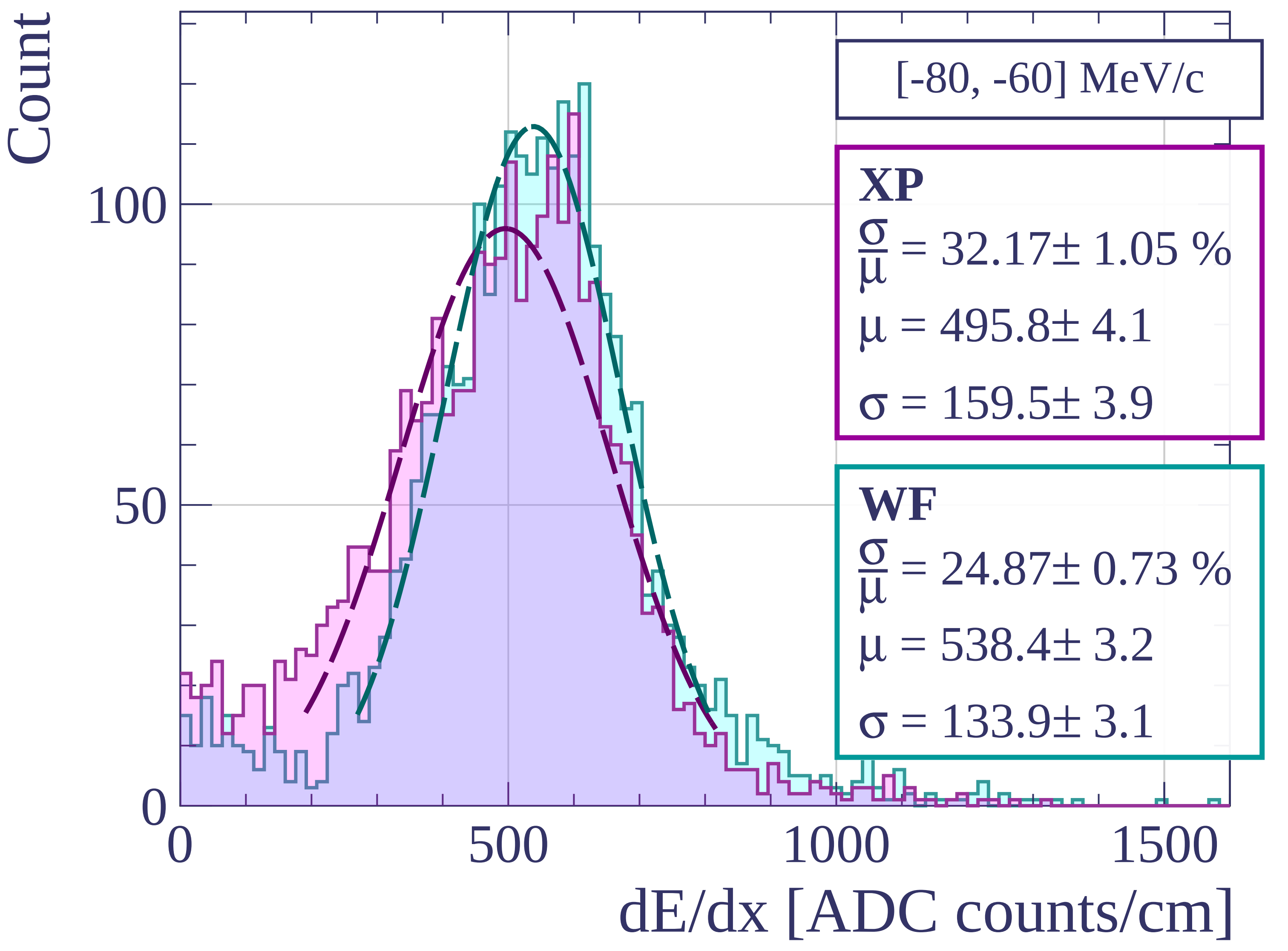

Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-160, -140] MeV/c
XP
σ/μ = 26.33± 1.14 %
μ = 486.2± 4.4
σ = 128.0± 4.4
WF
σ/μ = 21.64± 0.96 %
μ = 501.1± 3.9
σ = 108.4± 4.0

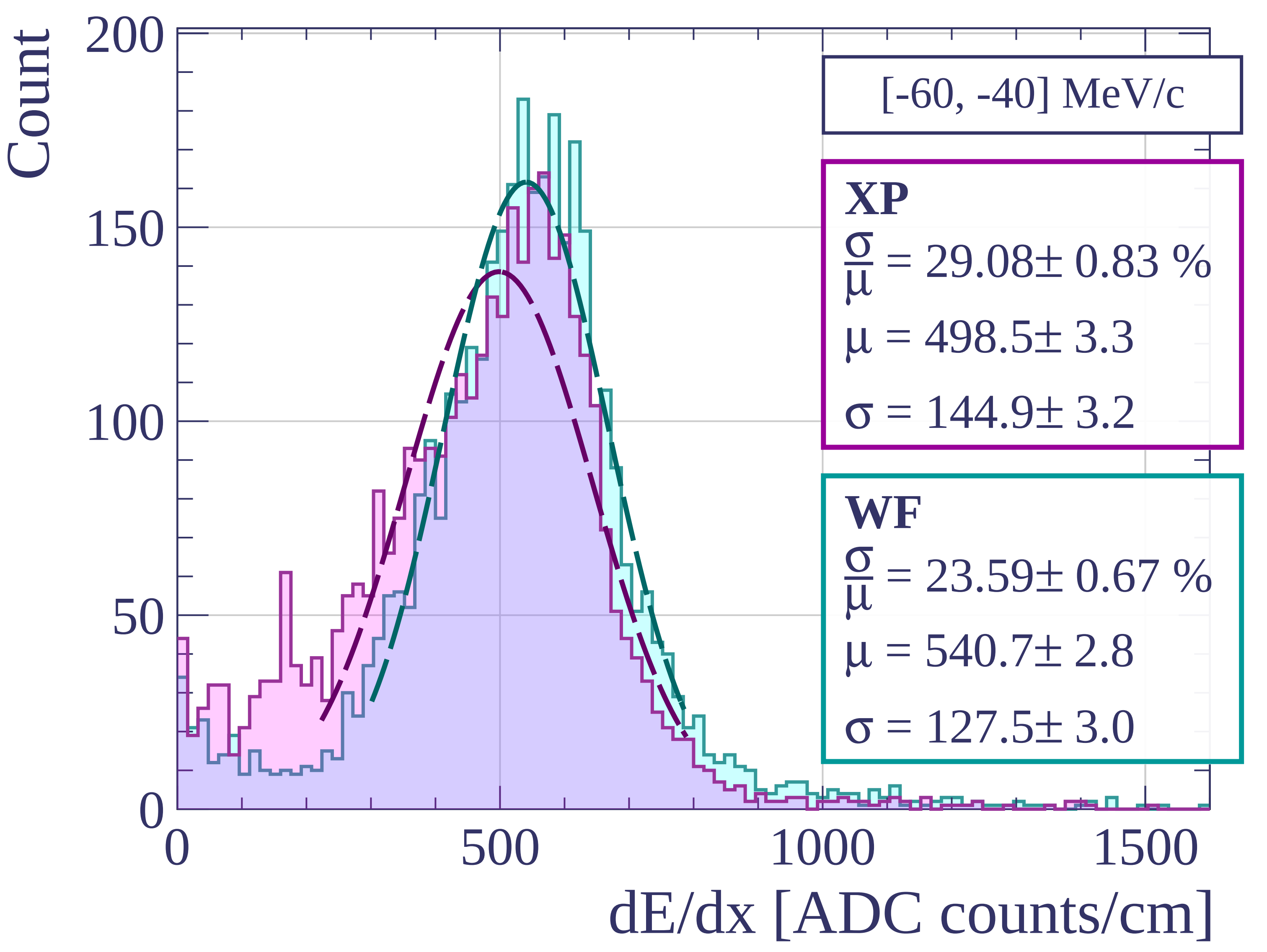


Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-140, -120] MeV/c
XP
$\frac{\sigma}{\mu}$ = 29.46± 1.17 %
μ = 484.7± 4.5
σ = 142.8± 4.3
WF
$\frac{\sigma}{\mu}$ = 23.87± 1.05 %
μ = 518.0± 4.1
σ = 123.6± 4.4

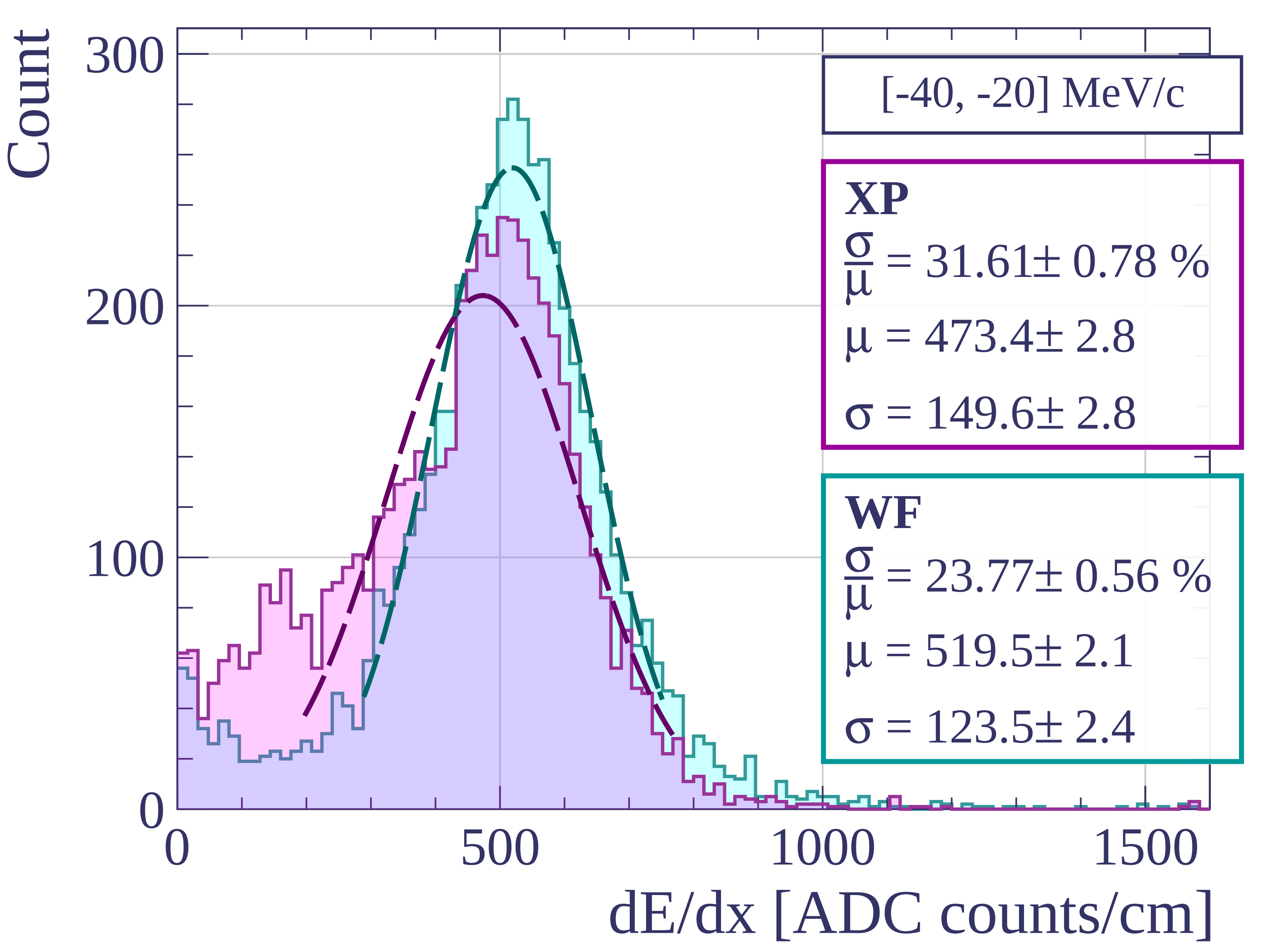

Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-120, -100] MeV/c
XP
σ/μ = 30.99± 1.04 %
μ = 499.1± 4.5
σ = 154.6± 3.8
WF
σ/μ = 25.56± 0.89 %
μ = 539.9± 4.0
σ = 138.0± 3.8


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-100, -80] MeV/c
XP
$\frac{\sigma}{\mu}$ = 32.39± 1.11 %
μ = 496.5± 4.4
σ = 160.8± 4.1
WF
$\frac{\sigma}{\mu}$ = 25.22± 0.83 %
μ = 543.8± 3.7
σ = 137.1± 3.6

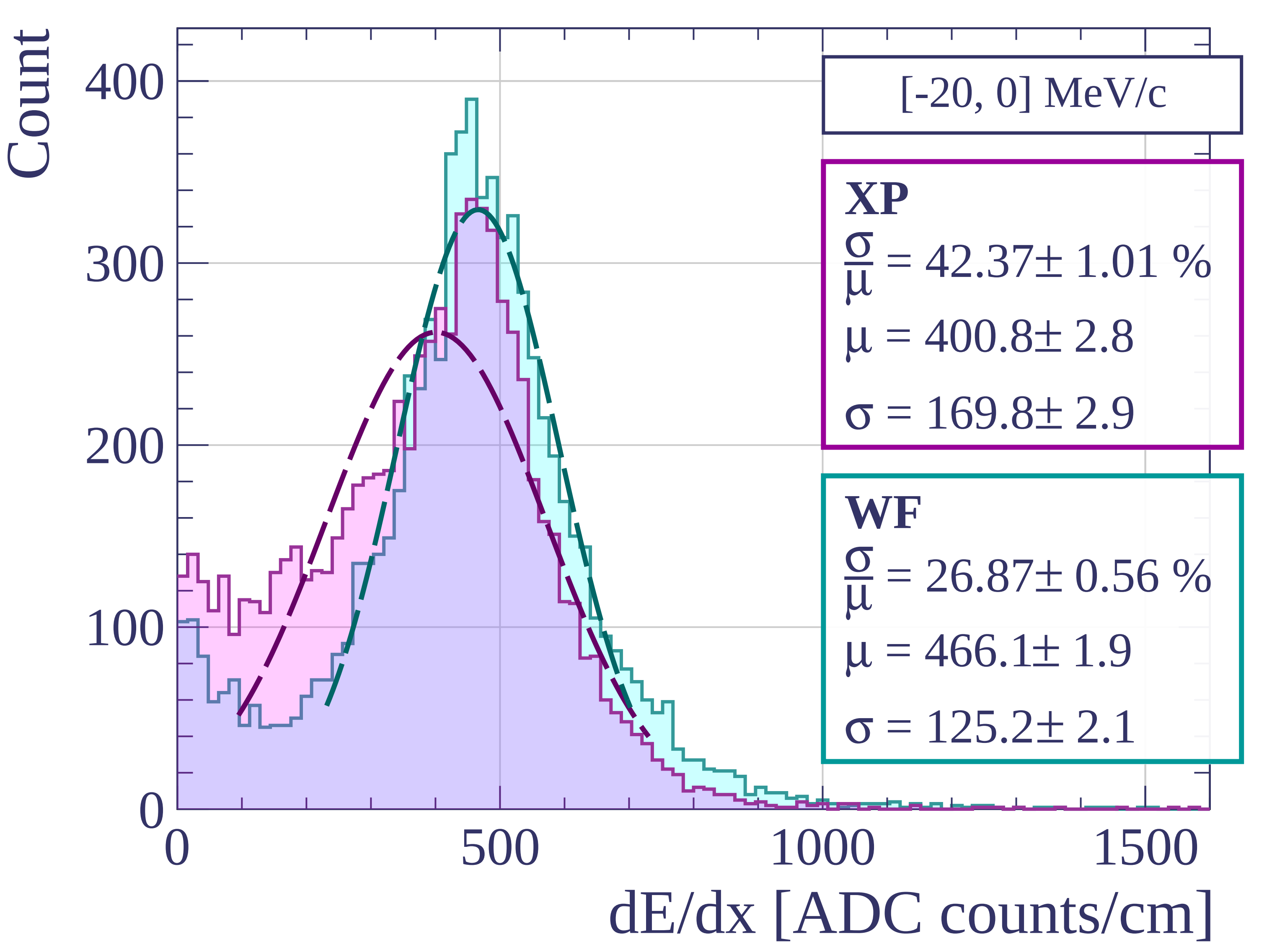

Count
0
50
100
0
500
1000
1500
dE/dx [ADC counts/cm]
[-80, -60] MeV/c
XP
$\frac{\sigma}{\mu}$ = 32.17± 1.05 %
μ = 495.8± 4.1
σ = 159.5± 3.9
WF
$\frac{\sigma}{\mu}$ = 24.87± 0.73 %
μ = 538.4± 3.2
σ = 133.9± 3.1



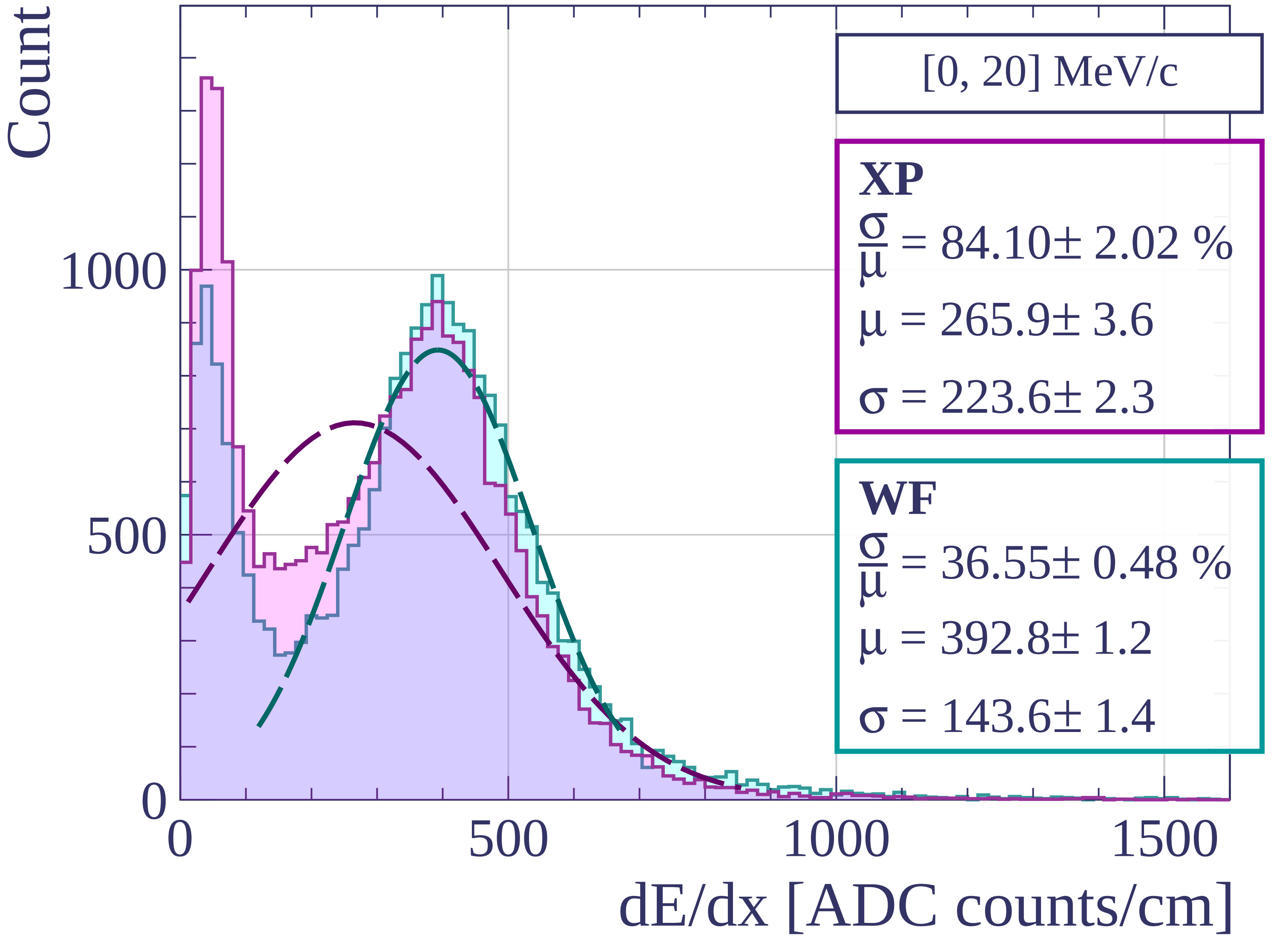
Count
200
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-60, -40] MeV/c
XP
σ/μ = 29.08± 0.83 %
μ = 498.5± 3.3
σ = 144.9± 3.2
WF
σ/μ = 23.59± 0.67 %
μ = 540.7± 2.8
σ = 127.5± 3.0



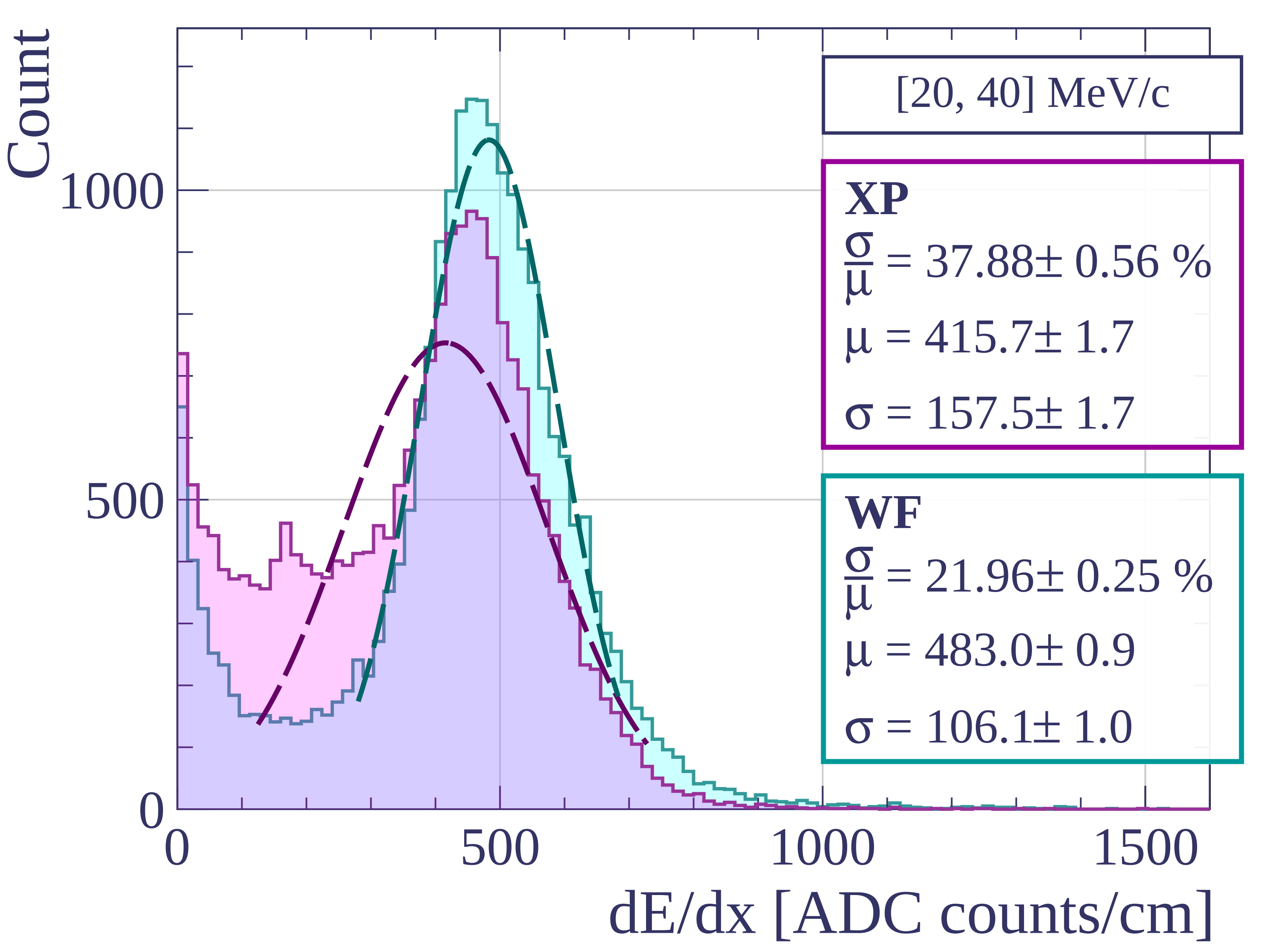
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-40, -20] MeV/c
XP
σ/μ = 31.61± 0.78 %
μ = 473.4± 2.8
σ = 149.6± 2.8
WF
σ/μ = 23.77± 0.56 %
μ = 519.5± 2.1
σ = 123.5± 2.4



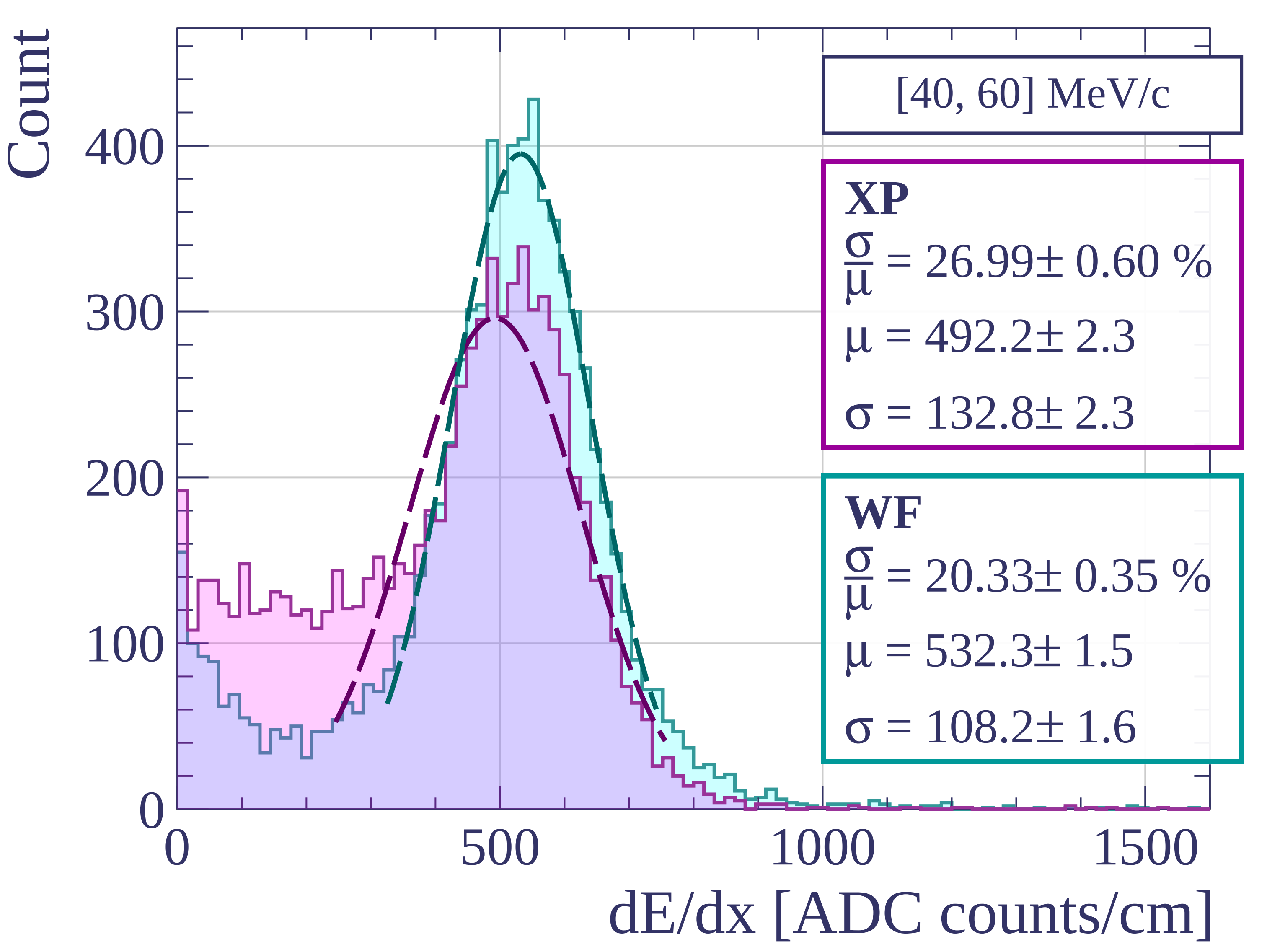
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[-20, 0] MeV/c
XP
$\frac{\sigma}{\mu}$ = 42.37± 1.01 %
μ = 400.8± 2.8
σ = 169.8± 2.9
WF
$\frac{\sigma}{\mu}$ = 26.87± 0.56 %
μ = 466.1± 1.9
σ = 125.2± 2.1



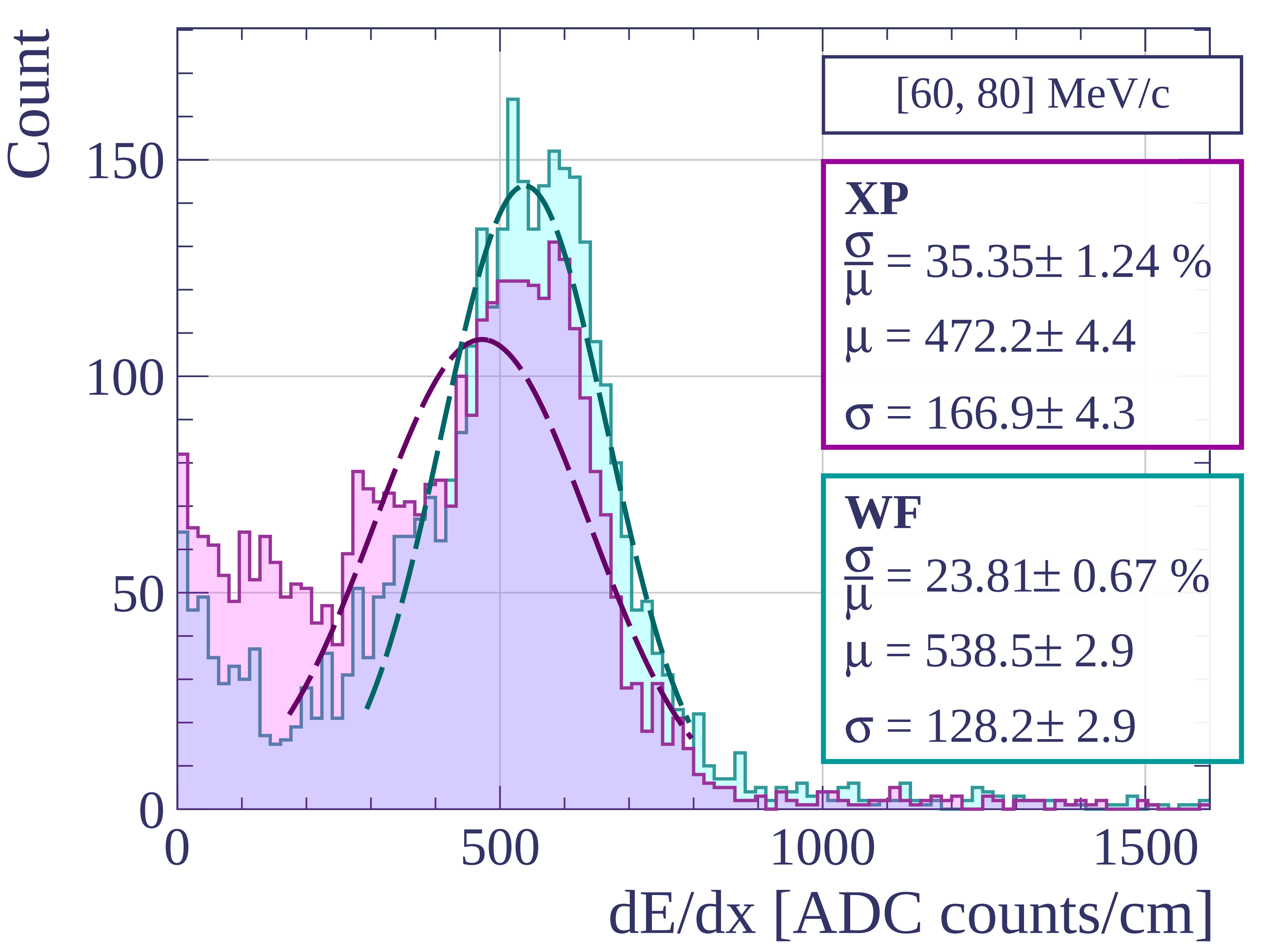
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[0, 20] MeV/c
XP
σ/μ = 84.10± 2.02 %
μ = 265.9± 3.6
σ = 223.6± 2.3
WF
σ/μ = 36.55± 0.48 %
μ = 392.8± 1.2
σ = 143.6± 1.4



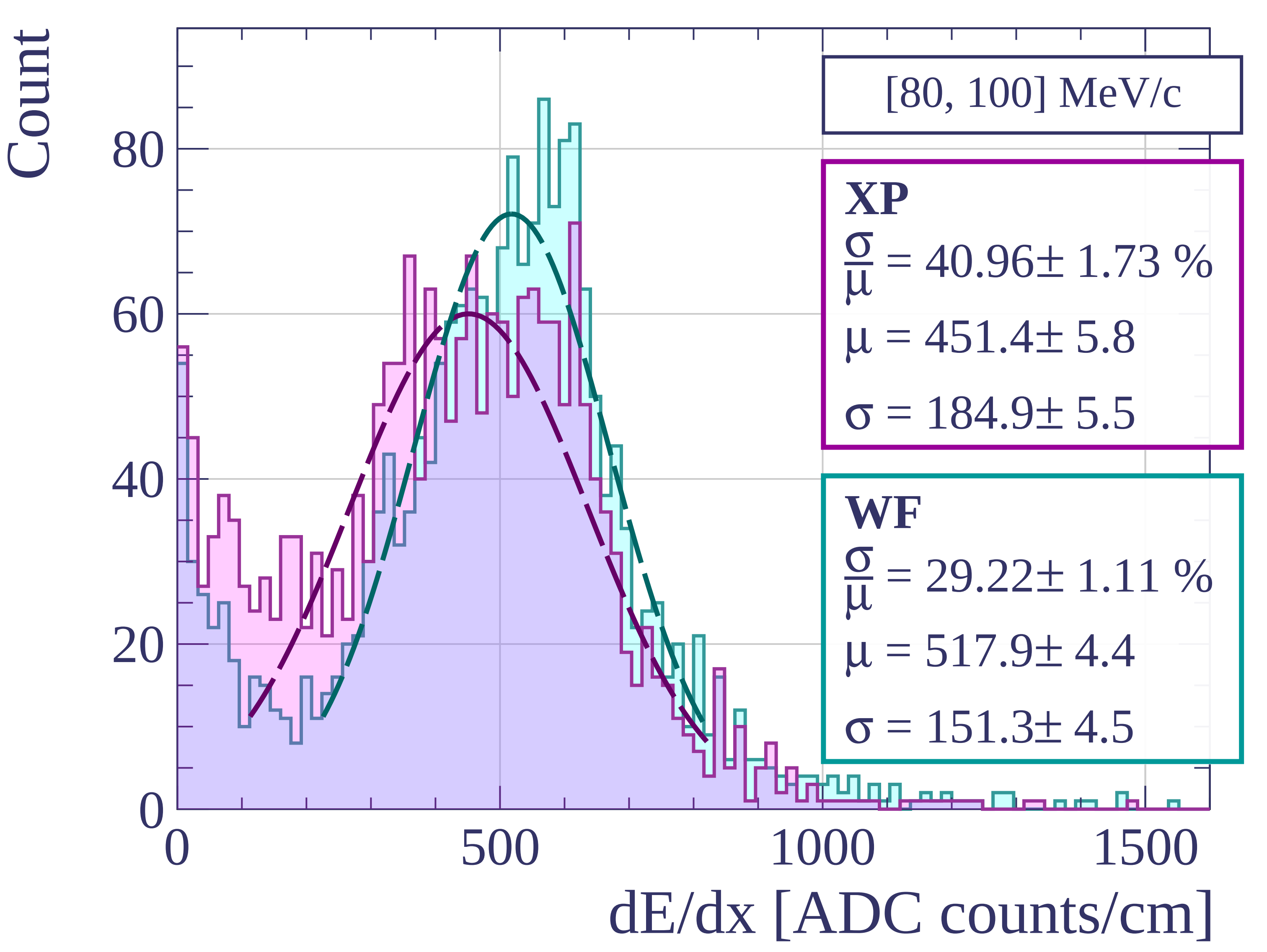
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[20, 40] MeV/c
XP
σ/μ = 37.88± 0.56 %
μ = 415.7± 1.7
σ = 157.5± 1.7
WF
σ/μ = 21.96± 0.25 %
μ = 483.0± 0.9
σ = 106.1± 1.0



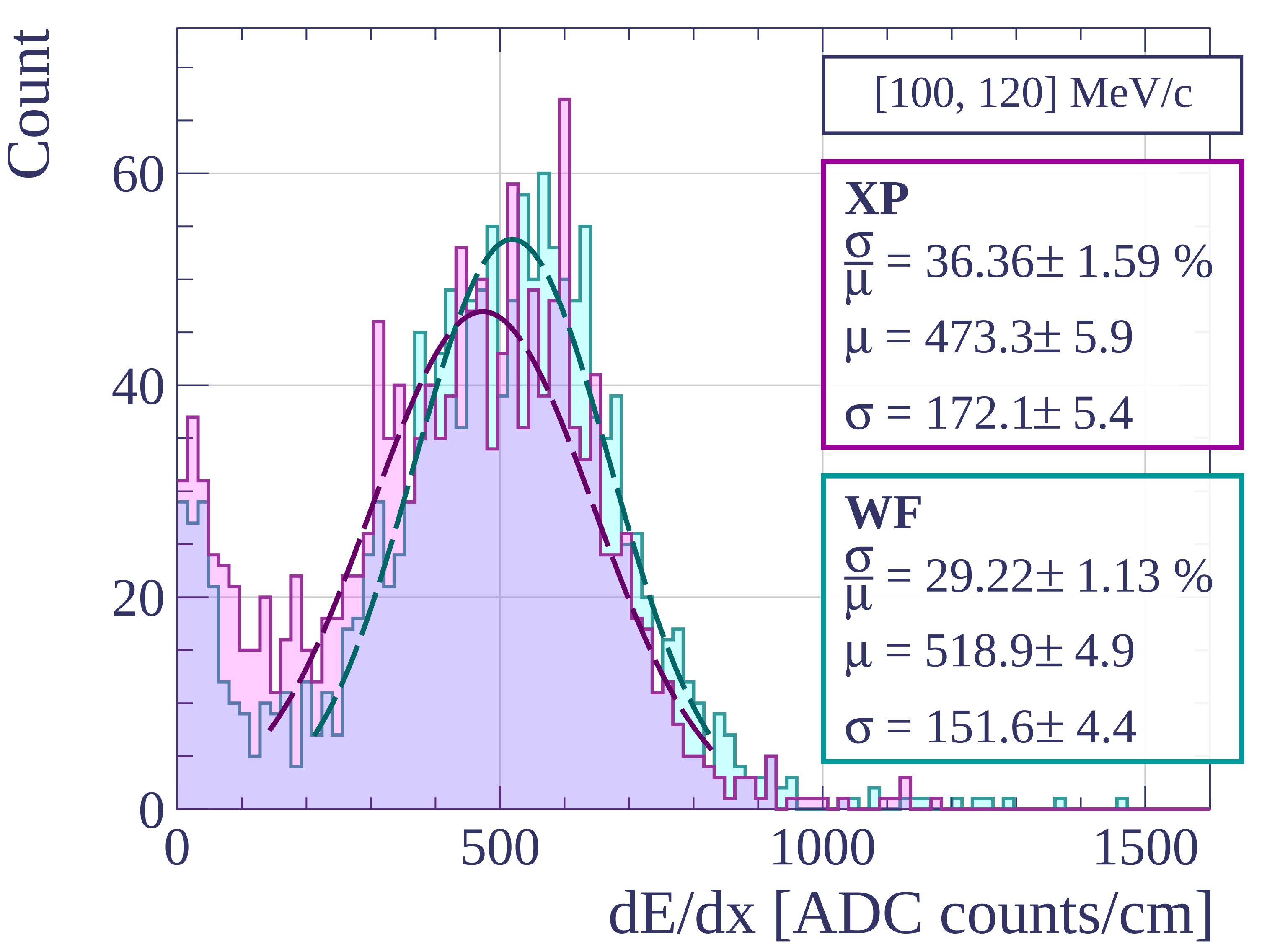
Count
0
100
200
300
400
0
500
1000
1500
dE/dx [ADC counts/cm]
[40, 60] MeV/c
XP
σ/μ = 26.99± 0.60 %
μ = 492.2± 2.3
σ = 132.8± 2.3
WF
σ/μ = 20.33± 0.35 %
μ = 532.3± 1.5
σ = 108.2± 1.6



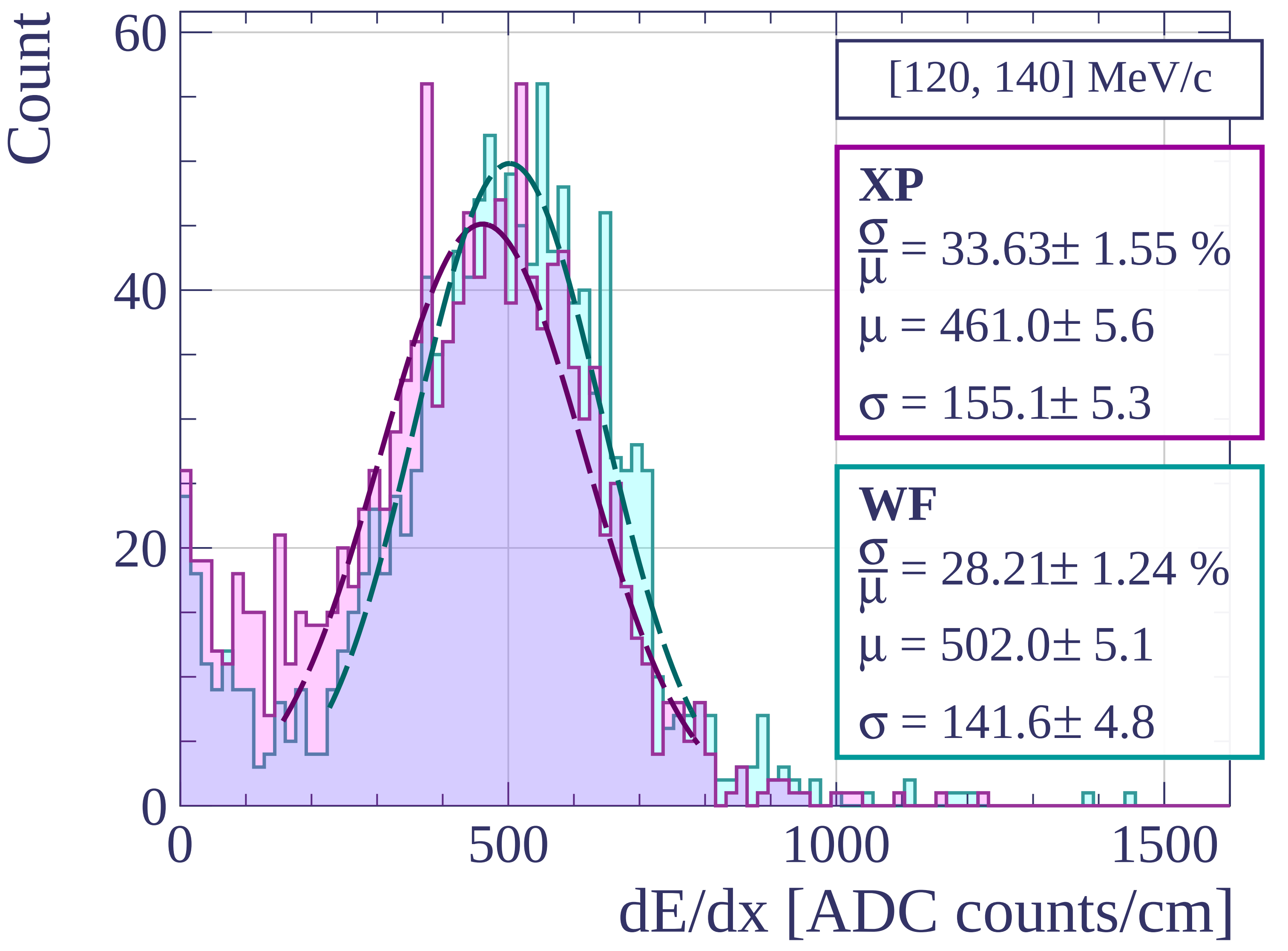
Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[60, 80] MeV/c
XP
$\frac{\sigma}{\mu}$ = 35.35± 1.24 %
μ = 472.2± 4.4
σ = 166.9± 4.3
WF
$\frac{\sigma}{\mu}$ = 23.81± 0.67 %
μ = 538.5± 2.9
σ = 128.2± 2.9



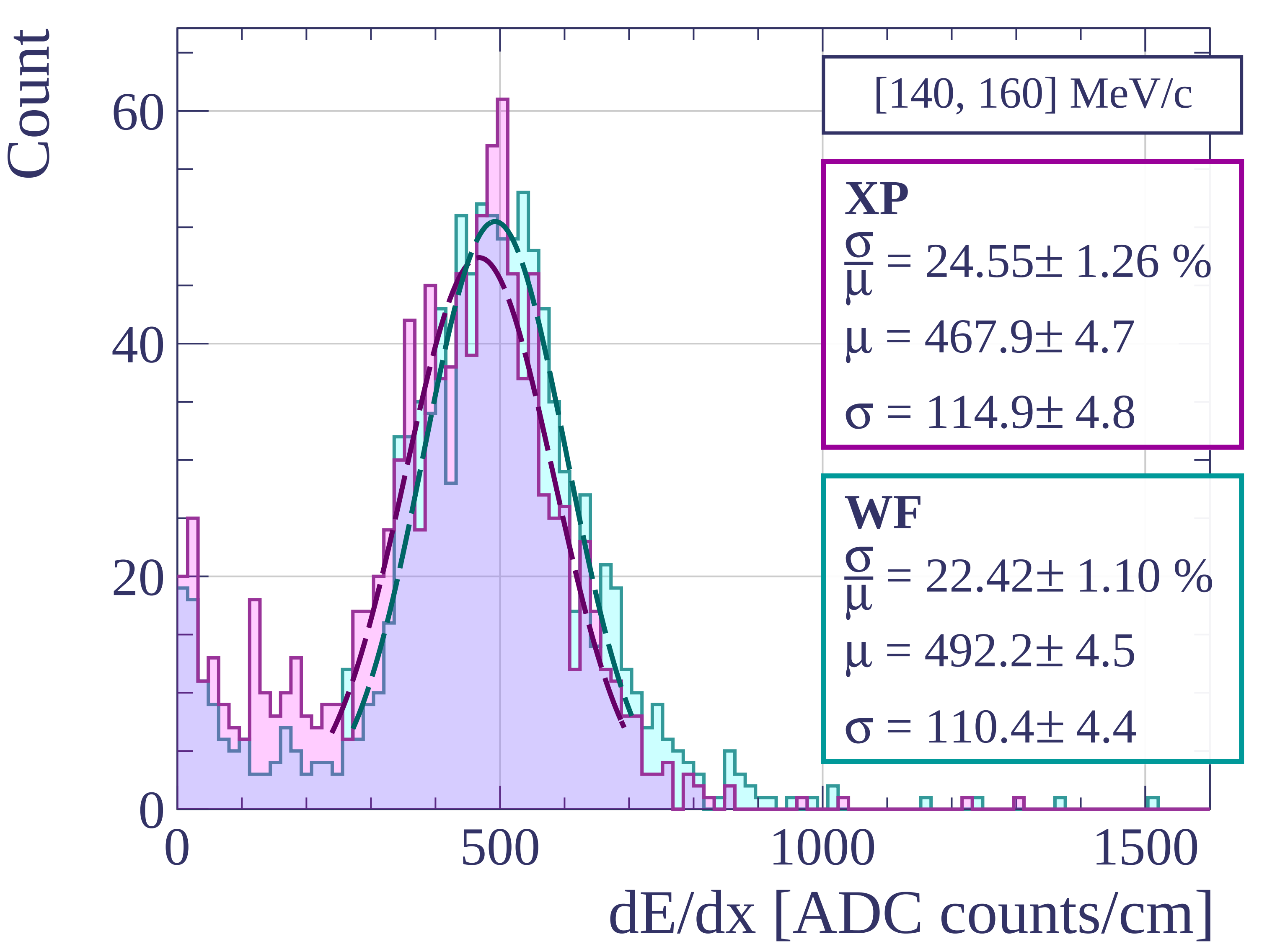
Count
dE/dx [ADC counts/cm]
0
20
40
60
80
0
500
1000
1500
[80, 100] MeV/c
XP
$\frac{\sigma}{\mu}$ = 40.96± 1.73 %
μ = 451.4± 5.8
σ = 184.9± 5.5
WF
$\frac{\sigma}{\mu}$ = 29.22± 1.11 %
μ = 517.9± 4.4
σ = 151.3± 4.5



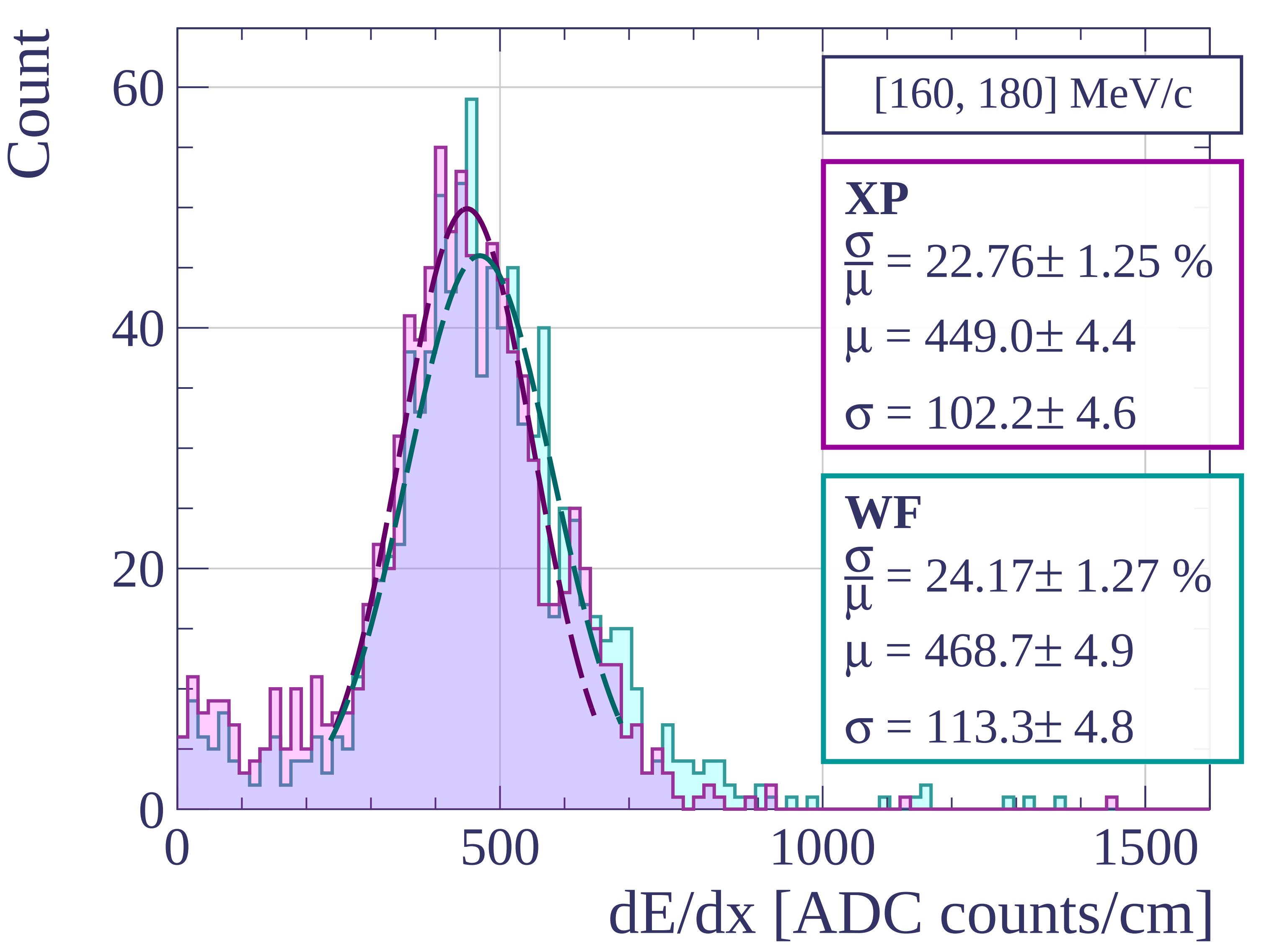
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[100, 120] MeV/c
XP
$\frac{\sigma}{\mu}$ = 36.36± 1.59 %
μ = 473.3± 5.9
σ = 172.1± 5.4
WF
$\frac{\sigma}{\mu}$ = 29.22± 1.13 %
μ = 518.9± 4.9
σ = 151.6± 4.4



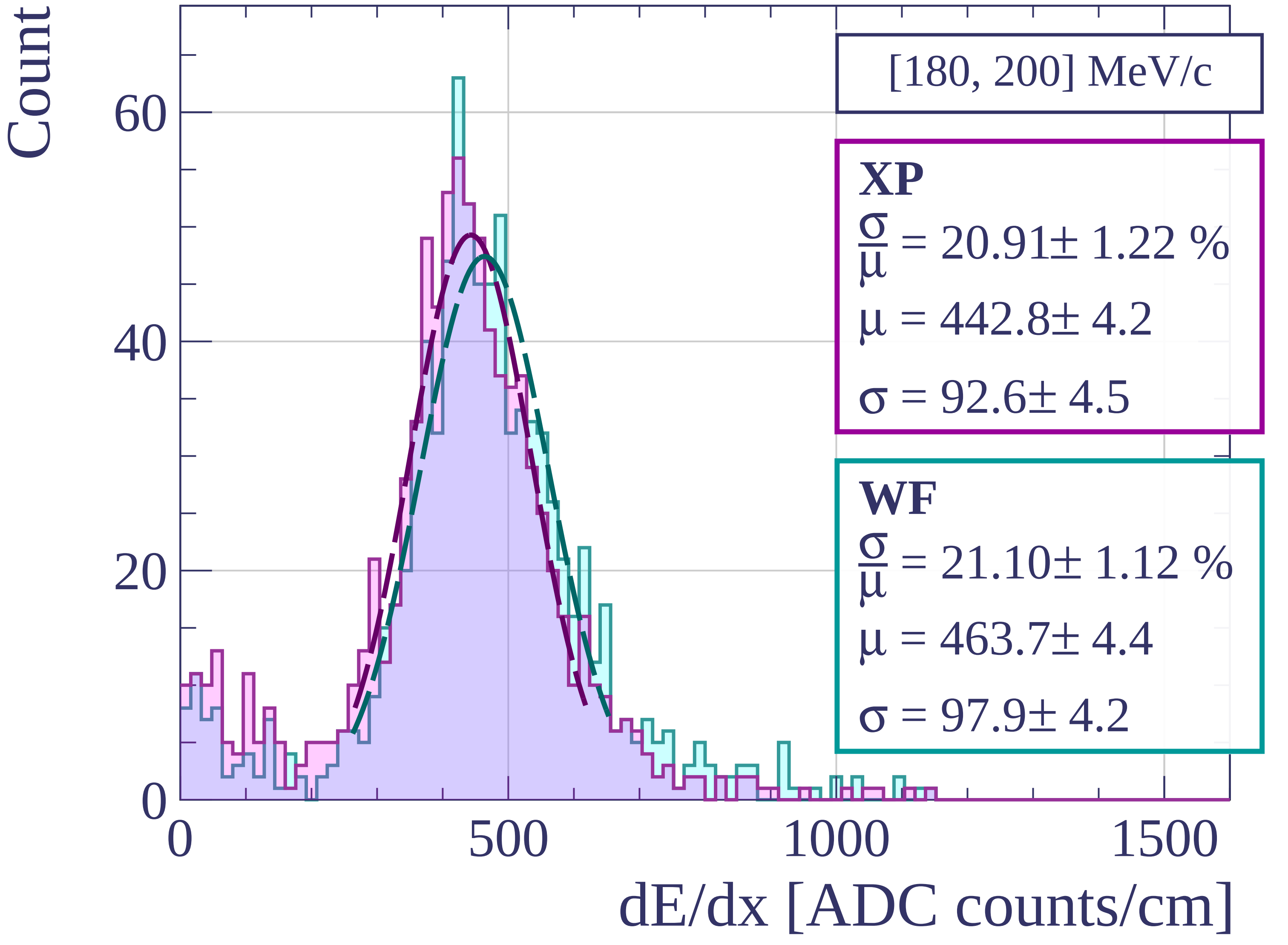
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[120, 140] MeV/c
XP
σ/μ = 33.63± 1.55 %
μ = 461.0± 5.6
σ = 155.1± 5.3
WF
σ/μ = 28.21± 1.24 %
μ = 502.0± 5.1
σ = 141.6± 4.8

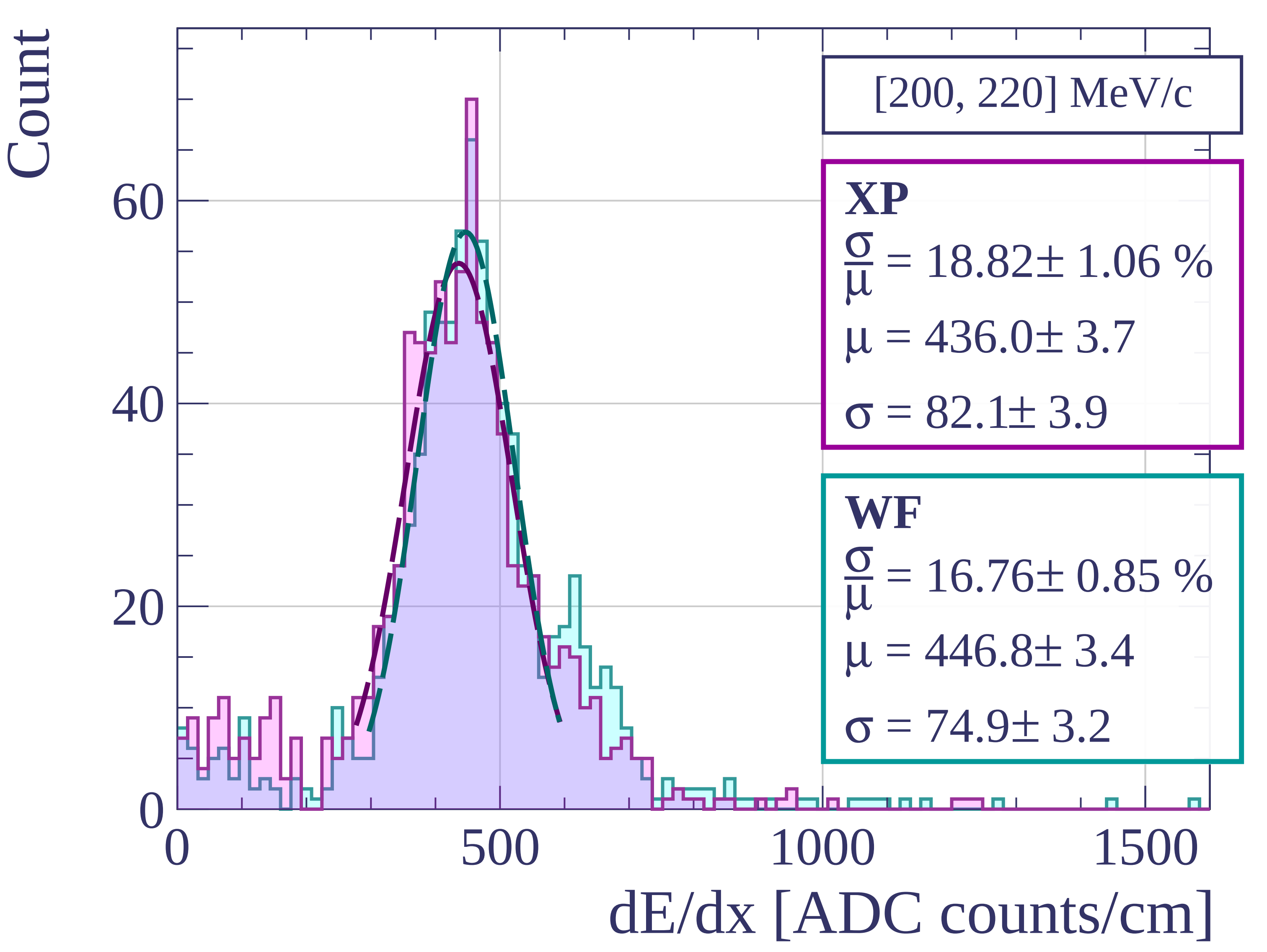


Count
dE/dx [ADC counts/cm]
[140, 160] MeV/c
XP
$\frac{\sigma}{\mu}$ = 24.55± 1.26 %
μ = 467.9± 4.7
σ = 114.9± 4.8
WF
$\frac{\sigma}{\mu}$ = 22.42± 1.10 %
μ = 492.2± 4.5
σ = 110.4± 4.4
0
20
40
60
0
500
1000
1500

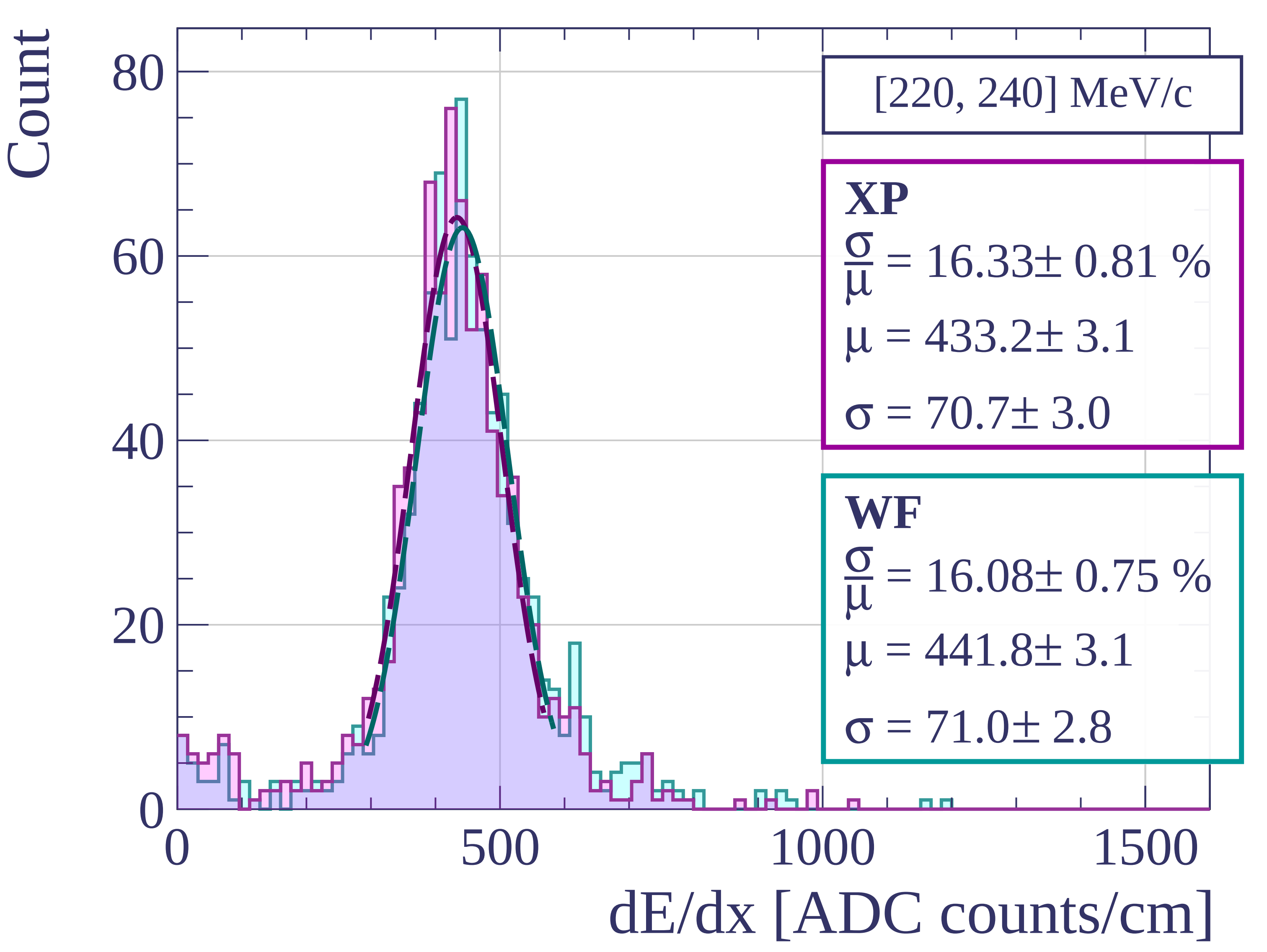


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[160, 180] MeV/c
XP
$\frac{\sigma}{\mu}$ = 22.76± 1.25 %
μ = 449.0± 4.4
σ = 102.2± 4.6
WF
$\frac{\sigma}{\mu}$ = 24.17± 1.27 %
μ = 468.7± 4.9
σ = 113.3± 4.8

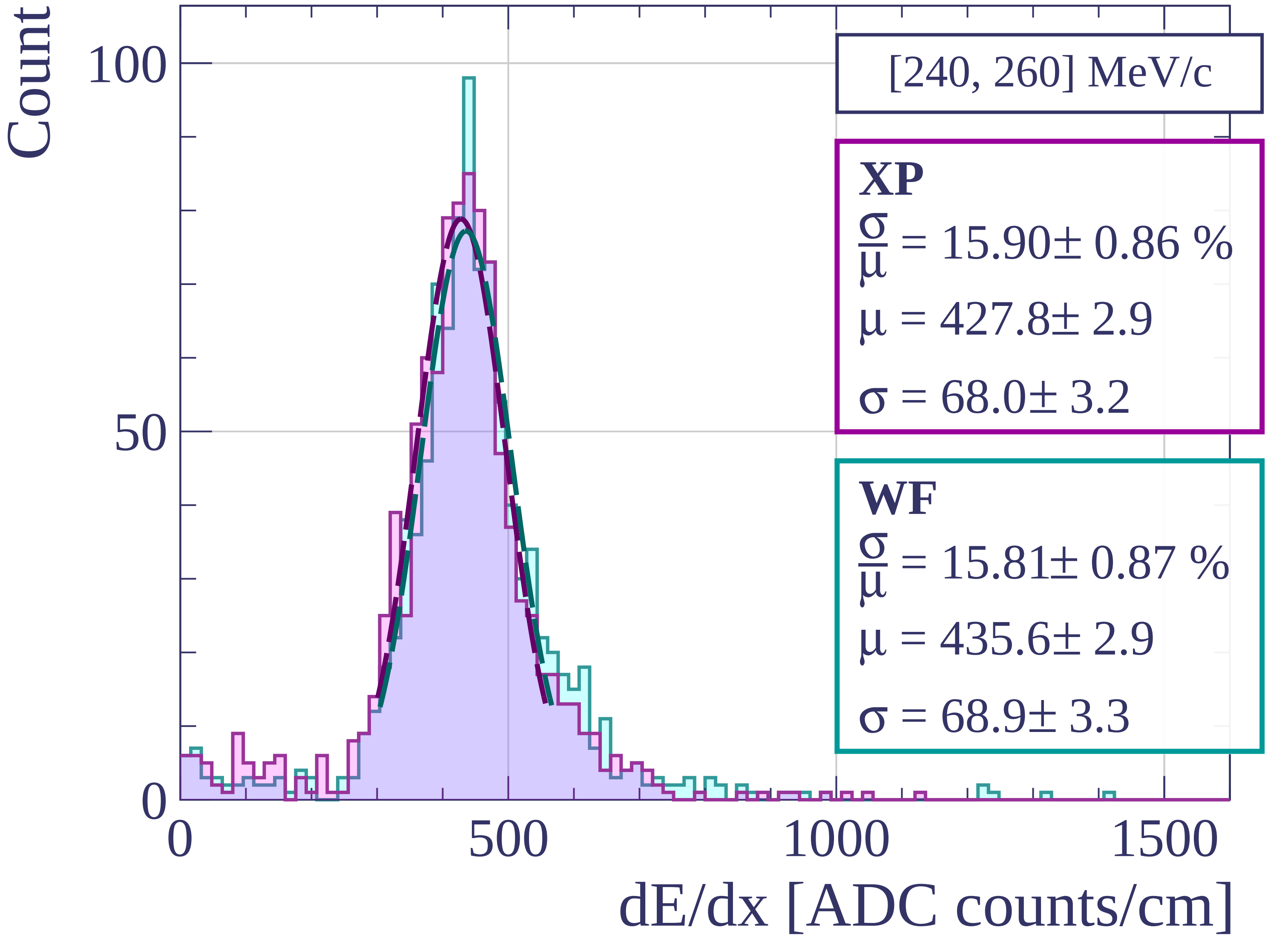

Count
0
20
40
60
0
500
1000
1500
dE/dx [ADC counts/cm]
[180, 200] MeV/c
XP
$\frac{\sigma}{\mu}$ = 20.91± 1.22 %
μ = 442.8± 4.2
σ = 92.6± 4.5
WF
$\frac{\sigma}{\mu}$ = 21.10± 1.12 %
μ = 463.7± 4.4
σ = 97.9± 4.2



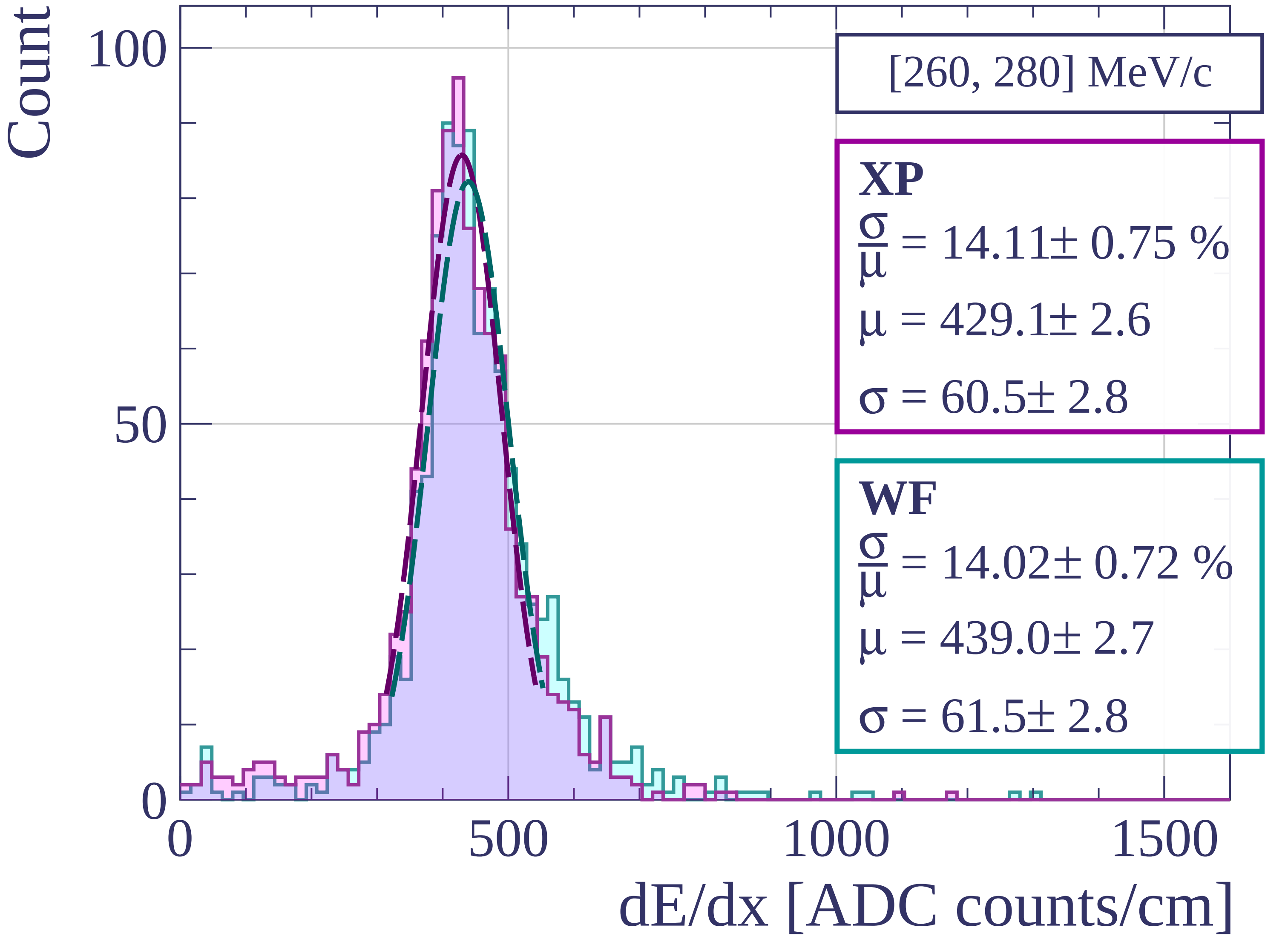
Count
dE/dx [ADC counts/cm]
[200, 220] MeV/c
XP
$\frac{\sigma}{\mu}$ = 18.82± 1.06 %
μ = 436.0± 3.7
σ = 82.1± 3.9
WF
$\frac{\sigma}{\mu}$ = 16.76± 0.85 %
μ = 446.8± 3.4
σ = 74.9± 3.2
0
20
40
60
0
500
1000
1500



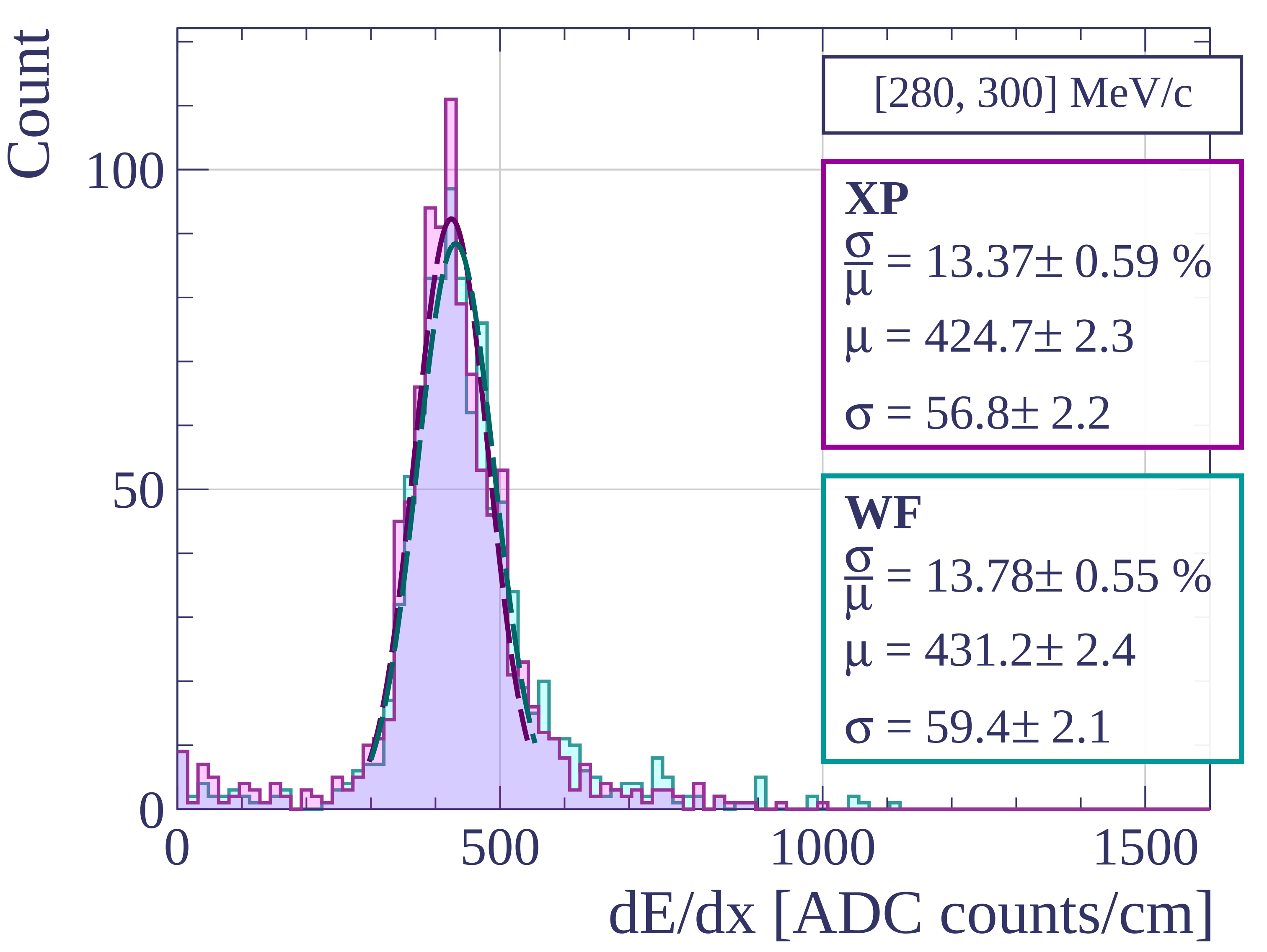
Count
dE/dx [ADC counts/cm]
[220, 240] MeV/c
XP
σ/μ = 16.33± 0.81 %
μ = 433.2± 3.1
σ = 70.7± 3.0
WF
σ/μ = 16.08± 0.75 %
μ = 441.8± 3.1
σ = 71.0± 2.8
0
20
40
60
80
0
500
1000
1500



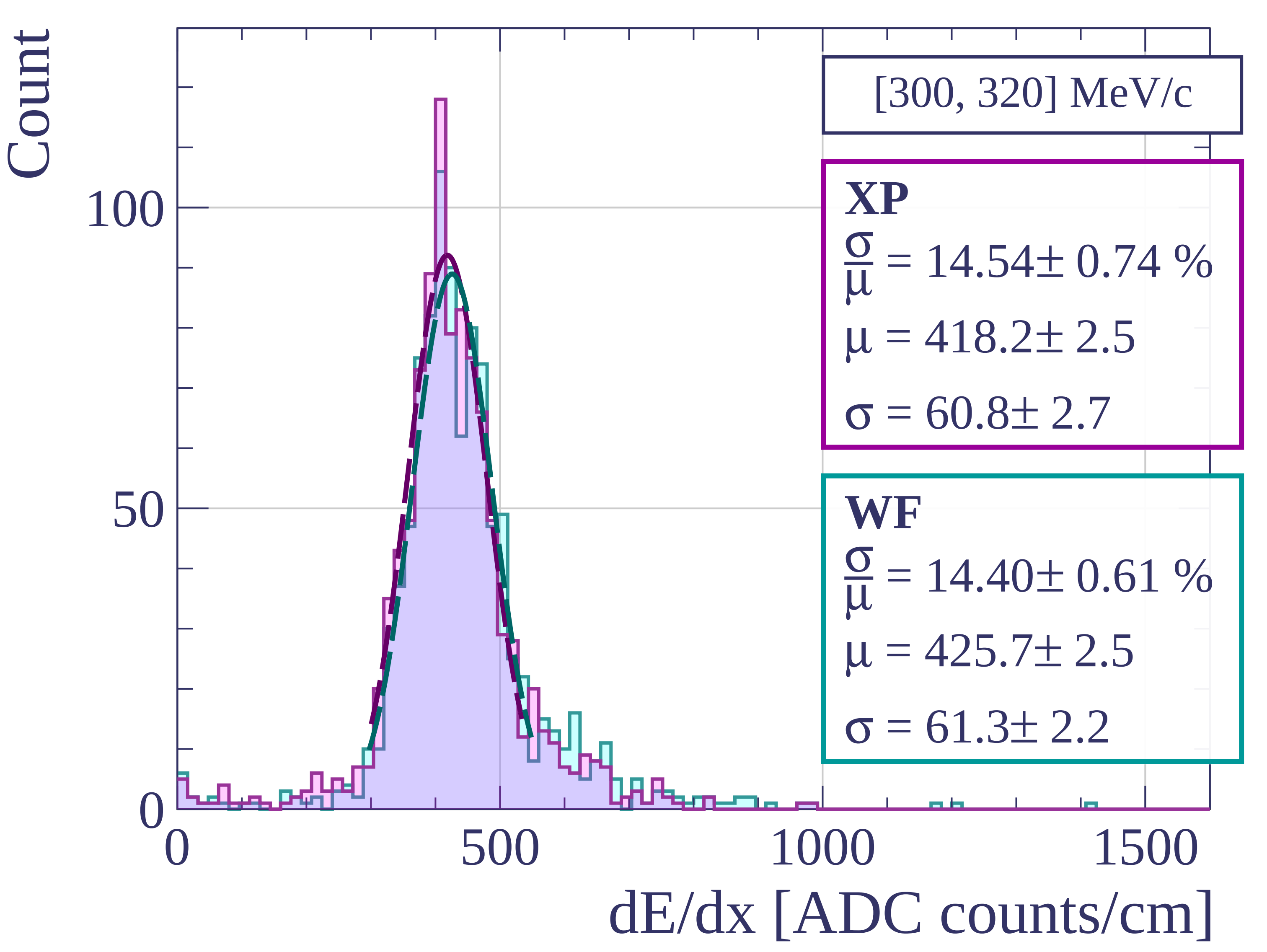
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[240, 260] MeV/c
XP
$\frac{\sigma}{\mu}$ = 15.90± 0.86 %
μ = 427.8± 2.9
σ = 68.0± 3.2
WF
$\frac{\sigma}{\mu}$ = 15.81± 0.87 %
μ = 435.6± 2.9
σ = 68.9± 3.3



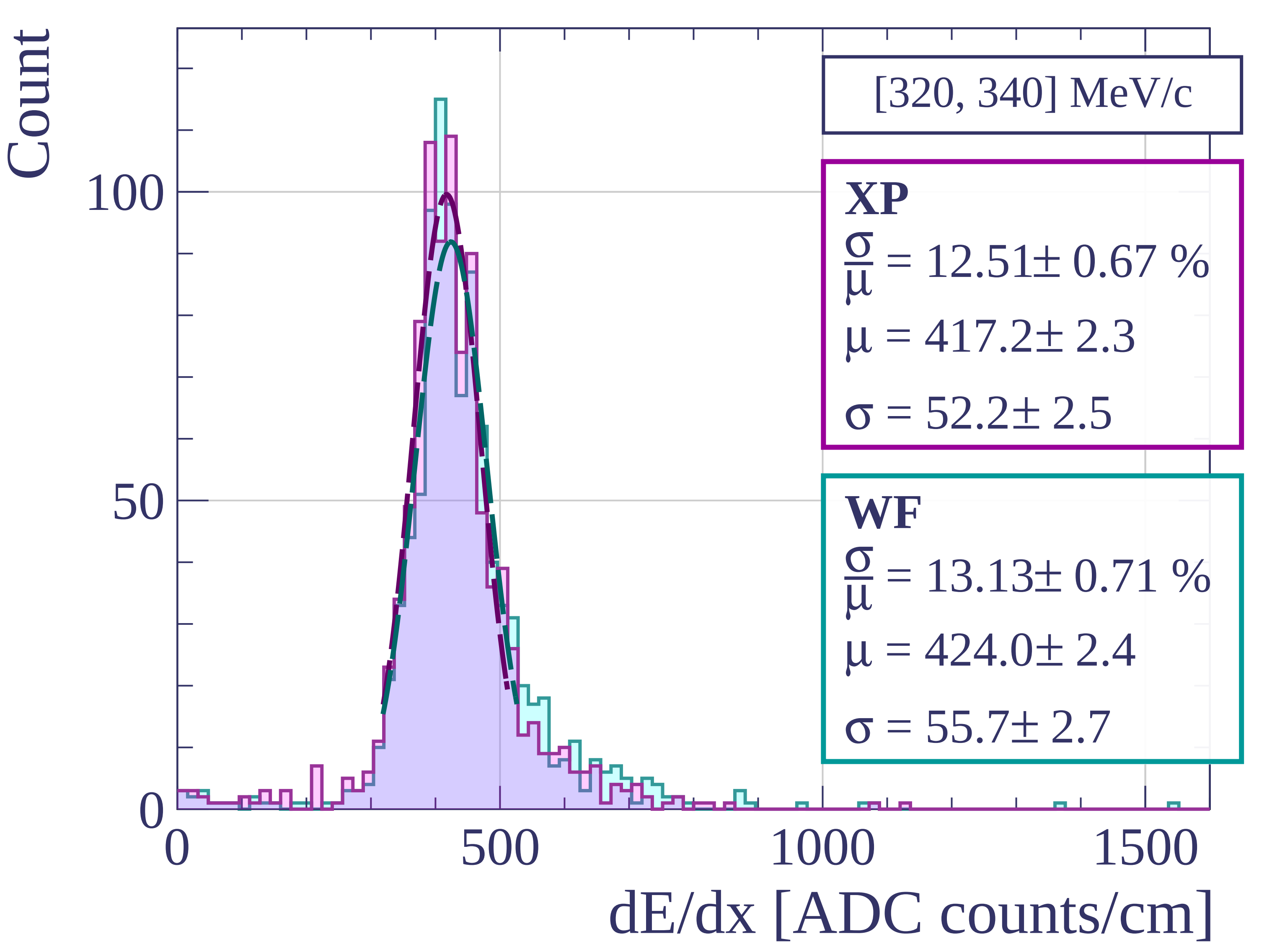
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[260, 280] MeV/c
XP
$\frac{\sigma}{\mu}$ = 14.11± 0.75 %
μ = 429.1± 2.6
σ = 60.5± 2.8
WF
$\frac{\sigma}{\mu}$ = 14.02± 0.72 %
μ = 439.0± 2.7
σ = 61.5± 2.8



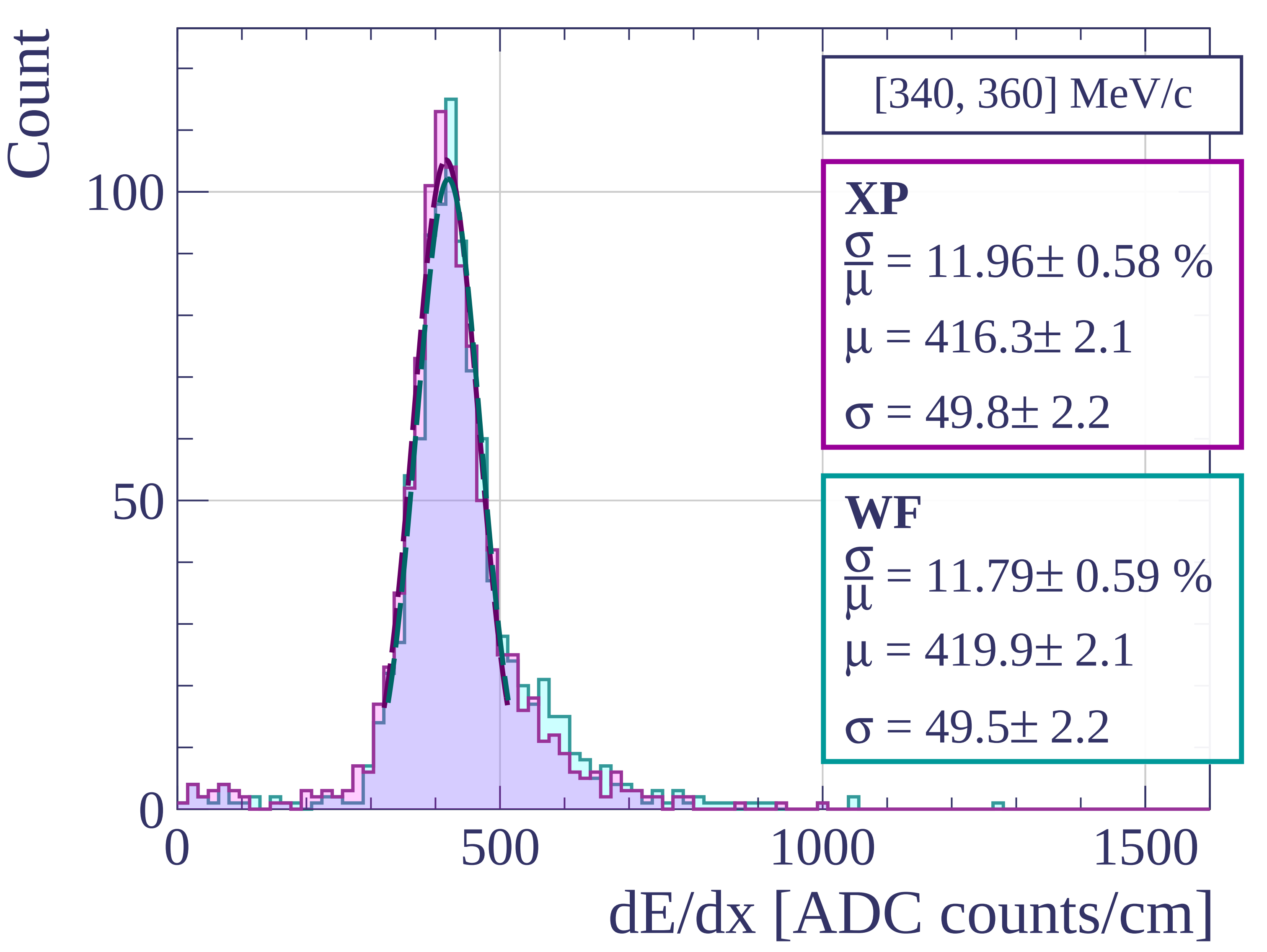
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[280, 300] MeV/c
XP
σ/μ = 13.37± 0.59 %
μ = 424.7± 2.3
σ = 56.8± 2.2
WF
σ/μ = 13.78± 0.55 %
μ = 431.2± 2.4
σ = 59.4± 2.1



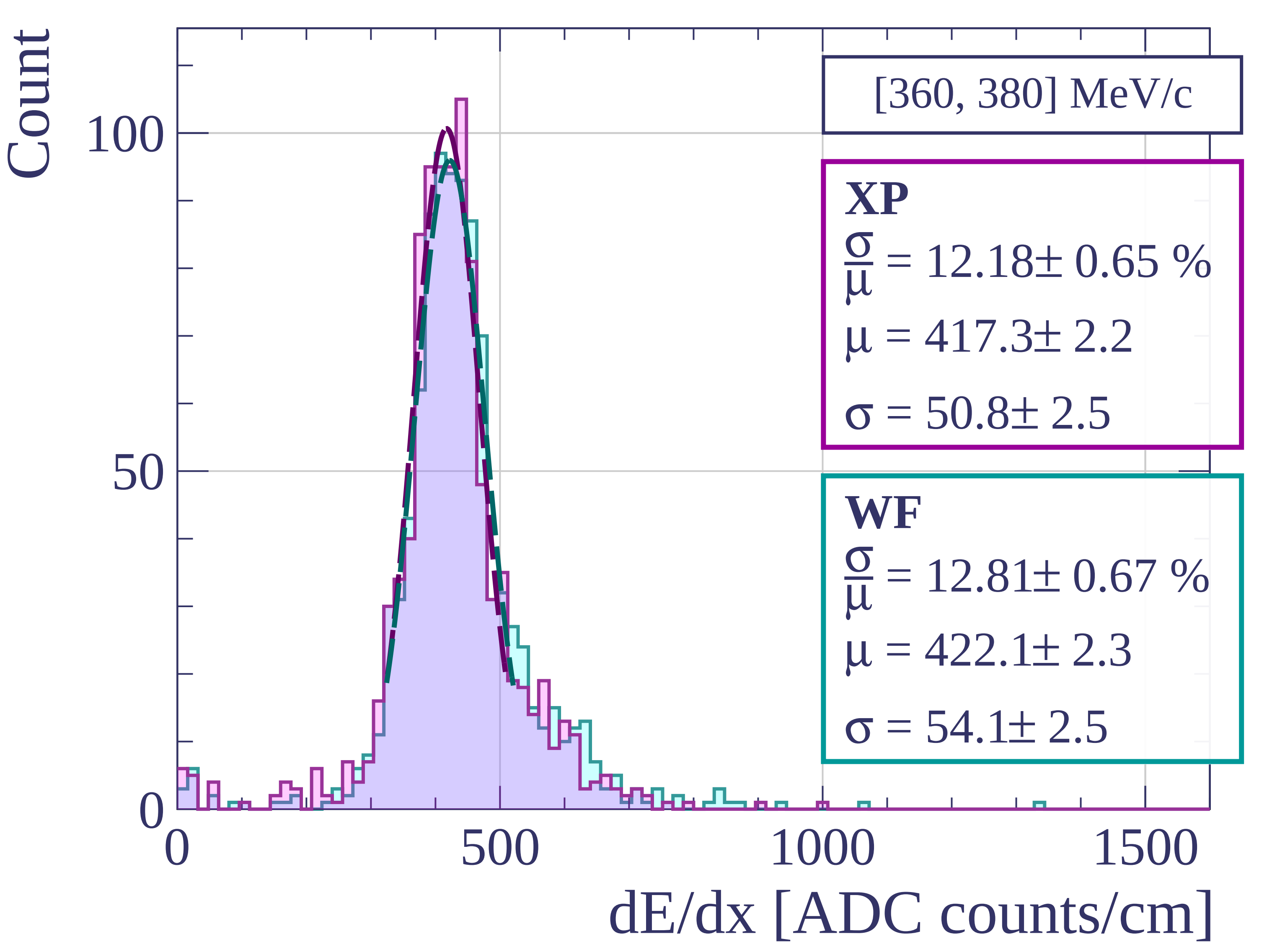
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[300, 320] MeV/c
XP
σ/μ = 14.54± 0.74 %
μ = 418.2± 2.5
σ = 60.8± 2.7
WF
σ/μ = 14.40± 0.61 %
μ = 425.7± 2.5
σ = 61.3± 2.2



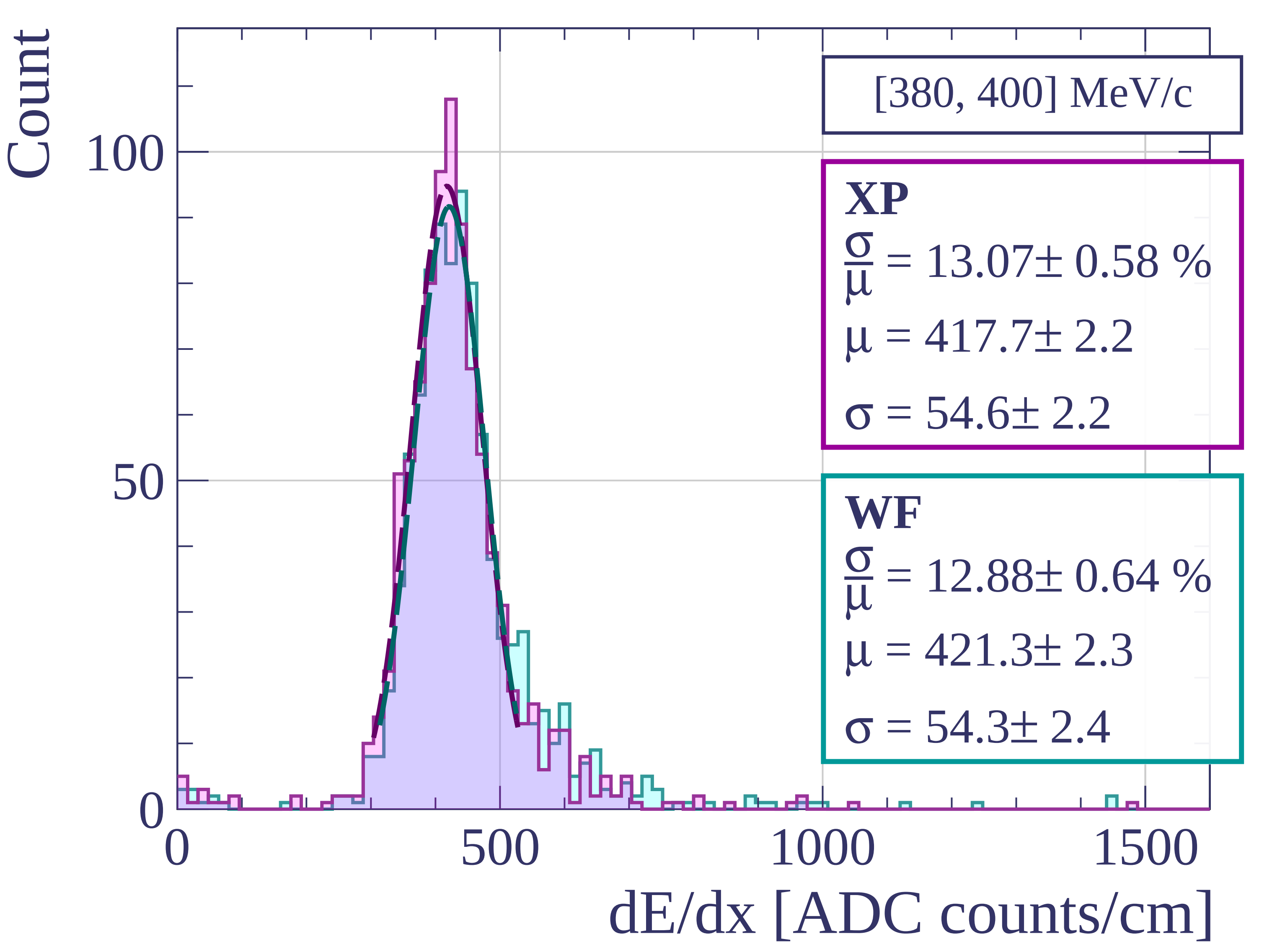
Count
dE/dx [ADC counts/cm]
[320, 340] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.51± 0.67 %
μ = 417.2± 2.3
σ = 52.2± 2.5
WF
$\frac{\sigma}{\mu}$ = 13.13± 0.71 %
μ = 424.0± 2.4
σ = 55.7± 2.7
0
50
100
0
500
1000
1500



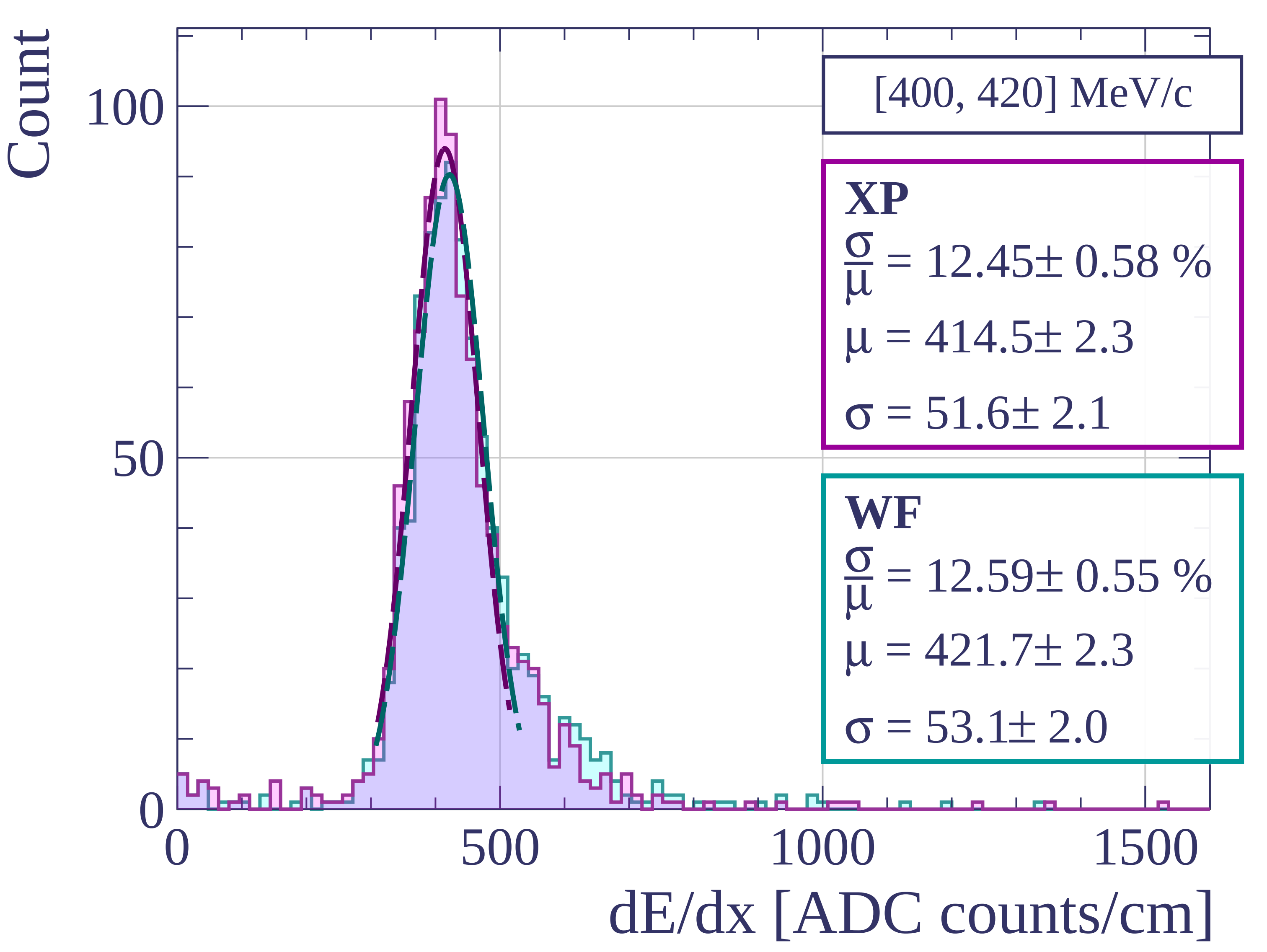
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[340, 360] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.96± 0.58 %
μ = 416.3± 2.1
σ = 49.8± 2.2
WF
$\frac{\sigma}{\mu}$ = 11.79± 0.59 %
μ = 419.9± 2.1
σ = 49.5± 2.2



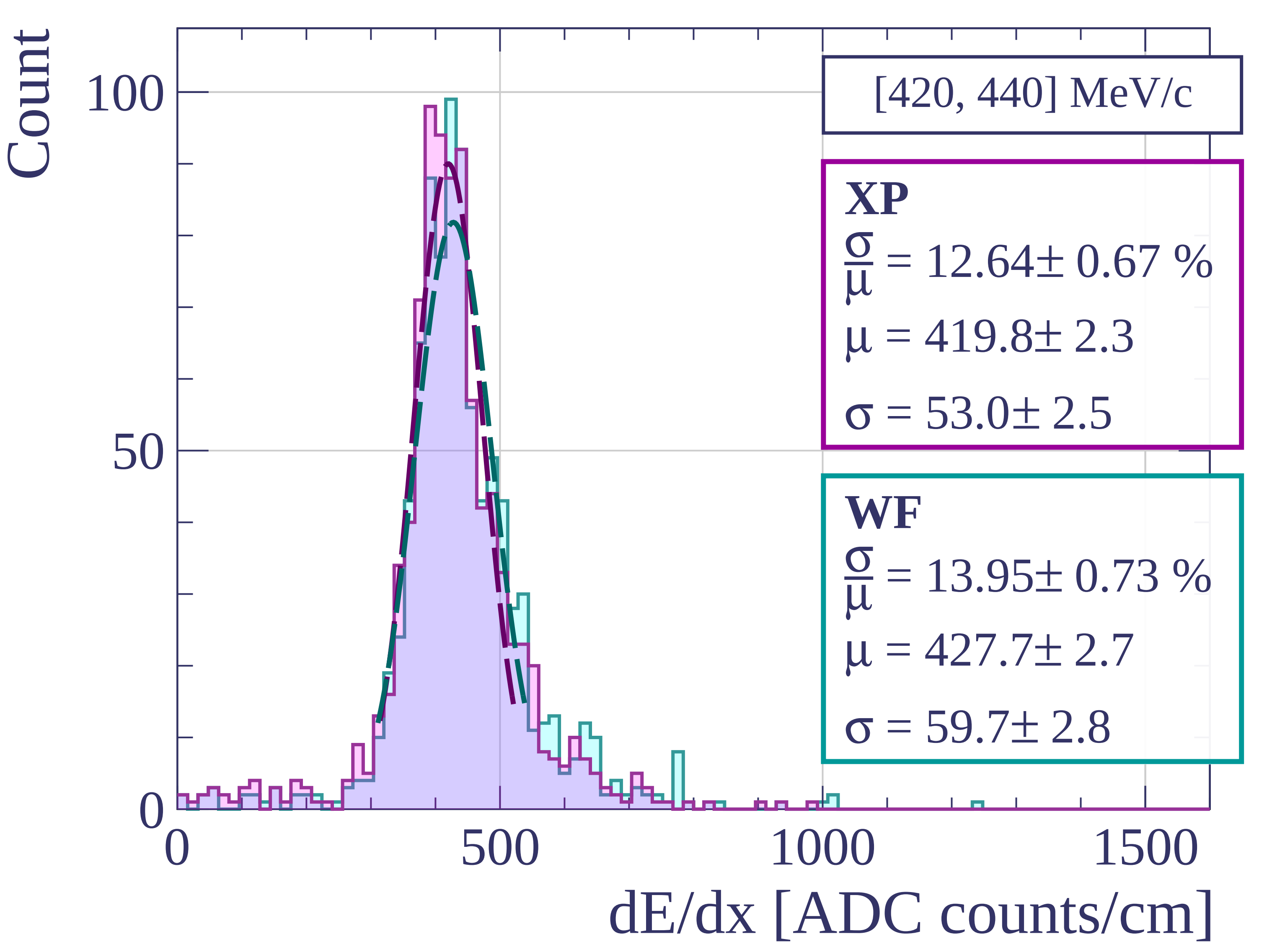
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[360, 380] MeV/c
XP
σ/μ = 12.18± 0.65 %
μ = 417.3± 2.2
σ = 50.8± 2.5
WF
σ/μ = 12.81± 0.67 %
μ = 422.1± 2.3
σ = 54.1± 2.5

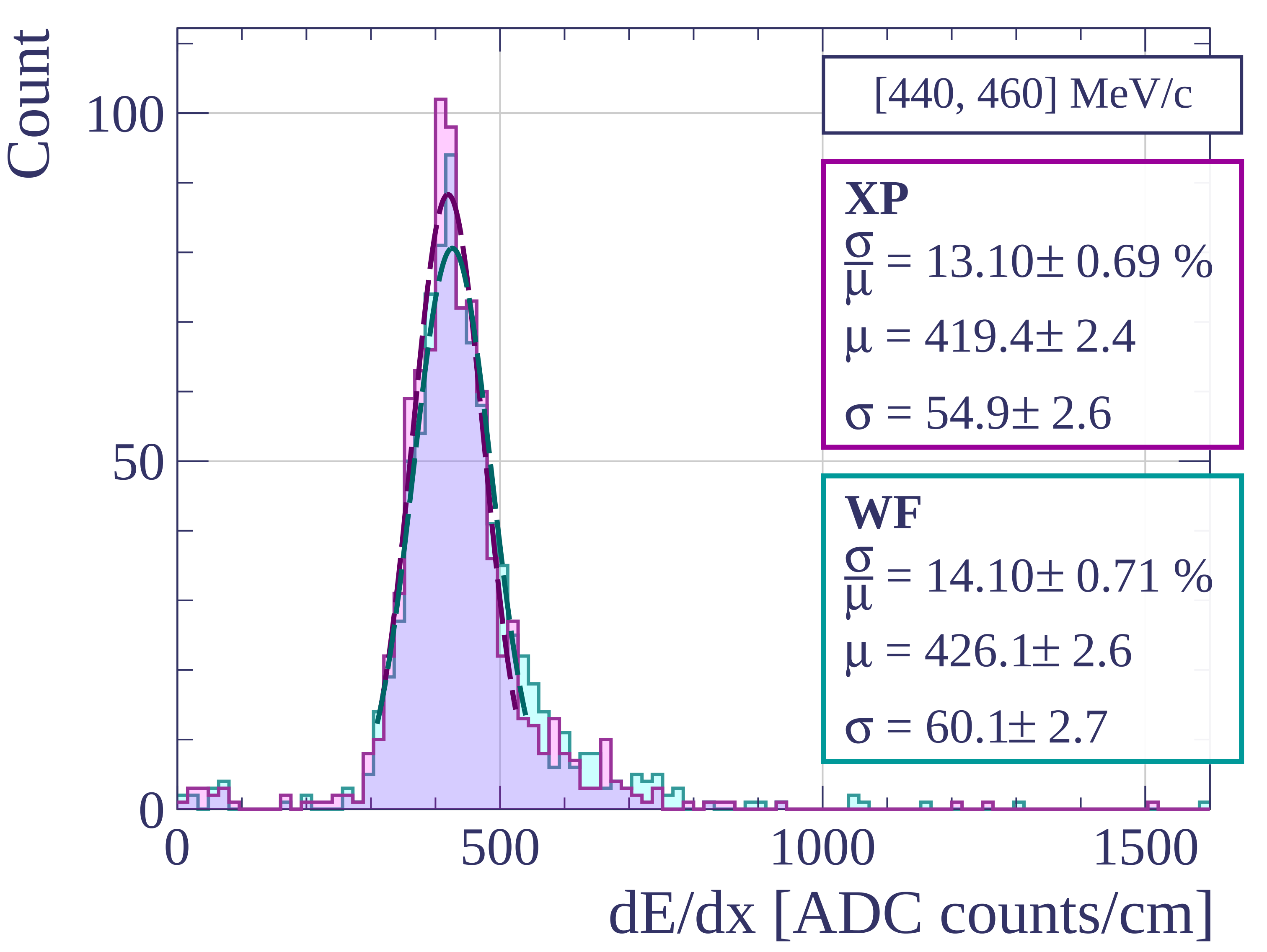


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[380, 400] MeV/c
XP
σ/μ = 13.07± 0.58 %
μ = 417.7± 2.2
σ = 54.6± 2.2
WF
σ/μ = 12.88± 0.64 %
μ = 421.3± 2.3
σ = 54.3± 2.4

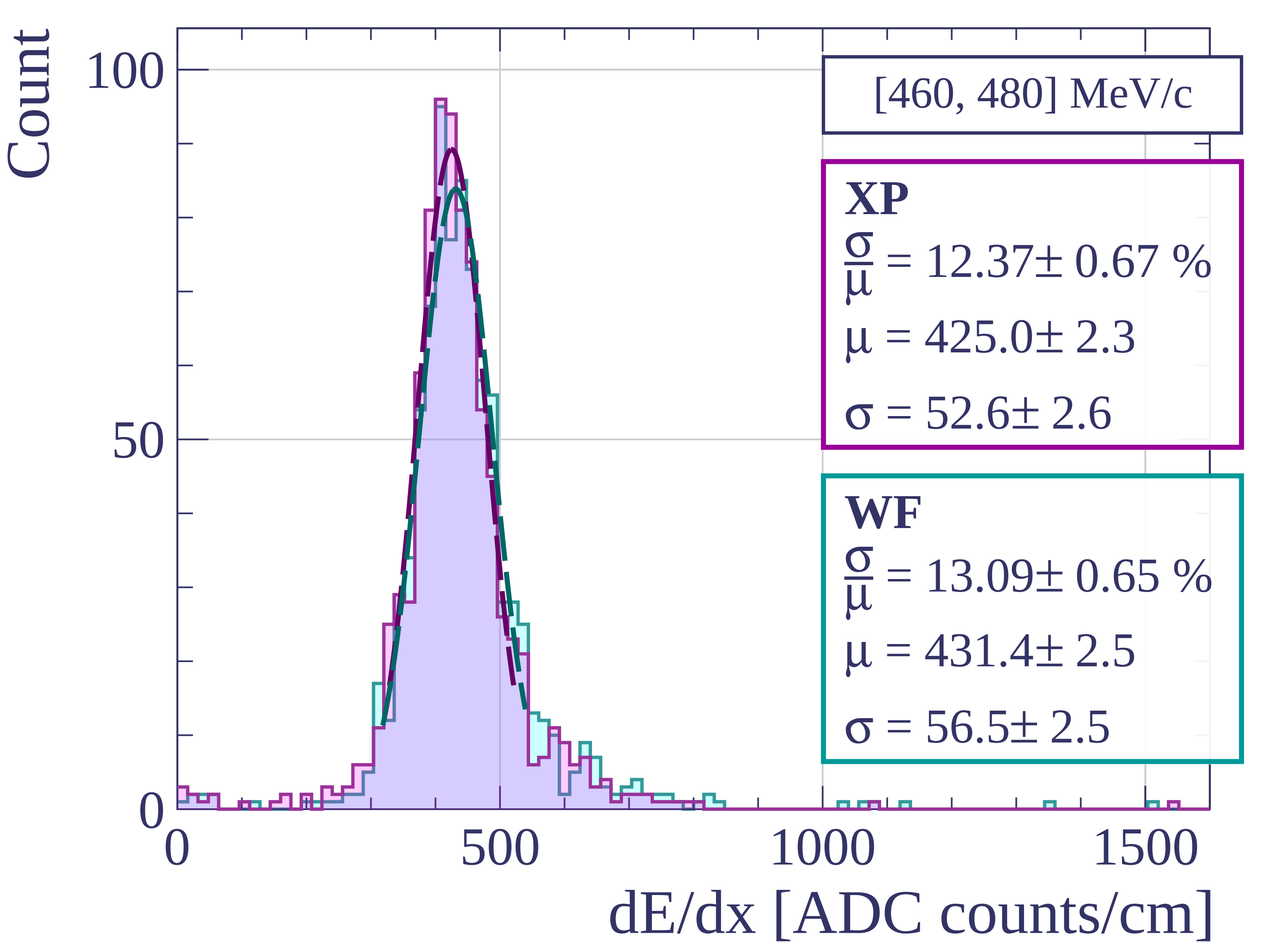


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[400, 420] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.45± 0.58 %
μ = 414.5± 2.3
σ = 51.6± 2.1
WF
$\frac{\sigma}{\mu}$ = 12.59± 0.55 %
μ = 421.7± 2.3
σ = 53.1± 2.0

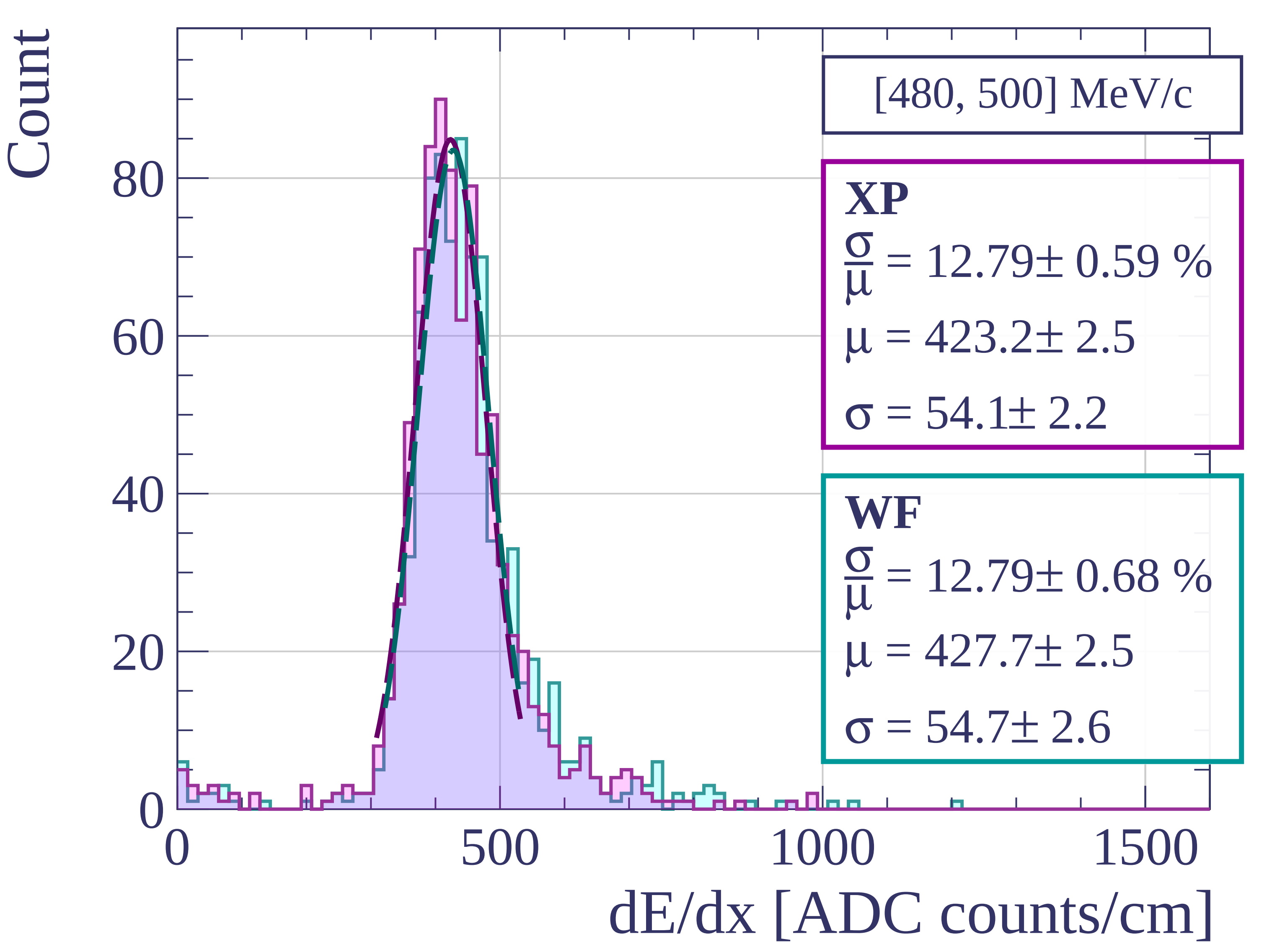

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[420, 440] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.64± 0.67 %
μ = 419.8± 2.3
σ = 53.0± 2.5
WF
$\frac{\sigma}{\mu}$ = 13.95± 0.73 %
μ = 427.7± 2.7
σ = 59.7± 2.8

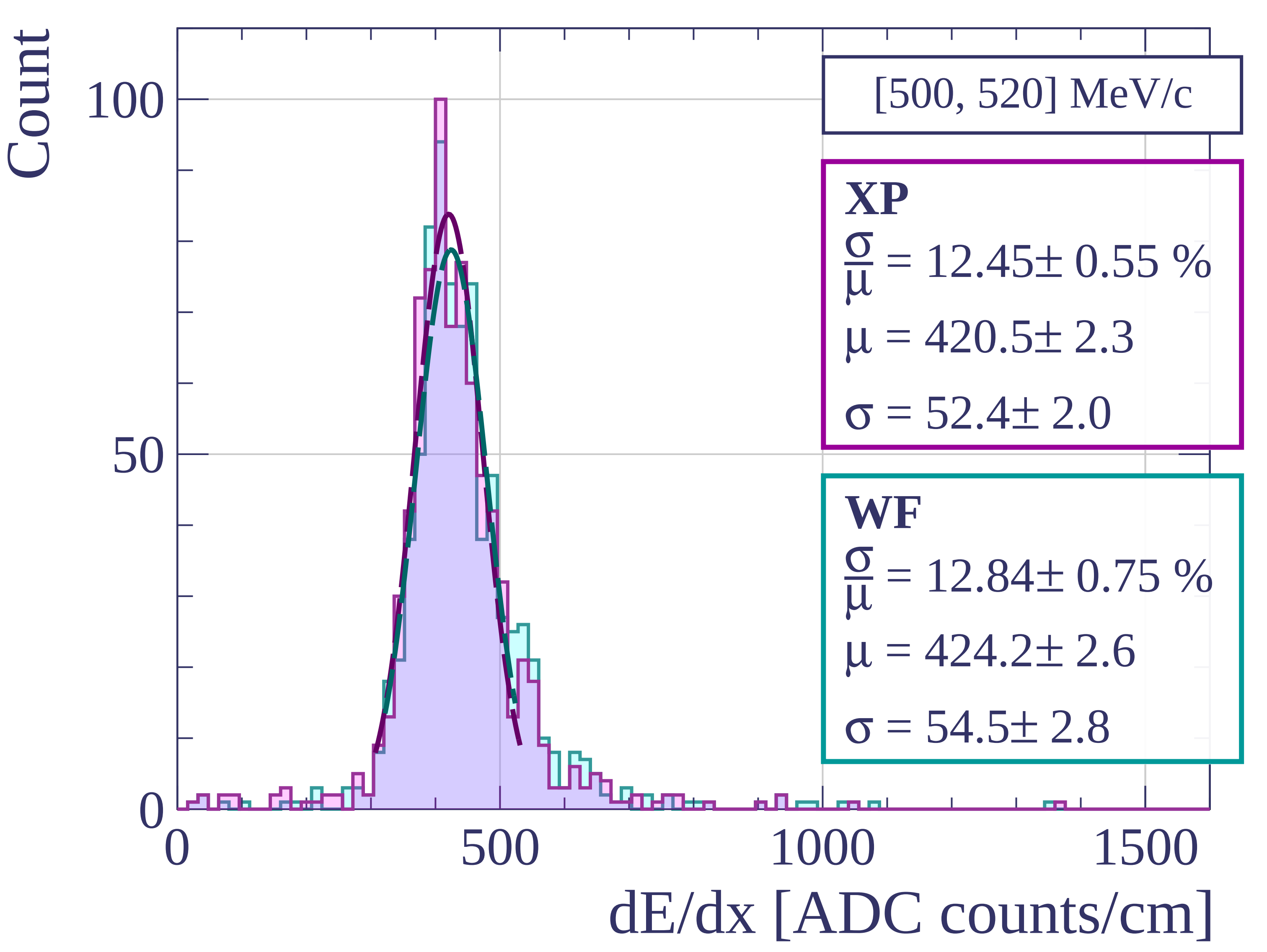


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[440, 460] MeV/c
XP
σ/μ = 13.10± 0.69 %
μ = 419.4± 2.4
σ = 54.9± 2.6
WF
σ/μ = 14.10± 0.71 %
μ = 426.1± 2.6
σ = 60.1± 2.7

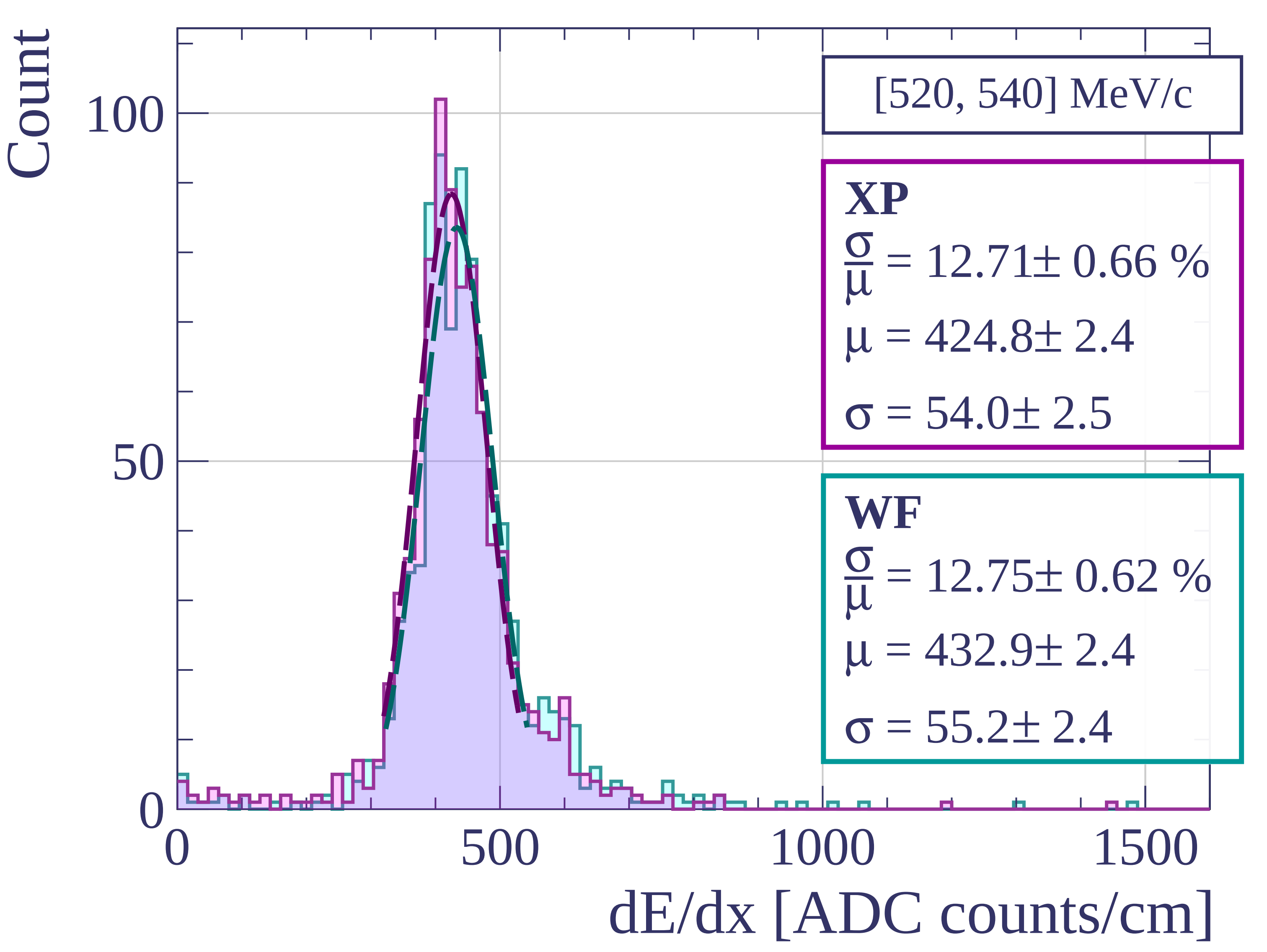


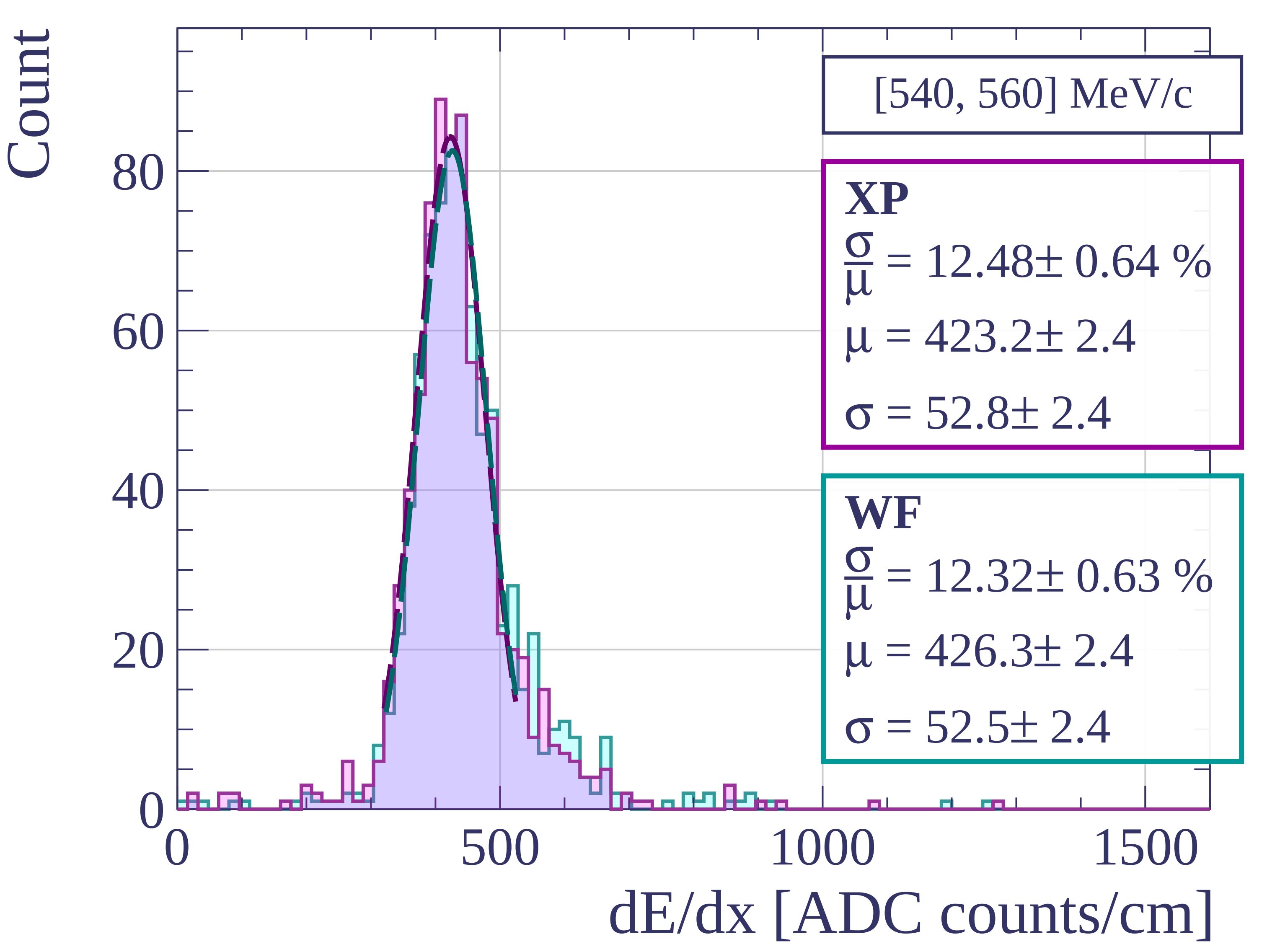
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[460, 480] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.37± 0.67 %
μ = 425.0± 2.3
σ = 52.6± 2.6
WF
$\frac{\sigma}{\mu}$ = 13.09± 0.65 %
μ = 431.4± 2.5
σ = 56.5± 2.5


Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[480, 500] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.79± 0.59 %
μ = 423.2± 2.5
σ = 54.1± 2.2
WF
$\frac{\sigma}{\mu}$ = 12.79± 0.68 %
μ = 427.7± 2.5
σ = 54.7± 2.6

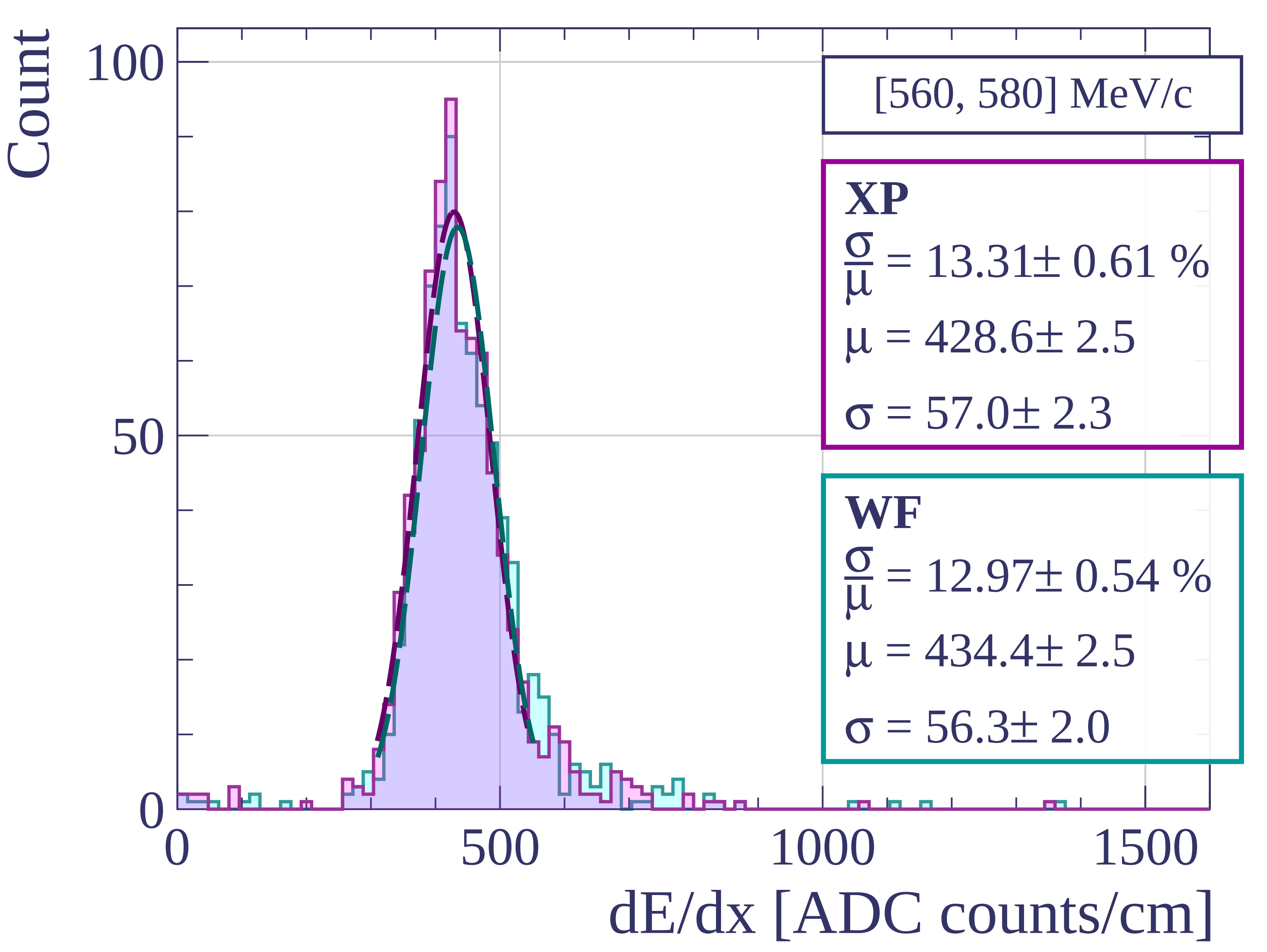


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[500, 520] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.45± 0.55 %
μ = 420.5± 2.3
σ = 52.4± 2.0
WF
$\frac{\sigma}{\mu}$ = 12.84± 0.75 %
μ = 424.2± 2.6
σ = 54.5± 2.8

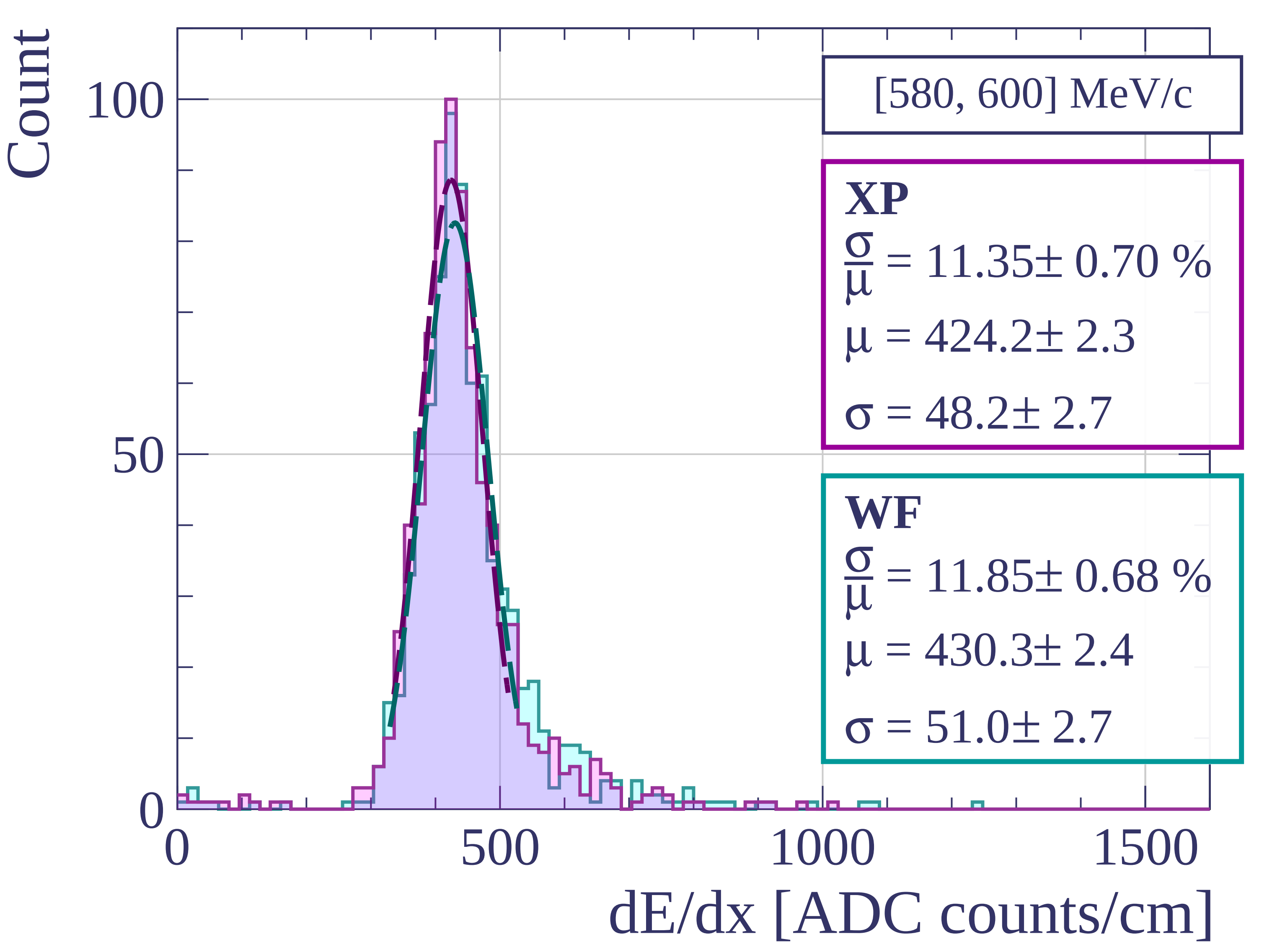

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[520, 540] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.71± 0.66 %
μ = 424.8± 2.4
σ = 54.0± 2.5
WF
$\frac{\sigma}{\mu}$ = 12.75± 0.62 %
μ = 432.9± 2.4
σ = 55.2± 2.4

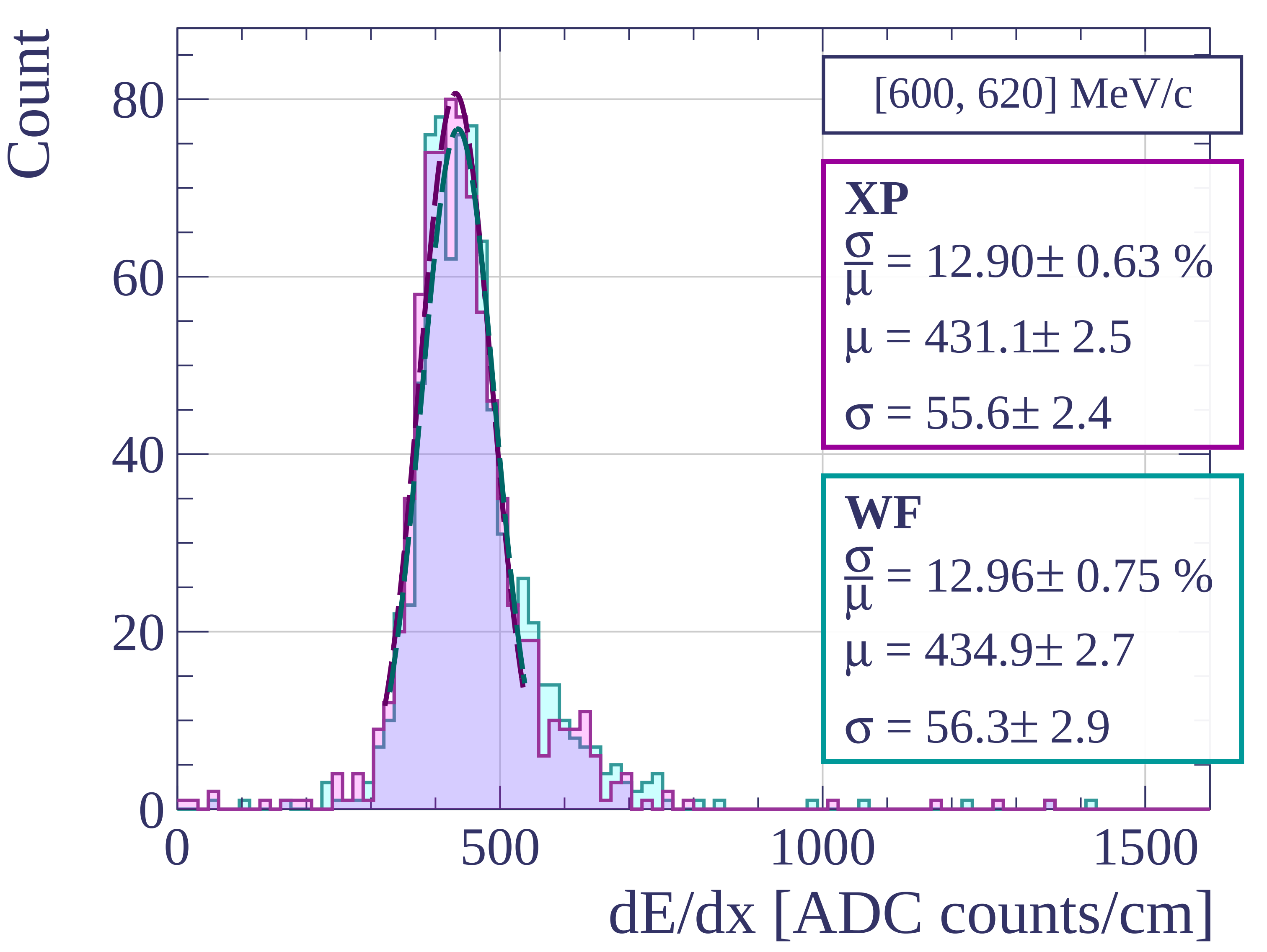


Count
dE/dx [ADC counts/cm]
0
20
40
60
80
0
500
1000
1500
[540, 560] MeV/c
XP
σ/μ = 12.48± 0.64 %
μ = 423.2± 2.4
σ = 52.8± 2.4
WF
σ/μ = 12.32± 0.63 %
μ = 426.3± 2.4
σ = 52.5± 2.4

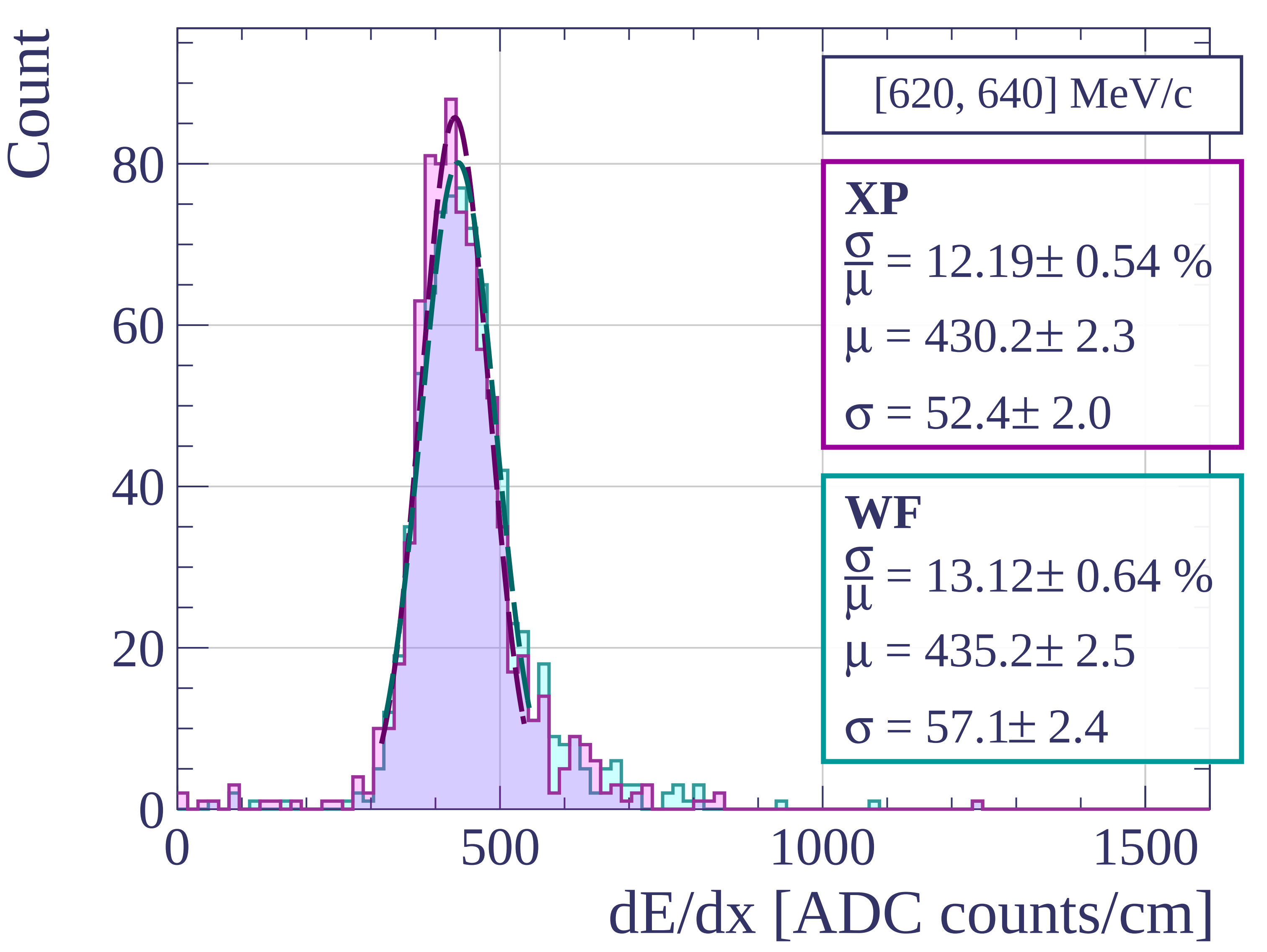


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[560, 580] MeV/c
XP
σ/μ = 13.31± 0.61 %
μ = 428.6± 2.5
σ = 57.0± 2.3
WF
σ/μ = 12.97± 0.54 %
μ = 434.4± 2.5
σ = 56.3± 2.0

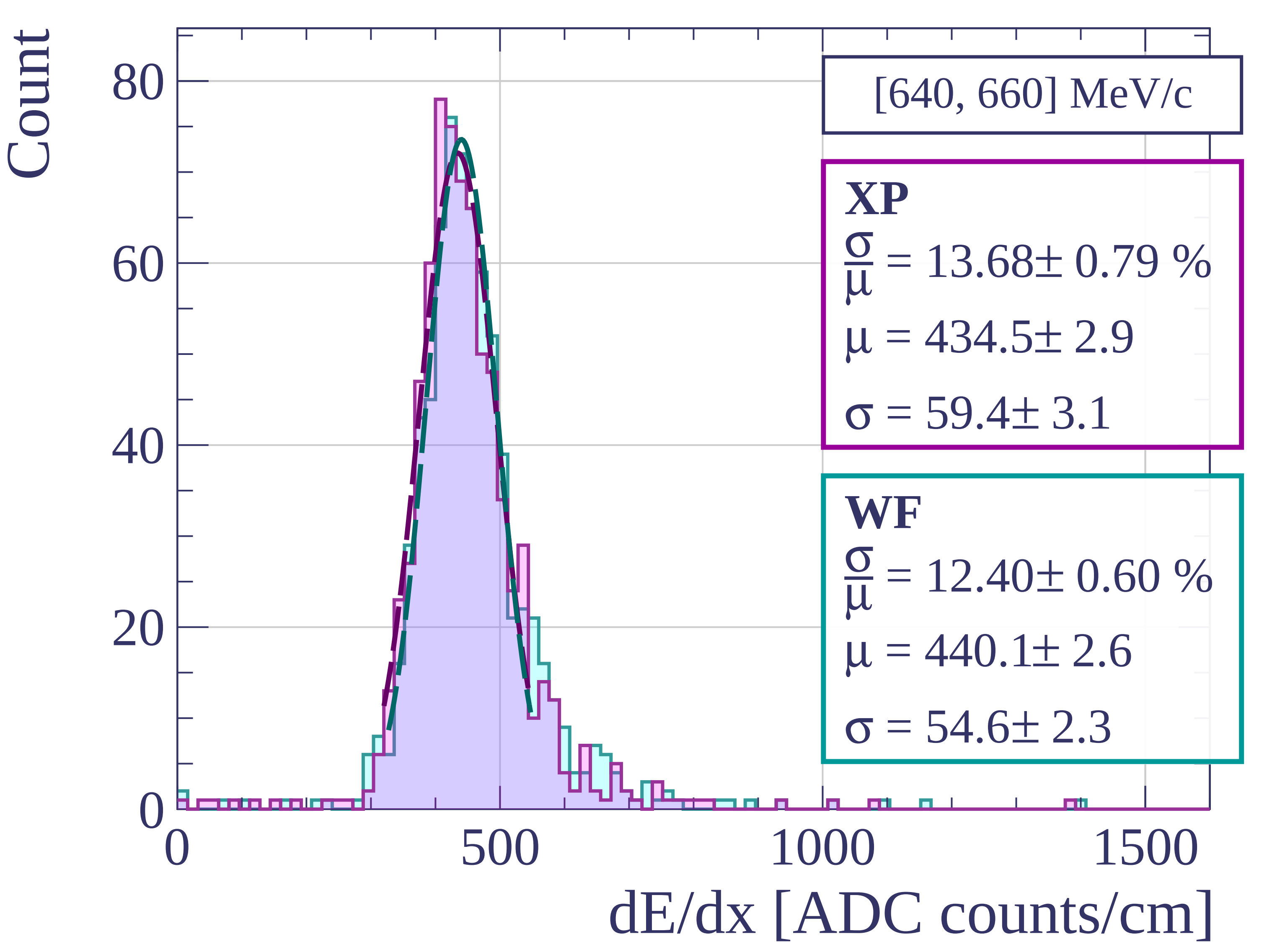

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[580, 600] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.35± 0.70 %
μ = 424.2± 2.3
σ = 48.2± 2.7
WF
$\frac{\sigma}{\mu}$ = 11.85± 0.68 %
μ = 430.3± 2.4
σ = 51.0± 2.7



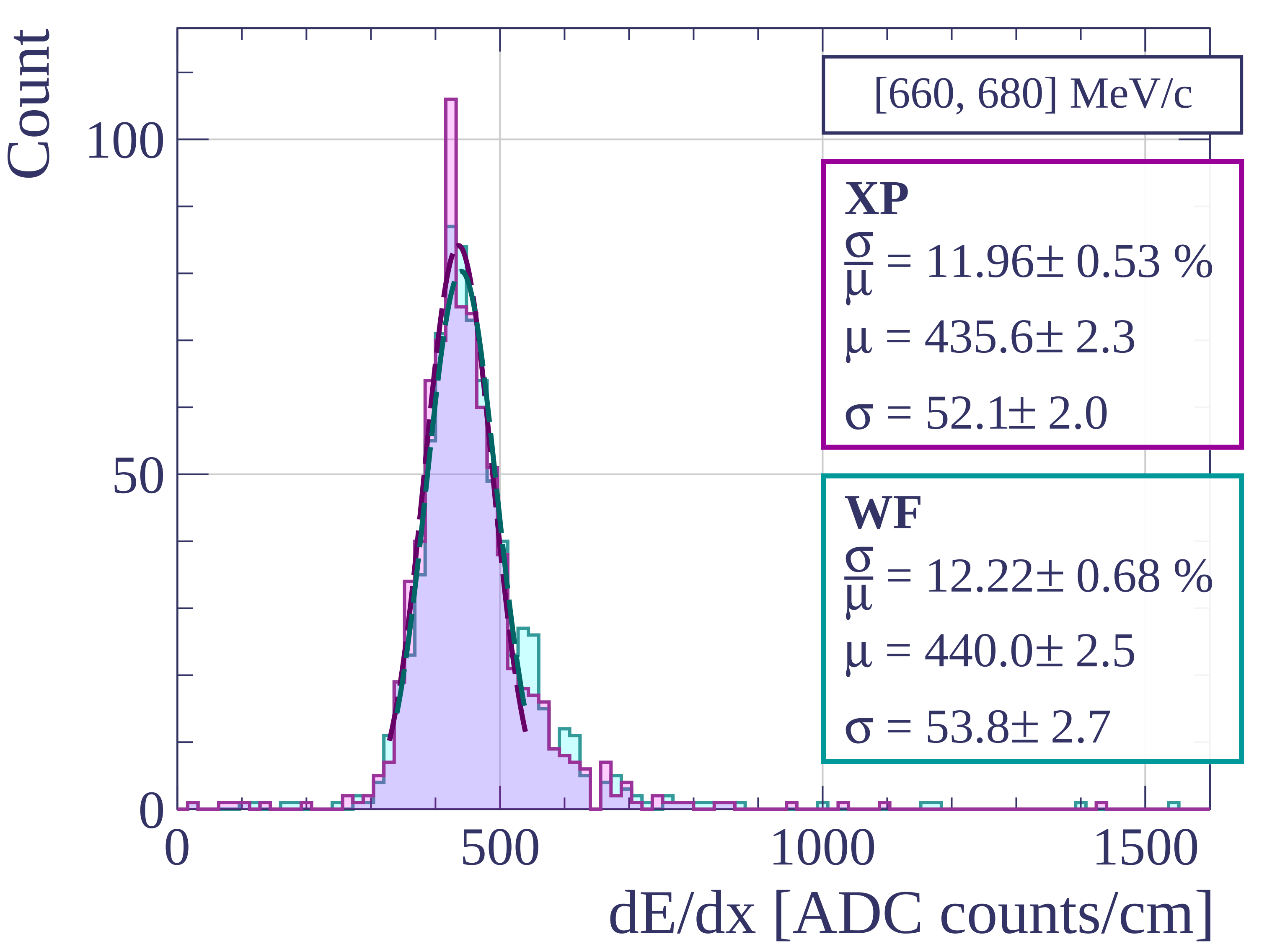
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[600, 620] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.90± 0.63 %
μ = 431.1± 2.5
σ = 55.6± 2.4
WF
$\frac{\sigma}{\mu}$ = 12.96± 0.75 %
μ = 434.9± 2.7
σ = 56.3± 2.9



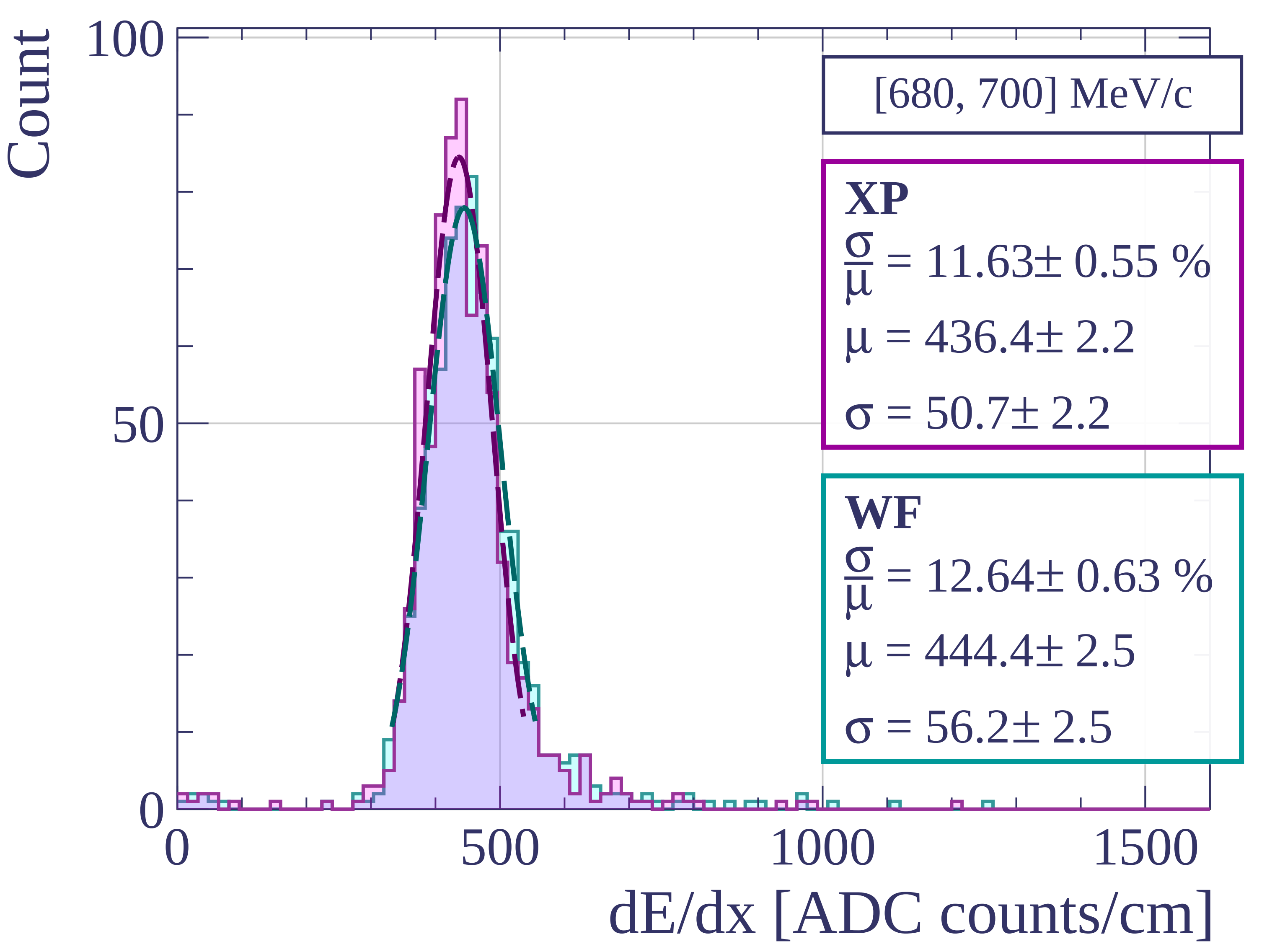
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[620, 640] MeV/c
XP
σ/μ = 12.19± 0.54 %
μ = 430.2± 2.3
σ = 52.4± 2.0
WF
σ/μ = 13.12± 0.64 %
μ = 435.2± 2.5
σ = 57.1± 2.4



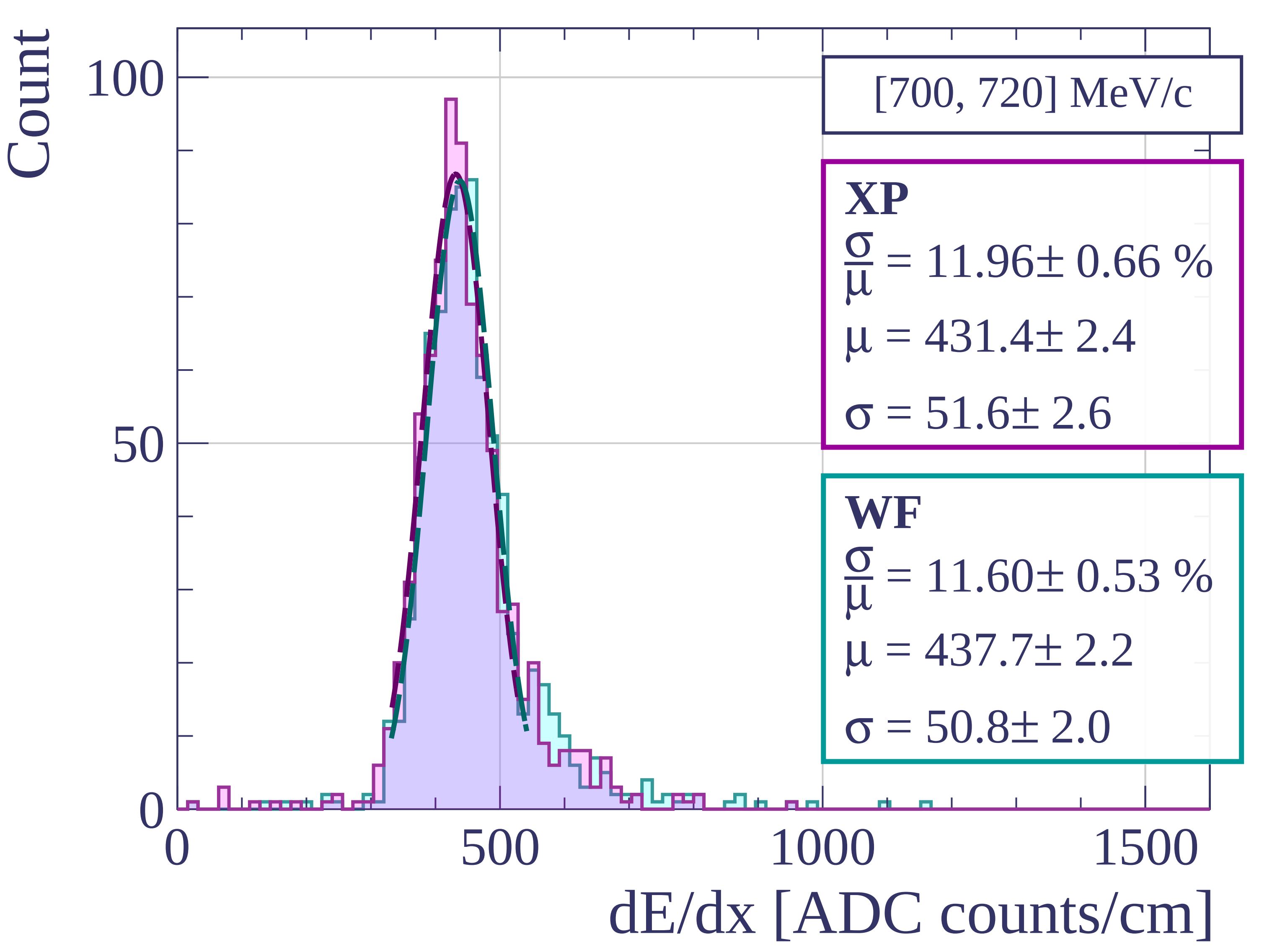
Count
dE/dx [ADC counts/cm]
0
20
40
60
80
0
500
1000
1500
[640, 660] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.68± 0.79 %
μ = 434.5± 2.9
σ = 59.4± 3.1
WF
$\frac{\sigma}{\mu}$ = 12.40± 0.60 %
μ = 440.1± 2.6
σ = 54.6± 2.3

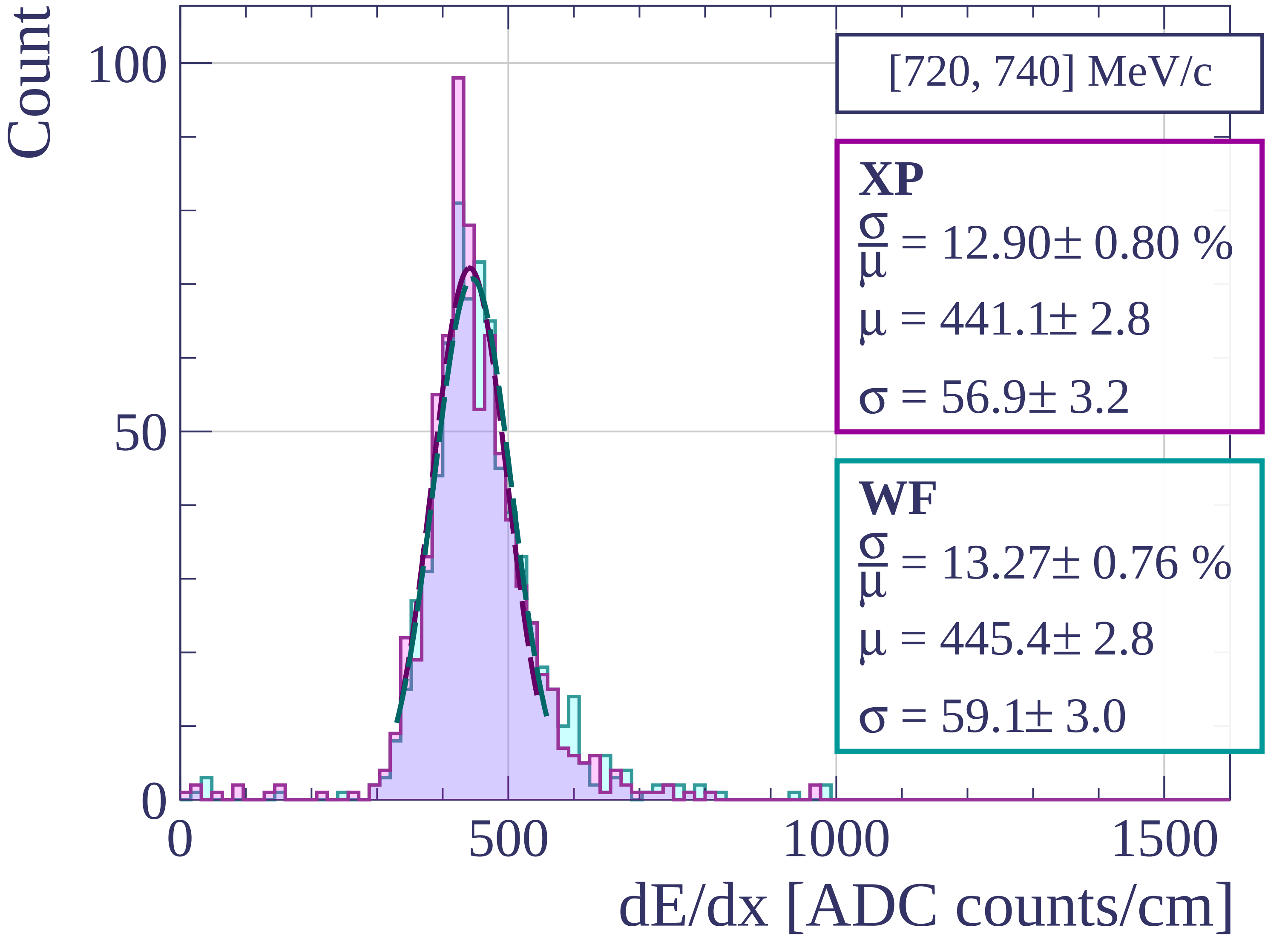


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[660, 680] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.96± 0.53 %
μ = 435.6± 2.3
σ = 52.1± 2.0
WF
$\frac{\sigma}{\mu}$ = 12.22± 0.68 %
μ = 440.0± 2.5
σ = 53.8± 2.7

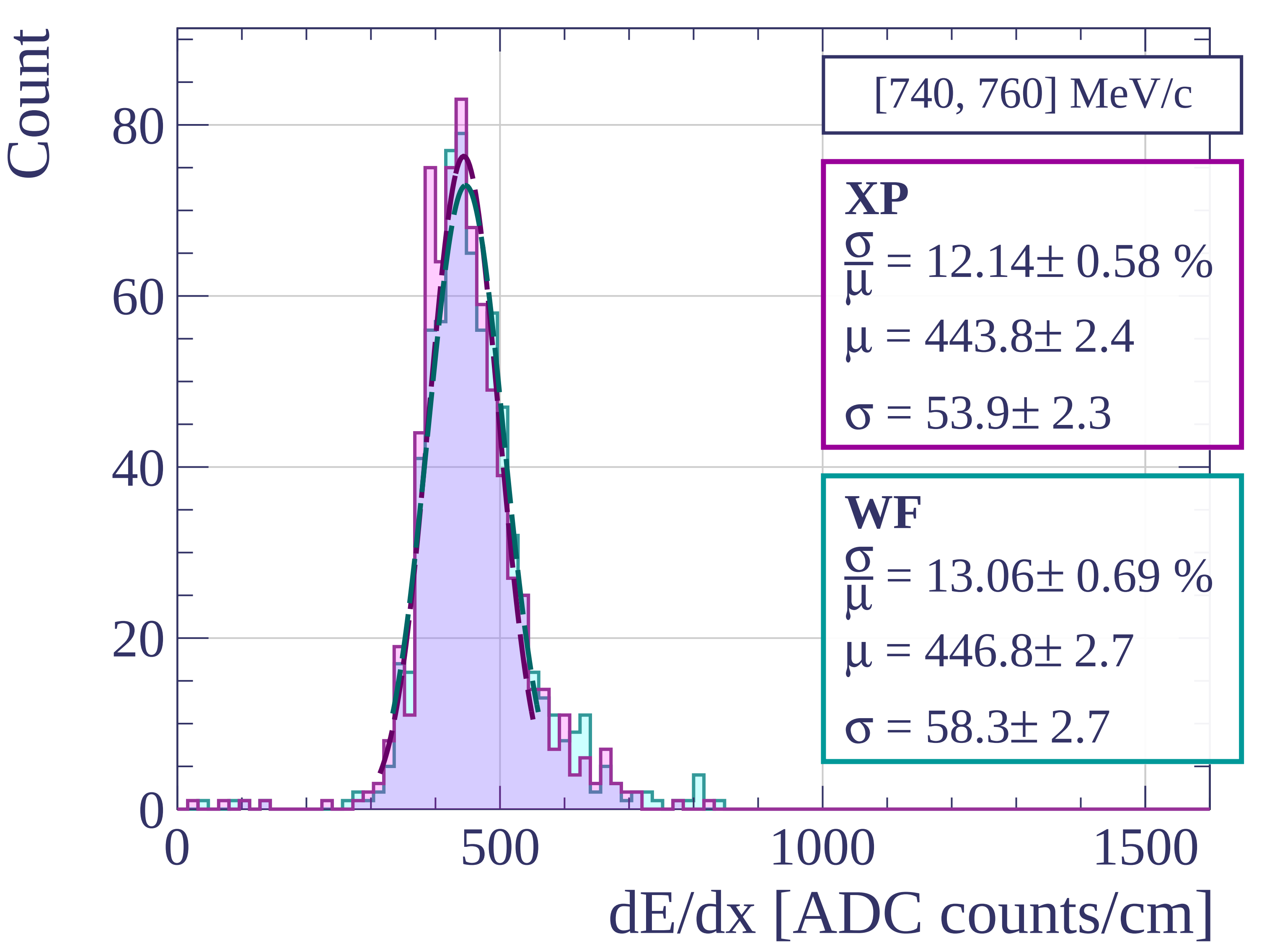


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[680, 700] MeV/c
XP
σ/μ = 11.63± 0.55 %
μ = 436.4± 2.2
σ = 50.7± 2.2
WF
σ/μ = 12.64± 0.63 %
μ = 444.4± 2.5
σ = 56.2± 2.5

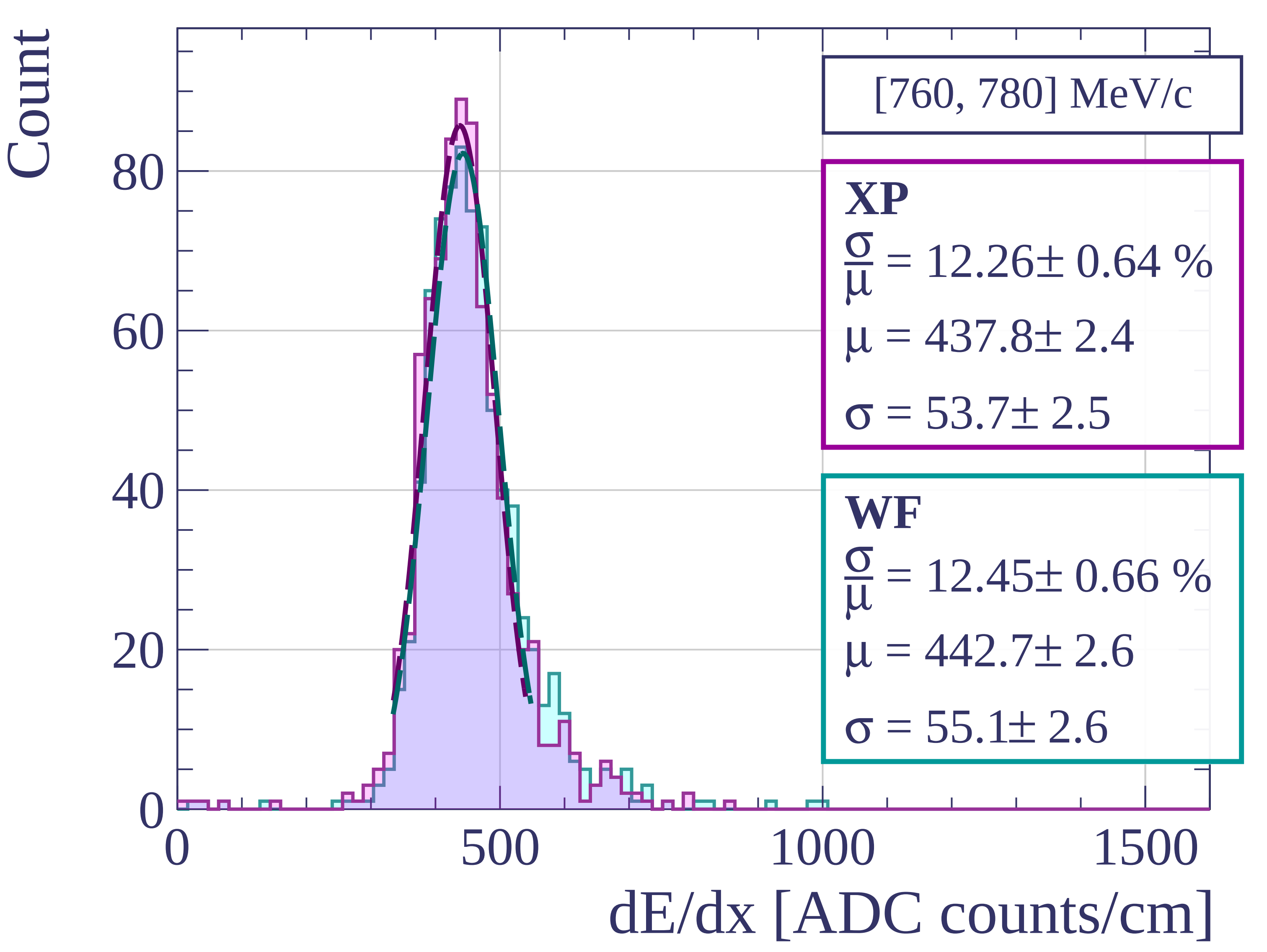

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[700, 720] MeV/c
XP
σ/μ = 11.96± 0.66 %
μ = 431.4± 2.4
σ = 51.6± 2.6
WF
σ/μ = 11.60± 0.53 %
μ = 437.7± 2.2
σ = 50.8± 2.0



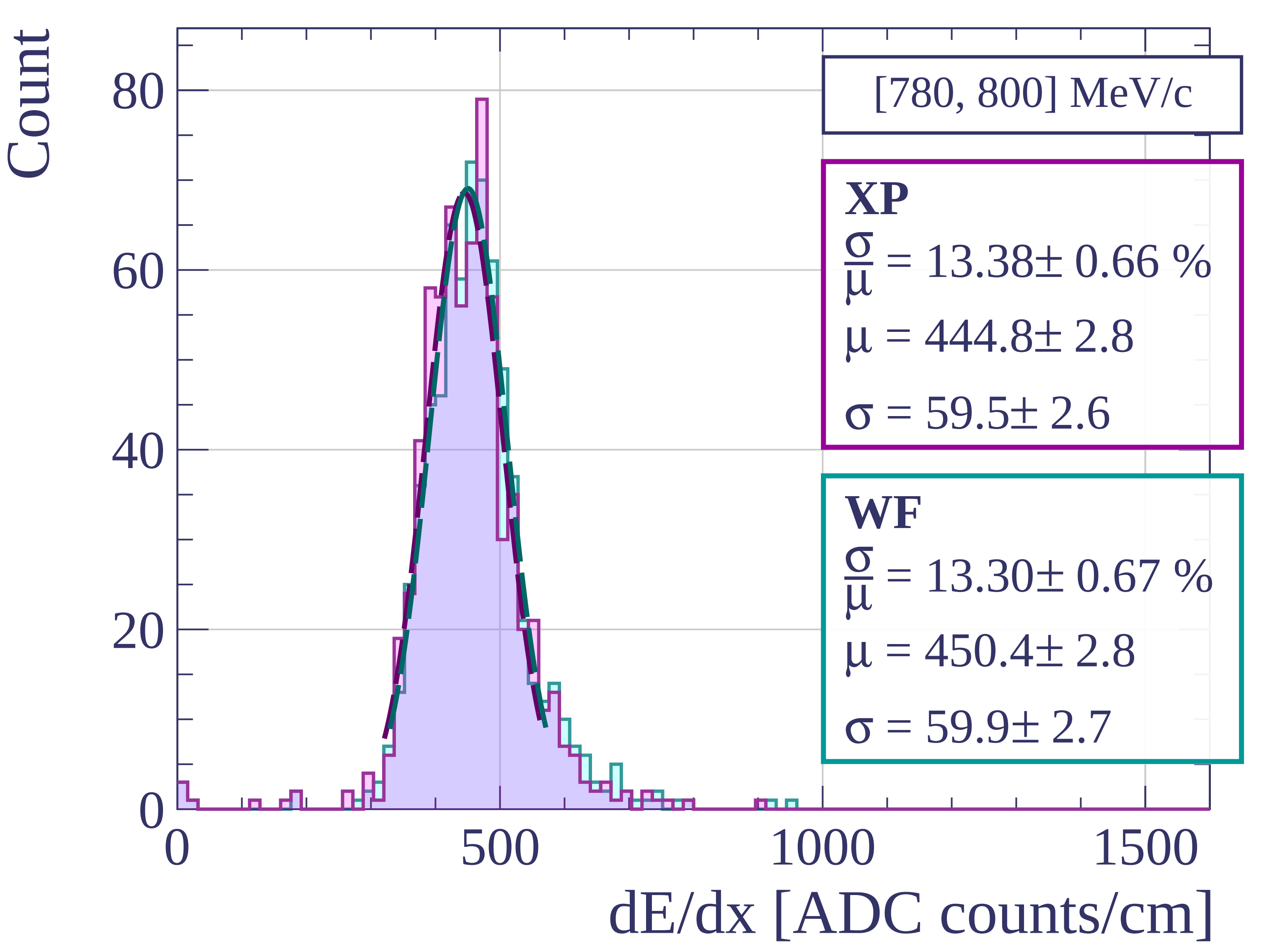
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[720, 740] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.90± 0.80 %
μ = 441.1± 2.8
σ = 56.9± 3.2
WF
$\frac{\sigma}{\mu}$ = 13.27± 0.76 %
μ = 445.4± 2.8
σ = 59.1± 3.0

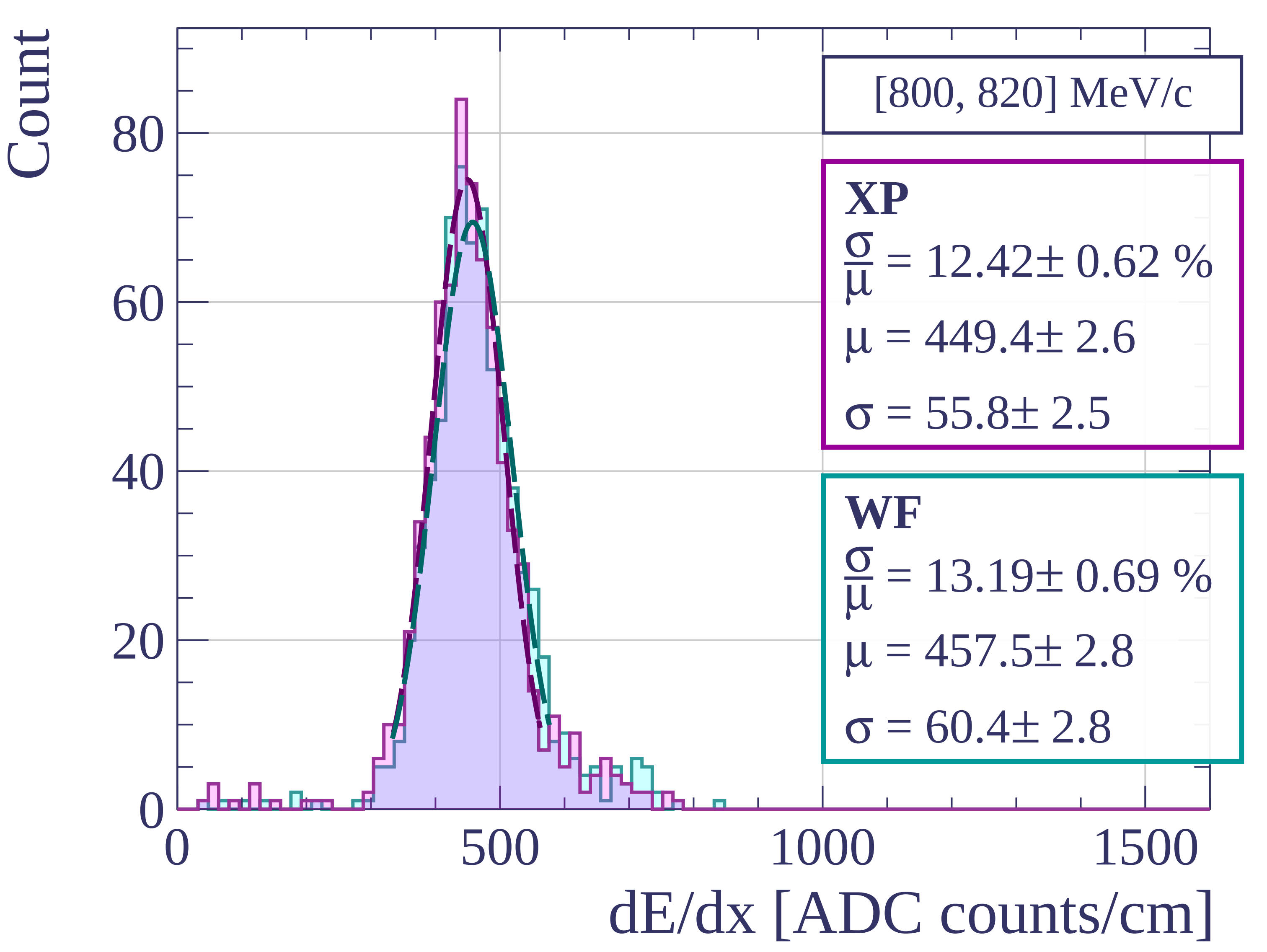


Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[740, 760] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.14± 0.58 %
μ = 443.8± 2.4
σ = 53.9± 2.3
WF
$\frac{\sigma}{\mu}$ = 13.06± 0.69 %
μ = 446.8± 2.7
σ = 58.3± 2.7

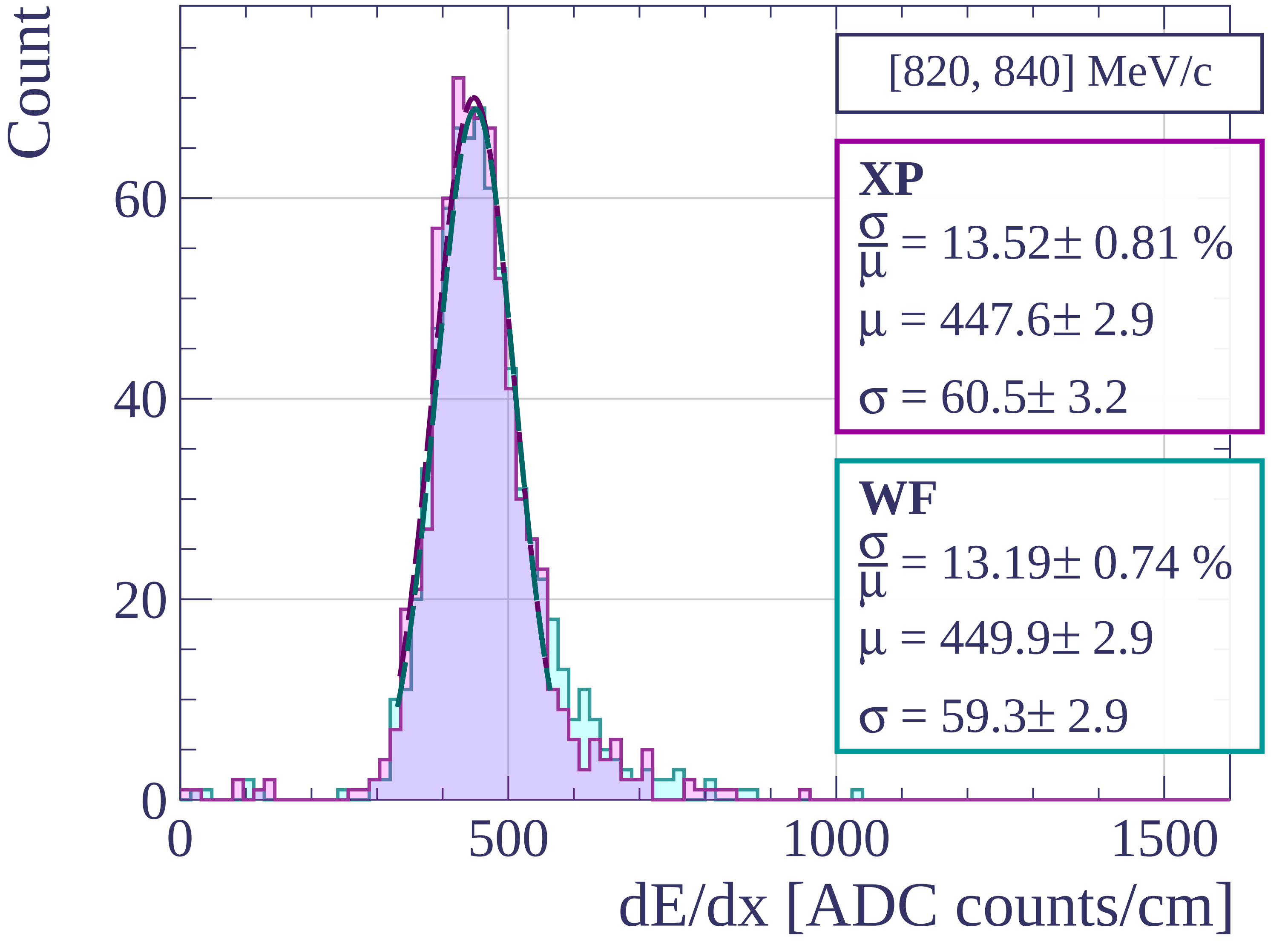


Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[760, 780] MeV/c
XP
σ/μ = 12.26± 0.64 %
μ = 437.8± 2.4
σ = 53.7± 2.5
WF
σ/μ = 12.45± 0.66 %
μ = 442.7± 2.6
σ = 55.1± 2.6

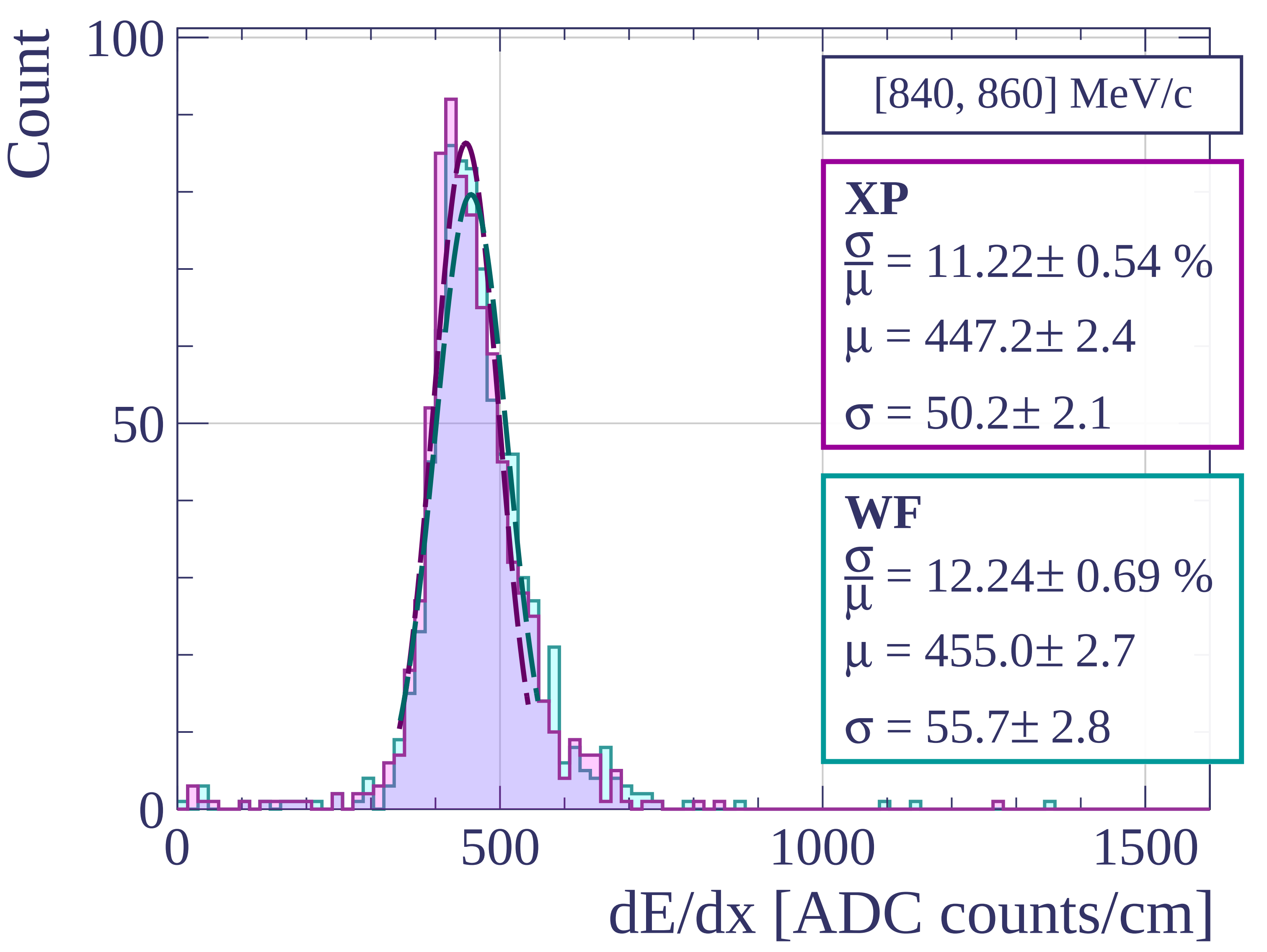

Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[780, 800] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.38± 0.66 %
μ = 444.8± 2.8
σ = 59.5± 2.6
WF
$\frac{\sigma}{\mu}$ = 13.30± 0.67 %
μ = 450.4± 2.8
σ = 59.9± 2.7



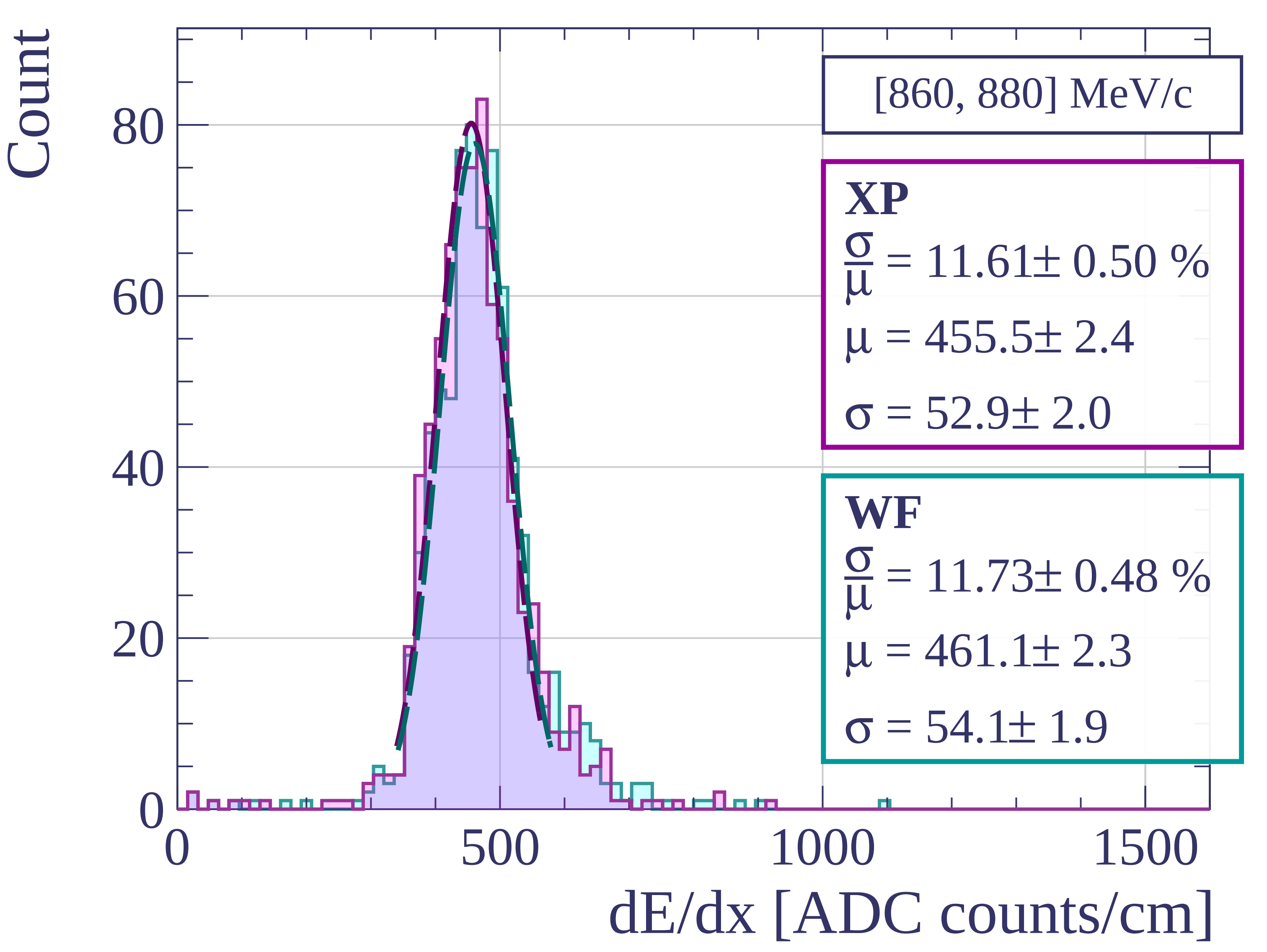
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[800, 820] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.42± 0.62 %
μ = 449.4± 2.6
σ = 55.8± 2.5
WF
$\frac{\sigma}{\mu}$ = 13.19± 0.69 %
μ = 457.5± 2.8
σ = 60.4± 2.8



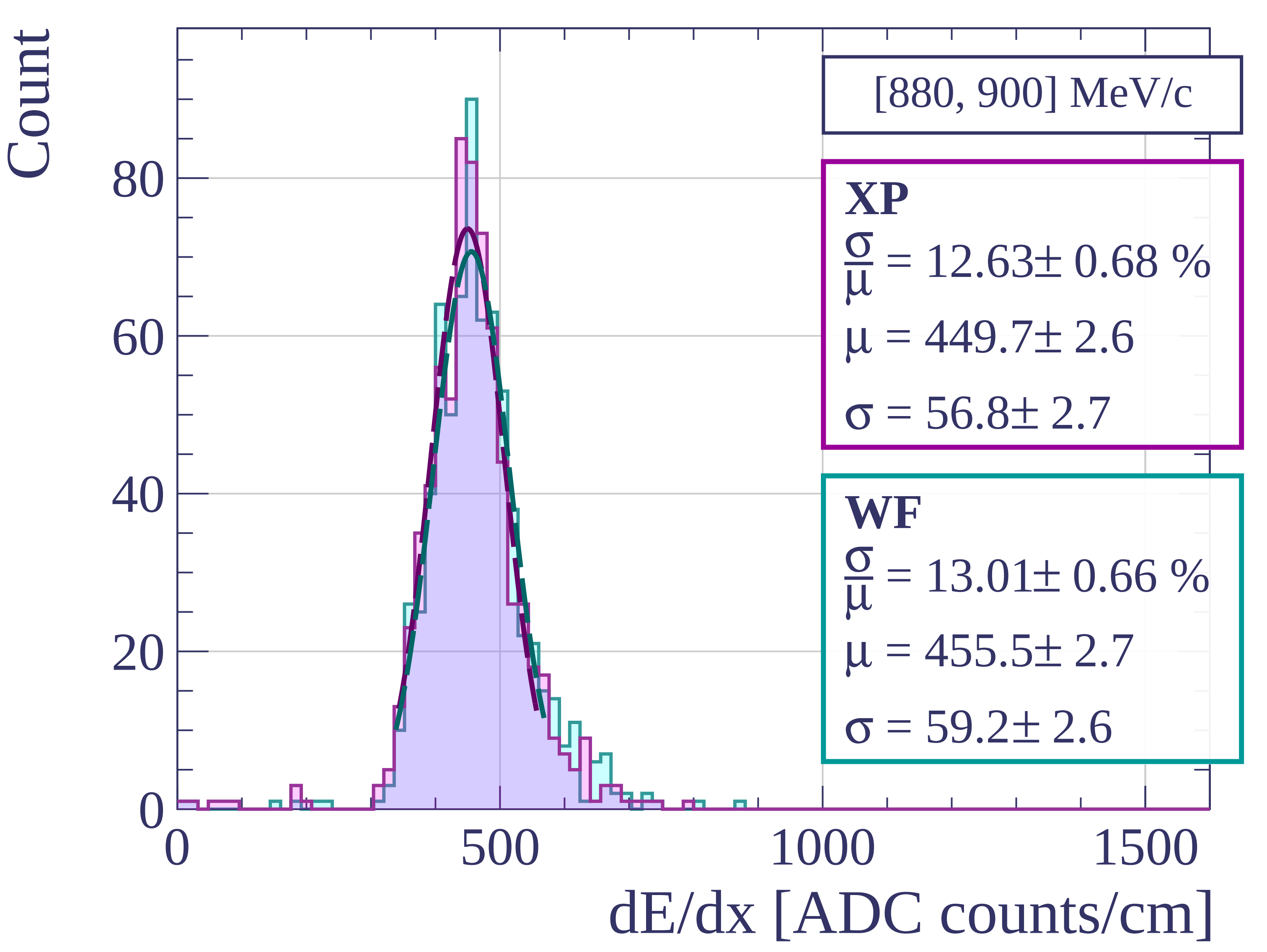
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[820, 840] MeV/c
XP
σ/μ = 13.52± 0.81 %
μ = 447.6± 2.9
σ = 60.5± 3.2
WF
σ/μ = 13.19± 0.74 %
μ = 449.9± 2.9
σ = 59.3± 2.9



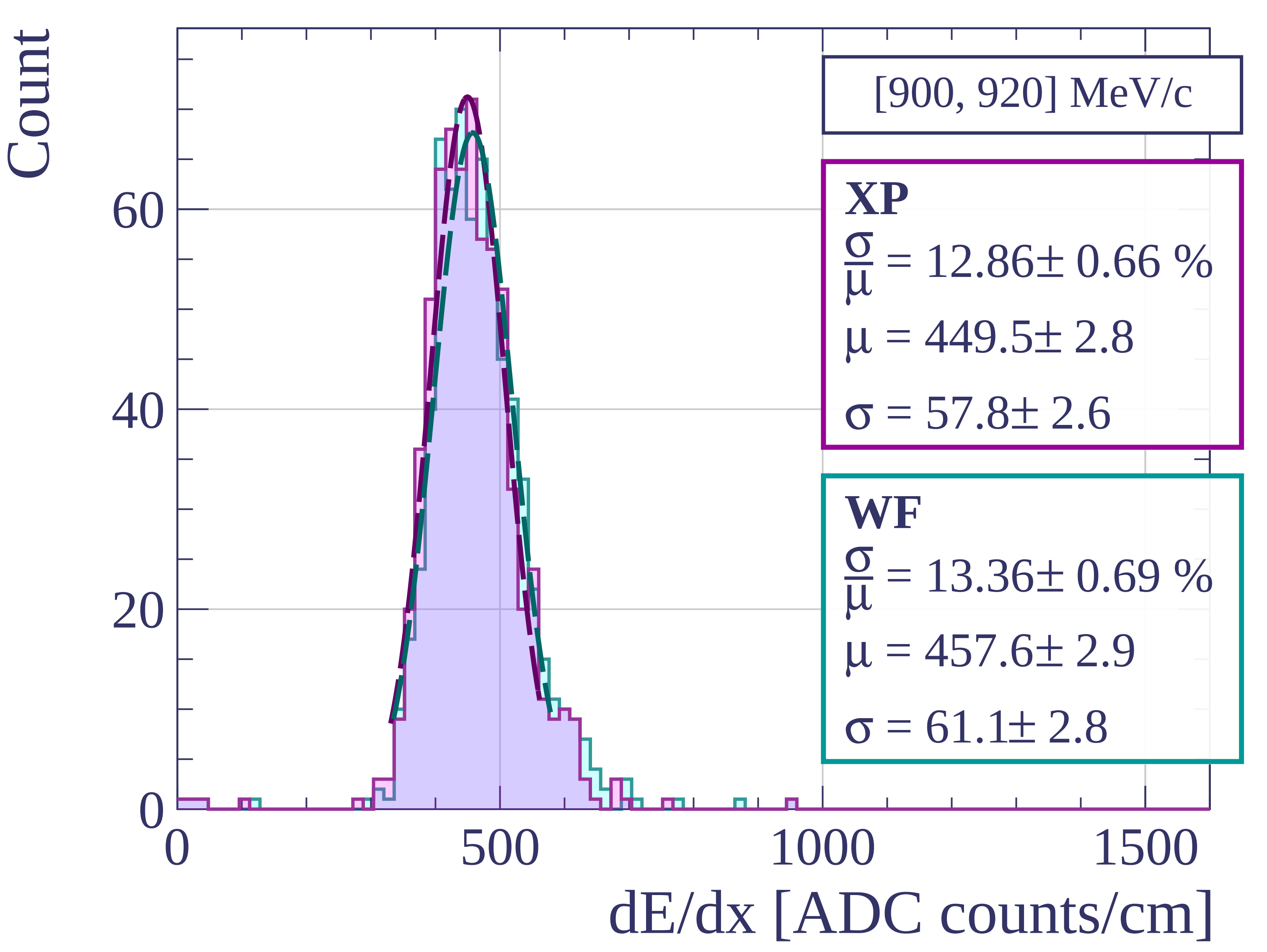
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[840, 860] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.22± 0.54 %
μ = 447.2± 2.4
σ = 50.2± 2.1
WF
$\frac{\sigma}{\mu}$ = 12.24± 0.69 %
μ = 455.0± 2.7
σ = 55.7± 2.8

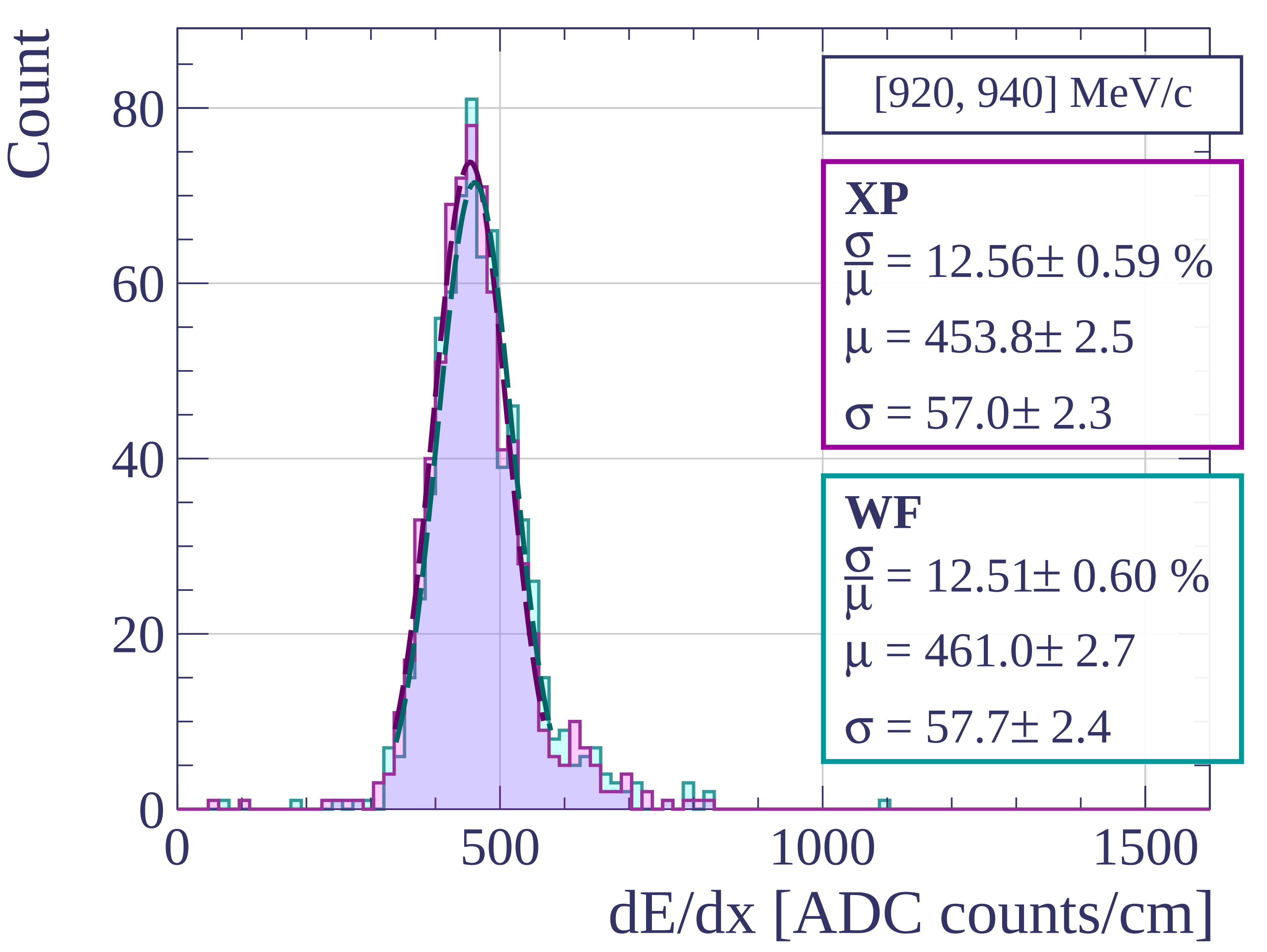


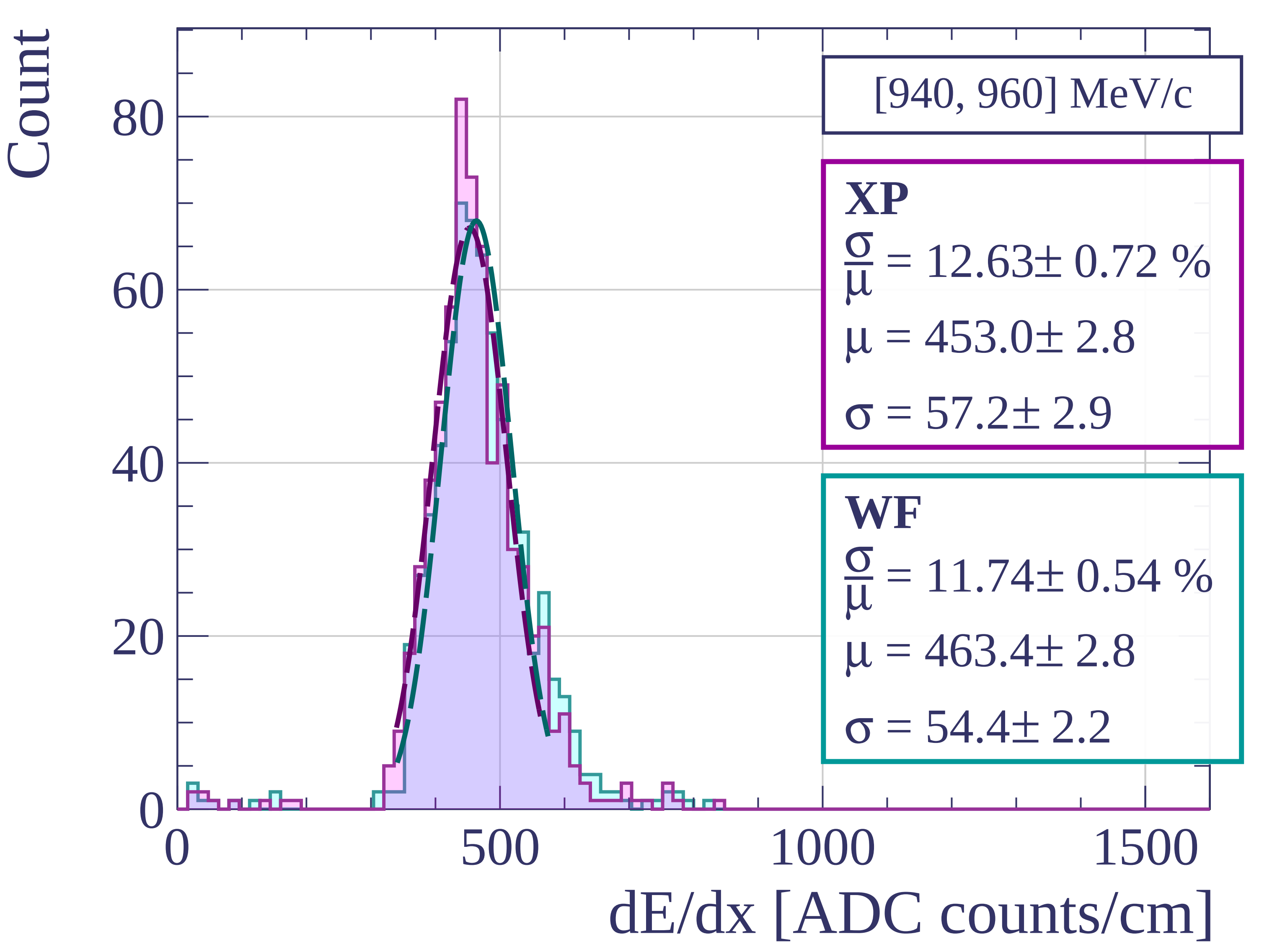
Count
dE/dx [ADC counts/cm]
0
20
40
60
80
0
500
1000
1500
[860, 880] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.61± 0.50 %
μ = 455.5± 2.4
σ = 52.9± 2.0
WF
$\frac{\sigma}{\mu}$ = 11.73± 0.48 %
μ = 461.1± 2.3
σ = 54.1± 1.9



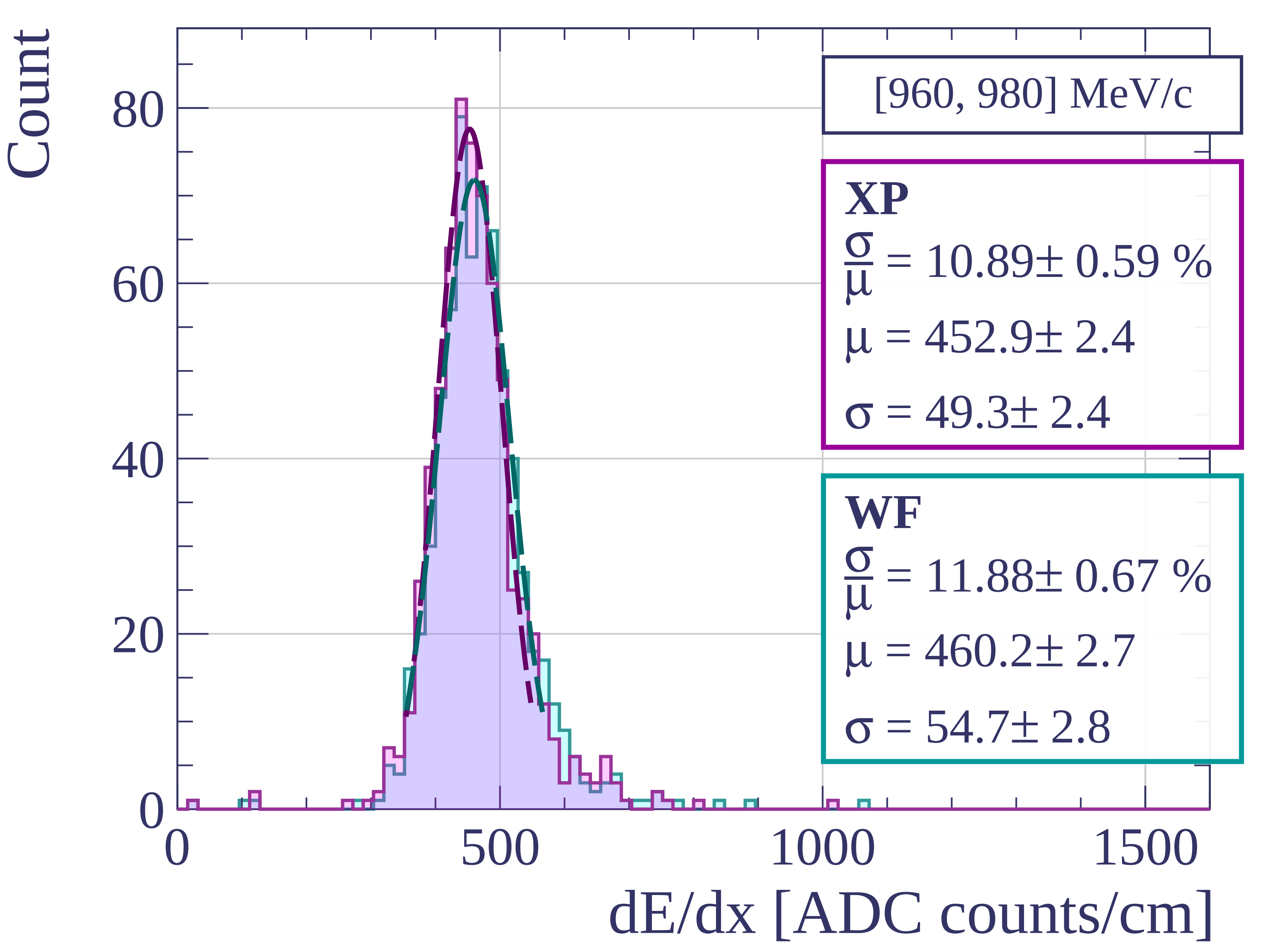
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[880, 900] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.63± 0.68 %
μ = 449.7± 2.6
σ = 56.8± 2.7
WF
$\frac{\sigma}{\mu}$ = 13.01± 0.66 %
μ = 455.5± 2.7
σ = 59.2± 2.6


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[900, 920] MeV/c
XP
σ/μ = 12.86± 0.66 %
μ = 449.5± 2.8
σ = 57.8± 2.6
WF
σ/μ = 13.36± 0.69 %
μ = 457.6± 2.9
σ = 61.1± 2.8

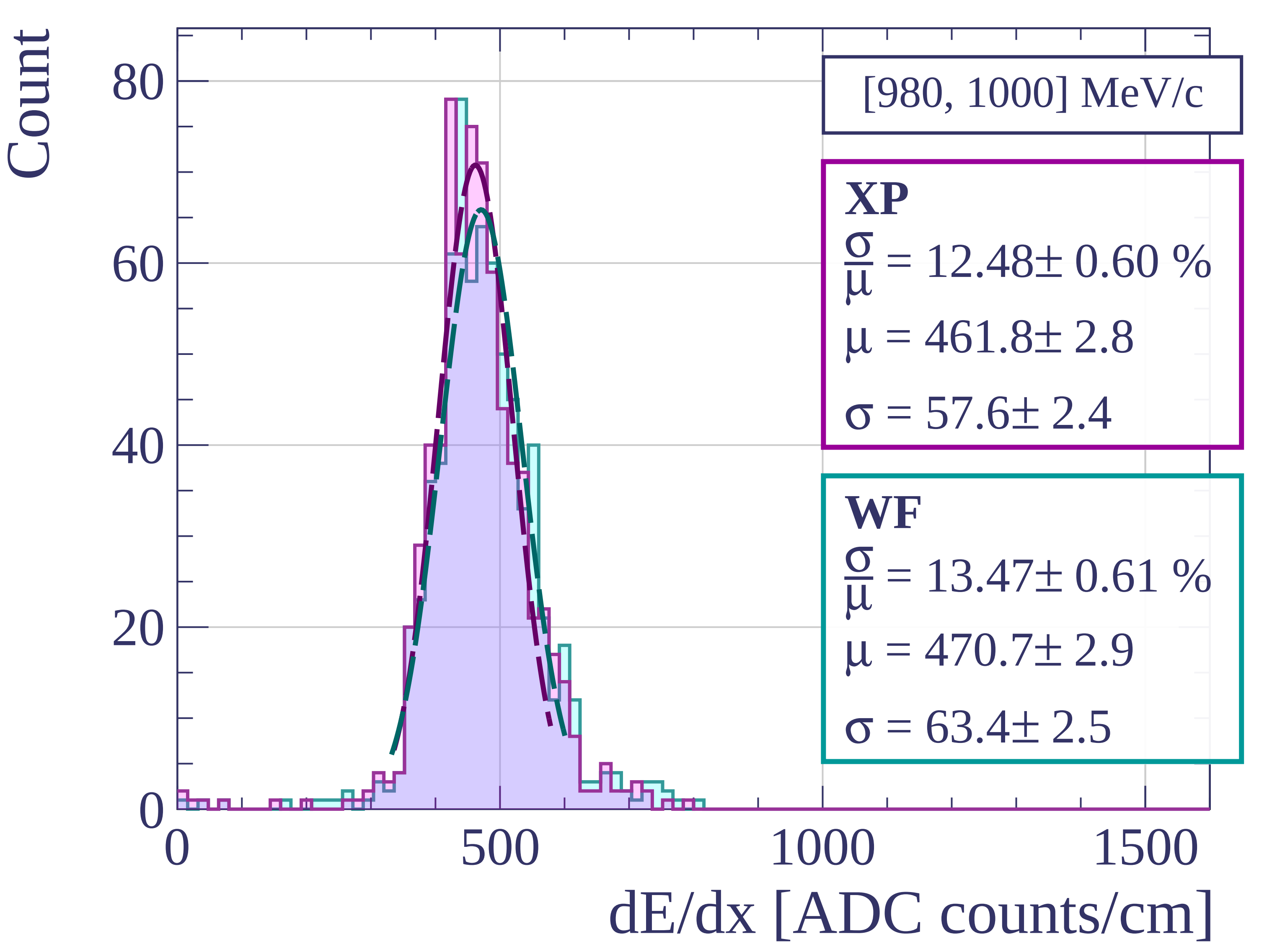

Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[920, 940] MeV/c
XP
σ/μ = 12.56± 0.59 %
μ = 453.8± 2.5
σ = 57.0± 2.3
WF
σ/μ = 12.51± 0.60 %
μ = 461.0± 2.7
σ = 57.7± 2.4



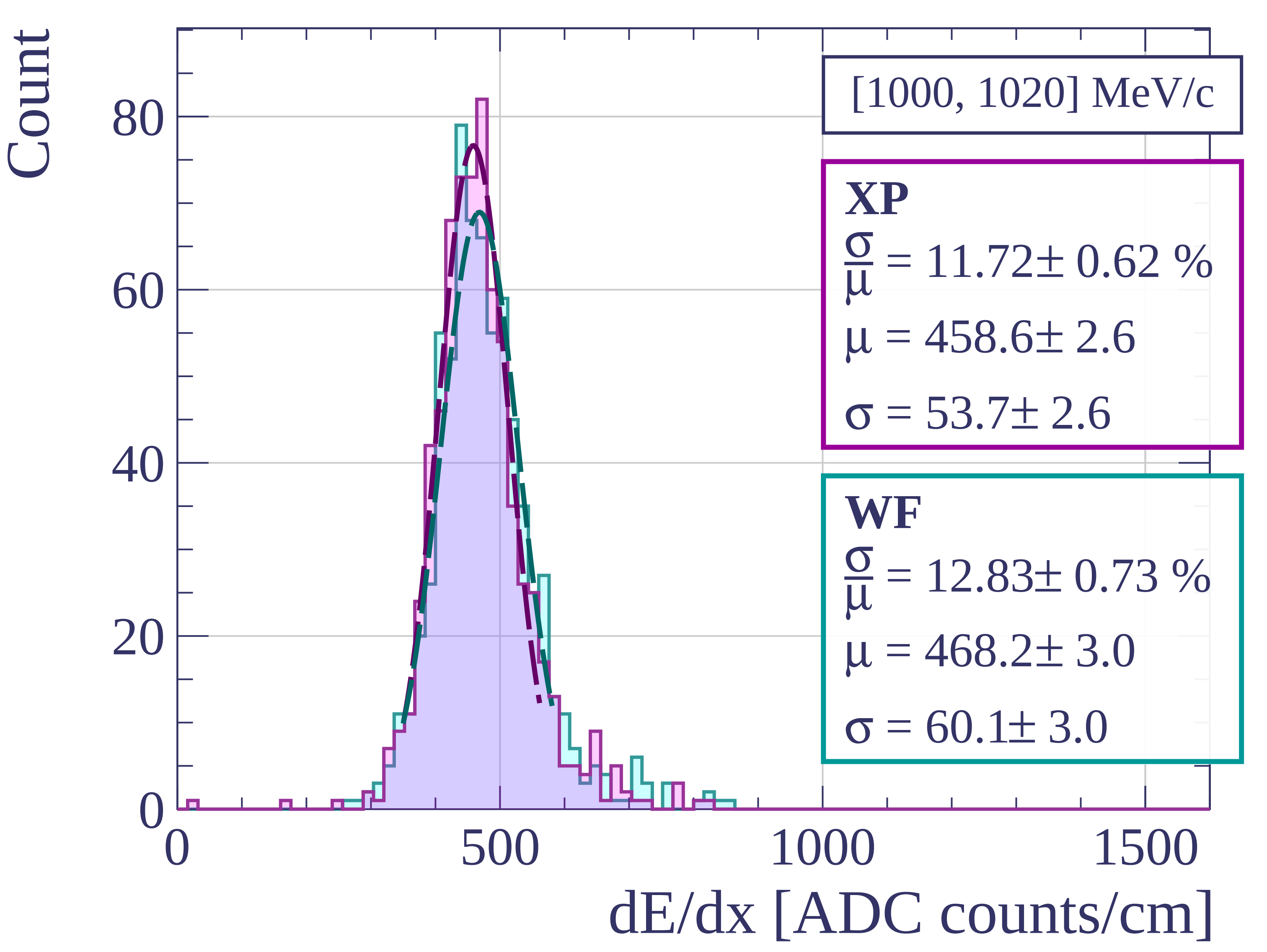
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[940, 960] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.63± 0.72 %
μ = 453.0± 2.8
σ = 57.2± 2.9
WF
$\frac{\sigma}{\mu}$ = 11.74± 0.54 %
μ = 463.4± 2.8
σ = 54.4± 2.2

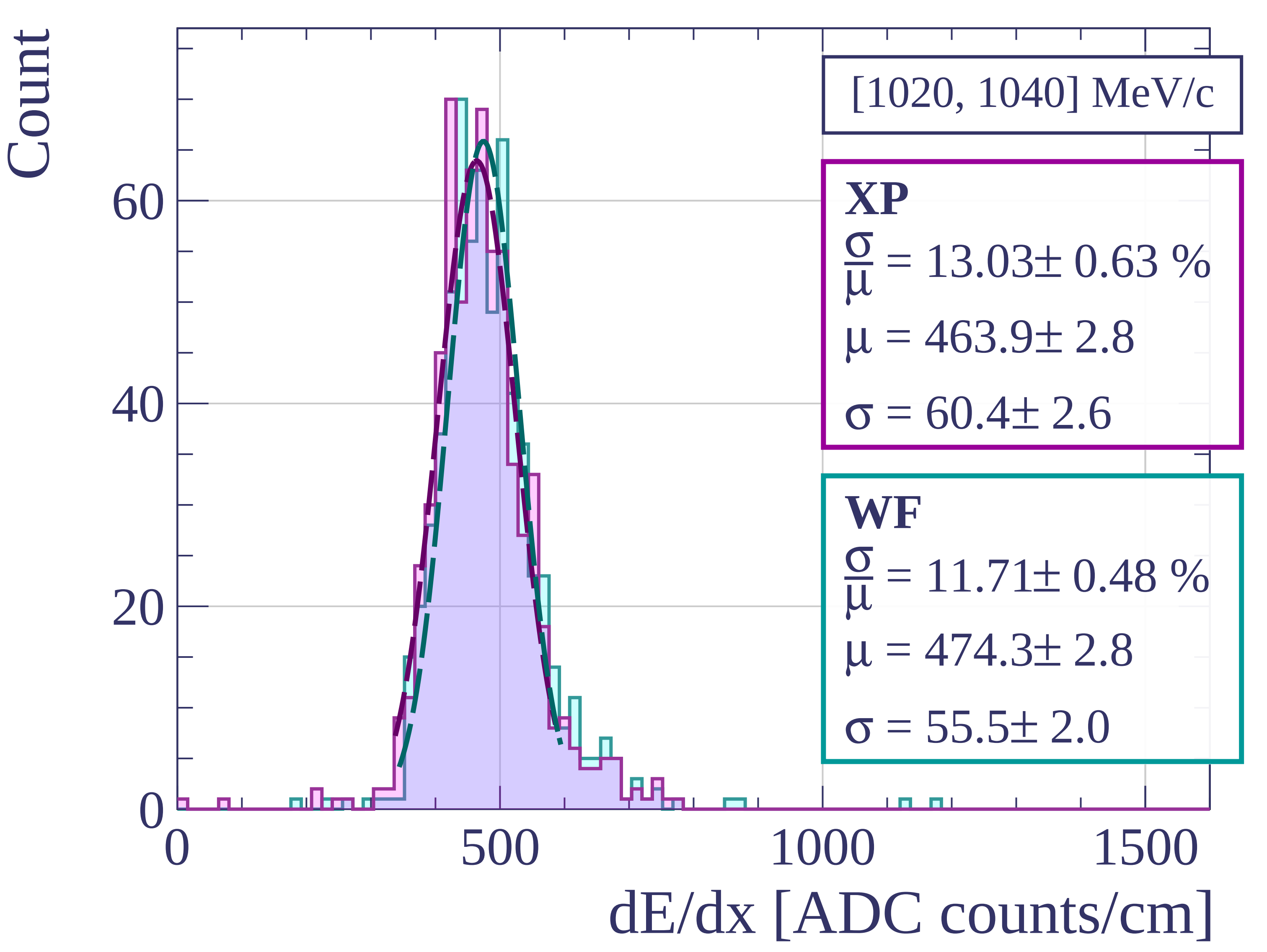


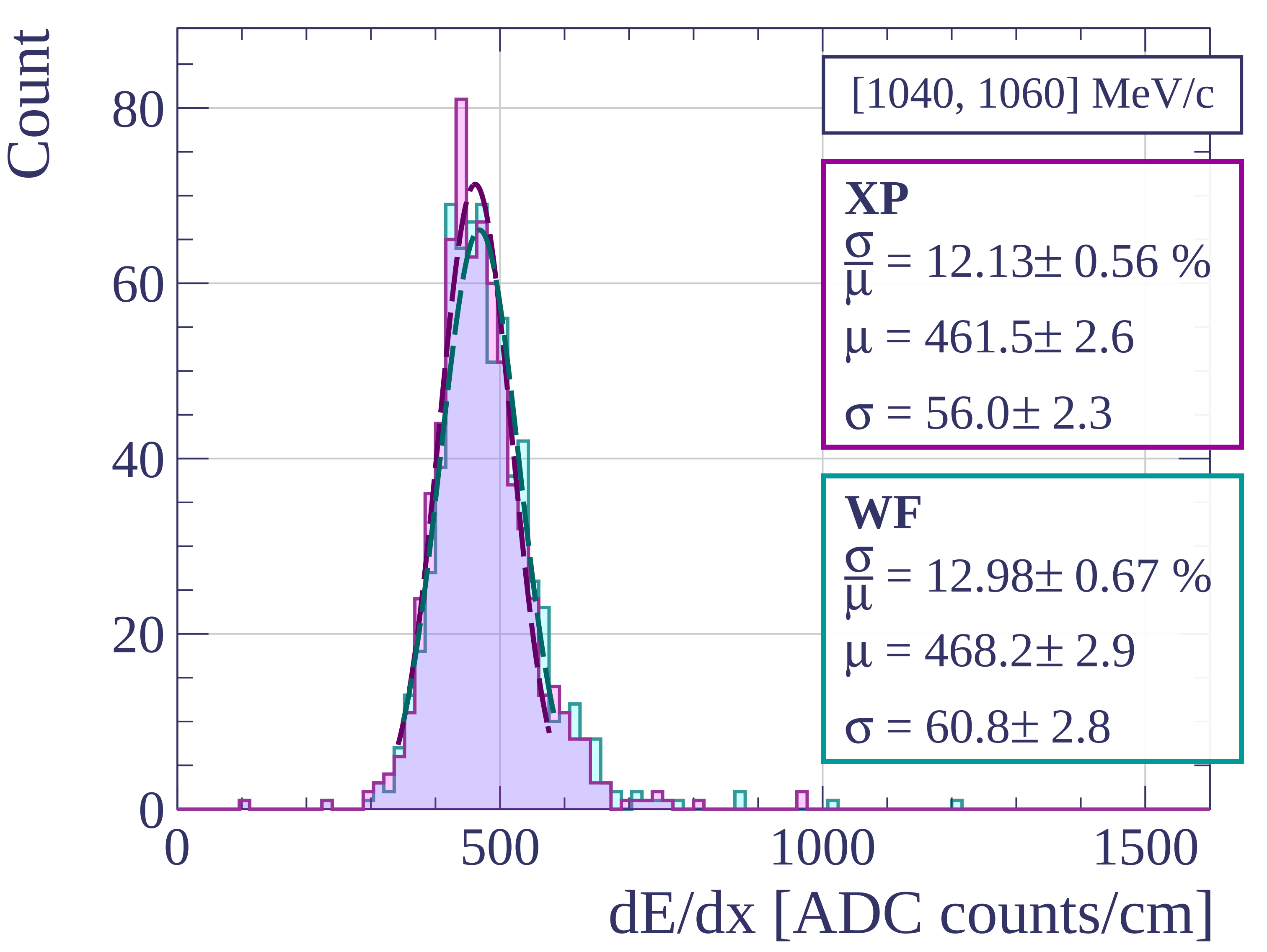
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[960, 980] MeV/c
XP
σ/μ = 10.89± 0.59 %
μ = 452.9± 2.4
σ = 49.3± 2.4
WF
σ/μ = 11.88± 0.67 %
μ = 460.2± 2.7
σ = 54.7± 2.8



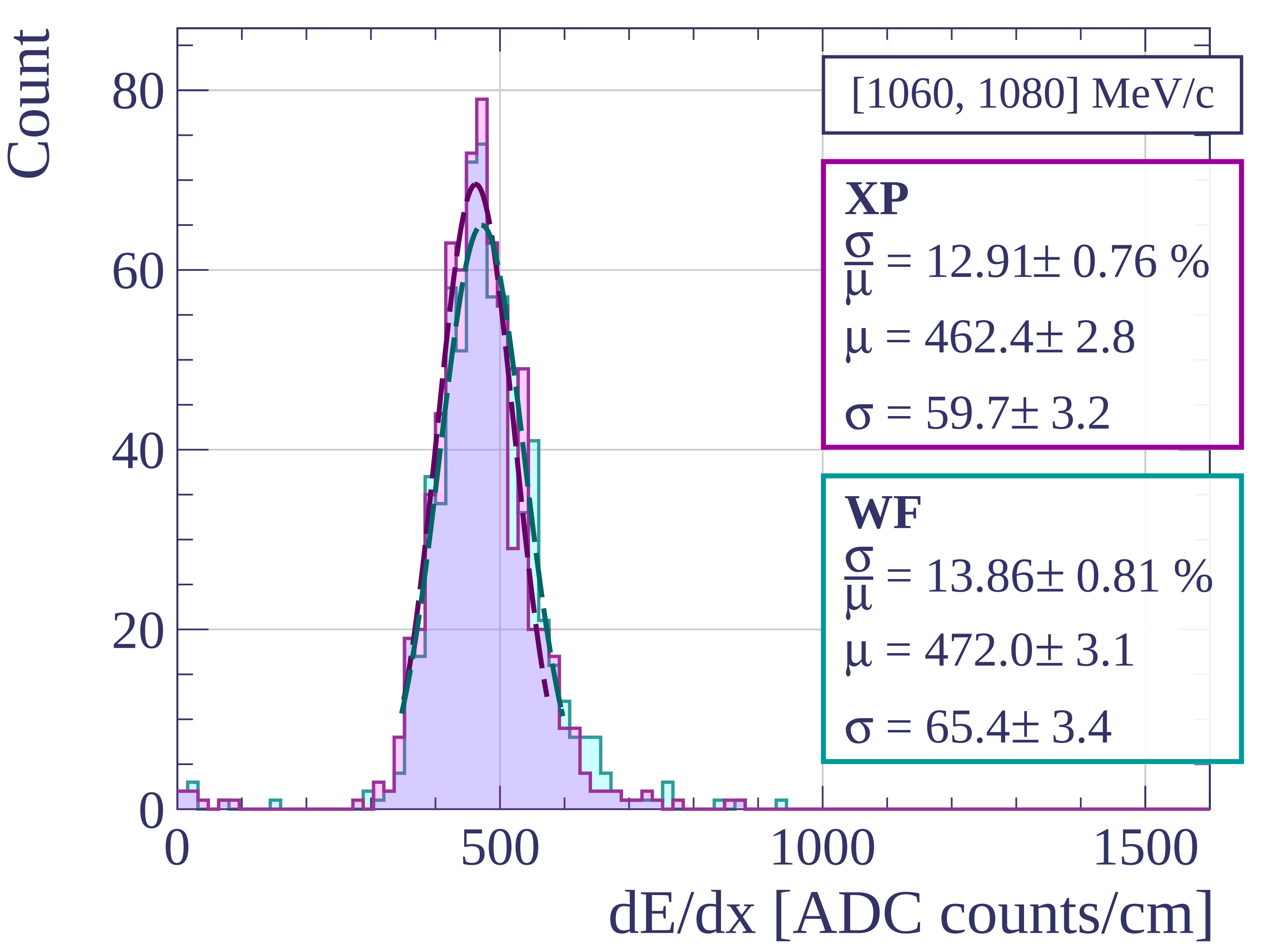
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[980, 1000] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.48± 0.60 %
μ = 461.8± 2.8
σ = 57.6± 2.4
WF
$\frac{\sigma}{\mu}$ = 13.47± 0.61 %
μ = 470.7± 2.9
σ = 63.4± 2.5


Count
dE/dx [ADC counts/cm]
0
20
40
60
80
0
500
1000
1500
[1000, 1020] MeV/c
XP
σ/μ = 11.72± 0.62 %
μ = 458.6± 2.6
σ = 53.7± 2.6
WF
σ/μ = 12.83± 0.73 %
μ = 468.2± 3.0
σ = 60.1± 3.0

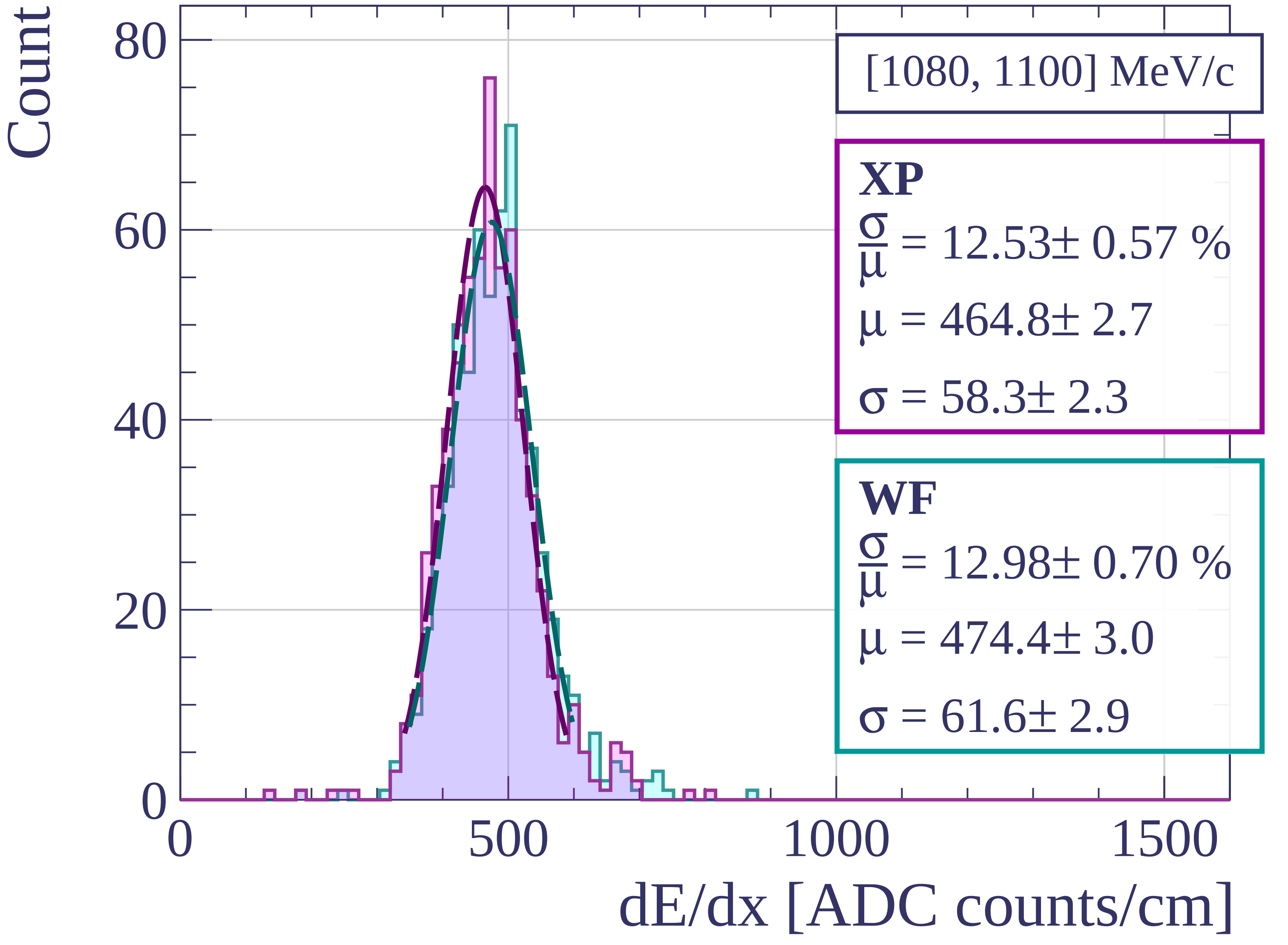

Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1020, 1040] MeV/c
XP
σ/μ = 13.03± 0.63 %
μ = 463.9± 2.8
σ = 60.4± 2.6
WF
σ/μ = 11.71± 0.48 %
μ = 474.3± 2.8
σ = 55.5± 2.0



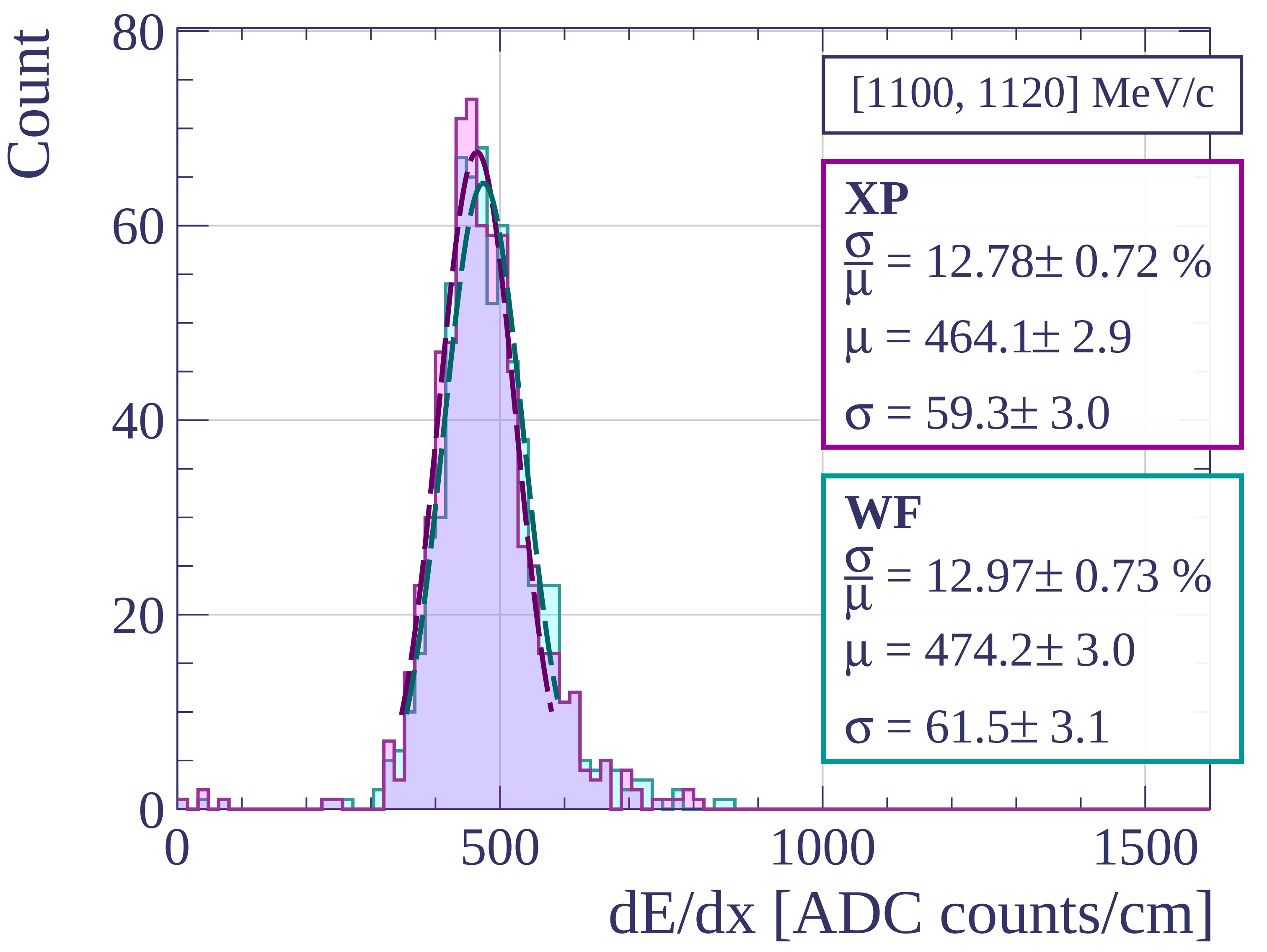
Count
0
20
40
60
80
0
500
1000
1500
dE/dx [ADC counts/cm]
[1040, 1060] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.13± 0.56 %
μ = 461.5± 2.6
σ = 56.0± 2.3
WF
$\frac{\sigma}{\mu}$ = 12.98± 0.67 %
μ = 468.2± 2.9
σ = 60.8± 2.8



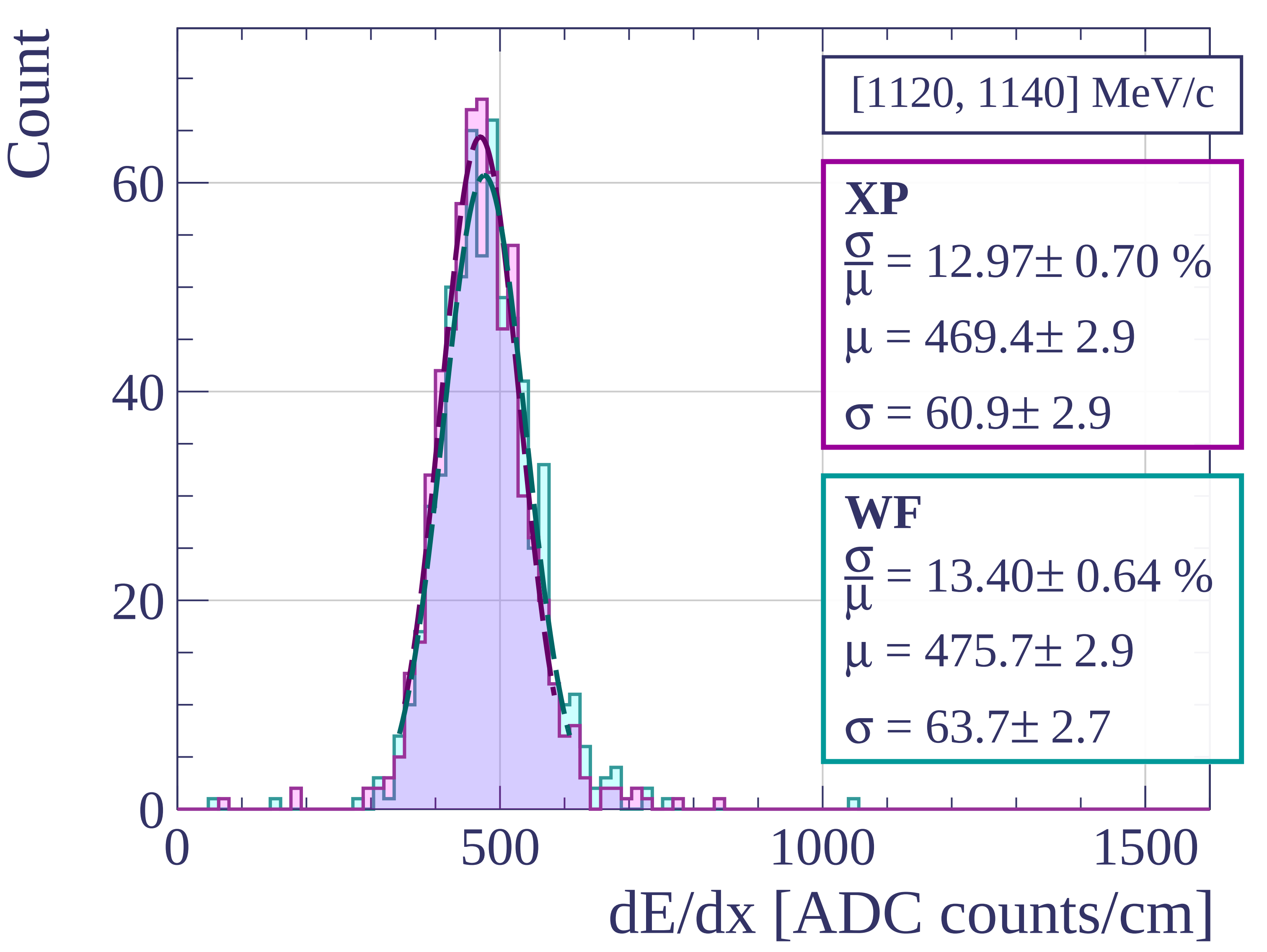
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1060, 1080] MeV/c
XP
σ/μ = 12.91± 0.76 %
μ = 462.4± 2.8
σ = 59.7± 3.2
WF
σ/μ = 13.86± 0.81 %
μ = 472.0± 3.1
σ = 65.4± 3.4



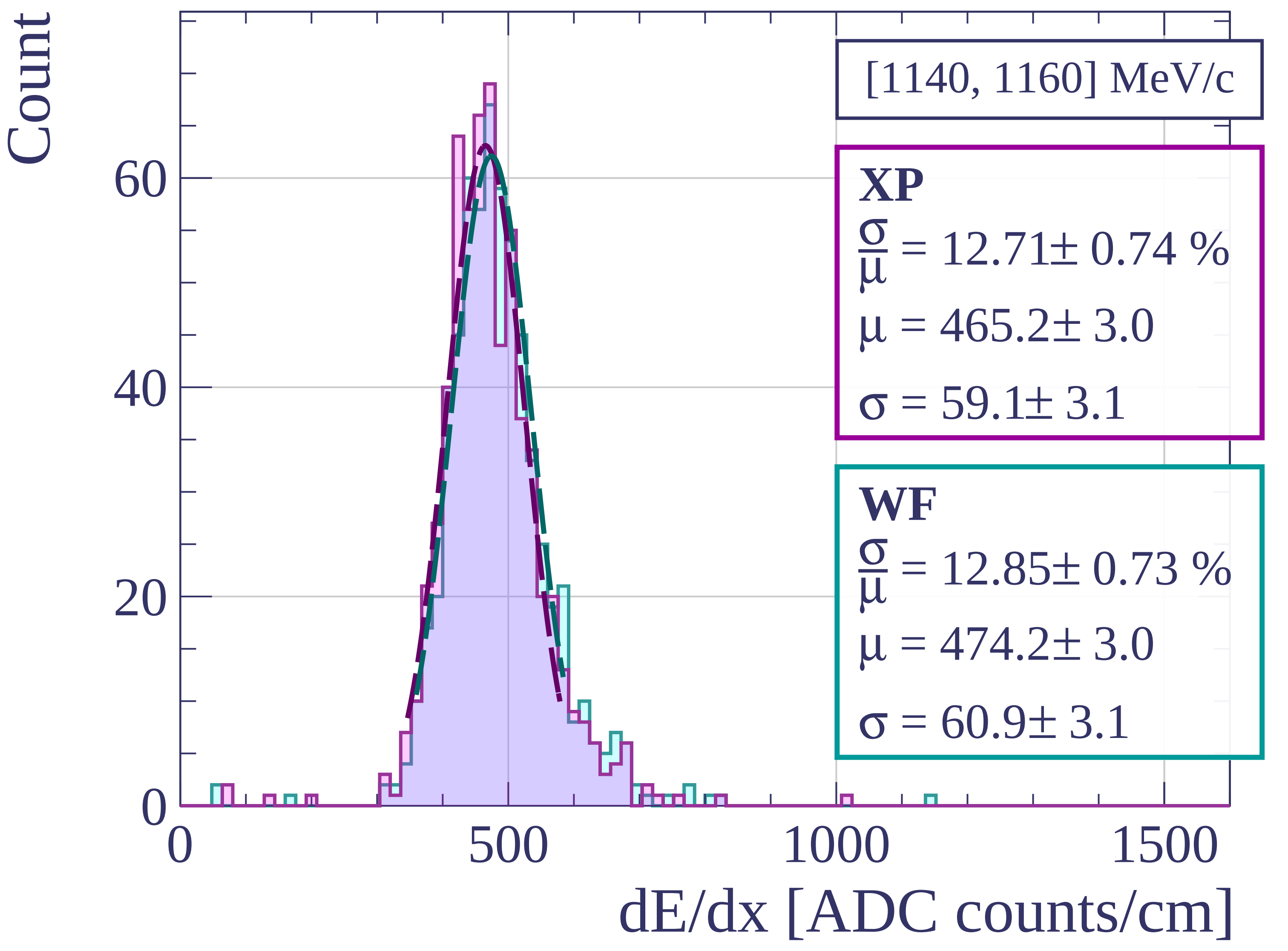
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
20
40
60
80
[1080, 1100] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.53± 0.57 %
μ = 464.8± 2.7
σ = 58.3± 2.3
WF
$\frac{\sigma}{\mu}$ = 12.98± 0.70 %
μ = 474.4± 3.0
σ = 61.6± 2.9



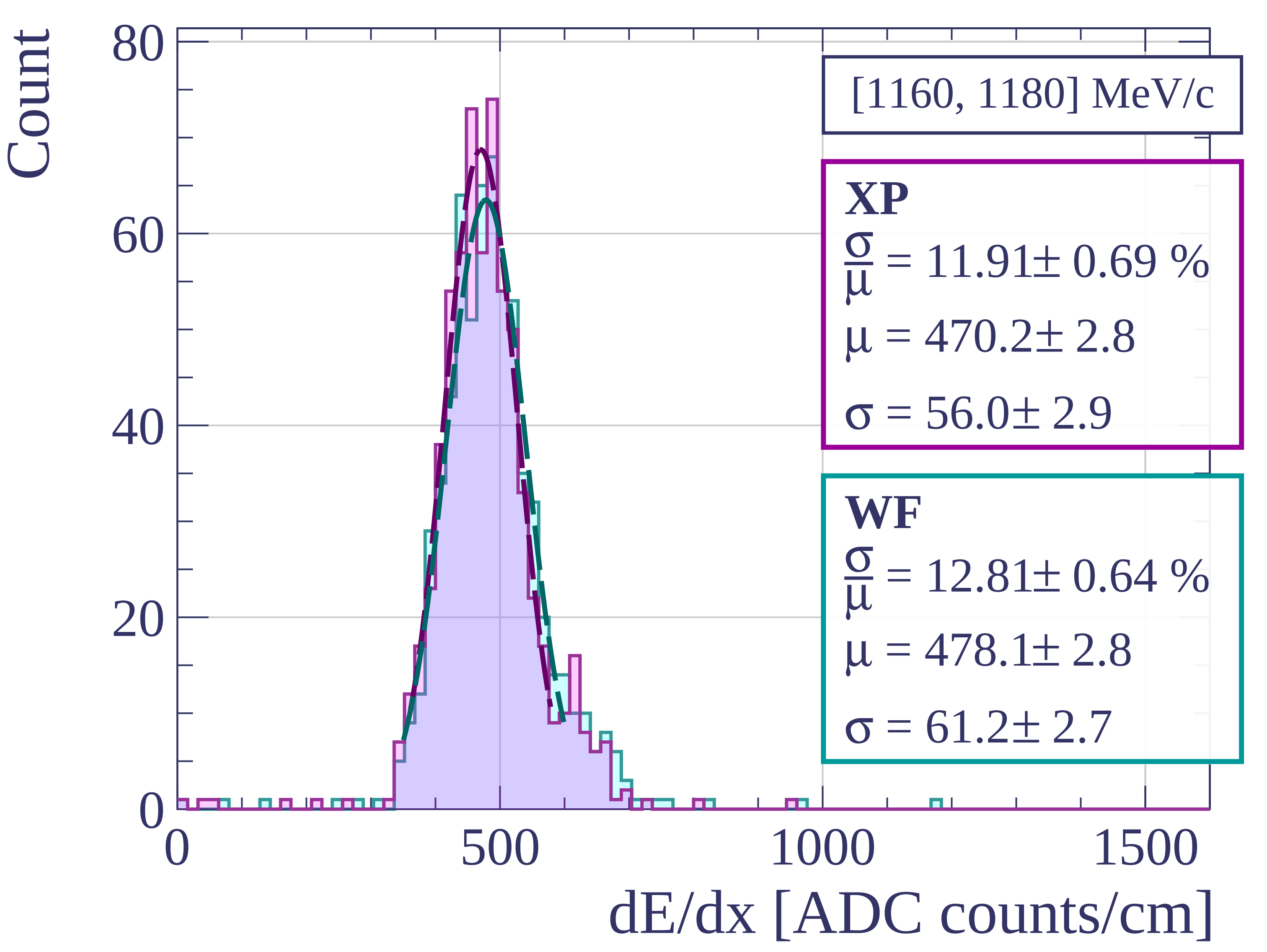
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1100, 1120] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.78± 0.72 %
μ = 464.1± 2.9
σ = 59.3± 3.0
WF
$\frac{\sigma}{\mu}$ = 12.97± 0.73 %
μ = 474.2± 3.0
σ = 61.5± 3.1



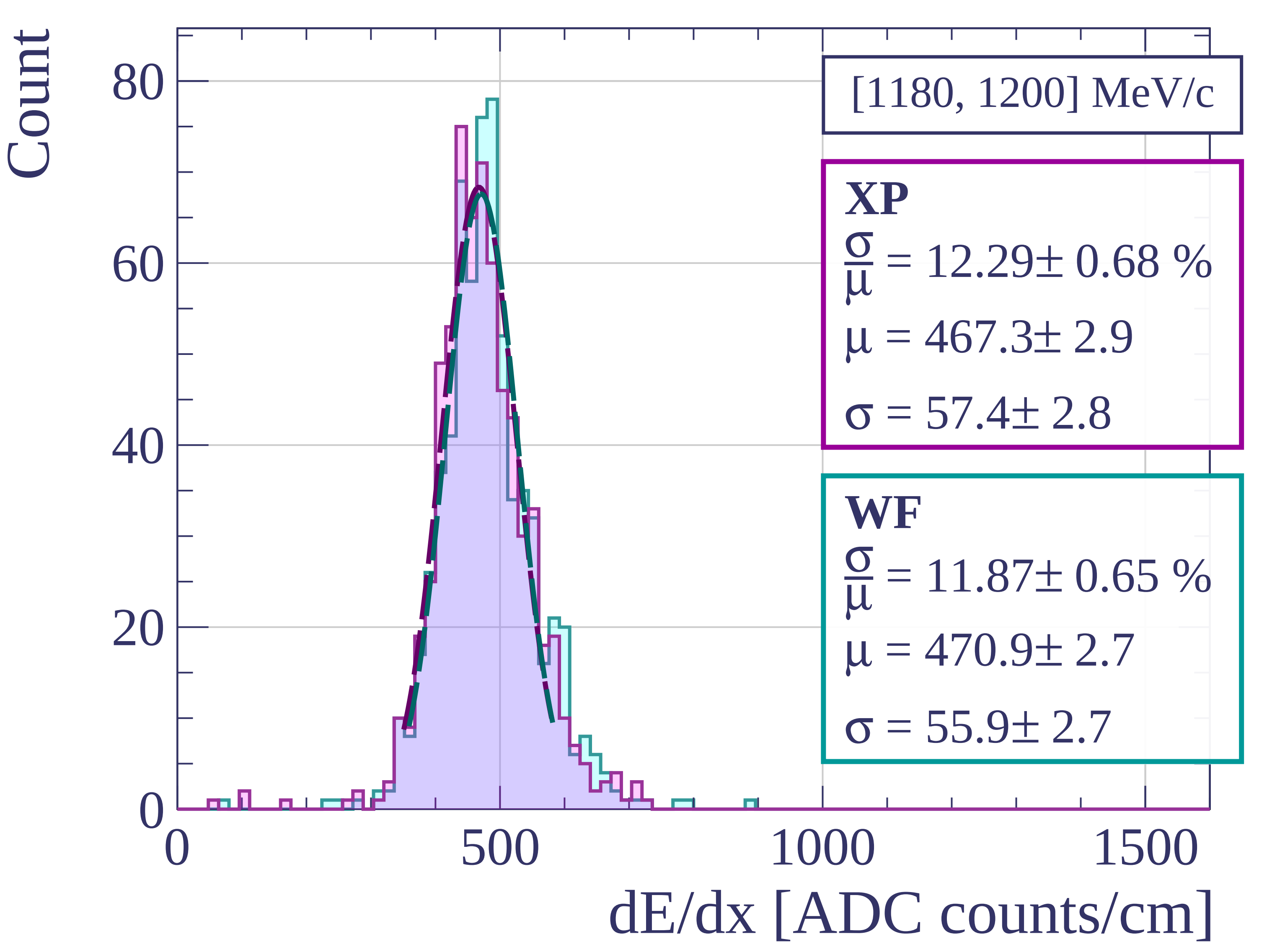
Count
dE/dx [ADC counts/cm]
[1120, 1140] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.97± 0.70 %
μ = 469.4± 2.9
σ = 60.9± 2.9
WF
$\frac{\sigma}{\mu}$ = 13.40± 0.64 %
μ = 475.7± 2.9
σ = 63.7± 2.7



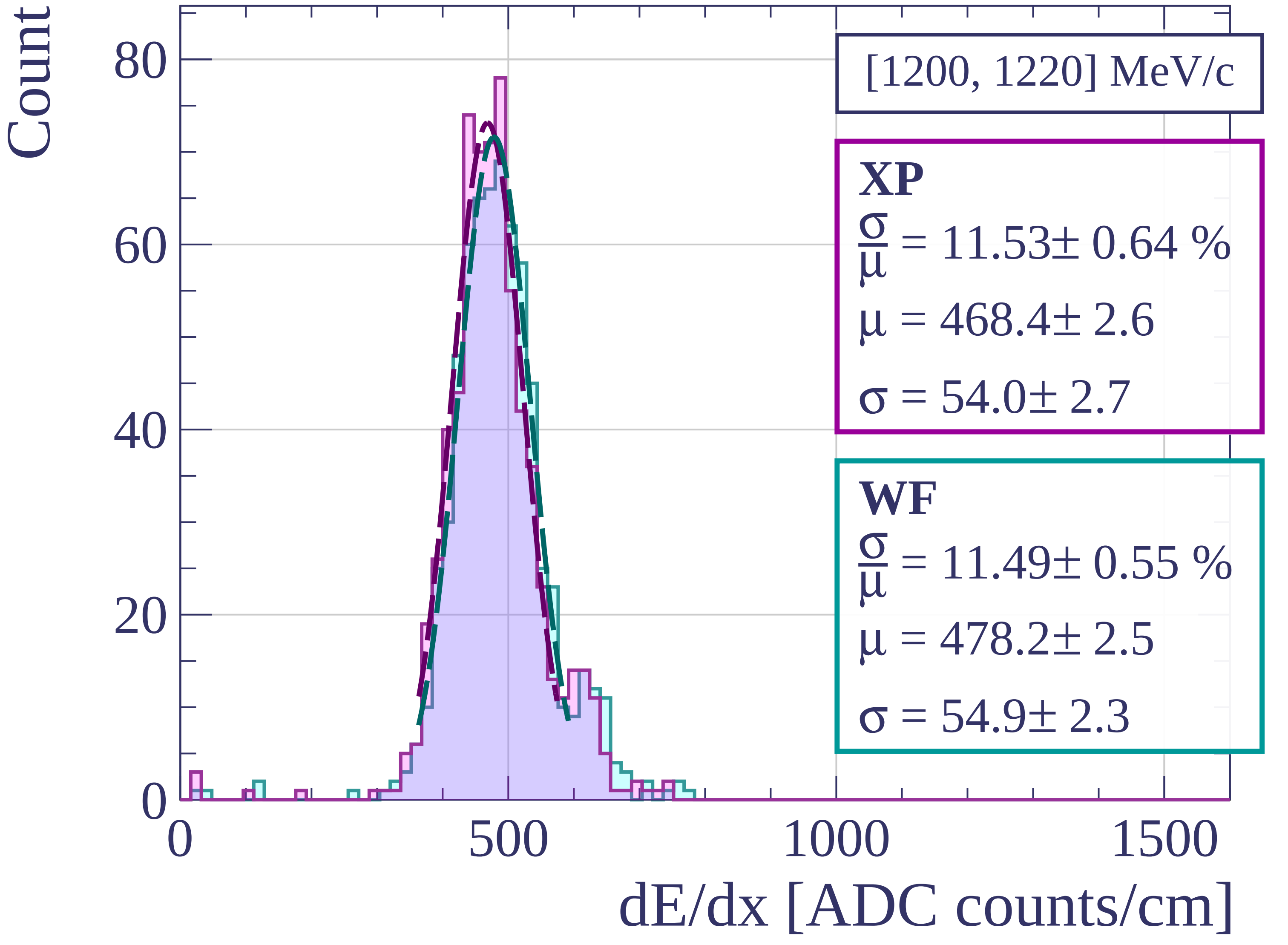
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1140, 1160] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.71± 0.74 %
μ = 465.2± 3.0
σ = 59.1± 3.1
WF
$\frac{\sigma}{\mu}$ = 12.85± 0.73 %
μ = 474.2± 3.0
σ = 60.9± 3.1



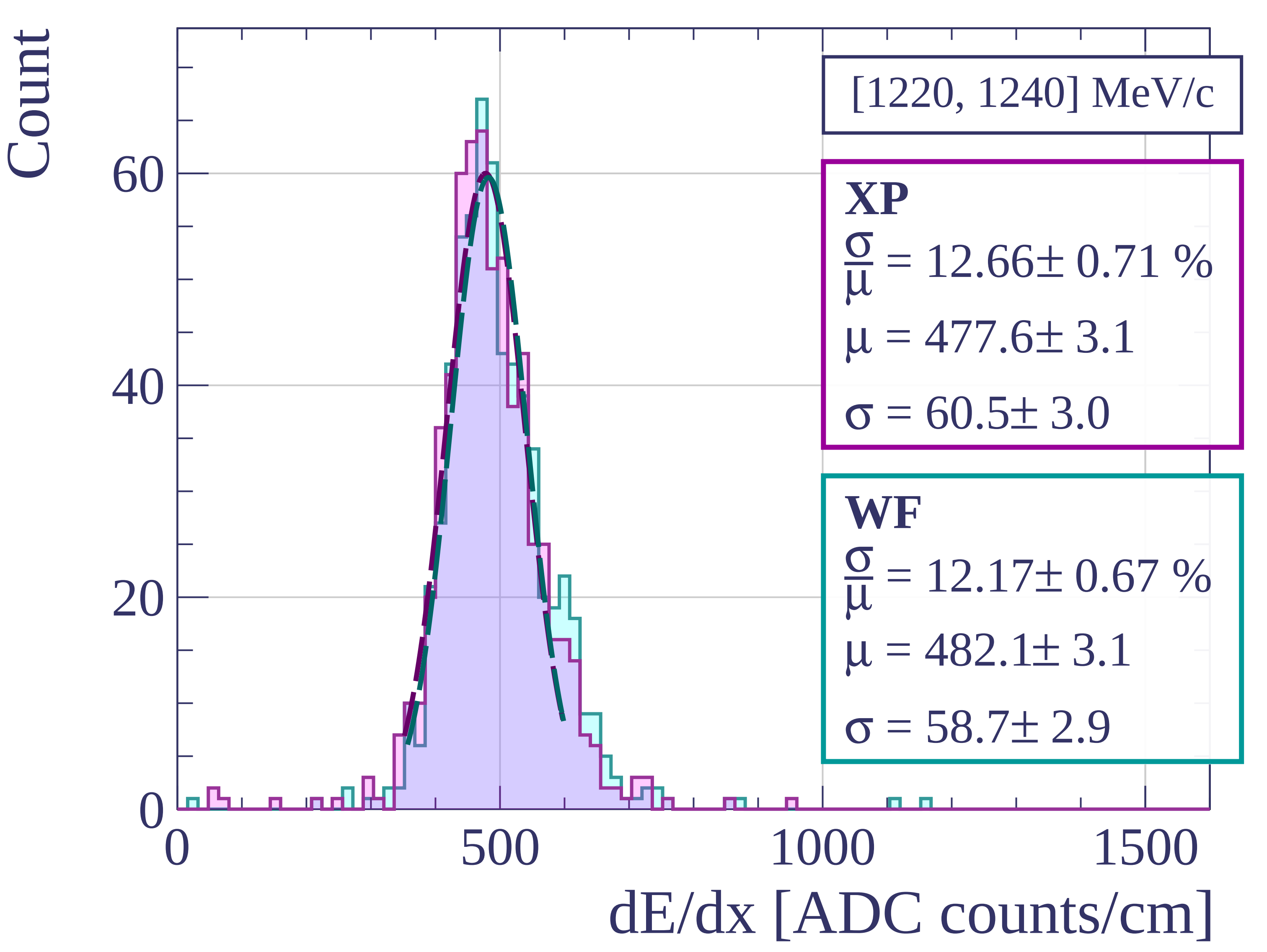
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1160, 1180] MeV/c
XP
σ/μ = 11.91± 0.69 %
μ = 470.2± 2.8
σ = 56.0± 2.9
WF
σ/μ = 12.81± 0.64 %
μ = 478.1± 2.8
σ = 61.2± 2.7

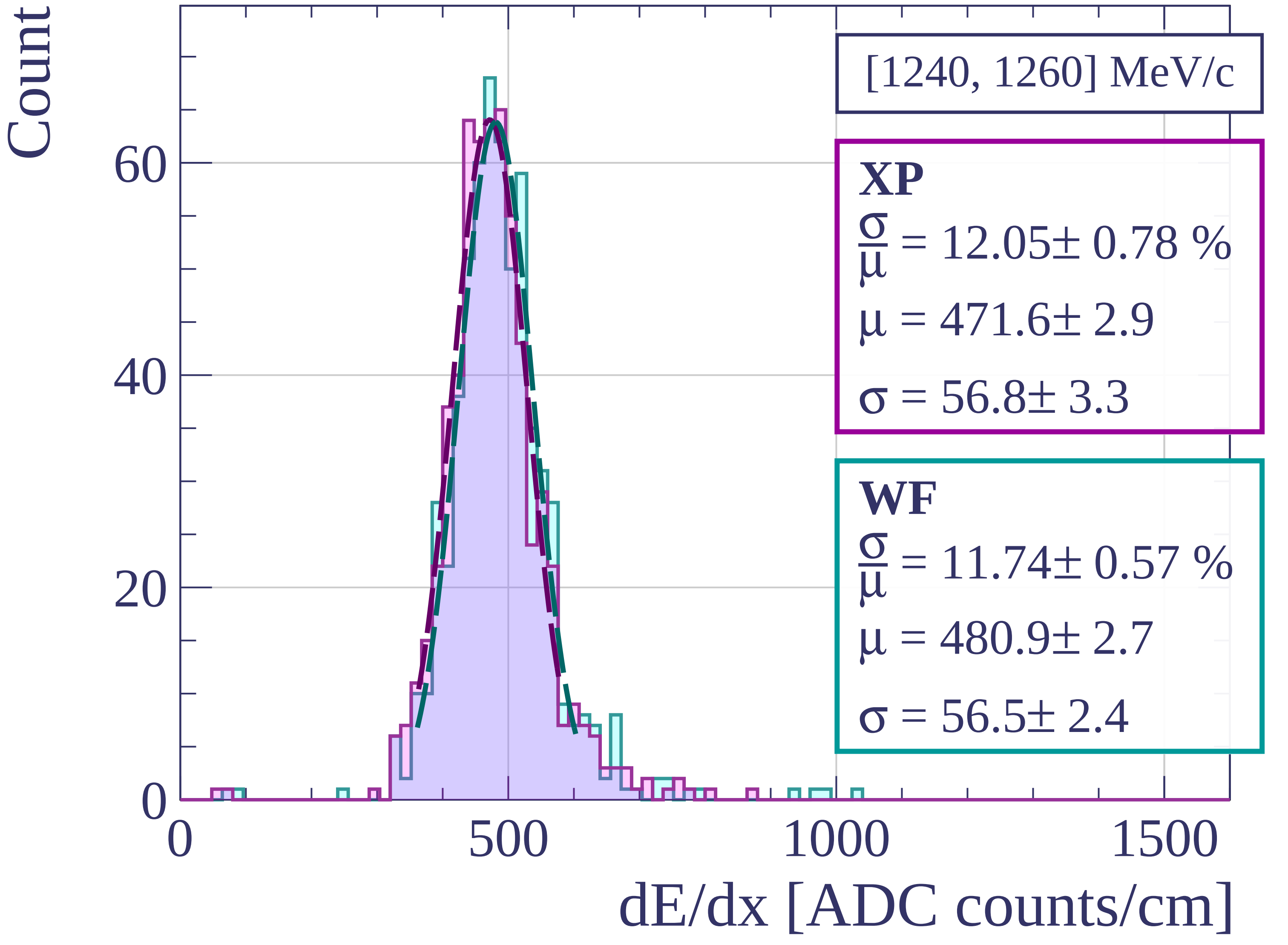


Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1180, 1200] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.29± 0.68 %
μ = 467.3± 2.9
σ = 57.4± 2.8
WF
$\frac{\sigma}{\mu}$ = 11.87± 0.65 %
μ = 470.9± 2.7
σ = 55.9± 2.7

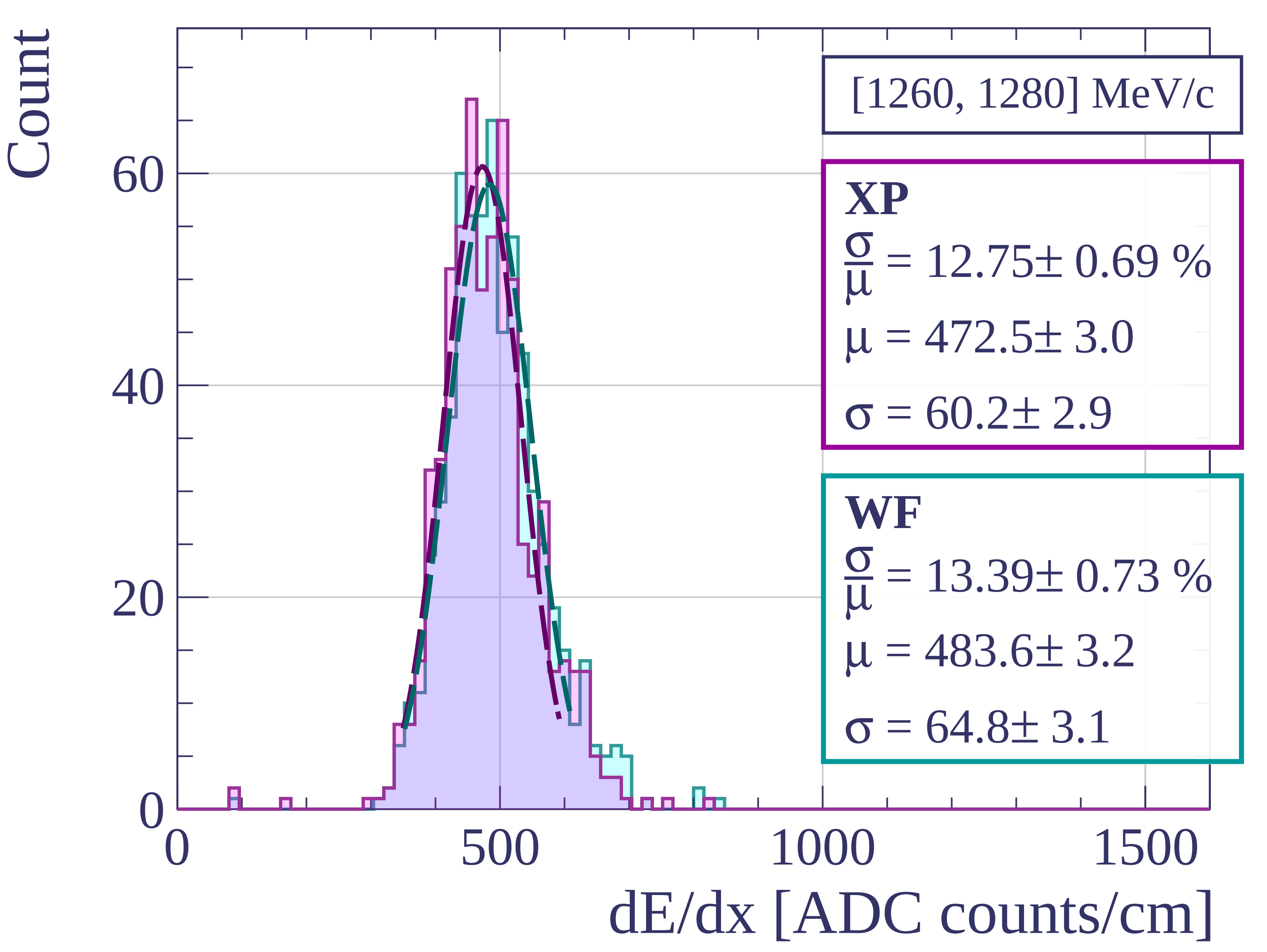


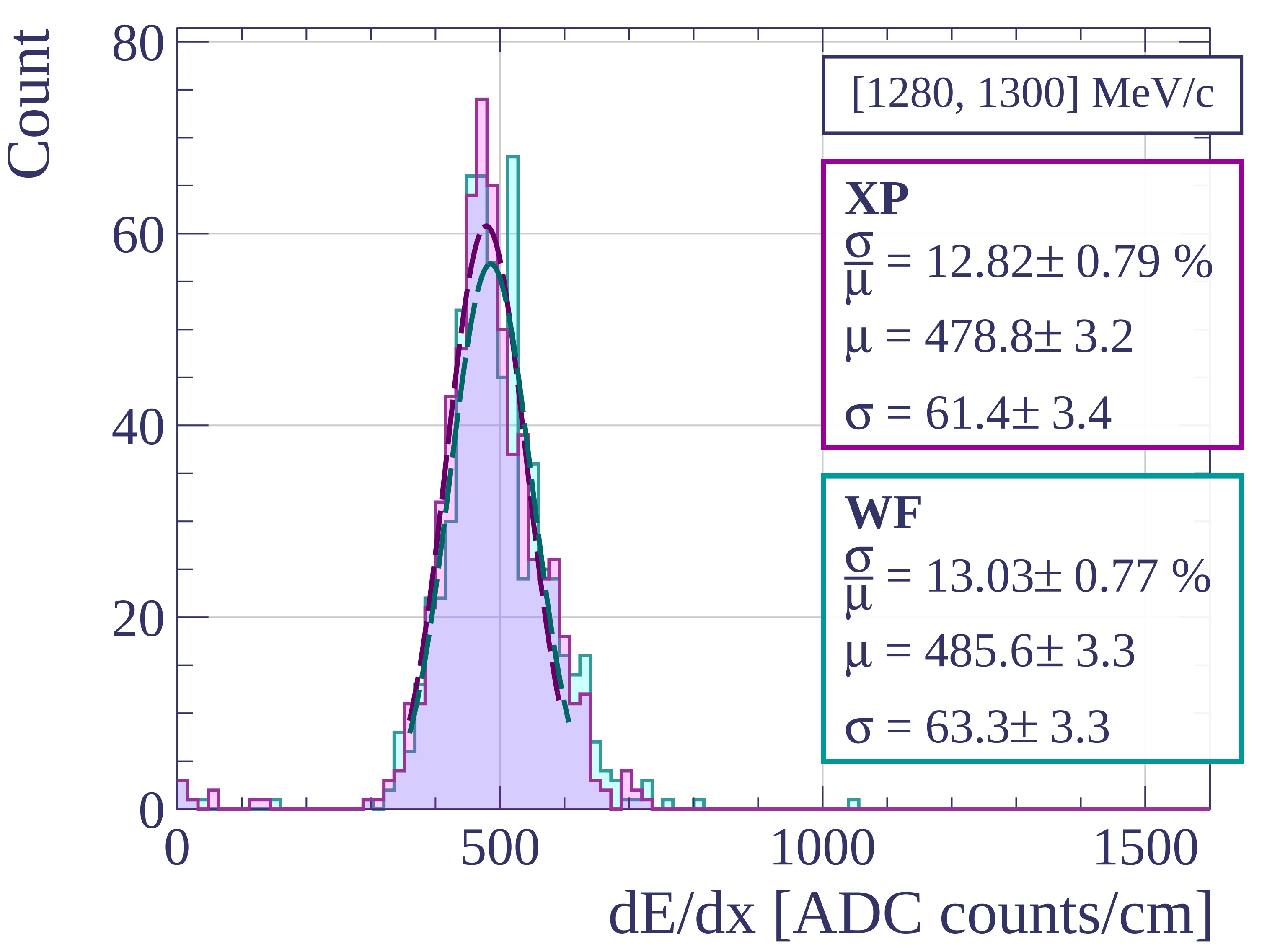
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1200, 1220] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.53± 0.64 %
μ = 468.4± 2.6
σ = 54.0± 2.7
WF
$\frac{\sigma}{\mu}$ = 11.49± 0.55 %
μ = 478.2± 2.5
σ = 54.9± 2.3

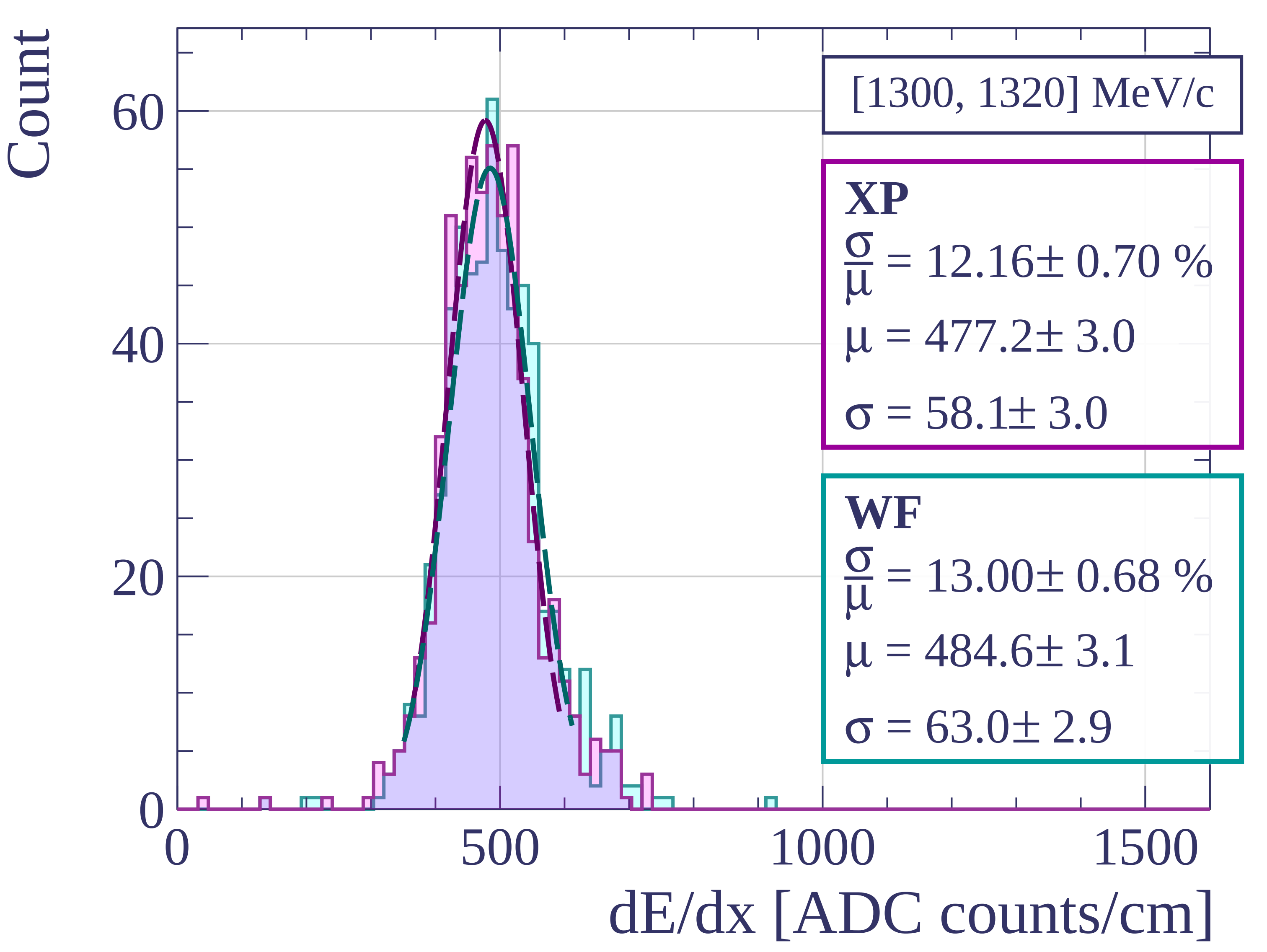

Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1220, 1240] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.66± 0.71 %
μ = 477.6± 3.1
σ = 60.5± 3.0
WF
$\frac{\sigma}{\mu}$ = 12.17± 0.67 %
μ = 482.1± 3.1
σ = 58.7± 2.9



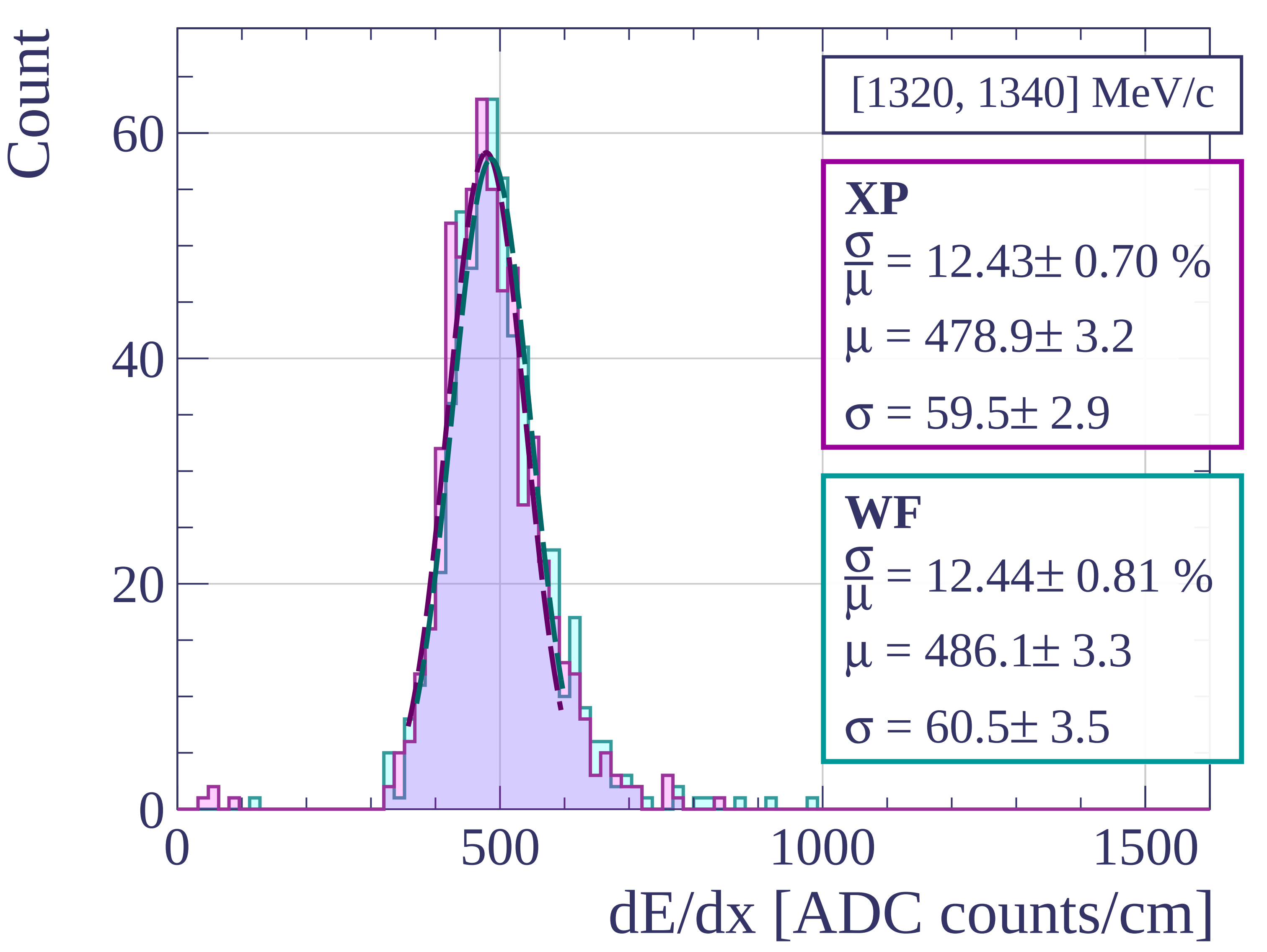
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1240, 1260] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.05± 0.78 %
μ = 471.6± 2.9
σ = 56.8± 3.3
WF
$\frac{\sigma}{\mu}$ = 11.74± 0.57 %
μ = 480.9± 2.7
σ = 56.5± 2.4


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1260, 1280] MeV/c
XP
σ/μ = 12.75± 0.69 %
μ = 472.5± 3.0
σ = 60.2± 2.9
WF
σ/μ = 13.39± 0.73 %
μ = 483.6± 3.2
σ = 64.8± 3.1

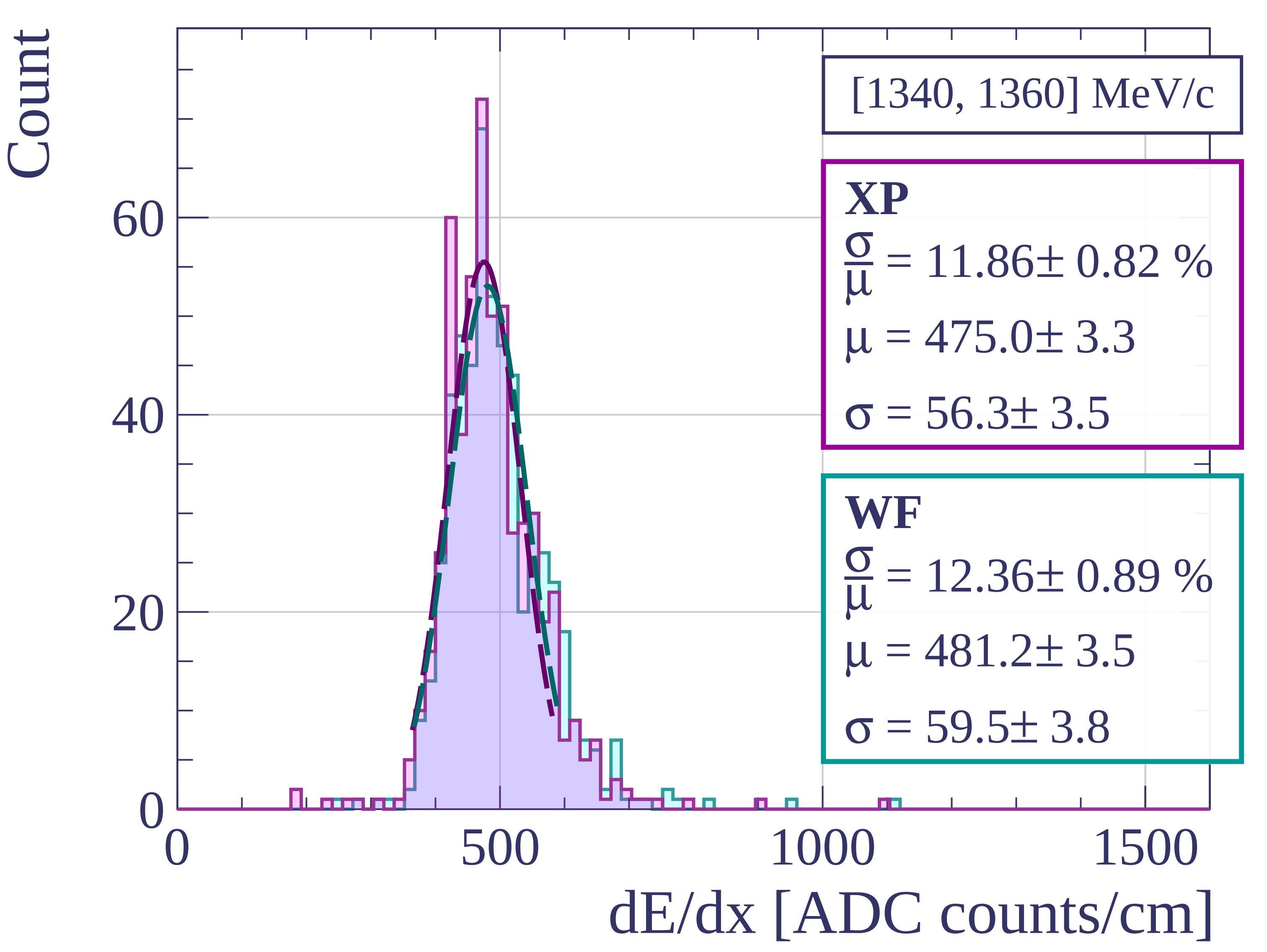

Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1280, 1300] MeV/c
XP
σ/μ = 12.82± 0.79 %
μ = 478.8± 3.2
σ = 61.4± 3.4
WF
σ/μ = 13.03± 0.77 %
μ = 485.6± 3.3
σ = 63.3± 3.3



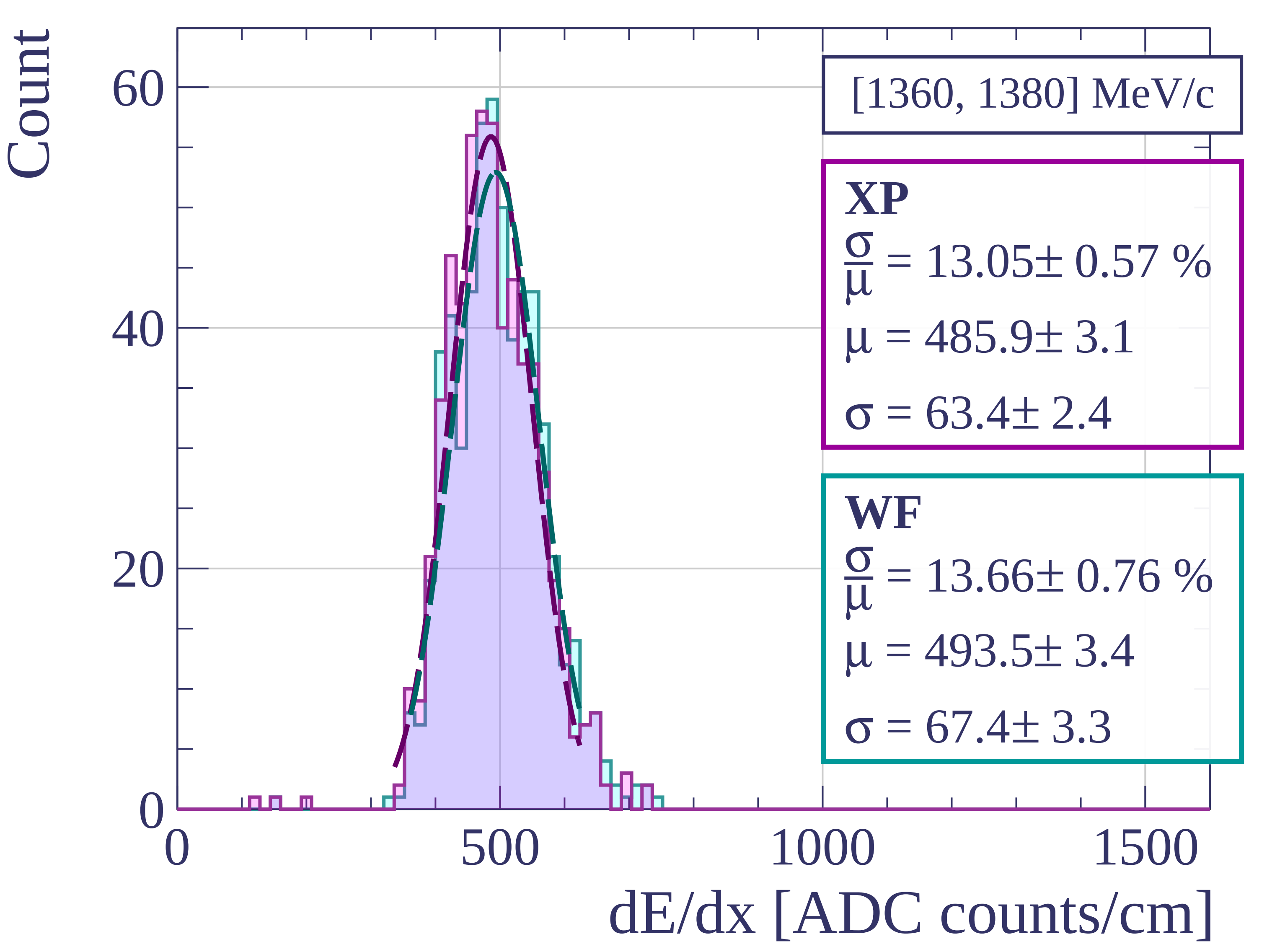
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1300, 1320] MeV/c
XP
σ/μ = 12.16± 0.70 %
μ = 477.2± 3.0
σ = 58.1± 3.0
WF
σ/μ = 13.00± 0.68 %
μ = 484.6± 3.1
σ = 63.0± 2.9

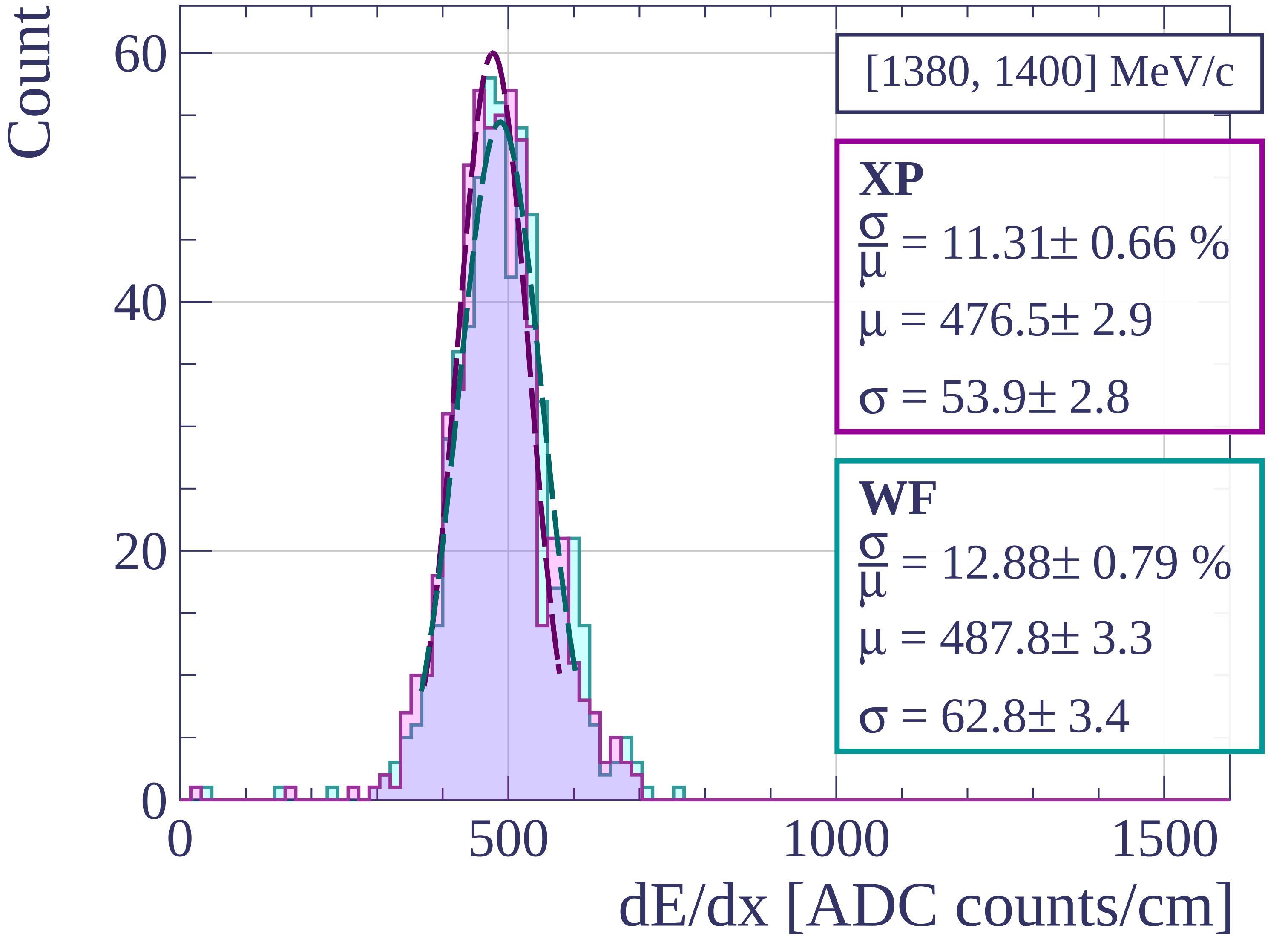


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1320, 1340] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.43± 0.70 %
μ = 478.9± 3.2
σ = 59.5± 2.9
WF
$\frac{\sigma}{\mu}$ = 12.44± 0.81 %
μ = 486.1± 3.3
σ = 60.5± 3.5

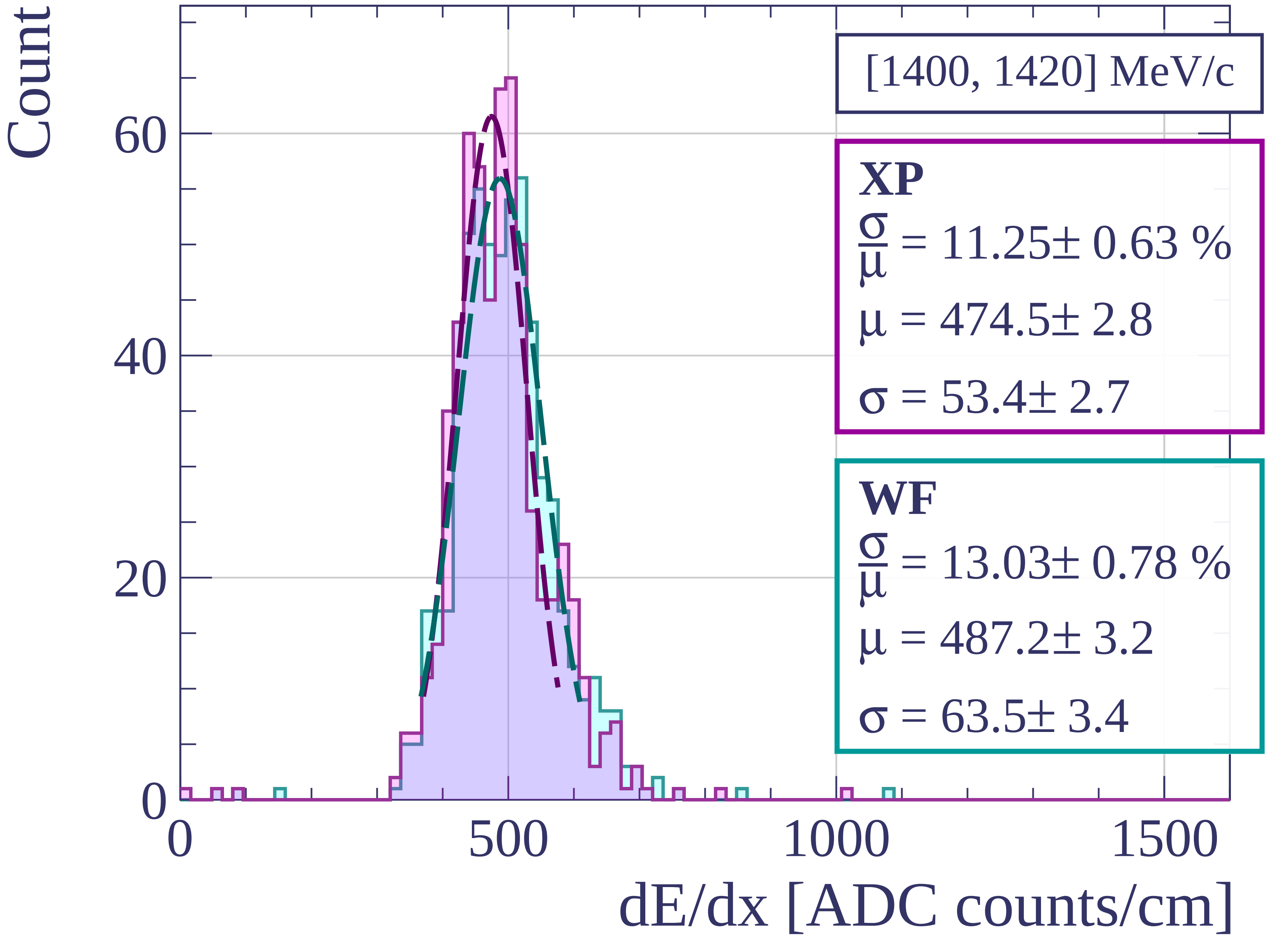


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1340, 1360] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.86± 0.82 %
μ = 475.0± 3.3
σ = 56.3± 3.5
WF
$\frac{\sigma}{\mu}$ = 12.36± 0.89 %
μ = 481.2± 3.5
σ = 59.5± 3.8

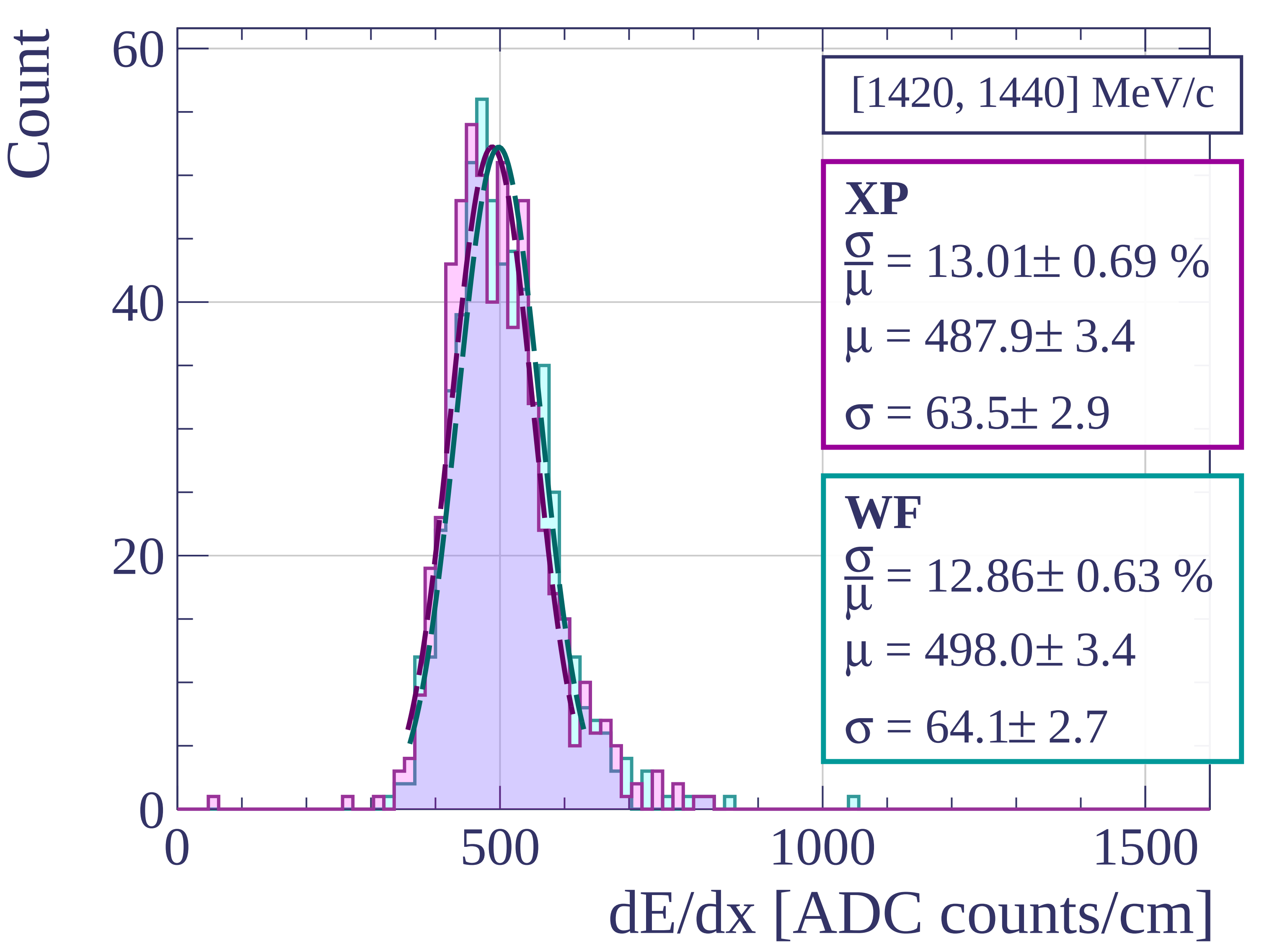

Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1360, 1380] MeV/c
XP
σ/μ = 13.05± 0.57 %
μ = 485.9± 3.1
σ = 63.4± 2.4
WF
σ/μ = 13.66± 0.76 %
μ = 493.5± 3.4
σ = 67.4± 3.3



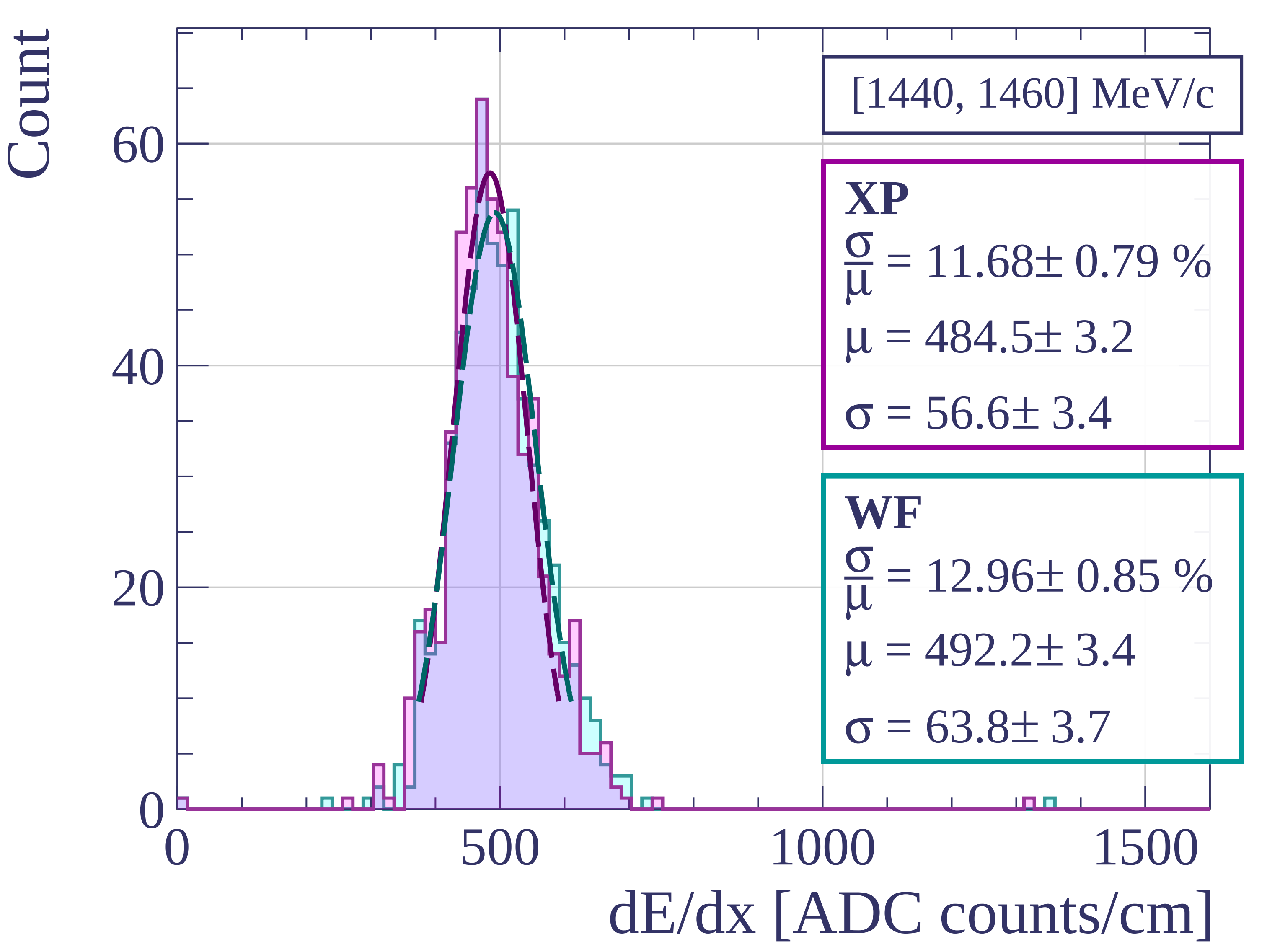
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1380, 1400] MeV/c
XP
σ/μ = 11.31± 0.66 %
μ = 476.5± 2.9
σ = 53.9± 2.8
WF
σ/μ = 12.88± 0.79 %
μ = 487.8± 3.3
σ = 62.8± 3.4

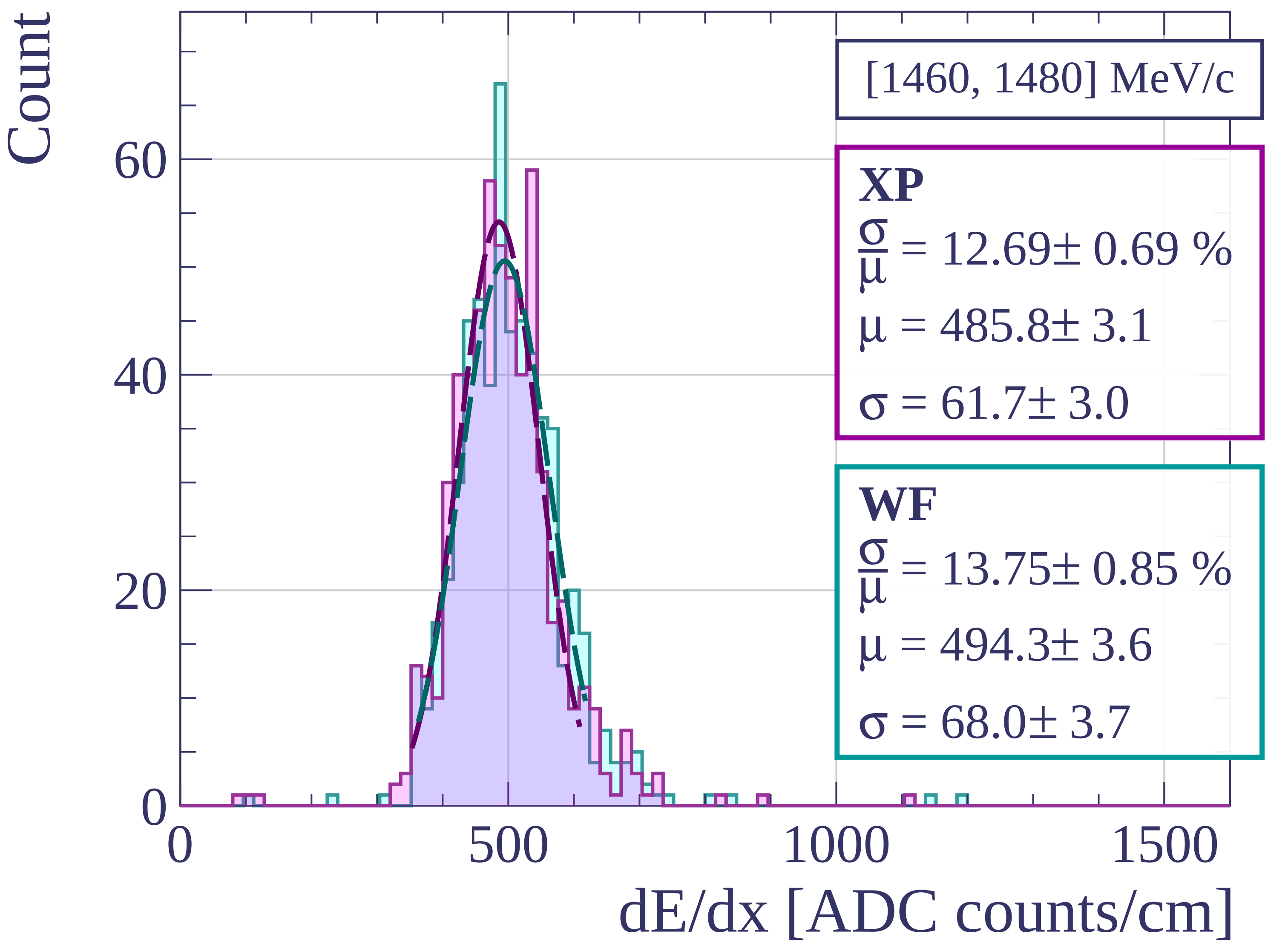


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1400, 1420] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.25± 0.63 %
μ = 474.5± 2.8
σ = 53.4± 2.7
WF
$\frac{\sigma}{\mu}$ = 13.03± 0.78 %
μ = 487.2± 3.2
σ = 63.5± 3.4

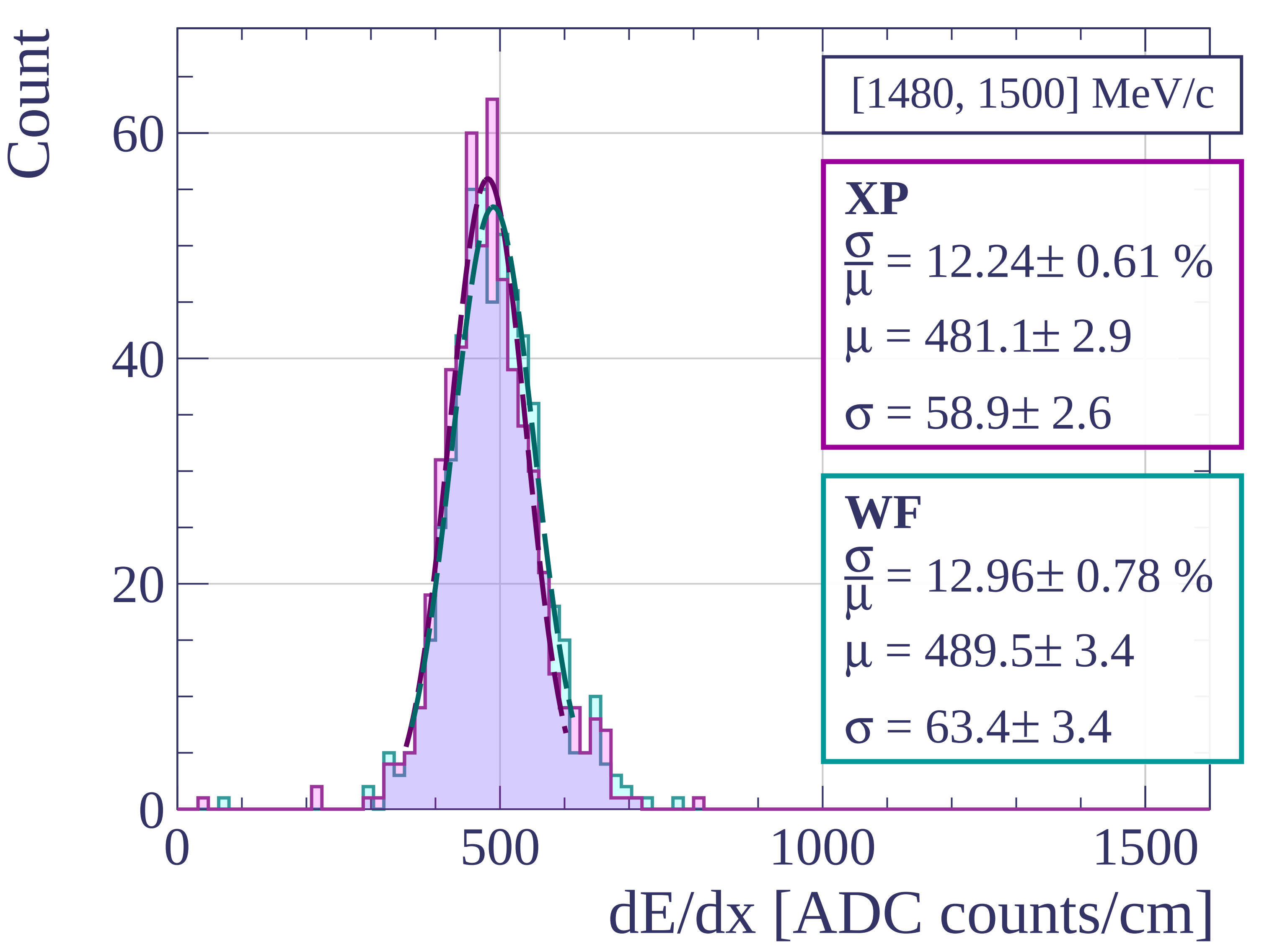


Count
dE/dx [ADC counts/cm]
[1420, 1440] MeV/c
XP
σ/μ = 13.01± 0.69 %
μ = 487.9± 3.4
σ = 63.5± 2.9
WF
σ/μ = 12.86± 0.63 %
μ = 498.0± 3.4
σ = 64.1± 2.7
0
20
40
60
0
500
1000
1500

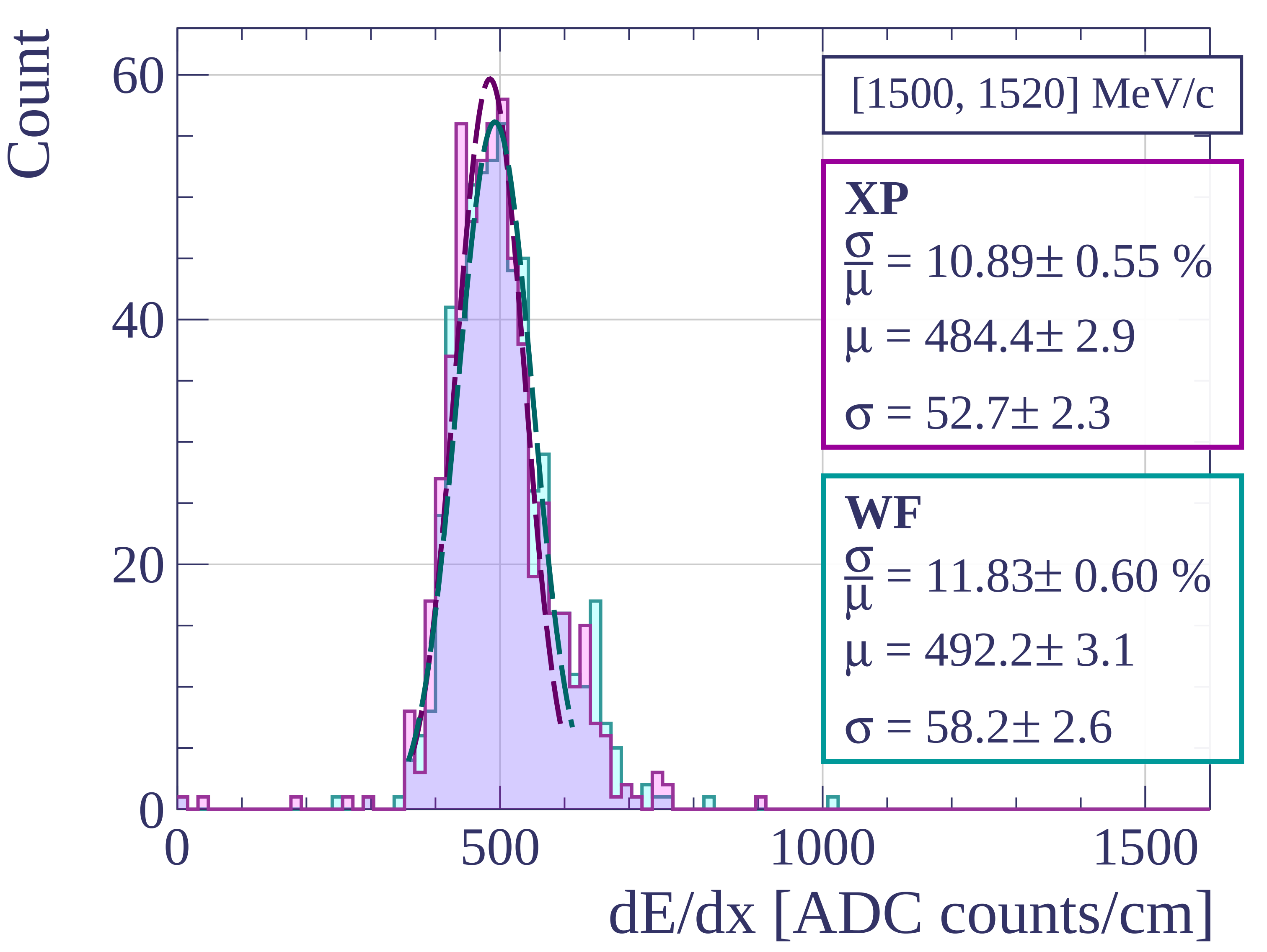

Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1440, 1460] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.68± 0.79 %
μ = 484.5± 3.2
σ = 56.6± 3.4
WF
$\frac{\sigma}{\mu}$ = 12.96± 0.85 %
μ = 492.2± 3.4
σ = 63.8± 3.7



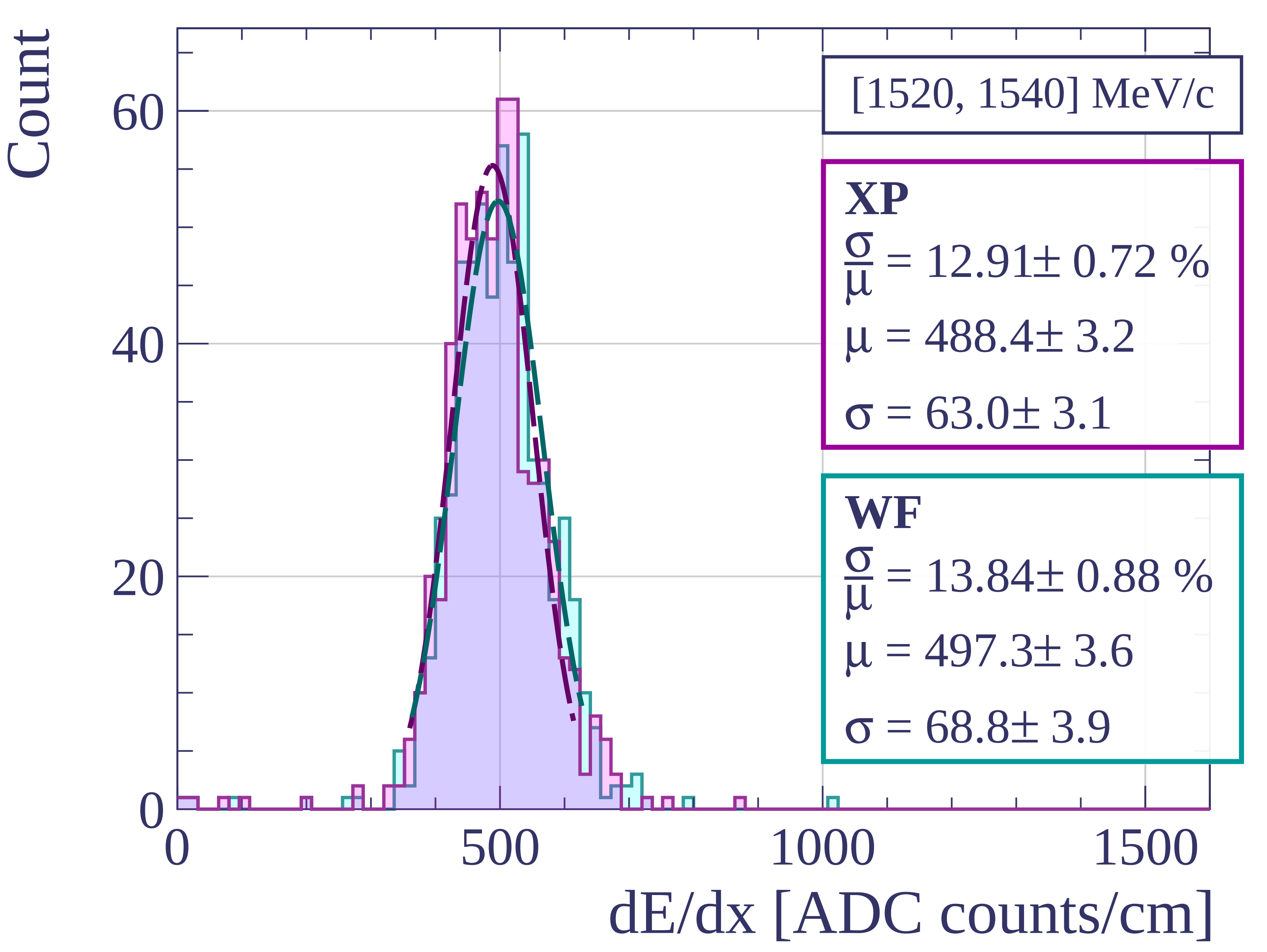
Count
0
20
40
60
0
500
1000
1500
dE/dx [ADC counts/cm]
[1460, 1480] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.69± 0.69 %
μ = 485.8± 3.1
σ = 61.7± 3.0
WF
$\frac{\sigma}{\mu}$ = 13.75± 0.85 %
μ = 494.3± 3.6
σ = 68.0± 3.7



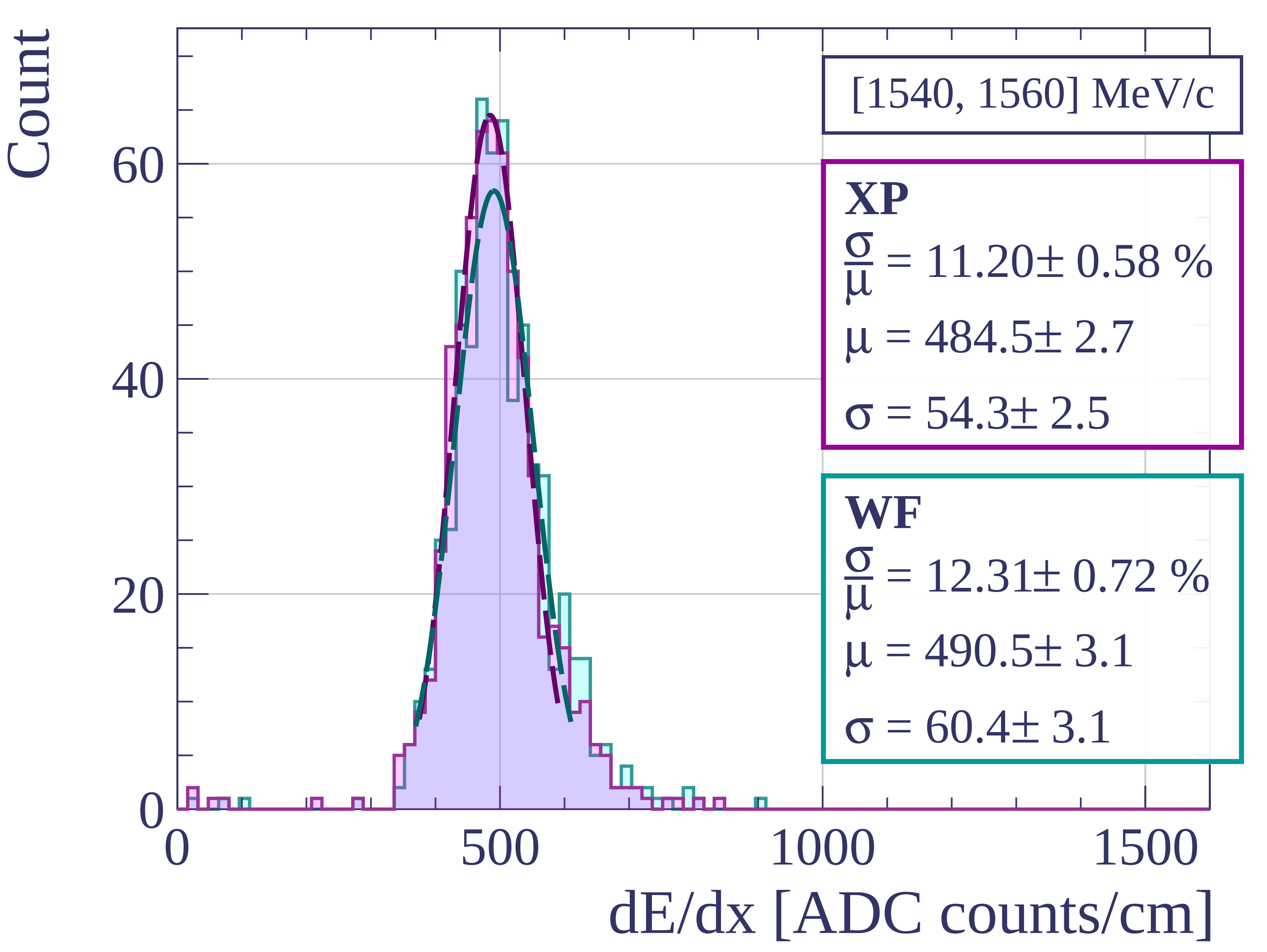
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1480, 1500] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.24± 0.61 %
μ = 481.1± 2.9
σ = 58.9± 2.6
WF
$\frac{\sigma}{\mu}$ = 12.96± 0.78 %
μ = 489.5± 3.4
σ = 63.4± 3.4



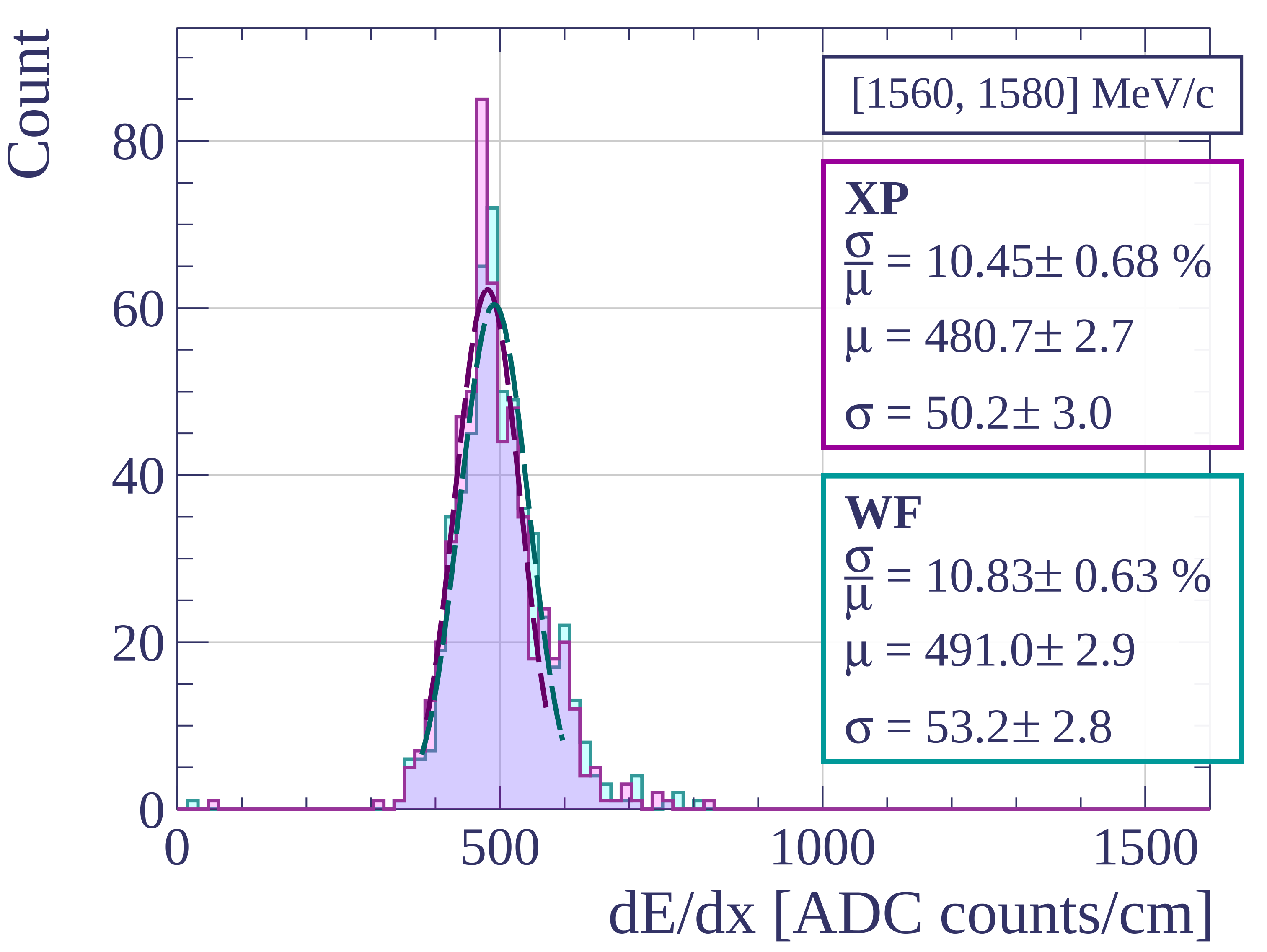
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1500, 1520] MeV/c
XP
σ/μ = 10.89± 0.55 %
μ = 484.4± 2.9
σ = 52.7± 2.3
WF
σ/μ = 11.83± 0.60 %
μ = 492.2± 3.1
σ = 58.2± 2.6



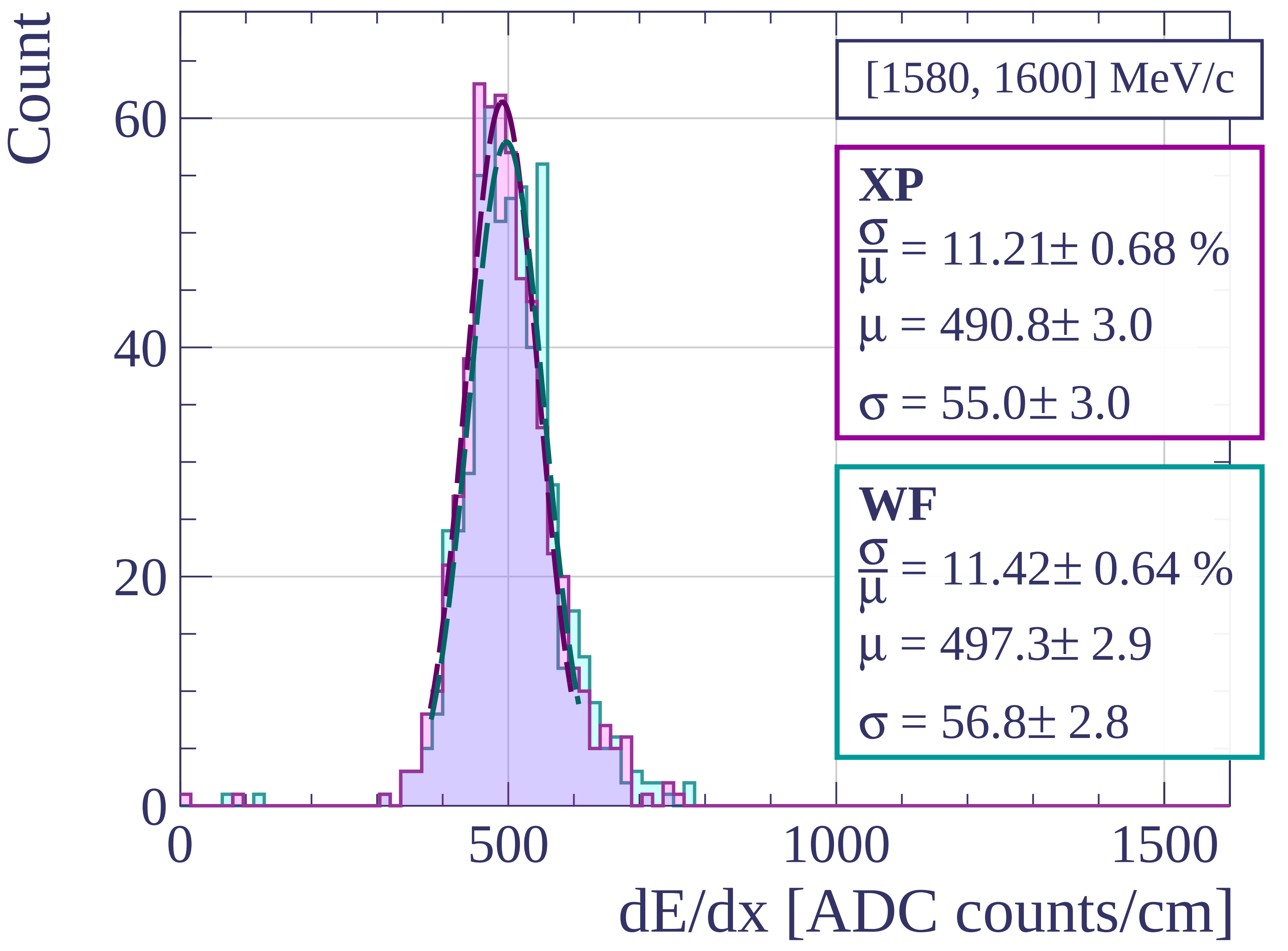
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1520, 1540] MeV/c
XP
σ/μ = 12.91± 0.72 %
μ = 488.4± 3.2
σ = 63.0± 3.1
WF
σ/μ = 13.84± 0.88 %
μ = 497.3± 3.6
σ = 68.8± 3.9



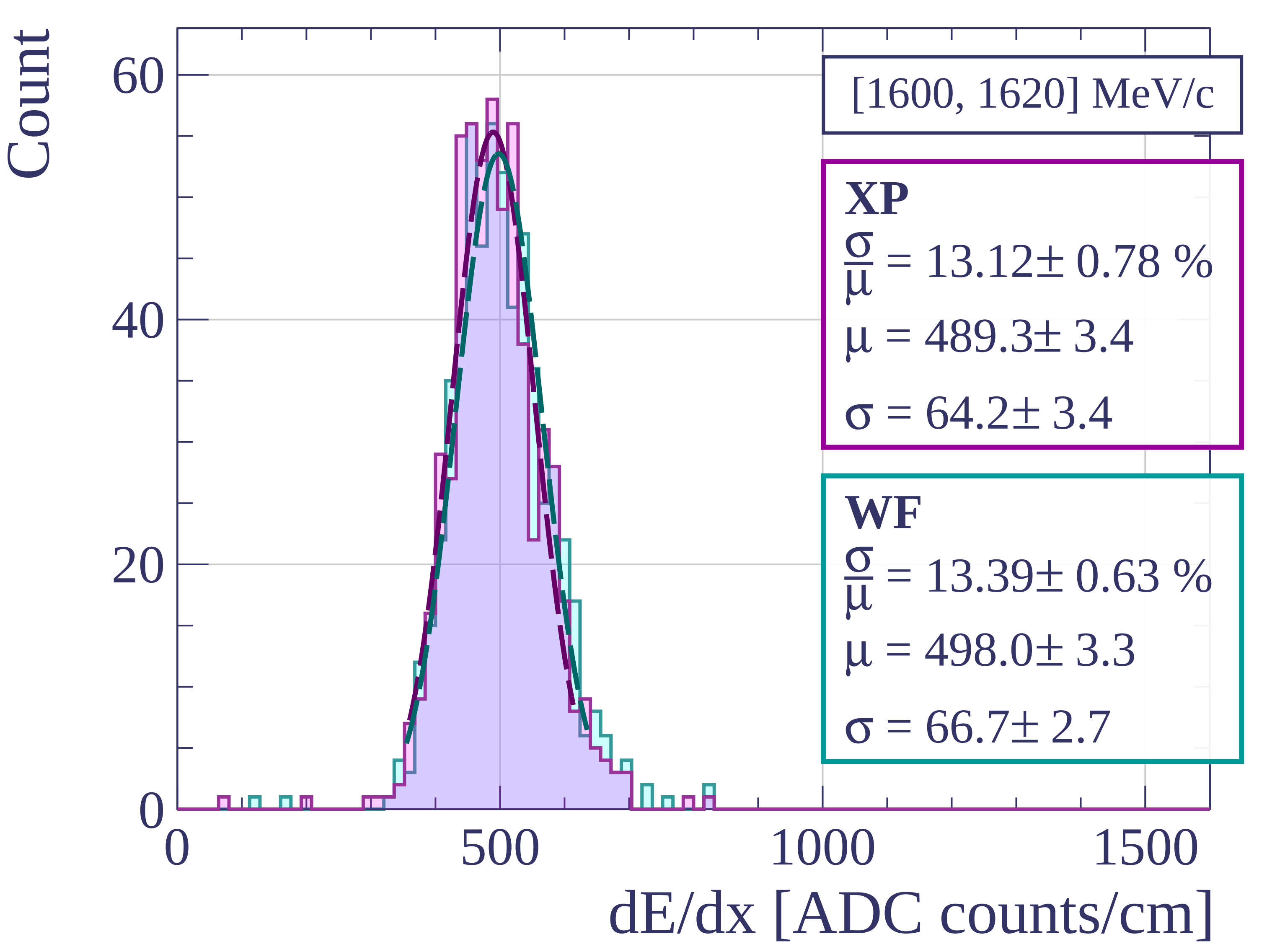
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1540, 1560] MeV/c
XP
σ/μ = 11.20± 0.58 %
μ = 484.5± 2.7
σ = 54.3± 2.5
WF
σ/μ = 12.31± 0.72 %
μ = 490.5± 3.1
σ = 60.4± 3.1



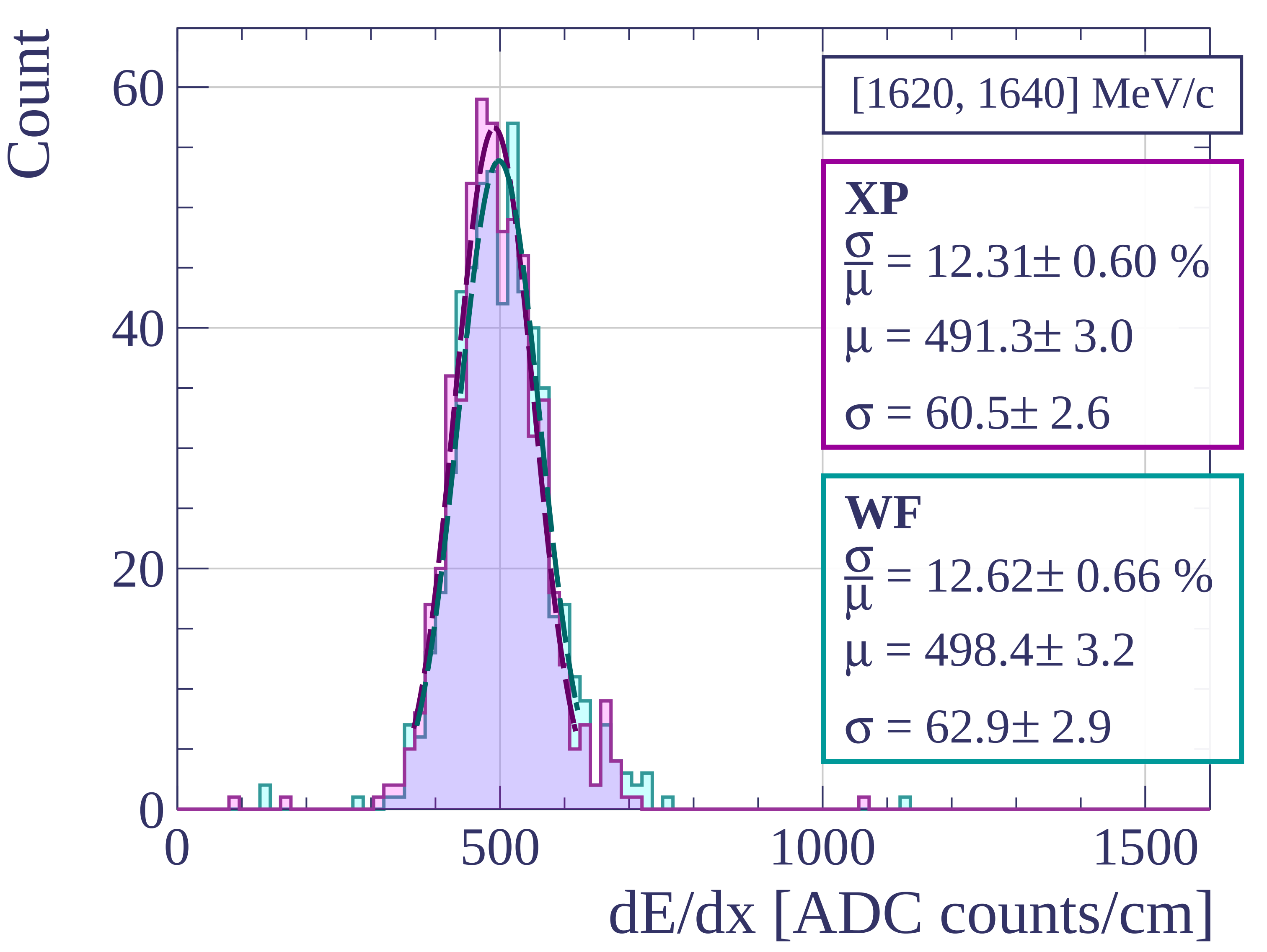
Count
80
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1560, 1580] MeV/c
XP
σ/μ = 10.45± 0.68 %
μ = 480.7± 2.7
σ = 50.2± 3.0
WF
σ/μ = 10.83± 0.63 %
μ = 491.0± 2.9
σ = 53.2± 2.8



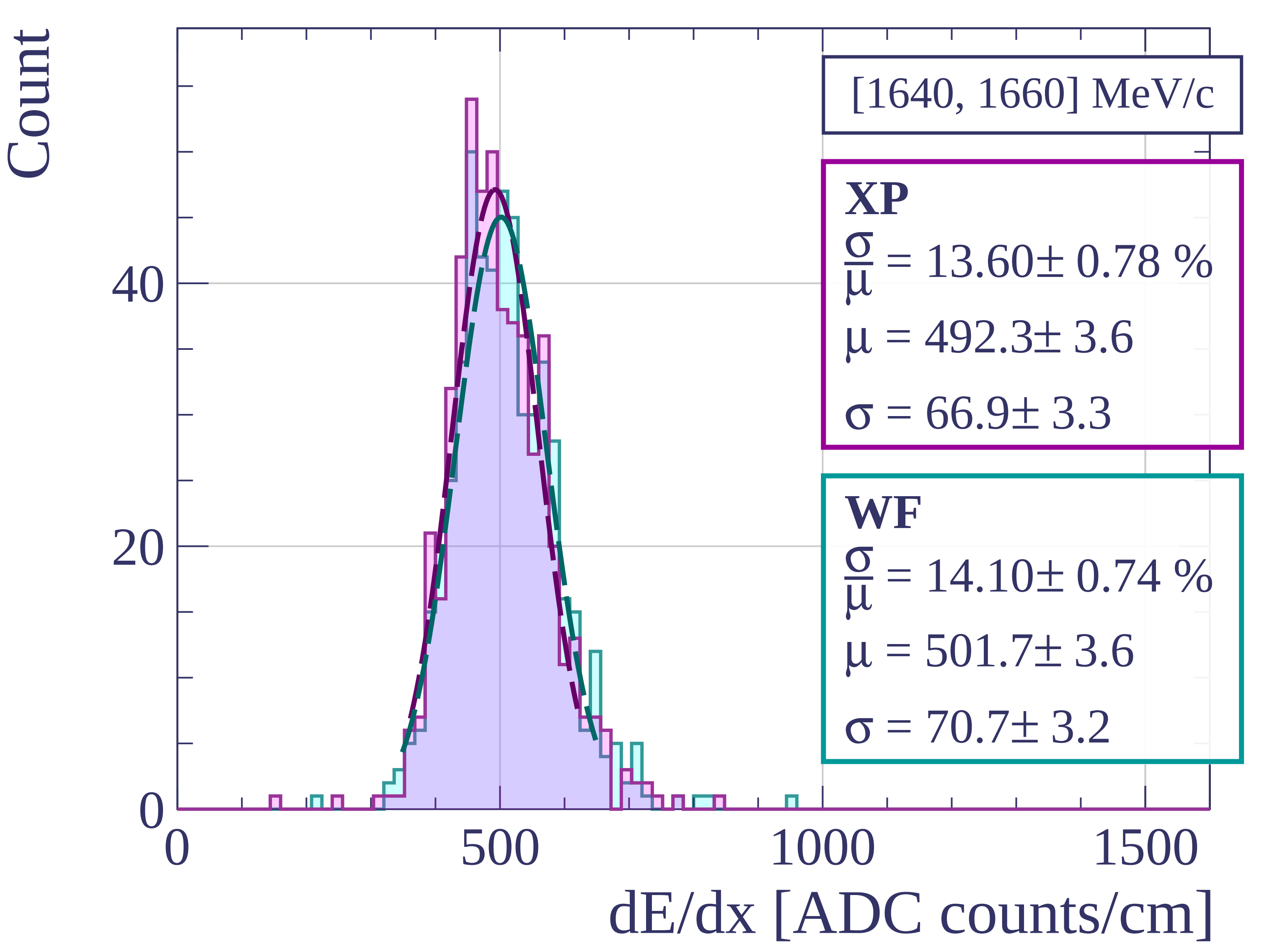
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1580, 1600] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.21± 0.68 %
μ = 490.8± 3.0
σ = 55.0± 3.0
WF
$\frac{\sigma}{\mu}$ = 11.42± 0.64 %
μ = 497.3± 2.9
σ = 56.8± 2.8



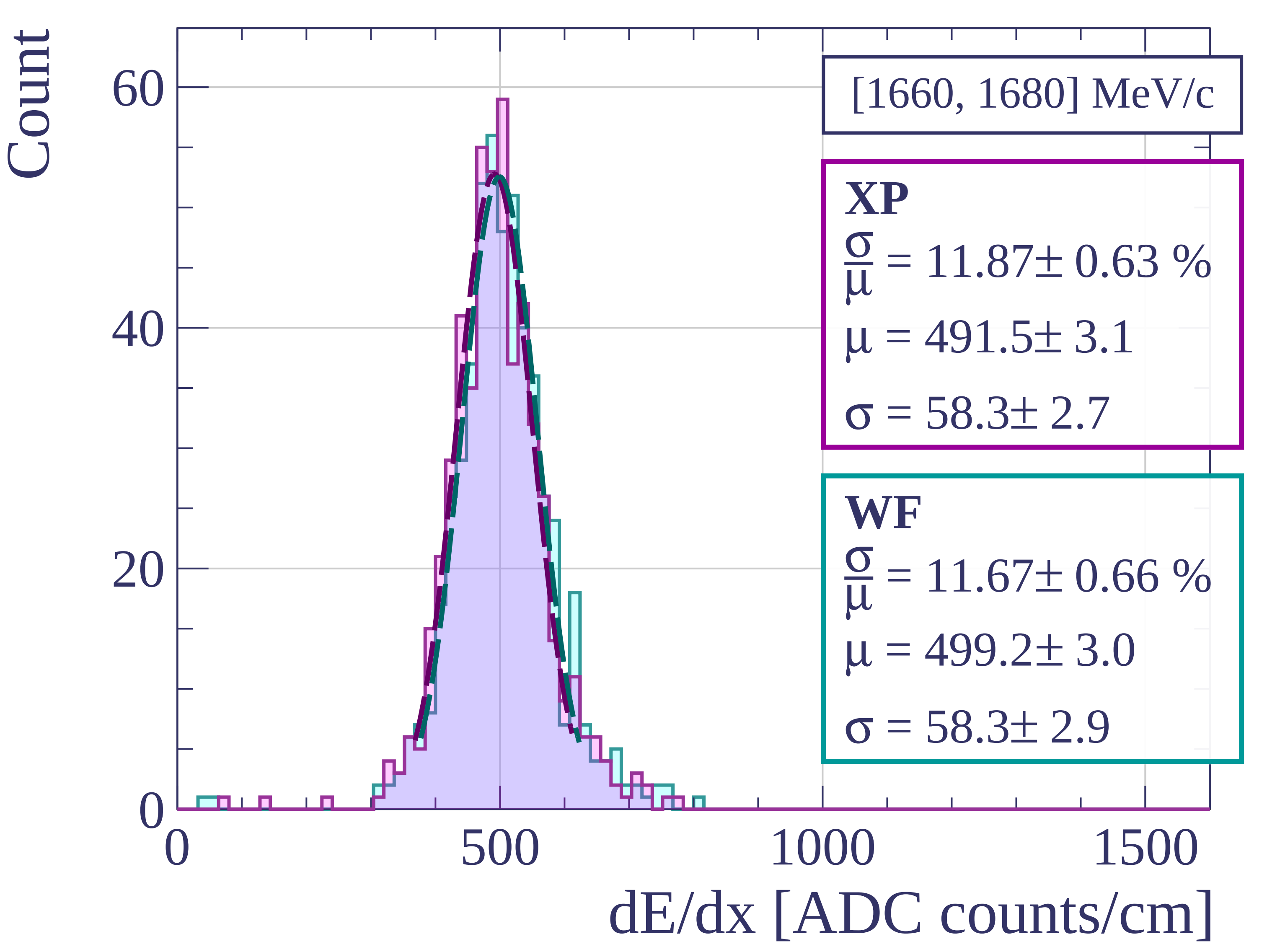
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1600, 1620] MeV/c
XP
σ/μ = 13.12± 0.78 %
μ = 489.3± 3.4
σ = 64.2± 3.4
WF
σ/μ = 13.39± 0.63 %
μ = 498.0± 3.3
σ = 66.7± 2.7

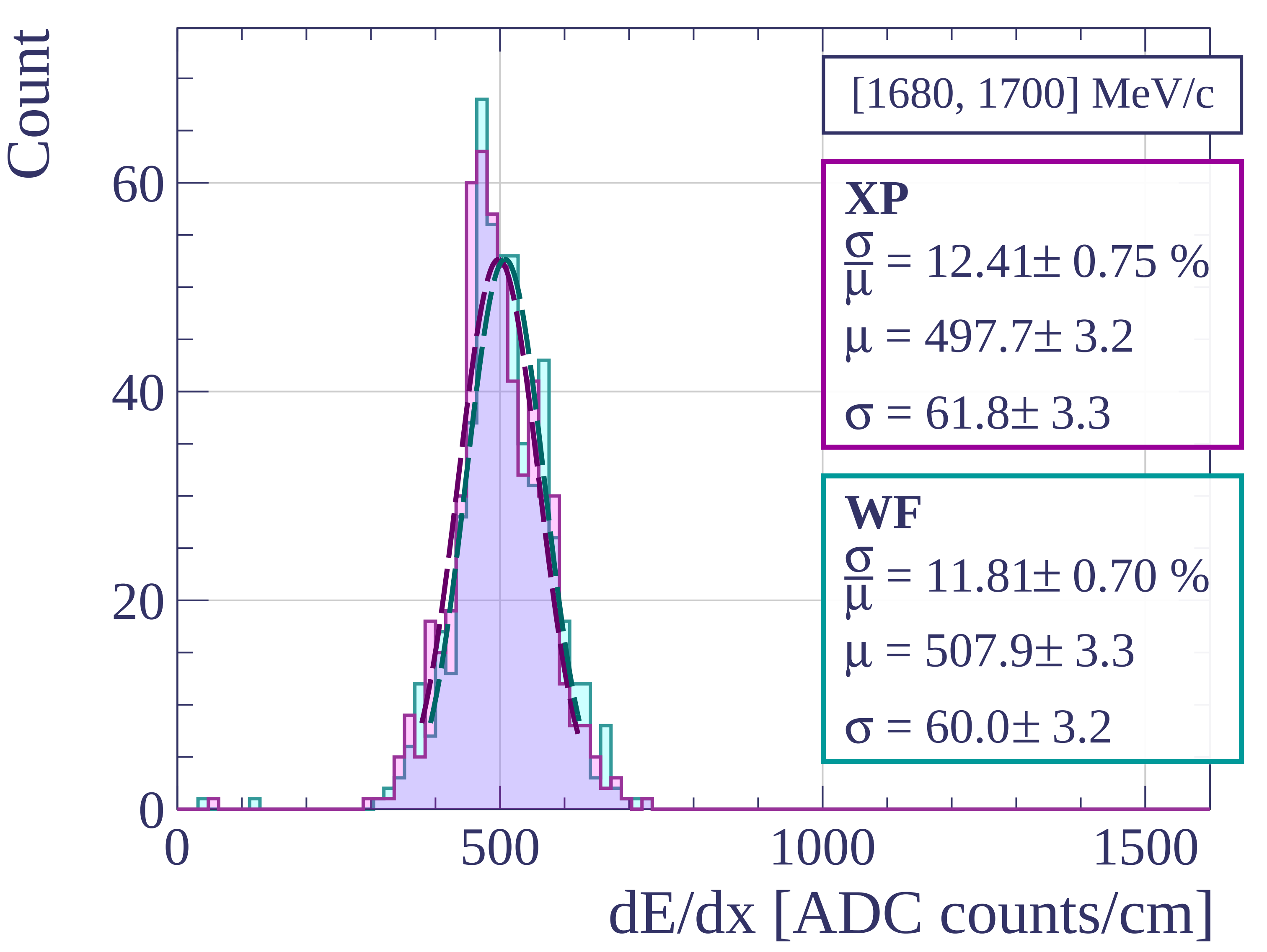


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1620, 1640] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.31± 0.60 %
μ = 491.3± 3.0
σ = 60.5± 2.6
WF
$\frac{\sigma}{\mu}$ = 12.62± 0.66 %
μ = 498.4± 3.2
σ = 62.9± 2.9

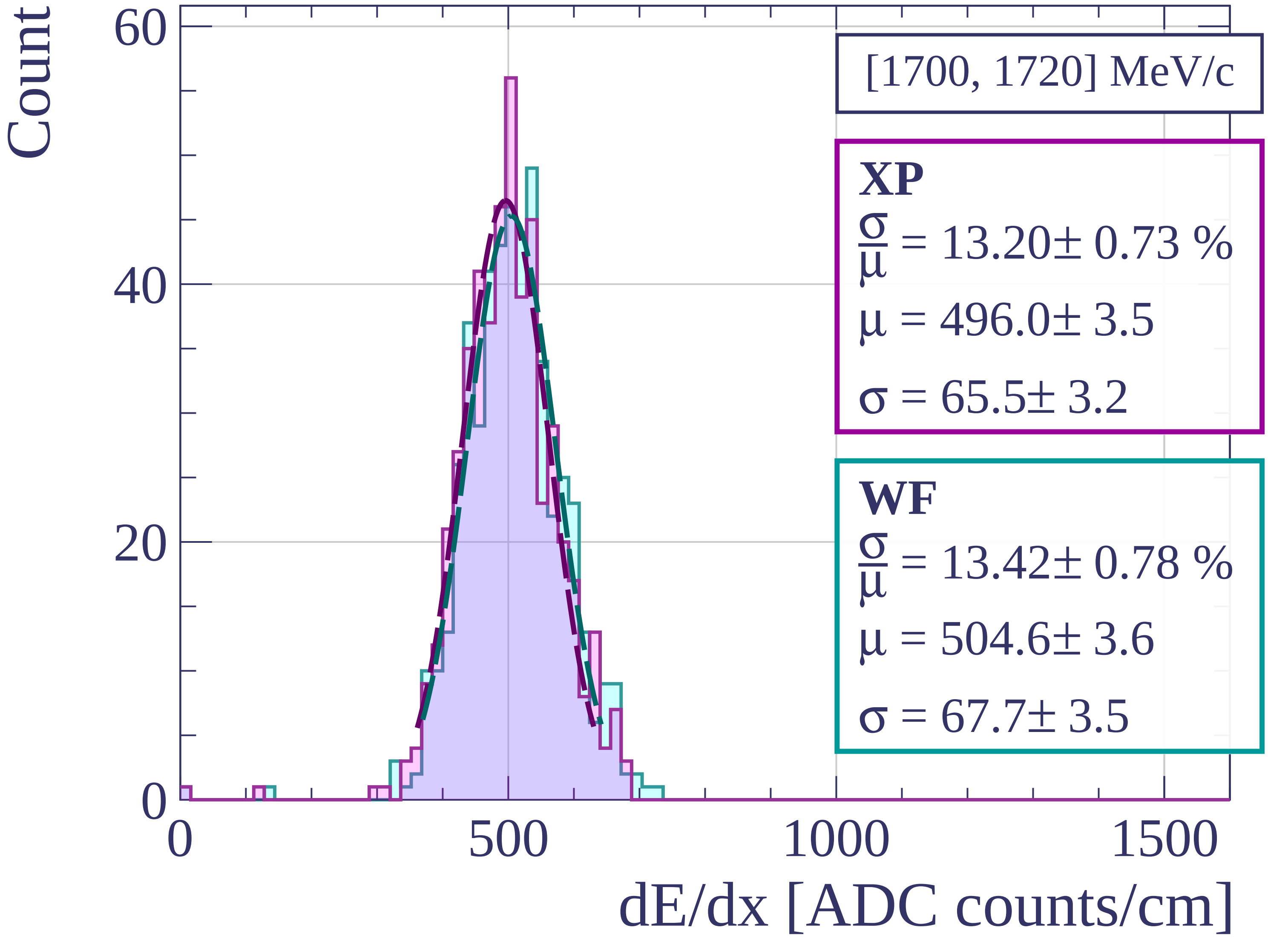


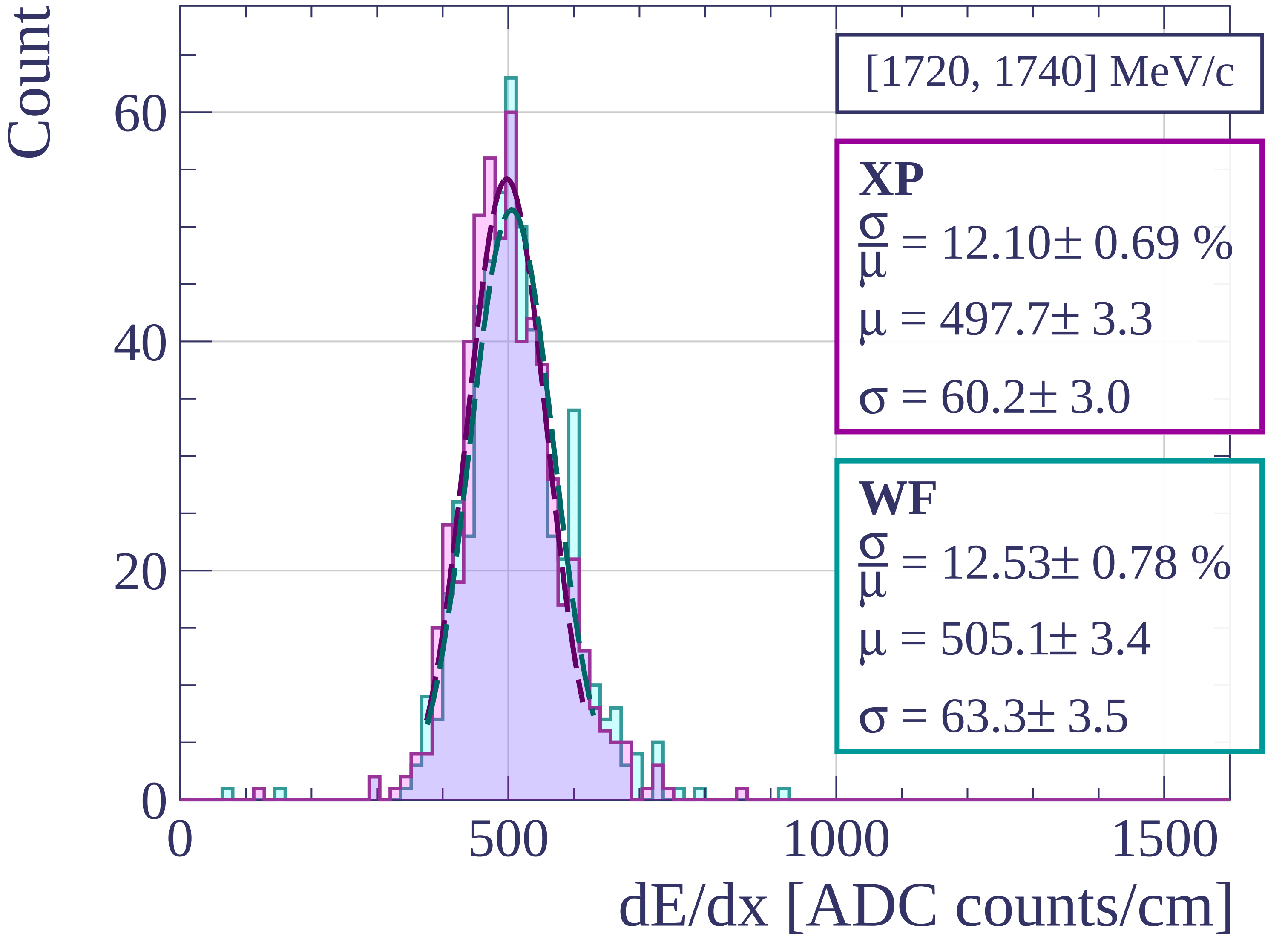
Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1640, 1660] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.60± 0.78 %
μ = 492.3± 3.6
σ = 66.9± 3.3
WF
$\frac{\sigma}{\mu}$ = 14.10± 0.74 %
μ = 501.7± 3.6
σ = 70.7± 3.2

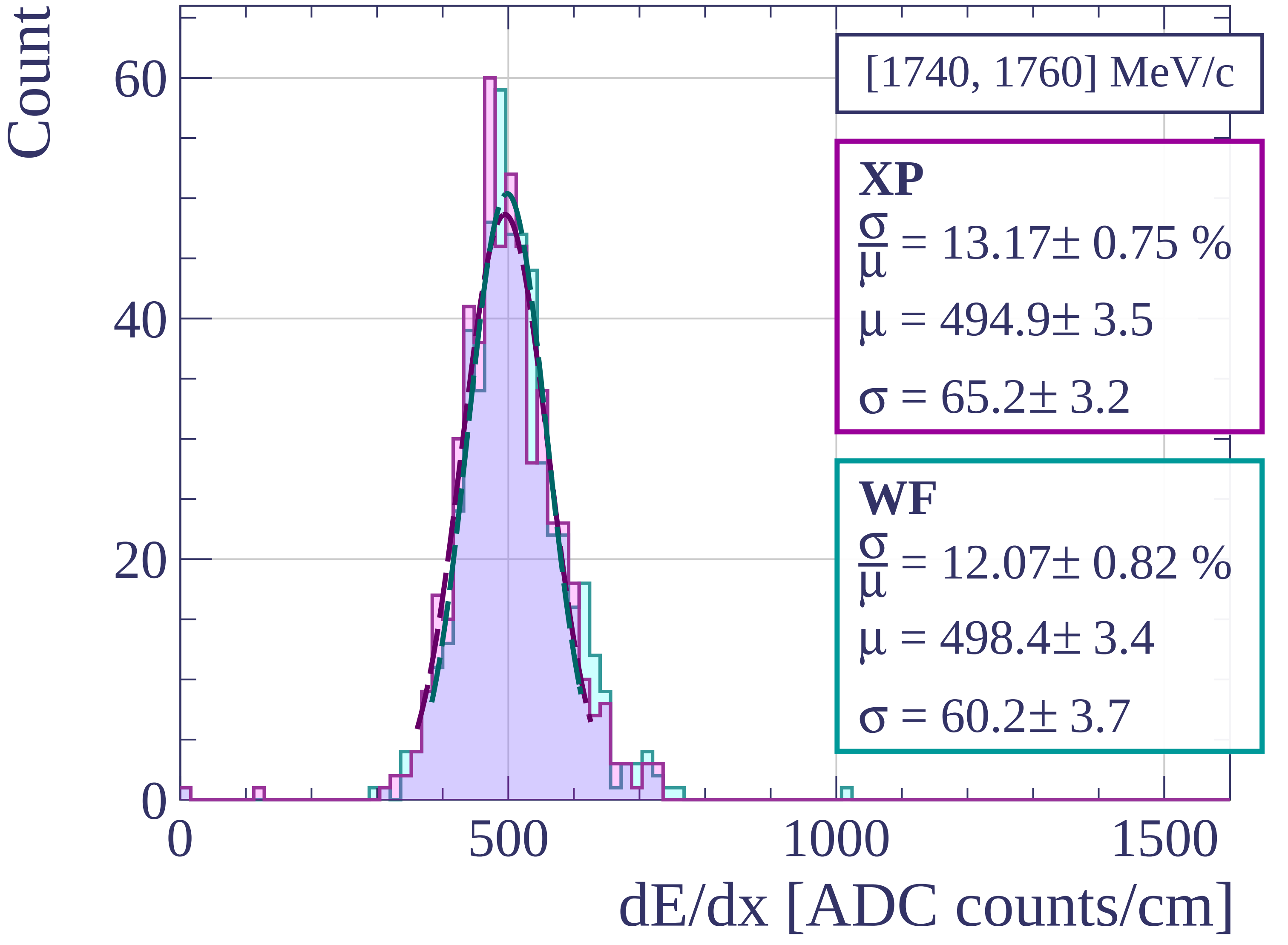

Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1660, 1680] MeV/c
XP
σ/μ = 11.87± 0.63 %
μ = 491.5± 3.1
σ = 58.3± 2.7
WF
σ/μ = 11.67± 0.66 %
μ = 499.2± 3.0
σ = 58.3± 2.9



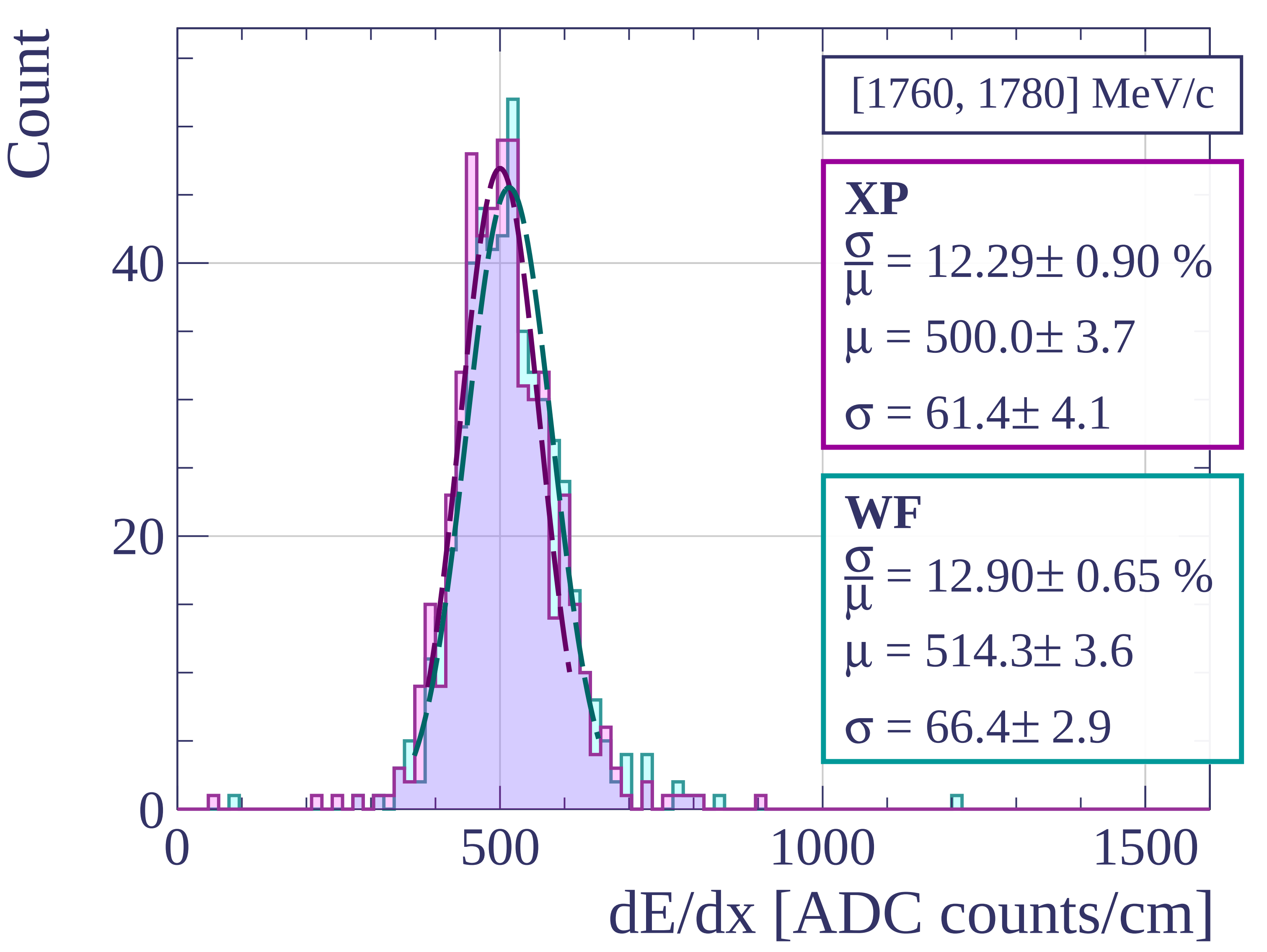
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1680, 1700] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.41± 0.75 %
μ = 497.7± 3.2
σ = 61.8± 3.3
WF
$\frac{\sigma}{\mu}$ = 11.81± 0.70 %
μ = 507.9± 3.3
σ = 60.0± 3.2


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1700, 1720] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.20± 0.73 %
μ = 496.0± 3.5
σ = 65.5± 3.2
WF
$\frac{\sigma}{\mu}$ = 13.42± 0.78 %
μ = 504.6± 3.6
σ = 67.7± 3.5

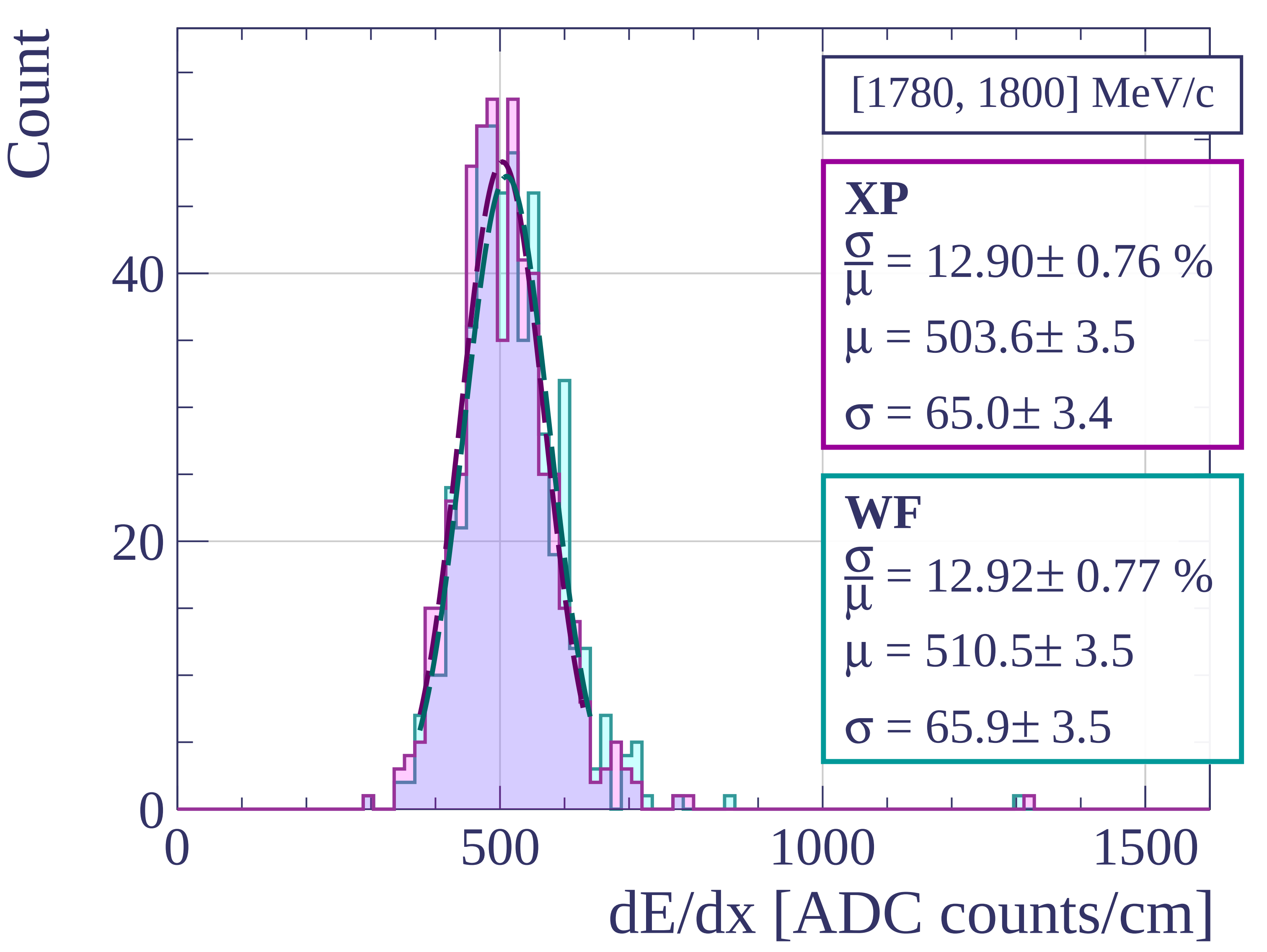

Count
0
20
40
60
0
500
1000
1500
dE/dx [ADC counts/cm]
[1720, 1740] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.10± 0.69 %
μ = 497.7± 3.3
σ = 60.2± 3.0
WF
$\frac{\sigma}{\mu}$ = 12.53± 0.78 %
μ = 505.1± 3.4
σ = 63.3± 3.5

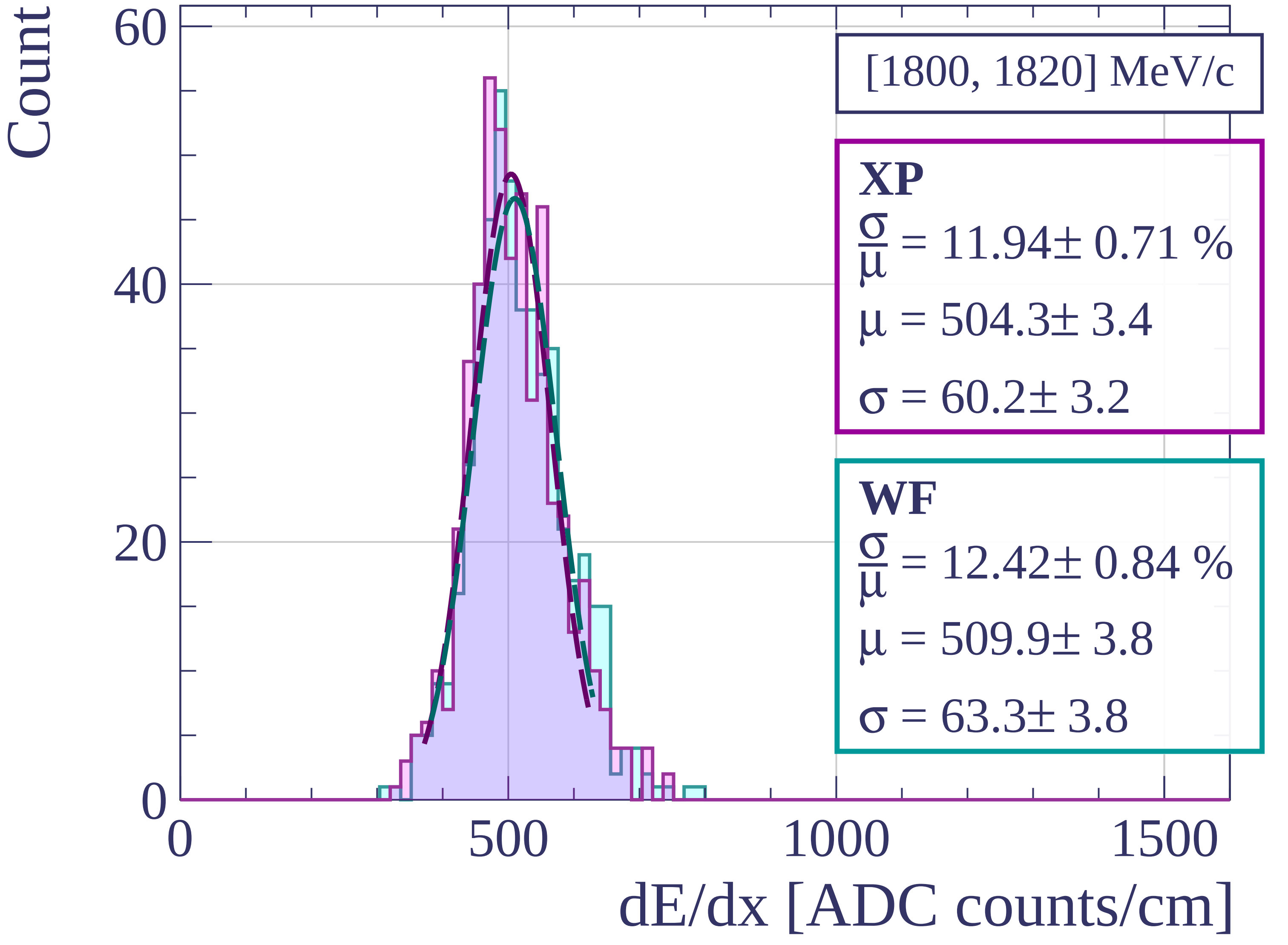


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1740, 1760] MeV/c
XP
σ/μ = 13.17± 0.75 %
μ = 494.9± 3.5
σ = 65.2± 3.2
WF
σ/μ = 12.07± 0.82 %
μ = 498.4± 3.4
σ = 60.2± 3.7

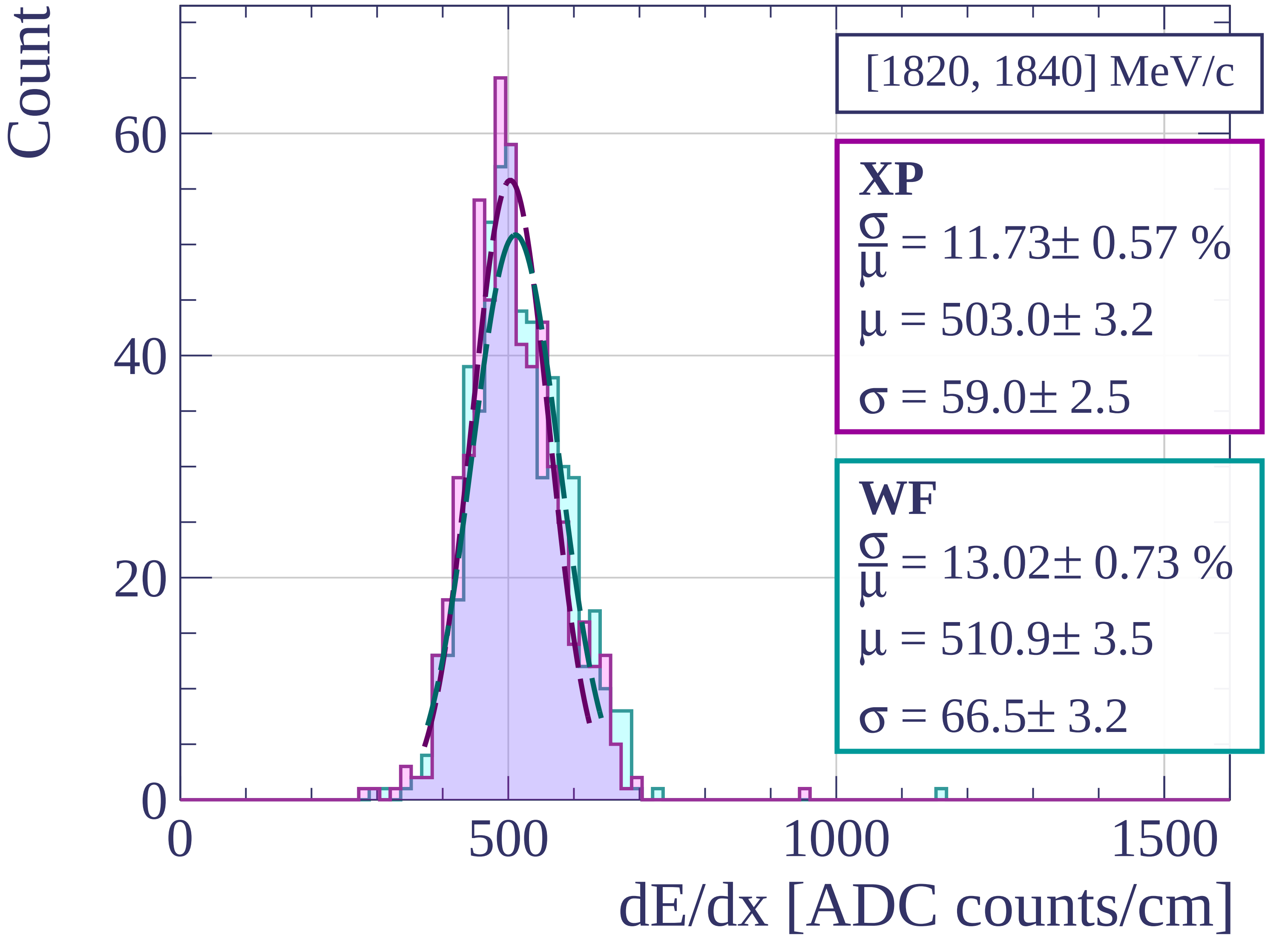


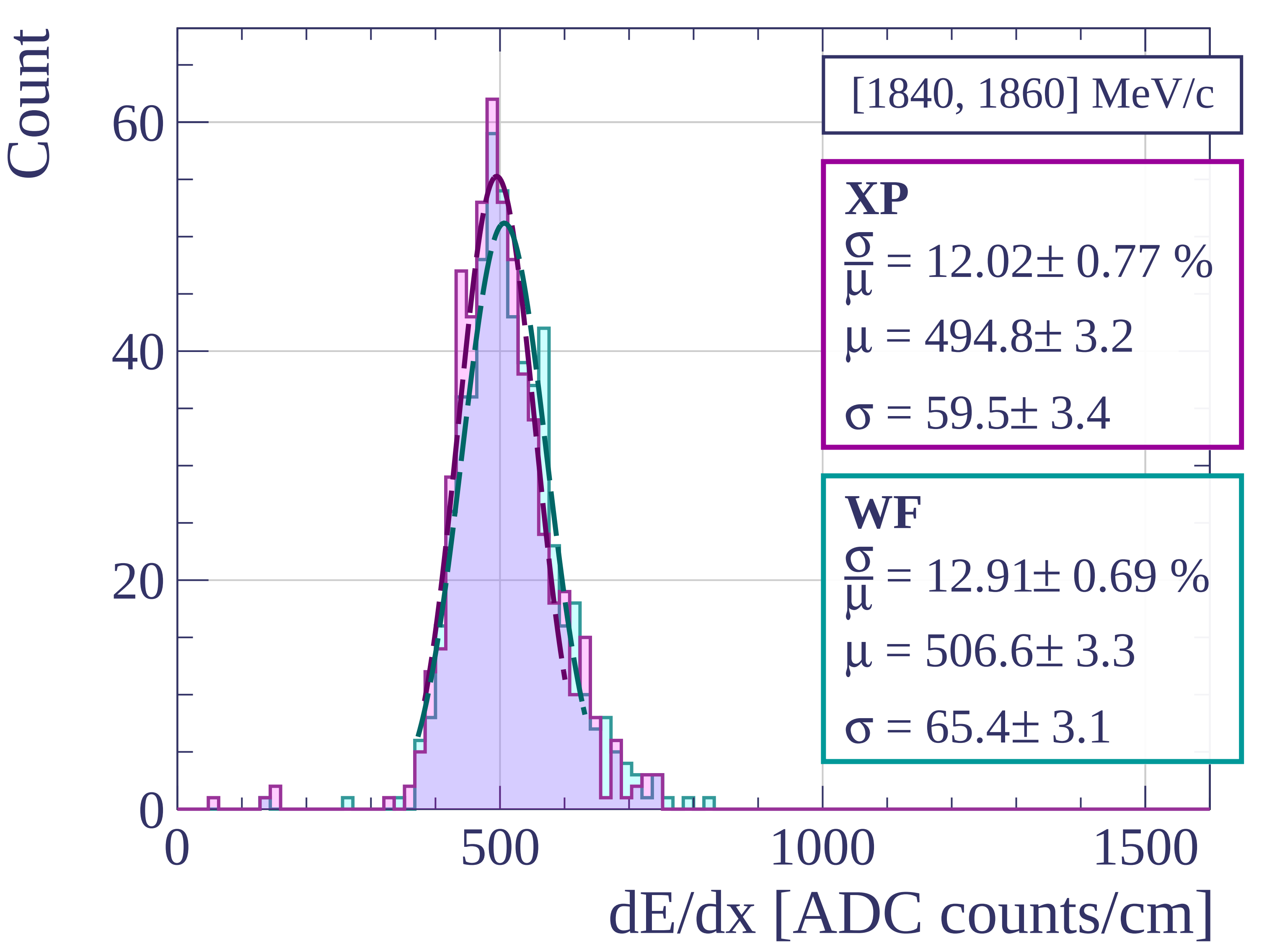
Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1760, 1780] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.29± 0.90 %
μ = 500.0± 3.7
σ = 61.4± 4.1
WF
$\frac{\sigma}{\mu}$ = 12.90± 0.65 %
μ = 514.3± 3.6
σ = 66.4± 2.9


Count
0
20
40
0
500
1000
1500
dE/dx [ADC counts/cm]
[1780, 1800] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.90± 0.76 %
μ = 503.6± 3.5
σ = 65.0± 3.4
WF
$\frac{\sigma}{\mu}$ = 12.92± 0.77 %
μ = 510.5± 3.5
σ = 65.9± 3.5

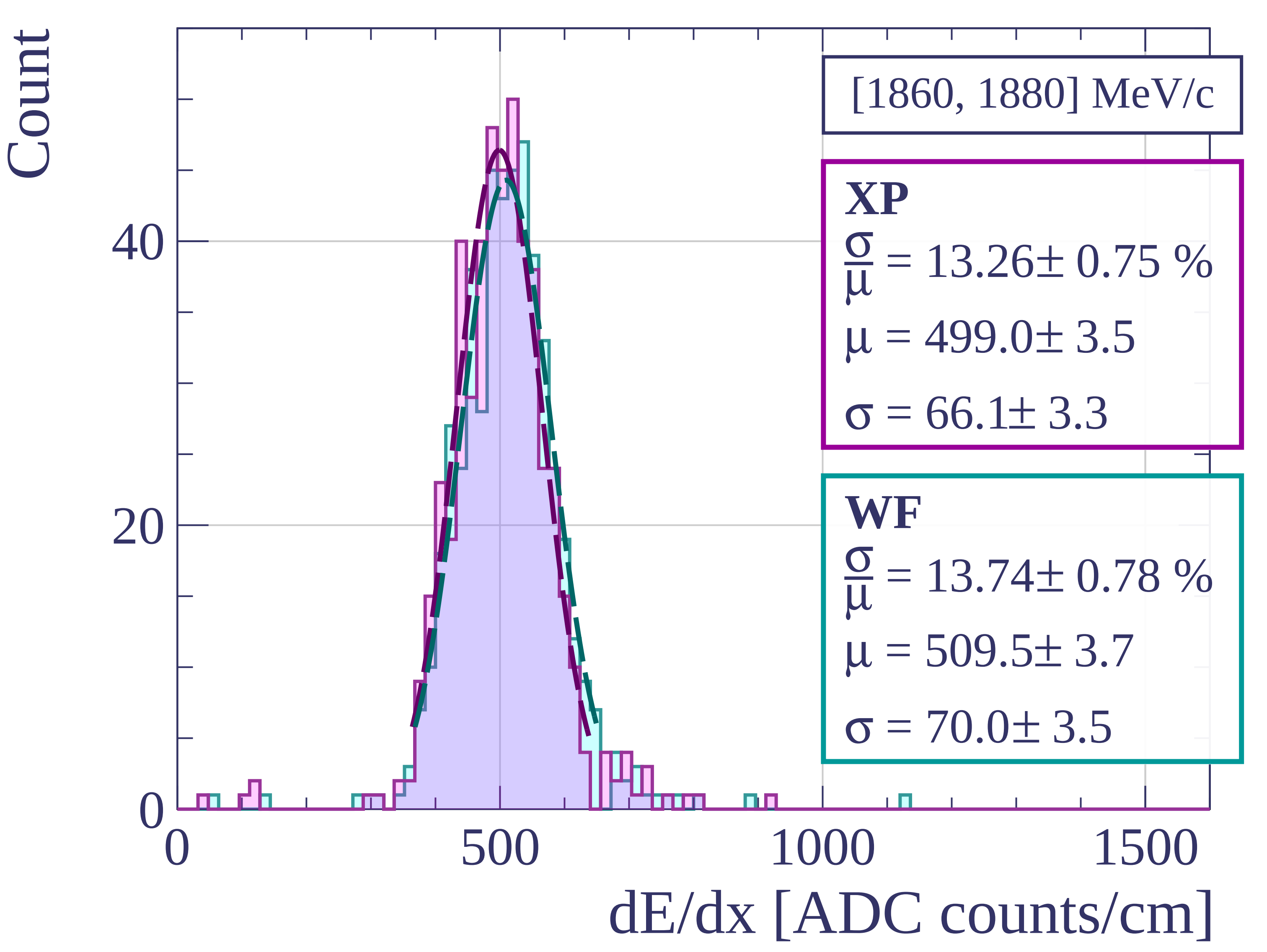


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1800, 1820] MeV/c
XP
σ/μ = 11.94 ± 0.71 %
μ = 504.3 ± 3.4
σ = 60.2 ± 3.2
WF
σ/μ = 12.42 ± 0.84 %
μ = 509.9 ± 3.8
σ = 63.3 ± 3.8

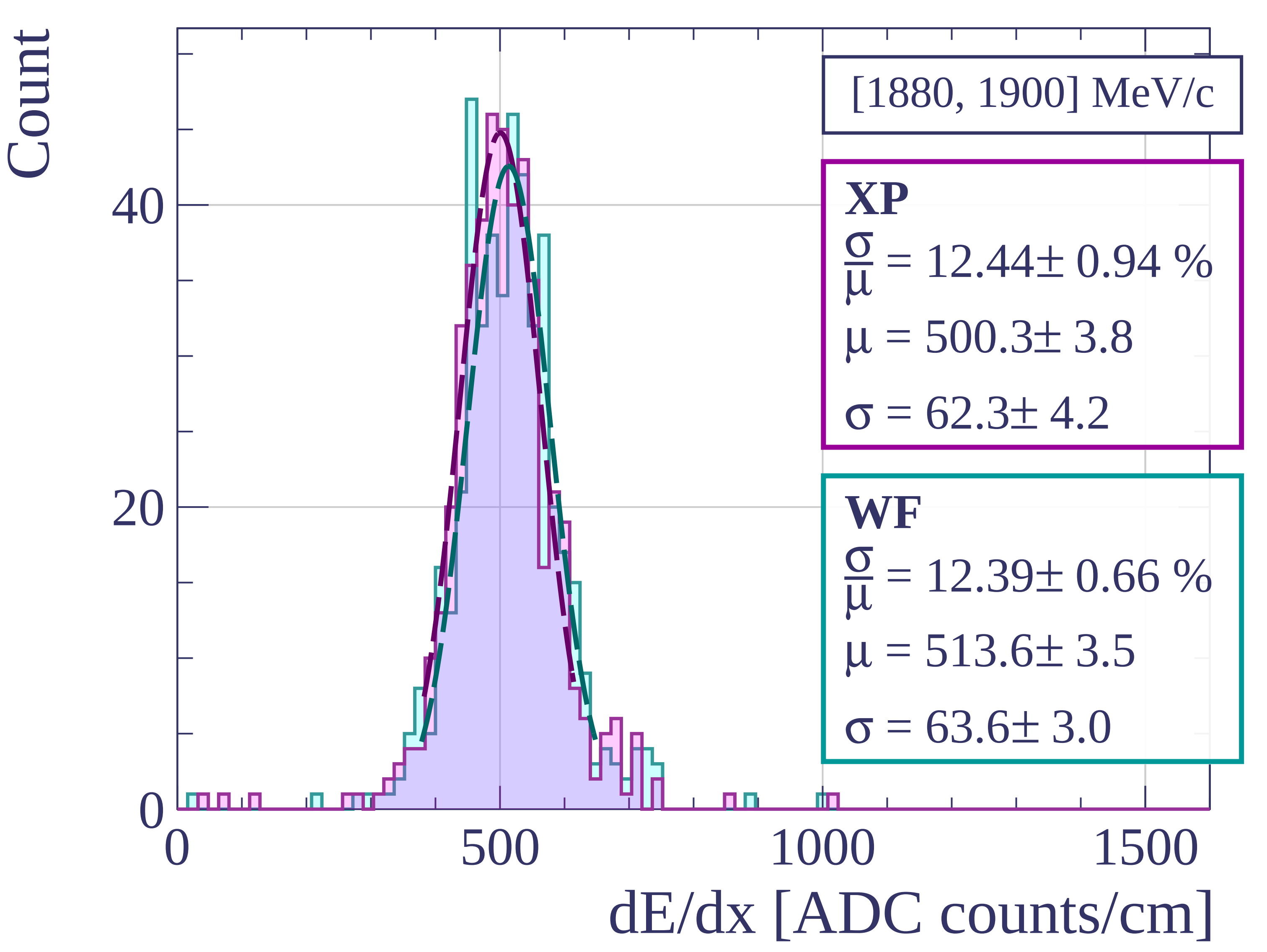

Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1820, 1840] MeV/c
XP
σ/μ = 11.73± 0.57 %
μ = 503.0± 3.2
σ = 59.0± 2.5
WF
σ/μ = 13.02± 0.73 %
μ = 510.9± 3.5
σ = 66.5± 3.2

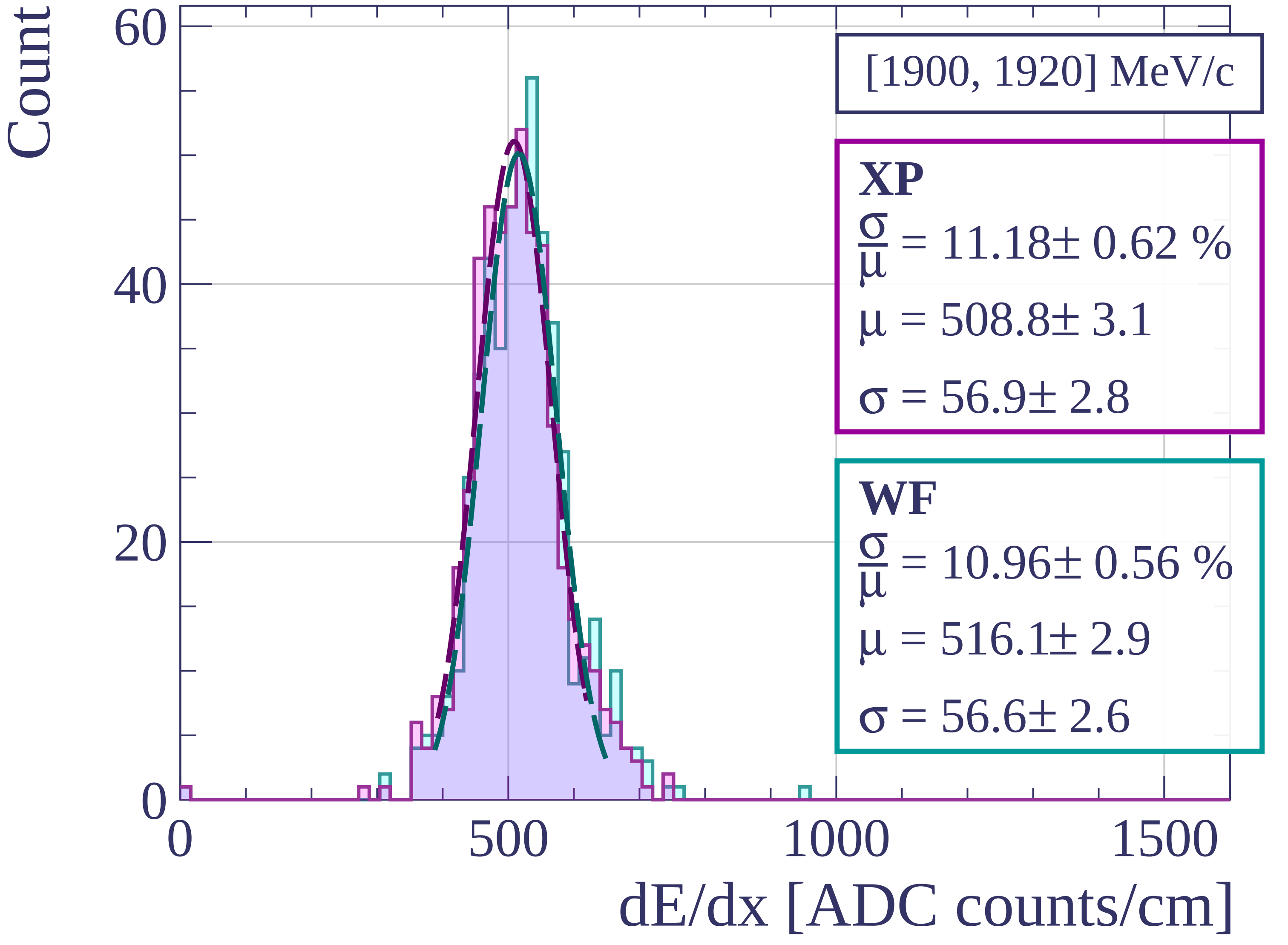


Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1840, 1860] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.02± 0.77 %
μ = 494.8± 3.2
σ = 59.5± 3.4
WF
$\frac{\sigma}{\mu}$ = 12.91± 0.69 %
μ = 506.6± 3.3
σ = 65.4± 3.1

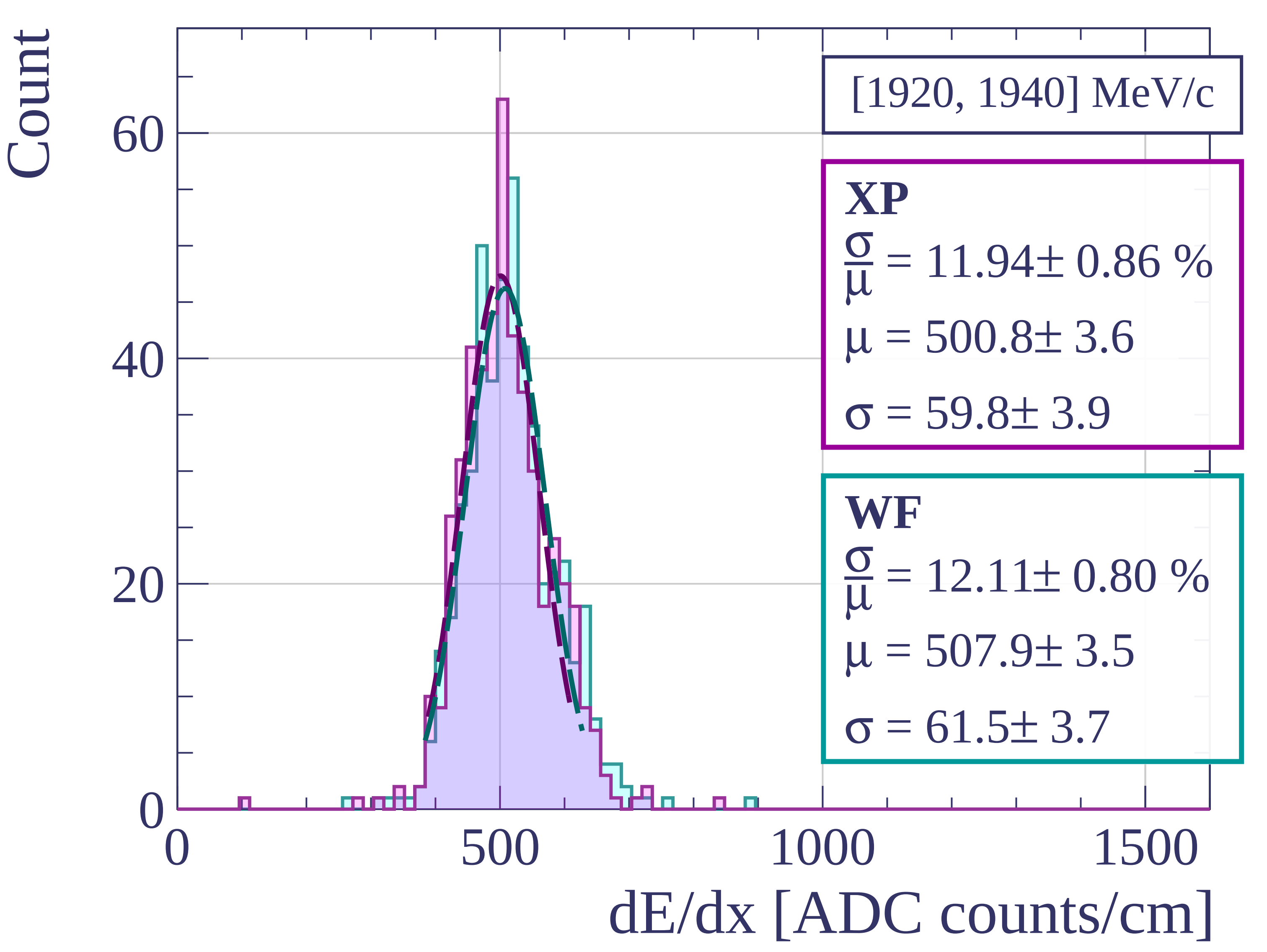


Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1860, 1880] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.26± 0.75 %
μ = 499.0± 3.5
σ = 66.1± 3.3
WF
$\frac{\sigma}{\mu}$ = 13.74± 0.78 %
μ = 509.5± 3.7
σ = 70.0± 3.5

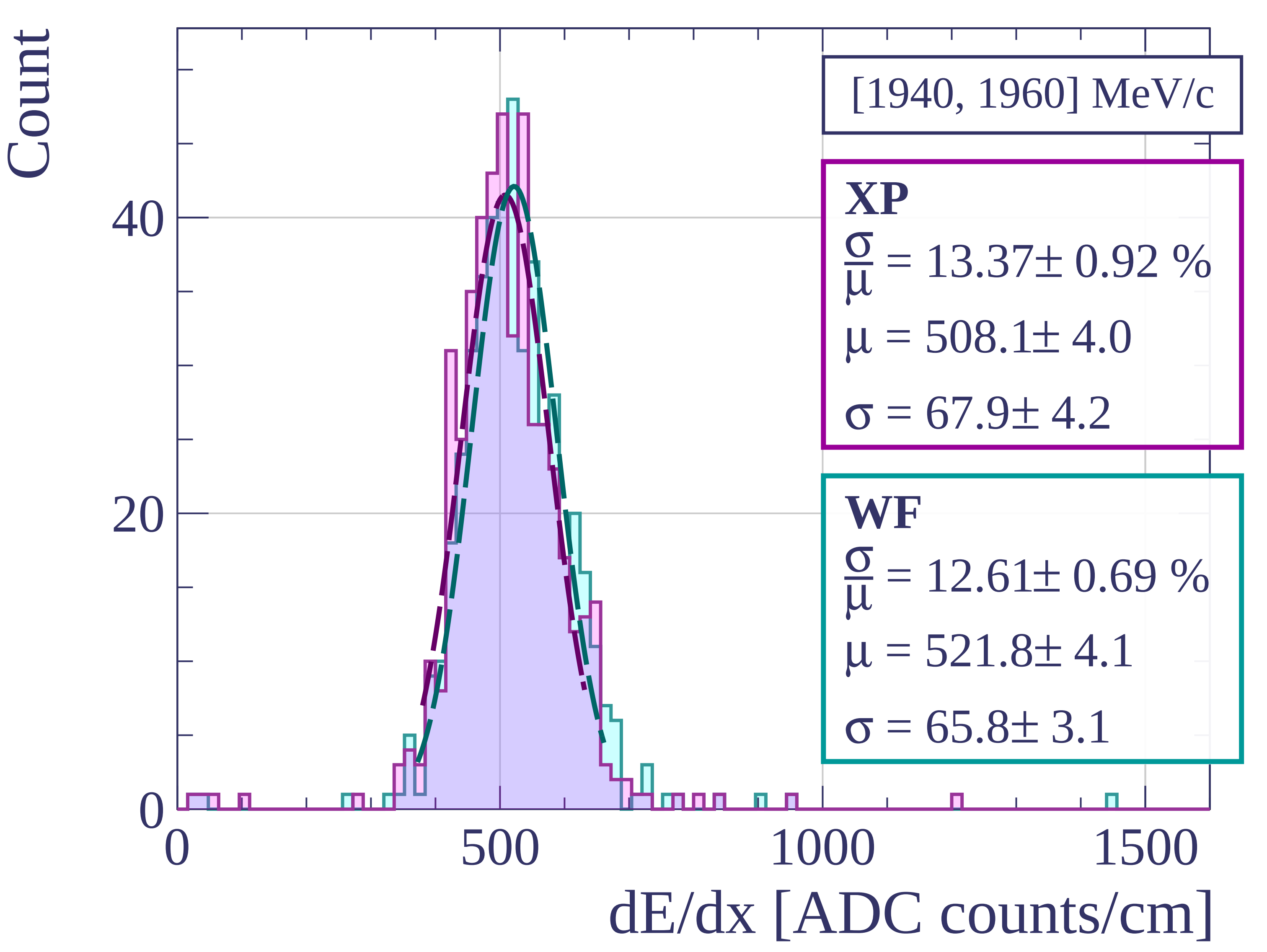

Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1880, 1900] MeV/c
XP
$\frac{\sigma}{\mu}$ = 12.44± 0.94 %
μ = 500.3± 3.8
σ = 62.3± 4.2
WF
$\frac{\sigma}{\mu}$ = 12.39± 0.66 %
μ = 513.6± 3.5
σ = 63.6± 3.0



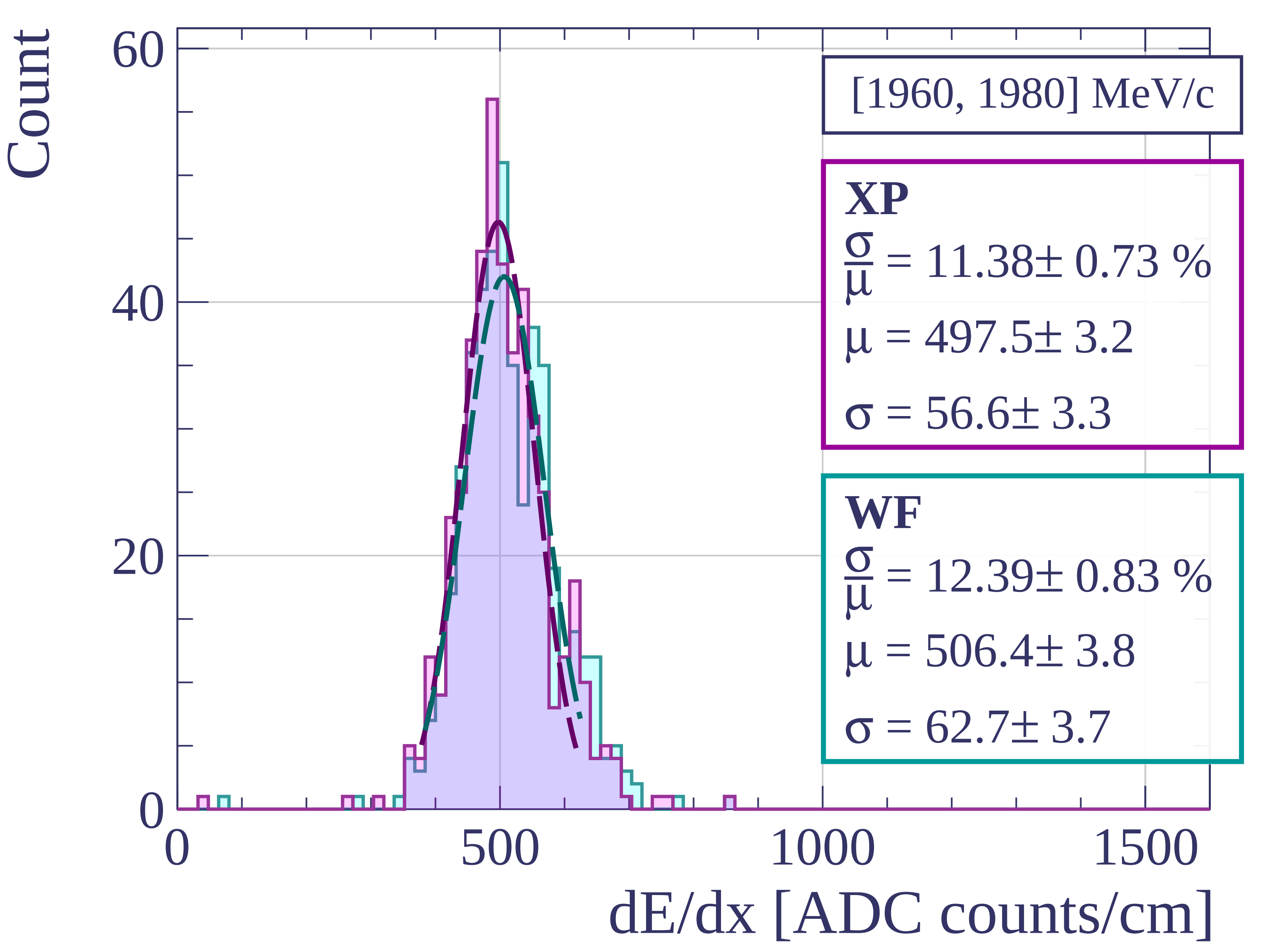
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1900, 1920] MeV/c
XP
σ/μ = 11.18± 0.62 %
μ = 508.8± 3.1
σ = 56.9± 2.8
WF
σ/μ = 10.96± 0.56 %
μ = 516.1± 2.9
σ = 56.6± 2.6



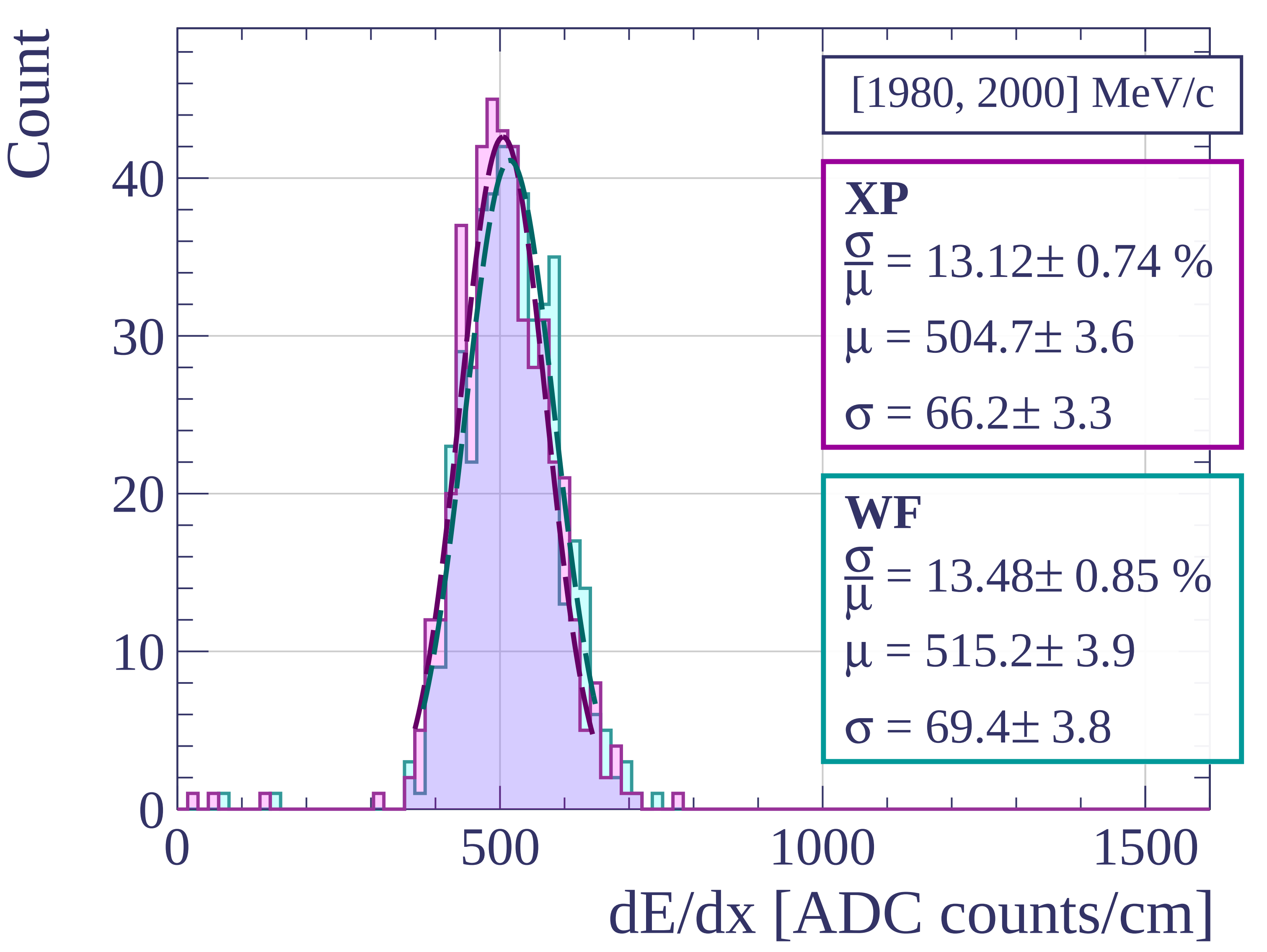
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1920, 1940] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.94± 0.86 %
μ = 500.8± 3.6
σ = 59.8± 3.9
WF
$\frac{\sigma}{\mu}$ = 12.11± 0.80 %
μ = 507.9± 3.5
σ = 61.5± 3.7



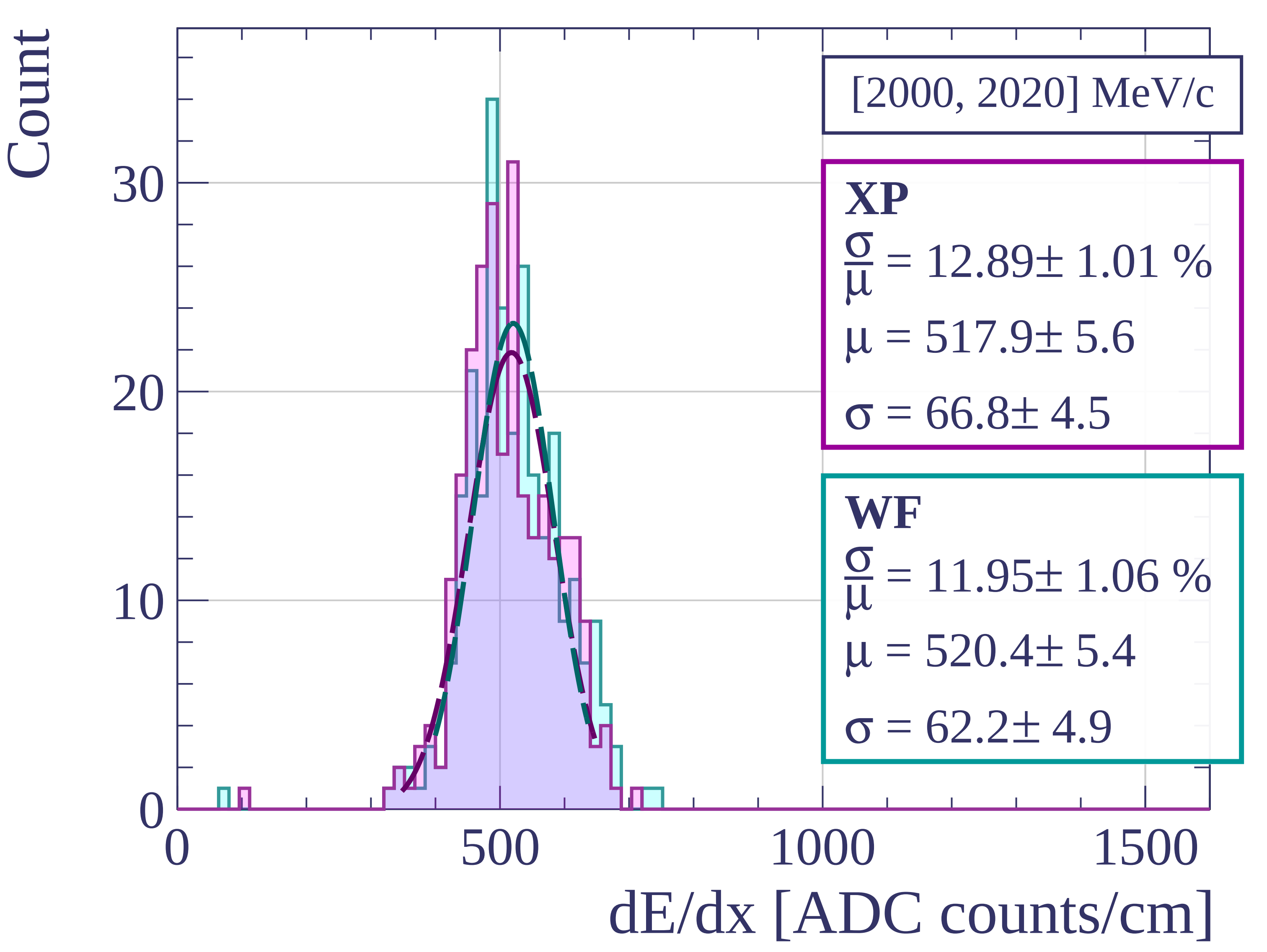
Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1940, 1960] MeV/c
XP
σ/μ = 13.37± 0.92 %
μ = 508.1± 4.0
σ = 67.9± 4.2
WF
σ/μ = 12.61± 0.69 %
μ = 521.8± 4.1
σ = 65.8± 3.1



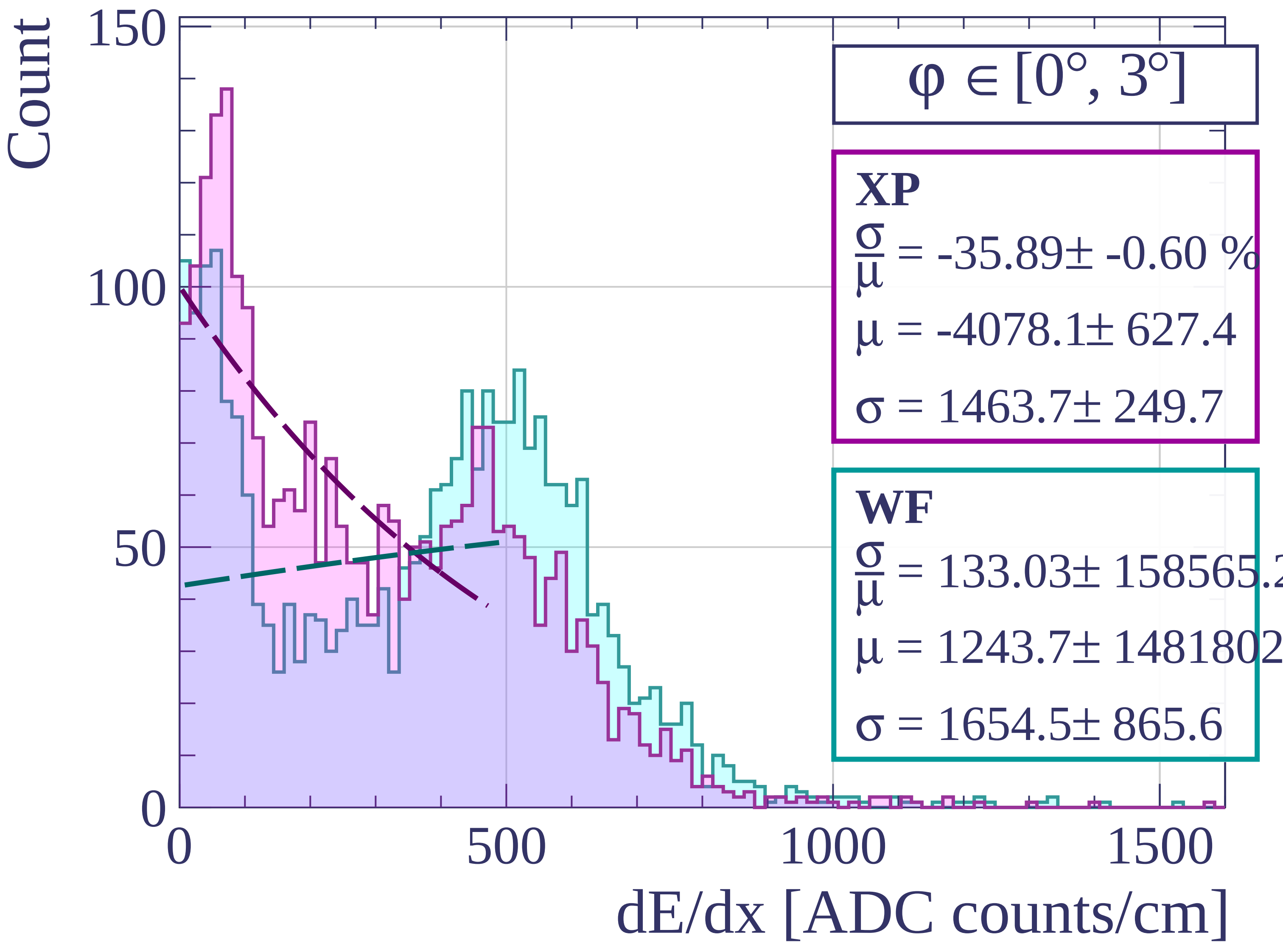
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[1960, 1980] MeV/c
XP
$\frac{\sigma}{\mu}$ = 11.38± 0.73 %
μ = 497.5± 3.2
σ = 56.6± 3.3
WF
$\frac{\sigma}{\mu}$ = 12.39± 0.83 %
μ = 506.4± 3.8
σ = 62.7± 3.7



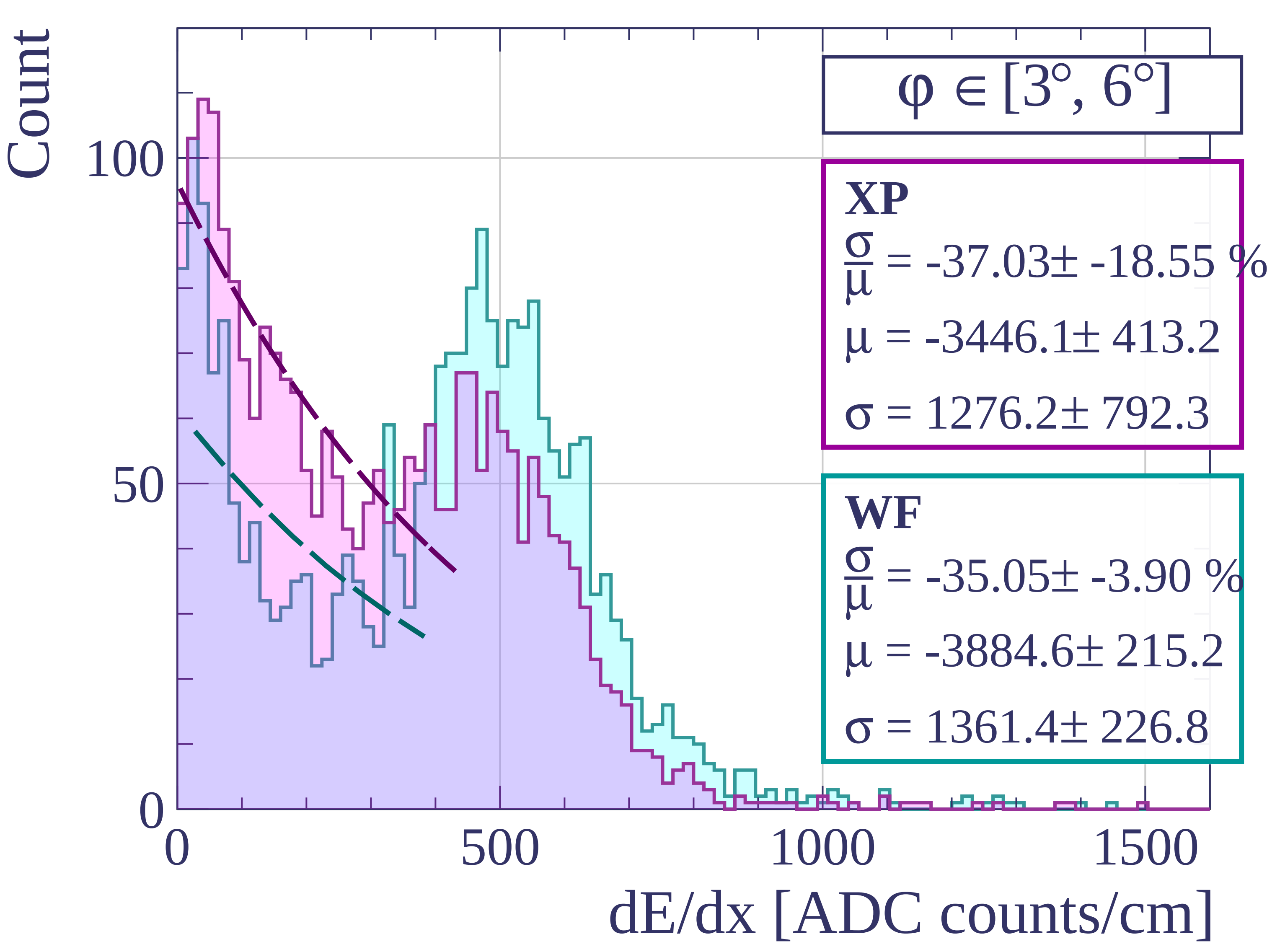
Count
0
10
20
30
40
0
500
1000
1500
dE/dx [ADC counts/cm]
[1980, 2000] MeV/c
XP
$\frac{\sigma}{\mu}$ = 13.12± 0.74 %
μ = 504.7± 3.6
σ = 66.2± 3.3
WF
$\frac{\sigma}{\mu}$ = 13.48± 0.85 %
μ = 515.2± 3.9
σ = 69.4± 3.8

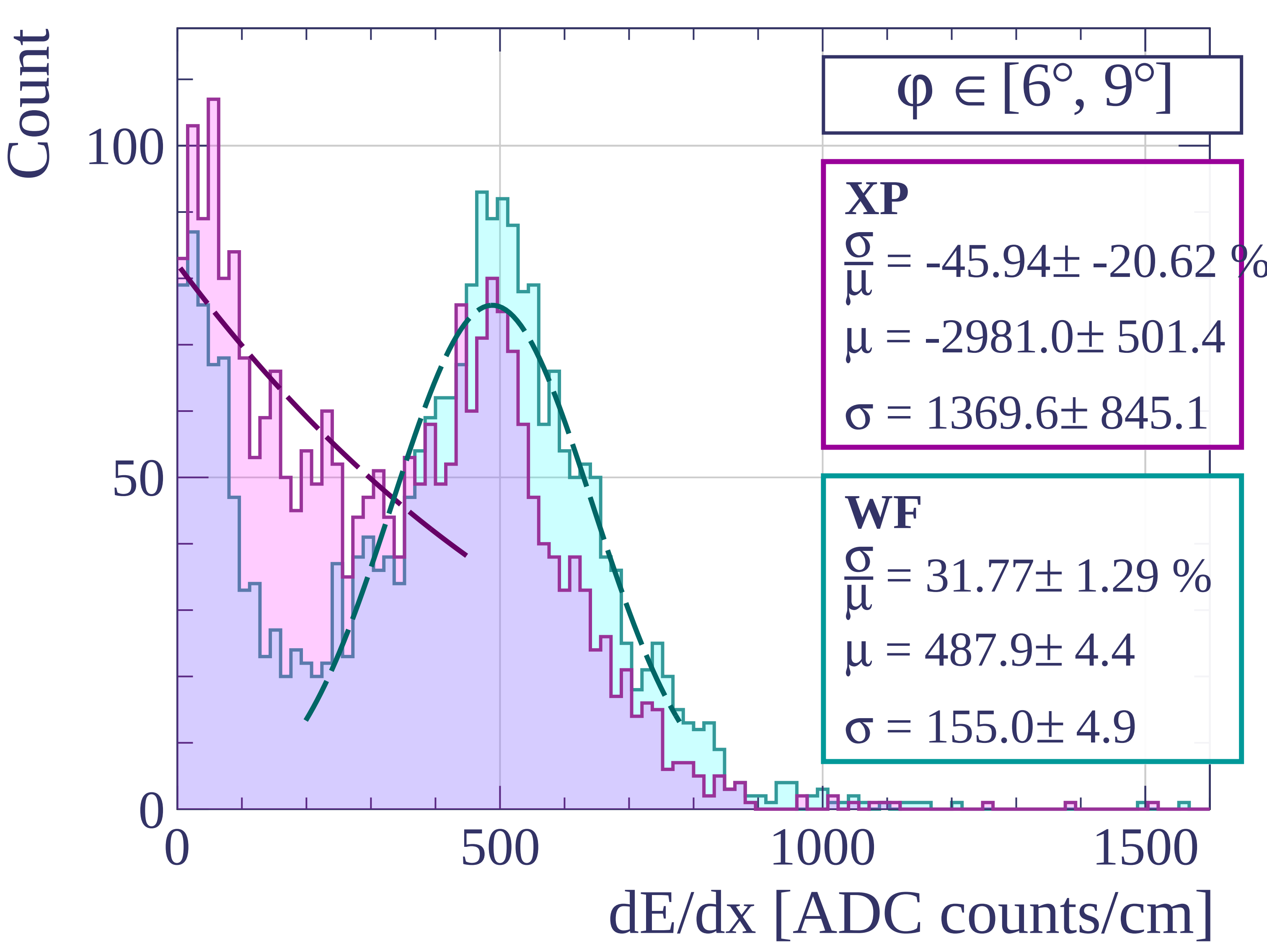


Count
30
20
10
0
0
500
1000
1500
dE/dx [ADC counts/cm]
[2000, 2020] MeV/c
XP
σ/μ = 12.89± 1.01 %
μ = 517.9± 5.6
σ = 66.8± 4.5
WF
σ/μ = 11.95± 1.06 %
μ = 520.4± 5.4
σ = 62.2± 4.9

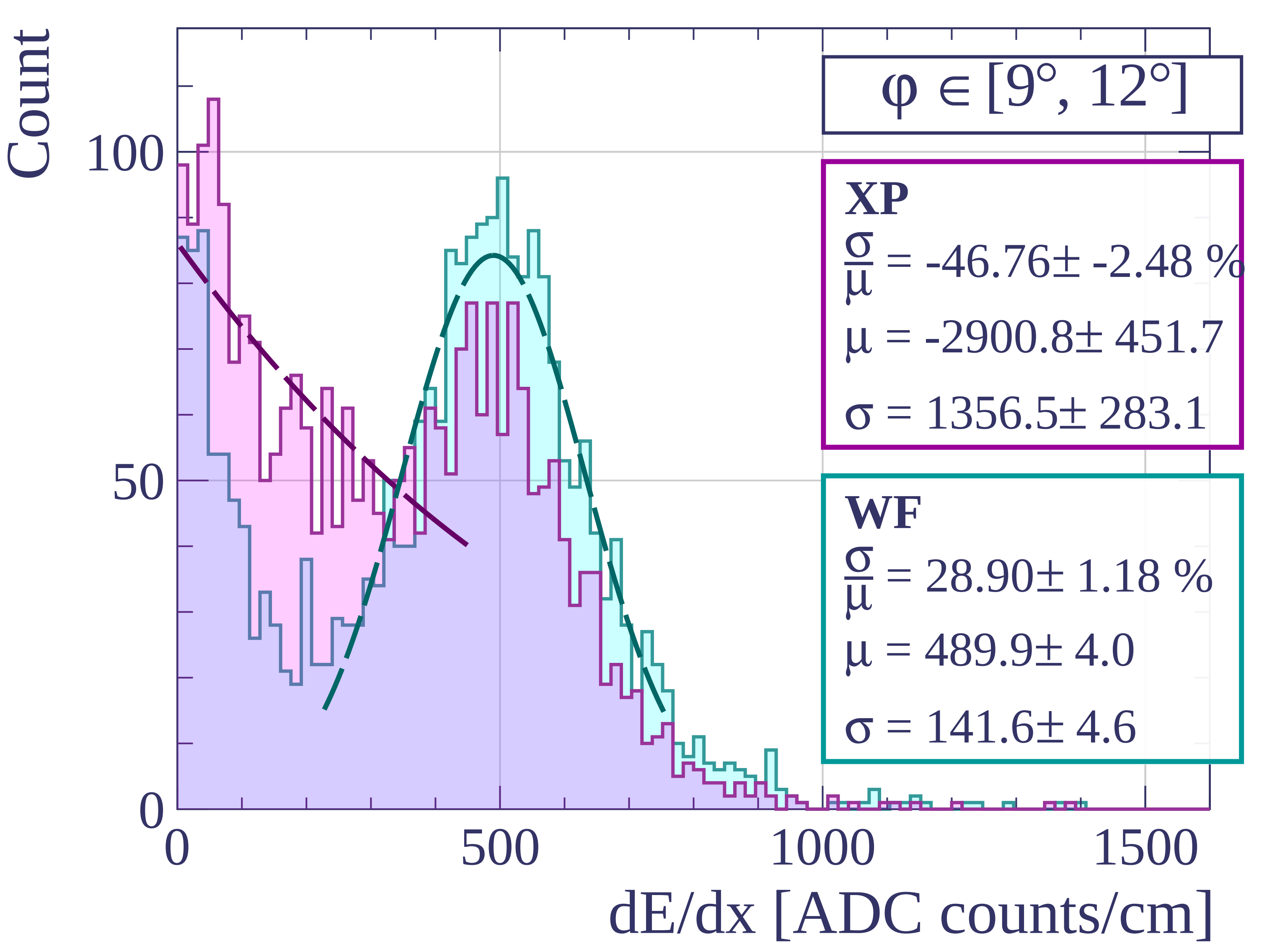


Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [0°, 3°]
XP
σ/μ = -35.89± -0.60 %
μ = -4078.1± 627.4
σ = 1463.7± 249.7
WF
σ/μ = 133.03± 158565.2
μ = 1243.7± 1481802
σ = 1654.5± 865.6

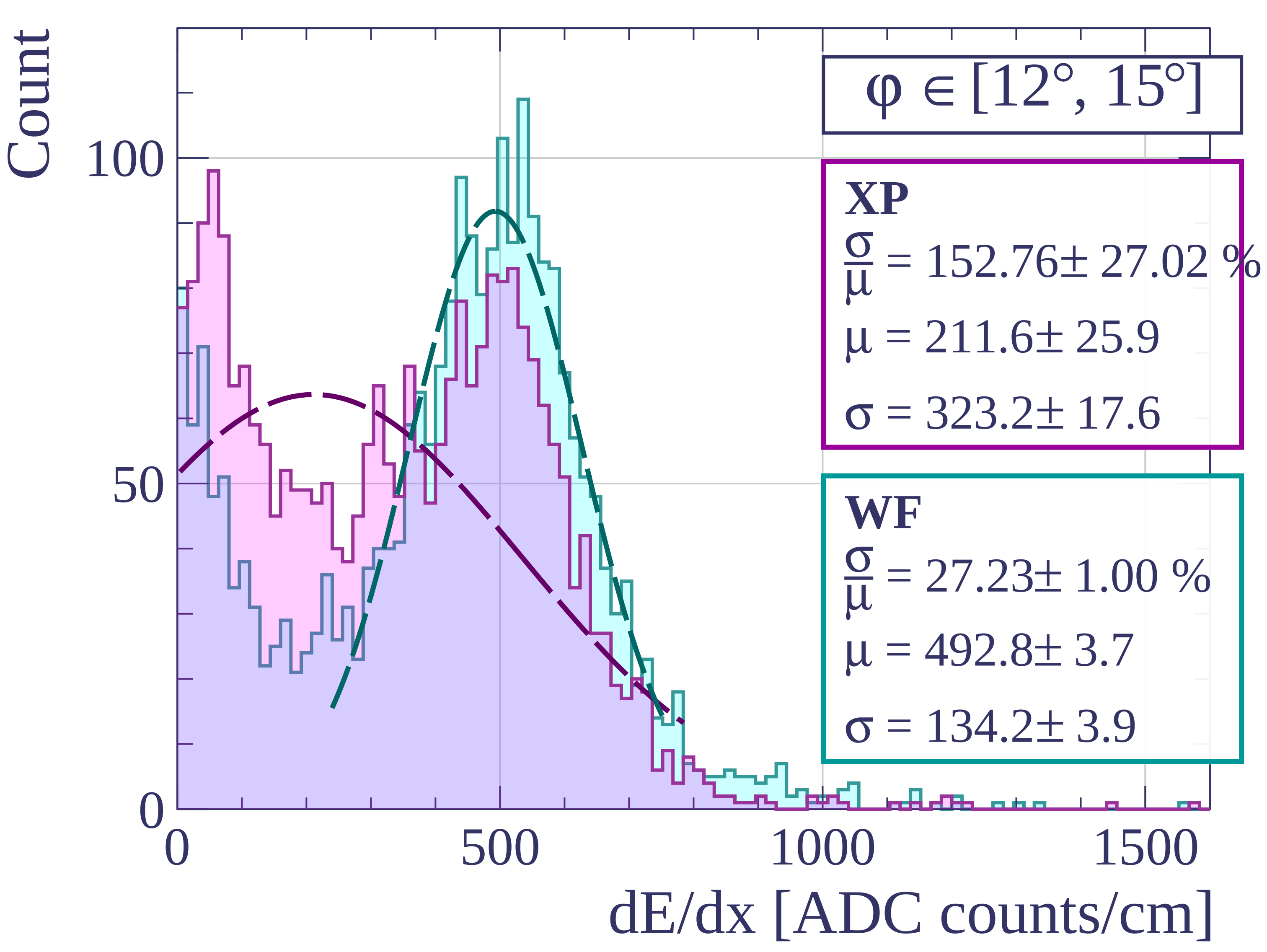

φ ∈ [3°, 6°]
XP
σ/μ = -37.03± -18.55 %
μ = -3446.1± 413.2
σ = 1276.2± 792.3
WF
σ/μ = -35.05± -3.90 %
μ = -3884.6± 215.2
σ = 1361.4± 226.8
Count
dE/dx [ADC counts/cm]
0
50
100
0
500
1000
1500



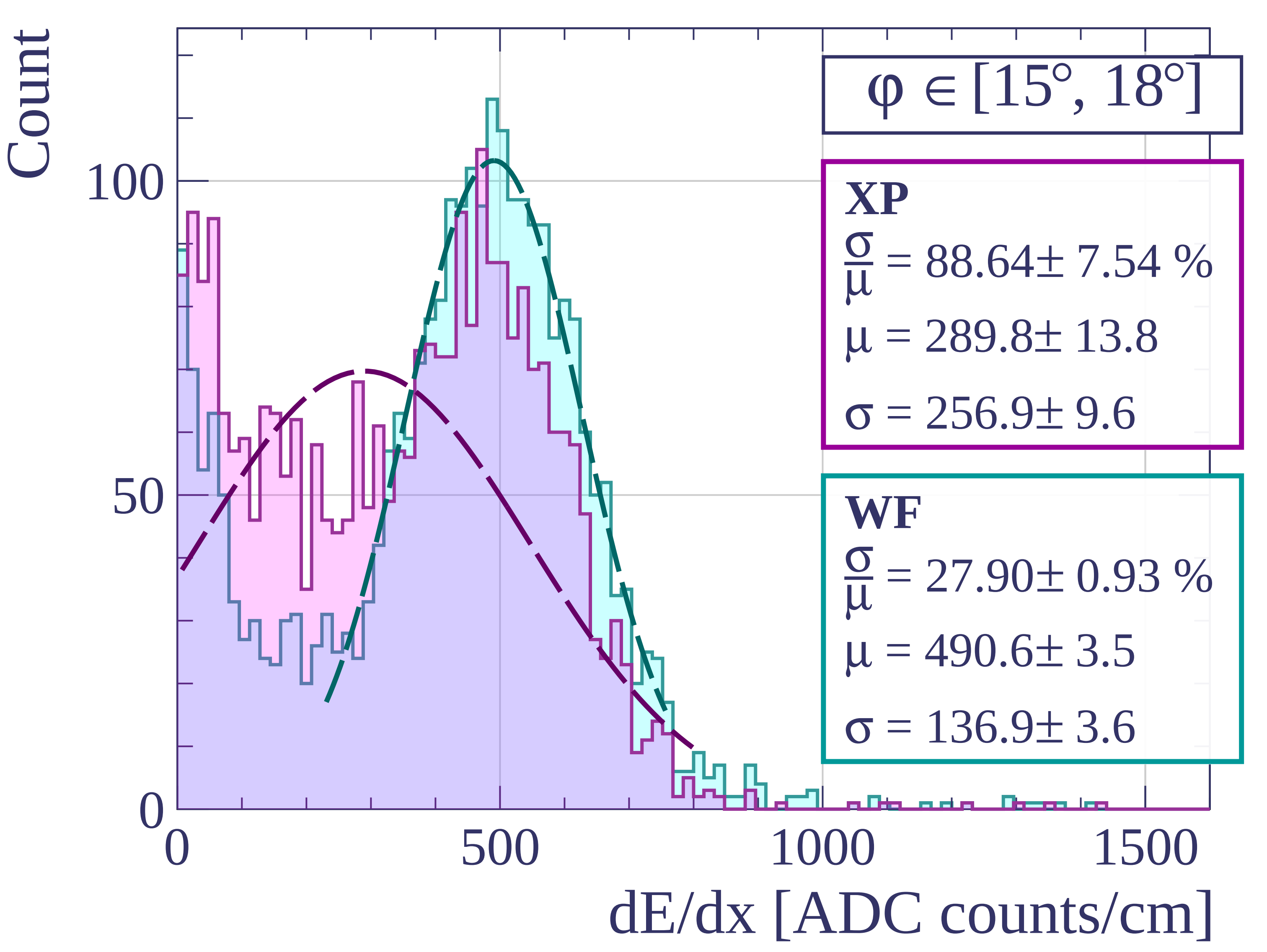
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [6°, 9°]
XP
σ/μ = -45.94± -20.62 %
μ = -2981.0± 501.4
σ = 1369.6± 845.1
WF
σ/μ = 31.77± 1.29 %
μ = 487.9± 4.4
σ = 155.0± 4.9

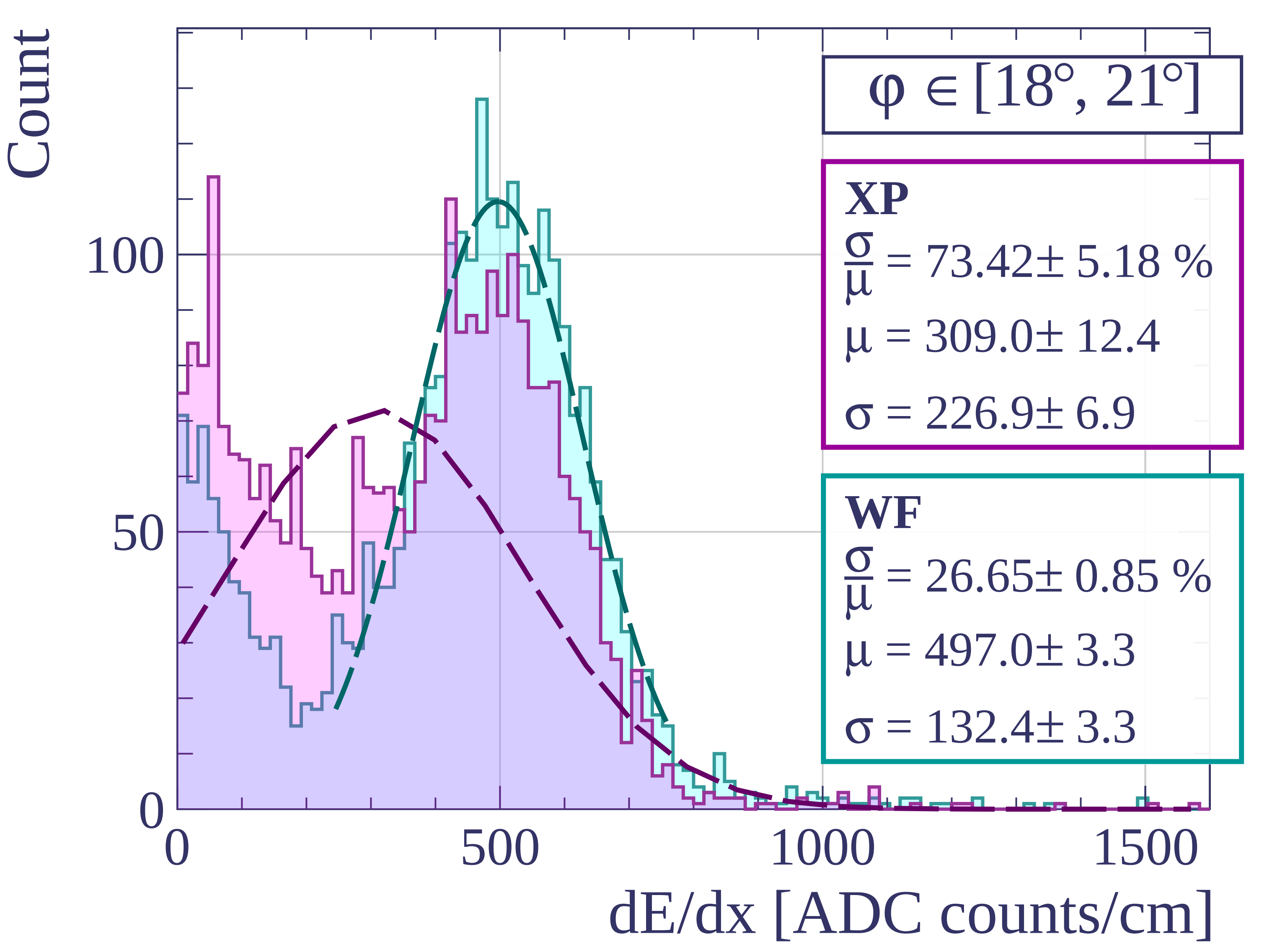


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [9°, 12°]
XP
σ/μ = -46.76± -2.48 %
μ = -2900.8± 451.7
σ = 1356.5± 283.1
WF
σ/μ = 28.90± 1.18 %
μ = 489.9± 4.0
σ = 141.6± 4.6

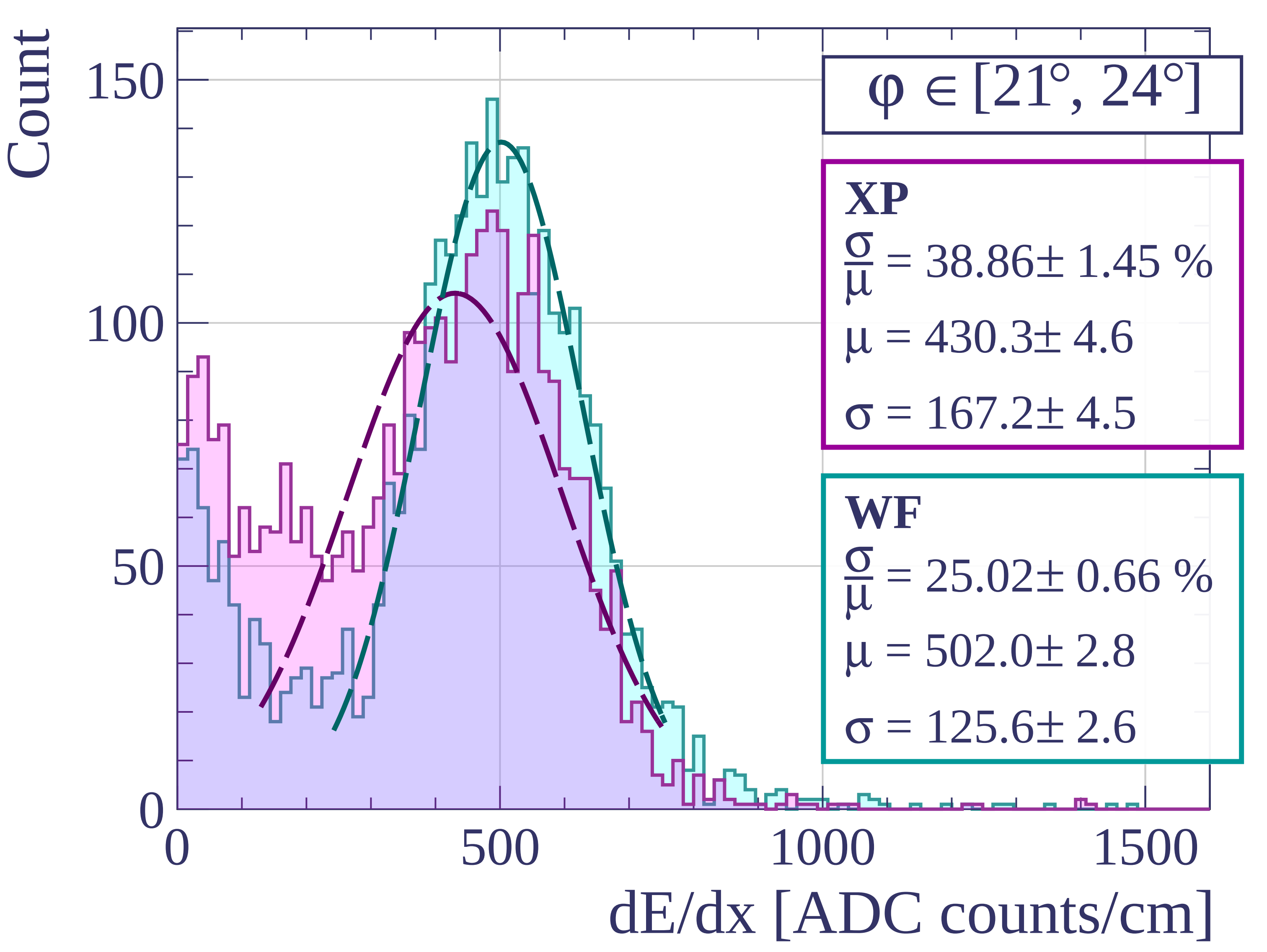


Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [12°, 15°]
XP
σ/μ = 152.76± 27.02 %
μ = 211.6± 25.9
σ = 323.2± 17.6
WF
σ/μ = 27.23± 1.00 %
μ = 492.8± 3.7
σ = 134.2± 3.9

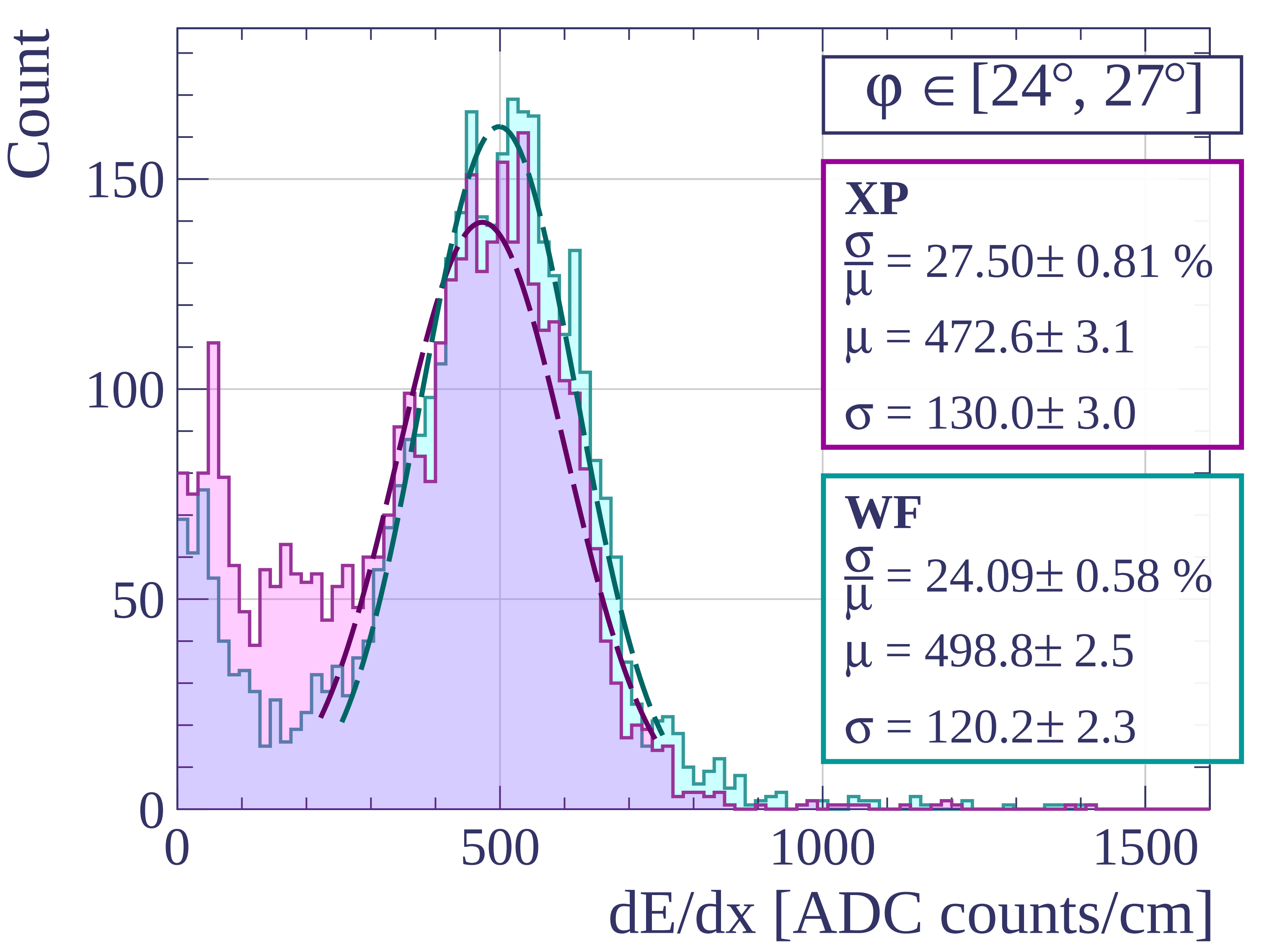

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [15°, 18°]
XP
σ/μ = 88.64± 7.54 %
μ = 289.8± 13.8
σ = 256.9± 9.6
WF
σ/μ = 27.90± 0.93 %
μ = 490.6± 3.5
σ = 136.9± 3.6



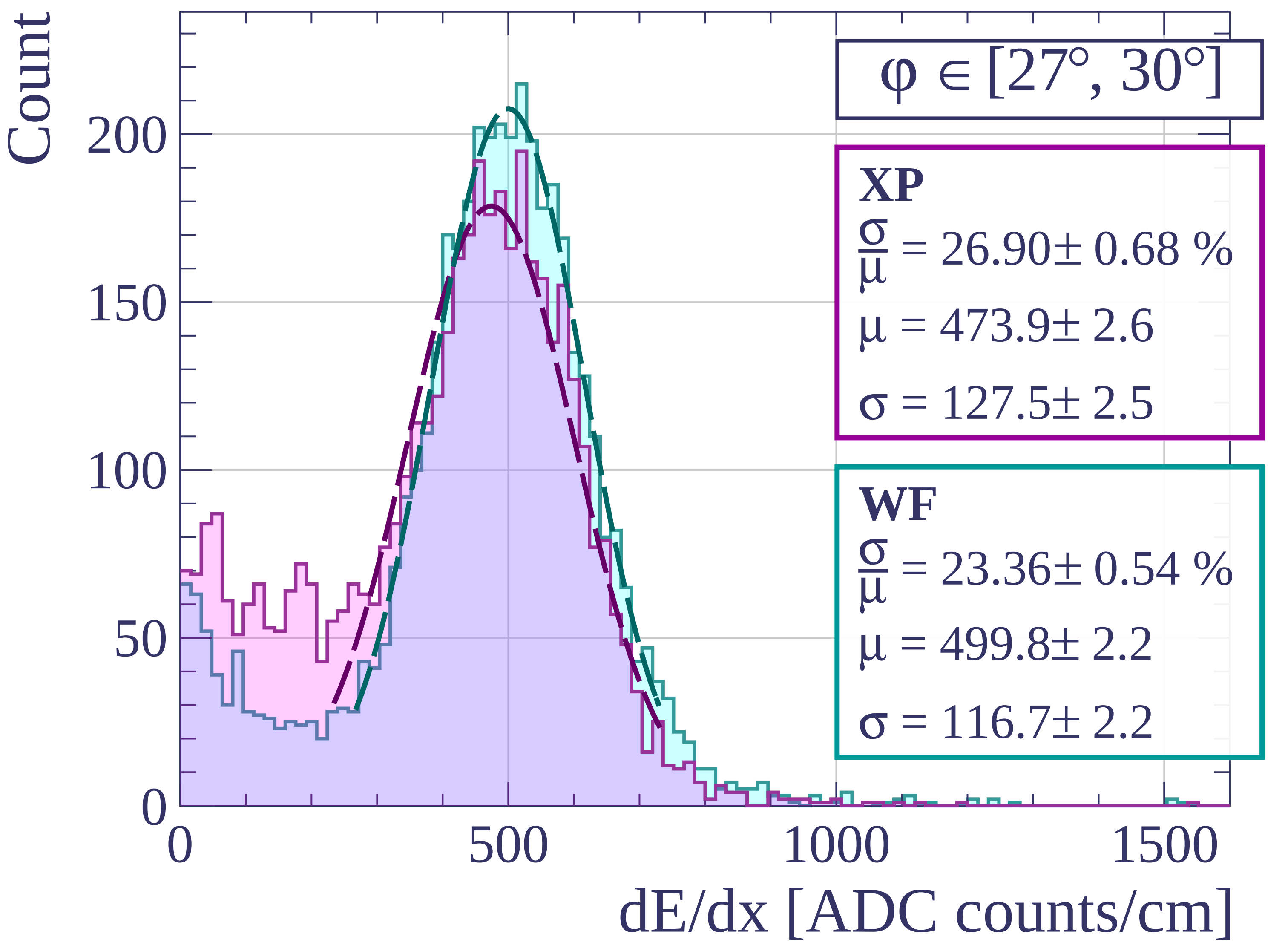
Count
dE/dx [ADC counts/cm]
0
50
100
0
500
1000
1500
φ ∈ [18°, 21°]
XP
σ/μ = 73.42± 5.18 %
μ = 309.0± 12.4
σ = 226.9± 6.9
WF
σ/μ = 26.65± 0.85 %
μ = 497.0± 3.3
σ = 132.4± 3.3



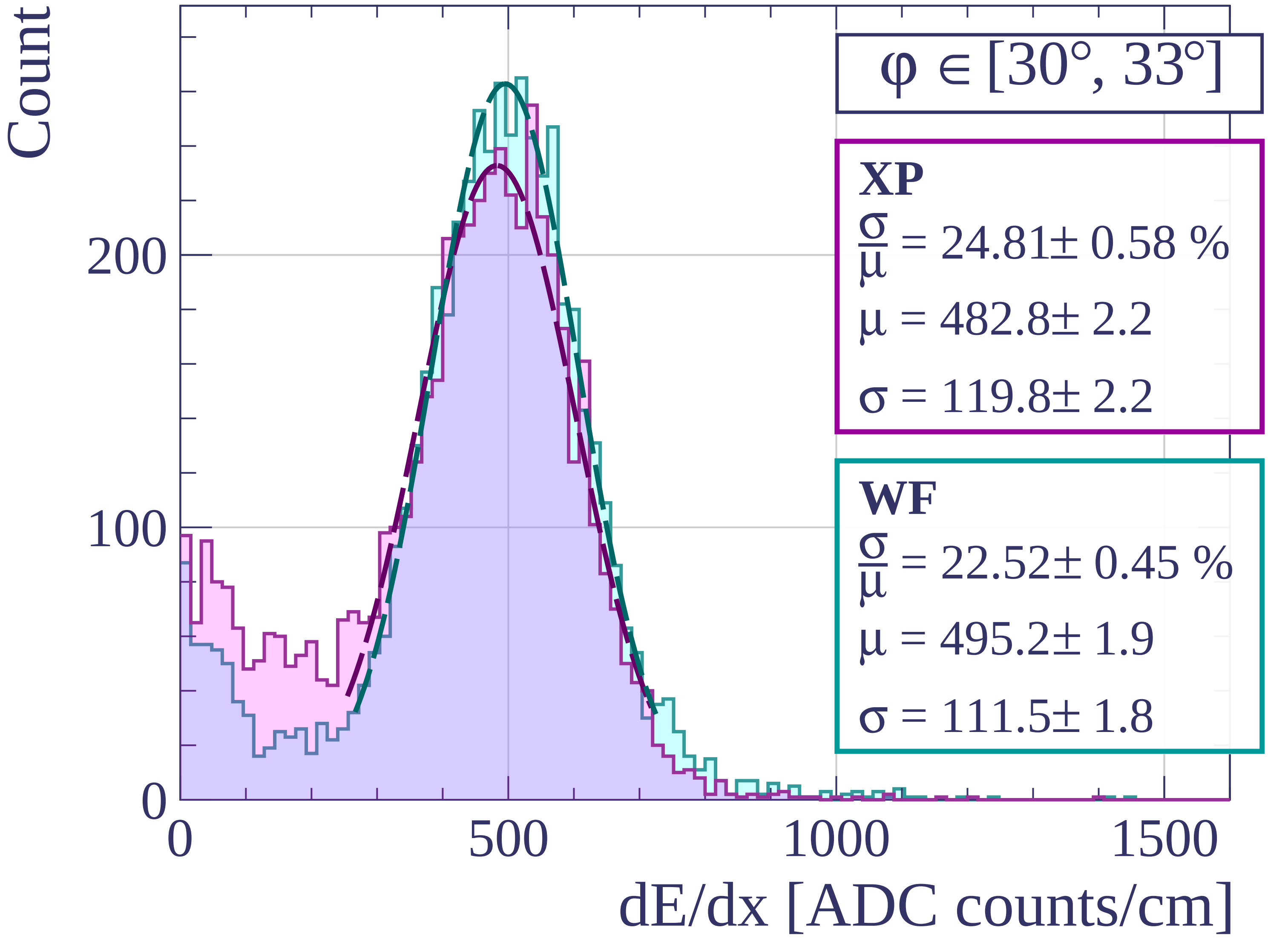
Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [21°, 24°]
XP
σ/μ = 38.86± 1.45 %
μ = 430.3± 4.6
σ = 167.2± 4.5
WF
σ/μ = 25.02± 0.66 %
μ = 502.0± 2.8
σ = 125.6± 2.6



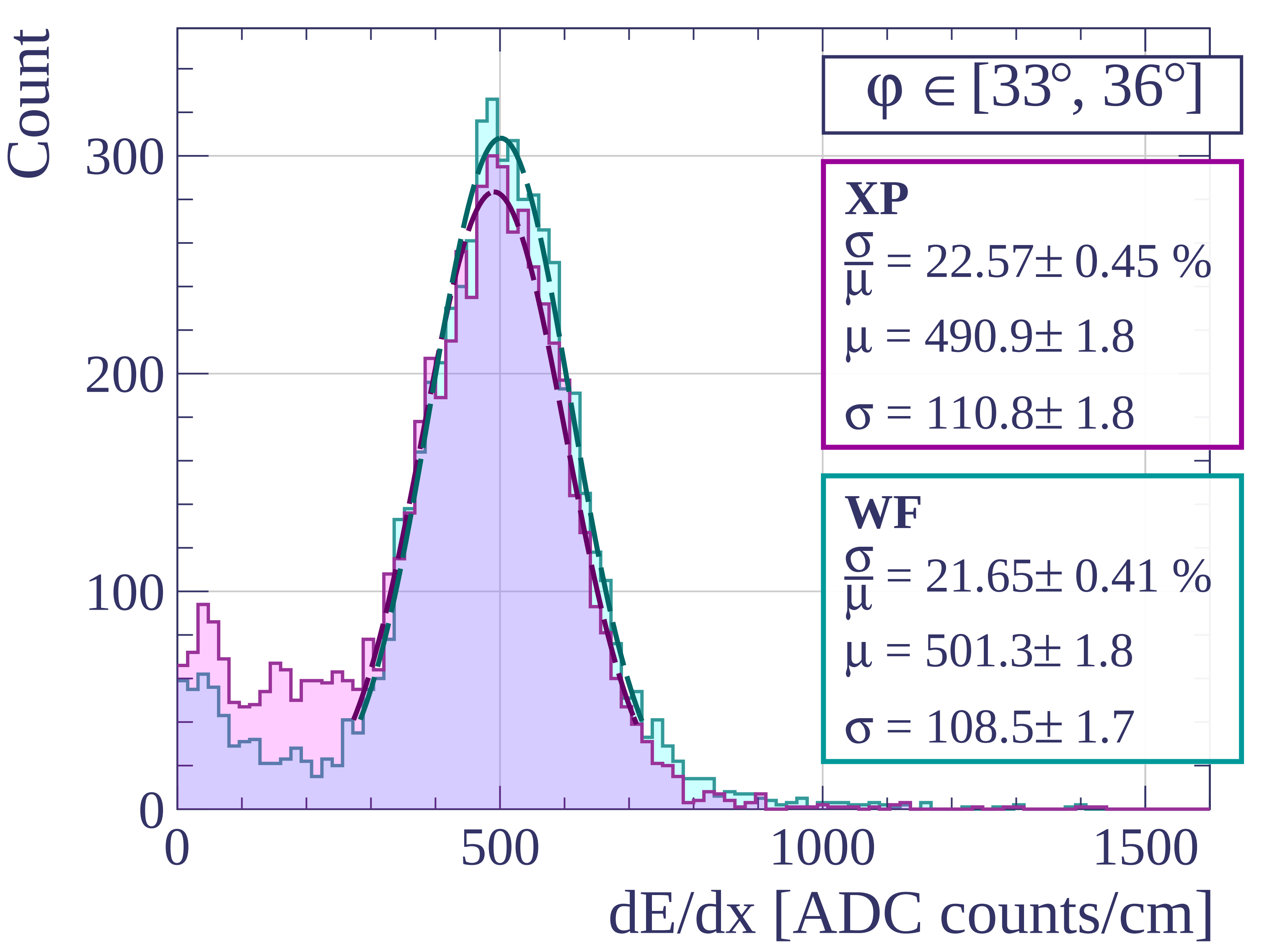
Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [24°, 27°]
XP
σ/μ = 27.50± 0.81 %
μ = 472.6± 3.1
σ = 130.0± 3.0
WF
σ/μ = 24.09± 0.58 %
μ = 498.8± 2.5
σ = 120.2± 2.3



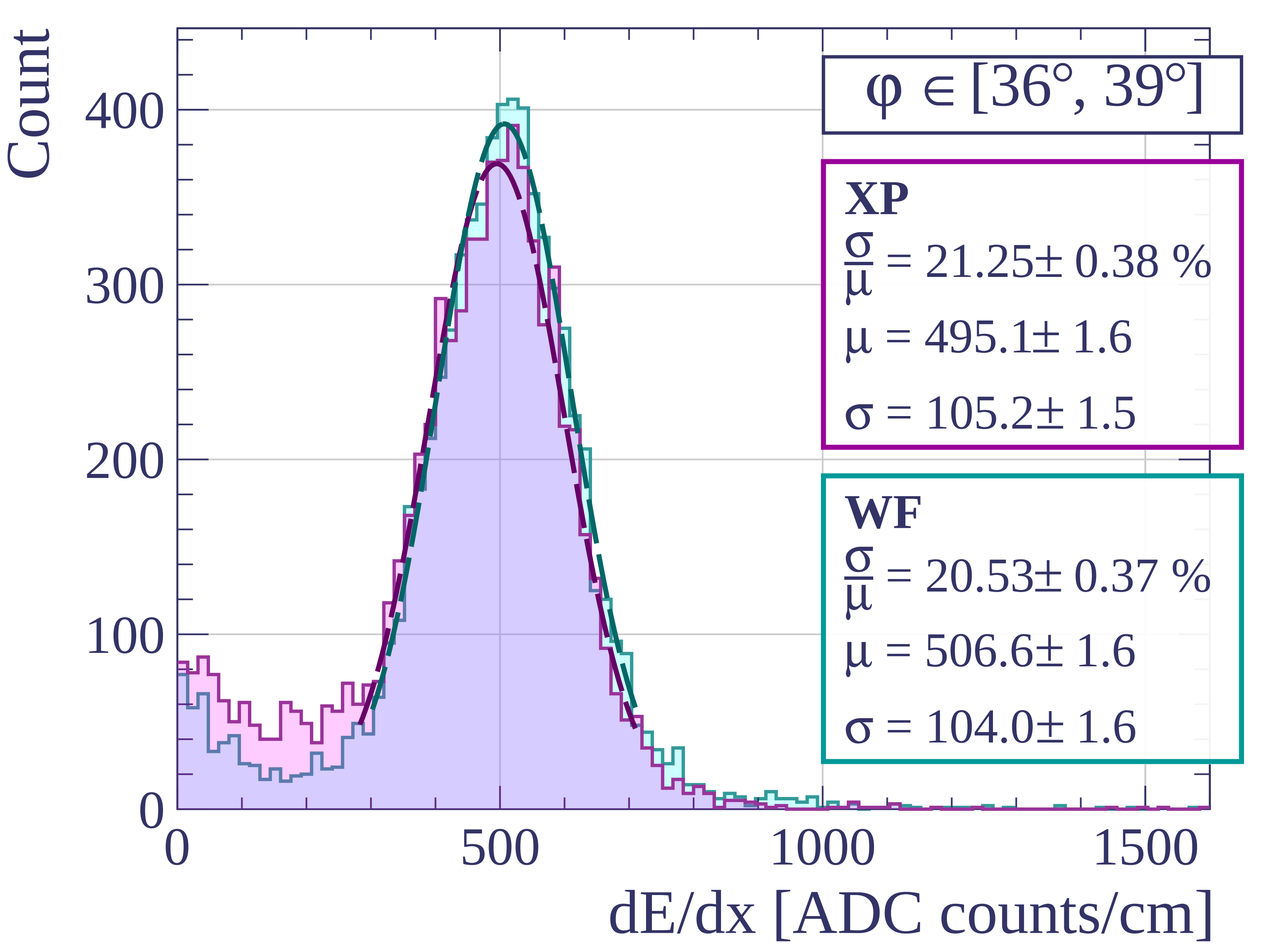
Count
0
50
100
150
200
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [27°, 30°]
XP
σ/μ = 26.90± 0.68 %
μ = 473.9± 2.6
σ = 127.5± 2.5
WF
σ/μ = 23.36± 0.54 %
μ = 499.8± 2.2
σ = 116.7± 2.2



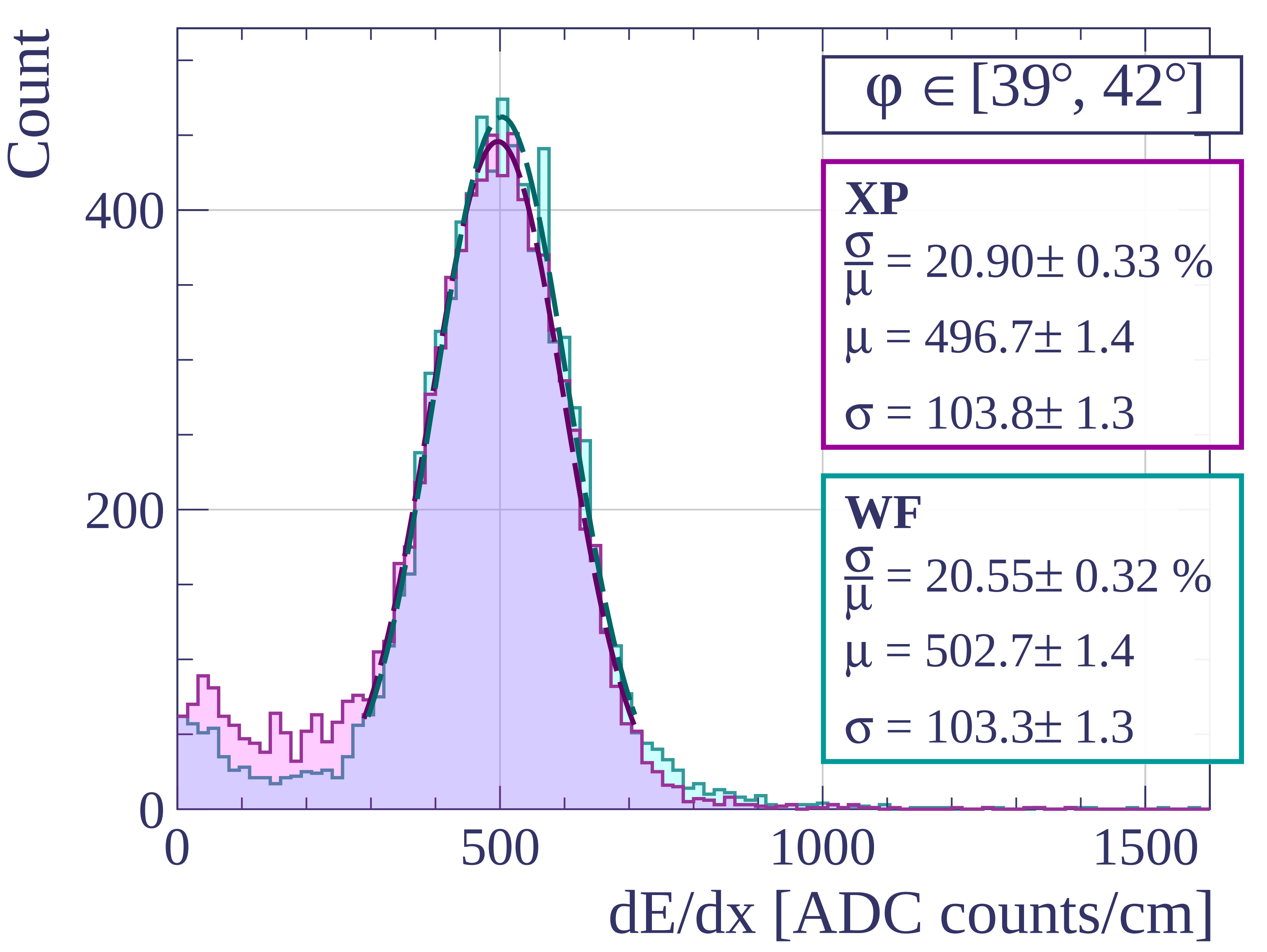
Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [30°, 33°]
XP
σ/μ = 24.81± 0.58 %
μ = 482.8± 2.2
σ = 119.8± 2.2
WF
σ/μ = 22.52± 0.45 %
μ = 495.2± 1.9
σ = 111.5± 1.8



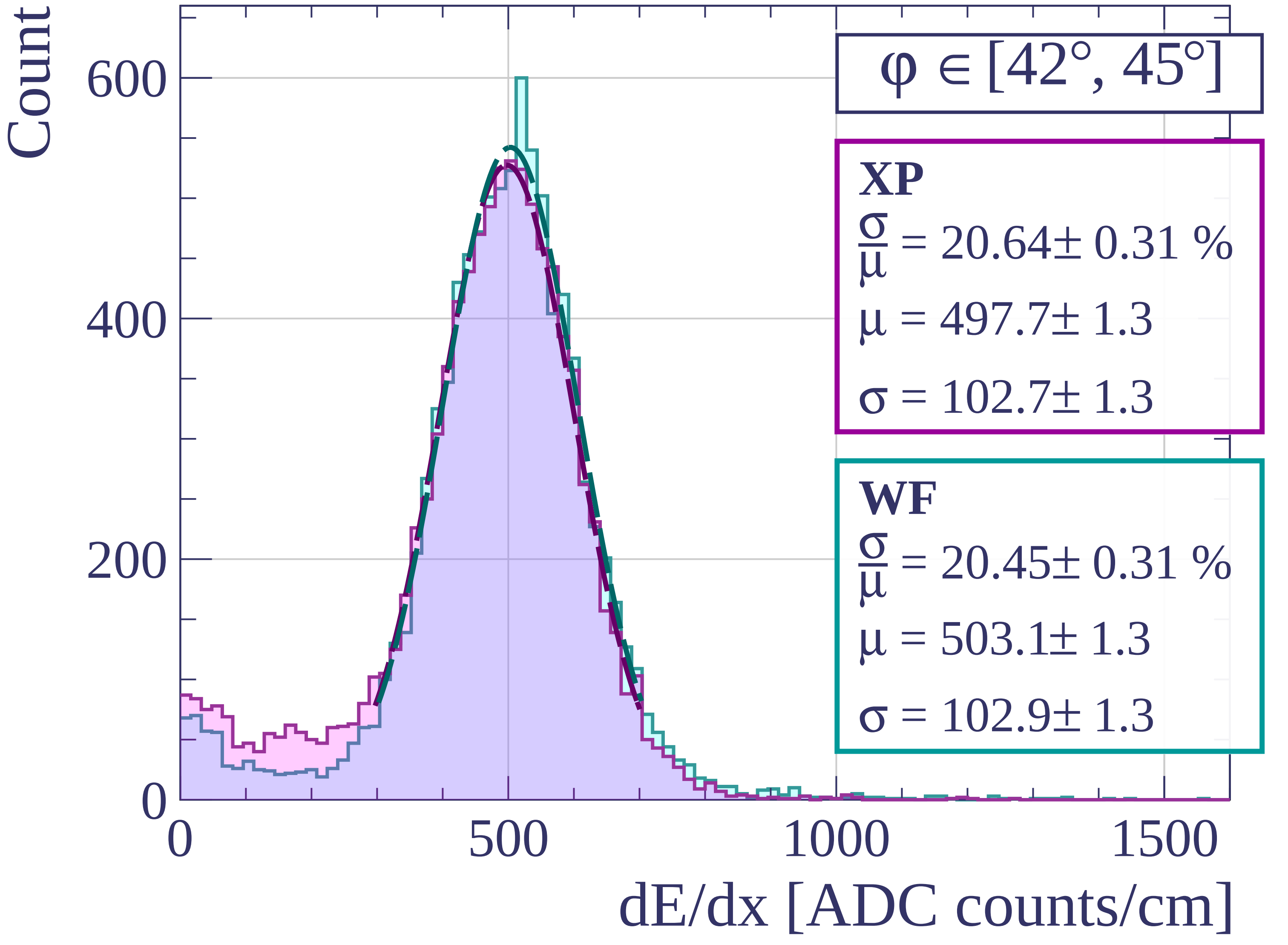
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [33°, 36°]
XP
σ/μ = 22.57± 0.45 %
μ = 490.9± 1.8
σ = 110.8± 1.8
WF
σ/μ = 21.65± 0.41 %
μ = 501.3± 1.8
σ = 108.5± 1.7



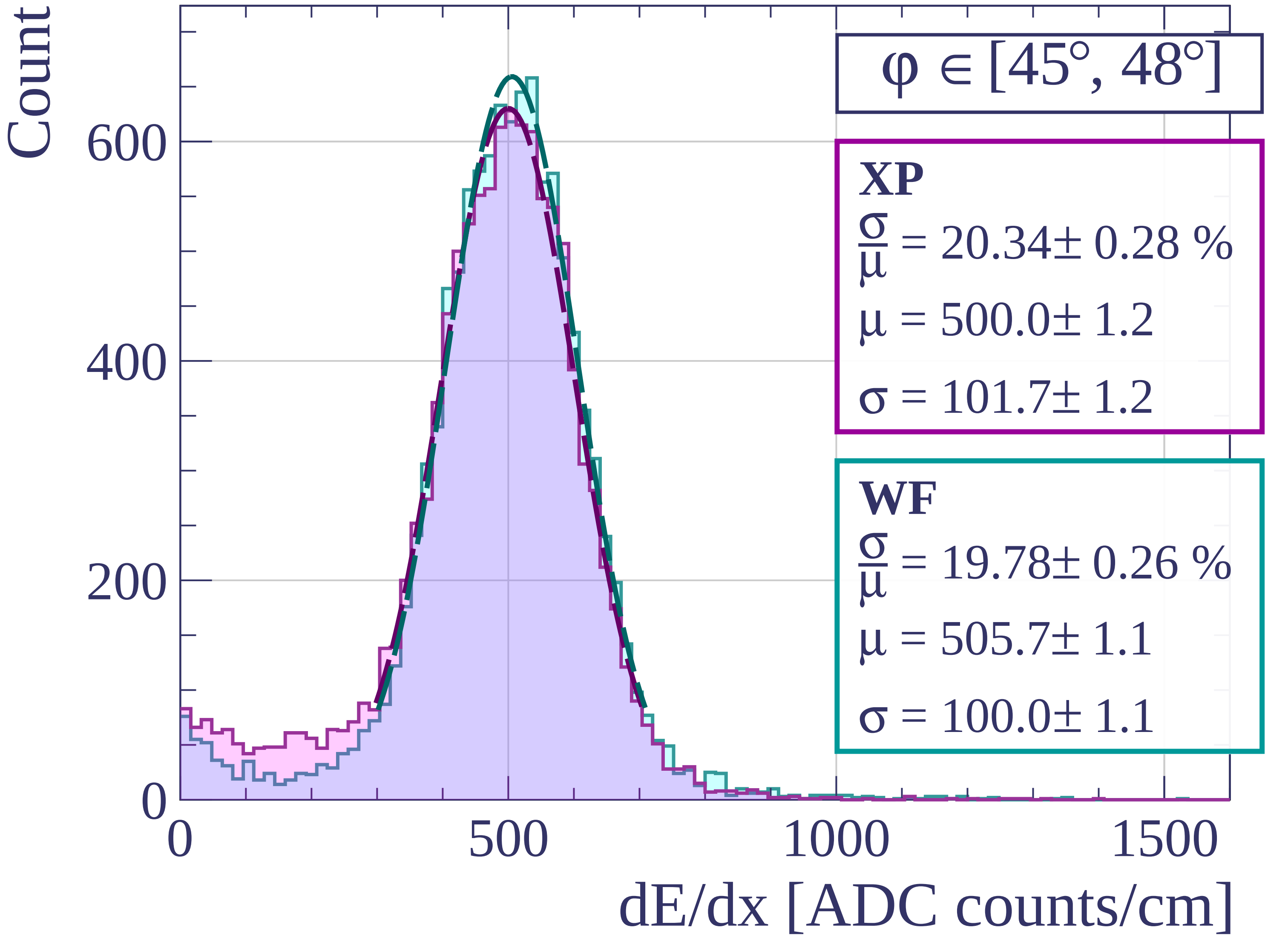
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [36°, 39°]
XP
σ/μ = 21.25± 0.38 %
μ = 495.1± 1.6
σ = 105.2± 1.5
WF
σ/μ = 20.53± 0.37 %
μ = 506.6± 1.6
σ = 104.0± 1.6



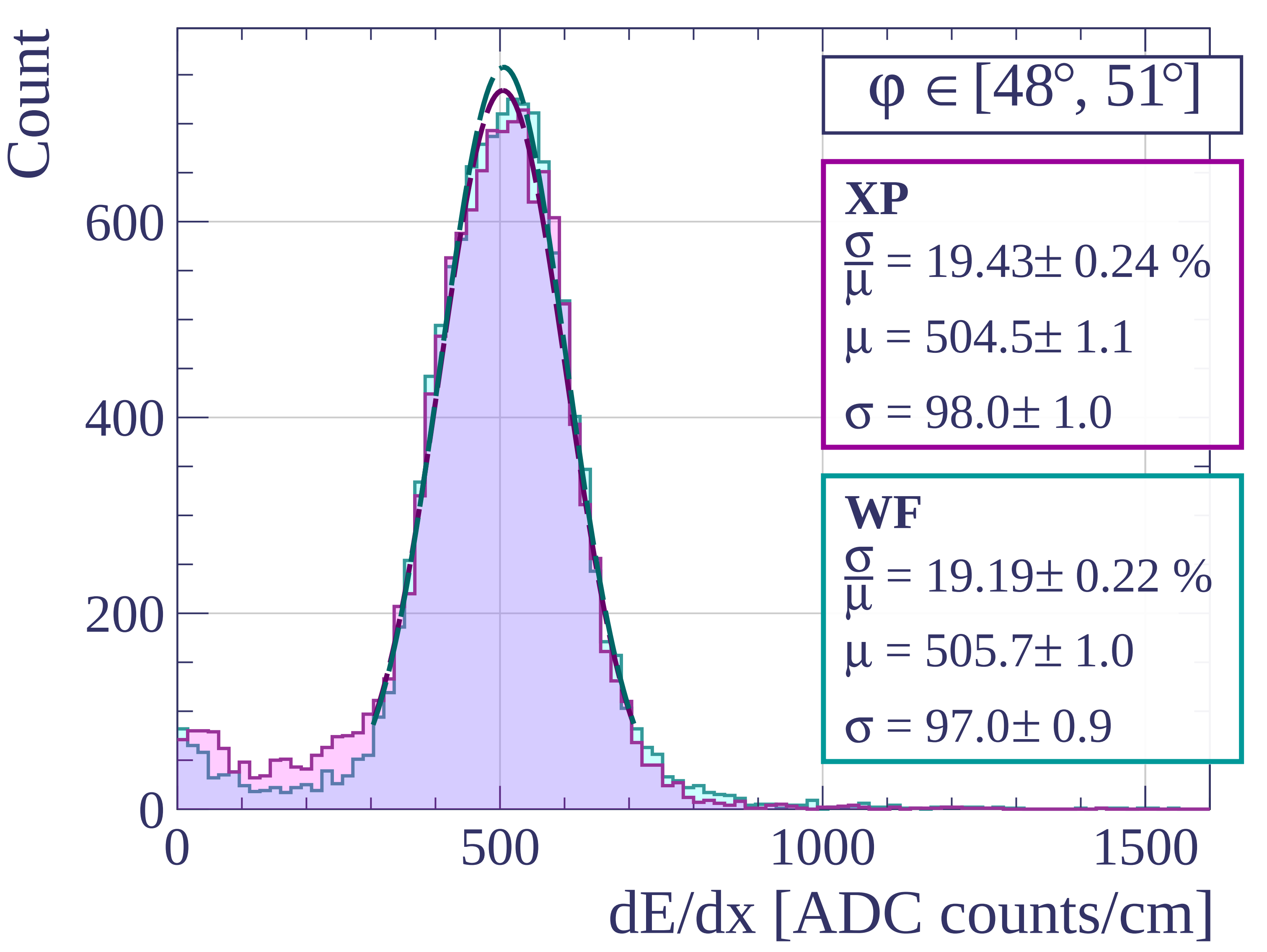
Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [39°, 42°]
XP
σ/μ = 20.90± 0.33 %
μ = 496.7± 1.4
σ = 103.8± 1.3
WF
σ/μ = 20.55± 0.32 %
μ = 502.7± 1.4
σ = 103.3± 1.3



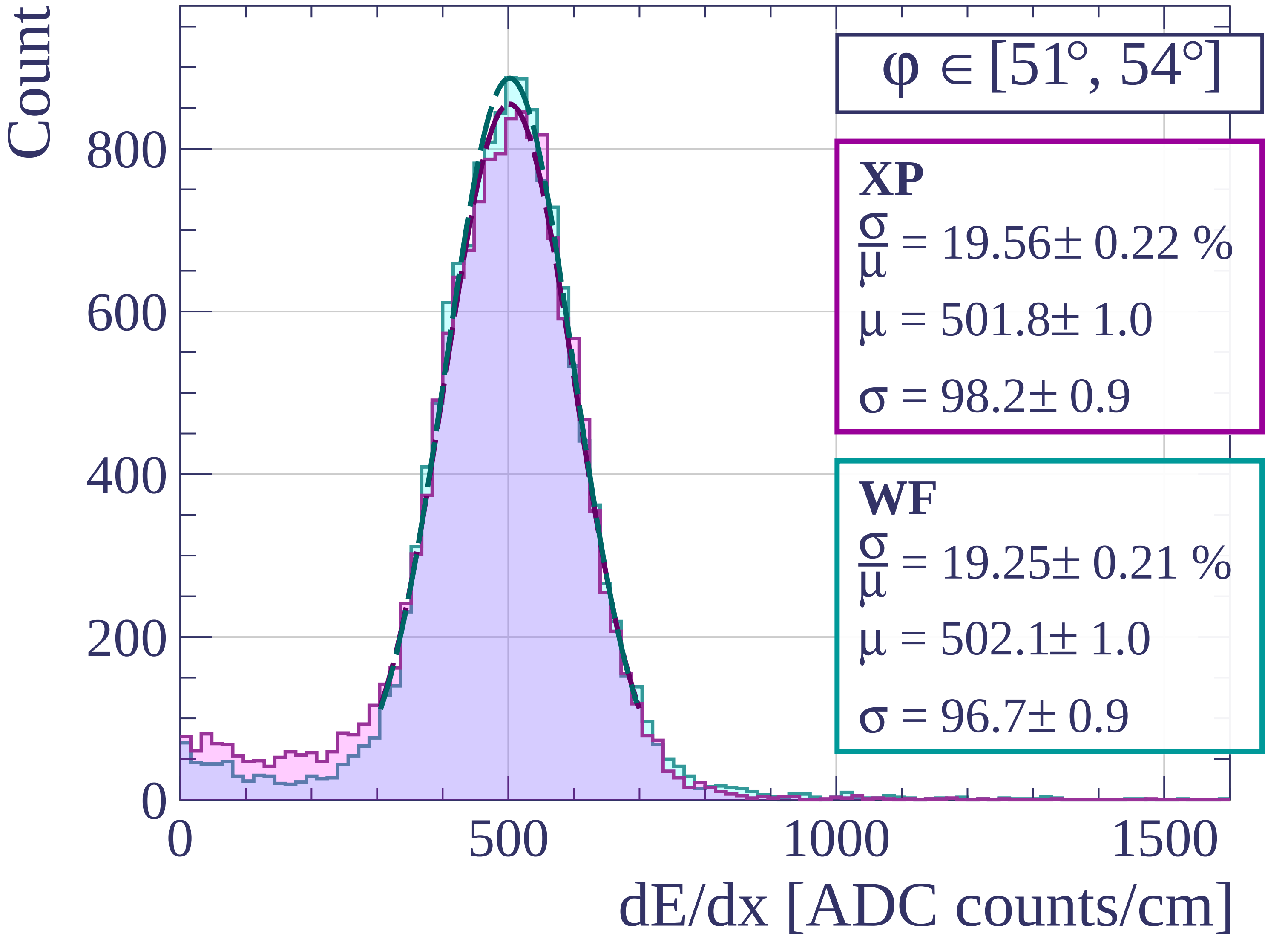
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [42°, 45°]
XP
σ/μ = 20.64± 0.31 %
μ = 497.7± 1.3
σ = 102.7± 1.3
WF
σ/μ = 20.45± 0.31 %
μ = 503.1± 1.3
σ = 102.9± 1.3



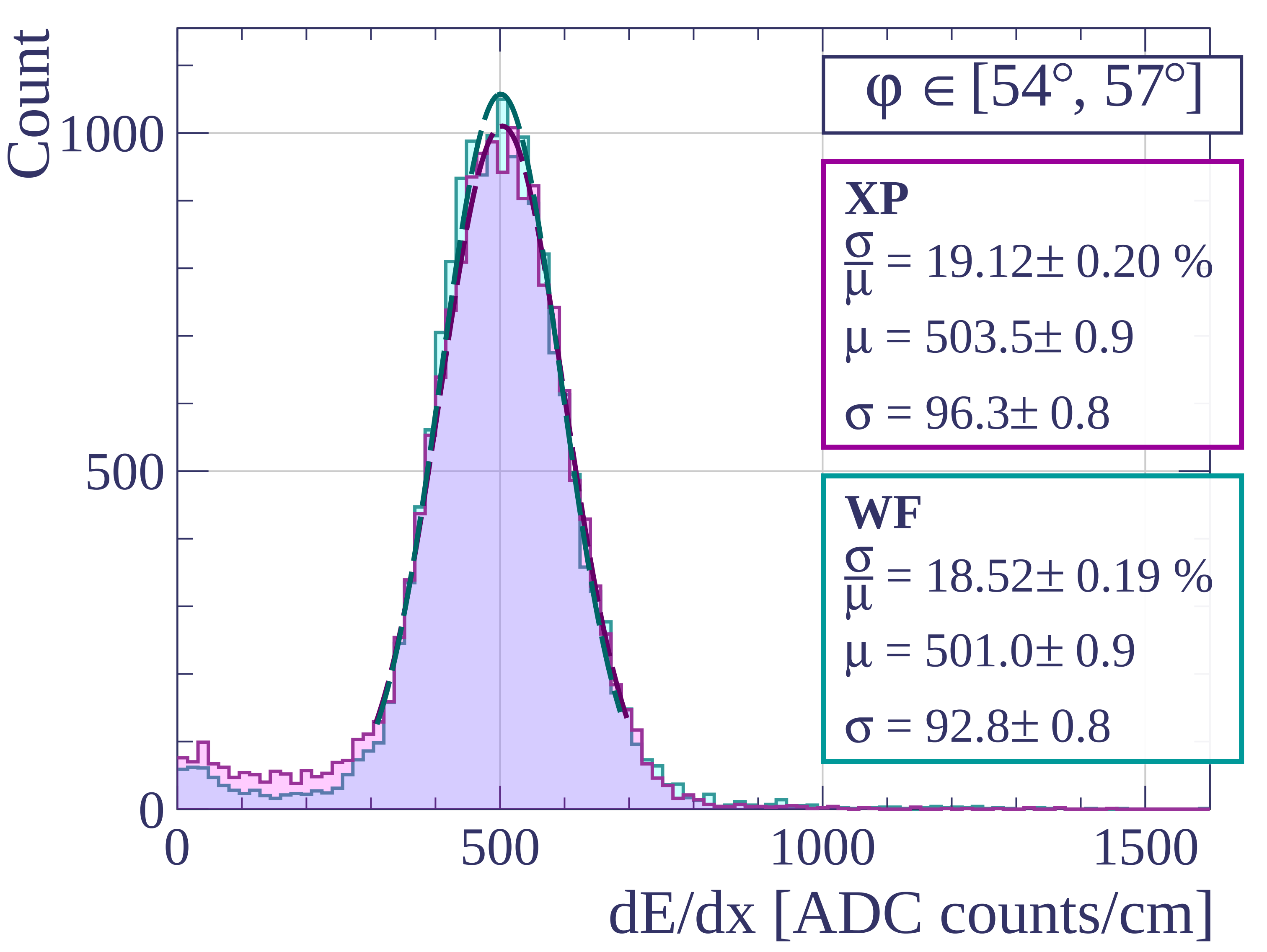
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [45°, 48°]
XP
σ/μ = 20.34± 0.28 %
μ = 500.0± 1.2
σ = 101.7± 1.2
WF
σ/μ = 19.78± 0.26 %
μ = 505.7± 1.1
σ = 100.0± 1.1



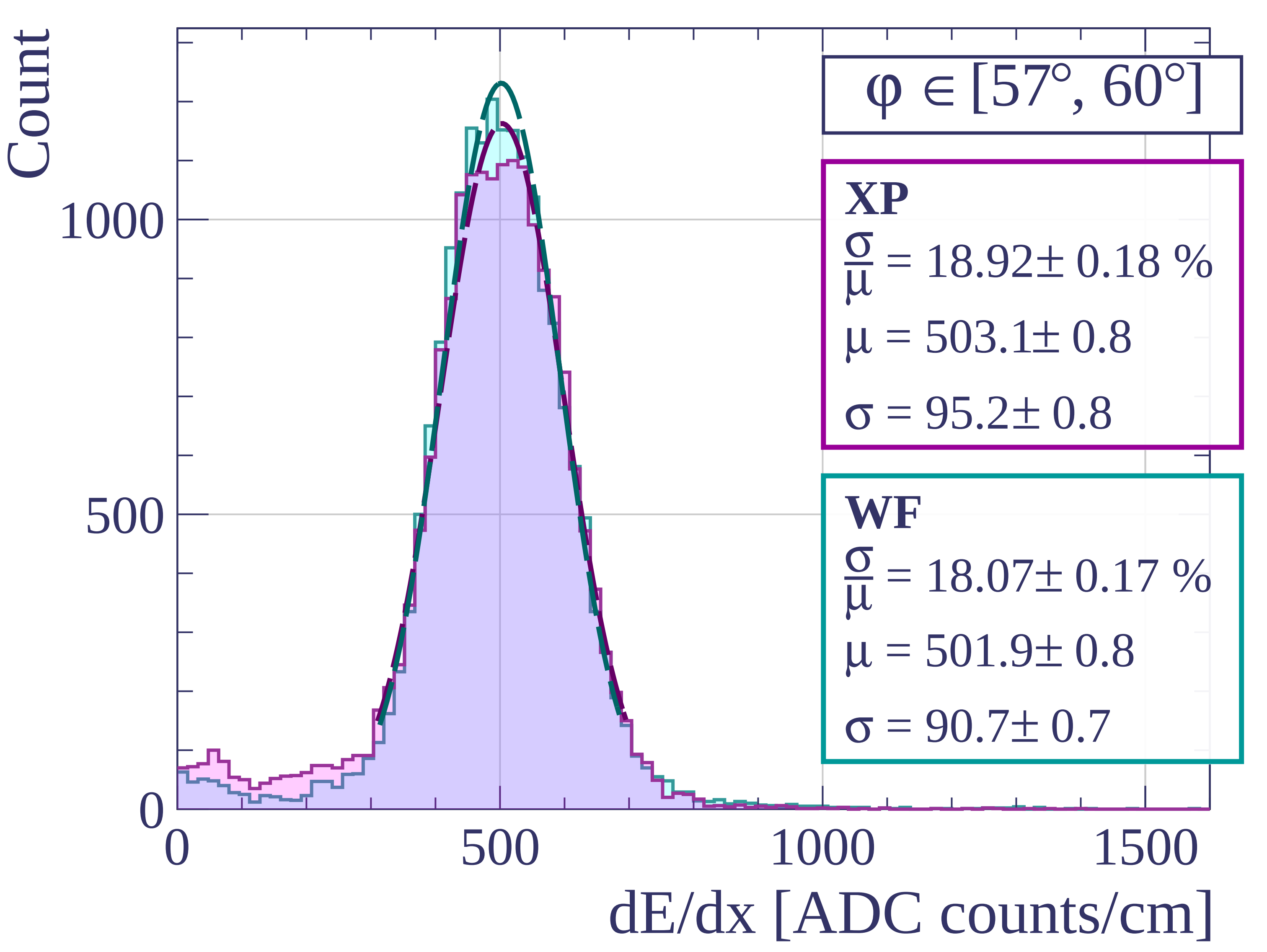
Count
dE/dx [ADC counts/cm]
0
200
400
600
0
500
1000
1500
φ ∈ [48°, 51°]
XP
σ/μ = 19.43± 0.24 %
μ = 504.5± 1.1
σ = 98.0± 1.0
WF
σ/μ = 19.19± 0.22 %
μ = 505.7± 1.0
σ = 97.0± 0.9



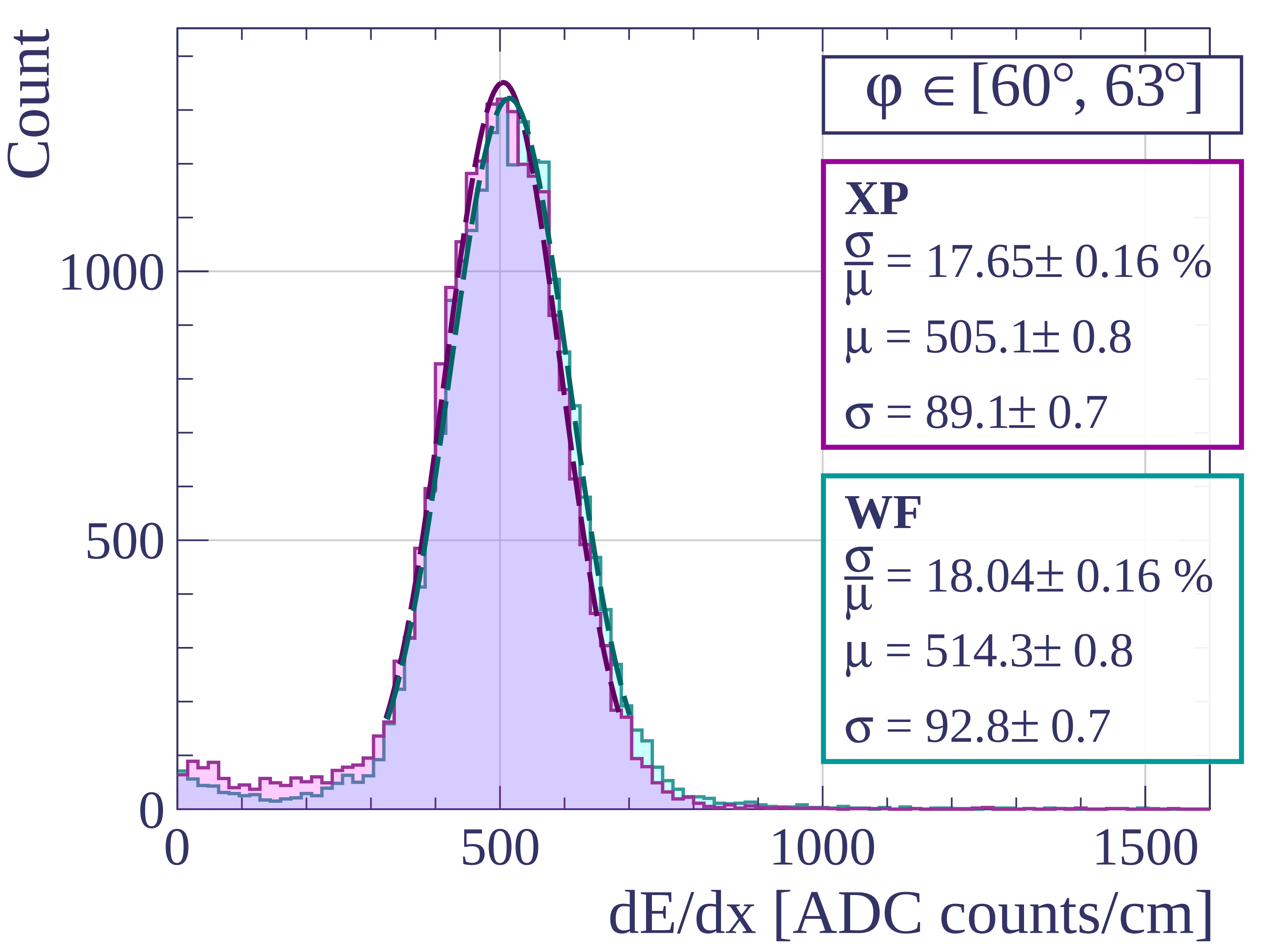
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [51°, 54°]
XP
σ/μ = 19.56± 0.22 %
μ = 501.8± 1.0
σ = 98.2± 0.9
WF
σ/μ = 19.25± 0.21 %
μ = 502.1± 1.0
σ = 96.7± 0.9



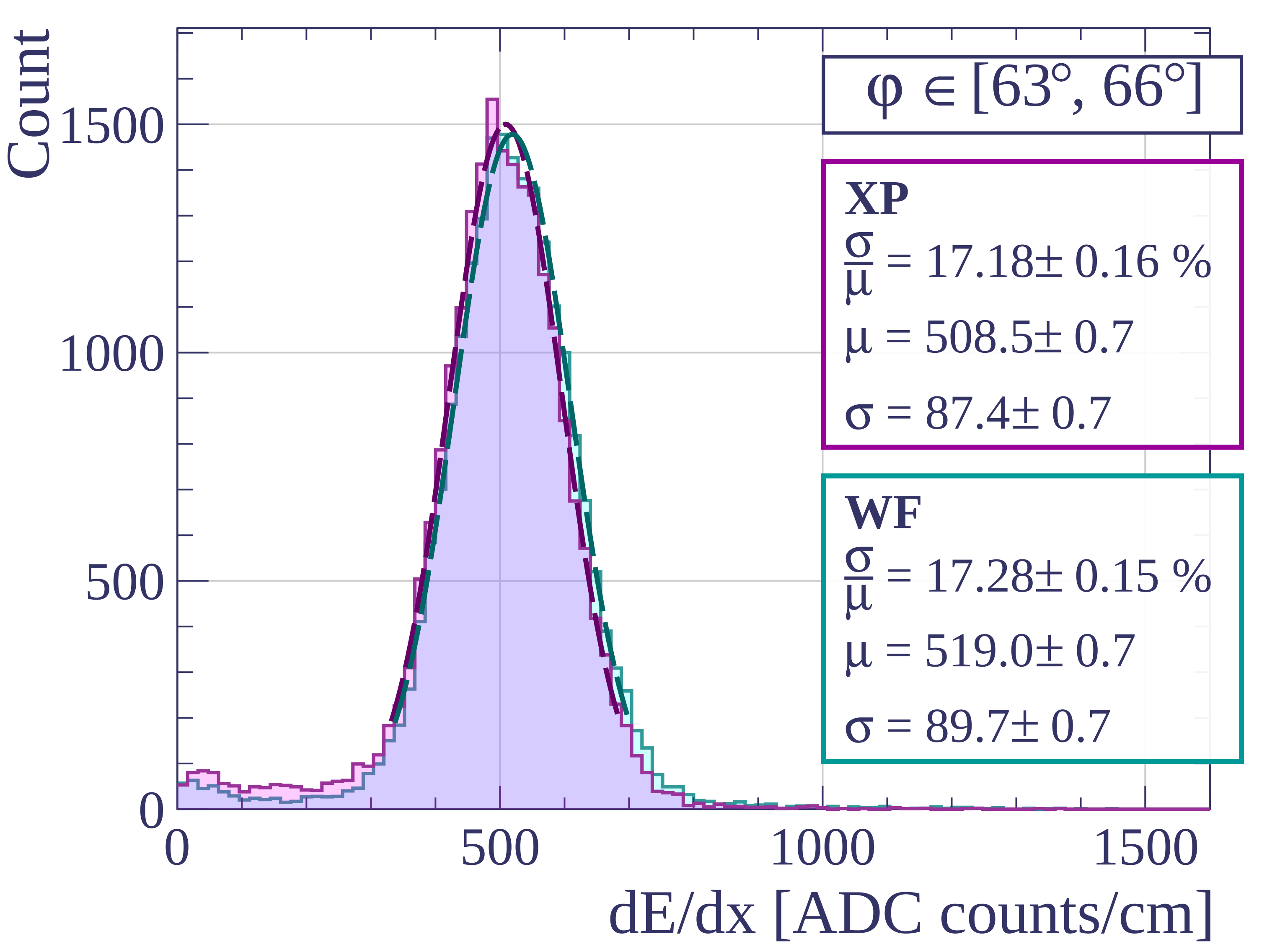
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [54°, 57°]
XP
σ/μ = 19.12± 0.20 %
μ = 503.5± 0.9
σ = 96.3± 0.8
WF
σ/μ = 18.52± 0.19 %
μ = 501.0± 0.9
σ = 92.8± 0.8



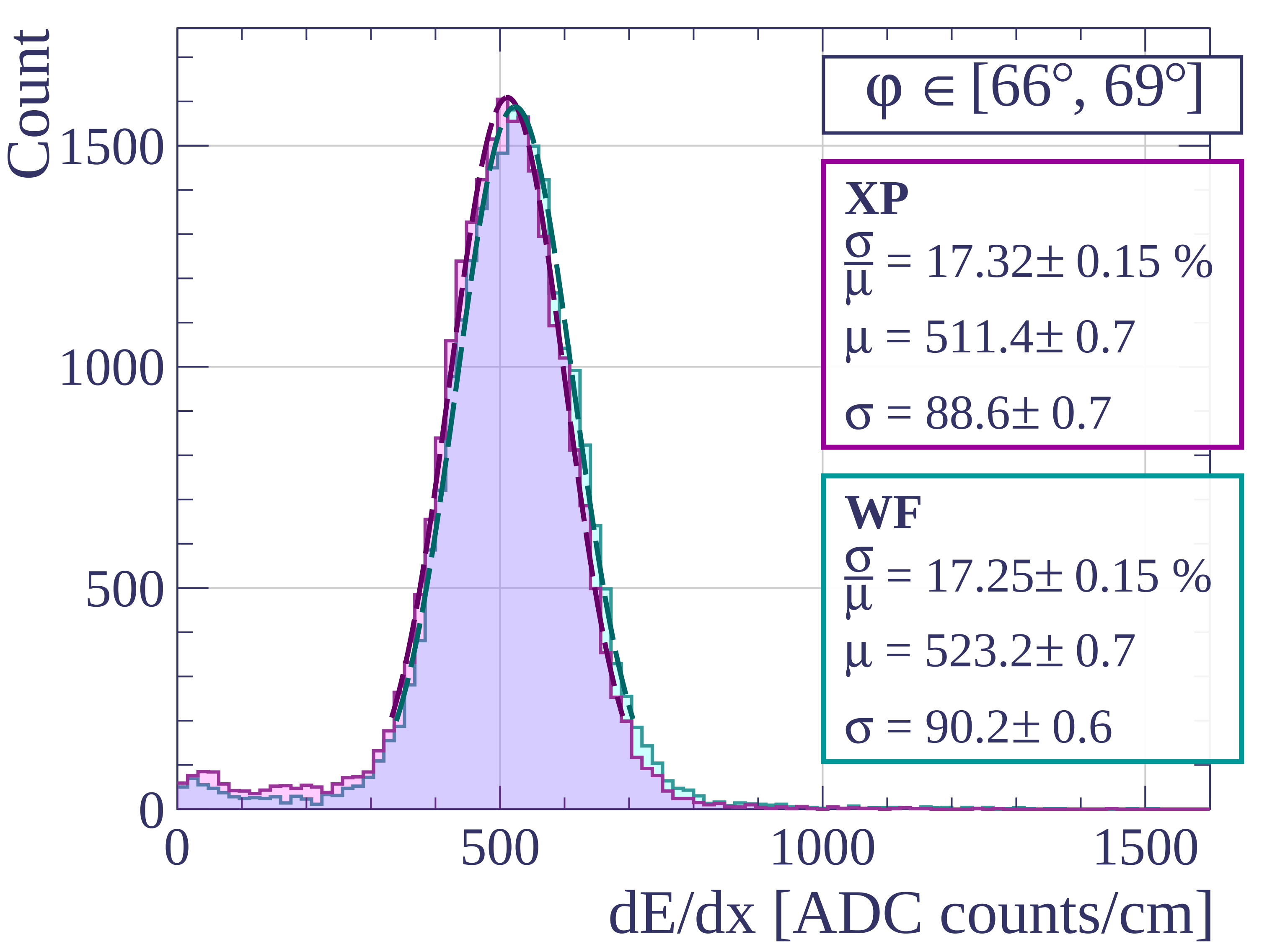
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [57°, 60°]
XP
σ/μ = 18.92± 0.18 %
μ = 503.1± 0.8
σ = 95.2± 0.8
WF
σ/μ = 18.07± 0.17 %
μ = 501.9± 0.8
σ = 90.7± 0.7



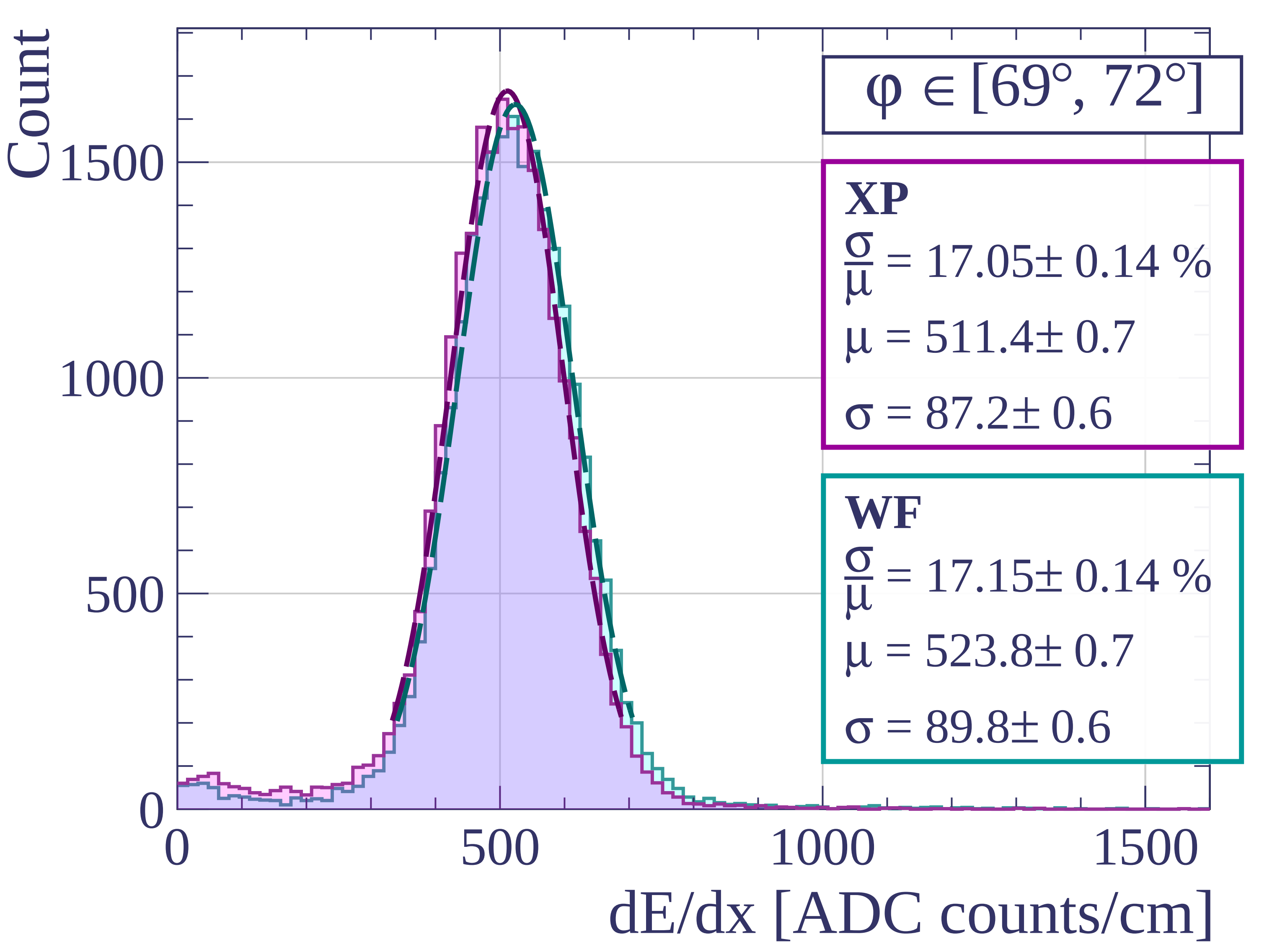
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [60°, 63°]
XP
σ/μ = 17.65± 0.16 %
μ = 505.1± 0.8
σ = 89.1± 0.7
WF
σ/μ = 18.04± 0.16 %
μ = 514.3± 0.8
σ = 92.8± 0.7



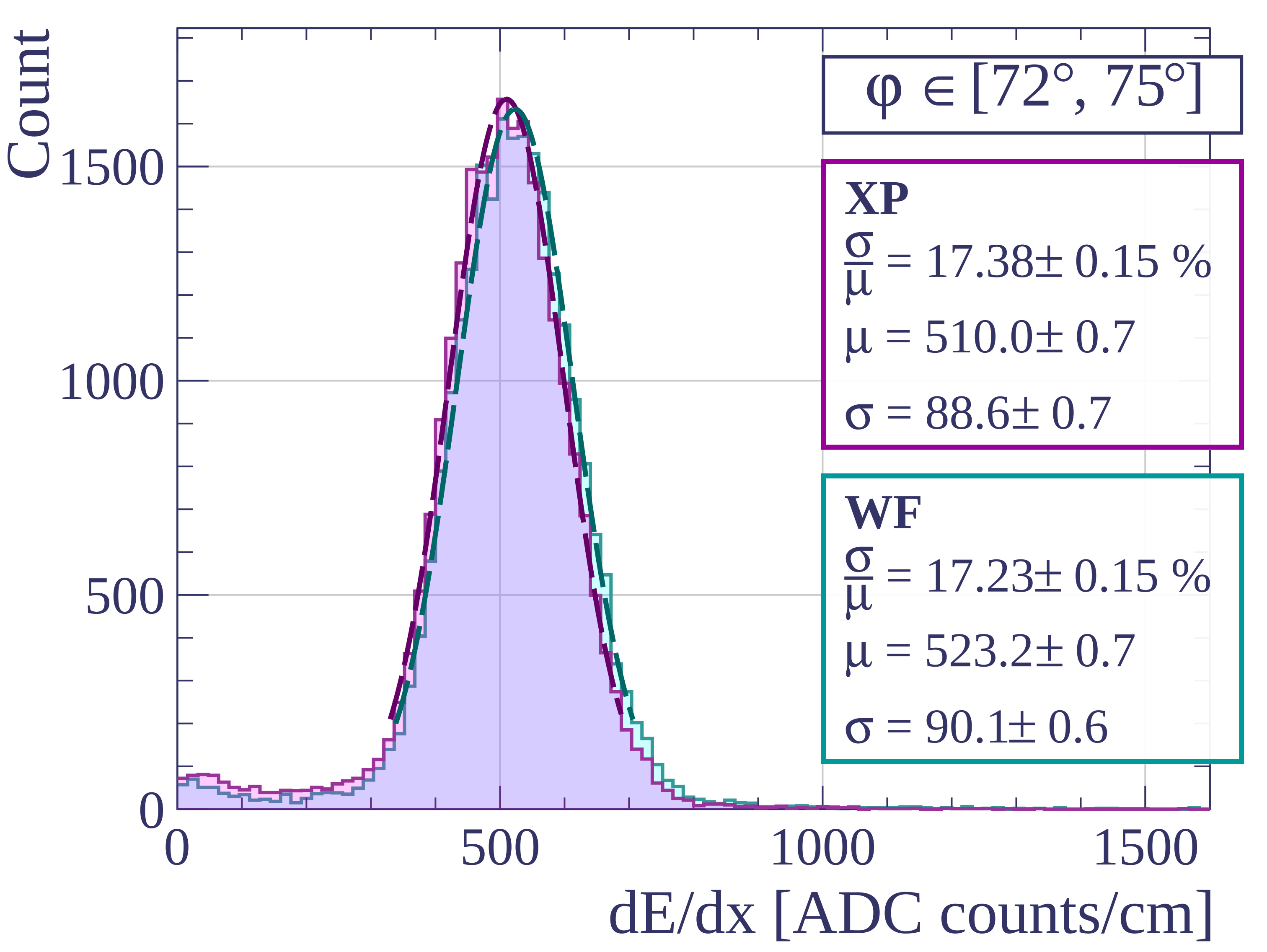
Count
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [63°, 66°]
XP
σ/μ = 17.18± 0.16 %
μ = 508.5± 0.7
σ = 87.4± 0.7
WF
σ/μ = 17.28± 0.15 %
μ = 519.0± 0.7
σ = 89.7± 0.7



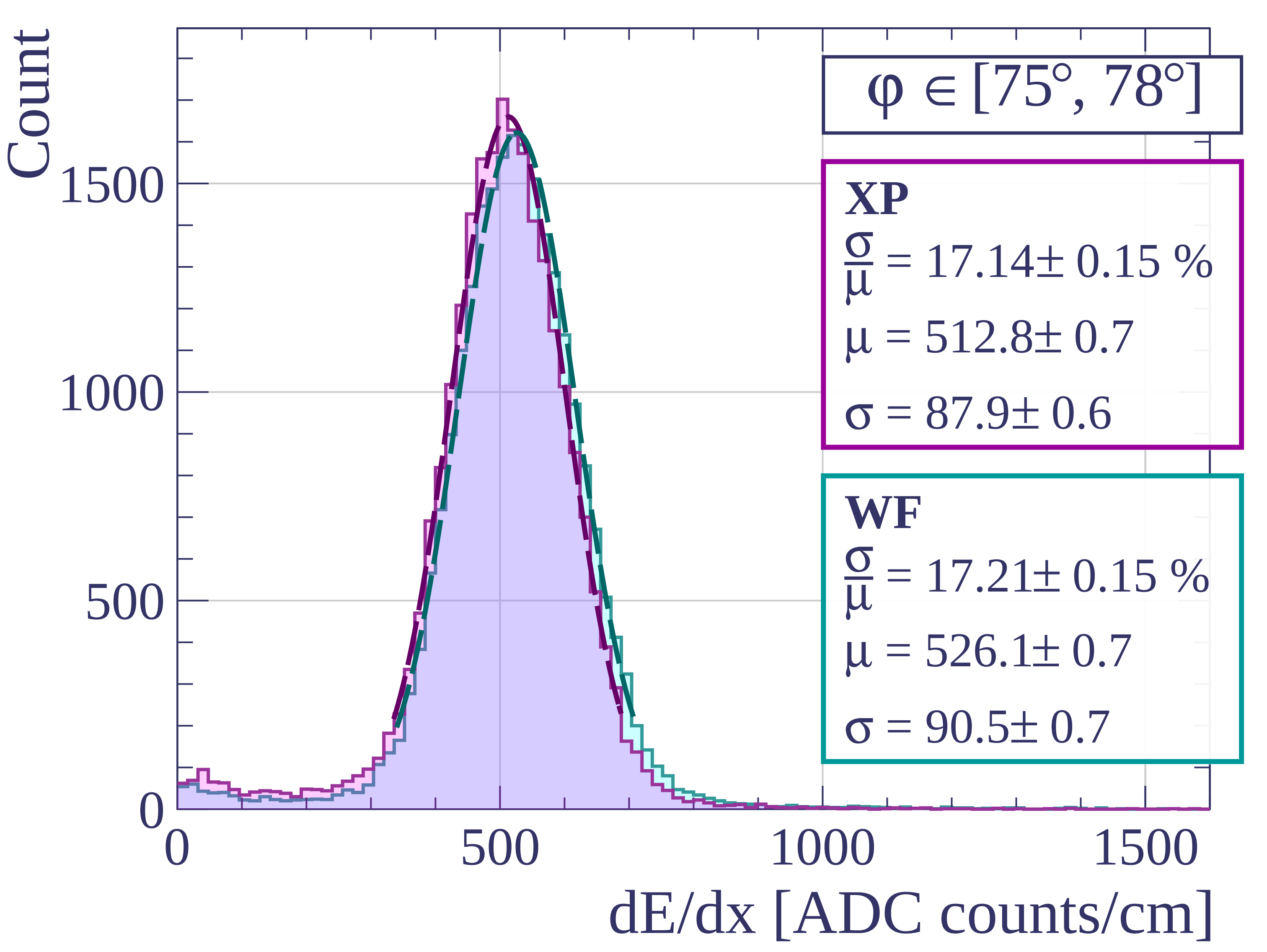
Count
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
$\varphi \in [66°, 69°]$
XP
$\frac{\sigma}{\mu} = 17.32 \pm 0.15$ %
$\mu = 511.4 \pm 0.7$
$\sigma = 88.6 \pm 0.7$
WF
$\frac{\sigma}{\mu} = 17.25 \pm 0.15$ %
$\mu = 523.2 \pm 0.7$
$\sigma = 90.2 \pm 0.6$



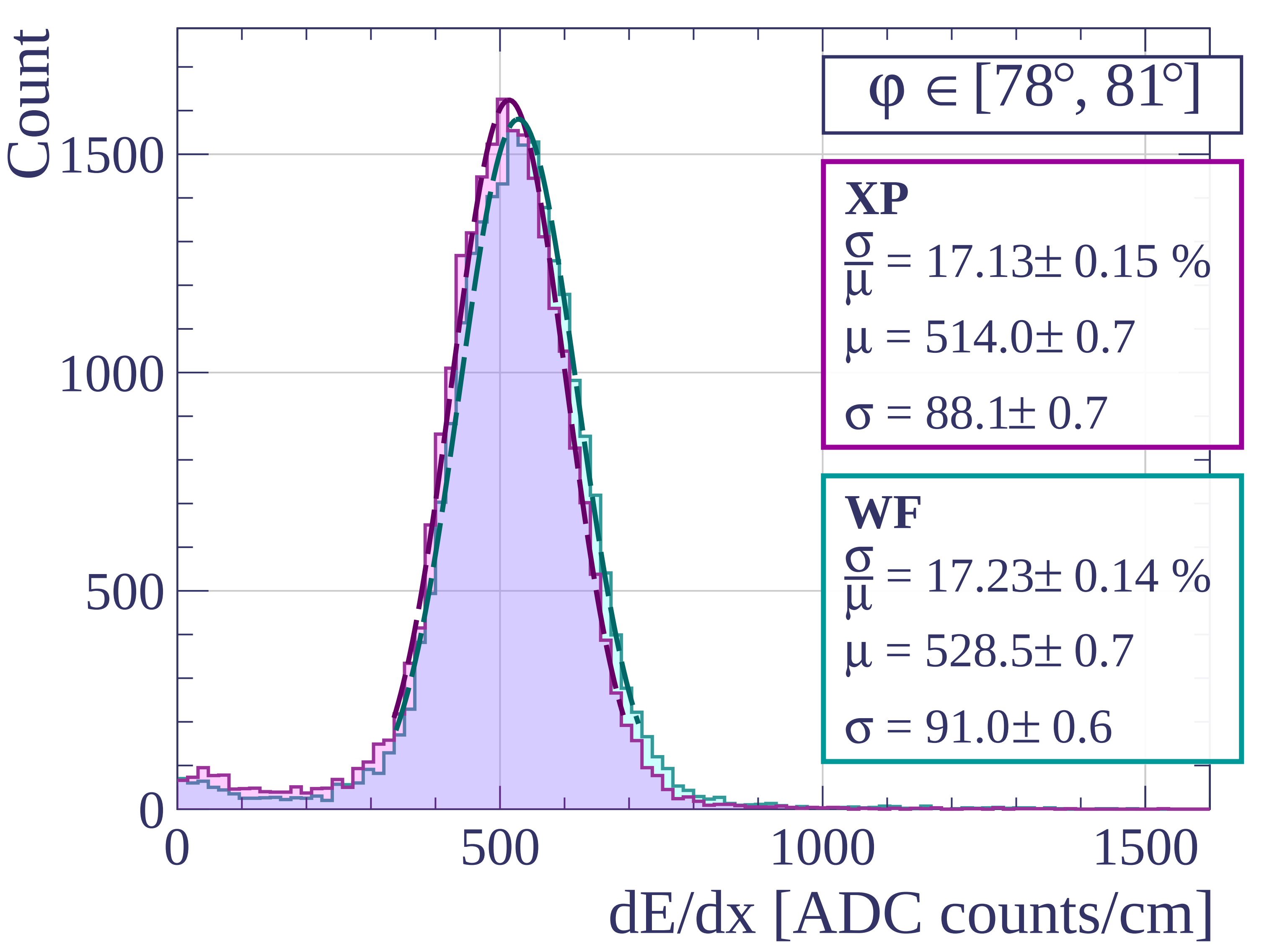
Count
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [69°, 72°]
XP
σ/μ = 17.05± 0.14 %
μ = 511.4± 0.7
σ = 87.2± 0.6
WF
σ/μ = 17.15± 0.14 %
μ = 523.8± 0.7
σ = 89.8± 0.6



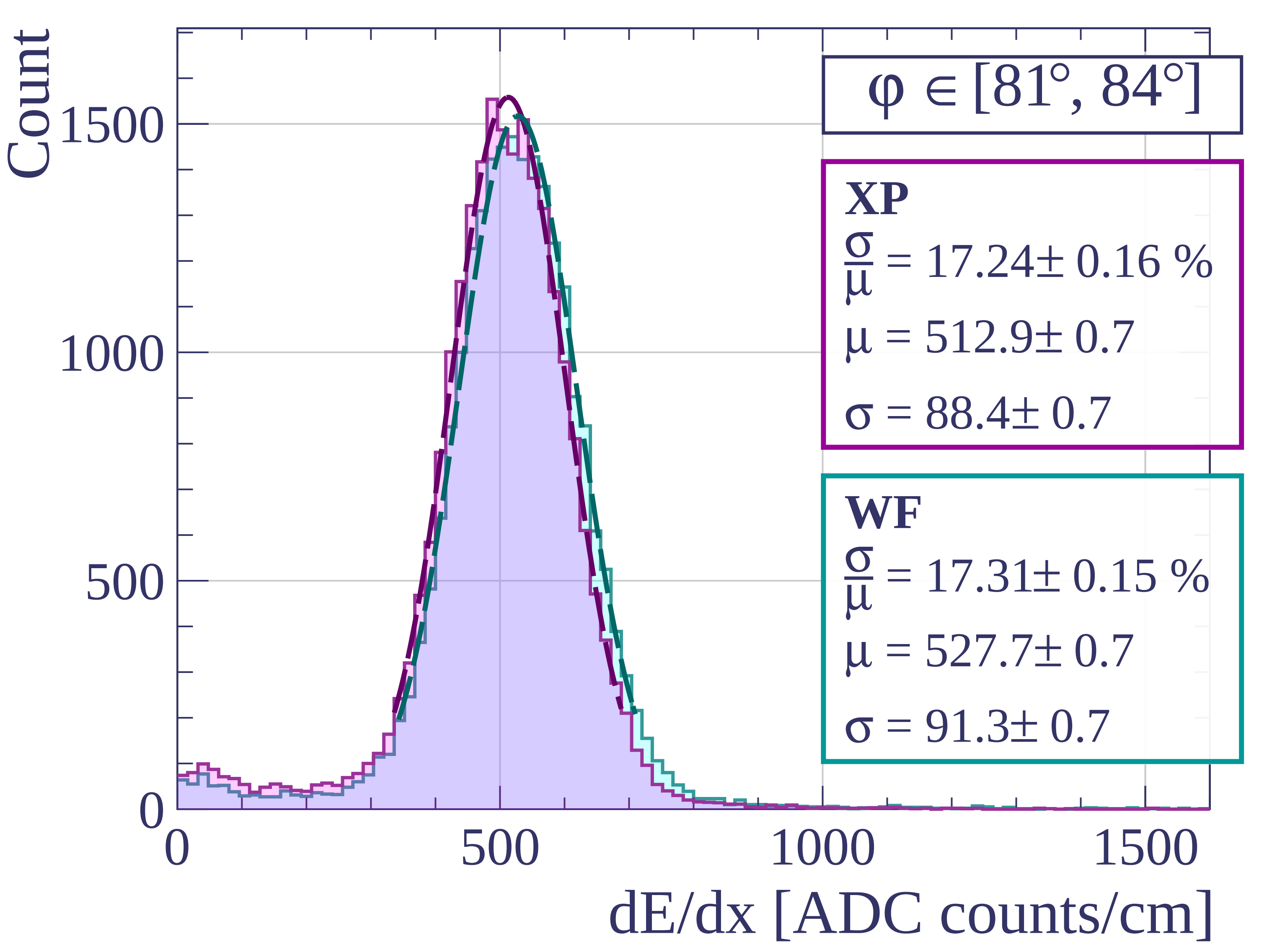
Count
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [72°, 75°]
XP
σ/μ = 17.38± 0.15 %
μ = 510.0± 0.7
σ = 88.6± 0.7
WF
σ/μ = 17.23± 0.15 %
μ = 523.2± 0.7
σ = 90.1± 0.6



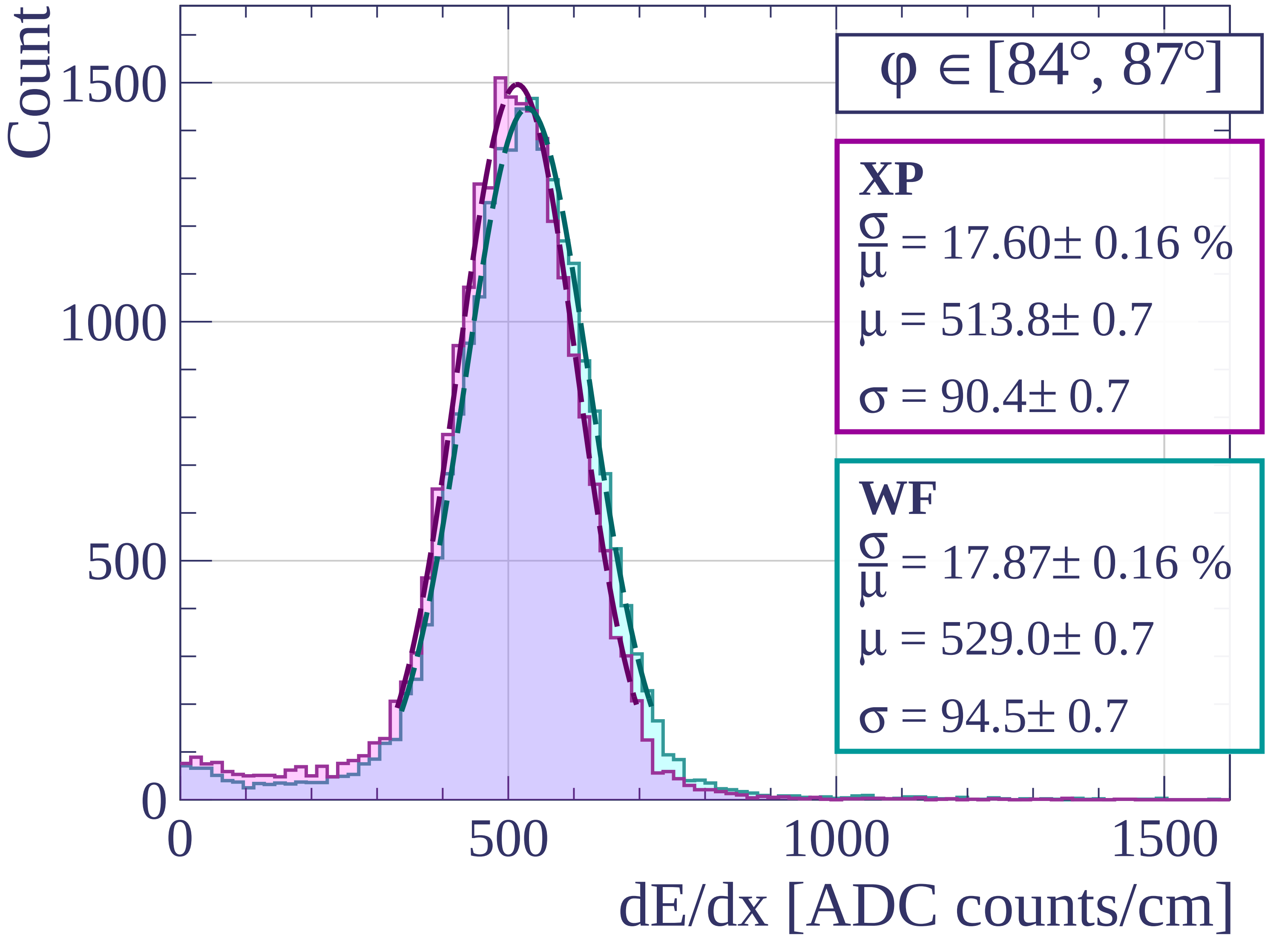
Count
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [75°, 78°]
XP
σ/μ = 17.14± 0.15 %
μ = 512.8± 0.7
σ = 87.9± 0.6
WF
σ/μ = 17.21± 0.15 %
μ = 526.1± 0.7
σ = 90.5± 0.7



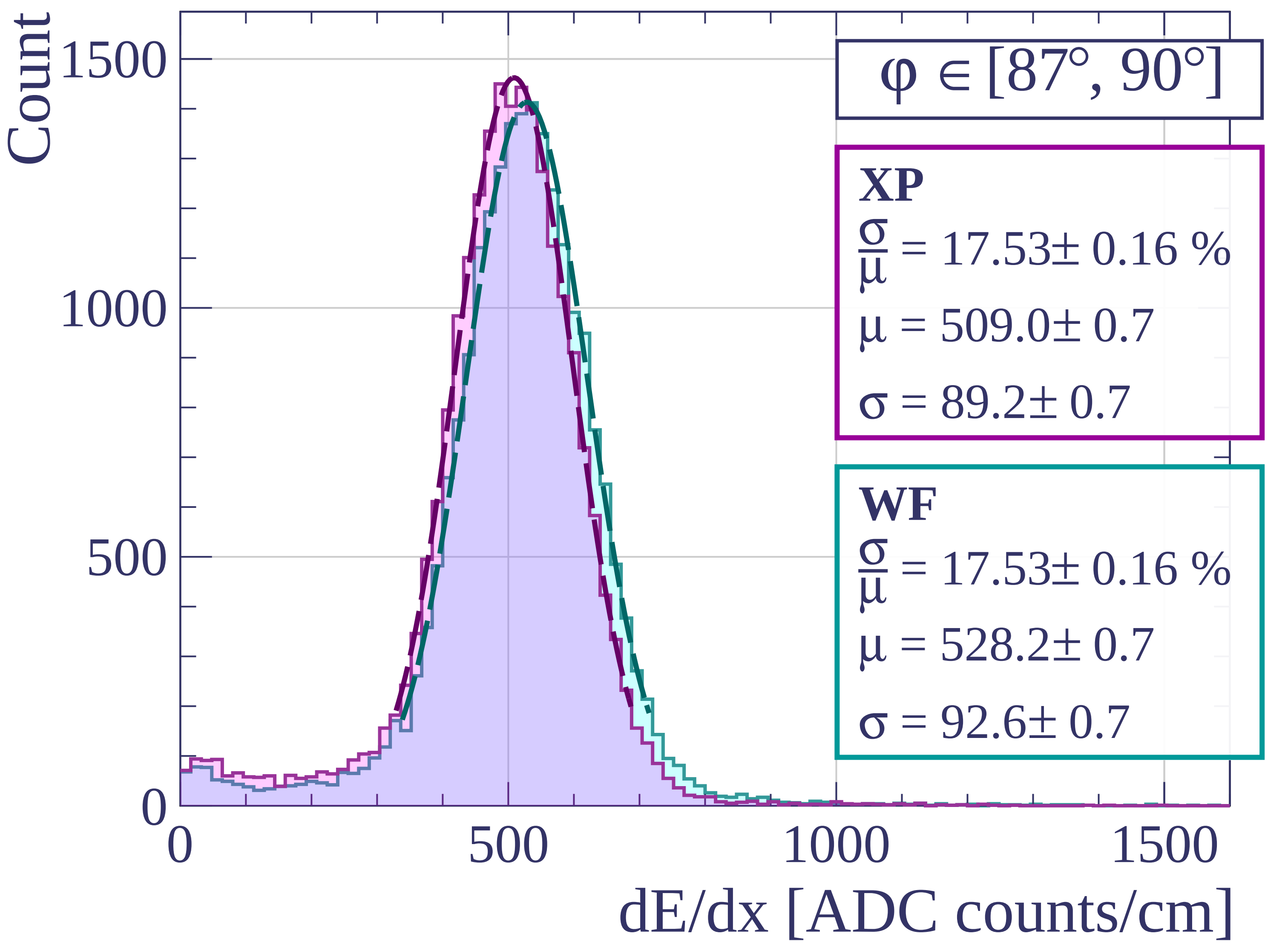
Count
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [78°, 81°]
XP
σ/μ = 17.13± 0.15 %
μ = 514.0± 0.7
σ = 88.1± 0.7
WF
σ/μ = 17.23± 0.14 %
μ = 528.5± 0.7
σ = 91.0± 0.6



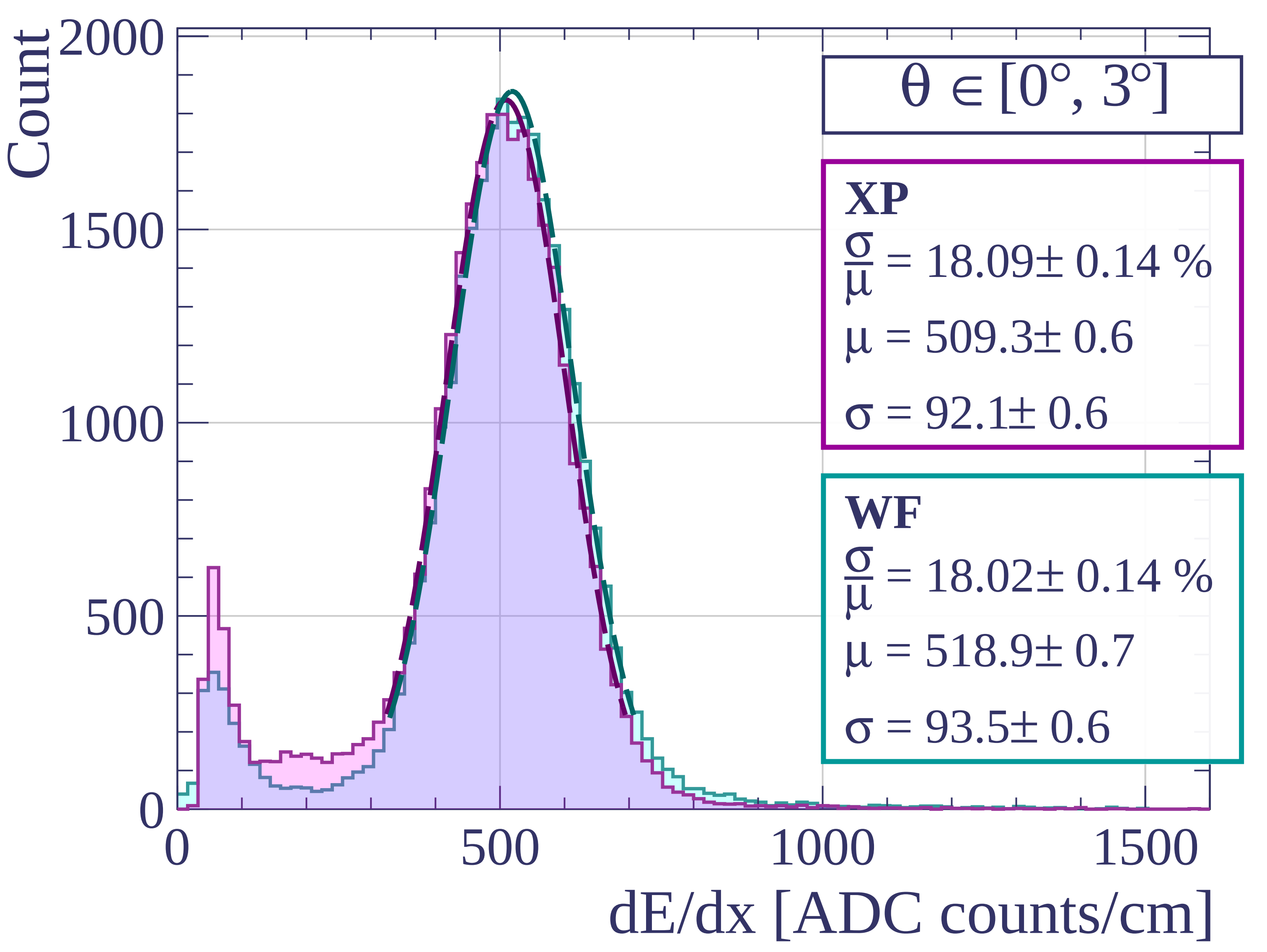
Count
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [81°, 84°]
XP
σ/μ = 17.24± 0.16 %
μ = 512.9± 0.7
σ = 88.4± 0.7
WF
σ/μ = 17.31± 0.15 %
μ = 527.7± 0.7
σ = 91.3± 0.7



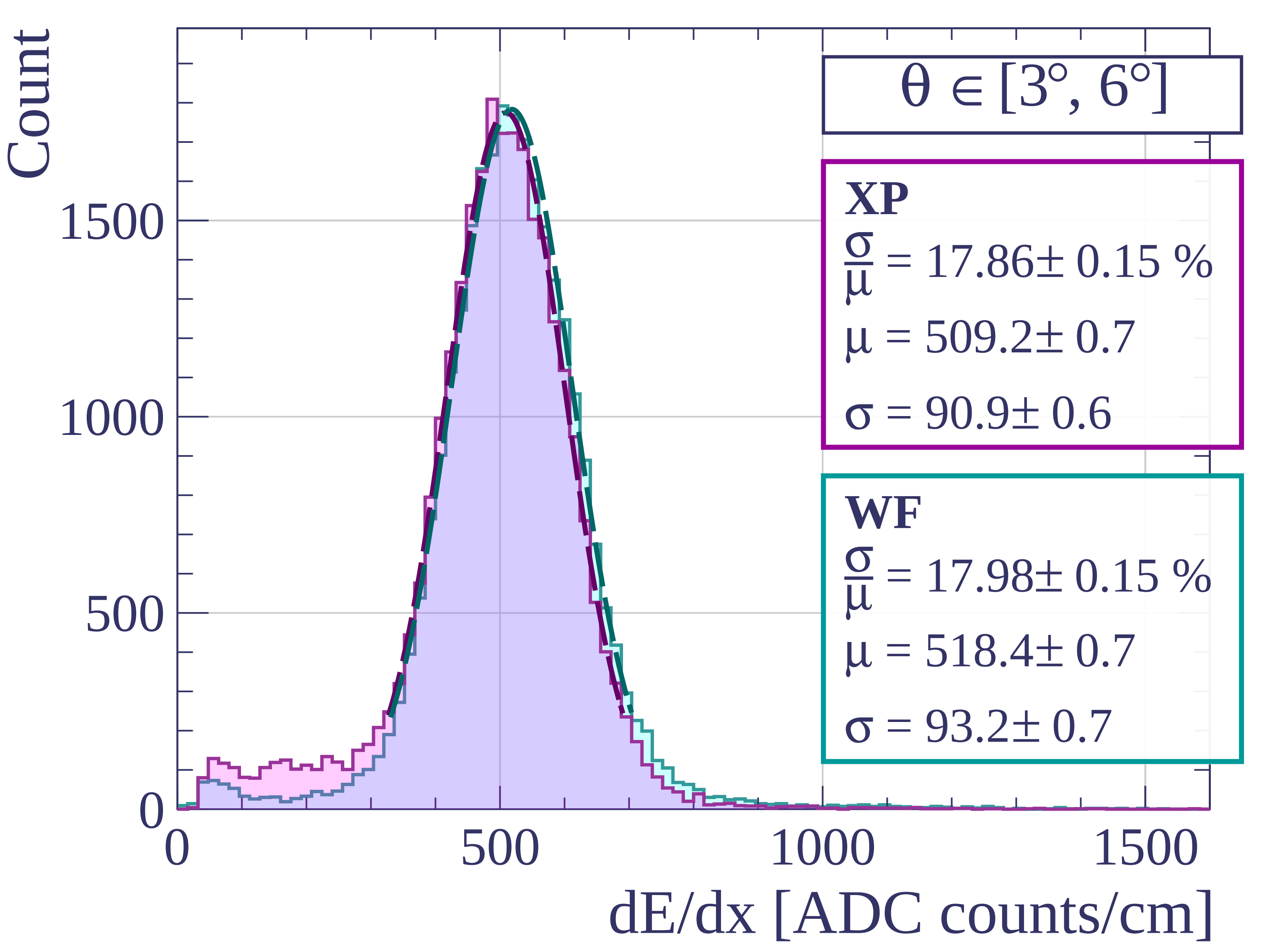
Count
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [84°, 87°]
XP
σ/μ = 17.60± 0.16 %
μ = 513.8± 0.7
σ = 90.4± 0.7
WF
σ/μ = 17.87± 0.16 %
μ = 529.0± 0.7
σ = 94.5± 0.7



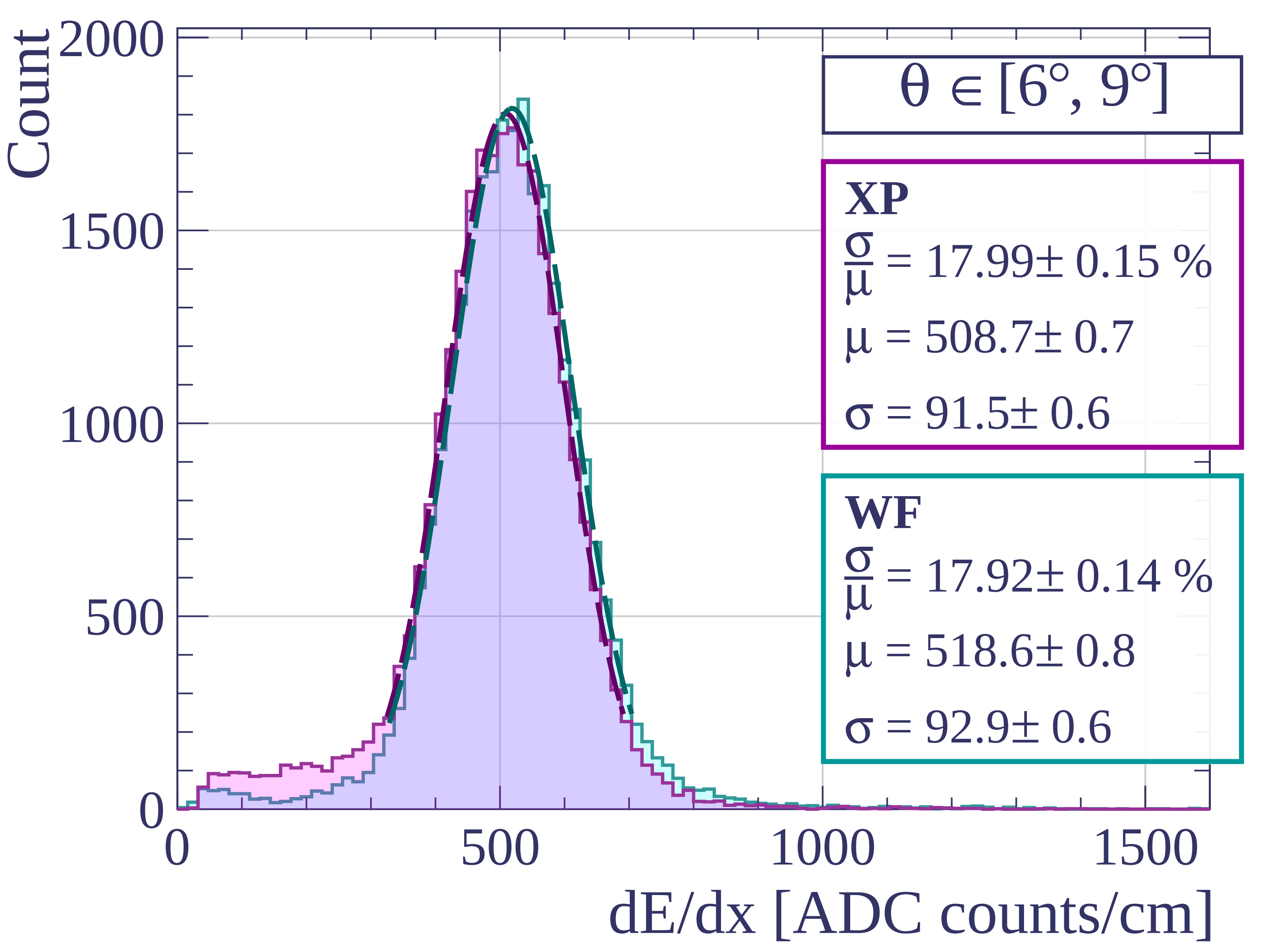
Count
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
φ ∈ [87°, 90°]
XP
σ/μ = 17.53± 0.16 %
μ = 509.0± 0.7
σ = 89.2± 0.7
WF
σ/μ = 17.53± 0.16 %
μ = 528.2± 0.7
σ = 92.6± 0.7

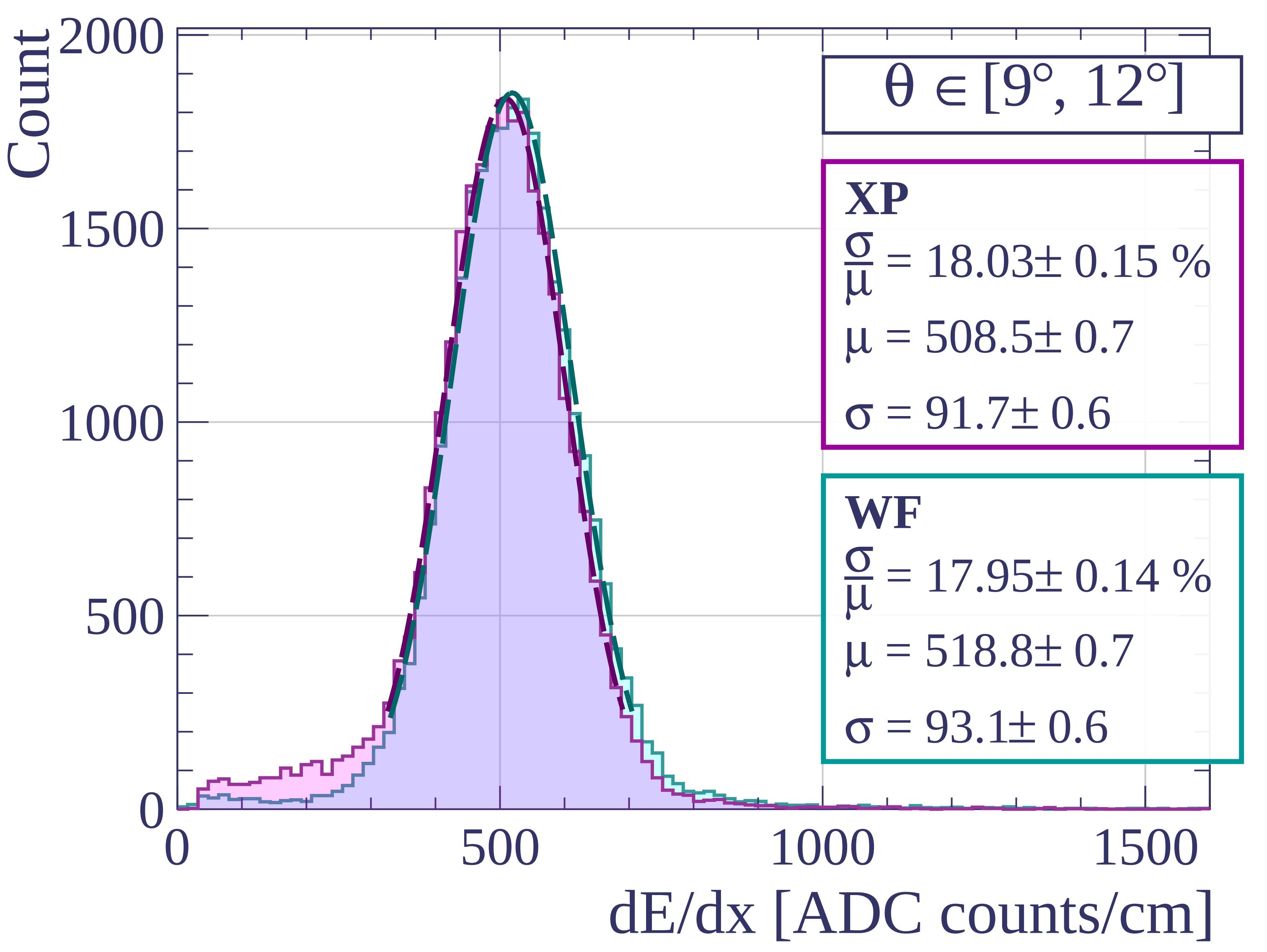


Count
2000
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [0°, 3°]
XP
σ/μ = 18.09± 0.14 %
μ = 509.3± 0.6
σ = 92.1± 0.6
WF
σ/μ = 18.02± 0.14 %
μ = 518.9± 0.7
σ = 93.5± 0.6

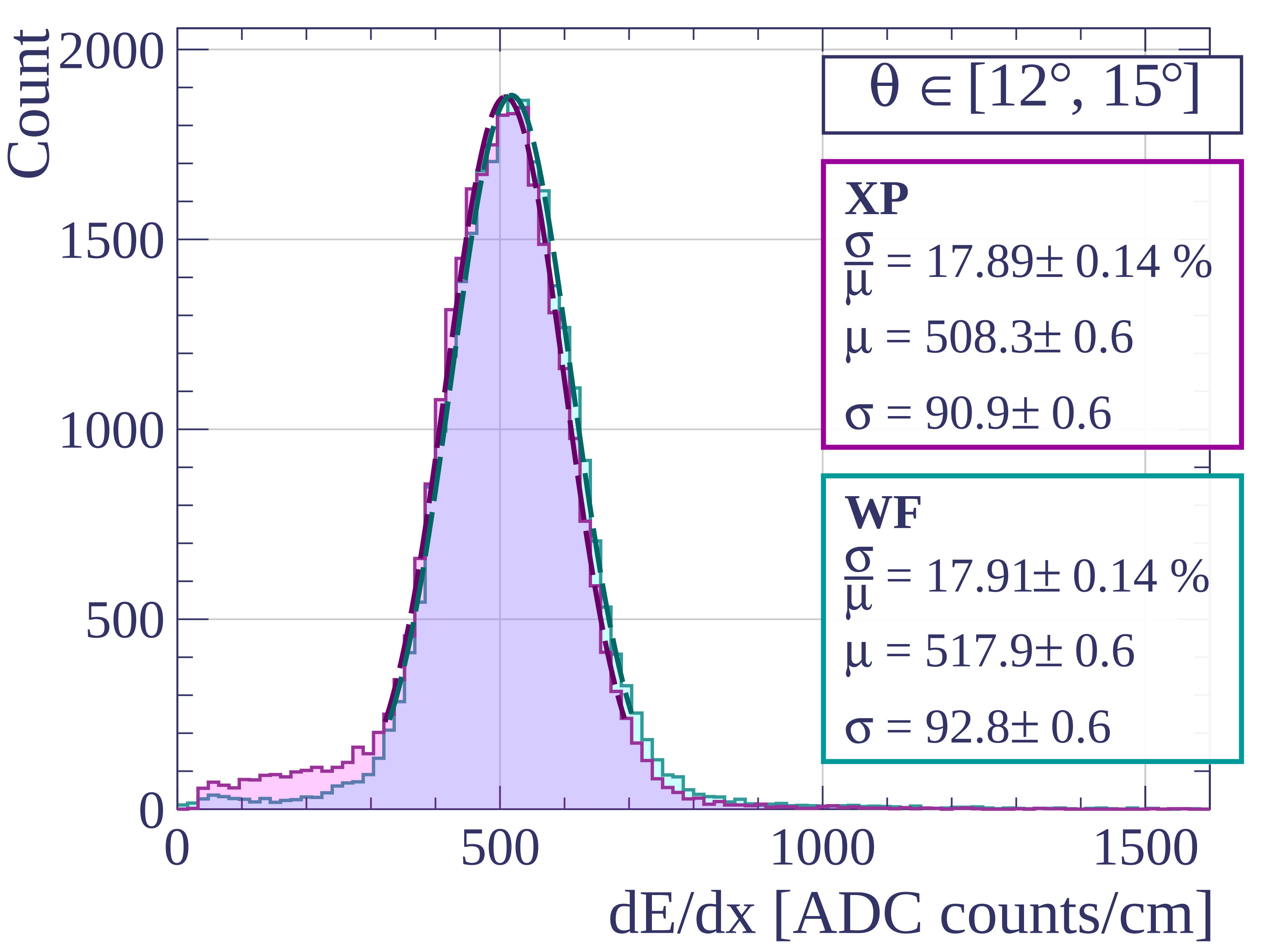


Count
0
500
1000
1500
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [3°, 6°]
XP
σ/μ = 17.86± 0.15 %
μ = 509.2± 0.7
σ = 90.9± 0.6
WF
σ/μ = 17.98± 0.15 %
μ = 518.4± 0.7
σ = 93.2± 0.7

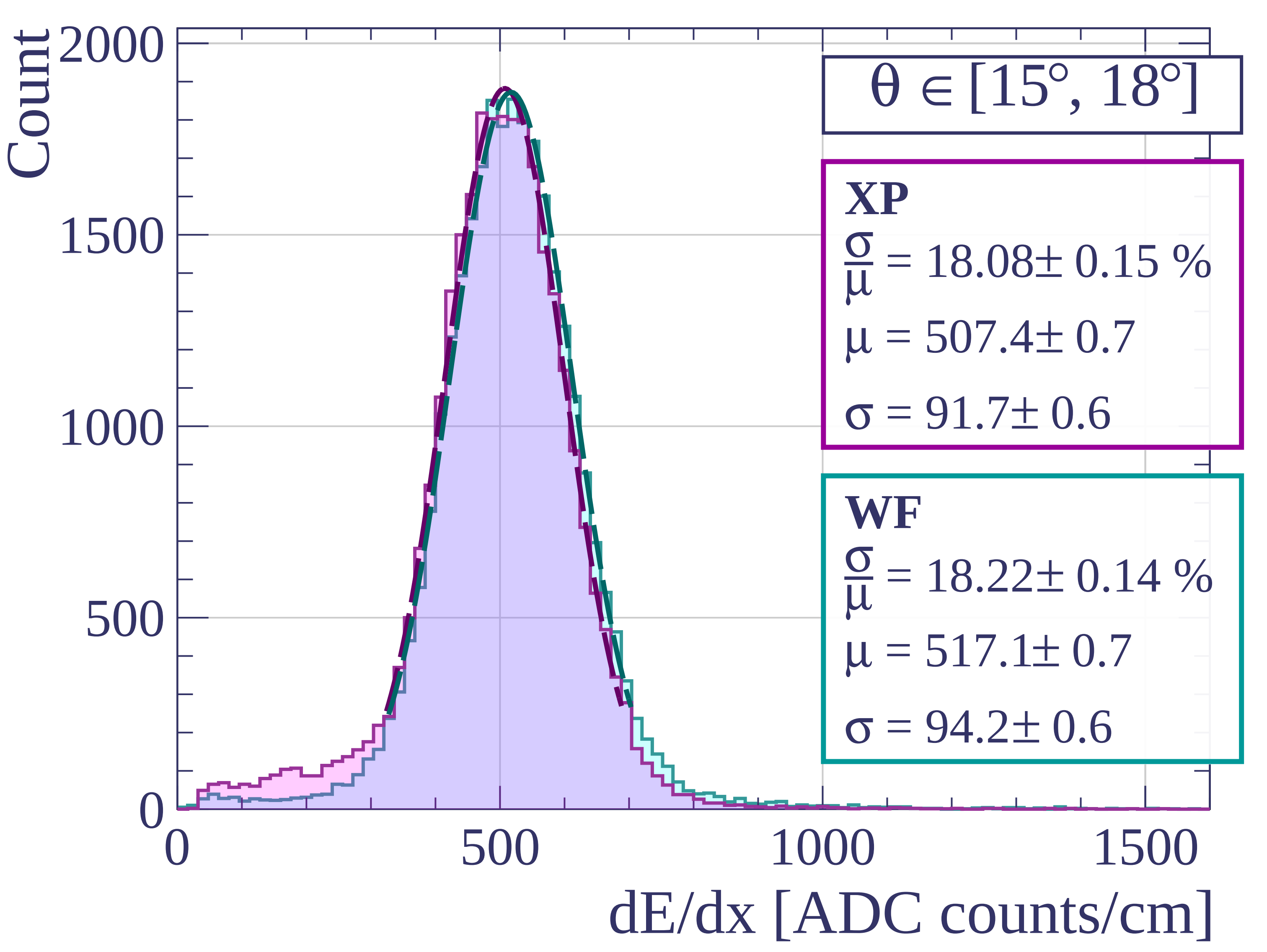

Count
2000
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [6°, 9°]
XP
σ/μ = 17.99± 0.15 %
μ = 508.7± 0.7
σ = 91.5± 0.6
WF
σ/μ = 17.92± 0.14 %
μ = 518.6± 0.8
σ = 92.9± 0.6



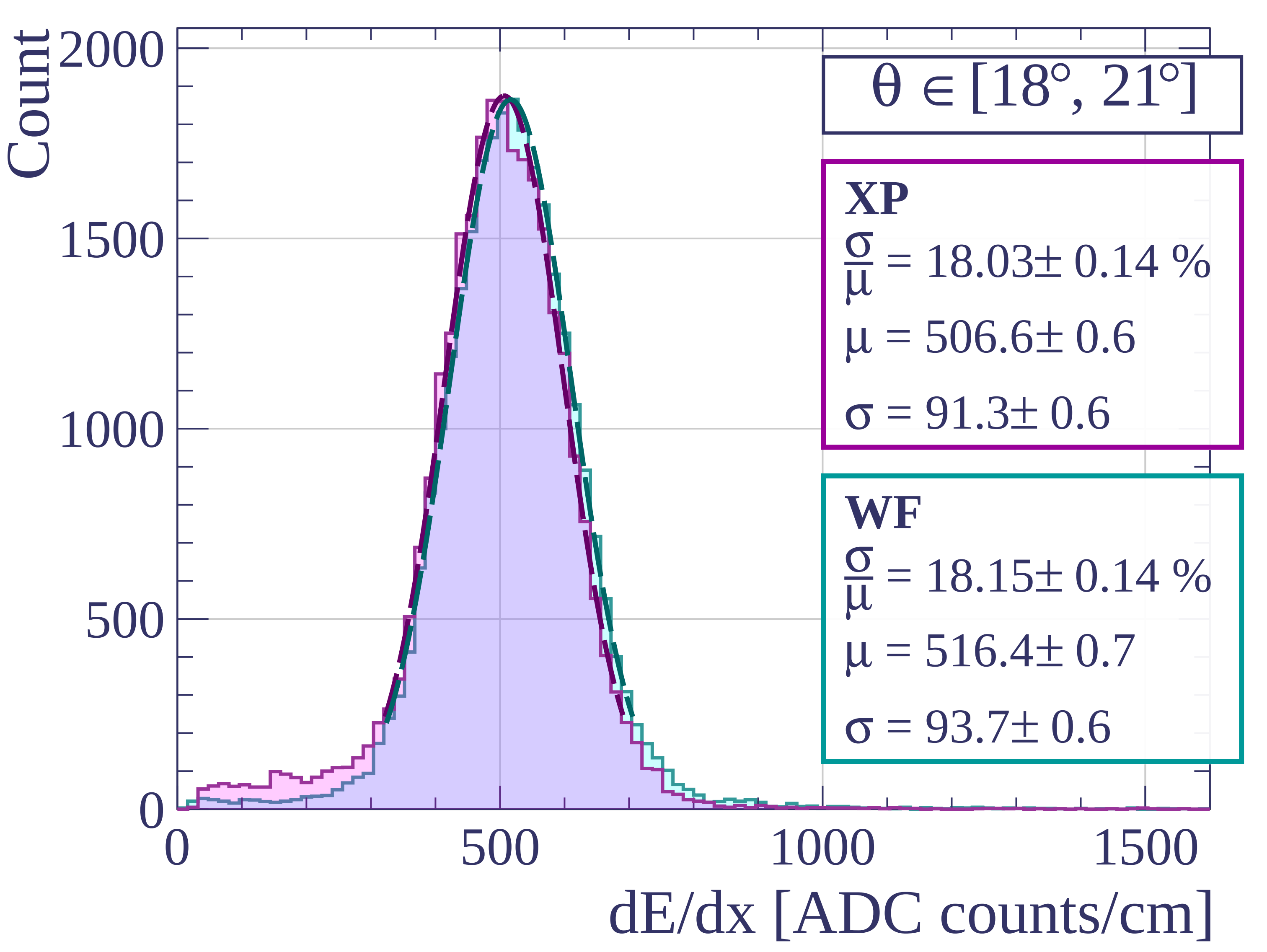
Count
2000
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [9°, 12°]
XP
σ/μ = 18.03± 0.15 %
μ = 508.5± 0.7
σ = 91.7± 0.6
WF
σ/μ = 17.95± 0.14 %
μ = 518.8± 0.7
σ = 93.1± 0.6



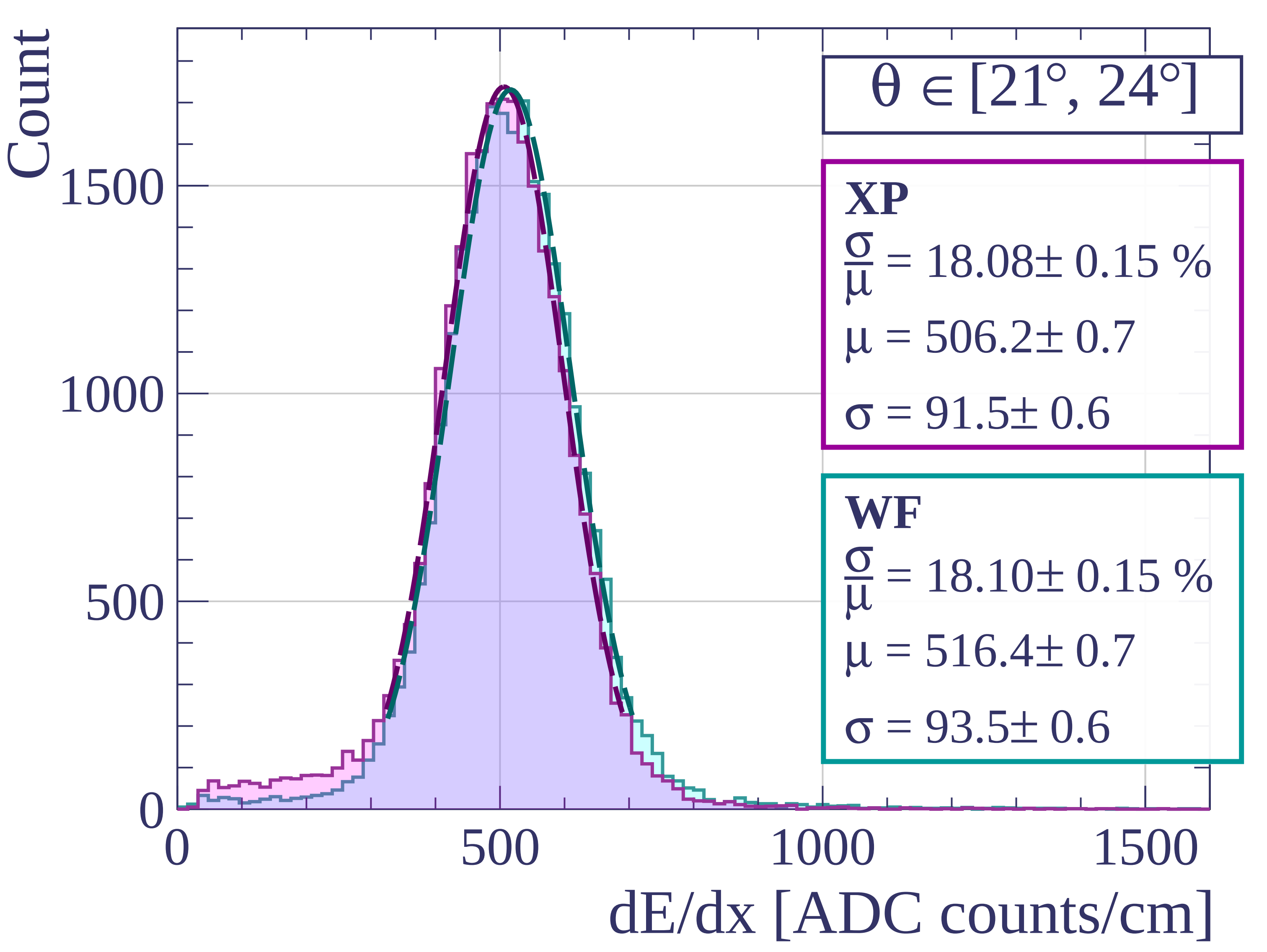
Count
2000
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [12°, 15°]
XP
σ/μ = 17.89± 0.14 %
μ = 508.3± 0.6
σ = 90.9± 0.6
WF
σ/μ = 17.91± 0.14 %
μ = 517.9± 0.6
σ = 92.8± 0.6



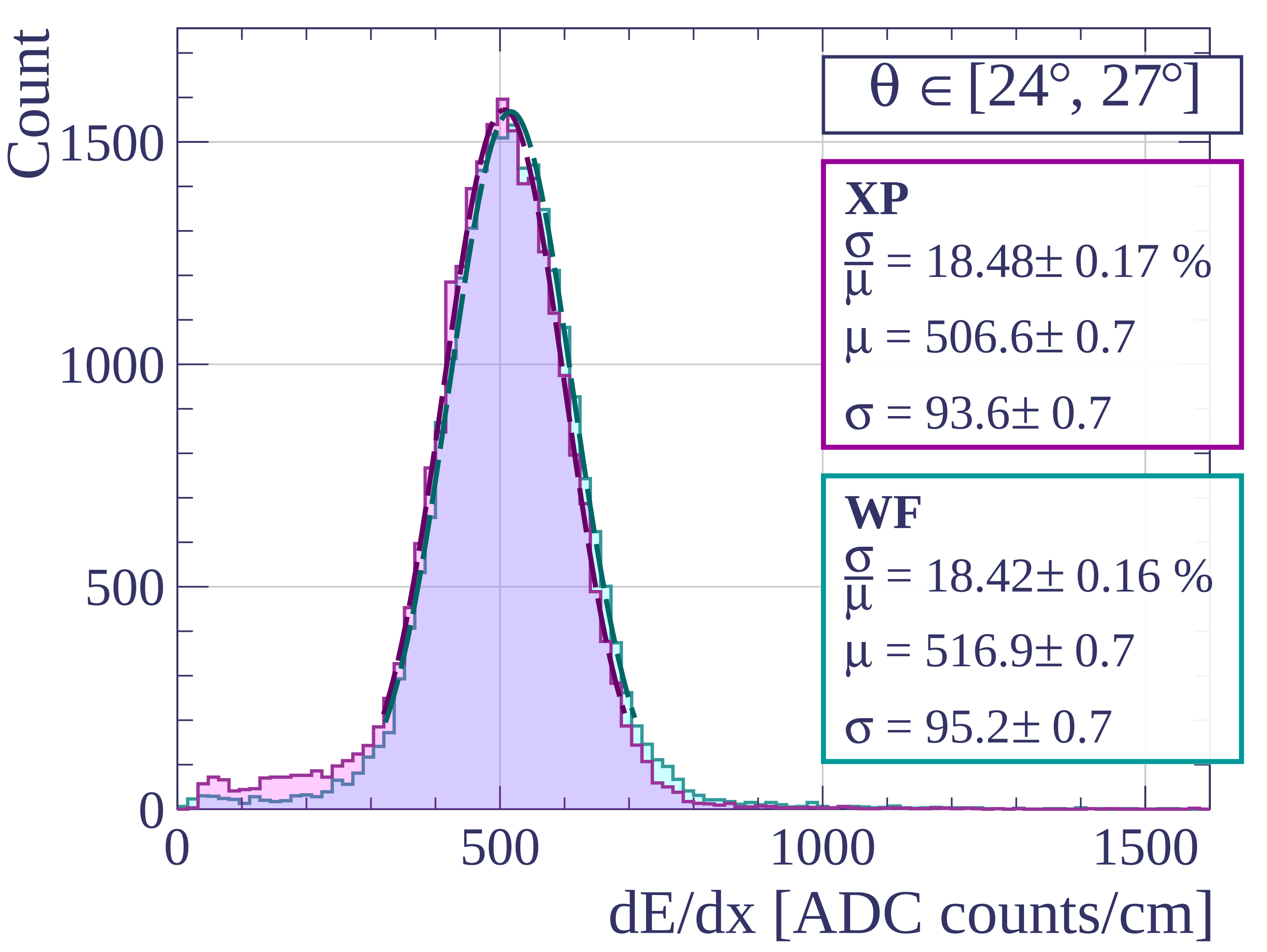
Count
2000
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [15°, 18°]
XP
σ/μ = 18.08± 0.15 %
μ = 507.4± 0.7
σ = 91.7± 0.6
WF
σ/μ = 18.22± 0.14 %
μ = 517.1± 0.7
σ = 94.2± 0.6



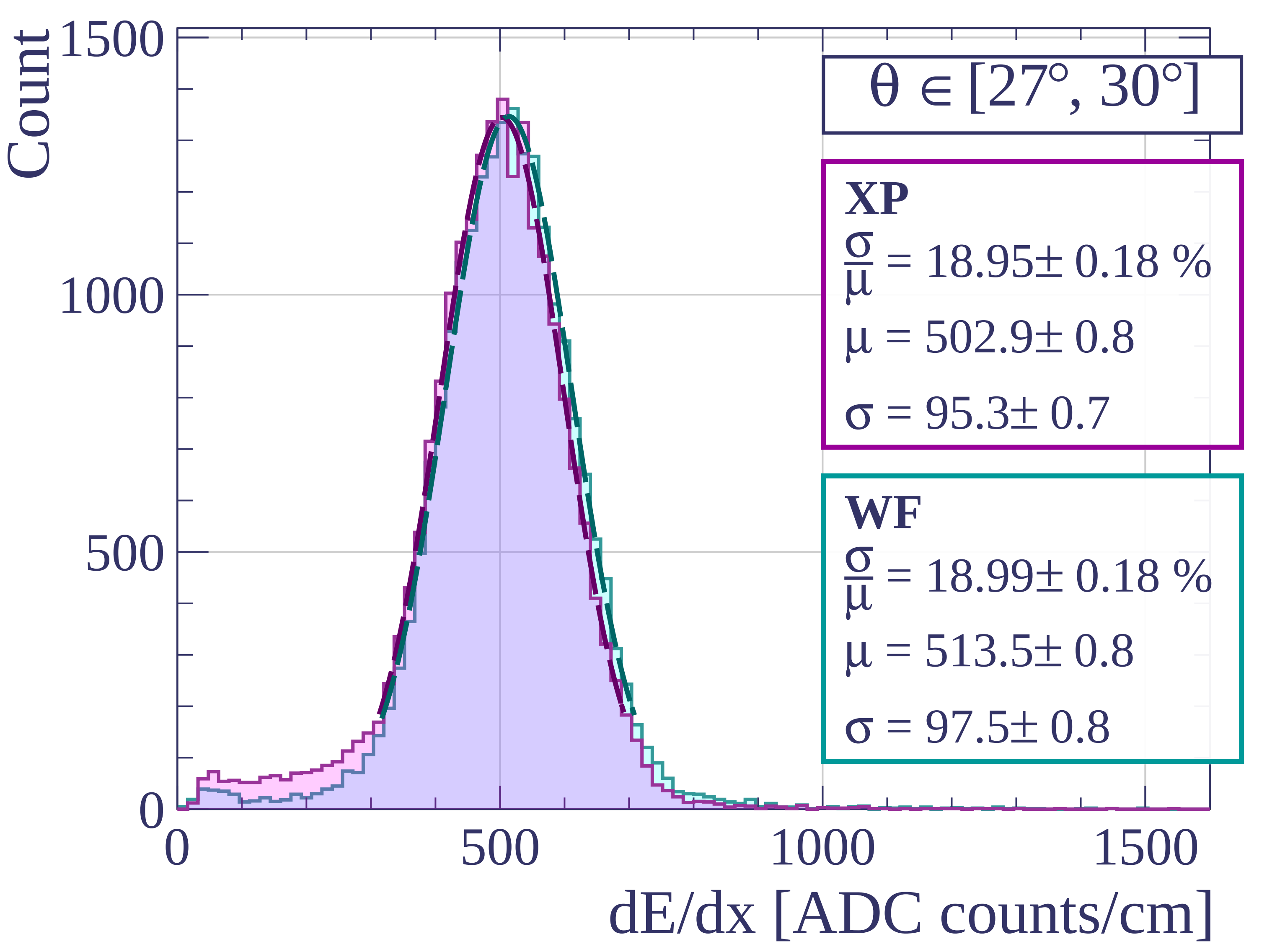
Count
2000
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [18°, 21°]
XP
σ/μ = 18.03± 0.14 %
μ = 506.6± 0.6
σ = 91.3± 0.6
WF
σ/μ = 18.15± 0.14 %
μ = 516.4± 0.7
σ = 93.7± 0.6



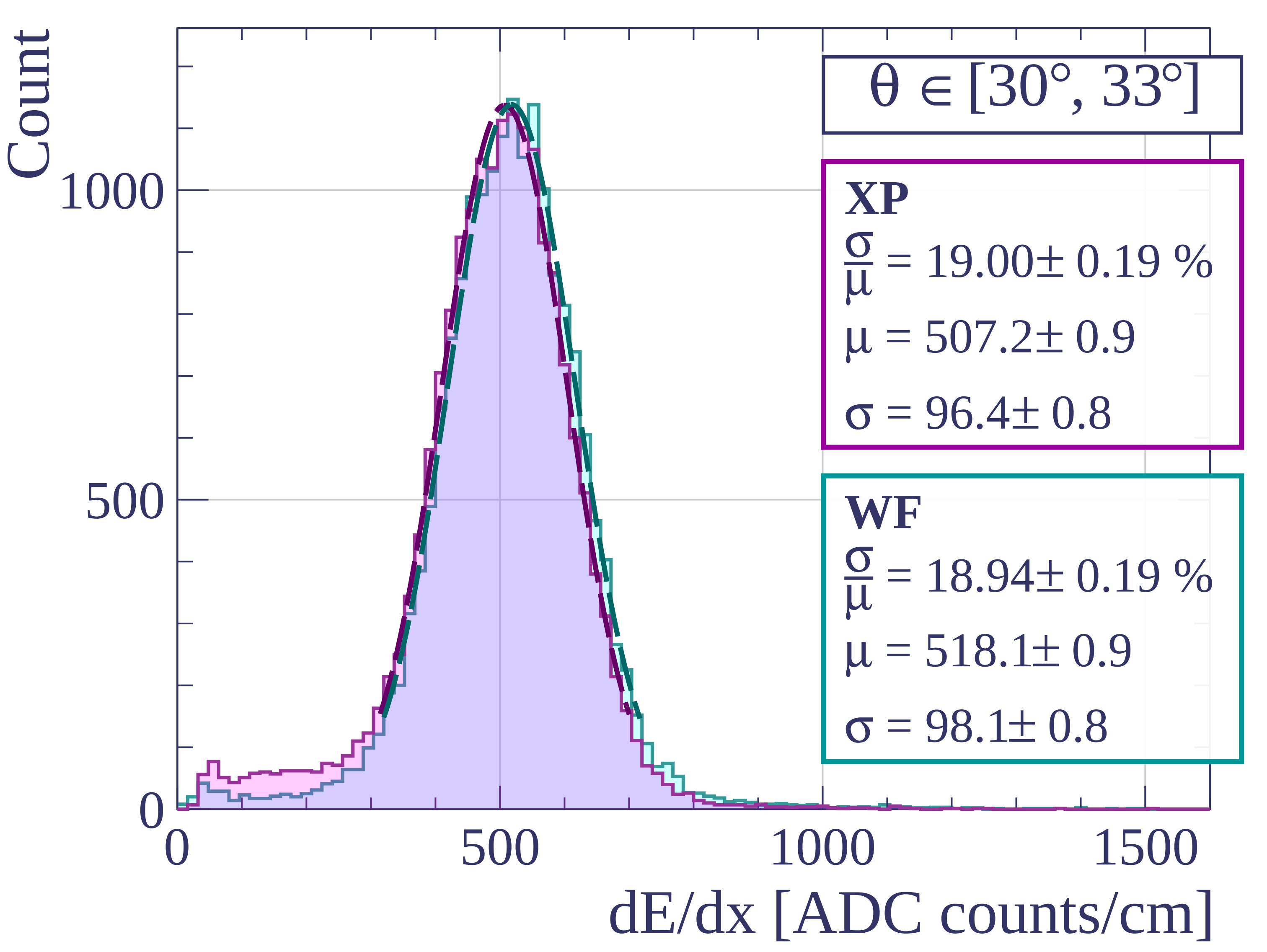
Count
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [21°, 24°]
XP
σ/μ = 18.08± 0.15 %
μ = 506.2± 0.7
σ = 91.5± 0.6
WF
σ/μ = 18.10± 0.15 %
μ = 516.4± 0.7
σ = 93.5± 0.6



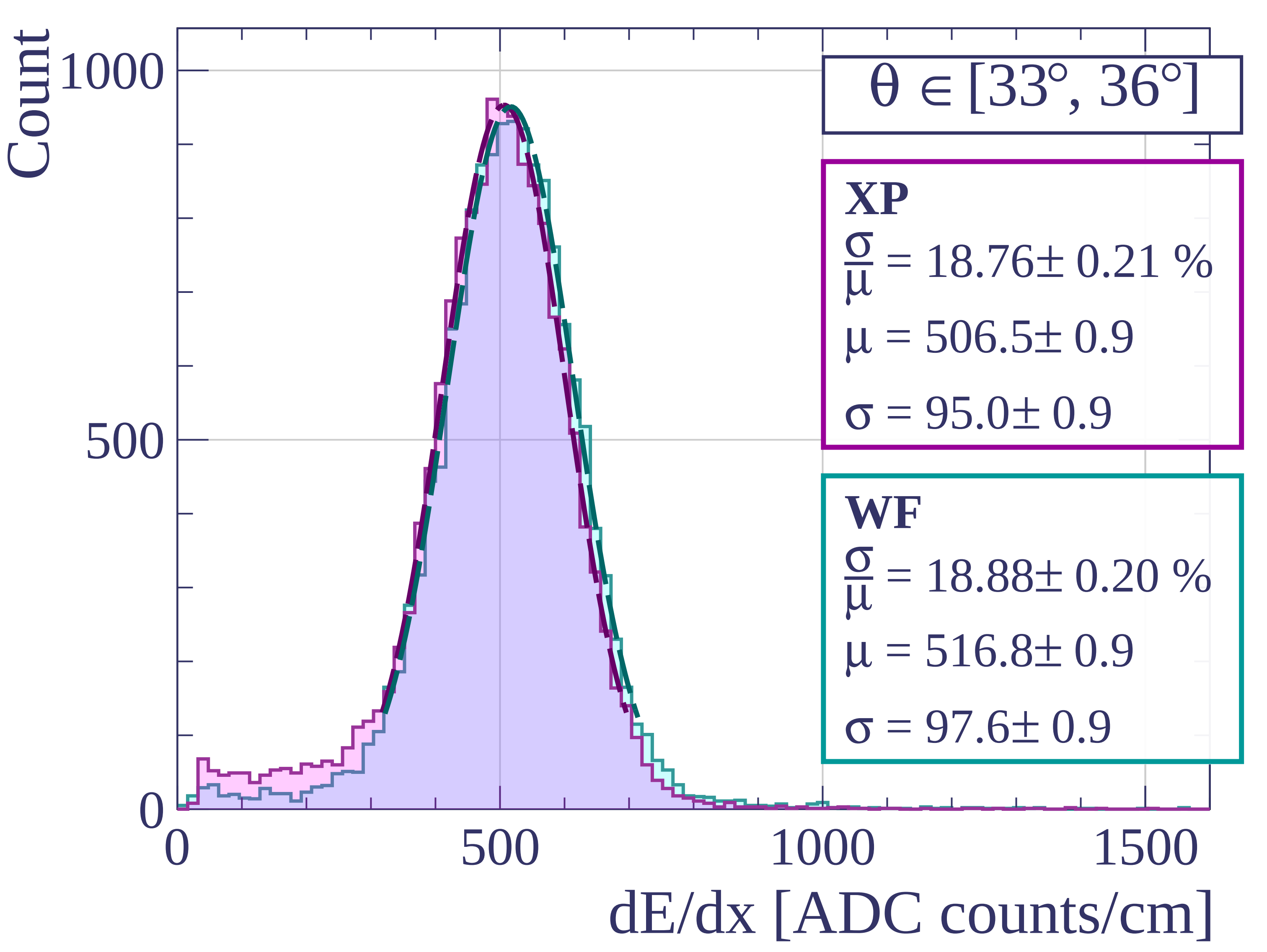
Count
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [24°, 27°]
XP
σ/μ = 18.48± 0.17 %
μ = 506.6± 0.7
σ = 93.6± 0.7
WF
σ/μ = 18.42± 0.16 %
μ = 516.9± 0.7
σ = 95.2± 0.7



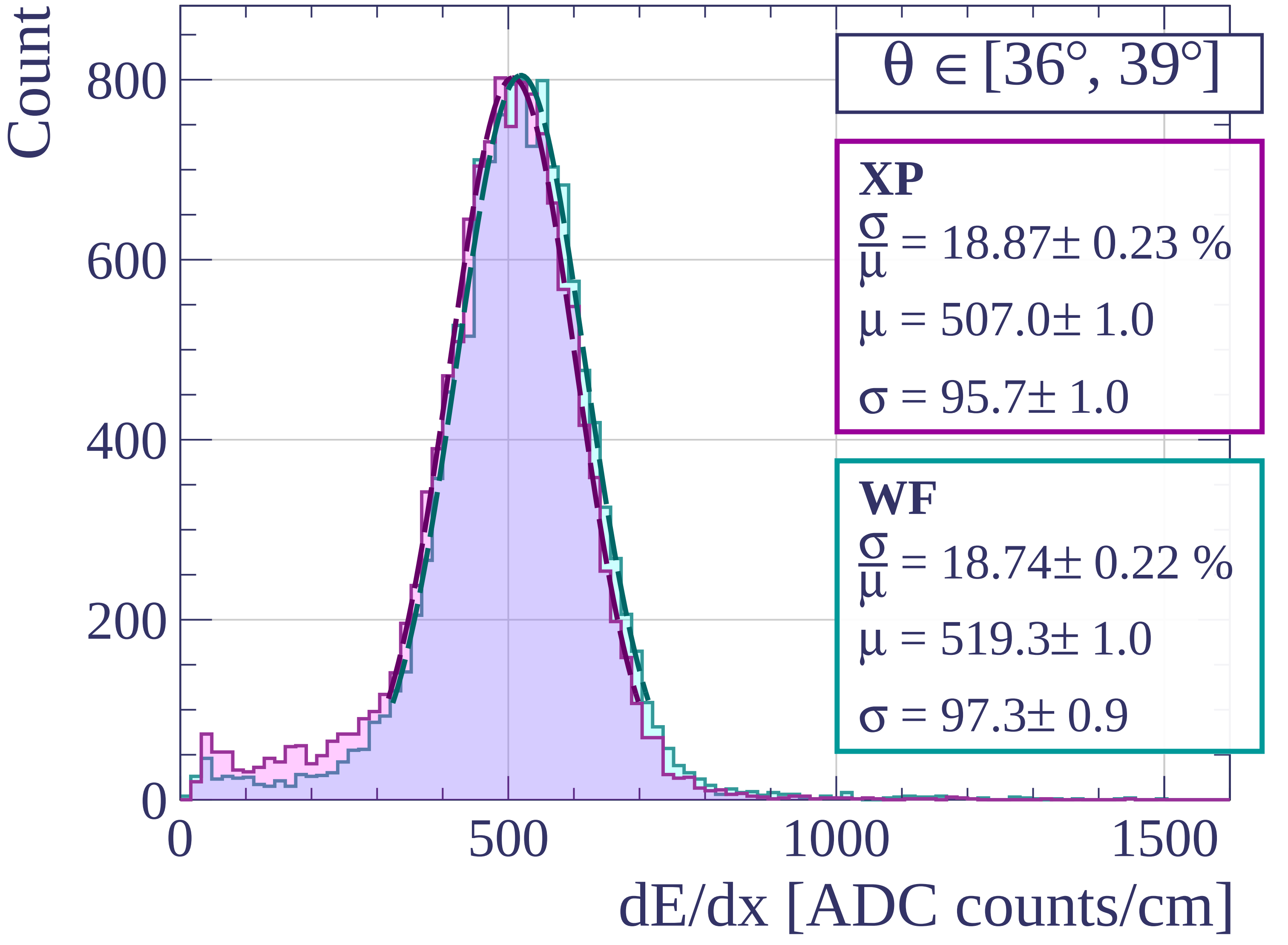
Count
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [27°, 30°]
XP
σ/μ = 18.95± 0.18 %
μ = 502.9± 0.8
σ = 95.3± 0.7
WF
σ/μ = 18.99± 0.18 %
μ = 513.5± 0.8
σ = 97.5± 0.8



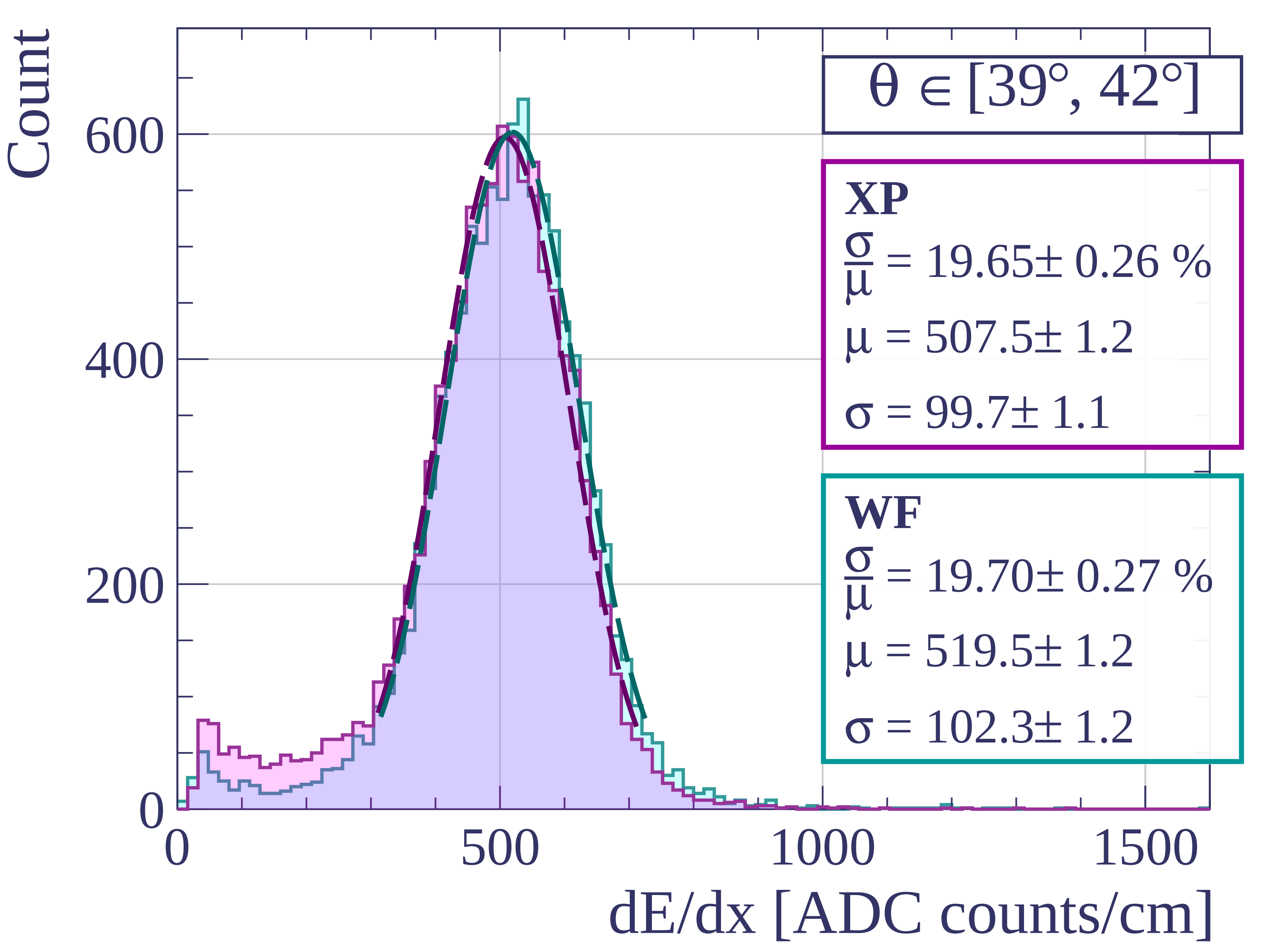
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [30°, 33°]
XP
σ/μ = 19.00± 0.19 %
μ = 507.2± 0.9
σ = 96.4± 0.8
WF
σ/μ = 18.94± 0.19 %
μ = 518.1± 0.9
σ = 98.1± 0.8



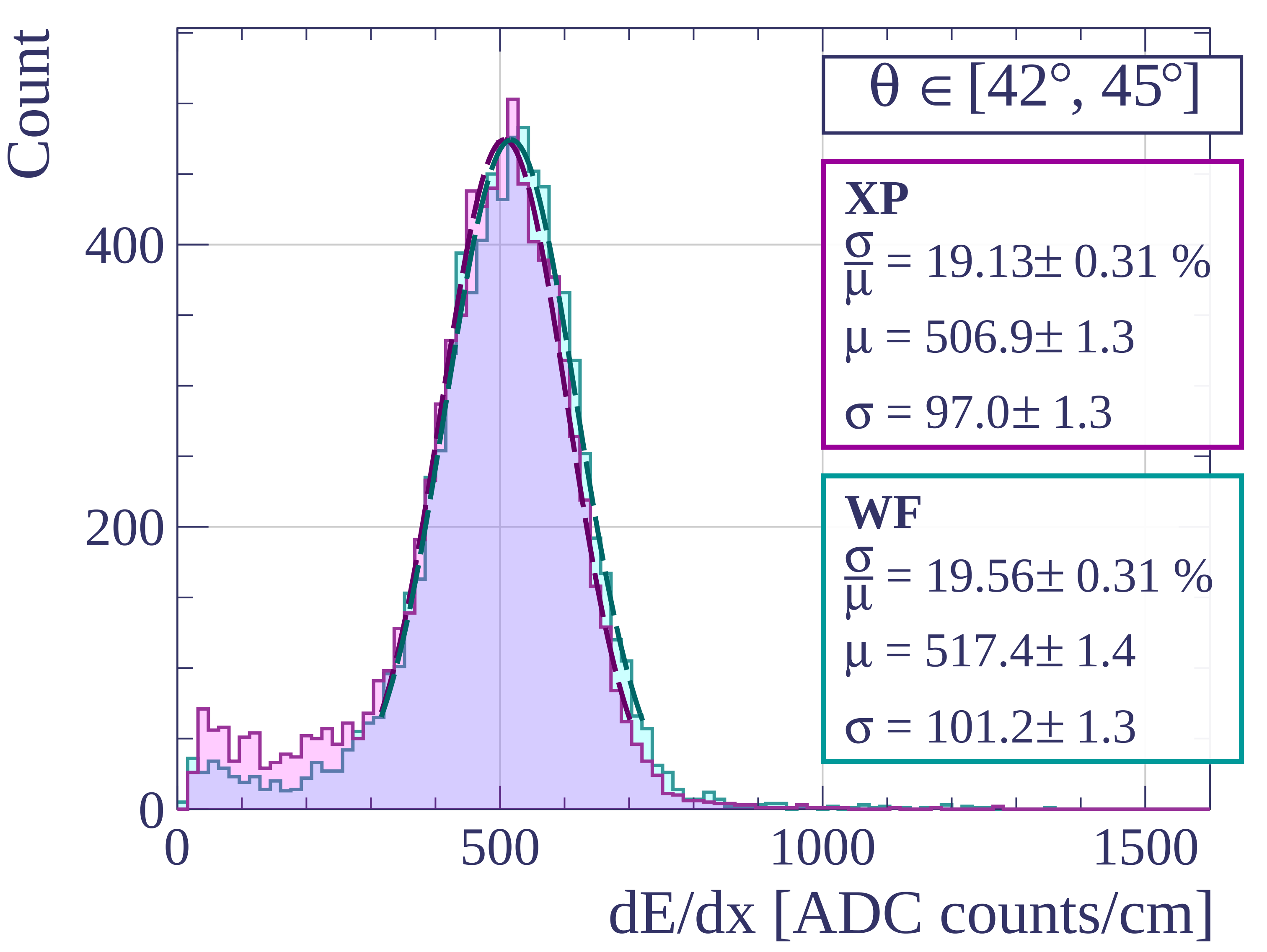
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [33°, 36°]
XP
σ/μ = 18.76± 0.21 %
μ = 506.5± 0.9
σ = 95.0± 0.9
WF
σ/μ = 18.88± 0.20 %
μ = 516.8± 0.9
σ = 97.6± 0.9



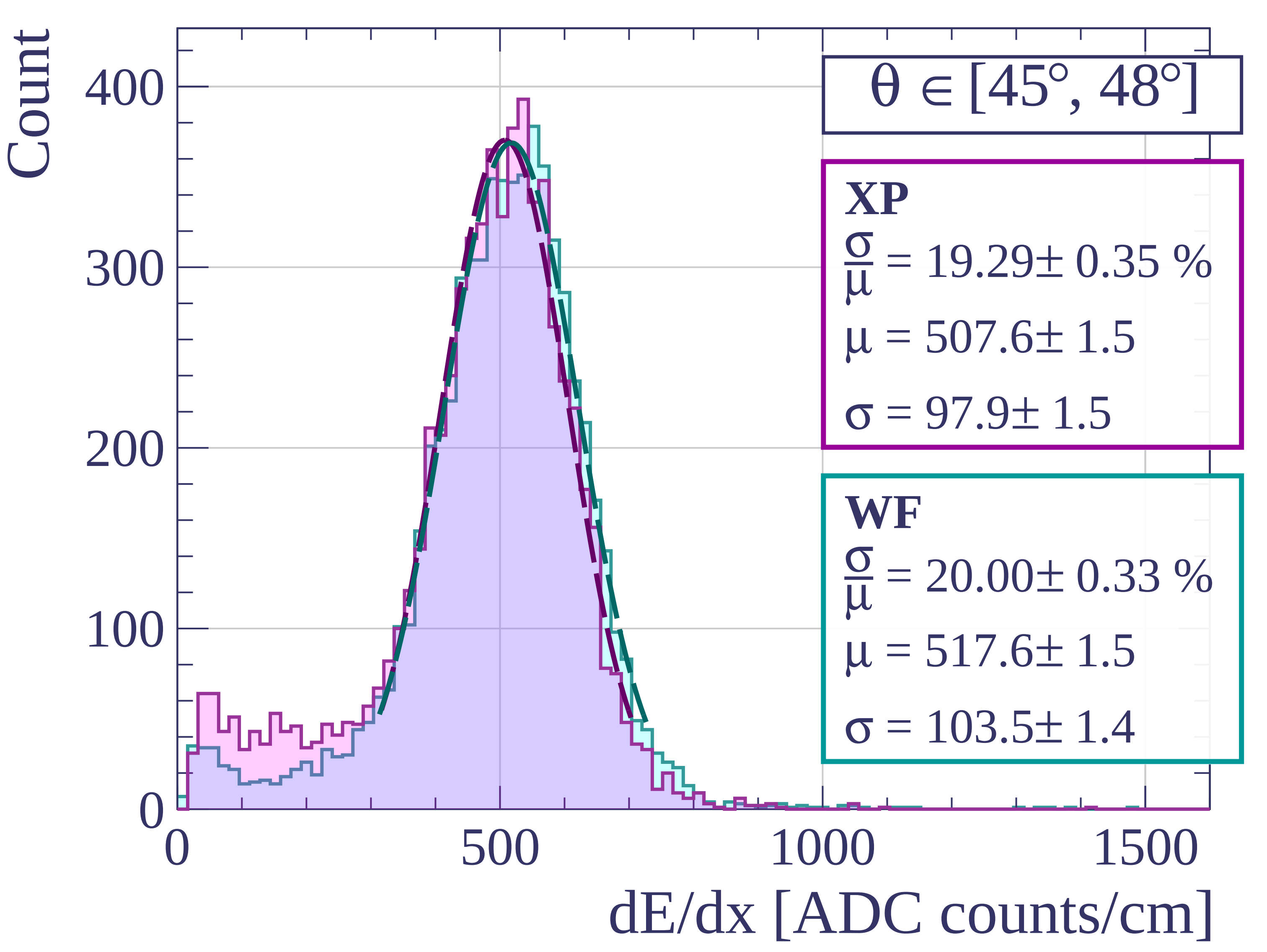
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [36°, 39°]
XP
σ/μ = 18.87± 0.23 %
μ = 507.0± 1.0
σ = 95.7± 1.0
WF
σ/μ = 18.74± 0.22 %
μ = 519.3± 1.0
σ = 97.3± 0.9



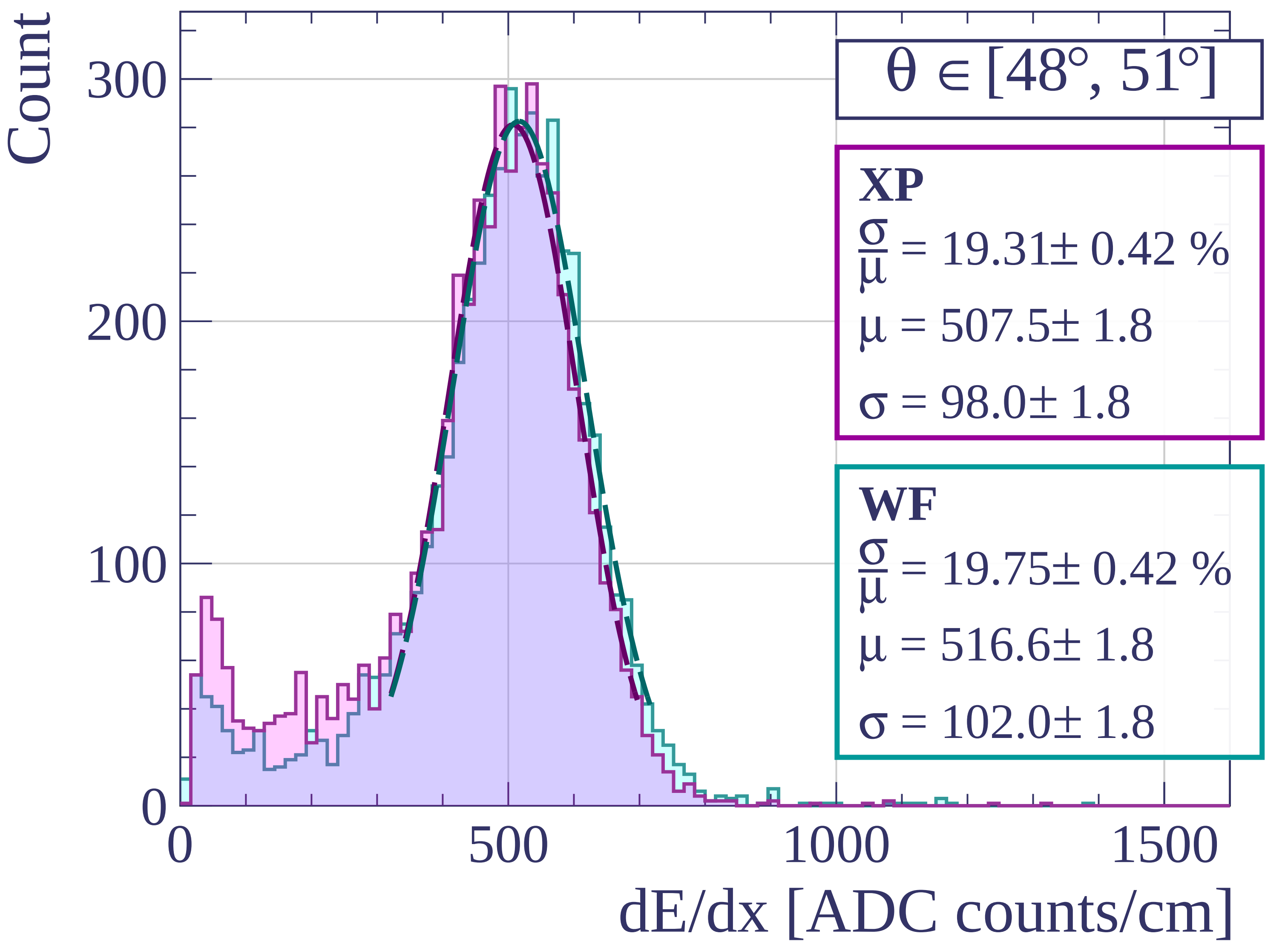
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [39°, 42°]
XP
σ/μ = 19.65± 0.26 %
μ = 507.5± 1.2
σ = 99.7± 1.1
WF
σ/μ = 19.70± 0.27 %
μ = 519.5± 1.2
σ = 102.3± 1.2



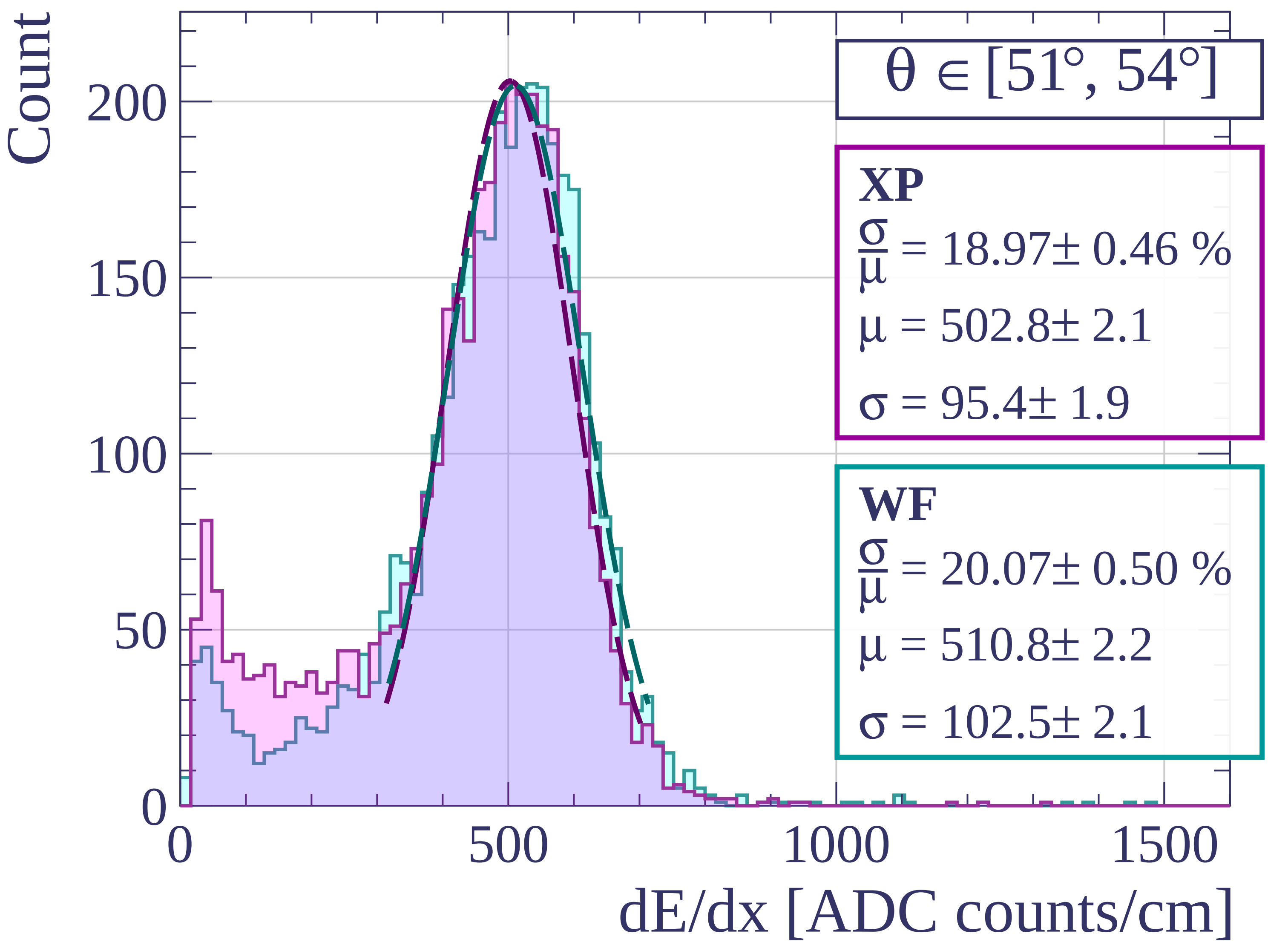
Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [42°, 45°]
XP
σ/μ = 19.13± 0.31 %
μ = 506.9± 1.3
σ = 97.0± 1.3
WF
σ/μ = 19.56± 0.31 %
μ = 517.4± 1.4
σ = 101.2± 1.3



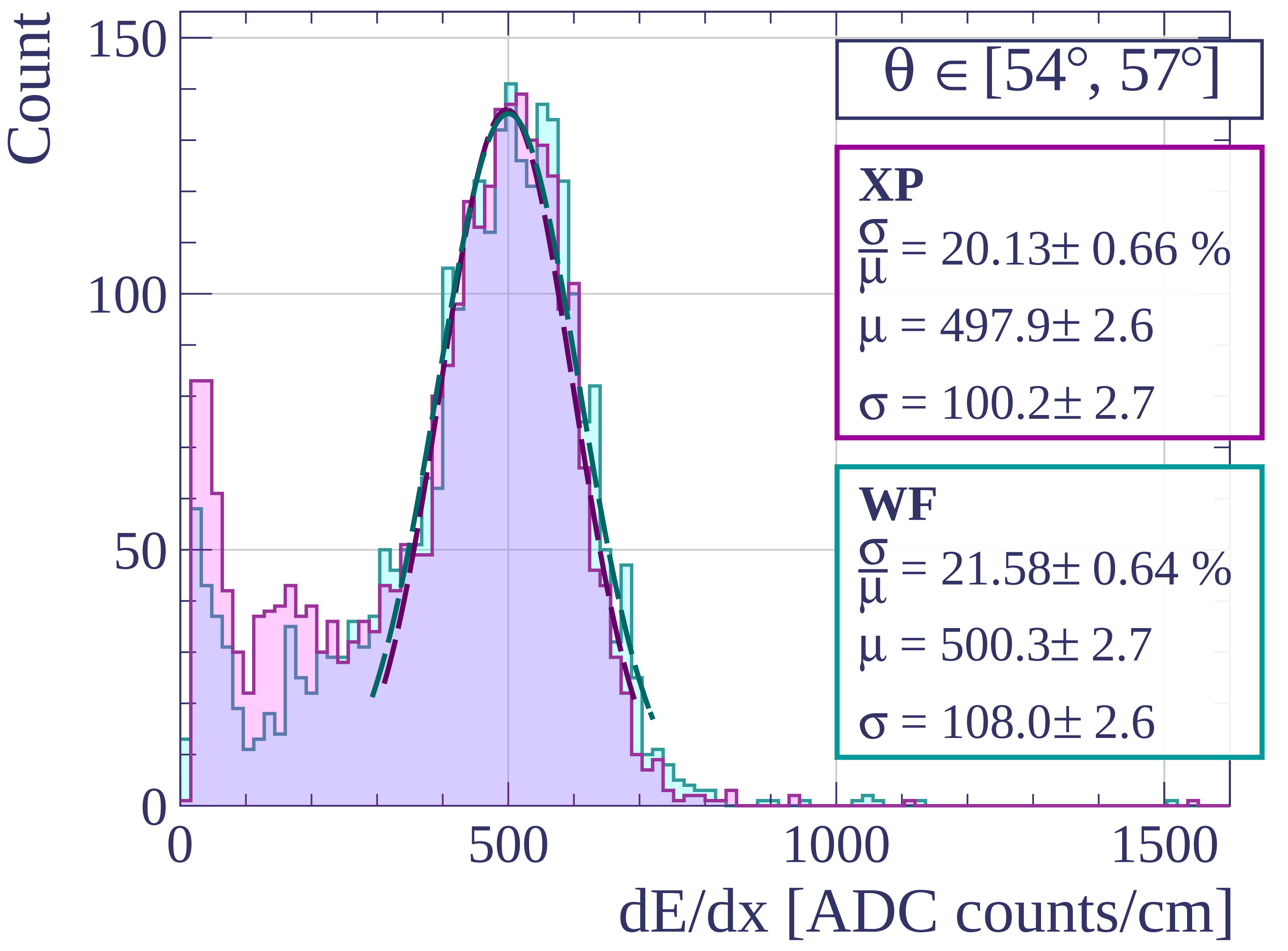
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [45°, 48°]
XP
σ/μ = 19.29± 0.35 %
μ = 507.6± 1.5
σ = 97.9± 1.5
WF
σ/μ = 20.00± 0.33 %
μ = 517.6± 1.5
σ = 103.5± 1.4

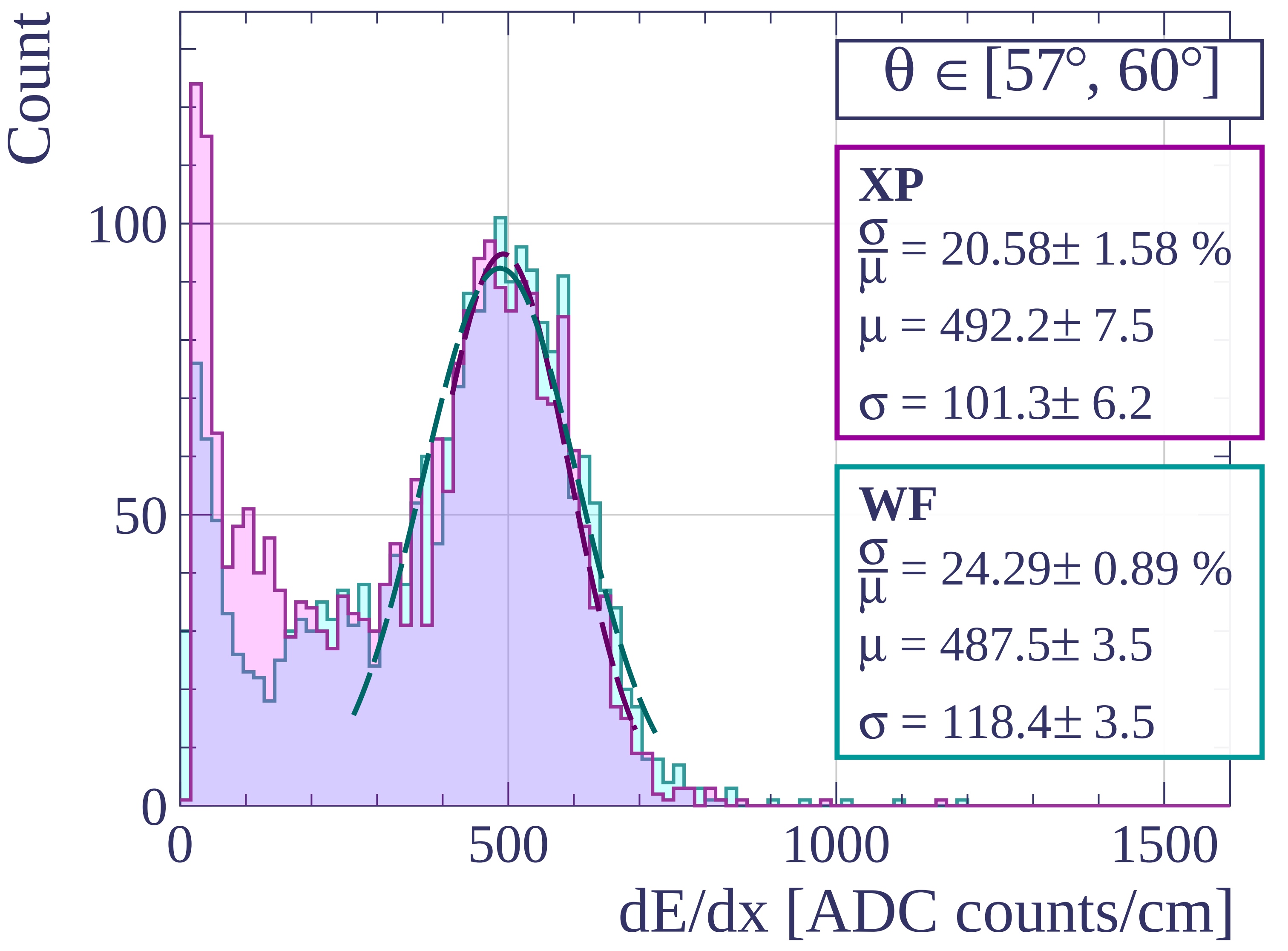


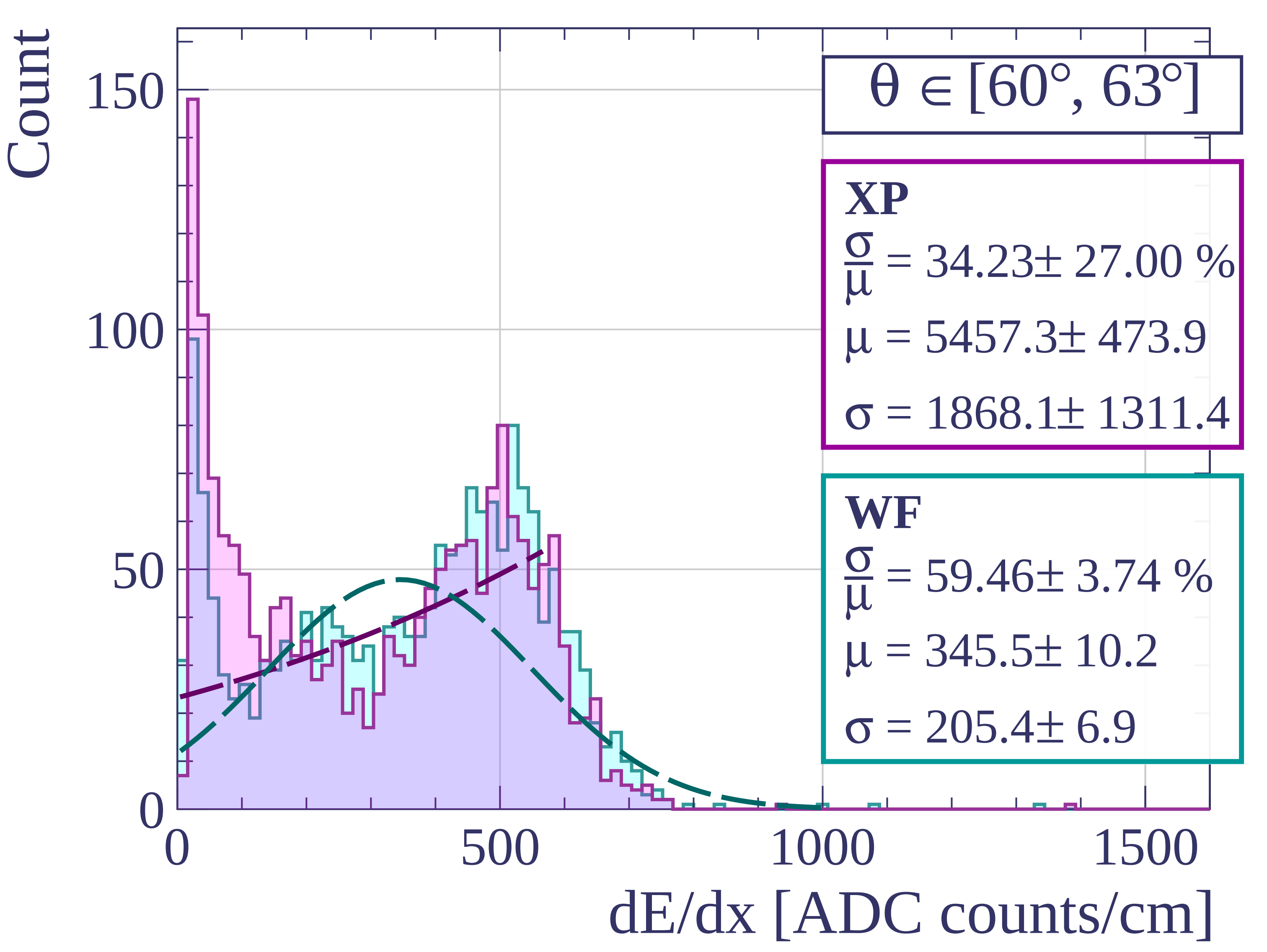
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [48°, 51°]
XP
σ/μ = 19.31± 0.42 %
μ = 507.5± 1.8
σ = 98.0± 1.8
WF
σ/μ = 19.75± 0.42 %
μ = 516.6± 1.8
σ = 102.0± 1.8



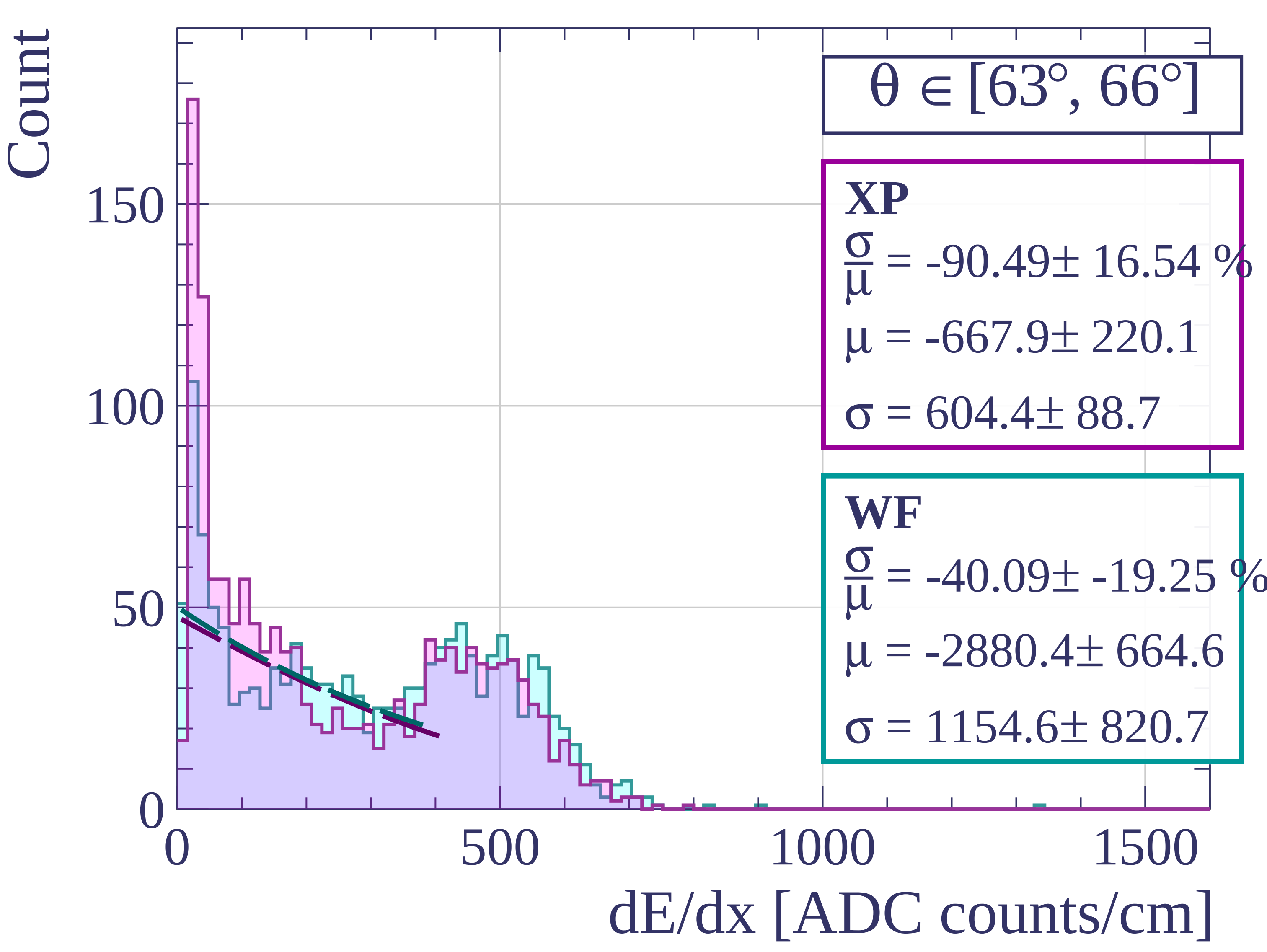
Count
dE/dx [ADC counts/cm]
θ ∈ [51°, 54°]
XP
σ/μ = 18.97± 0.46 %
μ = 502.8± 2.1
σ = 95.4± 1.9
WF
σ/μ = 20.07± 0.50 %
μ = 510.8± 2.2
σ = 102.5± 2.1
0
50
100
150
200
0
500
1000
1500


Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [54°, 57°]
XP
σ/μ = 20.13± 0.66 %
μ = 497.9± 2.6
σ = 100.2± 2.7
WF
σ/μ = 21.58± 0.64 %
μ = 500.3± 2.7
σ = 108.0± 2.6

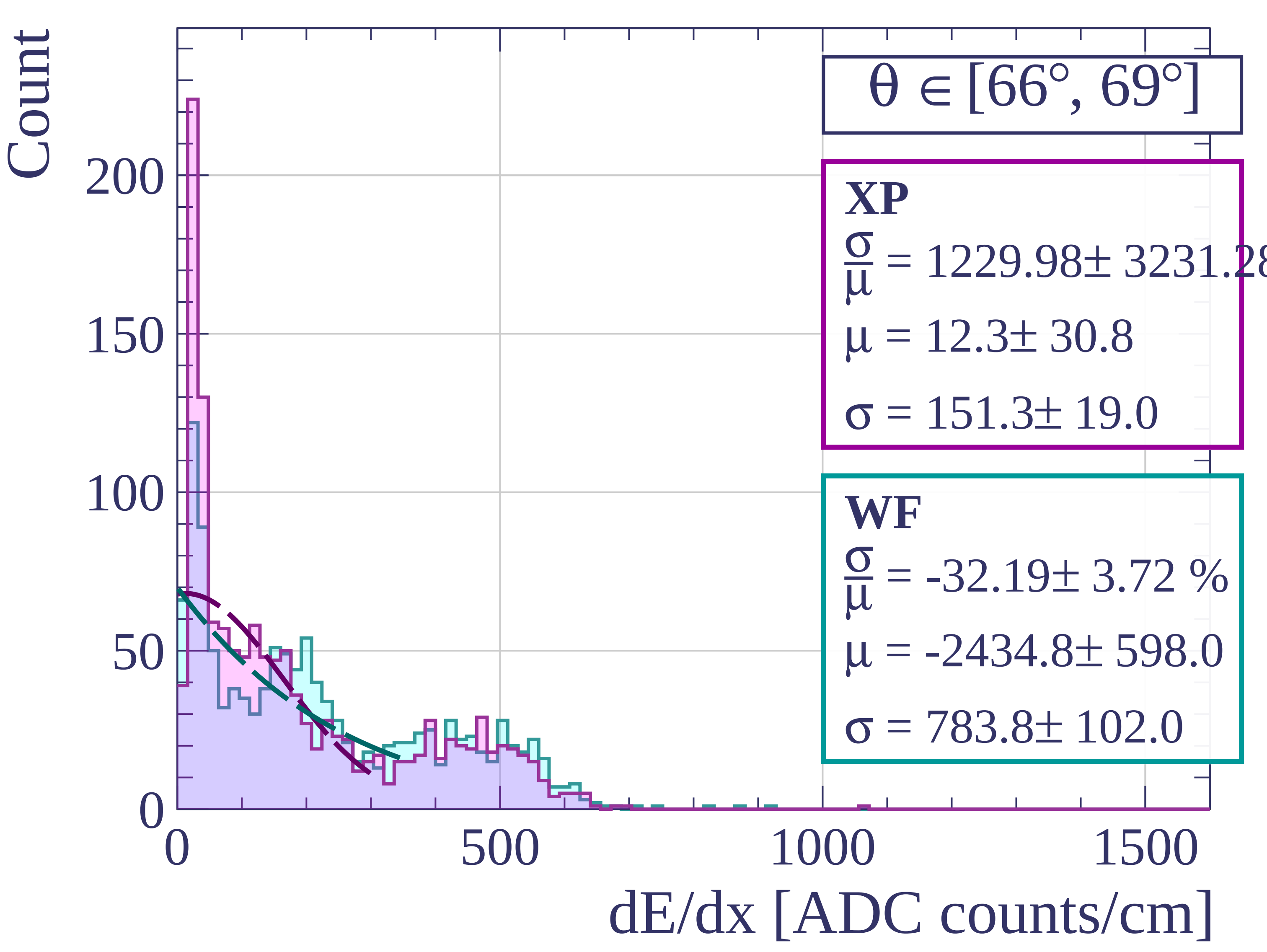

Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [57°, 60°]
XP
σ/μ = 20.58± 1.58 %
μ = 492.2± 7.5
σ = 101.3± 6.2
WF
σ/μ = 24.29± 0.89 %
μ = 487.5± 3.5
σ = 118.4± 3.5



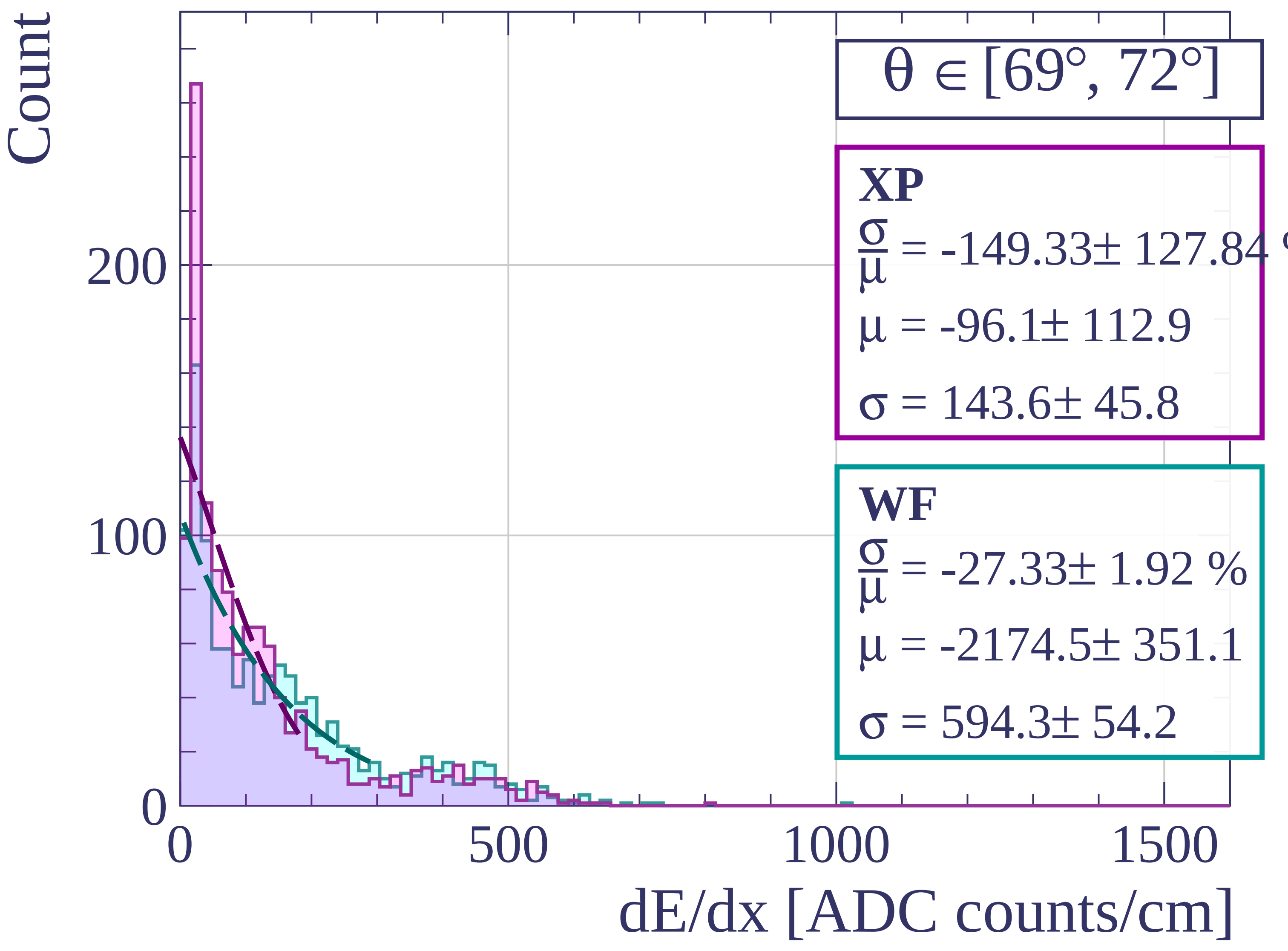
Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [60°, 63°]
XP
σ/μ = 34.23± 27.00 %
μ = 5457.3± 473.9
σ = 1868.1± 1311.4
WF
σ/μ = 59.46± 3.74 %
μ = 345.5± 10.2
σ = 205.4± 6.9



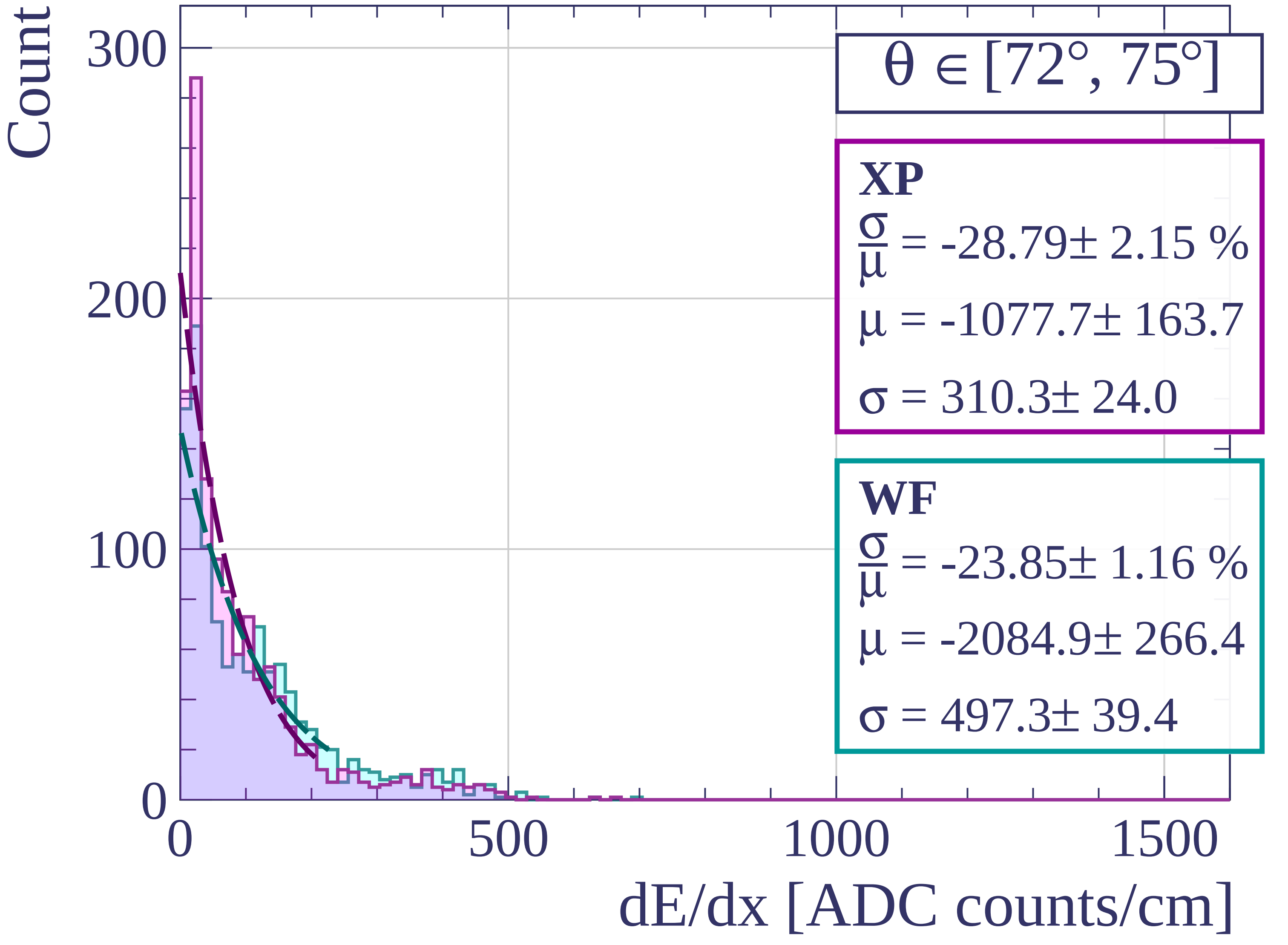
Count
dE/dx [ADC counts/cm]
0
50
100
150
0
500
1000
1500
θ ∈ [63°, 66°]
XP
σ/μ = -90.49± 16.54 %
μ = -667.9± 220.1
σ = 604.4± 88.7
WF
σ/μ = -40.09± -19.25 %
μ = -2880.4± 664.6
σ = 1154.6± 820.7



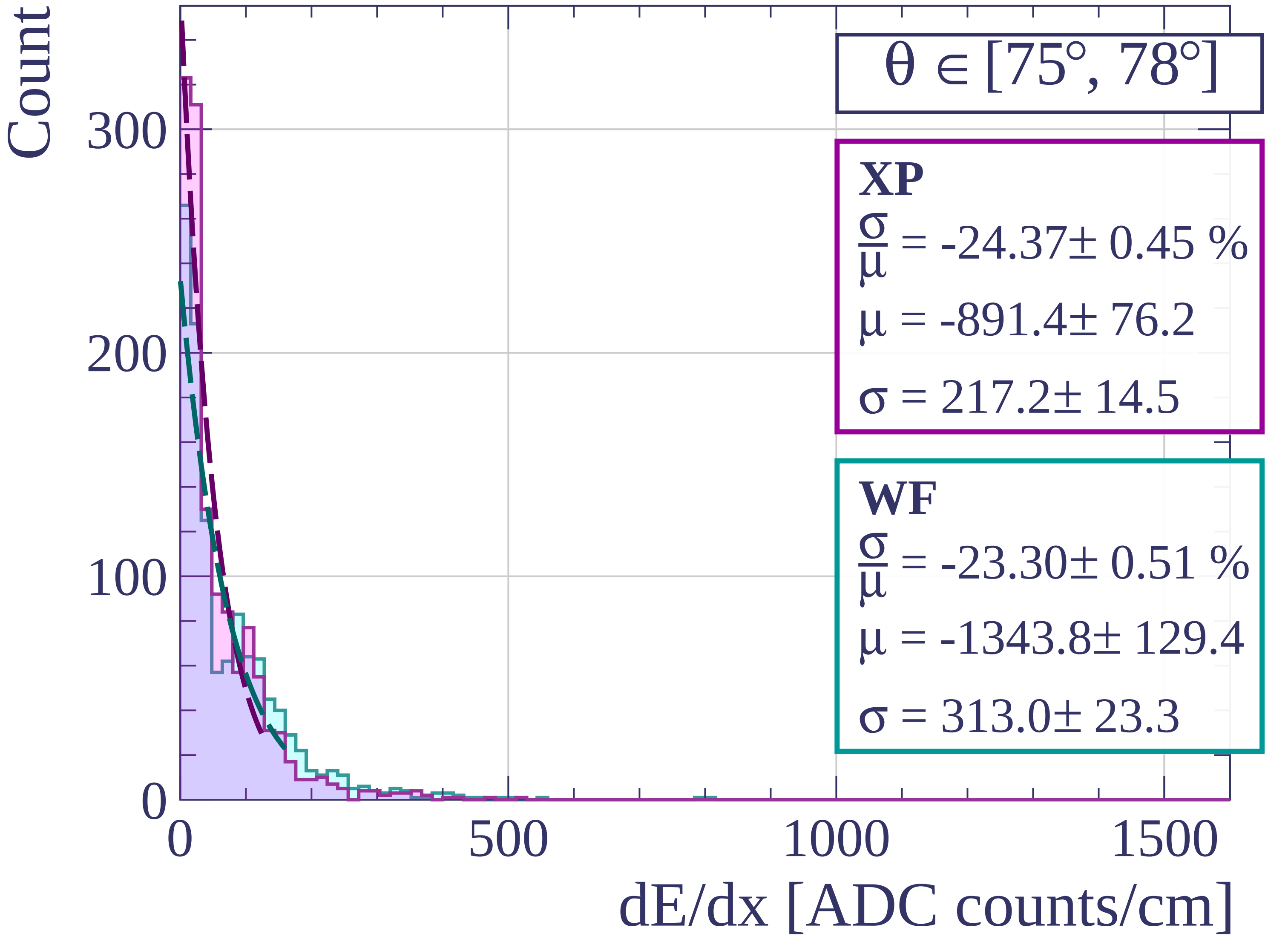
Count
dE/dx [ADC counts/cm]
0
50
100
150
200
0
500
1000
1500
θ ∈ [66°, 69°]
XP
σ/μ = 1229.98± 3231.28
μ = 12.3± 30.8
σ = 151.3± 19.0
WF
σ/μ = -32.19± 3.72 %
μ = -2434.8± 598.0
σ = 783.8± 102.0



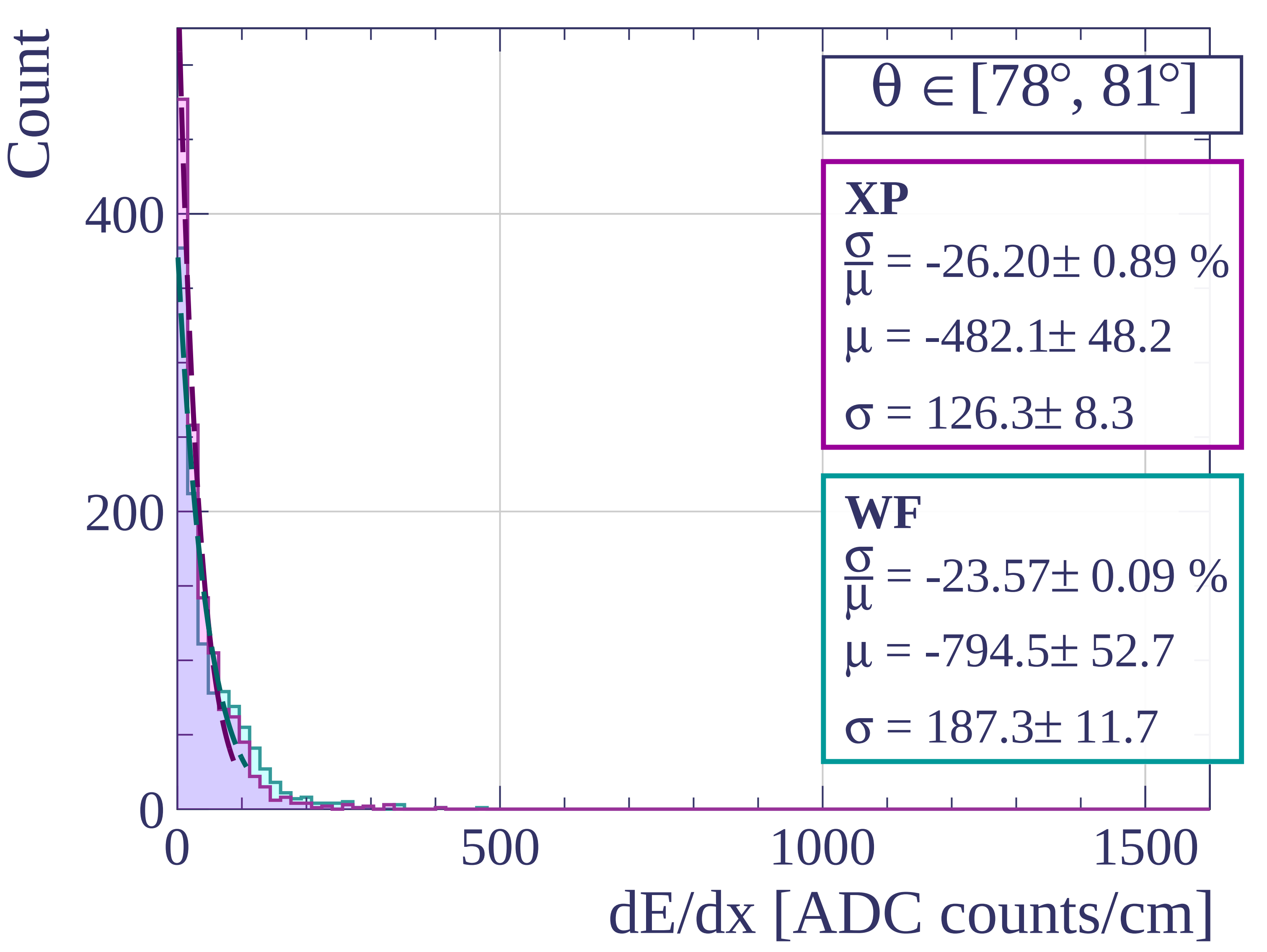
Count
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [69°, 72°]
XP
σ/μ = -149.33± 127.84 %
μ = -96.1± 112.9
σ = 143.6± 45.8
WF
σ/μ = -27.33± 1.92 %
μ = -2174.5± 351.1
σ = 594.3± 54.2

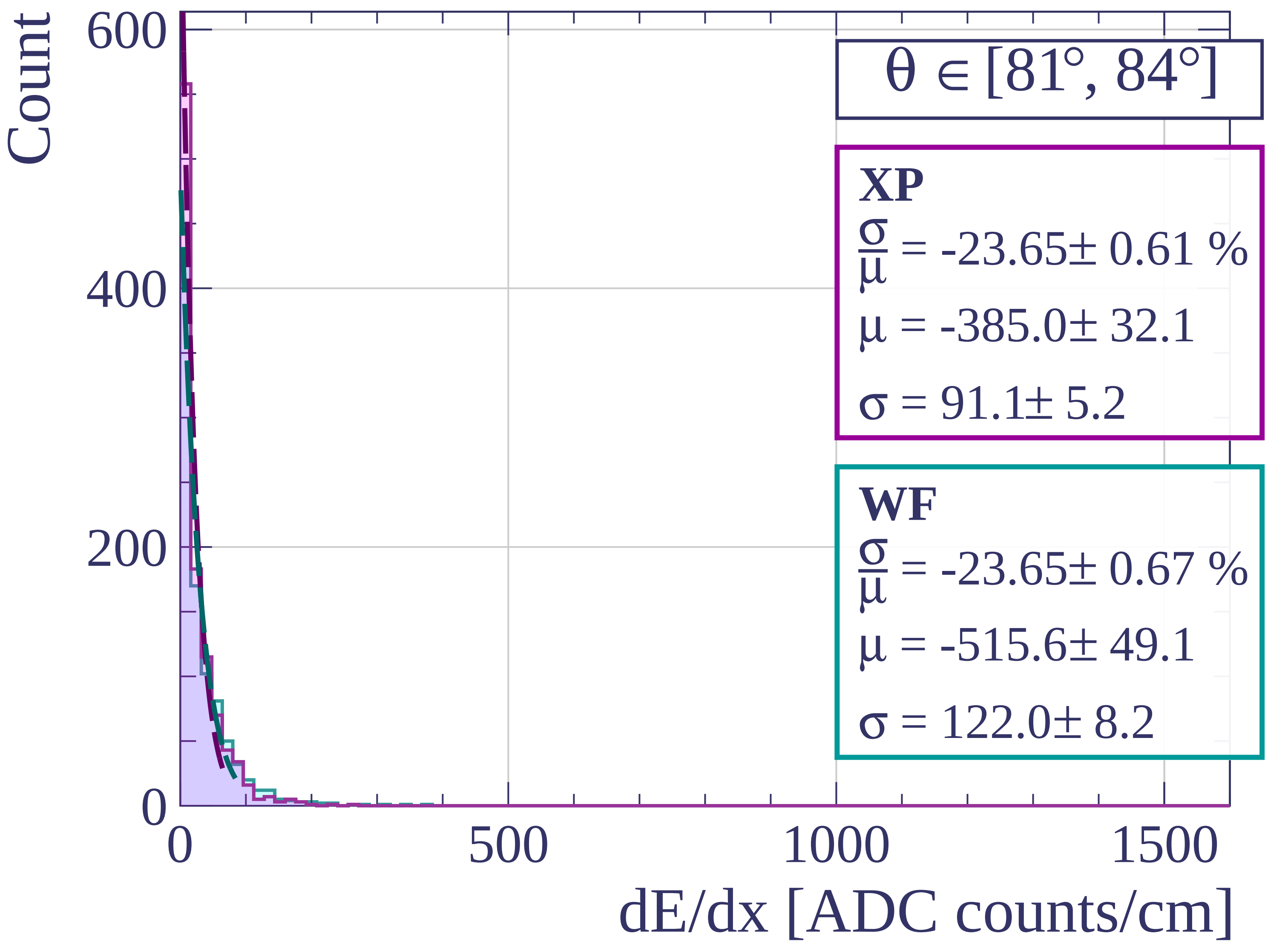


Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [72°, 75°]
XP
σ/μ = -28.79± 2.15 %
μ = -1077.7± 163.7
σ = 310.3± 24.0
WF
σ/μ = -23.85± 1.16 %
μ = -2084.9± 266.4
σ = 497.3± 39.4

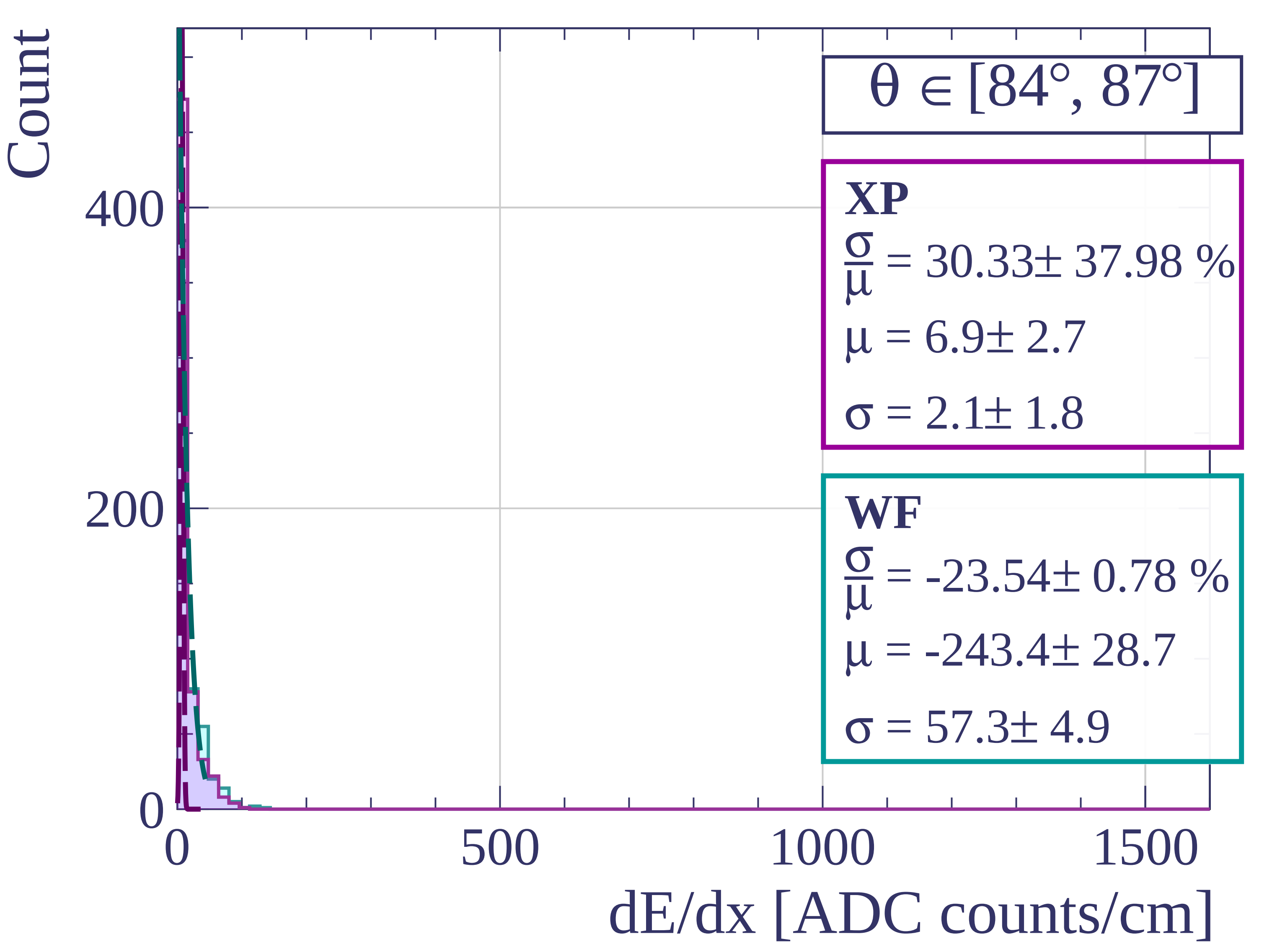


Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [75°, 78°]
XP
σ/μ = -24.37± 0.45 %
μ = -891.4± 76.2
σ = 217.2± 14.5
WF
σ/μ = -23.30± 0.51 %
μ = -1343.8± 129.4
σ = 313.0± 23.3

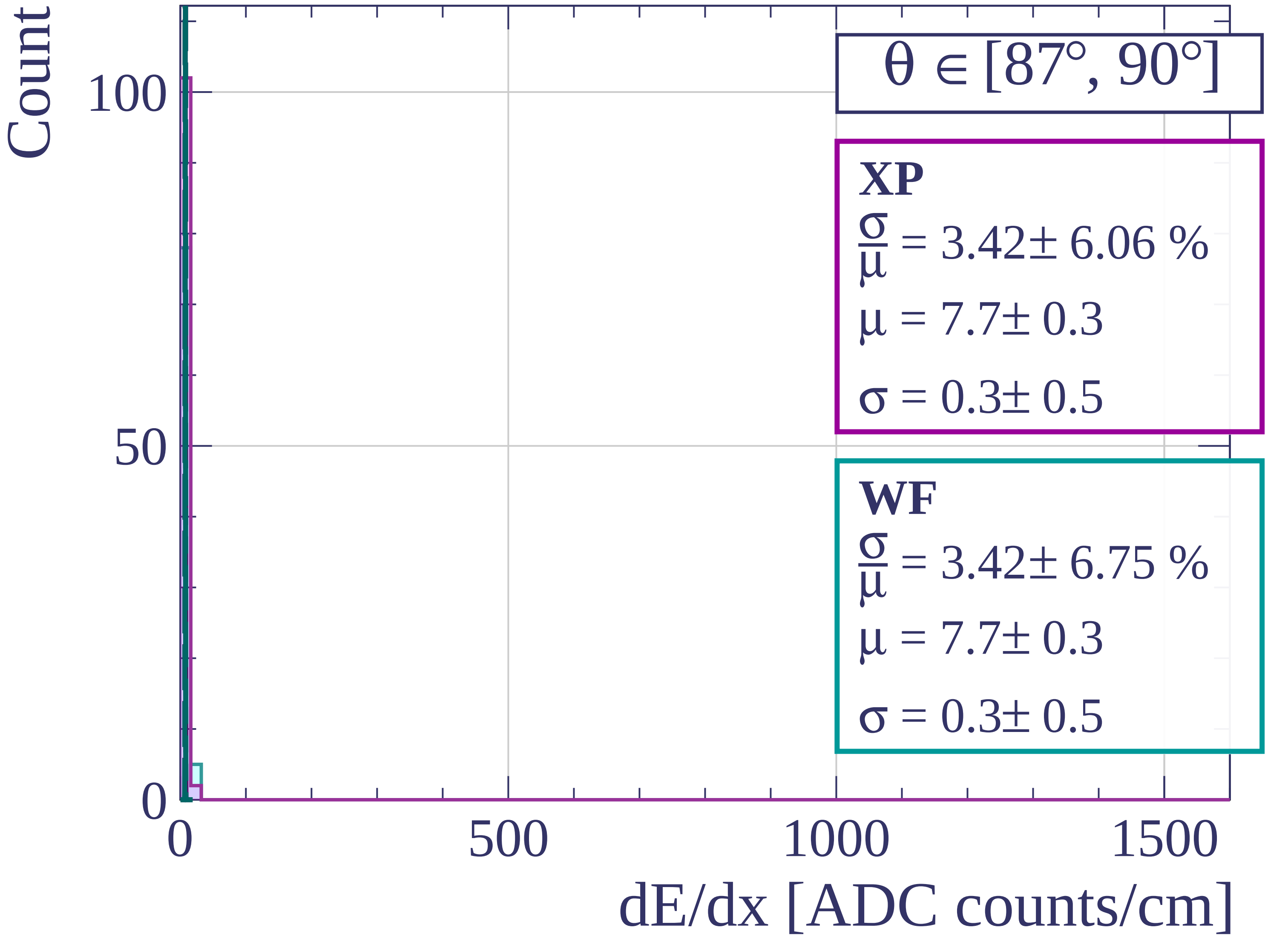

Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [78°, 81°]
XP
$\frac{\sigma}{\mu}$ = -26.20± 0.89 %
μ = -482.1± 48.2
σ = 126.3± 8.3
WF
$\frac{\sigma}{\mu}$ = -23.57± 0.09 %
μ = -794.5± 52.7
σ = 187.3± 11.7



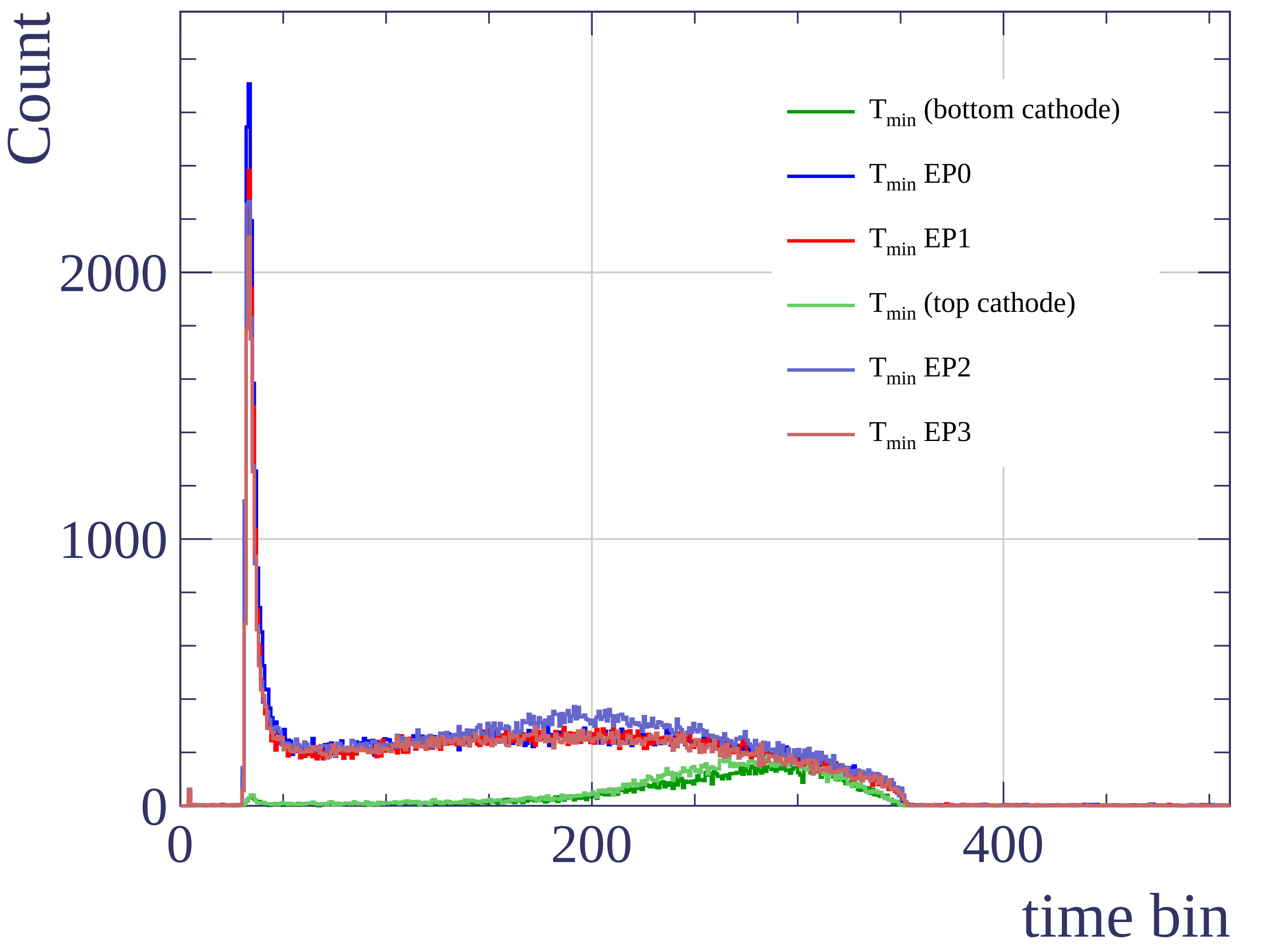
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [81°, 84°]
XP
σ/μ = -23.65± 0.61 %
μ = -385.0± 32.1
σ = 91.1± 5.2
WF
σ/μ = -23.65± 0.67 %
μ = -515.6± 49.1
σ = 122.0± 8.2



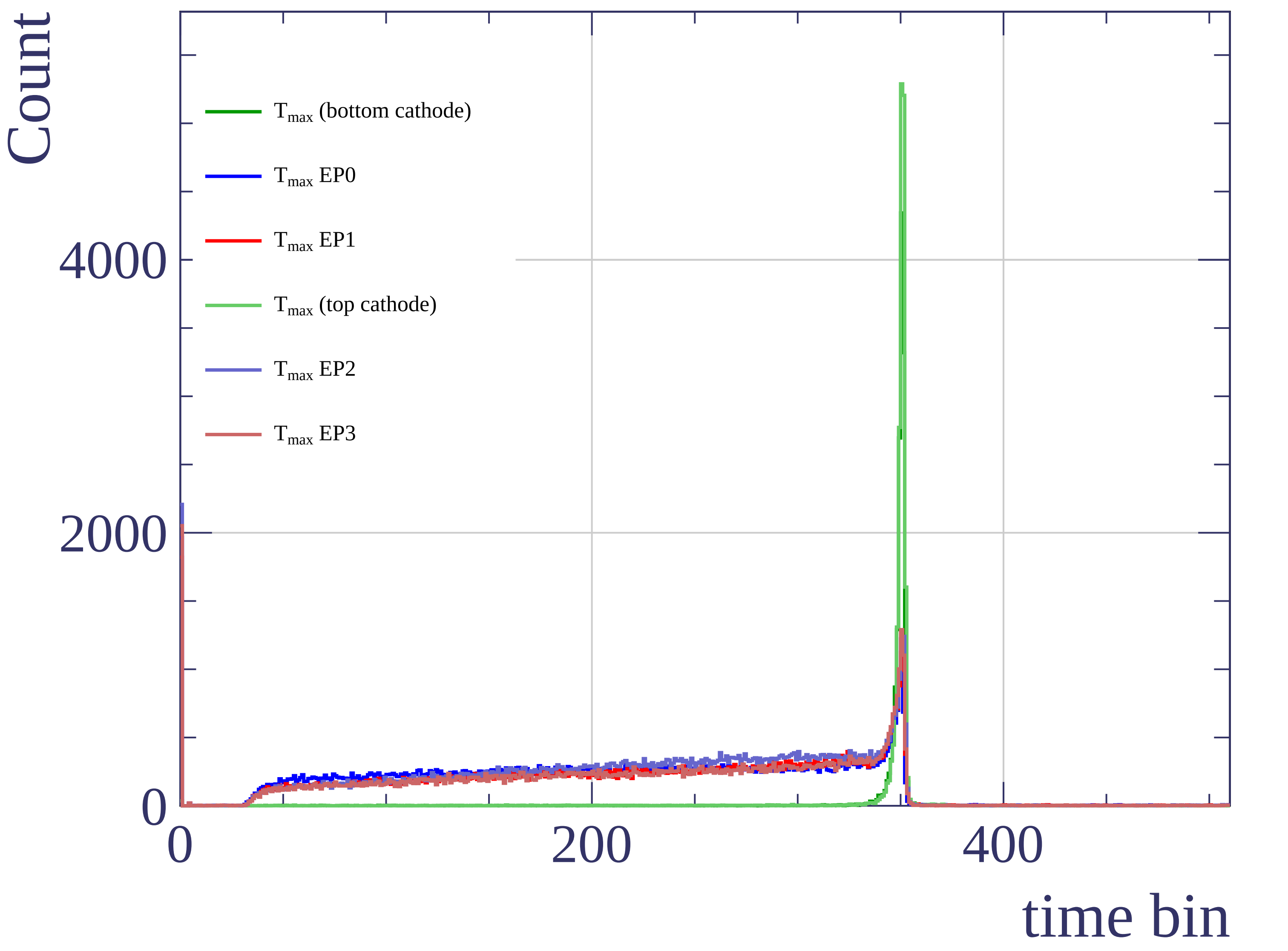
Count
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [84°, 87°]
XP
σ/μ = 30.33± 37.98 %
μ = 6.9± 2.7
σ = 2.1± 1.8
WF
σ/μ = -23.54± 0.78 %
μ = -243.4± 28.7
σ = 57.3± 4.9



Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
θ ∈ [87°, 90°]
XP
$\frac{\sigma}{\mu}$ = 3.42± 6.06 %
μ = 7.7± 0.3
σ = 0.3± 0.5
WF
$\frac{\sigma}{\mu}$ = 3.42± 6.75 %
μ = 7.7± 0.3
σ = 0.3± 0.5



Count
2000
1000
0
0
200
400
time bin
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3



Count
4000
2000
0
0
200
400
time bin
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3